Arithmetic Quantum Mechanics Lab Book

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Contents

1 Frontmatter, Status, and Scope

This lab book tracks a research programme investigating possible bridges between arithmetic zeta functions, arithmetic quantum mechanics, supersymmetric quantum mechanics, and topological lattice models such as Kitaev and Levin-Wen systems.

No mathematical or physical claim in this lab book is treated as established until it is marked as *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact.

1.1 Claim Status

The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record. They separate programme-level analogies from results.

1.2 Evidence Chain

Every substantive assertion should carry the provenance required by its status:

1. a *Question* or *Proposal* is allowed to name a gap, but must not be worded as an established result;
2. a *Source-local* claim must cite the relevant local source manifest and locator under `references/`;
3. a locally derived claim must point to the derivation and to any convention in `CONVENTIONS.md` on which the derivation depends;
4. a computational or generated-data claim must point to the producer, run bundle, schema entry, and `INDEX.md` row; and
5. a *Checked* claim must identify the local derivation, test, or run artifact that checks the invariant being used.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results.

2.1 Arithmetic And Zeta-Function Side

- Which zeta functions are in scope: local zeta functions over finite fields, Hasse-Weil zeta functions, motivic zeta functions, or other variants?
- Which Frobenius convention and cohomology theory will be used for trace formulas?
- Which finite examples are small enough for exact Sage or GAP checks?

2.2 Quantum-Mechanical Side

- What is meant locally by arithmetic quantum mechanics: a literal Hilbert-space model, a spectral dictionary, or a finite-dimensional approximation?
- Which supersymmetric-QM convention fixes the grading, supercharge, adjoint, Hamiltonian, and Witten-index normalization?
- Which invariants can be checked before proposing an analogy: nilpotence, positivity, trace identities, index stability, or spectral reconstruction?

2.3 Topological-Code Side

- Which lattice model is being used: toric code, quantum double, or Levin-Wen string-net?
- Which boundary conditions, lattice orientation, and star/plaquette conventions are fixed?
- Which category data are required: fusion rules, F-symbols, R-symbols, pivotal or spherical structure, and Drinfeld-center data?

2.4 First Decisions

The first technical session should not start with formulas. It should register sources and fix the minimum conventions needed to compare one finite arithmetic zeta example with one finite quantum or topological model invariant.

3 Lab Log

Chronology boundary: this shard is complete through the 2026-05-26 minimal-fermion entry below, which reaches Section 54; the master report now continues through AQM-91. For later status, use the shard text, report maps, `INDEX.md`, and the relevant source manifests until a log catch-up pass.

2026-05-24

Initial lab-book scaffold created.

- Established `report.tex` as the sharded LaTeX master.
- Added report source maps, convention and index files, source-manifest placeholders, run-bundle rules, and a tool runner.
- Recorded that no sources have been acquired and no scientific claims have been checked yet.

2026-05-24 Steane Clifford Ghost Gaussians

The report now includes sourced toric/CSS/Steane stabilizer-supercharge derivations. The latest addition is Section 14: arbitrary Clifford covariance is stated for signed Pauli stabilizer presentations, and presentation changes are lifted by number-preserving fermionic Gaussian operations on the ghosts. Generated evidence is registered through `scripts/lattice_codes/steane_all_clifford_ghost_gaussians.jl`, `runs/2026-05-24-steane-all-clifford-ghost-gaussians/`, and its CSVs.

2026-05-24 Arithmetic Quantum Fields

Section 15 adds the first explicit arithmetic-quantum-field proposal. A finite function space $E \leq \mathrm{Map}(X, V)$ receives a pointwise symplectic form; after quotienting its radical, its Weyl-Heisenberg displacements define the field observable algebra. Generated examples are registered through `scripts/bridges/arithmetic_quantum_fields.jl`, `runs/2026-05-24-arithmetic-quantum-fields/`, and AQM-14 CSV rows in `INDEX.md`.

2026-05-24 Projective Sheaf Fields

Sections 16 and 17 add the first projective version; Section 18 works out the rational-point stalks; Section 19 records the working dictionary "stalk equals local field germ, fiber equals sampled field value." Local Stacks Project TeX sources now ground the definitions of presheaf, sheaf, stalk, projective space, varieties, twists, and $H^0(\mathbf{P}^n, \mathcal{O}(d))$. Generated projective-line evidence is registered through `scripts/bridges/projective_line_sheaf_fields.jl`, `runs/2026-05-24-projective-line-sheaf-fi` and AQM-15–AQM-18 CSV rows in `INDEX.md`.

2026-05-25 Canonical Boson/Fermion Field Test

Section 20 adds a sourced finite-mode comparison with standard canonical fields. The bosonic test uses $\mathrm{Map}(X, \mathbb{R}_q \oplus \mathbb{R}_p)$ with the summed symplectic form and Stone–von Neumann uniqueness. The fermionic test uses CAR data on $\mathrm{Re}(Z \oplus \overline{Z})$ and the antisymmetric Fock space. The naive proposal that Grassmann numbers are a drop-in target replacement is rejected in this strict Weyl-projective sense; Grassmann variables remain a sourced GAR/fermionic phase-space calculus for later work.

Sections 21 and 22 correct the Grassmann displacement question using the active official arXiv source chain `cahill-glauber-density-fermions-1999` in `references/canonical_fields/SOURCES`.

md; its local TeX locator is `references/canonical_fields/cahill_glauber_physics_9808029v1_source/fxxx.tex`. The older APS PDF and `marker` artifacts are historical source-integrity records only, not active report locators. The abstract displacement symbols are represented by $D(\gamma) = \exp(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i))$ inside the Grassmann-extended finite CAR algebra. The derivation shard proves unitarity, the operator shifts $a_i \mapsto a_i + \gamma_i$, and the Grassmann-valued ray composition law, while retaining only the ordinary complex-Hilbert-space caveat as a boundary statement.

Section 23 returns to arithmetic quantum fields in the prime-field case. It separates $\operatorname{Spec} \mathbb{F}_p$, which has one point and gives one qudit, from the underlying set $\mathbb{F}_p$, which has p points. For $X = \mathbb{F}_p$ as a set and $E = \operatorname{Map}(X, \mathbb{F}_p^2)$, the shard proves that the arithmetic Weyl–Heisenberg group is exactly the standard odd-p qudit Pauli group, with the same projective labels and the $p = 2$ fourth-root phase caveat for qubits.

Section 24 then computes the first mixed finite affine spectrum, $R = \mathbb{Z}/6\mathbb{Z}$. The prime ideals are (2) and (3), with residue fields $\mathbb{F}_2$ and $\mathbb{F}_3$. Thus the local factors are a qubit and a qutrit. The label group is their orthogonal direct sum, but the Weyl representation lives on $\mathbb{C}^2 \otimes \mathbb{C}^3$, so the observable algebra is M_6, not $M_2 \oplus M_3$.

Section 25 adds the finite Zariski-open observable-net convention. The open-local assignment $U \mapsto \mathcal{A}(U)$ is proved covariant under inclusions of opens, while the closed-complement assignment $U \mapsto \mathcal{A}(X \setminus U)$ is recorded as a useful but contravariant vanishing-locus convention. The field edge case has only $\emptyset$ and X; for $\operatorname{Spec}(\mathbb{Z}/6\mathbb{Z})$, the singleton opens give the commuting local subalgebras $M_2 \otimes \mathbf{1}_3$ and $\mathbf{1}_2 \otimes M_3$, which generate M_6.

Section 26 records the directed-system choice needed for a quasi-local algebra. The primary construction is the filtered colimit over quasi-compact Zariski opens, $\varinjlim_U \mathcal{A}(U)$, with inclusions following open inclusion. The closed-complement construction is retained only after reindexing by closed supports ordered by inclusion, as a separate vanishing-locus or defect algebra.

Section 27 begins the presheaf/cosheaf layer. It defines presheaves, sheaves, precosheaves, and cosheaves from local Stacks and Curry sources. The open-local observable assignment is proved to be a precosheaf/local net, but not an ordinary cosheaf of unital $*$-algebras: for the $\mathbb{Z}/6\mathbb{Z}$ two-point cover, the cosheaf coproduct would be $M_2 *_{\mathbb{C}} M_3$, while the physical observable algebra is $M_2 \otimes M_3$. The Weyl label assignment $E(U) = \bigoplus_{x \in U} E_x$ is identified as the cosheaf-like layer.

Section 28 adds the state side. The local state assignment $U \mapsto \operatorname{State}(\mathcal{A}(U))$ is a presheaf of convex sets by restriction of functionals, equivalently by partial trace in finite tensor products. It is not generally a sheaf: the $\mathbb{Z}/6\mathbb{Z}$ two-point system gives two different global density matrices with the same local marginals, and a three-qubit overlap example shows that compatible marginals need not admit any global state. For the algebraic quasi-local algebra, global states are exactly compatible inverse limit families of local states.

Sections 29, 30, and 31 extend the arithmetic-field programme from finite spectra to affine schemes of finite type over $\mathbb{F}_q$. The new convention indexes finite Weyl/qudit sites by closed points, whose residue fields are finite by the Nullstellensatz. For $\operatorname{Spec} \mathbb{F}_q[t]$, closed points are monic irreducible polynomials and the point x_π carries a $q^{\deg \pi}$-level qudit. For $\operatorname{Spec} \mathbb{F}_q[x, y]$, rational points give a finite q^2-site truncation, but the full quasi-local algebra ranges over all closed points of all degrees. Generic and curve-generic points remain geometric/topological points rather than finite Weyl sites, and regular polynomial profiles are recorded as formal infinite-support fields rather than quasi-local operators.

Sections 32, 33, and 34 add the rational-function Weyl-Heisenberg layer. A new local source manifest registers LCA, locally compact ring, finite Frobenius-ring, finite Clifford/Weil, local-field, and adelic sources. The report now separates the algebraic $K = k(t)$ Heisenberg group, which has an explicit irreducible unitary Schrödinger representation on $\ell^2(K)$, from the full LCA Stone-von

Neumann setting $K_{\mathrm{disc}} \times \widehat{K_{\mathrm{disc}}}$ and from local or adelic completions. The $k = \mathbb{F}_3$ examples compute concrete ω-commutation relations for rational-function labels.

Sections 35 and 36 return to concrete finite affine examples. For $R_p = \mathbb{F}_3[t]/(p)$, the report computes the spectrum from the irreducible factorization of p, proves the Chinese-remainder and reduced-quotient descriptions, and gives the complete residue-field Weyl-Heisenberg representation on $\bigotimes_\pi \ell^2(\mathbb{F}_3[t]/(\pi))$. The $t^n = 1$ shard records the characteristic-3 formula $t^n - 1 = (t^m - 1)^{3^{v_3(n)}}$, the cyclotomic degree count via $\mathrm{ord}_d(3)$, and the resulting dimension $\dim \mathcal{H} = 3^m$, with nilpotent multiplicities retained as coordinate-ring data but invisible to the present residue-qudit labels.

Sections 37 and 38 add the nilpotent-sensitive layer. For a local factor $A = \mathbb{F}_3[t]/(\pi^e)$, the report defines the top-coefficient character, proves it is generating, and constructs the projective unitary Weyl representation on $\ell^2(A)$. The Weyl operators span the full matrix algebra, so the representation is irreducible. The $\mathfrak{m}$-adic filtration is interpreted as a jet filtration: position jets of order r pair with momentum jets of order $e-1-r$. For $t^n = 1$, this thickened layer has dimension 3^n, while the reduced residue layer has dimension 3^m for $m = n/3^{v_3(n)}$.

Sections 39 and 40 add the stabilizer local-to-global layer. A phased Lagrangian label subspace is proved to give a rank-one full stabilizer state, while vanishing stabilizer descent defect is the separate condition that the global stabilizer labels are generated from a chosen cover. The two-qutrit example over $\mathbb{F}_3$, generated by $(1, 0, -1, 0)$ and $(0, 1, 0, 1)$, is Lagrangian but has zero intersection with both one-site label spaces, so its descent defect has dimension 2. Thus the broad equivalence between choosing a full stabilizer and satisfying local-to-global gluing is rejected; the corrected statement requires both a global Lagrangian and zero descent defect, plus phase compatibility on overlaps.

Section 41 treats the missing finite product-field example. For $R = k^n$, with $k = \mathbb{F}_q$, the spectrum has n points $\mathfrak{m}_i = \ker(\mathrm{pr}_i)$, each with residue field k, and the idempotents e_i make the topology discrete. Hence the residue-qudit field on $X = \operatorname{Spec} R$ has

$$E(X) = k_X^n \oplus k_Z^n, \qquad \mathcal{H}(X) = \ell^2(k)^{\otimes n}, \qquad \mathcal{A}(X) \cong M_{q^n}(\mathbb{C}).$$

Thus $X = \mathrm{Spec}(k^n)$ gives exactly the ambient n q-level qudit Weyl/Pauli net. Stabilizer codes remain additional data: isotropic additive label subgroups and compatible phases, with the k-linear stabilizers as a subclass when q is not prime.

Section 42 starts the Cech layer. A new Stacks Cohomology source is registered for the standard Cech complex, H^0 gluing, refinements, and ordered Cech cochains, with Curry supplying the dual cosheaf homology convention. The report proves that the full residue-label cosheaf $E(U) = \bigoplus_{x\in U} E_x$ has $\check{H}_0(\mathcal{U}, E) \cong E(U)$ and no higher Cech homology on finite covers, by decomposing pointwise into full simplices. The quantum counterparts are then separated by layer: stabilizer descent is a genuine abelian cokernel of zeroth Cech generation, state descent is the marginal comparison map, and observable descent is the comparison from ordinary algebraic cover colimits to the physical local net.

Section 43 starts the dynamics layer. Gross's arXiv TeX source for finite-dimensional Hudson theory is registered locally. The report fixes the odd-characteristic half-Weyl normalization, then proves that every symplectic morphism of nondegenerate finite label spaces induces a unital $*$-monomorphism of Weyl observable algebras, and that symplectic automorphisms induce Gaussian/quasi-free automorphisms. Gross's Clifford theorem supplies projective unitary implementations, while Gross-Hudson identifies pure finite Gaussian states with stabilizer states.

Section 44 starts the finite-field scale or continuum-limit layer. For $X_0/\mathbb{F}_q$, the rank-r scale is $X_r = X_0 \times_{\mathbb{F}_q} \mathbb{F}_{q^r}$. The report proves that base change of schemes is canonical but naive diagonal Weyl-label embeddings multiply absolute-trace commutators by s/r. Trace-normalized embeddings repair this and are canonical on the prime-to-p subtower. For $\operatorname{Spec} \mathbb{F}_3$, the current canonical continuum algebra is the prime-to-3 filtered colimit of $M_{3^r}(\mathbb{C})$, and Frobenius acts by $W(q, p) \mapsto W(q^3, p^3)$.

Section 45 proves that the remaining full-tower trace choices are coherent gauge data rather than an obstruction. The key object is a trace-density system $a_r \in \mathbb{F}_{q^r}$ satisfying $\mathrm{Tr}_{\mathbb{F}_{q^s}/\mathbb{F}_{q^r}}(a_s) = a_r$ for every $r \mid s$. Such systems exist by a factorial-tower trace-surjectivity construction, and they define coherent normalizers $c_{s,r} = a_s/a_r$. Frobenius acts on this space of gauges; for $p \mid s/r$, no trace-one scalar fixed by the relative Frobenius group exists.

Section 46 builds the first arithmetic base-space field over $\mathrm{Spec}\,\mathbb{Z}$. The closed-prime layer has one ℓ-level qudit at each rational prime and an algebraic infinite tensor product over finite prime supports. The generic point is separated as a $\mathbb{Q}$-Weyl sector rather than a finite qudit. The report rejects $z \mapsto z+1$ as a scheme automorphism of $\mathrm{Spec}\,\mathbb{Z}$, then defines the internal residue-translation action $n \mapsto \alpha_n$ on the prime-qudit algebra, proves its profinite local periods, and shows that α_n for $n \neq 0$ is not implemented in the $|0\rangle$ incomplete tensor sector.

2026-05-26 Affine Line and Regular Functions

Sections 47–49 prove the base-preserving symmetry group $x \mapsto ax+b$, build its closed-point Weyl-net action, and separate the full $\mathbb{F}_3$ net from rational finite-support truncations. Sections 50–53 then identify regular functions as residue profiles, finite equations h as support selectors, and $W_h^{\mathrm{red}}(F,G)$ as the actual finite Weyl smearing. The affine action is $g_\# F = F \circ g^{-1}$, it transports reduced smearings by $\alpha_g W_h^{\mathrm{red}}(F,G) = W_{g_\# h}^{\mathrm{red}}(g_\# F, g_\# G)$, and repeated factors remain Artin-ring thickness rather than extra closed points.

2026-05-26 Minimal Fermion Catalogue

Section 54 registers sources for supergeometry, spin structures, and logarithmic forms, then fixes the first fermion catalogue. Ordinary $\mathrm{Spec}\,k$ has no forced fermion mode; the minimal sourced entries are an odd superpoint, $\Pi\Omega^1$ of an even dual-number point, and a log-spin curve such as $(\mathbf{P}^1_{\mathbb{F}_3}, \{0,\infty\}, \mathcal{O})$.

4 Reproducibility Map

Generated artifacts live in run bundles:

```
runs/<YYYY-MM-DD>-<slug>/
  README.md
  data/
  figures/
```

Each producer must be listed in `scripts/run_all.jl` and `INDEX.md`. Each CSV must be documented in `data/SCHEMA.md`. Each report claim that uses generated data must point back to the producer and run bundle.

4.1 Current Validation Commands

```
make check-report-shards
make report
julia --project=. -e 'using Pkg; Pkg.test()'
julia --project=. scripts/run_all.jl --fast
```

The registered producers are indexed in `INDEX.md`; the canonical local driver is `scripts/run_all.jl`.

5 Source Queue and Open Questions

5.1 Source Queue

Registered sources now cover toric-code stabilizers, symplectic QECC conventions, finite Hudson/Gaussian stabilizer theory, canonical CCR and CAR field conventions, Fock spaces, Weyl-Heisenberg and Weil representations over LCA groups, locally compact rings, finite Frobenius rings, finite Heisenberg groups, and local/adelic fields, and the algebraic-geometry definitions used for spectra, Zariski topology, affine-scheme functoriality, sheaves, cosheaves, Cech cohomology, projective space, separable trace pairing, and absolute/relative Frobenius. Before adding technical content in the remaining directions, register sources for:

- Weil conjectures, zeta functions, Frobenius traces, and cohomological normalization.
- Arithmetic quantum mechanics and spectral interpretations of arithmetic data.
- Adelic self-dual models, Bost–Connes-style arithmetic C^*-dynamical systems, and any zeta-producing time evolution proposed for the $\operatorname{Spec}\mathbb{Z}$ prime-qudit algebra.
- Supersymmetric quantum mechanics, Witten index, and Hodge/de Rham model conventions.
- Levin-Wen string-net models and categorical input data beyond the toric stabilizer/CSS layer.

5.2 Open Questions

1. What is the smallest finite arithmetic example whose zeta data can be computed exactly and traced through the proposed dictionary?
2. Which quantum-mechanical invariant is the first honest comparator: trace, index, spectrum, ground-state degeneracy, or projector rank?
3. Which parts of the comparison are literal constructions and which parts are analogy or bookkeeping?
4. What is the correct generic rational-function sector for $\mathbf{A}^1_{\mathbb{F}_q}$: a discrete $k(x)$ Weyl system with a dual-label affine action, a local-field completion at infinity, or an adelic self-dual model?

6 Toric Code Supercharge From Local Checks

6.1 Question

Can one formulate the Kitaev toric code concretely as

$$H = QQ^* + Q^*Q$$

for a supercharge Q, and can the code space be realized by the elementary "Q-cohomology" construction?

The answer recorded here is:

> Yes, after adding one auxiliary fermion, or ghost, for each local stabilizer check. This gives a local-check supersymmetric package. It is not yet a purely physical-qubit supercharge.

The toric-code stabilizer convention is sourced from the local TeX file `references/toric_code/kitaev_quant_ph_9707021_source/anyons.tex`. Lines 189–222 define qubits on edges, star operators A_s, plaquette operators B_p, and the protected subspace; lines 328–337 define Kitaev's Hamiltonian and state that its ground state is the protected subspace.

6.2 Elementary Construction

Let V be the physical Hilbert space of the toric-code qubits. For every star or plaquette stabilizer S_i, define the violated-check projector

$$P_i = \frac{1 - S_i}{2}.$$

Thus $P_i v = 0$ means that the state v satisfies check i, while $P_i v = v$ means that this component violates check i. The shifted positive toric-code Hamiltonian is

$$H_{\mathrm{TC}} = \sum_i P_i.$$

Now add one auxiliary fermion for each check. Concretely, this means we allow formal labels e_i, one per check, and impose the usual rule that swapping two labels changes the sign. Let c_i^* be the operation "attach the label e_i", and let c_i be the operation "remove the label e_i", with zero result if the label is absent. These operations satisfy

$$c_i c_j^* + c_j^* c_i = \delta_{ij}, \qquad c_i c_j + c_j c_i = 0, \qquad c_i^* c_j^* + c_j^* c_i^* = 0.$$

Define the cochain supercharge

$$Q = \sum_i c_i^* P_i.$$

This operator sends a physical state to the formal list of its violated local checks. Because the projectors P_i commute, and because two fermion creations anticommute, $Q^2 = 0$. The adjoint $Q^* = \sum_i c_i P_i$ removes one ghost label. The cross-terms in $QQ^* + Q^*Q$ cancel in pairs, again using commutation of the projectors and anticommutation of the fermions, so

$$QQ^* + Q^*Q = \sum_i P_i = H_{\mathrm{TC}}$$

on the physical ghost-vacuum sector, and $H_{\mathrm{TC}} \otimes I$ on the full ghost-extended space.

6.3 What The Cohomology Means

Here is the promised elementary definition. Degree zero means "no ghost label attached", so a degree-zero state is just a physical toric-code state $v \in V$. The degree-zero Q-cohomology is:

- closed degree-zero states: those v with $Qv = 0$;
- exact degree-zero states: those of the form Qw where w has degree -1.

There are no degree -1 states in this construction, so no nonzero exact degree-zero states. Also, $Qv = \sum_i e_i \otimes P_i v$, and the one-ghost labels e_i are independent. Therefore

$$Qv = 0 \iff P_i v = 0 \text{ for every check } i \iff S_i v = v \text{ for every check } i.$$

Thus the degree-zero Q-cohomology is exactly the toric-code code space. The qualifier "degree-zero" is essential: on a code state all P_i vanish, so the full ghost-extended cohomology also contains ghost-labelled copies of the code space in higher degrees. The physical code is the no-ghost part.

Equivalently, using the boundary operator $Q^* = \sum_i c_i P_i$, degree-zero homology is the quotient of all physical states by the span of states with at least one local-check violation. Since the P_i are commuting orthogonal projectors, this quotient has the protected subspace as its canonical representative.

6.4 Algebraic Validation

The validation script is `scripts/lattice_codes/toric_supercharge_validation.jl`. It checks the stabilizer data algebraically, not by building the full physical or ghost Hilbert matrices. For the $k = 4$ torus it verifies:

- $n = 2k^2 = 32$ physical qubits and 32 local checks;
- each check has weight 4;
- the binary symplectic commutator of every pair of checks is zero;
- the stabilizer rank is $2k^2 - 2 = 30$;
- the code dimension is $2^{32-30} = 4$;
- the stored `q_square_certificate` and `anticommutator_certificate` booleans are true because the checked commuting-projector and involutive-check hypotheses needed by the CAR/projector proof hold;
- the degree-zero Q-cohomology dimension is 4.

The generated CSV is `runs/2026-05-24-toric-supercharge/data/toric_supercharge_summary.csv`. Its first row is a sentinel noting that no full physical or ghost Hilbert matrices are built. Thus the run records the checked hypotheses and summary booleans used by the formal proof above; it is not a stored matrix-level certificate for $Q^2 = 0$ or the anticommutator.

6.5 Boundary Maps and the Same Construction

There is a second, more topological description of the toric code. It starts from the cellular chain complex over $\mathbb{F}_2$

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0,$$

where C_2 is spanned by faces, C_1 by edges, and C_0 by vertices. The local Bombin–Martin-Delgado TeX source listed in `references/toric_code/SOURCES.md` defines these maps for 2-complexes in lines 2050–2140 and records $\partial^2 = 0$. The local QECC Zoo source in the same manifest records the toric-code family as a CSS code with star and plaquette stabilizers in lines 20–39.

The cellular description does suggest a different supercharge first. Define the cochain maps

$$\delta^0 = \partial_1^T : C^0 \to C^1, \qquad \delta^1 = \partial_2^T : C^1 \to C^2.$$

On the finite graded vector space $C^0 \oplus C^1 \oplus C^2$, the cellular supercharge

$$Q_{\text{cell}} = \delta^0 + \delta^1$$

is nilpotent because

$$Q_{\text{cell}}^2 = \delta^1\delta^0 = (\partial_1\partial_2)^T = 0.$$

Its middle cohomology means the following elementary quotient: take all edge bit strings $a \in C^1$ with $\delta^1 a = 0$, and identify two such strings when they differ by $\delta^0 b$ for some vertex bit string $b \in C^0$. In symbols,

$$H^1_{\text{cell}} = \ker(\delta^1)/\operatorname{im}(\delta^0).$$

The quantum CSS unification is then:

$$H_X = \partial_1, \qquad H_Z = \partial_2^T.$$

The rows of H_X are the vertex-star supports for the X-checks. The rows of H_Z, equivalently the columns of ∂_2, are the face-boundary supports for the Z-checks. The CSS commutation condition is not extra:

$$H_X H_Z^T = \partial_1\partial_2 = 0.$$

The precise chain-to-ghost statement is proved in Section 7. It gives an explicit map from cellular cohomology classes to toric-code states and derives the ghost anticommutator from the CAR relations and the commuting stabilizer projectors. The operator-level comparison of Q_{cell} and Q_{gh} is in Section 8.

Concretely, the boundary map $\partial_2 : C_2 \to C_1$ and $\partial_1 : C_1 \to C_0$ act on classical $\mathbb{F}_2$-chains: formal sums of faces, edges, and vertices. They classify error strings, check supports, and logical operators. The boundary data determine the stabilizer checks S_i, and those checks determine the projectors $P_i = (1 - S_i)/2$ used in Q.

The chain complex also explains the number of encoded qubits. The first homology group is

$$H_1 = \ker(\partial_1)/\operatorname{im}(\partial_2).$$

For the torus this has dimension 2 over $\mathbb{F}_2$, so the code space has dimension $2^2 = 4$. In the Z-basis one sees the dual statement: the plaquette constraints select 1-cocycles, and the star operators identify cocycles differing by a coboundary. Hence the code basis is naturally labelled by H^1, while the logical string operators are labelled by the paired H_1/H^1 data.

The validation script is `scripts/lattice_codes/toric_chain_ghost_unification.jl`. For $k = 4$, it verifies $\partial_1 \partial_2 = 0$, checks that rows of ∂_1 match the star supports and columns of ∂_2 match the plaquette supports, and computes

$$\operatorname{rank} \partial_1 = 15, \quad \operatorname{rank} \partial_2 = 15, \quad \dim H_1 = 32 - 15 - 15 = 2.$$

It also records that $Q_{\mathrm{cell}}^2 = 0$, that the middle cellular cohomology has 2 binary generators, that the Z-basis cocycle space has $2^{17} = 131072$ elements, and that quotienting by the $2^{15} = 32768$ star-generated coboundaries leaves 4 classes. The run output is `runs/2026-05-24-toric-chain-ghost-unification/data/toric_chain_ghost_unification.csv`.

7 Proof of the Toric Chain-to-Ghost Lift

This section replaces the slogan "the chain complex supplies the supports" by an explicit finite-dimensional statement.

7.1 Data

Fix the $k \times k$ periodic square cellulation of the torus, $k \geq 2$. All chain groups in this subsection are vector spaces over $\mathbb{F}_2$:

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0.$$

The bases are faces, edges, and vertices. Thus

$$|C_0| = k^2, \qquad |C_1| = 2k^2, \qquad |C_2| = k^2.$$

Let

$$\delta^0 = \partial_1^T : C^0 \to C^1, \qquad \delta^1 = \partial_2^T : C^1 \to C^2.$$

The cellular supercharge is the degree $+1$ map

$$Q_{\text{cell}} = \delta^0 + \delta^1 \quad \text{on} \quad C^0 \oplus C^1 \oplus C^2.$$

Since $\partial_1 \partial_2 = 0$, one has

$$Q_{\text{cell}}^2 = \delta^1 \delta^0 = (\partial_1 \partial_2)^T = 0.$$

The degree-one cellular cohomology is the quotient

$$H^1_{\text{cell}} = \ker(\delta^1)/\operatorname{im}(\delta^0).$$

In elementary terms, its elements are edge-bit strings a whose parity around every plaquette is zero, with two such strings identified if they differ by the edge-bit string produced by applying star flips at some set of vertices.

7.2 From Boundary Matrices to Pauli Checks

The physical Hilbert space is

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes C_1}.$$

Its computational Z-basis is written $\{|a\rangle : a \in C^1\}$, where $a_e \in \mathbb{F}_2$ is the bit on edge e.

For an edge-bit string $r \in C^1$, define Pauli monomials by

$$X^r|a\rangle = |a + r\rangle, \qquad Z^r|a\rangle = (-1)^{r \cdot a}|a\rangle,$$

where $r \cdot a = \sum_e r_e a_e$ is computed in $\mathbb{F}_2$. They obey

$$X^r Z^s = (-1)^{r \cdot s} Z^s X^r.$$

For a vertex v, let $r_v = \delta^0 v \in C^1$, the row of ∂_1 supported on edges incident to v. Define the star check

$$A_v = X^{r_v}.$$

For a face f, let $s_f = \partial_2 f \in C_1$, identified with its edge-bit string in C^1. Define the plaquette check

$$B_f = Z^{s_f}.$$

Then

$$A_v B_f = (-1)^{r_v \cdot s_f} B_f A_v.$$

But $r_v \cdot s_f$ is exactly the coefficient of v in $\partial_1 \partial_2 f$, hence it is zero. Therefore every A_v commutes with every B_f. Checks of the same Pauli type commute automatically, and each check squares to the identity.

7.3 The Code Space as a Linearized Cohomology Set

The toric-code space is

$$\mathcal{C} = \{|\psi\rangle \in \mathcal{H} : A_v|\psi\rangle = |\psi\rangle,\ B_f|\psi\rangle = |\psi\rangle \text{ for all } v, f\}.$$

We now construct a basis of $\mathcal{C}$ from $\ker(\delta^1)/\operatorname{im}(\delta^0)$.

First, for a basis vector $|a\rangle$,

$$B_f|a\rangle = (-1)^{s_f \cdot a}|a\rangle.$$

The condition $B_f|a\rangle = |a\rangle$ for every f is therefore

$$s_f \cdot a = 0 \quad \text{for all } f,$$

which is precisely $\delta^1 a = 0$. Thus the simultaneous $+1$ eigenspace of all plaquettes is spanned by $|a\rangle$ with $a \in \ker(\delta^1)$.

Second, a star acts by

$$A_v|a\rangle = |a + \delta^0 v\rangle.$$

Thus the star group translates a by elements of $\operatorname{im}(\delta^0)$. Because $\delta^1\delta^0 = 0$, these translations preserve $\ker(\delta^1)$.

For each class $[a] \in H^1_{\text{cell}}$, define

$$|[a]\rangle = \frac{1}{\sqrt{|\operatorname{im}\delta^0|}} \sum_{u \in \operatorname{im}(\delta^0)} |a + u\rangle.$$

This vector is well-defined: replacing a by $a + u_0$ just reindexes the sum. It is fixed by every star A_v, because A_v translates the summation set by $\delta^0 v$. It is fixed by every plaquette B_f, because every $a + u$ in the sum lies in $\ker(\delta^1)$. Hence $|[a]\rangle \in \mathcal{C}$.

If $[a] \neq [b]$, the two sums have disjoint computational-basis supports, so $|[a]\rangle$ and $|[b]\rangle$ are orthogonal. Conversely, any vector in the plaquette $+1$ eigenspace is a linear combination of basis states with $a \in \ker(\delta^1)$, and imposing all star equations forces its coefficients to be constant on each coset of $\operatorname{im}(\delta^0)$. Therefore the vectors $|[a]\rangle$ form a basis of $\mathcal{C}$, and there is a concrete isomorphism

$$\mathbb{C}[H^1_{\text{cell}}] \cong \mathcal{C}, \qquad [a] \mapsto |[a]\rangle.$$

For the periodic square torus this gives four states. Indeed, suppose $\sum_v \alpha_v(\text{row } v \text{ of } \partial_1) = 0$. For each edge with endpoints v, w, the coefficient of that edge gives $\alpha_v + \alpha_w = 0$. The graph is connected, so all α_v are equal. Thus the only row relation is the sum of all rows, and $\operatorname{rank}\partial_1 = k^2 - 1$.

Similarly, suppose $\sum_f \beta_f \partial_2 f = 0$. For each edge separating adjacent faces f, g, the coefficient of that edge gives $\beta_f + \beta_g = 0$. The dual graph is connected, so all β_f are equal. Thus the only column relation is the sum of all face boundaries, and $\operatorname{rank}\partial_2 = k^2 - 1$. Hence

$$\dim_{\mathbb{F}_2} H^1_{\text{cell}} = 2k^2 - (k^2 - 1) - (k^2 - 1) = 2.$$

Thus $\dim_{\mathbb{C}} \mathcal{C} = |H^1_{\text{cell}}| = 2^2 = 4$.

7.4 The Ghost Supercharge Is the Koszul Lift of These Checks

Let the list of stabilizer checks be

$$\{S_i\}_{i\in I} = \{A_v : v \in C_0\} \cup \{B_f : f \in C_2\}.$$

The previous subsection proved that all S_i commute and that $S_i^2 = I$. Define the violated-check projectors

$$P_i = \frac{I - S_i}{2}.$$

They are self-adjoint idempotents, and they commute with each other.

Now attach one auxiliary fermion to each $i \in I$. Let ϵ_i create the i-ghost and ι_i annihilate it. These operations act on the exterior algebra on the ghost labels and satisfy

$$\iota_i\epsilon_j + \epsilon_j\iota_i = \delta_{ij}, \qquad \epsilon_i\epsilon_j + \epsilon_j\epsilon_i = 0, \qquad \iota_i\iota_j + \iota_j\iota_i = 0.$$

They commute with the physical projectors P_i. Define

$$Q_{\mathrm{gh}} = \sum_{i\in I} \epsilon_i P_i, \qquad Q^*_{\mathrm{gh}} = \sum_{i\in I} \iota_i P_i.$$

Nilpotence is immediate but worth writing out:

$$Q^2_{\mathrm{gh}} = \sum_{i,j} \epsilon_i\epsilon_j P_i P_j = \sum_{i<j} (\epsilon_i\epsilon_j + \epsilon_j\epsilon_i) P_i P_j = 0.$$

For the anticommutator,

$$\begin{aligned} Q_{\mathrm{gh}}Q^*_{\mathrm{gh}} + Q^*_{\mathrm{gh}}Q_{\mathrm{gh}} &= \sum_{i,j} (\epsilon_i\iota_j + \iota_j\epsilon_i) P_i P_j \\ &= \sum_i P_i^2 = \sum_i P_i. \end{aligned}$$

This is the shifted toric-code Hamiltonian on $\mathcal{H}$, tensored with the identity on the ghost exterior algebra.

Finally, degree-zero means "no ghost present". A degree-zero vector $|\psi\rangle \in \mathcal{H}$ is Q_{gh}-closed exactly when

$$Q_{\mathrm{gh}}|\psi\rangle = \sum_i \epsilon_i P_i |\psi\rangle = 0,$$

which, by independence of the one-ghost labels, is equivalent to $P_i|\psi\rangle = 0$ for every i. This is equivalent to $S_i|\psi\rangle = |\psi\rangle$ for every star and plaquette. There are no degree -1 states, so no degree-zero exact states. Therefore

$$H^0(Q_{\mathrm{gh}}) = \mathcal{C} \cong \mathbb{C}[H^1_{\mathrm{cell}}].$$

This is the concrete unification. Q_{cell} is the cellular differential on finite $\mathbb{F}_2$-cochains. Q_{gh} is a complex-linear operator on the physical Hilbert space tensored with ghosts. The first one supplies the quotient set H^1_{cell}; the second one has the toric-code Hamiltonian as its anticommutator and selects the complex vector space freely spanned by that quotient. The sharper operator-level relation is derived in Section 8.

8 Operator Relation Between Cellular and Ghost Supercharges

The honest operator-level answer is: Q_{cell} is not the restriction of Q_{gh} to a low-ghost-number subspace of the check-ghost Fock space. The precise relation is induced, not restrictive: the cellular maps δ^0, δ^1 define translations and syndrome characters on the physical edge-bit Hilbert space, and Q_{gh} is the Koszul differential for the corresponding commuting projector equations.

8.1 Why It Is Not a Low-Ghost Restriction

Let

$$E = C^1$$

be the $\mathbb{F}_2$-vector space of edge-bit strings, and let

$$\mathcal{H} = \mathbb{C}[E]$$

be the complex vector space with basis $\{|a\rangle : a \in E\}$. The check-ghost space used by Q_{gh} is

$$G_{\text{check}} = \mathbb{C}\{\eta_v : v \in C_0\} \oplus \mathbb{C}\{\theta_f : f \in C_2\}.$$

The full ghost-extended space is

$$\mathcal{H}_{\text{gh}} = \mathcal{H} \otimes \Lambda(G_{\text{check}}).$$

Its low ghost-number pieces are therefore

$$\mathcal{H} \otimes \Lambda^0 G_{\text{check}} = \mathcal{H},$$

$$\mathcal{H} \otimes \Lambda^1 G_{\text{check}} = \mathcal{H} \otimes \big(\mathbb{C}\{v\text{-ghosts}\} \oplus \mathbb{C}\{f\text{-ghosts}\}\big).$$

There is no edge one-particle sector here. But the cellular differential has the form

$$Q_{\text{cell}} : \ C^0 \to C^1 \to C^2.$$

Its middle degree is the edge space C^1. In contrast,

$$Q_{\text{gh}} : \ \mathcal{H} \otimes \Lambda^r G_{\text{check}} \longrightarrow \mathcal{H} \otimes \Lambda^{r+1} G_{\text{check}}$$

adds a vertex-check or face-check ghost and keeps the same physical Hilbert space. On ghost degree zero it maps

$$\mathcal{H} \longrightarrow \mathcal{H} \otimes (\mathbb{C}\{v\text{-ghosts}\} \oplus \mathbb{C}\{f\text{-ghosts}\}),$$

not $C^0 \to C^1$. On ghost degree one it maps to two-check ghosts, not to faces. Thus Q_{cell} is not literally Q_{gh} on any canonical low-ghost subspace of $\mathcal{H}_{\text{gh}}$.

If one wants a cell-fermion operator whose one-particle restriction is Q_{cell}, one must build a different Fock space. Choose an oriented signed cellular boundary over $\mathbb{Z}$,

$$C_2^{\mathbb{Z}} \xrightarrow{\partial_2^{\mathbb{Z}}} C_1^{\mathbb{Z}} \xrightarrow{\partial_1^{\mathbb{Z}}} C_0^{\mathbb{Z}}, \qquad \partial_1^{\mathbb{Z}} \partial_2^{\mathbb{Z}} = 0.$$

Let $W = W_0 \oplus W_1 \oplus W_2$ be the complex vector space with basis the oriented cells, and let a_σ^*, a_σ be creation and annihilation operators for cell fermions. The number-preserving CAR bilinear with this one-particle matrix is

$$\widehat{Q}_{\text{cell}} = \sum_{e,v} (\delta^0_{\mathbb{Z}})_{ev} a_e^* a_v + \sum_{f,e} (\delta^1_{\mathbb{Z}})_{fe} a_f^* a_e.$$

It preserves fermion number and raises the cell degree of each one-particle label. On the one-particle sector W, its restriction is exactly the signed cellular cochain differential and squares to zero because $\delta^1_{\mathbb{Z}}\delta^0_{\mathbb{Z}} = 0$. No full-Fock nilpotence claim is made for this number-preserving bilinear. Such a claim would require a separate graded exterior-chain derivation with Koszul signs, which is not fixed in the local conventions here. This is therefore only a one-particle cell-fermion comparison, not a check-ghost construction and not the check-ghost Fock space used by Q_{gh}. Reducing the signed incidence matrices modulo 2 gives the CSS supports used below.

8.2 The Induced Operator Construction

The actual bridge starts with the mod-2 cellular maps

$$C^0 \xrightarrow{\delta^0} E \xrightarrow{\delta^1} C^2, \qquad \delta^1\delta^0 = 0.$$

The edge-bit space E is an abelian group under addition. For each $u \in E$, define the translation operator on $\mathcal{H} = \mathbb{C}[E]$ by

$$T_u|a\rangle = |a+u\rangle.$$

For each $s \in E$, define the character multiplication operator by

$$M_s|a\rangle = (-1)^{s\cdot a}|a\rangle.$$

These obey

$$T_u M_s = (-1)^{s\cdot u} M_s T_u.$$

Now feed the boundary maps into these two representations of E. For a vertex v, set

$$A_v = T_{\delta^0 v}.$$

For a face f, set $s_f = \partial_2 f \in E$ and

$$B_f = M_{s_f}.$$

Then

$$A_v B_f = (-1)^{s_f\cdot\delta^0 v} B_f A_v.$$

The exponent $s_f \cdot \delta^0 v$ is the f-coordinate of $\delta^1\delta^0 v$, equivalently the v-coefficient of $\partial_1\partial_2 f$. It is zero. Hence all star and plaquette checks commute.

The violated-check projectors are therefore

$$P^X_v = \frac{I - T_{\delta^0 v}}{2}, \qquad P^Z_f = \frac{I - M_{s_f}}{2}.$$

Their action on the computational basis is concrete:

$$P^X_v|a\rangle = \frac{|a\rangle - |a+\delta^0 v\rangle}{2},$$

and

$$P^Z_f|a\rangle = \frac{1-(-1)^{s_f\cdot a}}{2}|a\rangle = (\delta^1 a)_f|a\rangle,$$

where the bit $(\delta^1 a)_f \in \mathbb{F}_2$ is embedded as 0 or 1 in $\mathbb{C}$. Thus the plaquette part of Q_{gh} reads off the cellular coboundary $\delta^1 a$ as a face-ghost syndrome, while the star part is a finite-difference operator along the $\operatorname{im}\delta^0$ orbit.

8.3 The Ghost Differential

Let exterior multiplication by η_v and θ_f be denoted by the same symbols. Define

$$Q_X = \sum_v \eta_v P_v^X, \qquad Q_Z = \sum_f \theta_f P_f^Z, \qquad Q_{\mathrm{gh}} = Q_X + Q_Z.$$

The projectors commute because the checks commute, and exterior multiplication by ghost labels anticommutes. Therefore

$$Q_{\mathrm{gh}}^2 = \sum_{i,j} \gamma_i \gamma_j P_i P_j = \sum_{i<j} (\gamma_i \gamma_j + \gamma_j \gamma_i) P_i P_j = 0,$$

where γ_i ranges over all η_v, θ_f. With ghost annihilation operators γ_i^* satisfying $\gamma_i^* \gamma_j + \gamma_j \gamma_i = \delta_{ij}$, one also has

$$Q_{\mathrm{gh}} Q_{\mathrm{gh}}^* + Q_{\mathrm{gh}}^* Q_{\mathrm{gh}} = \sum_i P_i.$$

Thus Q_{gh} is the Koszul differential for the equations

$$T_{\delta^0 v} \psi = \psi, \qquad M_{s_f} \psi = \psi.$$

Those equations are exactly the stabilizer equations.

8.4 Cohomology-Level Relation

Let

$$\psi = \sum_{a \in E} \psi_a |a\rangle \in \mathcal{H}$$

be a no-ghost state. The plaquette equations are:

$$P_f^Z \psi = 0 \text{ for all } f \quad \Longleftrightarrow \quad \psi_a = 0 \text{ unless } \delta^1 a = 0.$$

Indeed, P_f^Z multiplies the coefficient of $|a\rangle$ by the bit $(\delta^1 a)_f$. The star equations are:

$$P_v^X \psi = 0 \text{ for all } v \quad \Longleftrightarrow \quad \psi_a = \psi_{a+\delta^0 v} \text{ for all } a, v.$$

Thus a no-ghost state is Q_{gh}-closed exactly when its coefficients are supported on $\ker \delta^1$ and are constant on cosets of $\operatorname{im} \delta^0$. Since there are no degree -1 states, there are no no-ghost exact states. Therefore

$$H^0(Q_{\mathrm{gh}}) \cong \mathbb{C}[\ker \delta^1 / \operatorname{im} \delta^0] = \mathbb{C}[H^1(Q_{\mathrm{cell}})].$$

This is the concrete relation. Q_{cell} does not act inside the check-ghost Fock space. Instead, its two arrows induce two kinds of operators on $\mathbb{C}[C^1]$: δ^0 gives translations, and δ^1 gives syndrome multiplications. The ghost supercharge is the Koszul operator for the projector constraints built from those induced operators. The equality is therefore a cohomology-level identification, not an operator restriction:

$$H^0(Q_{\mathrm{gh}}) \simeq \mathbb{C}[H^1(Q_{\mathrm{cell}})].$$

9 Chain Complexes as Symplectic CSS Stabilizers

This section answers the symplectic-geometry question. The conclusion is:

> The chain/cochain CSS construction is not an alternative to the symplectic description. It is a functorial way to manufacture an isotropic stabilizer subspace L inside the Pauli symplectic vector space. The usual homology and cohomology groups appear as the two halves of the symplectic reduction $L^\perp/L$.

The local source anchors are as follows. Gottesman's TeX source records CSS commutation as row orthogonality in `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 1227–1246, and the binary symplectic stabilizer form in lines 1293–1341. Ashikhmin–Knill's local TeX source `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313, gives the nonbinary shift/phase commutator; lines 351–407 relate nonbinary stabilizers to self-orthogonal classical data. The Bombin–Martin-Delgado local source `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 1206–1259, writes qudit Pauli operators with a symplectic commutator; lines 2050–2136 define the chain/cochain pairing; lines 2383–2452 state the homological-code construction as a symplectic code.

9.1 The Symplectic Stabilizer Side

First take prime qudit dimension p. Put one qudit on each of n sites. The Pauli label space is

$$V = \mathbb{F}_p^n \oplus \mathbb{F}_p^n.$$

We write an element as (x, z), where x is the vector of X-exponents and z is the vector of Z-exponents. With $\zeta = \exp(2\pi i/p)$, define

$$X^x|a\rangle = |a + x\rangle, \qquad Z^z|a\rangle = \zeta^{z\cdot a}|a\rangle,$$

where $a, x, z \in \mathbb{F}_p^n$. Then

$$X^x Z^z X^{x'} Z^{z'} = \zeta^{z\cdot x' - x\cdot z'} X^{x'} Z^{z'} X^x Z^z.$$

Thus the commutator is controlled by the symplectic form

$$\omega((x, z), (x', z')) = z \cdot x' - x \cdot z'.$$

This is the local convention used throughout the section: the first component is the X-exponent and the ordered X-then-Z mixed pairing is negative.

A stabilizer specified by exponent labels $L \subset V$ is commuting exactly when

$$\omega(\ell, \ell') = 0 \quad \text{for all } \ell, \ell' \in L.$$

That is, L is isotropic. Exponent labels determine this centralizer and commutation condition; an actual stabilizer code also requires compatible phases/eigenvalue characters. For the canonical CSS lifts used below, if $\dim L = r$, the joint eigenspace has dimension

$$p^{n-r}.$$

The logical Pauli labels are not V/L, because Pauli operators must preserve the code space. They are

$$L^\perp/L, \qquad L^\perp = \{v \in V : \omega(v, \ell) = 0 \text{ for all } \ell \in L\}.$$

This is the finite symplectic reduction. Its dimension is $2(n - r)$.

9.2 The CSS Case Inside the Symplectic Module

A CSS stabilizer is specified by two subspaces

$$R_X, R_Z \subset \mathbb{F}_p^n.$$

The X-type labels are $(x, 0)$ with $x \in R_X$, and the Z-type labels are $(0, z)$ with $z \in R_Z$. Hence

$$L = (R_X \oplus 0) + (0 \oplus R_Z) \subset V.$$

The same-type commutators vanish automatically. The mixed commutator is

$$\omega((x, 0), (0, z)) = -x \cdot z.$$

Therefore

$$L \text{ isotropic} \quad \Longleftrightarrow \quad x \cdot z = 0 \text{ for all } x \in R_X,\ z \in R_Z.$$

If H_X and H_Z are matrices whose rows span R_X and R_Z, this condition is exactly

$$H_X H_Z^T = 0.$$

So the CSS row-orthogonality condition is literally the isotropy condition in the Pauli symplectic vector space.

9.3 Feed in a Chain Complex

Now let

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

be a finite chain complex over $\mathbb{F}_p$, so

$$\partial_1 \partial_2 = 0.$$

Put the physical qudits on the chosen basis of C_1, so $n = \dim C_1$. Let

$$\delta^0 = \partial_1^T : C^0 \to C^1, \qquad \delta^1 = \partial_2^T : C^1 \to C^2.$$

After identifying C^1 with coordinate row vectors on the edge basis, define

$$R_X = \operatorname{im} \delta^0, \qquad R_Z = \operatorname{im} \partial_2.$$

Thus the stabilizer subspace is

$$L = (\operatorname{im} \delta^0 \oplus 0) + (0 \oplus \operatorname{im} \partial_2) \subset C^1 \oplus C_1.$$

We now derive isotropy, not just state it. Take $\alpha \in C^0$ and $\beta \in C_2$. The corresponding X- and Z-type labels are $(\delta^0 \alpha, 0)$ and $(0, \partial_2 \beta)$. Their symplectic pairing is

$$\begin{aligned} \omega((\delta^0 \alpha, 0), (0, \partial_2 \beta)) &= -\langle \delta^0 \alpha, \partial_2 \beta \rangle \\ &= -\langle \alpha, \partial_1 \partial_2 \beta \rangle \\ &= 0. \end{aligned}$$

The first line is the Pauli symplectic form. The second line is the definition of the coboundary as the transpose of the boundary. The last line is exactly $\partial_1 \partial_2 = 0$. This is the clean bridge: the chain-complex identity is the CSS isotropy identity.

9.4 The Exact Sequence Hidden in Symplectic Geometry

The stronger statement identifies the symplectic reduction. Let

$$B^1 = \operatorname{im}\delta^0, \quad Z^1 = \ker\delta^1, \quad B_1 = \operatorname{im}\partial_2, \quad Z_1 = \ker\partial_1.$$

Then $L = B^1 \oplus B_1$, using the CSS embedding into $C^1 \oplus C_1$.

Compute $L^\perp$. A general Pauli label is $(x, z) \in C^1 \oplus C_1$. It is orthogonal to every $(u, 0)$ with $u \in B^1$ iff

$$0 = \omega((x, z), (u, 0)) = z \cdot u \quad \text{for all } u \in \operatorname{im}\delta^0.$$

This says z annihilates $\operatorname{im}\delta^0$, equivalently

$$z \in (\operatorname{im}\delta^0)^\perp = \ker\partial_1 = Z_1.$$

It is orthogonal to every $(0, b)$ with $b \in B_1$ iff

$$0 = \omega((x, z), (0, b)) = -x \cdot b \quad \text{for all } b \in \operatorname{im}\partial_2.$$

This says

$$x \in (\operatorname{im}\partial_2)^\perp = \ker\delta^1 = Z^1.$$

Therefore

$$L^\perp = Z^1 \oplus Z_1.$$

Quotienting by $L = B^1 \oplus B_1$ gives the short exact sequence

$$0 \longrightarrow B^1 \oplus B_1 \longrightarrow Z^1 \oplus Z_1 \longrightarrow H^1 \oplus H_1 \longrightarrow 0,$$

and hence

$$L^\perp / L \cong H^1 \oplus H_1.$$

This is the precise connection. The chain/cochain short exact sequence is the CSS presentation of the symplectic reduction of an isotropic stabilizer subspace.

9.5 Lagrangians and Logical Bases

If L has dimension $n - k$, then $L^\perp/L$ has dimension $2k$. The code encodes k qudits. A stabilizer code itself is therefore represented by an isotropic subspace L, not usually a Lagrangian subspace of V. It becomes Lagrangian only when $k = 0$, i.e. for a stabilizer state.

The Lagrangians appear one level down, in the logical symplectic module

$$L^\perp / L \cong H^1 \oplus H_1.$$

For example,

$$\Lambda_X = Z^1 \oplus B_1, \qquad \Lambda_Z = B^1 \oplus Z_1$$

are isotropic subspaces of $L^\perp$ containing L. Their quotients are

$$\Lambda_X / L \cong H^1, \qquad \Lambda_Z / L \cong H_1.$$

These are complementary Lagrangian subspaces of the logical Pauli module, and the induced symplectic pairing on the ordered X-then-Z logical labels is the negative of the natural evaluation pairing:

$$H^1 \times H_1 \longrightarrow \mathbb{F}_p, \qquad ([x], [z]) \mapsto -x(z).$$

Equivalently, if the arguments are ordered Z-then-X, the same symplectic form gives the positive evaluation $x(z)$. Choosing a logical computational basis is choosing one of these Lagrangian polarizations.

9.6 Qudit Generalization

For prime p, the derivation above is literal. For prime powers $q = p^m$, replace $\mathbb{F}_p$ by $\mathbb{F}_q$ and use the additive character

$$\chi(t) = \exp(2\pi i \ \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t)/p).$$

The Pauli commutator is then controlled by the trace-symplectic form

$$\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}\big(z \cdot x' - x \cdot z'\big).$$

The CSS chain-complex condition over $\mathbb{F}_q$ is still

$$\partial_1 \partial_2 = 0,$$

and the same calculation proves isotropy. Thus the chain/cochain construction does generalize cleanly to field-valued qudits.

The caveat is composite non-field dimension D. One can work over $\mathbb{Z}_D$, but then the clean vector-space statements about rank, dimension, and duality become module statements. Torsion can matter. This is why the field symplectic formulation is the robust one: the chain complex is safe when interpreted as producing the isotropic subspace L, while the logical information is recovered from the symplectic reduction $L^\perp/L$.

9.7 Exact Validation

The validation script is `scripts/lattice_codes/symplectic_css_bridge_validation.jl`. It builds oriented boundary matrices for the $k = 4$ periodic square torus over $\mathbb{F}_p$ for $p = 2, 3, 5$. It verifies:

$$\partial_1 \partial_2 = 0, \qquad L \subseteq L^\perp,$$

$$\mathrm{rank}\, \partial_1 = 15, \qquad \mathrm{rank}\, \partial_2 = 15, \qquad \dim H_1 = 32 - 15 - 15 = 2.$$

The corresponding stabilizer subspace has rank 30, so the symplectic count gives $32 - 30 = 2$ encoded qudits and Hilbert dimension p^2. The generated CSV is `runs/2026-05-24-symplectic-css-bridge/data/symplectic_css_bridge_summary.csv`.

10 Symplectic Dictionary for CSS Supercharges

This section relates the supercharge operators to the symplectic stabilizer geometry of Section 9. The result is not that the supercharge is itself a symplectic form. Rather, for any CSS stabilizer subspace L, the supercharge is the Koszul differential for the commuting projector equations defining the L-invariant subspace.

The source anchors are registered in the local symplectic-QECC and toric-code source manifests. The locators used here are Gottesman `Thesis.tex`, lines 1293–1341; Ashikhmin–Knill `n9.tex`, lines 249–375; and Bombin–Martin-Delgado `HomologicalErrorCorrection6.tex`, lines 1206–1259.

10.1 CSS Symplectic Data

Work first over a prime field $\mathbb{F}_p$. A CSS code on n qudits is given by two row spaces

$$R_X, R_Z \subset \mathbb{F}_p^n$$

with

$$x \cdot z = 0 \quad \text{for all } x \in R_X,\ z \in R_Z.$$

The Pauli label space is

$$V = \mathbb{F}_p^n \oplus \mathbb{F}_p^n, \qquad \omega((x, z), (x', z')) = z \cdot x' - x \cdot z'.$$

The CSS stabilizer label subspace is

$$L = (R_X \oplus 0) + (0 \oplus R_Z) \subset V.$$

The row-orthogonality condition is exactly $L \subseteq L^\perp$.

Let

$$G = (\ell_1, \ldots, \ell_r)$$

be an ordered $\mathbb{F}_p$-basis of L. This basis is extra data. It is needed to write a small generator-based supercharge, but the code subspace depends only on L.

10.2 Projectors From Symplectic Labels

For $v = (x, z) \in V$, write $W(v) = X^x Z^z$. In the whole Pauli label space,

$$W(v)W(w) = \zeta^{\omega(v,w)} W(w)W(v), \qquad \zeta = e^{2\pi i/p}.$$

For the CSS subspace $L = (R_X \oplus 0) + (0 \oplus R_Z)$, the canonical lifts are stronger than commuting. If $\ell = (x, z)$ and $\ell' = (x', z')$ lie in L, then $z \cdot x' = x \cdot z' = 0$, because $R_X \perp R_Z$. Hence

$$W(\ell)W(\ell') = W(\ell + \ell'), \qquad W(\ell)^* = W(-\ell).$$

Thus W restricts to an honest representation of the additive group L.

For each generator ℓ_i, define the stabilizer-line averaging projector

$$\Pi_i = \frac{1}{p} \sum_{a \in \mathbb{F}_p} W(a\ell_i).$$

This is the projector onto states fixed by the one-parameter stabilizer subgroup $\{W(a\ell_i) : a \in \mathbb{F}_p\}$. To check this directly,

$$\begin{aligned}\Pi_i^2 &= \frac{1}{p^2}\sum_{a,b} W((a+b)\ell_i)\\ &= \frac{1}{p}\sum_{c} W(c\ell_i) = \Pi_i,\end{aligned}$$

since for each c there are p pairs (a,b) with $a+b=c$. Also $\Pi_i^* = \Pi_i$, because $W(a\ell_i)^* = W(-a\ell_i)$. Set

$$P_i = I - \Pi_i.$$

Then P_i is a self-adjoint projector onto the subspace violating the ℓ_i-stabilizer equation. The projectors P_i commute with each other because the line averages commute.

For qubits, $p = 2$, this reduces to the previous formula:

$$\Pi_i = \frac{I + W(\ell_i)}{2}, \qquad P_i = \frac{I - W(\ell_i)}{2}.$$

10.3 The Generator-Based Supercharge

Attach one fermionic ghost ϵ_i to each chosen generator ℓ_i. Exterior multiplication by ϵ_i and contraction ι_i satisfy

$$\iota_i\epsilon_j + \epsilon_j\iota_i = \delta_{ij}, \qquad \epsilon_i\epsilon_j + \epsilon_j\epsilon_i = 0, \qquad \iota_i\iota_j + \iota_j\iota_i = 0.$$

They commute with all physical Pauli projectors. Define

$$Q_G = \sum_{i=1}^{r} \epsilon_i P_i, \qquad Q_G^* = \sum_{i=1}^{r} \iota_i P_i.$$

Now compute:

$$\begin{aligned}Q_G^2 &= \sum_{i,j} \epsilon_i\epsilon_j P_i P_j\\ &= \sum_{i<j} (\epsilon_i\epsilon_j + \epsilon_j\epsilon_i) P_i P_j = 0.\end{aligned}$$

The anticommutator is

$$\begin{aligned}Q_G Q_G^* + Q_G^* Q_G &= \sum_{i,j} (\epsilon_i\iota_j + \iota_j\epsilon_i) P_i P_j\\ &= \sum_i P_i^2 = \sum_i P_i.\end{aligned}$$

Thus the supersymmetric Hamiltonian is the stabilizer-violation penalty for the chosen generator basis.

Degree zero means no ghost. A physical vector ψ is degree-zero closed iff

$$Q_G\psi = \sum_i \epsilon_i P_i \psi = 0.$$

The one-ghost labels are independent, so this is equivalent to $P_i\psi = 0$ for every i, equivalently $\Pi_i\psi = \psi$ for every i. Since the ℓ_i form a basis of L, this means

$$W(\ell)\psi = \psi \quad \text{for every } \ell \in L.$$

There are no degree -1 states, so no degree-zero exact states. Therefore

$$H^0(Q_G) = \{\psi : W(\ell)\psi = \psi \text{ for all } \ell \in L\},$$

the CSS stabilizer code space. If $r = \dim L$, its dimension is p^{n-r}.

10.4 Logical Paulis Commute With the Supercharge

Let $v \in L^{\perp}$. Then $\omega(v, \ell_i) = 0$ for every i, so $W(v)$ commutes with every $W(a\ell_i)$, hence with every Π_i, P_i, and

$$Q_G = \sum_i \epsilon_i P_i.$$

Therefore $W(v)$ acts on $H^0(Q_G)$.

If $v \in L$, then $W(v)$ acts as the identity on $H^0(Q_G)$ in the trivial-character sector. Hence the action factors through

$$L^{\perp}/L.$$

This is the exact point where symplectic geometry enters the supercharge cohomology: the degree-zero cohomology carries the logical Pauli action of the symplectic reduction.

If $v \notin L^{\perp}$, then some ℓ_i has $\omega(\ell_i, v) \neq 0$. For $\psi \in H^0(Q_G)$,

$$W(a\ell_i)W(v)\psi = \zeta^{a\omega(\ell_i,v)}W(v)\psi.$$

Averaging over a gives $\Pi_i W(v)\psi = 0$. Thus

$$P_i W(v)\psi = W(v)\psi, \qquad Q_G W(v)\psi \text{ has a nonzero } \epsilon_i\text{-component.}$$

So Pauli labels outside $L^{\perp}$ are precisely those detected by the supercharge as nonzero stabilizer syndrome.

10.5 Basis Dependence and Basis-Free Variant

The operator Q_G depends on the chosen basis G of L. Changing generators changes the individual projectors P_i and therefore changes the positive Hamiltonian $\sum_i P_i$. The degree-zero cohomology does not change: it is always the common invariant subspace for the whole L.

There is also a basis-free, but redundant, version. Let $\mathbb{P}(L)$ be the set of one-dimensional subspaces of L. For a line $\lambda = \mathbb{F}_p\ell$, define

$$\Pi_\lambda = \frac{1}{p}\sum_{a\in\mathbb{F}_p} W(a\ell), \qquad P_\lambda = I - \Pi_\lambda.$$

This is independent of the nonzero representative ℓ. Attach one ghost ϵ_λ to each line and set

$$Q_L^{\mathrm{all}} = \sum_{\lambda\in\mathbb{P}(L)} \epsilon_\lambda P_\lambda.$$

The same CAR/projector proof gives $(Q_L^{\mathrm{all}})^2 = 0$. Its degree-zero cohomology is again the CSS code space. The price is redundancy: if $\dim L = r$, then

$$|\mathbb{P}(L)| = \frac{p^r - 1}{p - 1}.$$

The generator-based Q_G is smaller; the projective-line Q_L^{all} is intrinsic to L.

10.6 General CSS Dictionary

For a CSS code with check matrices H_X, H_Z over $\mathbb{F}_p$, the dictionary is summarized by the table below. In the cellular toric-code case, this means

$$Q_{\text{cell}} = \delta^0 + \delta^1 \quad \rightsquigarrow \quad R_X = \operatorname{im}\delta^0, \quad R_Z = \operatorname{im}\partial_2 \quad \rightsquigarrow \quad L \quad \rightsquigarrow \quad Q_G.$$

The cellular differential supplies the two CSS support spaces. The symplectic step packages them as L. The ghost/Koszul step turns a chosen basis of L into commuting projectors and the operator Q_G.

CSS/symplectic object	supercharge object
R_X, R_Z	X- and Z-type stabilizer equations
$L = (R_X \oplus 0) + (0 \oplus R_Z)$	common invariant equations
$L \subseteq L^\perp$	commuting projectors P_i
basis G of L	minimal ghost list
$P_i = I - \frac{1}{p}\sum_a W(a\ell_i)$	violated-check projector
$Q_G = \sum_i \epsilon_i P_i$	Koszul supercharge
$L^\perp$	Paulis commuting with Q_G
$L^\perp/L$	logical Paulis on $H^0(Q_G)$

Thus the supercharge is a homological-algebra package built on top of an isotropic subspace. Symplectic geometry supplies L, its normalizer $L^\perp$, and the logical quotient; the supercharge resolves the common stabilizer equations defining the code space.

10.7 Prime-Power Note

For $q = p^m$, replace $\mathbb{F}_p$ by $\mathbb{F}_q$, and use the additive character

$$\chi(t) = \exp(2\pi i \ \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t)/p).$$

For a line $\mathbb{F}_q\ell$, the invariant projector is

$$\Pi_\ell = \frac{1}{q}\sum_{a\in\mathbb{F}_q} W(a\ell).$$

The same calculation proves $\Pi_\ell^2 = \Pi_\ell$, commuting projectors for isotropic L, $Q^2 = 0$, and

$$H^0(Q) \cong \text{the } L\text{-stabilizer code}.$$

For composite non-field qudit dimension, one must replace vector spaces by modules over $\mathbb{Z}_D$, and the clean count $p^{n-\dim L}$ need not be imported without checking torsion.

10.8 Exact Validation

The script `scripts/lattice_codes/css_supercharge_symplectic_dictionary.jl` validates the dictionary without building Hilbert matrices. It checks the binary Steane CSS code and a qutrit CSS toy code. The CSV `runs/2026-05-24-css-supercharge-symplectic-dictionary/data/css_supercharge_symplectic_dictionary.csv` records CSS isotropy, stabilizer rank, code dimension, logical symplectic dimension, ghost counts, and the Q^2 and anticommutator certificates.

11 Steane Code Symplectic Data in Molecular Detail

This section fixes one concrete Steane-code presentation and writes all finite-vector-space objects explicitly. The local source anchor is Gottesman's `Thesis.tex`: lines 1249–1267 give the seven-qubit CSS stabilizer table, and lines 1269–1284 give the two displayed encoded codewords.

11.1 Binary Check Space

Work over $\mathbb{F}_2$ with qubits ordered $1, \ldots, 7$. The three binary rows read from Gottesman's X-check table are

$$\begin{aligned} g_1 &= 1111000, \\ g_2 &= 1100110, \\ g_3 &= 1010101. \end{aligned}$$

Equivalently,

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

Let

$$D = \mathrm{row}(H) = \langle g_1, g_2, g_3 \rangle \subset \mathbb{F}_2^7.$$

The eight elements of D are

$$\begin{gathered} 0000000,\ 1111000,\ 1100110,\ 1010101, \\ 0011110,\ 0101101,\ 0110011,\ 1001011. \end{gathered}$$

The row dot products are

$$HH^T = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_2.$$

Thus $D \subseteq D^\perp$.

Put

$$C = D^\perp = \{a \in \mathbb{F}_2^7 : Ha^T = 0\}.$$

Since $\operatorname{rank} H = 3$, $\dim C = 4$. Let

$$u = 0000111.$$

Then $u \in C$, but $u \notin D$. Therefore

$$C = D \sqcup (u + D).$$

The second coset is

$$\begin{gathered} 0000111,\ 1111111,\ 1100001,\ 1010010, \\ 0011001,\ 0101010,\ 0110100,\ 1001100. \end{gathered}$$

These are exactly the two cosets supporting Gottesman's displayed $|\overline{0}\rangle$ and $|\overline{1}\rangle$ states.

11.2 The Stabilizer Subspace

The Pauli label space is

$$V = \mathbb{F}_2^7 \oplus \mathbb{F}_2^7.$$

We write a label as (x, z), corresponding to $X^x Z^z$. The binary symplectic form is

$$\omega((x, z), (x', z')) = z \cdot x' + x \cdot z',$$

where the sign from the odd-prime formula disappears in characteristic 2.

The six stabilizer labels are

$$\begin{aligned} &\ell_1 = (g_1, 0), \quad \ell_2 = (g_2, 0), \quad \ell_3 = (g_3, 0), \\ &\ell_4 = (0, g_1), \quad \ell_5 = (0, g_2), \quad \ell_6 = (0, g_3). \end{aligned}$$

Their span is

$$L = (D \oplus 0) + (0 \oplus D) \subset V.$$

This is an isotropic subspace. Indeed, same-type pairings vanish automatically. For a mixed pair,

$$\omega((d, 0), (0, d')) = d \cdot d' = 0 \qquad (d, d' \in D),$$

because $D \subseteq D^\perp$. Therefore $L \subseteq L^\perp$.

The dimensions and sizes are

$$\dim L = 6, \qquad |L| = 2^6 = 64.$$

11.3 The Centralizer

Let $(x, z) \in V$. It is orthogonal to every $(d, 0) \in D \oplus 0$ exactly when

$$z \cdot d = 0 \quad \text{for all } d \in D,$$

that is, $z \in C$. It is orthogonal to every $(0, d) \in 0 \oplus D$ exactly when

$$x \cdot d = 0 \quad \text{for all } d \in D,$$

that is, $x \in C$. Hence

$$L^\perp = C \oplus C.$$

Consequently

$$\dim L^\perp = 8, \qquad |L^\perp| = 2^8 = 256.$$

11.4 The Logical Symplectic Quotient

The logical Pauli label space is the finite symplectic reduction

$$L^\perp / L = (C \oplus C)/(D \oplus D) \cong (C/D) \oplus (C/D).$$

Since $C = D \sqcup (u + D)$, the quotient C/D is one-dimensional. Thus

$$\dim(L^\perp / L) = 2.$$

Choose the representatives

$$\overline{X} = (u, 0), \qquad \overline{Z} = (0, u).$$

Their symplectic pairing is

$$\omega(\overline{X}, \overline{Z}) = u \cdot u = 1 + 1 + 1 = 1 \quad \text{in } \mathbb{F}_2.$$

Thus the corresponding Pauli operators anticommute. The four logical Pauli classes are represented by

$$(0, 0), \quad (u, 0), \quad (0, u), \quad (u, u).$$

This is the full symplectic anatomy of the code: D supplies the CSS check space, $L = D \oplus D$ is the isotropic stabilizer subspace, $L^{\perp} = C \oplus C$ is the Pauli centralizer, and $L^{\perp}/L$ is the one-qubit logical symplectic module.

12 Steane Code Fermions, Supercharge, and Cohomology

Keep the notation of Section 11. This section constructs the actual fermions and supercharge for the same six Gottesman-table stabilizer generators.

12.1 Physical Stabilizer Projectors

The physical Hilbert space is

$$\mathcal{H} = \mathbb{C}\{|a\rangle : a \in \mathbb{F}_2^7\}.$$

For $d \in \mathbb{F}_2^7$, define

$$X(d)|a\rangle = |a+d\rangle, \qquad Z(d)|a\rangle = (-1)^{d\cdot a}|a\rangle.$$

The six stabilizer operators are

$$\begin{aligned} S_1 &= X(g_1), \quad S_2 = X(g_2), \quad S_3 = X(g_3), \\ S_4 &= Z(g_1), \quad S_5 = Z(g_2), \quad S_6 = Z(g_3). \end{aligned}$$

The violated-check projectors are

$$P_i = \frac{I - S_i}{2}, \qquad \Pi_i = \frac{I + S_i}{2}.$$

Since the labels span the isotropic subspace L, all S_i, hence all P_i, commute.

12.2 The Two Codeword Cosets

The two normalized Steane basis states in this convention are

$$|\overline{0}\rangle = \frac{1}{\sqrt{8}} \sum_{d\in D} |d\rangle, \qquad |\overline{1}\rangle = \frac{1}{\sqrt{8}} \sum_{d\in D} |u+d\rangle.$$

For an X-type check $X(g_j)$, adding $g_j \in D$ permutes D and also permutes $u + D$. Hence

$$X(g_j)|\overline{0}\rangle = |\overline{0}\rangle, \qquad X(g_j)|\overline{1}\rangle = |\overline{1}\rangle.$$

For a Z-type check $Z(g_j)$, every vector in D is orthogonal to g_j, and u is also orthogonal to g_j. Therefore

$$Z(g_j)|d\rangle = |d\rangle, \qquad Z(g_j)|u+d\rangle = |u+d\rangle.$$

So both displayed states are fixed by all six stabilizers.

Conversely, let

$$|\psi\rangle = \sum_{a\in\mathbb{F}_2^7} \psi_a |a\rangle.$$

The equations $Z(g_j)|\psi\rangle = |\psi\rangle$ force $\psi_a = 0$ unless $a \in C = D^\perp$. The equations $X(g_j)|\psi\rangle = |\psi\rangle$ force $\psi_{a+g_j} = \psi_a$, so the amplitudes are constant on D-cosets inside C. There are exactly two such cosets, D and $u + D$. Thus the common stabilizer $+1$ space is

$$\mathcal{C} = \operatorname{span}\{|\overline{0}\rangle, |\overline{1}\rangle\}, \qquad \dim \mathcal{C} = 2.$$

12.3 Six Fermions

Let

$$E = \mathrm{span}_{\mathbb{C}}\{\epsilon_1, \ldots, \epsilon_6\}.$$

The ghost Fock space is the exterior algebra ΛE. Exterior multiplication by ϵ_i and contraction ι_i obey

$$\iota_i\epsilon_j + \epsilon_j\iota_i = \delta_{ij}, \qquad \epsilon_i\epsilon_j + \epsilon_j\epsilon_i = 0, \qquad \iota_i\iota_j + \iota_j\iota_i = 0.$$

The ghost-extended Hilbert space is

$$\mathcal{K} = \mathcal{H} \otimes \Lambda E, \qquad \dim \mathcal{K} = 2^7\, 2^6 = 8192.$$

Define

$$Q = \sum_{i=1}^{6} \epsilon_i P_i, \qquad Q^* = \sum_{i=1}^{6} \iota_i P_i.$$

Written without abbreviations,

$$\begin{aligned} Q = \epsilon_1 \frac{I - X(g_1)}{2} + \epsilon_2 \frac{I - X(g_2)}{2} + \epsilon_3 \frac{I - X(g_3)}{2} \\ + \epsilon_4 \frac{I - Z(g_1)}{2} + \epsilon_5 \frac{I - Z(g_2)}{2} + \epsilon_6 \frac{I - Z(g_3)}{2}. \end{aligned}$$

Because the P_i commute and the ϵ_i anticommute,

$$Q^2 = \sum_{i<j} (\epsilon_i\epsilon_j + \epsilon_j\epsilon_i) P_i P_j = 0.$$

The supersymmetric Hamiltonian is

$$\begin{aligned} QQ^* + Q^*Q &= \sum_{i,j} (\epsilon_i\iota_j + \iota_j\epsilon_i) P_i P_j \\ &= \sum_i P_i. \end{aligned}$$

Thus it is the Steane stabilizer-violation Hamiltonian, shifted and scaled so that the code sector has energy 0.

12.4 Cohomology, Defined and Computed

Ghost degree is exterior degree. The degree-q subspace is

$$\mathcal{K}^q = \mathcal{H} \otimes \Lambda^q E.$$

The degree-q cohomology of Q is

$$H^q(Q) = \frac{\{\phi \in \mathcal{K}^q : Q\phi = 0\}}{\{Q\eta : \eta \in \mathcal{K}^{q-1}\}}.$$

For $q = 0$, the denominator is zero because $\mathcal{K}^{-1} = 0$.

Decompose the physical Hilbert space into stabilizer-syndrome sectors

$$\mathcal{H} = \bigoplus_{\lambda \in \mathbb{F}_2^6} \mathcal{H}_\lambda, \qquad S_i\psi = (-1)^{\lambda_i}\psi.$$

On $\mathcal{H}_\lambda$, the projector P_i acts as multiplication by λ_i. Therefore

$$Q|_{\mathcal{H}_\lambda \otimes \Lambda E} = \epsilon(\alpha_\lambda), \qquad \alpha_\lambda = \sum_i \lambda_i \epsilon_i,$$

where $\epsilon(\alpha)$ means exterior multiplication by α.

If $\lambda \neq 0$, choose a contraction h with $h(\alpha_\lambda) = 1$. Then

$$hQ + Qh = I$$

on $\mathcal{H}_\lambda \otimes \Lambda E$, so that syndrome block is acyclic: it contributes no cohomology. If $\lambda = 0$, then $Q = 0$, and $\mathcal{H}_0 = \mathcal{C}$ has dimension 2. Hence

$$H^q(Q) \cong \mathcal{C} \otimes \Lambda^q E, \qquad \dim H^q(Q) = 2\binom{6}{q}.$$

The dimensions are

q	0	1	2	3	4	5	6
$\dim H^q(Q)$	2	12	30	40	30	12	2.

The physical Steane code is the no-ghost part:

$$H^0(Q) = \mathcal{C} = \operatorname{span}\{|\bar{0}\rangle, |\bar{1}\rangle\}.$$

The full Q-cohomology has ghost-labelled copies of the same code space, with total dimension $2 \cdot 2^6 = 128$. The ground space of $\{Q, Q^*\}$ has the same ghost degeneracy. The no-ghost sector is therefore the physical stabilizer code; the higher ghost-degree classes are auxiliary Koszul bookkeeping.

12.5 What the Higher Ghost Classes Mean

The word "bookkeeping" in the previous paragraph should be read narrowly. It means that higher ghost degree does not add new Steane logical qubits. It does not mean that the Koszul cohomology is meaningless.

The six equations defining the code are

$$P_1\psi = \cdots = P_6\psi = 0.$$

The Koszul differential remembers these as six separate equations. On a nonzero syndrome sector $\lambda \neq 0$, at least one equation is violated and the complex is contractible by the homotopy h above. On the zero-syndrome sector, all six functions P_i vanish. Therefore Q becomes the zero operator on

$$\mathcal{C} \otimes \Lambda E,$$

and every exterior ghost monomial survives.

Thus the higher cohomology is exactly

$$H^{>0}(Q) = \mathcal{C} \otimes \Lambda^{>0} E.$$

It records the exterior algebra on the six chosen stabilizer equations over the constraint surface. Equivalently, it is the finite-dimensional analogue of conormal or derived-intersection data for the equations $P_i = 0$. This data is useful if the question is about the Koszul resolution itself, or about how the constraint presentation changes when generators are added, removed, or made redundant.

It is not, however, an invariant of the Steane code alone. If we choose a different independent generating set for the same L, the ghost basis changes. If we add redundant stabilizer generators, the higher exterior algebra gets larger while the physical code space is unchanged. The intrinsic QECC content is therefore the no-ghost code $\mathcal{C} = H^0(Q)$ together with the logical symplectic quotient $L^\perp/L$. The full Q-cohomology is a presentation-dependent derived enhancement of the stabilizer equations.

12.6 Exact Validation

The producer `scripts/lattice_codes/steane_supercharge_molecular.jl` writes the CSV files `runs/2026-05-24-steane-supercharge-molecular/data/steane_molecular_summary.csv`, `runs/2026-05-24-steane-supercharge-molecular/data/steane_molecular_vectors.csv`, and `runs/2026-05-24-steane-supercharge-molecular/data/steane_cohomology_by_degree.csv`. The summary CSV records $\dim D = 3$, $\dim C = 4$, $\dim L = 6$, $\dim L^{\perp} = 8$, $|\mathcal{K}| = 8192$, $\dim H^0(Q) = 2$, and full cohomology dimension 128. The vector CSV records the two codeword cosets and the logical representatives $(u, 0)$, $(0, u)$. The cohomology CSV records the seven dimensions displayed above.

13 Steane Clifford Morphisms of the Koszul Supercharge

Here "symplectic morphism" means a Clifford operation: a unitary U whose conjugation action on Paulis induces a symplectic automorphism F_U of the binary label space. Gottesman's local source anchors are `Thesis.tex`: lines 2271–2305 for valid encoded Clifford operations preserving the stabilizer, lines 2307–2348 for the Hadamard, phase, and CNOT Pauli-label actions, and lines 2439–2445 for bitwise Hadamard and phase on the seven-qubit code.

13.1 General Clifford Lift

Let $G = (\ell_1, \dots, \ell_6)$ be the Steane generator list from Section 11, and let

$$Q_G = \sum_i \epsilon_i P_{\ell_i}, \qquad P_{\ell_i} = \frac{I - W(\ell_i)}{2}.$$

If a Clifford U sends $W(\ell)$ to $W(F_U\ell)$, up to the stabilizer phase convention, then

$$U P_{\ell_i} U^\dagger = P_{F_U \ell_i}.$$

Let $G' = F_U(G)$, give G' ghosts ϵ'_i, and define

$$Q_{G'} = \sum_i \epsilon'_i P_{F_U \ell_i}.$$

The map

$$\Phi_U = U \otimes \Lambda T, \qquad T(\epsilon_i) = \epsilon'_i,$$

satisfies

$$\Phi_U Q_G = Q_{G'} \Phi_U.$$

This is a chain isomorphism, hence it induces isomorphisms

$$H^q(Q_G) \cong H^q(Q_{G'}).$$

If G' is just a permutation of G, and T is the same permutation of ghosts, this is an automorphism of the same written operator Q_G. If G' is a different independent basis of the same L, it is instead an isomorphism to a different presentation of the same stabilizer constraints.

13.2 Transversal Hadamard

For bitwise Hadamard $R^{\otimes 7}$,

$$F_H(x, z) = (z, x).$$

Therefore

$$(g_i, 0) \mapsto (0, g_i), \qquad (0, g_i) \mapsto (g_i, 0).$$

So

$$\ell_1 \leftrightarrow \ell_4, \quad \ell_2 \leftrightarrow \ell_5, \quad \ell_3 \leftrightarrow \ell_6.$$

With the ghost swap

$$\epsilon_1 \leftrightarrow \epsilon_4, \quad \epsilon_2 \leftrightarrow \epsilon_5, \quad \epsilon_3 \leftrightarrow \epsilon_6,$$

we get the exact chain automorphism

$$(R^{\otimes 7} \otimes \Lambda T_H) Q_G = Q_G (R^{\otimes 7} \otimes \Lambda T_H).$$

On the logical symplectic quotient this sends

$$\overline{X} = (u, 0) \mapsto (0, u) = \overline{Z}, \qquad \overline{Z} = (0, u) \mapsto (u, 0) = \overline{X}.$$

On the no-ghost code basis,

$$R^{\otimes 7}|\overline{0}\rangle = \frac{|\overline{0}\rangle + |\overline{1}\rangle}{\sqrt{2}}, \qquad R^{\otimes 7}|\overline{1}\rangle = \frac{|\overline{0}\rangle - |\overline{1}\rangle}{\sqrt{2}}.$$

Thus the induced map on $H^0(Q_G)$ is the encoded Hadamard, and on $H^q(Q_G) \cong \mathcal{C} \otimes \Lambda^q E$ it is encoded Hadamard tensored with the ghost swap.

13.3 Transversal Phase

For bitwise phase $P^{\otimes 7}$, ignoring Pauli phases in the binary label,

$$F_P(x, z) = (x, z + x).$$

Hence

$$(g_i, 0) \mapsto (g_i, g_i) = \ell_i + \ell_{i+3}, \qquad (0, g_i) \mapsto (0, g_i) = \ell_{i+3}.$$

These six images are an independent basis of the same L, but they are not the original generator list and not a permutation of it. Therefore

$$P^{\otimes 7} \otimes \Lambda T_P$$

is a chain isomorphism

$$(\mathcal{H} \otimes \Lambda E, Q_G) \cong (\mathcal{H} \otimes \Lambda E', Q_{F_P(G)}),$$

not literally an automorphism of the original written Q_G.

To compare $Q_{F_P(G)}$ back with Q_G, decompose into the same physical syndrome sectors. The zero-syndrome sector is the Steane code. Every nonzero syndrome sector is contractible, as in Section 12. Both complexes therefore deformation retract to their zero-syndrome cohomology model

$$\mathcal{C} \otimes \Lambda E$$

after choosing the corresponding ghost basis. This is the precise homotopy equivalence: it is presentation-dependent on ghosts, but identical on the physical code sector. On logical labels,

$$\overline{X} = (u, 0) \mapsto (u, u) = \overline{X} + \overline{Z}, \qquad \overline{Z} = (0, u) \mapsto (0, u) = \overline{Z}.$$

13.4 Conclusion

For Steane, Clifford symmetry gives more than an abstract equivalence relation. It gives a chain isomorphism from the Koszul complex of one stabilizer presentation to the Koszul complex of the transformed presentation. When the Clifford permutes the chosen generators, as transversal Hadamard does here, this is an exact automorphism of Q_G. When the Clifford changes the independent generator basis, as transversal phase does here, the complexes are homotopy equivalent through their common zero-syndrome deformation retract, but the full higher-ghost cohomology remains presentation-labelled.

13.5 Run Artifact Record

The producer `scripts/lattice_codes/steane_clifford_koszul_morphisms.jl` writes the Steane Clifford/Koszul run bundle `runs/2026-05-24-steane-clifford-koszul-morphisms/`. Its summary CSV records Boolean and string fields for the Hadamard generator permutation, the phase generator-basis change, the binary logical-label images, the count of 63 nonzero syndrome sectors, the summary contractibility flag, and the cohomology dimensions listed in Section 12. The companion generator-map CSV lists the six stabilizer-label images and their coordinates in the original generator basis. These artifacts record checked hypotheses and summary consequences used by the local derivation above; they do not contain full chain-map matrices, contraction homotopies, or residual identities.

14 Clifford Group Ghost-Gaussian Theorem

The previous section treated two transversal Steane Cliffords. The correct general statement is stronger, but it needs one extra convention: binary Pauli labels alone are not enough. Gottesman's local source says that Clifford normalizer elements conjugate Pauli products to Pauli products `Thesis.tex`, lines 2292–2305 and 2307–2348; the binary stabilizer language drops phases at lines 1303–1327. The source manifest records the full local path. The supercharge projectors do not allow that drop.

14.1 Signed Pauli Convention

For n qubits define the Hermitian representative

$$W(x,z) = i^{x\cdot z}\bigotimes_{j=1}^{n} X_j^{x_j} Z_j^{z_j}, \qquad x,z\in\mathbb{F}_2^n.$$

A signed Pauli stabilizer generator is

$$S = \tau W(x,z), \qquad \tau\in\{\pm 1\}.$$

The violated-check projector is

$$P_S = \frac{I-S}{2}.$$

Thus if a Clifford gives

$$UW(x,z)U^\dagger = \tau W(x',z'),$$

then the transported projector is

$$UP_{W(x,z)}U^\dagger = \frac{I-\tau W(x',z')}{2}.$$

When $\tau=-1$, this is the opposite syndrome projector for the unsigned binary label (x',z'). This is why the theorem below is stated for signed Pauli operators, not only for symplectic binary labels.

14.2 The Koszul Complex Attached to a Signed Presentation

Let

$$G = (S_1,\ldots,S_r)$$

be an ordered independent list of commuting signed Hermitian Pauli operators, with no product equal to $-I$. Let $E=\mathrm{span}_{\mathbb{C}}\{\epsilon_1,\ldots,\epsilon_r\}$, and let $c_i^\dagger$ denote exterior multiplication by ϵ_i on ΛE. The ghost-extended complex is

$$\mathcal{K}(G) = \mathcal{H}\otimes\Lambda E, \qquad Q_G = \sum_{i=1}^{r} c_i^\dagger P_{S_i}.$$

This is exactly the earlier supercharge notation $\sum_i \epsilon_i P_{S_i}$, now with the creation operator written explicitly. Since the P_{S_i} commute and the $c_i^\dagger$ anticommute, $Q_G^2 = 0$.

14.3 Theorem: Arbitrary Clifford Covariance

Let U be any n-qubit Clifford. Define the transported signed presentation

$$U(G) = \big(US_1U^\dagger, \dots, US_rU^\dagger\big)$$

and give it ghosts $E^U = \mathrm{span}\{\epsilon_1^U, \dots, \epsilon_r^U\}$. Let $T_U : E \to E^U$ be the linear map $T_U(\epsilon_i) = \epsilon_i^U$. Then

$$\Phi_U = U \otimes \Lambda T_U$$

is a chain isomorphism

$$\Phi_U Q_G = Q_{U(G)} \Phi_U.$$

Equivalently,

$$Q_{U(G)} = (U \otimes \Lambda T_U) Q_G (U^\dagger \otimes \Lambda T_U^{-1}).$$

Proof. For each i,

$$UP_{S_i}U^\dagger = U\frac{I - S_i}{2}U^\dagger = \frac{I - US_iU^\dagger}{2} = P_{US_iU^\dagger}.$$

Now apply both sides to a pure tensor $\psi \otimes \omega$:

$$\begin{aligned}
\Phi_U Q_G(\psi \otimes \omega) &= \sum_i U(P_{S_i}\psi) \otimes \epsilon_i^U \wedge (\Lambda T_U)\omega \\
&= \sum_i P_{US_iU^\dagger}(U\psi) \otimes \epsilon_i^U \wedge (\Lambda T_U)\omega \\
&= Q_{U(G)} \Phi_U(\psi \otimes \omega).
\end{aligned}$$

The map is invertible because U and T_U are invertible. This proves the chain isomorphism and hence gives canonical isomorphisms

$$H^q(Q_G) \cong H^q(Q_{U(G)})$$

for every ghost degree q.

14.4 Where the Ghost Gaussian Enters

The covariance theorem uses the transported ordered list $U(G)$, so the ghost map is just $\epsilon_i \mapsto \epsilon_i^U$. The genuinely nontrivial ghost operation appears when we rewrite a signed stabilizer group in a different independent generator basis.

First consider a row swap: exchange S_a and S_b. The supercharge is unchanged after the same swap of ghost creation operators,

$$c_a^\dagger \leftrightarrow c_b^\dagger.$$

This is a number-preserving fermionic Gaussian operation on the ghost Fock space.

Now consider the elementary row shear

$$S_a \longmapsto S_aS_b, \qquad S_i \longmapsto S_i \quad (i \neq a).$$

The key projector identity is not linear over $\mathbb{C}$ in the syndrome bits; it contains the old stabilizer operator:

$$\begin{aligned}
P_{S_aS_b} &= \frac{I - S_aS_b}{2} \\
&= \frac{I - S_a}{2} + S_a\frac{I - S_b}{2} \\
&= P_{S_a} + S_aP_{S_b}.
\end{aligned}$$

Therefore the sheared supercharge is

$$\begin{aligned} Q' &= c_a^\dagger P_{S_a S_b} + c_b^\dagger P_{S_b} + \sum_{i \neq a,b} c_i^\dagger P_{S_i} \\ &= c_a^\dagger P_{S_a} + (c_b^\dagger + S_a c_a^\dagger) P_{S_b} + \sum_{i \neq a,b} c_i^\dagger P_{S_i}. \end{aligned}$$

Thus the ghost creation operators transform by

$$c_b^\dagger \longmapsto c_b^\dagger + S_a c_a^\dagger, \qquad c_i^\dagger \longmapsto c_i^\dagger \quad (i \neq b).$$

This is a number-preserving fermionic Gaussian with an even operator-valued coefficient:

$$\Gamma_{ab} = \exp(S_a c_a^\dagger c_b).$$

Indeed S_a commutes with all projectors and ghost operators, and the commutator gives

$$\Gamma_{ab} c_b^\dagger \Gamma_{ab}^{-1} = c_b^\dagger + S_a c_a^\dagger, \qquad \Gamma_{ab} c_i^\dagger \Gamma_{ab}^{-1} = c_i^\dagger \quad (i \neq b).$$

Consequently

$$Q' = \Gamma_{ab} Q_G \Gamma_{ab}^{-1}.$$

Row swaps and row shears generate $\mathrm{GL}_r(\mathbb{F}_2)$. Therefore every independent presentation

$$S'_a = \prod_i S_i^{A_{ai}}, \qquad A \in \mathrm{GL}_r(\mathbb{F}_2),$$

is obtained by composing these elementary ghost Gaussians. The closed projector formula for a row $I = (i_1, \ldots, i_m)$ is the telescoping identity

$$P_{\prod_{\nu=1}^m S_{i_\nu}} = \sum_{\mu=1}^m \left(\prod_{\nu<\mu} S_{i_\nu} \right) P_{S_{i_\mu}}.$$

This formula is the concrete reason that a scalar exterior $\mathrm{GL}_r(\mathbb{F}_2)$ operation on ghosts is not enough on the full physical Hilbert space. The syndrome parity $\sigma_a \oplus \sigma_b$ is represented over $\mathbb{C}$ by

$$\sigma_a + (-1)^{\sigma_a} \sigma_b,$$

not by $\sigma_a + \sigma_b$. The coefficient $(-1)^{\sigma_a}$ is exactly the eigenvalue of S_a.

On the code subspace all $S_i = +1$. Hence every row shear reduces there to the ordinary exterior linear map

$$c_b^\dagger \mapsto c_b^\dagger + c_a^\dagger.$$

So the higher-ghost cohomology is functorial under the exterior representation of the chosen presentation change on the zero-syndrome sector, while the full chain-level comparison remembers operator-valued syndrome signs.

14.5 Steane: All Clifford Generators

For the Steane code, $r = 6$. The standard Clifford generating gates are the seven single-qubit H's, seven single-qubit P's, and the 42 ordered CNOT gates. This is the generating set described in the local Gottesman source at lines 2292–2348. The producer `scripts/lattice_codes/steane_all_clifford_ghost_gaussians.jl` applies these 56 gates to the six sourced Steane stabilizer generators from Section 11.

The run records 336 signed generator images. Every one of the 56 image lists has binary rank 6 and is isotropic for the binary symplectic form. Since the listed H, P, and CNOT gates generate the Clifford group, any Clifford word is handled by composing these signed Pauli-label updates and then applying the covariance theorem above.

The same run records the 45 elementary ghost-Gaussian presentation moves for the Steane six-generator group: 15 row swaps and 30 directed row shears. For each shear it checks all 64 Steane syndrome sectors, i.e.

$$30 \cdot 64 = 1920$$

instances of

$$P(S_a S_b) = P(S_a) + S_a P(S_b).$$

If a Clifford U lies in the Steane stabilizer normalizer, then

$$U S_i U^\dagger = \prod_j S_j^{A_{ij}}$$

for some $A \in \mathrm{GL}_6(\mathbb{F}_2)$. In that case

$$(U \otimes I) Q_G (U^\dagger \otimes I) = Q_{A(G)} = \Gamma_A Q_G \Gamma_A^{-1},$$

where Γ_A is any product of the elementary ghost Gaussians above that realizes the row-reduction path for A. If U is an arbitrary Clifford not preserving the Steane stabilizer group, the first equality still gives a strict supercharge for the transported code $U\mathcal{C}$, but no ghost-only operation can turn its physical projectors back into the original Steane projectors.

14.6 Conclusion

The precise answer is therefore Clifford conjugation on physical qubits plus a fermionic Gaussian change of ghost presentation. No Bogoliubov mixing of creation and annihilation operators is needed for the cohomological complex; the transformations preserve ghost degree. The only non-scalar feature is essential: when a product stabilizer is rewritten in terms of old violated-check projectors, the Gaussian coefficients are even operators generated by the old stabilizers themselves.

15 Arithmetic Quantum Fields

15.1 Status and Source Anchors

This section introduces a proposal. The finite-field Weyl facts are source-local: Ashikhmin–Knill define p^m-ary shift/phase operators and their commutator in `n9.tex`, lines 249–313, and Bombin–Martin-Delgado record the qudit Pauli symplectic commutator in `HomologicalErrorCorrection6.tex`, lines 1206–1259. The new content below is the checked construction of field label spaces from finite function spaces.

15.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & X \text{ is a finite physical space.} \\
D2 & : & (V, \omega_V) \text{ is a finite symplectic } F\text{-vector space.} \\
D3 & : & E \leq \mathrm{Map}(X, V) \text{ is a chosen } F\text{-linear field space.} \\
D4 & : & \Omega_E(\phi, \psi) = \sum_{x \in X} w_x \omega_V(\phi(x), \psi(x)). \\
L1 & : & \mathrm{Map}(X, V) \text{ is an } F\text{-vector space pointwise.} \\
L2 & : & \Omega_E \text{ is alternating and bilinear.} \\
L3 & : & \Omega \text{ is nondegenerate on all maps if } \omega_V \text{ is and } w_x \neq 0. \\
L4 & : & E \text{ is symplectic iff } \mathrm{rad}(E) = 0. \\
L5 & : & E_{\mathrm{red}} = E/\,\mathrm{rad}(E) \text{ is symplectic.} \\
L6 & : & W : E_{\mathrm{red}} \to U(\mathcal{H}) \text{ is a projective Weyl representation.} \\
T1 & : & \mathcal{A}_{X,E} = \mathrm{span}_{\mathbb{C}}\{W(e)\} \text{ is the field-observable algebra.}
\end{array}$$

The numbering is used only to check that no later construction is invoked before its inputs are defined.

15.3 Definitions D1–D4

Let $F = \mathbb{F}_q$, $q = p^m$. A finite physical space is first only a finite set X. If X has additional structure, for example an F-vector-space or variety structure, then $\mathrm{Hom}(X, V)$ means a specified class of structure-preserving maps. Without that extra choice the neutral object is

$$\mathrm{Map}(X, V) = \{\phi : X \to V\}.$$

The proposed arithmetic field label space is not automatically all maps. It is a chosen F-linear subspace

$$E \leq \mathrm{Map}(X, V).$$

Examples include all functions, linear maps, affine maps, and polynomial functions up to a degree bound, all interpreted after evaluation on the finite point set.

Fix nonzero weights $w_x \in F^\times$. In this shard and in the validation run all weights are $w_x = 1$. Define

$$\Omega_E(\phi, \psi) = \sum_{x \in X} w_x \, \omega_V(\phi(x), \psi(x)).$$

15.4 Pointwise Linear Algebra L1–L5

Lemma $L1$. $\mathrm{Map}(X, V)$ is an F-vector space under pointwise operations:

$$(a\phi + b\psi)(x) = a\phi(x) + b\psi(x).$$

Proof. The right side lies in V, so it defines a map $X \to V$. Associativity, commutativity of addition, distributivity, scalar identity, and additive inverses hold after evaluating at each x, where they are exactly the vector space axioms in V. Thus the function set is a vector space. Any subset closed under these operations is an F-linear field subspace E.

Lemma *L*2**.** Ω_E is F-bilinear and alternating. *Proof.* For $a, b \in F$,

$$\begin{aligned}\Omega_E(a\phi + b\psi, \eta) &= \sum_x w_x \omega_V(a\phi(x) + b\psi(x), \eta(x)) \\ &= a\Omega_E(\phi, \eta) + b\Omega_E(\psi, \eta).\end{aligned}$$

Linearity in the second argument is identical. Also

$$\Omega_E(\phi, \phi) = \sum_x w_x \omega_V(\phi(x), \phi(x)) = 0$$

because ω_V is alternating. Hence Ω_E is an alternating bilinear form.

Lemma *L*3**.** If $E = \mathrm{Map}(X, V)$, ω_V is nondegenerate, and every $w_x \neq 0$, then Ω_E is nondegenerate. *Proof.* Let $\phi \neq 0$. Choose x_0 with $\phi(x_0) \neq 0$. Since ω_V is nondegenerate, choose $v \in V$ with $\omega_V(\phi(x_0), v) \neq 0$. Define $\psi(x_0) = w_{x_0}^{-1} v$ and $\psi(x) = 0$ for $x \neq x_0$. Then

$$\Omega_E(\phi, \psi) = \omega_V(\phi(x_0), v) \neq 0.$$

Thus no nonzero ϕ is orthogonal to all maps.

Lemma *L*4**.** For a subspace E, define

$$\mathrm{rad}(E) = \{\phi \in E : \Omega_E(\phi, \psi) = 0 \text{ for all } \psi \in E\}.$$

Then (E, Ω_E) is symplectic iff $\mathrm{rad}(E) = 0$. *Proof.* By definition, nondegeneracy means the only vector pairing to zero with every element of E is 0. That set is exactly $\mathrm{rad}(E)$. Together with Lemma $L2$, this is precisely the definition of a symplectic vector space.

Lemma *L*5**.** The quotient $E_{\mathrm{red}} = E/\mathrm{rad}(E)$ is symplectic. *Proof.* If $r \in \mathrm{rad}(E)$, then $\Omega_E(\phi + r, \psi) = \Omega_E(\phi, \psi)$, so the form descends to the quotient. If $[\phi]$ pairs to zero with every $[\psi]$, then ϕ lies in the radical, so $[\phi] = 0$. Hence the descended form is nondegenerate.

15.5 Weyl-Heisenberg Field Displacements L6

Choose a symplectic basis of E_{red}, so

$$E_{\mathrm{red}} \cong F^n \oplus F^n, \qquad e = (q, p).$$

Let $\chi(t) = \exp(2\pi i \, \mathrm{Tr}_{F/\mathbb{F}_p}(t)/p)$. On $\mathcal{H} = \mathbb{C}^{F^n}$, set

$$T_q|a\rangle = |a + q\rangle, \qquad R_p|a\rangle = \chi(p \cdot a)|a\rangle, \qquad W(q, p) = T_q R_p.$$

Then

$$W(q, p)W(q', p') = \chi(p \cdot q')\, W(q + q', p + p')$$

and therefore

$$W(e)W(f) = \chi(\Omega(e, f))W(f)W(e).$$

The operators $W(e)$ are unitary because T_q permutes the basis and R_p is diagonal with unit-modulus entries. The multiplication law shows that $e \mapsto W(e)$ is a projective unitary representation of the additive group of E_{red}, with cocycle $c(e, f) = \chi(p \cdot q')$.

15.6 Definition: Arithmetic Quantum Field

An arithmetic quantum field datum is

$$(X, V, E, \Omega_E, W_E)$$

where X is finite, V is finite symplectic over F_q, $E \leq \mathrm{Map}(X, V)$ is a chosen finite field-label space, and W_E is the Weyl projective representation of E_{red}. The field displacement operators are

$$\mathsf{A}_{X,E} = \{W_E(e) : e \in E_{\mathrm{red}}\}.$$

The observable algebra of this finite arithmetic quantum field is

$$\mathcal{A}_{X,E} = \mathrm{span}_{\mathbb{C}} \mathsf{A}_{X,E} \subseteq \mathrm{End}(\mathcal{H}).$$

When $\dim_F E_{\mathrm{red}} = 2n$, the local Weyl source says these operators form an operator basis, so $\mathcal{A}_{X,E} \cong M_{q^n}(\mathbb{C})$.

15.7 Molecular F3 Examples

For the run we take $V = F_3^2$ with $\omega((q,p),(q',p')) = pq' - qp'$. A scalar function space $\mathcal{U} \leq \mathrm{Map}(X, F_3)$ gives

$$E = \mathcal{U}q \oplus \mathcal{U}p, \qquad B(f,g) = \sum_{x \in X} f(x)g(x),$$

$$\Omega((q,p),(q',p')) = B(p,q') - B(q,p').$$

If $m = \dim \mathcal{U}$ and $r = \mathrm{rank}\, B$, then

$$\dim E = 2m, \quad \mathrm{rank}\, \Omega = 2r, \quad \dim \mathrm{rad}(E) = 2(m - r).$$

example	$\lvert X\rvert$	$\dim \mathcal{U}$	r	$\dim \mathrm{rad}$	$\lvert E_{\mathrm{red}}\rvert$
two points, all maps	2	2	2	0	81
three points, all maps	3	3	3	0	729
F_3, $\langle t \rangle$	3	1	1	0	9
F_3, $\langle 1, t \rangle$	3	2	1	2	9
F_3, $\langle 1, t, t^2 \rangle$	3	3	3	0	729
F_3^2, $\langle x, y \rangle$	9	2	0	4	1
F_3^2, all maps	9	9	9	0	387420489
$y = x^2$, $\langle 1 \rangle$	3	1	0	2	1
$y = x^2$, $\langle x, y \rangle$	3	2	2	0	81
$y = x^2$, $\langle 1, x, y \rangle$	3	3	3	0	729
$G_m(F_3)$, $\langle 1, t \rangle$	2	2	2	0	81

The negative controls are meaningful. For affine fields on F_3, the constant function has zero norm because $1 + 1 + 1 = 0$ in F_3, so constants lie in the radical. For $\langle x, y \rangle$ on F_3^2, every sum $\sum x^2$, $\sum xy$, and $\sum y^2$ vanishes, so the entire field space is radical and the reduced Weyl label group is trivial.

For the parabola $X = \{(0,0),(1,1),(2,1)\}$, the checked basis values are

$$1 = (1,1,1), \qquad x = (0,1,2), \qquad y = (0,1,1),$$

with Gram matrix
$$\begin{pmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix},$$
which has rank 3. Thus degree-one ambient coordinate functions already give the full nondegenerate three-dimensional scalar function algebra on these three points.

15.8 Exact Validation

The producer is `scripts/bridges/arithmetic_quantum_fields.jl`. It writes `runs/2026-05-24-arithmetic-qua` `data/arithmetic_quantum_field_examples.csv` and `runs/2026-05-24-arithmetic-quantum-fields/` `data/arithmetic_quantum_field_bases.csv`. The summary CSV records point counts, scalar Gram ranks, symplectic ranks, radicals, reduced Weyl label counts, Hilbert dimensions, and observable-algebra dimensions. The basis CSV records every scalar basis vector and every scalar Gram row used in the examples above.

16 Projective Sheaf Field Definitions

16.1 Status and Ground Truth

This shard is definition-building, not a classification theorem. It is source-local where it recalls standard algebraic geometry and checked where it proves the finite linear-algebra statements. The local ground truth is the Stacks Project TeX snapshot registered in `references/algebraic_geometry/SOURCES.md`. We use:

- presheaves and sections: `stacks_project_sheaves.tex`, lines 73–123;
- sheaves: `stacks_project_sheaves.tex`, lines 500–525;
- Proj and its sheaves: `stacks_project_constructions.tex`, lines 939–1225;
- twists $\mathcal{O}(n)$: `stacks_project_constructions.tex`, lines 1695–1705;
- projective space: `stacks_project_constructions.tex`, lines 2775–2886;
- varieties and projective varieties: `stacks_project_varieties.tex`, lines 53–58 and 4843–4865;
- $H^0(\mathbf{P}^n, \mathcal{O}(d))$: `stacks_project_coherent.tex`, lines 1557–1622.

16.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & X \text{ is a topological space.} \\
D2 & : & \mathcal{F} \text{ is a presheaf on } X. \\
D3 & : & \mathcal{F} \text{ is a sheaf if compatible local sections glue uniquely.} \\
D4 & : & \Gamma(U,\mathcal{F}) = \mathcal{F}(U) = H^0(U,\mathcal{F}). \\
D5 & : & X \text{ is a variety, projective variety, or projective space over } F_q. \\
D6 & : & \mathcal{O}_X(d) \text{ is the } d\text{th twist on a projective space.} \\
D7 & : & (V,\omega_V) \text{ is a finite symplectic } F_q\text{-vector space.} \\
D8 & : & E(X,\mathcal{L},V) = H^0(X,\mathcal{L}) \otimes_{F_q} V. \\
D9 & : & \Omega \text{ is the finite rational-point pairing after choosing evaluations.} \\
L1 & : & E(X,\mathcal{L},V) \text{ is an } F_q\text{-vector space.} \\
L2 & : & \Omega \text{ is bilinear and alternating.} \\
L3 & : & E/\operatorname{rad}(E) \text{ is the honest Weyl label space.}
\end{array}$$

16.3 Presheaves

Let X be a topological space. A presheaf of sets $\mathcal{F}$ on X consists of two pieces of data.

First, every open subset $U \subset X$ is assigned a set $\mathcal{F}(U)$. Its elements are called sections over U. Second, every inclusion of open sets $V \subset U$ is assigned a restriction map

$$\rho^U_V : \mathcal{F}(U) \longrightarrow \mathcal{F}(V).$$

The maps must satisfy two identities:

$$\rho^U_U = \mathrm{id}_{\mathcal{F}(U)}, \qquad \rho^U_W = \rho^V_W \circ \rho^U_V \quad \text{when } W \subset V \subset U.$$

For $s \in \mathcal{F}(U)$, write

$$s|_V = \rho^U_V(s).$$

Thus $s|_W = (s|_V)|_W$. The notation

$$\Gamma(U, \mathcal{F}) = \mathcal{F}(U) = H^0(U, \mathcal{F})$$

means exactly the same set of sections over U; the last two symbols do not add a new construction in degree zero.

For a fixed field F_q, a presheaf of F_q-vector spaces is the same data with each $\mathcal{F}(U)$ an F_q-vector space and each restriction map F_q-linear. Equivalently, it is a presheaf of sets whose section sets carry compatible vector-space operations.

16.4 Sheaves

A sheaf is a presheaf satisfying a locality-and-gluing axiom. Let

$$U = \bigcup_{i \in I} U_i$$

be an open cover. Suppose $s_i \in \mathcal{F}(U_i)$ is a section on each member of the cover. The family is compatible if for every pair i, j,

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}.$$

The presheaf $\mathcal{F}$ is a sheaf if every compatible family has one and only one glued section $s \in \mathcal{F}(U)$ such that

$$s|_{U_i} = s_i \qquad \text{for all } i.$$

This says precisely: a sheaf is an object whose sections are determined locally, and whose locally compatible data assemble globally.

16.5 Varieties, Projective Space, and Twists

Let k be a field. In the Stacks convention, a variety over k is a scheme X over k such that X is integral and the structure morphism

$$X \longrightarrow \mathrm{Spec}(k)$$

is separated and of finite type. A projective variety is a variety for which this structure morphism is projective; equivalently in the cited Stacks source, it embeds as a closed subscheme of some $\mathbf{P}^n_k$.

Projective n-space over a ring R is

$$\mathbf{P}^n_R = \mathrm{Proj}(R[T_0, \dots, T_n]),$$

where every T_i has degree 1. Its standard opens are

$$D_+(T_i), \qquad i = 0, \dots, n,$$

and these cover $\mathbf{P}^n_R$. For $n = 1$, the two standard opens

$$D_+(X), \qquad D_+(Y)$$

are affine lines with coordinates Y/X and X/Y, respectively.

For $X = \mathrm{Proj}(S)$, the dth twist of the structure sheaf is

$$\mathcal{O}_X(d) = \widetilde{S(d)}.$$

In the special case $X = \mathbf{P}^n_R$, the local source computes

$$H^0(\mathbf{P}^n_R, \mathcal{O}(d)) = \begin{cases} R[T_0, \ldots, T_n]_d, & d \geq 0, \\ 0, & d < 0, \end{cases}$$

where $R[T_0, \ldots, T_n]_d$ is the degree-d homogeneous part. Thus

$$H^0(\mathbf{P}^1_{F_q}, \mathcal{O}(d)) = F_q\text{-span}\{X^{d-i}Y^i : 0 \leq i \leq d\}$$

for $d \geq 0$.

16.6 Why Naive Hom Is Not the Projective Default

For a finite set X, the previous shard used $E \leq \mathrm{Map}(X, V)$. For a projective variety this is too naive if $\mathrm{Hom}(X, V)$ is read as regular maps to an affine vector space. A regular map to affine space is controlled by global regular functions, and the Stacks source records that a proper geometrically integral variety has

$$H^0(X, \mathcal{O}_X) = k.$$

So for connected projective spaces this would see only constants.

The projective replacement is therefore sheaf-theoretic:

$$E(X, \mathcal{L}, V) = H^0(X, \mathcal{L}) \otimes_{F_q} V,$$

where $\mathcal{L}$ is an explicitly chosen sheaf on X. The sheaf is part of the quantum-field datum. Different choices $\mathcal{L} = \mathcal{O}(d)$ give different finite field-label spaces.

16.7 Finite Symplectic Target and Pointwise Form

Let V be a finite-dimensional F_q-vector space. A symplectic form on V is an F_q-bilinear pairing

$$\omega_V : V \times V \to F_q$$

such that $\omega_V(v, v) = 0$ for all v and the only vector pairing to zero with every vector is 0. The standard two-dimensional example is

$$V = F_q^2, \qquad \omega_V((a, b), (c, d)) = bc - ad.$$

To turn $E(X, \mathcal{L}, V)$ into a finite Weyl label space we also need a finite scalar pairing on $H^0(X, \mathcal{L})$. Let $S \subset X(F_q)$ be a finite set of rational points, let $w_P \in F_q$, and choose an evaluation coordinate

$$\mathrm{ev}_P : H^0(X, \mathcal{L}) \to F_q$$

for every $P \in S$. Then define

$$B(s, t) = \sum_{P \in S} w_P \; \mathrm{ev}_P(s) \, \mathrm{ev}_P(t).$$

This evaluation choice is automatic for honest scalar functions, but for sections of a nontrivial line bundle it is a convention: one has chosen local trivializations or fixed projective representatives at the points.

For pure tensors set

$$\Omega(s \otimes v, t \otimes w) = B(s, t)\omega_V(v, w),$$

and extend bilinearly to all of $E(X, \mathcal{L}, V)$.

Lemma $L1$. $E(X, \mathcal{L}, V)$ is an F_q-vector space.

Proof. The source definition of a sheaf of vector spaces gives $H^0(X, \mathcal{L}) = \mathcal{L}(X)$ as an F_q-vector space. The tensor product of two F_q-vector spaces is an F_q-vector space, with addition and scalar multiplication inherited from the tensor construction. Hence $H^0(X, \mathcal{L}) \otimes_{F_q} V$ is an F_q-vector space.

Lemma $L2$. Ω is F_q-bilinear and alternating.

Proof. The map B is bilinear because it is a sum of products of linear evaluation maps. It is also symmetric from the displayed formula: $B(s,t) = B(t,s)$. The form ω_V is bilinear by definition. Therefore their tensor product pairing is bilinear on pure tensors and hence bilinear after extension. For a pure tensor,

$$\Omega(s \otimes v, s \otimes v) = B(s,s)\omega_V(v,v) = 0,$$

but arbitrary sums require the mixed terms. Let

$$\phi = \sum_i s_i \otimes v_i.$$

Then

$$\Omega(\phi,\phi) = \sum_i B(s_i,s_i)\omega_V(v_i,v_i) + \sum_{i<j} \big(B(s_i,s_j)\omega_V(v_i,v_j) + B(s_j,s_i)\omega_V(v_j,v_i)\big).$$

The first sum is zero because $\omega_V(v_i,v_i) = 0$. For the mixed terms, symmetry of B gives the common scalar $B(s_i,s_j)$, and the alternating property of ω_V gives

$$0 = \omega_V(v_i + v_j, v_i + v_j) = \omega_V(v_i,v_j) + \omega_V(v_j,v_i).$$

Thus every mixed pair cancels, so $\Omega(\phi,\phi) = 0$. This last step uses the diagonal alternating identity itself, not an inference from skew-symmetry; in characteristic 2, skew-symmetry and alternation are not interchangeable.

Lemma $L3$. Let

$$\operatorname{rad}(E) = \{\phi \in E : \Omega(\phi,\psi) = 0 \text{ for every } \psi \in E\}.$$

Then $E_{\text{red}} = E/\operatorname{rad}(E)$ has a nondegenerate alternating form induced by Ω.

Proof. If $r \in \operatorname{rad}(E)$, then $\Omega(\phi + r, \psi) = \Omega(\phi,\psi)$, so the form descends to the quotient. If a quotient class $[\phi]$ pairs to zero with every quotient class, then ϕ pairs to zero with every element of E, hence $\phi \in \operatorname{rad}(E)$ and $[\phi] = 0$. Thus the descended form is nondegenerate.

The Weyl-Heisenberg field displacement algebra is built from E_{red}, not from unreduced E, exactly as in Shard *AQM*-14.

17 Projective Line Sheaf Field Calculation

17.1 Status and Local Evidence

This shard is self-contained for the example

$$X = \mathbf{P}^1_{\mathbb{F}_3}, \qquad V = \mathbb{F}_3^2.$$

The algebraic geometry inputs are the local Stacks TeX files cited in Shard AQM-15. The exact finite calculation is checked by `scripts/bridges/projective_line_sheaf_fields.jl`. The AQM-16 output files are `runs/2026-05-24-projective-line-sheaf-fields/data/projective_line_sheaf_field_summary.csv` and `runs/2026-05-24-projective-line-sheaf-fields/data/projective_line_sheaf_field_basis_rows.csv`.

17.2 The Field and the Points

The field is

$$\mathbb{F}_3 = \{0, 1, 2\}, \qquad 2 = -1, \qquad 1 + 1 + 1 = 0.$$

The projective line has rational points given by nonzero pairs $(a, b) \in \mathbb{F}_3^2 \setminus \{(0, 0)\}$ modulo nonzero scalar multiplication:

$$[a : b] = [\lambda a : \lambda b] \quad (\lambda \in \mathbb{F}_3^\times).$$

There are four classes. The chosen representatives and order are

$$P_\infty = [1 : 0], \quad P_0 = [0 : 1], \quad P_1 = [1 : 1], \quad P_2 = [2 : 1].$$

17.3 The Sheaves and Their Sections

For every $d \geq 0$, the source computation gives

$$H^0(\mathbf{P}^1_{\mathbb{F}_3}, \mathcal{O}(d)) = \mathbb{F}_3[X, Y]_d.$$

Thus a basis is

$$X^d,\ X^{d-1}Y,\ X^{d-2}Y^2,\ \ldots,\ Y^d.$$

This is a statement about sections of $\mathcal{O}(d)$, not about global regular functions when $d > 0$.

For this finite rational-point calculation, evaluate $X^{d-i}Y^i$ on the fixed representatives above:

$$X^{d-i}Y^i(P_\infty) = \begin{cases} 1, & i = 0, \\ 0, & i > 0, \end{cases}$$

and

$$X^{d-i}Y^i(P_t) = t^{d-i} \qquad (t = 0, 1, 2).$$

This is the explicit chart-coordinate evaluation convention from CONVENTIONS (r). It is not silently canonical for every line bundle.

17.4 Scalar Pairing and Symplectic Extension

For scalar sections s, t, define

$$B(s,t) = \sum_{P \in \mathbf{P}^1(\mathbb{F}_3)} s(P)t(P).$$

All weights are 1. The target is $V = \mathbb{F}_3^2$ with $\omega((q,p),(q',p')) = pq' - qp'$. The field space is

$$E_d = H^0(\mathbf{P}^1, \mathcal{O}(d)) \otimes_{\mathbb{F}_3} V.$$

If $U_d = H^0(\mathbf{P}^1, \mathcal{O}(d))$, write

$$E_d = U_d q \oplus U_d p.$$

For $(q,p),(q',p') \in U_d q \oplus U_d p$,

$$\Omega_d((q,p),(q',p')) = B(p,q') - B(q,p').$$

If the scalar Gram matrix $G_d = (B(e_i, e_j))$ has rank r, then

$$\dim E_d = 2(d+1), \quad \operatorname{rank} \Omega_d = 2r, \quad \dim \operatorname{rad}(E_d) = 2(d+1-r).$$

The Weyl label space is $E_{d,\mathrm{red}} = E_d / \operatorname{rad}(E_d)$.

17.5 Degree 0: Constants

The basis is 1. Its value row is

$$1: \quad (1,1,1,1).$$

Therefore

$$B(1,1) = 1+1+1+1 = 4 = 1 \in \mathbb{F}_3.$$

So

$$G_0 = [1].$$

The scalar rank is 1, hence Ω_0 has rank 2. The reduced field has dimension 2, the Weyl Hilbert dimension is 3, and the Weyl operator basis has 9 elements.

17.6 Degree 1: Linear Sections

The basis is X, Y. The value rows are

$$X: \ (1,0,1,2), \qquad Y: \ (0,1,1,1).$$

Compute every scalar product:

$$\begin{aligned} B(X,X) &= 1^2 + 0^2 + 1^2 + 2^2 = 1+0+1+4 = 6 = 0, \\ B(X,Y) &= 1\cdot 0 + 0 \cdot 1 + 1 \cdot 1 + 2 \cdot 1 = 3 = 0, \\ B(Y,Y) &= 0^2 + 1^2 + 1^2 + 1^2 = 3 = 0. \end{aligned}$$

Thus

$$G_1 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The two sections evaluate independently at the four points, but the chosen all-weights-one scalar pairing is totally zero on their span. Consequently

$$\operatorname{rank} \Omega_1 = 0, \qquad \operatorname{rad}(E_1) = E_1.$$

The reduced Weyl label space is zero-dimensional. This is the first concrete warning that evaluation rank and symplectic rank are different questions.

17.7 Degree 2: Quadratic Sections

The basis is X^2, XY, Y^2. The value rows are

	P_∞	P_0	P_1	P_2
X^2	1	0	1	1
XY	0	0	1	2
Y^2	0	1	1	1

The scalar products give

$$G_2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{pmatrix}.$$

For example,

$$B(X^2, Y^2) = 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 2,$$

and

$$B(XY, XY) = 0 + 0 + 1 + 4 = 5 = 2.$$

The determinant is

$$\det(G_2) = 1 \in \mathbb{F}_3,$$

so G_2 has rank 3. Hence Ω_2 has rank 6, no radical, and

$$\dim E_2 = 6, \qquad \dim \mathcal{H} = 3^3 = 27, \qquad |\{W(e)\}| = 3^6 = 729.$$

17.8 Degree 3: Cubic Sections

The basis rows and Gram matrix are

	P_∞	P_0	P_1	P_2
X^3	1	0	1	2
X^2Y	0	0	1	1
XY^2	0	0	1	2
Y^3	0	1	1	1

and

$$G_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \end{pmatrix}.$$

The validator computes rank 4, equivalently $\det(G_3) = 1$. Thus

$$\dim E_3 = 8, \quad \operatorname{rank} \Omega_3 = 8, \quad \dim \mathcal{H} = 3^4 = 81.$$

17.9 Degree 4: First Evaluation Kernel

The scalar section dimension is 5, but there are only four rational points. The value rows include

$$X^3Y : \ (0, 0, 1, 2), \qquad XY^3 : \ (0, 0, 1, 2),$$

so

$$X^3Y - XY^3$$

evaluates to zero at all four chosen representatives. Therefore the scalar evaluation map has a one-dimensional kernel. The run confirms

$$\mathrm{rank}(\mathrm{ev}) = 4, \qquad \mathrm{rank}(G_4) = 4.$$

The scalar radical is exactly one-dimensional, and after tensoring with $V = \mathbb{F}_3^2$,

$$\dim \mathrm{rad}(E_4) = 2.$$

Thus

$$\dim E_4 = 10, \qquad \dim E_{4,\mathrm{red}} = 8, \qquad \dim \mathcal{H} = 3^4 = 81.$$

17.10 Run Summary

d	$\dim U_d$	rank(ev)	$\mathrm{rank}\, G_d$	$\dim \mathrm{rad}\, E_d$	$\dim E_{d,\mathrm{red}}$	$\dim \mathcal{H}$
0	1	1	1	0	2	3
1	2	2	0	4	0	1
2	3	3	3	0	6	27
3	4	4	4	0	8	81
4	5	4	4	2	8	81

The lessons are concrete: the projective sheaf field is controlled by $H^0(\mathcal{O}(d))$, the finite rational-point pairing is an additional choice and can introduce radicals, and the Weyl-Heisenberg observable algebra is formed after quotienting by this radical.

18 Projective Line Stalks

18.1 Status and Local Evidence

This shard continues Shard *AQM*-16. The phrase "each point in the example" means the four rational points used by the finite pairing: [1 : 0], [0 : 1], [1 : 1], and [2 : 1]. The scheme has more points than these four; a general formula for those other stalks is recorded at the end.

The source definition of a stalk is local: `references/algebraic_geometry/stacks_project_sheaves.tex`, lines 894–944. The source formula for stalks on $\mathrm{Proj}(S)$ is `references/algebraic_geometry/stacks_project_constructions.tex`, lines 1091–1116 and 1182–1207. The checked rows are produced by `scripts/bridges/projective_line_sheaf_fields.jl`.

18.2 What a Stalk Is

Let X be a topological space, $x \in X$, and $\mathcal{F}$ a presheaf. The stalk at x is

$$\mathcal{F}_x = \operatorname*{colim}_{x \in U} \mathcal{F}(U),$$

where U runs over open neighbourhoods of x. Concretely, an element is represented by (U, s), with $x \in U$ and $s \in \mathcal{F}(U)$, and $(U, s) \sim (U', s')$ when some open neighbourhood $U'' \subset U \cap U'$ of x satisfies $s|_{U''} = s'|_{U''}$. Thus a stalk element is a germ: it remembers the section only arbitrarily near the point.

18.3 The Proj Stalk Formula

Let

$$S = \mathbb{F}_3[X, Y], \qquad \deg X = \deg Y = 1, \qquad X_{\mathrm{sch}} = \mathrm{Proj}(S) = \mathbf{P}^1_{\mathbb{F}_3}.$$

A point of X_{sch} is a homogeneous prime $\mathfrak{p} \subset S$ not containing the irrelevant ideal (X, Y). For a graded S-module M, the associated sheaf $\widetilde{M}$ has stalk

$$(\widetilde{M})_{\mathfrak{p}} = M_{(\mathfrak{p})}.$$

The right hand side consists of homogeneous fractions

$$m/f, \qquad m \in M,\ f \in S \text{ homogeneous}, \quad \deg_M m = \deg_S f, \quad f \notin \mathfrak{p}.$$

For a shifted module such as $S(d)$, the equality of degrees is in the shifted grading of $S(d)$, while the denominator remains an element of S. For the structure sheaf, this gives

$$\mathcal{O}_{X_{\mathrm{sch}},\mathfrak{p}} = S_{(\mathfrak{p})}.$$

For the twist $\mathcal{O}(d) = \widetilde{S(d)}$,

$$\mathcal{O}(d)_{\mathfrak{p}} = S(d)_{(\mathfrak{p})}.$$

18.4 The Two Standard Charts

On the chart $D_+(X)$, set $u = Y/X$. Then

$$D_+(X) = \mathrm{Spec}\, \mathbb{F}_3[u].$$

On this chart we use the local frame $e_X^{(d)}$ of $\mathcal{O}(d)$ corresponding to X^d. The section

$$X^{d-i}Y^i$$

has local expression

$$u^i e_X^{(d)}.$$

On the chart $D_+(Y)$, set $v = X/Y$. Then

$$D_+(Y) = \operatorname{Spec} \mathbb{F}_3[v].$$

On this chart we use the local frame $e_Y^{(d)}$ corresponding to Y^d. The same section has local expression

$$v^{d-i} e_Y^{(d)}.$$

On the overlap $D_+(XY)$, $u = 1/v$, and the two frames satisfy

$$e_X^{(d)} = v^d e_Y^{(d)}, \qquad e_Y^{(d)} = u^d e_X^{(d)}.$$

This is why both local expressions describe the same germ where both charts are available.

18.5 Point P-Infinity: [1:0]

The homogeneous prime is

$$\mathfrak{p}_\infty = (Y).$$

Since $X \notin \mathfrak{p}_\infty$, this point lies in $D_+(X)$. In the coordinate $u = Y/X$, the corresponding prime of $\mathbb{F}_3[u]$ is

$$(u).$$

Therefore

$$\mathcal{O}_{P_\infty} = \mathbb{F}_3[u]_{(u)}, \qquad \mathfrak{m}_{P_\infty} = (u), \qquad \kappa(P_\infty) = \mathbb{F}_3[u]_{(u)}/(u) = \mathbb{F}_3.$$

The $\mathcal{O}(d)$-stalk is free of rank one:

$$\mathcal{O}(d)_{P_\infty} = \mathbb{F}_3[u]_{(u)} e_X^{(d)}.$$

For the monomial $X^{d-i}Y^i$,

$$(X^{d-i}Y^i)_{P_\infty} = u^i e_X^{(d)}.$$

Reducing modulo (u) gives

$$u^i e_X^{(d)} \longmapsto \begin{cases} 1 \cdot \overline{e}_X^{(d)}, & i = 0, \\ 0, & i > 0. \end{cases}$$

This is exactly the evaluation rule used earlier at $[1:0]$.

18.6 Point P0: [0:1]

The homogeneous prime is

$$\mathfrak{p}_0 = (X).$$

Since $Y \notin \mathfrak{p}_0$, use $D_+(Y)$ and $v = X/Y$. The local prime is (v). Hence

$$\mathcal{O}_{P_0} = \mathbb{F}_3[v]_{(v)}, \qquad \mathfrak{m}_{P_0} = (v), \qquad \kappa(P_0) = \mathbb{F}_3.$$

The twist stalk is

$$\mathcal{O}(d)_{P_0} = \mathbb{F}_3[v]_{(v)} e_Y^{(d)}.$$

The monomial $X^{d-i}Y^i$ has germ

$$v^{d-i} e_Y^{(d)}.$$

Modulo (v),

$$v^{d-i} e_Y^{(d)} \longmapsto \begin{cases} 1 \cdot \bar{e}_Y^{(d)}, & i = d, \\ 0, & i < d. \end{cases}$$

18.7 Point P1: [1:1]

The homogeneous prime is

$$\mathfrak{p}_1 = (X - Y).$$

On $D_+(Y)$, it becomes

$$(v - 1) \subset \mathbb{F}_3[v].$$

Therefore

$$\mathcal{O}_{P_1} = \mathbb{F}_3[v]_{(v-1)}, \qquad \mathfrak{m}_{P_1} = (v - 1), \qquad \kappa(P_1) = \mathbb{F}_3.$$

The twist stalk is

$$\mathcal{O}(d)_{P_1} = \mathbb{F}_3[v]_{(v-1)} e_Y^{(d)}.$$

The germ of $X^{d-i}Y^i$ is

$$v^{d-i} e_Y^{(d)}.$$

Modulo $(v - 1)$, we set $v = 1$, so

$$v^{d-i} e_Y^{(d)} \longmapsto 1 \cdot \bar{e}_Y^{(d)}$$

for every i. This explains why every monomial evaluates to 1 at $[1 : 1]$.

18.8 Point P2: [2:1]

The homogeneous prime is

$$\mathfrak{p}_2 = (X - 2Y) = (X + Y).$$

On $D_+(Y)$, it becomes

$$(v - 2) \subset \mathbb{F}_3[v].$$

Thus

$$\mathcal{O}_{P_2} = \mathbb{F}_3[v]_{(v-2)}, \qquad \mathfrak{m}_{P_2} = (v - 2), \qquad \kappa(P_2) = \mathbb{F}_3.$$

The twist stalk is

$$\mathcal{O}(d)_{P_2} = \mathbb{F}_3[v]_{(v-2)} e_Y^{(d)}.$$

The germ of $X^{d-i}Y^i$ is

$$v^{d-i}e_Y^{(d)}.$$

Modulo $(v-2)$, we set $v=2=-1$. Therefore

$$v^{d-i}e_Y^{(d)} \longmapsto 2^{d-i}\bar{e}_Y^{(d)}.$$

The residue is 1 when $d-i$ is even and 2 when $d-i$ is odd.

18.9 Target-Valued Stalks

The arithmetic field sheaf used in the example is

$$\mathcal{O}(d) \otimes_{\mathbb{F}_3} V, \qquad V = \mathbb{F}_3^2.$$

At any rational point P, its stalk is

$$(\mathcal{O}(d) \otimes V)_P = \mathcal{O}(d)_P \otimes_{\mathbb{F}_3} V.$$

Because $\mathcal{O}(d)_P$ is a free rank-one $\mathcal{O}_P$-module, this is two copies of the local ring:

$$(\mathcal{O}(d) \otimes V)_P \cong \mathcal{O}_P q \oplus \mathcal{O}_P p.$$

The residue at P is

$$(\mathcal{O}(d) \otimes V)_P \longrightarrow \kappa(P) \otimes_{\mathbb{F}_3} V = V,$$

and this is the pointwise value used in the finite symplectic pairing.

18.10 Other Scheme Points

The finite field algebra in Shard AQM-16 uses only rational points, but the scheme has more stalks. The generic point has $\mathcal{O}_\eta = \mathbb{F}_3(v)$ and $\mathcal{O}(d)_\eta = \mathbb{F}_3(v)e_Y^{(d)}$. A non-rational closed point on $D_+(Y)$ is given by an irreducible $h(v) \in \mathbb{F}_3[v]$ of degree > 1; then

$$\mathcal{O}_h = \mathbb{F}_3[v]_{(h)}, \quad \mathcal{O}(d)_h = \mathbb{F}_3[v]_{(h)}e_Y^{(d)}, \quad \kappa(h) = \mathbb{F}_3[v]/(h).$$

These stalks are not sampled by the current rational-point pairing.

19 Stalks as Field Germs

19.1 Answer to the Terminology Question

It is useful, but slightly dangerous, to say that a stalk is the "field at the point." The precise dictionary used here is:

$$\text{stalk} = \text{local field germ at the point}, \qquad \text{fiber} = \text{actual value after residue reduction.}$$

The source definition of a stalk as germs is `references/algebraic_geometry/stacks_project_sheaves.tex`, lines 894–944. The source Proj stalk formula is `references/algebraic_geometry/stacks_project_constructions.tex`, lines 1091–1116 and 1182–1207. The checked finite table is `runs/2026-05-24-projective-line-sheaf-fields/data/projective_line_stalk_rows.csv`.

Let $x \in X$ and let $\mathcal{E}$ be a sheaf of $\mathcal{O}_X$-modules. The stalk $\mathcal{E}_x$ consists of germs of sections near x. If X is a scheme, $\mathcal{O}_x$ is a local ring, with maximal ideal $\mathfrak{m}_x$, and residue field

$$\kappa(x) = \mathcal{O}_x/\mathfrak{m}_x.$$

The fiber of $\mathcal{E}$ at x is

$$\mathcal{E}(x) = \mathcal{E}_x \otimes_{\mathcal{O}_x} \kappa(x).$$

Thus $\mathcal{E}_x$ remembers infinitesimal or algebraic variation near x, while $\mathcal{E}(x)$ is the value obtained after setting the local coordinates equal to their value at x.

19.2 The Projective-Line Field Sheaf

In the current example

$$X = \mathbf{P}^1_{\mathbb{F}_3}, \qquad V = \mathbb{F}_3 q \oplus \mathbb{F}_3 p, \qquad \mathcal{E}_d = \mathcal{O}(d) \otimes_{\mathbb{F}_3} V.$$

The rational points are ordered as

$$P_\infty = [1:0], \quad P_0 = [0:1], \quad P_1 = [1:1], \quad P_2 = [2:1].$$

For every rational point P, the residue field is $\kappa(P) = \mathbb{F}_3$. Hence

$$\mathcal{E}_d(P) = \mathcal{E}_{d,P} \otimes_{\mathcal{O}_P} \kappa(P) \cong \mathbb{F}_3 q \oplus \mathbb{F}_3 p = V.$$

This is the exact sense in which the arithmetic quantum field has a point-value in V.

The stalks themselves are larger local modules. With

$$u = Y/X \quad \text{on } D_+(X), \qquad v = X/Y \quad \text{on } D_+(Y),$$

and local frames $e_X^{(d)}$, $e_Y^{(d)}$, the four rational-point stalks are:

P	$\mathcal{O}_P$	$\mathcal{O}(d)_P$	$\mathcal{E}_{d,P}$
P_∞	$\mathbb{F}_3[u]_{(u)}$	$\mathbb{F}_3[u]_{(u)} e_X^{(d)}$	$\mathbb{F}_3[u]_{(u)} e_X^{(d)} \otimes V$
P_0	$\mathbb{F}_3[v]_{(v)}$	$\mathbb{F}_3[v]_{(v)} e_Y^{(d)}$	$\mathbb{F}_3[v]_{(v)} e_Y^{(d)} \otimes V$
P_1	$\mathbb{F}_3[v]_{(v-1)}$	$\mathbb{F}_3[v]_{(v-1)} e_Y^{(d)}$	$\mathbb{F}_3[v]_{(v-1)} e_Y^{(d)} \otimes V$
P_2	$\mathbb{F}_3[v]_{(v-2)}$	$\mathbb{F}_3[v]_{(v-2)} e_Y^{(d)}$	$\mathbb{F}_3[v]_{(v-2)} e_Y^{(d)} \otimes V.$

19.3 Germ-to-Value Maps

Let

$$s_{d,i} = X^{d-i}Y^i, \qquad 0 \le i \le d,$$

and let $w = aq + bp \in V$. The vector-valued section is

$$\Phi_{d,i,w} = s_{d,i} \otimes w.$$

Its stalk germ and fiber value at the four rational points are:

P	germ in $\mathcal{E}_{d,P}$	fiber value in V
P_∞	$u^i e_X^{(d)} \otimes w$	$\begin{cases} w, & i = 0, \\ 0, & i > 0, \end{cases}$
P_0	$v^{d-i} e_Y^{(d)} \otimes w$	$\begin{cases} w, & i = d, \\ 0, & i < d, \end{cases}$
P_1	$v^{d-i} e_Y^{(d)} \otimes w$	w
P_2	$v^{d-i} e_Y^{(d)} \otimes w$	$2^{d-i} w.$

The last line uses $v = 2 = -1$ in $\mathbb{F}_3$. If $w = (a, b)$ in the ordered basis (q, p), multiplication by the scalar $\lambda \in \mathbb{F}_3$ means

$$\lambda w = (\lambda a, \lambda b).$$

These maps are not extra choices: they are the quotient maps $\mathcal{E}_{d,P} \to \mathcal{E}_d(P)$ induced by $\mathcal{O}_P \to \kappa(P)$.

19.4 All Scalar Residues for Degrees 0 Through 4

The following table gives the scalar $\lambda_{d,i}(P)$ such that

$$\Phi_{d,i,w}(P) = \lambda_{d,i}(P)\, w.$$

It is exactly the stalk-residue layer underlying the evaluation matrices in Shard AQM-16.

d	$s_{d,i}$	P_∞	P_0	P_1	P_2
0	1	1	1	1	1
1	X	1	0	1	2
1	Y	0	1	1	1
2	X^2	1	0	1	1
2	XY	0	0	1	2
2	Y^2	0	1	1	1
3	X^3	1	0	1	2
3	X^2Y	0	0	1	1
3	XY^2	0	0	1	2
3	Y^3	0	1	1	1
4	X^4	1	0	1	1
4	X^3Y	0	0	1	2
4	X^2Y^2	0	0	1	1
4	XY^3	0	0	1	2
4	Y^4	0	1	1	1.

19.5 Concrete Reading of the Table

For $d = 4$, the section $X^3Y \otimes (q + 2p)$ has:

P	germ	value
P_∞	$ue_X^{(4)} \otimes (q + 2p)$	0
P_0	$v^3 e_Y^{(4)} \otimes (q + 2p)$	0
P_1	$v^3 e_Y^{(4)} \otimes (q + 2p)$	$q + 2p$
P_2	$v^3 e_Y^{(4)} \otimes (q + 2p)$	$2q + p.$

The P_2 value is $2(q + 2p) = 2q + 4p = 2q + p$ in $\mathbb{F}_3$.

Thus a stalk is the correct local object if one wants to discuss the field near a point. The finite Weyl label construction samples the residue/fiber values, not the whole local stalk module.

20 Canonical Boson and Fermion Fields as Field-Label Tests

20.1 Status and Source Anchors

This shard is a kinematic comparison. It does not introduce dynamics, Hamiltonians, continuum limits, locality cones, or equations of motion. The question is only whether the field-label construction from Shard AQM-14 reproduces the standard finite-mode canonical fields.

The CCR and CAR ground truth is local; paths and hashes are registered in `references/canonical_fields/SOURCES.md`. The line anchors used here are: Derezinski's `derezinski.tex`, lines 1078–1135 for Weyl CCR, 1197–1284 for Schrodinger/Stone–von Neumann, 1437–1524 for CAR, 1612–1696 for finite CAR representations, 1803–1964 and 2098–2131 for Fock and creation operators, 2251–2301 for bosonic Fock CCR fields, and 2629–2662 for fermionic Fock CAR fields; Bekka's `CCR-GeneralRings-v5.tex`, lines 220–231, 253–273, 287–350, and 603–633; and Keyl–Schlingemann's `fidelity_arXiv.tex`, lines 253–260, 270–355, 411–492, and 496–531.

20.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & X \text{ is a finite set of sites.}\\
D2 & : & V_B = \mathbb{R}_q \oplus \mathbb{R}_p, \quad \omega_B((q,p),(q',p')) = pq' - p'q.\\
D3 & : & E_B = \mathrm{Map}(X, V_B), \quad \Omega_B = \sum_{x\in X} \omega_B.\\
D4 & : & W_B : E_B \to U(\mathcal{H}) \text{ satisfies Weyl-form CCR.}\\
L1 & : & (E_B, \Omega_B) \text{ is a real symplectic vector space.}\\
L2 & : & \text{Irreducible regular } W_B \text{ is Schrodinger by Stone–von Neumann.}\\
T1 & : & D1\text{–}D4 \text{ reproduce the finite standard bosonic field.}\\
D5 & : & Z = \mathrm{Map}(X, K) \text{ is a finite complex one-particle space.}\\
D6 & : & Y_F = \mathrm{Re}(Z \oplus \overline{Z}), \quad \alpha((z,\bar z),(w,\bar w)) = \mathrm{Re}\langle z, w\rangle.\\
D7 & : & \Gamma_a(Z) = \bigoplus_k \wedge^k Z, \quad \phi(z,\bar z) = a^*(z) + a(z).\\
L3 & : & a, a^* \text{ satisfy the finite exterior-algebra CAR.}\\
L4 & : & \{\phi(y), \phi(y')\} = 2\alpha(y,y').\\
T2 & : & D5\text{–}D7 \text{ reproduce the finite standard fermionic field.}\\
L5 & : & \text{Grassmann target replacement is not the same construction as } T2.
\end{array}
$$

20.3 Bosonic Field Labels D1–D4

Let X be finite. Write

$$\mathcal{Q}_X = \mathrm{Map}(X, \mathbb{R})$$

for the real configuration space. Its algebraic dual is identified with $\mathrm{Map}(X, \mathbb{R})$ by the finite sum

$$p(q) = \sum_{x\in X} p_x q_x.$$

Thus

$$E_B = \mathrm{Map}(X, \mathbb{R}_q \oplus \mathbb{R}_p) \cong \mathcal{Q}_X \oplus \mathcal{Q}_X^*.$$

For $\phi = (q,p)$ and $\psi = (q',p')$, define

$$\Omega_B(\phi,\psi) = \sum_{x\in X} \bigl(p_x q'_x - p'_x q_x\bigr).$$

A Weyl-form CCR representation of E_B is a map

$$W_B : E_B \to U(\mathcal{H})$$

such that

$$W_B(\phi)W_B(\psi) = \exp\left(-\frac{i}{2}\Omega_B(\phi,\psi)\right) W_B(\phi+\psi).$$

This is Derezinski's convention with $\eta = p$ and $q = q$.

Lemma $L1$. (E_B, Ω_B) is a finite-dimensional real symplectic vector space. *Proof.* Pointwise addition and scalar multiplication make E_B a real vector space. The formula for Ω_B is a finite sum of bilinear expressions, hence is bilinear. Also

$$\Omega_B(\phi,\phi) = \sum_x (p_x q_x - p_x q_x) = 0,$$

so it is alternating. If $\phi = (q,p) \neq 0$, then either some $q_{x_0} \neq 0$ or some $p_{x_0} \neq 0$. In the first case take ψ supported at x_0 with $p'_{x_0} = 1$ and $q'_{x_0} = 0$; then $\Omega_B(\phi,\psi) = -q_{x_0} \neq 0$. In the second case take ψ supported at x_0 with $q'_{x_0} = 1$ and $p'_{x_0} = 0$; then $\Omega_B(\phi,\psi) = p_{x_0} \neq 0$. Thus the radical is zero.

Lemma $L2$. Every irreducible regular representation of the Weyl-form CCR over (E_B, Ω_B) is unitarily equivalent to the Schrodinger representation on

$$L^2(\mathcal{Q}_X) \cong L^2(\mathbb{R}^{|X|}).$$

Proof. Choose an ordering of X. Then $\mathcal{Q}_X \cong \mathbb{R}^N$ for $N = |X|$, and Lemma $L1$ identifies E_B with $\mathcal{Q}_X^* \oplus \mathcal{Q}_X$ carrying the canonical symplectic form. Derezinski's finite-dimensional Stone–von Neumann theorem states that a regular CCR representation over such a finite symplectic space is a Schrodinger representation up to multiplicity, and irreducibility is exactly the one-copy case. Bekka's source records the same local-field uniqueness principle and the irreducibility of the Schrodinger pair.

Theorem $T1$. For finite X, the AQM-14 field-label recipe with real target V_B recovers the standard finite bosonic canonical field. *Proof.* The input is exactly $D1$–$D3$: a finite physical set, a target vector space with a scalar-valued alternating form, and a field-label space of maps. Lemma $L1$ checks that the summed pointwise form is symplectic. Definition $D4$ is the Weyl quantization step. By Lemma $L2$, the irreducible regular output is, up to unitary equivalence, the Schrodinger representation on $L^2(\mathbb{R}^{|X|})$. Derezinski's bosonic Fock source gives the equivalent finite-mode Fock realization by creation and annihilation operators. Therefore the finite standard bosonic field is a direct real analogue of the arithmetic finite-field Weyl construction, with $\mathbb{F}_q$ replaced by $\mathbb{R}$.

20.4 Fermionic Field Labels D5–D7

Now let K be a finite-dimensional complex internal one-particle Hilbert space and set

$$Z = \mathrm{Map}(X, K).$$

Its inner product is the finite sum

$$\langle z, w\rangle_Z = \sum_{x\in X} \langle z_x, w_x\rangle_K.$$

The real CAR label space is

$$Y_F = \mathrm{Re}(Z \oplus \overline{Z}) = \{(z, \bar{z}) : z \in Z\},$$

equipped with

$$\alpha((z,\bar{z}),(w,\bar{w})) = \mathrm{Re}\langle z,w\rangle_Z.$$

The fermionic Hilbert space is the antisymmetric Fock space

$$\Gamma_a(Z) = \bigoplus_{k=0}^{\dim Z} \wedge^k Z.$$

For $z \in Z$, define creation by $a^*(z)\Psi = z \wedge \Psi$. Define annihilation $a(z)$ as the adjoint of $a^*(z)$, equivalently on a decomposable wedge by

$$a(z)(w_1 \wedge \cdots \wedge w_k) = \sum_{j=1}^{k} (-1)^{j-1} \langle z, w_j\rangle\, w_1 \wedge \cdots \widehat{w_j} \cdots \wedge w_k.$$

The real fermion field operator is

$$\phi(z,\bar{z}) = a^*(z) + a(z).$$

Lemma $L3$. The exterior creation and annihilation operators satisfy

$$\{a^*(z), a^*(w)\} = 0, \qquad \{a(z), a(w)\} = 0, \qquad \{a(z), a^*(w)\} = \langle z, w\rangle I.$$

Proof. The first identity is the antisymmetry of the wedge product:

$$z \wedge w \wedge \Psi + w \wedge z \wedge \Psi = 0.$$

The second follows from the displayed contraction formula, since deleting two vectors in opposite orders gives opposite signs. For the mixed relation, use the contraction identity

$$a(z)(w \wedge \Psi) = \langle z, w\rangle \Psi - w \wedge a(z)\Psi.$$

Therefore

$$\begin{aligned} a(z)a^*(w)\Psi + a^*(w)a(z)\Psi &= a(z)(w \wedge \Psi) + w \wedge a(z)\Psi \\ &= \langle z, w\rangle \Psi. \end{aligned}$$

This proves the three CAR identities.

Lemma $L4$. For $y = (z,\bar{z})$ and $y' = (w,\bar{w})$,

$$\{\phi(y), \phi(y')\} = 2\alpha(y,y')I.$$

Proof. Expand the anticommutator and apply Lemma $L3$:

$$\begin{aligned} \{\phi(y), \phi(y')\} &= \{a^*(z) + a(z), a^*(w) + a(w)\} \\ &= \langle z, w\rangle I + \langle w, z\rangle I \\ &= 2\,\mathrm{Re}\langle z, w\rangle I = 2\alpha(y,y')I. \end{aligned}$$

Theorem $T2$. For finite X, the standard finite fermionic canonical field is the CAR construction $D5$–$D7$, not a Weyl-Heisenberg quantization of a Grassmann-valued target.

Proof. Derezinski's CAR definition uses a real Hilbert label space with positive scalar product and field operators satisfying an anticommutator relation. Definitions $D5$–$D7$ instantiate this with $Z = \mathrm{Map}(X,K)$. Lemma $L4$ proves the CAR relation on the antisymmetric Fock space, and Derezinski's Fock source identifies this as the fermionic Fock representation of the CAR. Thus the finite fermionic field arises from the same first step "choose site fields", but the quantization functor changes: symmetric symplectic/Weyl data give bosons, while positive-symmetric Clifford/CAR data give fermions.

20.5 Grassmann Target Test L5

Lemma *L*5. Replacing V_B by a Grassmann algebra is not a direct construction of the standard finite fermion field in the AQM-14 sense.

Proof. The AQM-14 Weyl construction requires an ordinary vector-space label group, a scalar-valued alternating form, and a projective unitary representation whose cocycle is a complex phase or finite-field character. The standard fermion field in $T2$ instead satisfies the CAR relation $\{\phi(y), \phi(y')\} = 2\alpha(y, y')I$, with a symmetric positive scalar product. This is already a different algebraic input.

Keyl–Schlingemann show that Grassmann variables do enter fermionic phase-space methods inside GAR. Cahill–Glauber give the sharper finite-mode object: Grassmann-valued displacement operators built from concrete CAR generators. Their multiplier is valued in the even Grassmann algebra, not generally in the complex phases. Therefore this is not, without extending the scalar algebra, the same as the ordinary projective unitary representation of the additive group $\mathrm{Map}(X, V)$ used in the bosonic and arithmetic Weyl constructions. The naive target-replacement conjecture is rejected only in that strict scalar-projective sense.

20.6 Current Dictionary

The comparison is therefore compact:

- finite boson: labels $\mathrm{Map}(X, \mathbb{R}_q \oplus \mathbb{R}_p)$, alternating symplectic form, CCR/Weyl quantization;
- finite arithmetic field: labels $E \leq \mathrm{Map}(X, V_{\mathbb{F}_q})$, alternating symplectic form after radical reduction, finite Weyl quantization;
- finite fermion: labels $\mathrm{Re}(Z \oplus \overline{Z})$ with $Z = \mathrm{Map}(X, K)$, positive symmetric form, CAR/Clifford quantization.

Thus the bosonic conjecture is checked at finite kinematic level. The fermionic conjecture is repaired: standard fermions first enter this programme through CAR/Clifford data on an antisymmetric Fock space, and Grassmann variables then give the Cahill–Glauber displacement calculus in the Grassmann-extended CAR algebra. Section 21 proves that positive representation theorem and records the ordinary complex-Hilbert-space caveat.

21 Grassmann-Weyl Fermion Displacements

21.1 Status and Source Anchors

This shard corrects the Grassmann question left open by Section 20. The intended object is not a homomorphism sending a Grassmann generator to a CAR annihilator. The intended object is the fermionic displacement operator of Cahill–Glauber:

$$D(\gamma) = \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i)\right),$$

where the $a_i, a_i^\dagger$ are concrete CAR operators and the γ_i, γ_i^* are Grassmann coefficients.

The canonical local source is the official arXiv source `references/canonical_fields/cahill_glauber_physics_9808029v1_source/fxxx.tex`. The source line anchors are: lines 323–335 for the fermion CAR and vacuum; 337–365 for Grassmann variables, cross anticommutation, and the adjoint convention; 603–643 for nilpotency and even/odd Grassmann functions; 367–390 for the definition and finite expansion of $D(\gamma)$; 391–414 for the displacement of a_n and $a_n^\dagger$; 416–464 for normal ordering and the ray-composition law; 815–833 for displacement operator completeness; and 835–854 for the caveat that physical quantum operators themselves do not involve Grassmann variables.

21.2 Lamport Dependency Skeleton

$D1$: I is a finite mode set.
$D2$: $\mathrm{CAR}(I)$ is generated by odd $a_i, a_i^\dagger$ with $\{a_i, a_j^\dagger\} = \delta_{ij}$.
$D3$: Λ_I is generated by odd γ_i, γ_i^* with all Grassmann anticommutators zero.
$D4$: $\mathcal{A}_I = \mathrm{CAR}(I)\widehat{\otimes}\Lambda_I$ is the graded tensor product.
$D5$: $\sigma(\alpha,\beta) = \sum_i (\beta_i^* \alpha_i - \alpha_i^* \beta_i)$.
$L1$: σ is an even central skew-adjoint Grassmann cocycle form.
$D6$: D_I is the abstract Grassmann displacement ray algebra.
$D7$: $K(\gamma) = \sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i)$, $\quad D(\gamma) = e^{K(\gamma)}$.
$L2$: $K(\gamma)^\dagger = -K(\gamma)$, hence $D(\gamma)$ is unitary.
$L3$: $D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n$ and similarly for $a_n^\dagger$.
$L4$: $[K(\alpha), K(\beta)] = \sigma(\alpha,\beta)$.
$T1$: $\gamma \mapsto D(\gamma)$ is a unitary representation of D_I in $\mathcal{A}_I$.
$T2$: A faithful ordinary complex-scalar Hilbert representation of this Grassmann cocycle is not the theorem.

21.3 The Concrete Target Algebra D1–D4

Let $I = \{1,\ldots,N\}$. The finite CAR algebra $\mathrm{CAR}(I)$ is the complex unital $*$-algebra generated by odd symbols $a_i, a_i^\dagger$, $i \in I$, subject to

$$\{a_i, a_j^\dagger\} = \delta_{ij} 1, \qquad \{a_i, a_j\} = 0, \qquad \{a_i^\dagger, a_j^\dagger\} = 0.$$

Here $\{x,y\} = xy + yx$. The adjoint satisfies $(a_i)^\dagger = a_i^\dagger$ and $(a_i^\dagger)^\dagger = a_i$.

Let Λ_I be the Grassmann $*$-algebra generated by odd symbols γ_i, γ_i^*, $i \in I$, with

$$\{\gamma_i, \gamma_j\} = 0, \qquad \{\gamma_i^*, \gamma_j\} = 0, \qquad \{\gamma_i^*, \gamma_j^*\} = 0.$$

The symbols γ_i and γ_i^* are independent generators related by the chosen adjoint. Following the source, adjunction reverses the order of all odd quantities:

$$(x_1 x_2 \cdots x_k)^\dagger = x_k^\dagger \cdots x_2^\dagger x_1^\dagger$$

when the x_r are fermionic operators or Grassmann variables.

Define the Grassmann-extended CAR algebra

$$\mathcal{A}_I = \mathrm{CAR}(I) \widehat{\otimes} \Lambda_I$$

as the unital $*$-algebra generated by the two preceding sets, with the additional cross anticommutation rules

$$\{\gamma_i, a_j\} = \{\gamma_i, a_j^\dagger\} = \{\gamma_i^*, a_j\} = \{\gamma_i^*, a_j^\dagger\} = 0.$$

An element is even if it is a sum of monomials with even total fermionic degree, and odd if it has odd total degree. Even Grassmann elements commute with all CAR generators. This is the algebra in which Cahill–Glauber's unitary displacement operators live.

21.4 The Abstract Ray Algebra D5–D6

A Grassmann displacement parameter is an I-tuple $\gamma = (\gamma_i)_{i \in I}$ of odd Grassmann elements, with adjoint tuple $\gamma^* = (\gamma_i^*)_{i \in I}$. The additive law is componentwise.

For two parameters α, β, define the even Grassmann form

$$\sigma(\alpha, \beta) = \sum_{i \in I} (\beta_i^* \alpha_i - \alpha_i^* \beta_i)$$

and its exponential

$$c(\alpha, \beta) = \exp\left(\frac{1}{2}\sigma(\alpha, \beta)\right).$$

The exponential is a finite polynomial because every Grassmann monomial is nilpotent.

Lemma *L*1. σ is even, central in $\mathcal{A}_I$, skew-adjoint, and gives a multiplicative two-cocycle c:

$$c(\alpha, \beta) c(\alpha + \beta, \delta) = c(\beta, \delta) c(\alpha, \beta + \delta).$$

Proof. Each summand $\beta_i^* \alpha_i - \alpha_i^* \beta_i$ is a product of two odd Grassmann elements, hence is even. Even Grassmann elements commute with all Grassmann generators and with all CAR generators, so σ is central. The adjoint convention gives

$$(\beta_i^* \alpha_i)^\dagger = \alpha_i^* \beta_i, \qquad (\alpha_i^* \beta_i)^\dagger = \beta_i^* \alpha_i,$$

so $\sigma(\alpha, \beta)^\dagger = -\sigma(\alpha, \beta)$. Bilinearity gives

$$\begin{aligned}
&\sigma(\alpha, \beta) + \sigma(\alpha + \beta, \delta) \\
&\qquad = \sigma(\alpha, \beta) + \sigma(\alpha, \delta) + \sigma(\beta, \delta) \\
&\qquad = \sigma(\beta, \delta) + \sigma(\alpha, \beta + \delta).
\end{aligned}$$

Because the exponents are central, exponentials multiply by adding exponents. This proves the displayed two-cocycle identity.

Define the abstract Grassmann displacement ray algebra D_I to be the unital $*$-algebra over the even Grassmann scalars generated by symbols $\mathsf{D}(\gamma)$, one for each displacement parameter, with relations

$$\mathsf{D}(0) = 1, \qquad \mathsf{D}(\alpha)\mathsf{D}(\beta) = \mathsf{D}(\alpha + \beta) c(\alpha, \beta), \qquad \mathsf{D}(\gamma)^\dagger = \mathsf{D}(-\gamma).$$

Lemma *L*1 is exactly the associativity check for the product relation.

21.5 The Cahill–Glauber Representation D7–T1

For a parameter γ, define

$$K(\gamma) = \sum_{i \in I} (a_i^\dagger \gamma_i - \gamma_i^* a_i), \qquad D(\gamma) = e^{K(\gamma)} \in \mathcal{A}_I.$$

Each term $a_i^\dagger \gamma_i$ and $\gamma_i^* a_i$ is even, so $K(\gamma)$ is even and the ordinary commutator is the relevant bracket.

Theorem $T1$: representation to be proved. The assignment

$$\Pi : \mathsf{D}_I \longrightarrow \mathcal{A}_I^\times, \qquad \Pi(\mathsf{D}(\gamma)) = D(\gamma) = \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i) \right),$$

is a unitary representation of the abstract Grassmann displacement ray algebra in the Grassmann-extended finite CAR algebra. Its product law is

$$D(\alpha) D(\beta) = D(\alpha + \beta) \exp\left(\frac{1}{2} \sum_i (\beta_i^* \alpha_i - \alpha_i^* \beta_i) \right).$$

21.6 Conclusion

The unitary representation of the abstract fermion displacement objects is:

$$\boxed{\mathsf{D}(\gamma) \mapsto \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i) \right)}$$

inside the Grassmann-extended finite CAR algebra. It is unitary, it displaces the concrete CAR generators by $a_i \mapsto a_i + \gamma_i$ and $a_i^\dagger \mapsto a_i^\dagger + \gamma_i^*$, and it obeys the Grassmann-valued ray product of Cahill–Glauber. The derivations of unitarity, the displacement identities, the product law, and the ordinary Hilbert-space boundary are in Section 22.

22 Grassmann-Weyl Fermion Displacement Derivations

22.1 Status and Dependencies

This shard proves the Lamport items $L2$–$T2$ stated in Section 21. It uses the same Cahill–Glauber source anchors from `references/canonical_fields/cahill_glauber_physics_9808029v1_source/fxxx.tex`: lines 367–390 for $D(\gamma)$, lines 391–414 for the displacement identities, lines 416–464 for the Baker–Hausdorff and ray-composition formula, and lines 835–854 for the ordinary-operator caveat.

$D1$–$D7$	:	definitions from Section 21.
$L1$	:	σ is the central Grassmann two-cocycle form.
$L2$	:	$K(\gamma)^\dagger = -K(\gamma)$, so $D(\gamma)$ is unitary.
$L3$	:	$D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n$ and similarly for $a_n^\dagger$.
$L4$	:	$[K(\alpha), K(\beta)] = \sigma(\alpha, \beta)$.
$T1$	:	$D(\gamma)$ represents the abstract displacement ray algebra.
$T2$	:	The theorem is Grassmann-extended CAR-unitary, not ordinary scalar-projective Hilbert-unitary.

22.2 Unitarity L2

Lemma $L2$.

$$K(\gamma)^\dagger = -K(\gamma), \qquad D(\gamma)^\dagger D(\gamma) = D(\gamma)D(\gamma)^\dagger = 1.$$

Proof. Using the adjoint convention from $D3$,

$$(a_i^\dagger \gamma_i)^\dagger = \gamma_i^* a_i, \qquad (\gamma_i^* a_i)^\dagger = a_i^\dagger \gamma_i.$$

Therefore

$$K(\gamma)^\dagger = \sum_i (\gamma_i^* a_i - a_i^\dagger \gamma_i) = -K(\gamma).$$

Hence $D(\gamma)^\dagger = e^{-K(\gamma)} = D(-\gamma)$, and multiplying $e^{-K(\gamma)}e^{K(\gamma)}$ or $e^{K(\gamma)}e^{-K(\gamma)}$ gives 1.

22.3 Displacement of the CAR Generators L3

Lemma $L3$. For every $n \in I$,

$$D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n, \qquad D(\gamma)^\dagger a_n^\dagger D(\gamma) = a_n^\dagger + \gamma_n^*.$$

Proof. The conjugation identity $e^{-K}Be^{K} = B + [B, K] + \frac{1}{2}[[B, K], K] + \cdots$ applies because K is even. For $i \neq n$, the i-summand of $K(\gamma)$ commutes with a_n: the sign from moving a_n through the CAR generator is cancelled by the sign from moving it through the Grassmann coefficient. For $i = n$,

$$\begin{aligned}
[a_n, a_n^\dagger \gamma_n - \gamma_n^* a_n] &= a_n a_n^\dagger \gamma_n - a_n^\dagger \gamma_n a_n \\
&= (1 - a_n^\dagger a_n)\gamma_n - a_n^\dagger \gamma_n a_n \\
&= \gamma_n - a_n^\dagger a_n \gamma_n + a_n^\dagger a_n \gamma_n \\
&= \gamma_n.
\end{aligned}$$

The element γ_n commutes with $K(\gamma)$, so all higher commutators vanish. Thus $D(\gamma)^\dagger a_n D(\gamma) = a_n + \gamma_n$. The same calculation with $a_n^\dagger$ gives

$$\begin{aligned}
[a_n^\dagger, a_n^\dagger \gamma_n - \gamma_n^* a_n] &= -a_n^\dagger \gamma_n^* a_n + \gamma_n^* a_n a_n^\dagger \\
&= \gamma_n^* a_n^\dagger a_n + \gamma_n^*(1 - a_n^\dagger a_n) \\
&= \gamma_n^*,
\end{aligned}$$

and again higher commutators vanish.

22.4 The Grassmann Commutator L4

Lemma *L*4. For all parameters α, β,

$$[K(\alpha), K(\beta)] = \sigma(\alpha, \beta).$$

Proof. It is enough to compute one mode, since different modes commute after the same double sign cancellation used in Lemma $L3$. Put

$$p = a^\dagger \alpha, \quad q = \alpha^* a, \quad r = a^\dagger \beta, \quad s = \beta^* a.$$

Then $K(\alpha) = p - q$ and $K(\beta) = r - s$. The products pr, rp, qs, sq vanish because $(a^\dagger)^2 = a^2 = 0$. The mixed commutators are

$$\begin{aligned}[p, s] &= a^\dagger \alpha \beta^* a - \beta^* a a^\dagger \alpha \\ &= -\beta^* \alpha\, a^\dagger a - \beta^* (1 - a^\dagger a) \alpha \\ &= -\beta^* \alpha,\end{aligned}$$

and

$$\begin{aligned}[q, r] &= \alpha^* a a^\dagger \beta - a^\dagger \beta \alpha^* a \\ &= \alpha^* (1 - a^\dagger a) \beta + \alpha^* \beta\, a^\dagger a \\ &= \alpha^* \beta.\end{aligned}$$

Therefore

$$[K(\alpha), K(\beta)] = [p - q, r - s] = -[p, s] - [q, r] = \beta^* \alpha - \alpha^* \beta.$$

Summing over $i \in I$ gives the displayed formula.

22.5 Representation Theorem T1

Theorem *T*1. The assignment

$$\Pi(\mathsf{D}(\gamma)) = \exp\left(\sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i) \right)$$

is a unitary representation of the abstract Grassmann displacement ray algebra in the Grassmann-extended finite CAR algebra.

Proof. Lemma $L2$ proves unitarity and the $*$-relation $D(\gamma)^\dagger = D(-\gamma)$. Lemma $L4$ says that the commutator $[K(\alpha), K(\beta)]$ is the central element $\sigma(\alpha, \beta)$. The Baker–Hausdorff identity for central commutator gives

$$\begin{aligned}e^{K(\alpha)} e^{K(\beta)} &= e^{K(\alpha)+K(\beta)} e^{\frac{1}{2}[K(\alpha), K(\beta)]} \\ &= e^{K(\alpha+\beta)} e^{\frac{1}{2}\sigma(\alpha, \beta)} \\ &= D(\alpha + \beta) c(\alpha, \beta).\end{aligned}$$

This is exactly the defining product relation of D_I. Lemma $L1$ supplies associativity of the abstract relation, so Π is a well-defined representation.

22.6 Boundary of the Statement T2

Theorem *T*2. The positive theorem $T1$ is not an ordinary projective representation $\gamma \mapsto U_\gamma \in U(\mathcal{H})$ with complex $U(1)$ multiplier. It is a unitary representation in $\mathcal{A}_I$, the CAR algebra extended by Grassmann coefficients.

Proof. The multiplier in $T1$ is

$$c(\alpha, \beta) = \exp\left(\frac{1}{2}\sum_i(\beta_i^*\alpha_i - \alpha_i^*\beta_i)\right),$$

which is an even Grassmann unit. Cahill–Glauber's notation records that no nonzero Grassmann monomial is an ordinary complex number, and it later separates physical quantum operators from Grassmann variables. Therefore unless the Grassmann coefficient algebra is included in the target, the multiplier is not a complex phase.

More explicitly, if $\rho : \Lambda_I \to B(\mathcal{H})$ is a Hilbert-space $*$-representation of the Grassmann coefficient algebra, then $\rho(\gamma_i) = 0$ for each i. Indeed, with $b = \rho(\gamma_i)$,

$$b^2 = 0, \qquad bb^\dagger + b^\dagger b = 0,$$

and hence for every $v \in \mathcal{H}$,

$$0 = \langle v, (bb^\dagger + b^\dagger b)v\rangle = \|b^\dagger v\|^2 + \|bv\|^2,$$

so $b = 0$. Thus the ordinary complex Hilbert representation collapses the Grassmann cocycle. The representation found in $T1$ avoids this collapse by keeping the Grassmann variables as formal odd coefficients anticommuting with the concrete CAR generators.

23 Prime-Field Arithmetic Fields and Qudit Paulis

23.1 Status and Source Anchors

This shard stays below schemes and sheaves. The only commutative rings are prime fields $\mathbb{F}_p$. The local Pauli/Weyl sources are Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313; Gottesman's thesis, `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 1627–1681; and Bombin–Martin–Delgado, `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 1208–1259.

Source-sign bridge. The local convention is `CONVENTIONS.md`, convention (t): $W(x,z) = T_x R_z$ with $T_x|s\rangle = |s+x\rangle$. The cited source formulas use the opposite commutator exponent if their displayed shift exponent is identified directly with this local X-coordinate; see Ashikhmin–Knill `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 265–268 and 307–313; Gottesman `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 1677–1680; and Bombin–Martin–Delgado `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 1247–1254. This shard uses the conversion $x_{\text{source}} = -x_{\text{local}}$, $z_{\text{source}} = z_{\text{local}}$, so the source exponent $xz' - zx'$ becomes the local exponent $zx' - xz'$. Agreement with the cited Pauli group is therefore after inverting the source shift generator, not identical displayed sign conventions.

23.2 Lamport Dependency Skeleton

$D1$: p is prime and $R = \mathbb{F}_p$.
$L1$: R has one prime ideal and $\operatorname{Frac}(R) = R$.
$D2$: $X_{\text{sp}} = \operatorname{Spec} R, \quad X_{\text{el}} = R$ as a finite set.
$D3$: $V = R_X \oplus R_Z, \quad \omega_V((x,z),(x',z')) = zx' - xz'$.
$D4$: $E_{\text{el}} = \operatorname{Map}(X_{\text{el}}, V)$.
$D5$: $\Omega((\xi,\zeta),(\xi',\zeta')) = \sum_{u\in X_{\text{el}}}(\zeta(u)\xi'(u) - \xi(u)\zeta'(u))$.
$D6$: $\mathcal{H}_X = \ell^2(R^{X_{\text{el}}}), \quad W(\xi,\zeta) = T_\xi R_\zeta$.
$L2$: (V,ω_V) and (E_{el},Ω) are symplectic.
$L3$: $W(\xi,\zeta)W(\xi',\zeta') = \chi(\zeta\cdot\xi')W(\xi+\xi',\zeta+\zeta')$.
$T1$: $\langle \chi(c)I, W(E_{\text{el}})\rangle$ is the standard odd-p qudit Pauli group on p qudits after the source-sign bridge.
$T2$: Stabilizers require an extra isotropic label subspace and phases.
$L4$: $p = 2$ has the same projective labels but needs the usual μ_4 Pauli phases for Hermitian Y.

23.3 Prime Field Points L1–D2

Lemma $L1$. The ring $R = \mathbb{F}_p$ has exactly one prime ideal, namely (0), and $\operatorname{Frac}(R) = R$.

Proof. A field has only two ideals, (0) and the whole field. The whole field is not prime because prime ideals are proper. The zero ideal is prime because the quotient $R/(0) = R$ is an integral domain. Thus $\operatorname{Spec} R = \{(0)\}$. The fraction field consists of symbols a/b with $b \neq 0$, and the map $a/b \mapsto ab^{-1}$ is a well-defined field isomorphism $\operatorname{Frac}(R) \cong R$.

There are therefore two concrete point sets in play. The spectrum

$$X_{\text{sp}} = \operatorname{Spec} \mathbb{F}_p$$

has one point. The underlying set of field elements

$$X_{\text{el}} = \mathbb{F}_p$$

has p points. The one-point choice gives one qudit. The p-point choice is the one that can give the p-qudit Pauli group.

23.4 The All-Map Arithmetic Field D3–L2

Let

$$V = R_X \oplus R_Z$$

with symplectic form

$$\omega_V((x, z), (x', z')) = zx' - xz'.$$

For $X = X_{\mathrm{el}}$, define

$$E_{\mathrm{el}} = \mathrm{Map}(X, V) \cong R^X \oplus R^X.$$

Write an element of E_{el} as (ξ, ζ), where $\xi, \zeta : X \to R$. Define

$$\Omega((\xi, \zeta), (\xi', \zeta')) = \sum_{u \in X} \big(\zeta(u)\xi'(u) - \xi(u)\zeta'(u)\big).$$

Lemma *L2.* (V, ω_V) and $(E_{\mathrm{el}}, \Omega)$ are symplectic R-vector spaces. Moreover $E_{\mathrm{el}} \cong R^p \oplus R^p$.

Proof. The form ω_V is bilinear. It is alternating because

$$\omega_V((x, z), (x, z)) = zx - xz = 0.$$

If $(x, z) \neq 0$, then $\omega_V((x, z), (1, 0)) = z$ and $\omega_V((x, z), (0, 1)) = -x$, so some pairing is nonzero. Thus V is symplectic.

Since X has p elements, choosing any ordering of X identifies R^X with R^p. Hence $E_{\mathrm{el}} \cong R^p \oplus R^p$. The form Ω is the direct sum of the pointwise symplectic forms. If $(\xi, \zeta) \neq 0$, choose u_0 with $(\xi(u_0), \zeta(u_0)) \neq 0$ and use the previous paragraph to choose a value at u_0 pairing nontrivially, with all other values zero. Thus Ω is nondegenerate and alternating.

Map versus Hom. The p-qudit statement uses all maps. If instead one chooses the structure-preserving space

$$\mathrm{Hom}_R(R, V),$$

where R is viewed as a one-dimensional R-vector space, then evaluation at 1 gives $\mathrm{Hom}_R(R, V) \cong V$. That smaller choice is the one-qudit label space, not the p-qudit label space.

23.5 Weyl Operators L3

Let $\chi(t) = \exp(2\pi i\, \tilde{t}/p)$, where $\tilde{t} \in \{0, \ldots, p-1\}$ represents $t \in R$. On the Hilbert space

$$\mathcal{H}_X = \ell^2(R^X),$$

with basis $\{|s\rangle : s : X \to R\}$, define

$$T_\xi |s\rangle = |s + \xi\rangle, \qquad R_\zeta |s\rangle = \chi\left(\sum_{u \in X} \zeta(u) s(u)\right) |s\rangle,$$

and

$$W(\xi, \zeta) = T_\xi R_\zeta.$$

Lemma $L3$. For $e = (\xi, \zeta)$ and $f = (\xi', \zeta')$,

$$W(e)W(f) = \chi(\zeta \cdot \xi')W(e + f),$$

where $\zeta \cdot \xi' = \sum_u \zeta(u)\xi'(u)$. Hence

$$W(e)W(f) = \chi(\Omega(e, f))W(f)W(e).$$

Proof. The translation operators add and the phase operators add:

$$T_\xi T_{\xi'} = T_{\xi+\xi'}, \qquad R_\zeta R_{\zeta'} = R_{\zeta+\zeta'}.$$

Also

$$\begin{aligned} R_\zeta T_{\xi'}|s\rangle &= R_\zeta|s + \xi'\rangle \\ &= \chi(\zeta \cdot s + \zeta \cdot \xi')|s + \xi'\rangle \\ &= \chi(\zeta \cdot \xi')T_{\xi'} R_\zeta|s\rangle. \end{aligned}$$

Therefore

$$T_\xi R_\zeta T_{\xi'} R_{\zeta'} = \chi(\zeta \cdot \xi')T_{\xi+\xi'} R_{\zeta+\zeta'}.$$

Swapping e and f gives the commutator factor

$$\chi(\zeta \cdot \xi' - \zeta' \cdot \xi) = \chi(\Omega(e, f)).$$

23.6 Identification with Qudit Paulis T1–T2

For each $u \in X$, let $\delta_u \in R^X$ be the delta function. Define single-site operators

$$X_u = W(\delta_u, 0), \qquad Z_u = W(0, \delta_u).$$

Let $\mathcal{P}_p^\chi(X)$ be the group generated by all X_u, Z_u and the central character phases $\chi(c)I$, $c \in R$. For odd p, this is the standard prime-dimensional qudit Pauli group after the source-sign conversion above: one p-dimensional qudit for each $u \in X$, shifts generated by X_u, and phases generated by Z_u. For $p = 2$, it is the unphased Weyl group whose projective labels are the usual qubit Pauli labels; the conventional Hermitian Y convention is recorded in $L4$.

Theorem $T1$. For $X = \mathbb{F}_p$ as a set of elements,

$$\langle \chi(c)I,\ W(e) : c \in R,\ e \in E_{\mathrm{el}}\rangle = \mathcal{P}_p^\chi(X).$$

Thus, for odd p, the all-map arithmetic quantum field on $X = \mathbb{F}_p$ with target $\mathrm{Frac}(\mathbb{F}_p)^2 = \mathbb{F}_p^2$ gives the ambient Gottesman/Ashikhmin–Knill p-qudit Pauli group for p qudits, after inverting the source shift generator as stated in the source-sign bridge. For $p = 2$, the projective statement is the same and the full conventional Pauli phase convention is $L4$.

Proof. The operators $W(\delta_u, 0)$ and $W(0, \delta_u)$ are among the arithmetic field Weyl operators, so $\mathcal{P}_p^\chi(X)$ is contained in the generated arithmetic Weyl group. Conversely, for any (ξ, ζ),

$$T_\xi = \prod_{u \in X} X_u^{\xi(u)}, \qquad R_\zeta = \prod_{u \in X} Z_u^{\zeta(u)},$$

with any fixed site order, because different X_u's commute and different Z_u's commute. Thus $W(\xi, \zeta) = T_\xi R_\zeta$ lies in $\mathcal{P}_p^\chi(X)$. The two generated groups are equal.

Theorem $T2$**.** The arithmetic field datum gives the ambient Pauli/Weyl group. A stabilizer code is extra data: an isotropic label subspace $L \leq E_{\mathrm{el}}$, plus a compatible choice of phases/eigenvalue character.

Proof. If $S \leq \mathcal{P}_p^{\chi}(X)$ is Abelian, its image L in the projective label space E_{el} satisfies $\Omega(L, L) = 0$ by Lemma $L3$. Conversely, if p is odd and $L \leq E_{\mathrm{el}}$ is isotropic, define

$$\widetilde{W}(\xi, \zeta) = \chi\left(\frac{1}{2}\zeta \cdot \xi\right) W(\xi, \zeta).$$

Using Lemma $L3$,

$$\widetilde{W}(e)\widetilde{W}(f) = \chi\left(\frac{1}{2}\Omega(e, f)\right) \widetilde{W}(e + f).$$

On an isotropic L, this becomes $\widetilde{W}(e)\widetilde{W}(f) = \widetilde{W}(e + f)$. Multiplying this lift by a character $L \to \mathbb{C}^{\times}$ changes the stabilizer eigenvalues. Thus the arithmetic field supplies the ambient labels and commutation form; the choice of stabilizer is a later choice of isotropic subspace and phases.

23.7 The p=2 Edge Case L4

For $p = 2$, the same label space E_{el}, the same Hilbert space, and the same commutator criterion

$$W(e)W(f) = (-1)^{\Omega(e,f)} W(f)W(e)$$

remain valid. The edge is the phase convention. There is no element $1/2 \in \mathbb{F}_2$, and the unphased operator $W(1, 1) = XZ$ is not the Hermitian Pauli Y. The standard qubit Pauli group uses fourth roots of unity and the relabeling

$$\widehat{W}(\xi, \zeta) = i^{\sum_u \xi(u)\zeta(u)} W(\xi, \zeta),$$

so $\widehat{W}(\delta_u, \delta_u) = Y_u$. Hence $p = 2$ has the same projective arithmetic field labels and the same isotropy criterion, but the full conventional Pauli group has center μ_4, not only the character phases μ_2.

23.8 Conclusion

The conjecture is correct with one precision. If $X = \operatorname{Spec} \mathbb{F}_p$, the construction gives one qudit. If $X = \mathbb{F}_p$ is the underlying p-element set and $E = \mathrm{Map}(X, \mathbb{F}_p^2)$, the Weyl–Heisenberg arithmetic field gives exactly the standard odd-p qudit Pauli group, and for $p = 2$ gives the same projective Pauli labels with the usual fourth-root phase adjustment. In every prime case the projective labels are

$$E \cong \mathbb{F}_p^p \oplus \mathbb{F}_p^p.$$

Actual stabilizer codes are not automatic fields; they are isotropic subspaces of this field label space together with phase/eigenvalue choices.

24 Spec Z/6Z and Residue Qudit Factorisation

24.1 Status and Source Anchors

This shard records the morally preferred convention suggested by Section 23: for a field $\mathbb{F}_p$, the arithmetic point space is $\operatorname{Spec}\mathbb{F}_p$, so the local quantum system is one p-level qudit. For a non-field finite ring there is no single quotient field. The concrete replacement used here is one local phase space over each residue field.

The ring calculation for $\mathbb{Z}/6\mathbb{Z}$ is proved directly below. The Weyl/Pauli facts use the same local sources as AQM-22: Ashikhmin–Knill `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313 for finite-field shift/phase bases and product laws, and Bombin–Martin–Delgado `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrect` `tex`, lines 1208–1259 for qudit Paulis, symplectic commutators, and projective label maps. The sign convention is also the one bridged explicitly in Section 23: relative to those source formulas, the local X-coordinate is the negative of the source shift exponent and the local Z-coordinate is unchanged.

24.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & R = \mathbb{Z}/6\mathbb{Z}. \\
L1 & : & R \cong \mathbb{F}_2 \times \mathbb{F}_3 \text{ by Chinese remainders.} \\
L2 & : & \operatorname{Spec} R = \{\mathfrak{p}_2 = (2), \mathfrak{p}_3 = (3)\}. \\
D2 & : & \kappa(\mathfrak{p}_2) = \mathbb{F}_2, \quad \kappa(\mathfrak{p}_3) = \mathbb{F}_3. \\
D3 & : & E_R = \kappa(\mathfrak{p}_2)^2 \oplus \kappa(\mathfrak{p}_3)^2. \\
D4 & : & \Omega_R = \Omega_2 \oplus \Omega_3, \text{ with each } \Omega_i \text{ the standard } zx' - xz'. \\
L3 & : & E_R \text{ is an orthogonal direct sum of local symplectic groups.} \\
D5 & : & \mathcal{H}_R = \ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3). \\
L4 & : & W_R(e_2, e_3) = W_2(e_2) \otimes W_3(e_3) \text{ has the product Weyl law.} \\
T1 & : & \operatorname{span}\{W_R(e) : e \in E_R\} \cong M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}). \\
T2 & : & \text{The Hilbert space is a tensor product, not } \mathbb{C}^2 \oplus \mathbb{C}^3.
\end{array}
$$

24.3 The Spectrum L1–D2

Let $R = \mathbb{Z}/6\mathbb{Z}$. Define

$$\Phi : R \longrightarrow \mathbb{F}_2 \times \mathbb{F}_3, \qquad \Phi(\bar{n}) = (\bar{n} \bmod 2, \bar{n} \bmod 3).$$

Lemma $L1$. Φ is a ring isomorphism.

Proof. It is a well-defined ring homomorphism because congruence modulo 6 implies congruence modulo 2 and modulo 3. Its kernel consists of residue classes divisible by both 2 and 3, hence by 6, so the kernel is zero. Both rings have six elements, so an injective map is bijective.

Lemma $L2$.

$$\operatorname{Spec} R = \{\mathfrak{p}_2, \mathfrak{p}_3\}, \qquad \mathfrak{p}_2 = (2) = \{0, 2, 4\}, \quad \mathfrak{p}_3 = (3) = \{0, 3\}.$$

Moreover $R/\mathfrak{p}_2 \cong \mathbb{F}_2$ and $R/\mathfrak{p}_3 \cong \mathbb{F}_3$.

Proof. Under Φ, the ideal $\mathfrak{p}_2 = (2)$ maps to $0 \times \mathbb{F}_3$, and $\mathfrak{p}_3 = (3)$ maps to $\mathbb{F}_2 \times 0$. The quotients are therefore $\mathbb{F}_2$ and $\mathbb{F}_3$, respectively, so both ideals are prime.

Every ideal J of $\mathbb{F}_2 \times \mathbb{F}_3$ is $I_2 \times I_3$: if $(a, b) \in J$, then $(a, 0) = (a, b)(1, 0) \in J$ and $(0, b) = (a, b)(0, 1) \in J$. Thus the proper ideals are 0×0, $0 \times \mathbb{F}_3$, and $\mathbb{F}_2 \times 0$. The quotient by 0×0 is

$\mathbb{F}_2 \times \mathbb{F}_3$, which is not a domain because $(1,0)(0,1) = 0$. Hence 0×0 is not prime, and the two ideals listed above are the only prime ideals.

The residue fields are

$$\kappa(\mathfrak{p}_2) = R/\mathfrak{p}_2 \cong \mathbb{F}_2, \qquad \kappa(\mathfrak{p}_3) = R/\mathfrak{p}_3 \cong \mathbb{F}_3.$$

24.4 Residue-Field Phase Space D3–L3

For a point $\mathfrak{p}$, define its local phase space by

$$V_{\mathfrak{p}} = \kappa(\mathfrak{p})_X \oplus \kappa(\mathfrak{p})_Z, \qquad \Omega_{\mathfrak{p}}((x,z),(x',z')) = zx' - xz'.$$

For $R = \mathbb{Z}/6\mathbb{Z}$, the global label group is

$$E_R = V_{\mathfrak{p}_2} \oplus V_{\mathfrak{p}_3} = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2.$$

This is a direct sum of finite abelian groups, not a vector space over one field. Define the alternating bicharacter

$$B_R(e,f) = (-1)^{\Omega_{\mathfrak{p}_2}(e_2,f_2)} \exp\left(\frac{2\pi i}{3}\Omega_{\mathfrak{p}_3}(e_3,f_3)\right),$$

where $e = (e_2, e_3)$ and $f = (f_2, f_3)$.

Lemma *L*3. The two local summands are orthogonal and nondegenerate: if $B_R(e,f) = 1$ for all $f \in E_R$, then $e = 0$.

Proof. Orthogonality is built into the formula: e_2 only pairs with f_2, and e_3 only pairs with f_3. The local form is nondegenerate because a nonzero (x,z) pairs nontrivially with either $(1,0)$ or $(0,1)$. If $B_R(e,f) = 1$ for all f, first take $f_3 = 0$, forcing $e_2 = 0$. Then take $f_2 = 0$, forcing $e_3 = 0$. Thus $e = 0$.

24.5 Weyl Representation L4–T1

Let

$$\mathcal{H}_2 = \ell^2(\mathbb{F}_2), \qquad \mathcal{H}_3 = \ell^2(\mathbb{F}_3), \qquad \mathcal{H}_R = \mathcal{H}_2 \otimes \mathcal{H}_3.$$

For $i = 2, 3$, let $W_i(x,z) = T_i(x)R_i(z)$ be the usual local shift-phase Weyl operator on $\ell^2(\mathbb{F}_i)$. Define

$$W_R(e_2, e_3) = W_2(e_2) \otimes W_3(e_3).$$

The $\mathbb{F}_2$ factor uses the unphased projective Weyl representatives from Section 23. Adjoining the conventional μ_4 phases changes the Hermitian representatives, for example $Y = iXZ$, but not the label group, operator span, or tensor-product Hilbert-space conclusion below.

Lemma *L*4. For $e = (e_2, e_3)$ and $f = (f_2, f_3)$,

$$W_R(e)W_R(f) = c_R(e,f)W_R(e+f),$$

where

$$c_R(e,f) = (-1)^{z_2 x_2'} \exp\left(\frac{2\pi i}{3} z_3 x_3'\right)$$

if $e_i = (x_i, z_i)$ and $f_i = (x_i', z_i')$. Consequently

$$W_R(e)W_R(f) = B_R(e,f)W_R(f)W_R(e).$$

Proof. The local Weyl laws give

$$W_i(e_i)W_i(f_i) = c_i(e_i, f_i)W_i(e_i + f_i).$$

Tensoring the two equations gives the displayed product law with $c_R = c_2c_3$. Dividing the product law for e, f by the one for f, e gives the displayed commutator bicharacter.

Theorem $T1$**.**

$$\mathrm{span}_{\mathbb{C}}\{W_R(e) : e \in E_R\} = \mathrm{End}(\mathcal{H}_2) \otimes \mathrm{End}(\mathcal{H}_3) \cong M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}).$$

Proof. For a single p-level qudit, the local source records that the p^2 operators T^xR^z form an operator basis. Thus $\{W_2(e_2)\}$ is a basis of $M_2(\mathbb{C})$, and $\{W_3(e_3)\}$ is a basis of $M_3(\mathbb{C})$. Tensor products of bases form a basis of the tensor product vector space, so the $4 \cdot 9 = 36$ operators $W_R(e_2, e_3)$ span $M_2(\mathbb{C}) \otimes M_3(\mathbb{C})$. The last isomorphism is the standard Kronecker product identification.

24.6 Direct Sum or Tensor Product? T2

Theorem $T2$**.** The quantum system attached to $\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$ by the residue-field Weyl rule is a qubit tensor a qutrit,

$$\mathcal{H}_R \cong \mathbb{C}^2 \otimes \mathbb{C}^3 \cong \mathbb{C}^6,$$

not the Hilbert direct sum $\mathbb{C}^2 \oplus \mathbb{C}^3$.

Proof. The direct sum occurs in the label group:

$$E_R = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2.$$

Weyl quantization sends an orthogonal direct sum of independent phase spaces to tensor products of the local Weyl representations, as shown explicitly in Lemma $L4$.

A Hilbert direct sum would have operator algebra

$$\mathrm{End}(\mathbb{C}^2) \oplus \mathrm{End}(\mathbb{C}^3) \cong M_2(\mathbb{C}) \oplus M_3(\mathbb{C}),$$

whose complex dimension is $4 + 9 = 13$ and whose center is two-dimensional. The Weyl operators from E_R instead give 36 independent operators and the full matrix algebra $M_6(\mathbb{C})$, whose center is one-dimensional. Thus the direct-sum Hilbert interpretation is incompatible with the Weyl label count and with Theorem $T1$.

24.7 Conclusion

The expectation "qutrit plus qubit" is correct only at the level of local factors. The precise statement is

$$\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z}) = \{\mathfrak{p}_2, \mathfrak{p}_3\}, \qquad \kappa(\mathfrak{p}_2) = \mathbb{F}_2, \quad \kappa(\mathfrak{p}_3) = \mathbb{F}_3,$$

so the local systems are one qubit and one qutrit. The global Weyl label space is their orthogonal direct sum, but the Hilbert space is their tensor product. Equivalently, the observable algebra is $M_2 \otimes M_3 \cong M_6$, not $M_2 \oplus M_3$.

25 Zariski Opens and Observable Nets

25.1 Status and Source Anchors

This shard stays at the concrete finite-spectrum level. The Zariski topology definitions are sourced from the Stacks Project snapshot `references/algebraic_geometry/stacks_project_algebra.tex`: the spectrum, $V(T)$, and $D(f)$ are defined in lines 2972–2983; the Zariski topology and standard opens are defined in lines 3104–3117. The finite Weyl local systems are those of AQM-22 and AQM-23.

The main conclusion is a convention choice. If the objective is a net whose inclusions follow inclusions of open sets, then an open U should carry the algebra acting nontrivially on U itself. The assignment $U \mapsto \mathcal{A}(X \setminus U)$ is also meaningful, but it is a closed-support or vanishing-locus assignment and its inclusions run in the opposite direction.

25.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & X = \operatorname{Spec} R, \quad V(T) = \{\mathfrak{p} : T \subset \mathfrak{p}\}, \quad D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}. \\
D2 & : & \text{For each } x \in X \text{ choose } \mathcal{H}_x = \ell^2(\kappa(x)),\ \mathcal{A}_x = \operatorname{End}(\mathcal{H}_x). \\
D3 & : & E(S) = \bigoplus_{x \in S} \kappa(x)^2, \quad \mathcal{H}(S) = \bigotimes_{x \in S} \mathcal{H}_x, \quad \mathcal{A}(S) = \bigotimes_{x \in S} \mathcal{A}_x. \\
L1 & : & S \subset T \Rightarrow \iota_{S,T} : \mathcal{A}(S) \hookrightarrow \mathcal{A}(T) \text{ is a functorial unital } *\text{-inclusion.} \\
L2 & : & \iota_{S,T} \text{ is the inclusion of Weyl algebras induced by } E(S) \hookrightarrow E(T). \\
T1 & : & U \mapsto \mathcal{A}(U) \text{ is the covariant open-local net.} \\
T2 & : & U \mapsto \mathcal{A}(X \setminus U) \text{ is contravariant in opens.} \\
E1 & : & X = \operatorname{Spec} \mathbb{F}_p \text{ has only } \emptyset \text{ and } X. \\
E2 & : & X = \operatorname{Spec}(\mathbb{Z}/6\mathbb{Z}) \text{ has two clopen points and algebras } M_2, M_3, M_6.
\end{array}
$$

25.3 Definitions D1–D3

Let R be a commutative ring and let

$$X = \operatorname{Spec} R$$

be the set of prime ideals of R. For a subset $T \subset R$, define

$$V(T) = \{\mathfrak{p} \in X : T \subset \mathfrak{p}\}.$$

The Zariski closed subsets are the subsets $V(T)$. The open subsets are their complements. For $f \in R$, define the standard open

$$D(f) = \{\mathfrak{p} \in X : f \notin \mathfrak{p}\} = X \setminus V(\{f\}).$$

In this shard X is finite. Fix an ordering of X only to make tensor products strict. For $x = \mathfrak{p} \in X$, set

$$\mathcal{H}_x = \ell^2(\kappa(x)), \qquad \mathcal{A}_x = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_x),$$

where $\kappa(x)$ is the residue field at x. The local phase label group is $\kappa(x)^2$ with the standard alternating form

$$\omega_x((q,p),(q',p')) = pq' - qp'.$$

For any subset $S \subset X$, define

$$E(S) = \bigoplus_{x \in S} \kappa(x)^2, \qquad \mathcal{H}(S) = \bigotimes_{x \in S} \mathcal{H}_x, \qquad \mathcal{A}(S) = \bigotimes_{x \in S} \mathcal{A}_x.$$

The empty direct sum is 0, and the empty tensor product is $\mathbb{C}$.

25.4 Functorial Inclusions L1–L2

Lemma $L1$. If $S \subset T \subset X$, then

$$\iota_{S,T} : \mathcal{A}(S) \longrightarrow \mathcal{A}(T), \qquad a \longmapsto a \otimes \mathbf{1}_{\mathcal{H}(T \setminus S)}$$

is an injective unital $*$-homomorphism. If $S \subset T \subset U$, then

$$\iota_{T,U} \circ \iota_{S,T} = \iota_{S,U}.$$

Proof. The map preserves multiplication, adjoints, and the unit because tensoring with an identity operator preserves these operations. It is injective: if $a \otimes \mathbf{1} = 0$, then applying the operator to a simple tensor $v \otimes w$ with $w \neq 0$ gives $av \otimes w = 0$ for every v, hence $av = 0$ for every v, and $a = 0$. The composition identity is

$$(a \otimes \mathbf{1}_{T \setminus S}) \otimes \mathbf{1}_{U \setminus T} = a \otimes \mathbf{1}_{U \setminus S}.$$

Lemma $L2$. Let $W_S(e)$ denote the tensor product of local Weyl operators for $e \in E(S)$. Under the extension-by-zero inclusion $E(S) \hookrightarrow E(T)$,

$$\iota_{S,T}(W_S(e)) = W_T(e).$$

Consequently the finite Weyl-Heisenberg observable algebra generated by $E(S)$ is included in the one generated by $E(T)$.

Proof. For $x \in S$, the x-factor of $W_T(e)$ is the same local Weyl operator as in $W_S(e)$. For $x \in T \setminus S$, extension by zero gives the local label 0, and $W_x(0) = \mathbf{1}_{\mathcal{H}_x}$. Thus

$$W_T(e) = W_S(e) \otimes \mathbf{1}_{\mathcal{H}(T \setminus S)} = \iota_{S,T}(W_S(e)).$$

Taking complex spans gives the stated Weyl algebra inclusion.

25.5 Open-Local Versus Closed-Complement T1–T2

Theorem $T1$. The assignment

$$U \longmapsto \mathcal{A}_{\text{open}}(U) := \mathcal{A}(U)$$

from Zariski open subsets $U \subset X$ to finite-dimensional $*$-algebras is covariant under inclusion: if $U \subset V$, then

$$\mathcal{A}_{\text{open}}(U) \hookrightarrow \mathcal{A}_{\text{open}}(V)$$

by $\iota_{U,V}$. This is the natural choice when one wants observable algebras with inclusions following open inclusions.

Proof. An inclusion of opens $U \subset V$ is an inclusion of subsets of the finite set X. Lemma $L1$ gives the unital $*$-inclusion, and Lemma $L2$ identifies it with the corresponding inclusion of Weyl-Heisenberg algebras.

Theorem $T2$**.** The assignment

$$U \longmapsto \mathcal{A}_{\mathrm{cl}}(U) := \mathcal{A}(X \setminus U)$$

acts nontrivially only on the closed complement of U, but it is contravariant in opens. If $U \subset V$, then

$$X \setminus V \subset X \setminus U,$$

so the natural inclusion is

$$\mathcal{A}_{\mathrm{cl}}(V) \hookrightarrow \mathcal{A}_{\mathrm{cl}}(U).$$

Proof. The complement of an open set is closed by definition of open. Complementing the inclusion $U \subset V$ reverses it, giving $X \setminus V \subset X \setminus U$. Lemma $L1$ then gives the displayed inclusion. Thus this construction is valid as a closed-support or vanishing-locus algebra, but it is not the open-local net of Theorem $T1$.

25.6 Edge Case: A Field E1

Let $R = \mathbb{F}_p$. AQM-22 proves that

$$X = \operatorname{Spec} \mathbb{F}_p = \{\eta\}, \qquad \eta = (0).$$

The only Zariski closed subsets are $V(0) = X$ and $V(1) = \emptyset$, hence the only opens are $\emptyset$ and X. The open-local algebras are

$$\mathcal{A}_{\mathrm{open}}(\emptyset) = \mathbb{C}, \qquad \mathcal{A}_{\mathrm{open}}(X) = \mathrm{End}(\ell^2(\mathbb{F}_p)) \cong M_p(\mathbb{C}),$$

with the scalar inclusion $\lambda \mapsto \lambda \mathbf{1}$. The closed-complement assignment reverses these:

$$\mathcal{A}_{\mathrm{cl}}(\emptyset) = M_p(\mathbb{C}), \qquad \mathcal{A}_{\mathrm{cl}}(X) = \mathbb{C}.$$

Thus the one-point field case has no nontrivial intermediate localization; it only tests the direction of the convention.

25.7 The Two-Point Case $\mathbb{Z}/6\mathbb{Z}$ E2

Let $R = \mathbb{Z}/6\mathbb{Z}$. AQM-23 proves

$$X = \operatorname{Spec} R = \{\mathfrak{p}_2, \mathfrak{p}_3\}, \qquad \mathfrak{p}_2 = (2), \quad \mathfrak{p}_3 = (3),$$

with residue fields $\kappa(\mathfrak{p}_2) = \mathbb{F}_2$ and $\kappa(\mathfrak{p}_3) = \mathbb{F}_3$. Since $2 \in \mathfrak{p}_2$ and $2 \notin \mathfrak{p}_3$,

$$D(2) = \{\mathfrak{p}_3\}, \qquad V(2) = \{\mathfrak{p}_2\}.$$

Since $3 \notin \mathfrak{p}_2$ and $3 \in \mathfrak{p}_3$,

$$D(3) = \{\mathfrak{p}_2\}, \qquad V(3) = \{\mathfrak{p}_3\}.$$

The standard opens therefore contain both singletons, so the topology is discrete:

$$\emptyset, \quad \{\mathfrak{p}_2\} = D(3), \quad \{\mathfrak{p}_3\} = D(2), \quad X.$$

With the order $\mathfrak{p}_2, \mathfrak{p}_3$,

$$\mathcal{H}(X) = \mathbb{C}^2 \otimes \mathbb{C}^3, \qquad \mathcal{A}(X) = M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}).$$

The open-local and closed-complement algebras are:

U	$\mathcal{A}_{\mathrm{open}}(U) \subset \mathcal{A}(X)$	$\mathcal{A}_{\mathrm{cl}}(U) = \mathcal{A}(X \setminus U)$
$\emptyset$	$\mathbb{C}\,\mathbf{1}$	$M_2 \otimes M_3$
$D(3) = \{\mathfrak{p}_2\}$	$M_2 \otimes \mathbf{1}_3$	$\mathbf{1}_2 \otimes M_3$
$D(2) = \{\mathfrak{p}_3\}$	$\mathbf{1}_2 \otimes M_3$	$M_2 \otimes \mathbf{1}_3$
X	$M_2 \otimes M_3$	$\mathbb{C}\,\mathbf{1}$

Thus $D(3)$, the open where 3 does not vanish, carries the qubit algebra in the open-local convention and the qutrit algebra in the closed-complement convention. Likewise $D(2)$ carries the qutrit algebra open-locally and the qubit algebra as its closed complement.

Finally,

$$[M_2 \otimes \mathbf{1}_3, \mathbf{1}_2 \otimes M_3] = 0,$$

and the two singleton open algebras generate

$$(M_2 \otimes \mathbf{1}_3)(\mathbf{1}_2 \otimes M_3) = M_2 \otimes M_3.$$

This is the finite-spectrum analogue of locality: disjoint point supports commute, and the union of the supports generates the tensor-product algebra.

25.8 Conclusion

The best correspondence for the present programme is

$$\boxed{U \text{ open} \quad \rightsquigarrow \quad \mathcal{A}(U) \text{ acting nontrivially on } U.}$$

This gives isotony of observable algebras:

$$U \subset V \quad \Rightarrow \quad \mathcal{A}(U) \subset \mathcal{A}(V).$$

The alternative

$$U \rightsquigarrow \mathcal{A}(X \setminus U)$$

should be named explicitly as a closed-complement or vanishing-locus assignment. It is useful when one wants observables supported where a function vanishes, but its functorial direction is opposite to open inclusion.

26 Quasi-Local Zariski Algebra

26.1 Status and Source Anchors

This shard records the finite-spectrum answer to the directed-system question left by AQM-24. It is restricted to the same residue-qudit setting as AQM-24: finite spectra with finite local Weyl factors and tensor-by-identity inclusions. It does not define a general infinite affine quasi-local algebra by taking a colimit over quasi-compact opens.

The Zariski-topology source remains `references/algebraic_geometry/stacks_project_algebra.tex`: standard opens form a basis in lines 3104–3117, $\operatorname{Spec} R$ is quasi-compact in lines 3251–3273, and the topology has a quasi-compact open basis with quasi-compact intersections in lines 3275–3298.

No analytic completion convention is fixed here. The object constructed below is the algebraic finite-support $*$-algebra. Because X is finite in this shard, that algebra has the global finite tensor product $\mathcal{A}(X)$ as a terminal value. The later infinite affine finite-region construction is AQM-28: finite closed supports inside $U \cap |X|_{\mathrm{cl}}$.

26.2 Lamport Dependency Skeleton

$D1$: A directed set is a preorder in which any two elements have a common upper bound.
$D2$: X is a finite residue-qudit spectrum as in AQM-24.
$D3$: $\mathsf{F}(X) = \{S \subset X : S \text{ finite}\}, \quad \mathcal{A}(S) = \bigotimes_{x\in S} \mathcal{A}_x$.
$L1$: $\mathsf{F}(X)$ is directed under inclusion.
$L2$: $S \subset T \Rightarrow \mathcal{A}(S) \hookrightarrow \mathcal{A}(T)$ functorially by tensoring identities.
$T1$: $\varinjlim_{S\in\mathsf{F}(X)} \mathcal{A}(S) = \mathcal{A}(X)$ for finite X.
$T2$: Quasi-compact opens in affine spectra have a global terminal open and are not the infinite finite-region i
$T3$: Infinite affine locality is the finite closed-support system of AQM-28.

26.3 Finite-Support Poset D1–L2

A preorder I is called directed if for every $i, j \in I$ there exists $k \in I$ such that $i \le k$ and $j \le k$. A directed system of $*$-algebras over I is a family $(A_i)_{i\in I}$ with structure maps $\alpha_{ij} : A_i \to A_j$ for $i \le j$, satisfying

$$\alpha_{ii} = \mathrm{id}, \qquad \alpha_{jk}\alpha_{ij} = \alpha_{ik}.$$

Let $X = \operatorname{Spec} R$ be one of the finite residue-qudit spectra of AQM-24. For a finite support $S \subset X$, AQM-24 defines

$$\mathcal{A}(S) = \bigotimes_{x\in S} \mathcal{A}_x$$

with empty tensor product $\mathbb{C}$. Write

$$\mathsf{F}(X) = \{S \subset X : S \text{ finite}\}$$

ordered by inclusion. Since X is finite here, every subset is finite.

Lemma $L1$. $\mathsf{F}(X)$ is directed under inclusion.

Proof. If $S, T \subset X$ are finite, then $S \cup T$ is finite and contains both. Thus $S \cup T$ is a common upper bound.

Lemma *L*2. The finite-support algebras form a directed system.

Proof. For $S \subset T$, use AQM-24's inclusion

$$\iota_{S,T} : \mathcal{A}(S) \hookrightarrow \mathcal{A}(T), \qquad a \mapsto a \otimes \mathbf{1}_{T \setminus S}.$$

AQM-24 proves that these maps are injective unital $*$-homomorphisms and that $\iota_{T,U}\iota_{S,T} = \iota_{S,U}$ for $S \subset T \subset U$. Together with Lemma $L1$, these are exactly the directed-system axioms.

26.4 Finite-Spectrum Quasi-Local System T1

Define the finite-spectrum algebraic quasi-local algebra by

$$\mathcal{A}_{\mathrm{finqloc}}(X) := \varinjlim_{S \in \mathsf{F}(X)} \mathcal{A}(S).$$

For an open $U \subset X$, use the same construction inside U:

$$\mathcal{A}_{\mathrm{finqloc}}(U) := \varinjlim_{S \in \mathsf{F}(U)} \mathcal{A}(S).$$

Since X and U are finite in this shard, X and U are terminal objects in their finite-support posets.

Theorem *T*1.

$$\mathcal{A}_{\mathrm{finqloc}}(X) \cong \mathcal{A}(X), \qquad \mathcal{A}_{\mathrm{finqloc}}(U) \cong \mathcal{A}(U).$$

Thus AQM-25 supplies the finite-spectrum analogue of a quasi-local algebra, not an infinite affine finite-region construction.

Proof. The colimit over a directed system with terminal object is the terminal value. Here X is terminal in $\mathsf{F}(X)$, and U is terminal in $\mathsf{F}(U)$. The comparison maps are the AQM-24 inclusions into the terminal tensor product.

26.5 Quasi-Compact Opens Boundary T2

The following tempting construction is not adopted here:

$$\varinjlim_{U \in \mathsf{K}(X)} \mathcal{A}(U), \qquad \mathsf{K}(X) = \{\text{quasi-compact Zariski opens of } X\}.$$

For an affine scheme $X = \operatorname{Spec} R$, the Stacks source cited above records that X is quasi-compact. Hence $X \in \mathsf{K}(X)$, and it is a terminal object under inclusion. Whenever all the algebras $\mathcal{A}(U)$ are already defined, this colimit is just the global algebra $\mathcal{A}(X)$. That terminal-object fact is harmless in the finite-spectrum examples below, but it does not model finite-region quasi-local behavior on an infinite affine space.

Therefore quasi-compact opens may be used here only as topological control data or in finite/global examples. They are not the general affine region poset for the programme until an additional nonterminal region convention, infinite tensor-product convention, or C^*-completion convention is fixed.

26.6 Closed Supports and AQM-28 T3

The closed-complement assignment from AQM-24 still sends an open U to

$$\mathcal{A}(X \setminus U).$$

With opens ordered by inclusion, this assignment is contravariant. Therefore it is not the primary open-local system if "local region grows" is meant to match $U \subset V$.

In finite spectra, the same tensor-factor construction becomes covariant if it is indexed directly by finite supports, and by finite closed supports when the intended support is a vanishing locus. If Z, Z' are closed subsets of the finite X, then $Z \cup Z'$ is closed and finite, and $Z, Z' \subset Z \cup Z'$. The assignment

$$Z \longmapsto \mathcal{A}(Z)$$

with the same tensor-by-identity inclusions is then a directed system. This is best read as a closed-support or vanishing-locus quasi-local algebra, not as the open-local observable algebra.

For infinite affine schemes of finite type over $\mathbb{F}_q$, this shard does not use all points of all quasi-compact opens as Weyl sites. AQM-28 gives the current convention: closed finite-residue sites only, finite supports $S \Subset U \cap |X|_{\mathrm{cl}}$, and

$$\mathcal{A}(U) = \varinjlim_{S \Subset U \cap |X|_{\mathrm{cl}}} \mathcal{A}(S).$$

That is the finite closed-support spine for later infinite affine locality.

26.7 Finite Checks

For $X = \operatorname{Spec} \mathbb{F}_p$, the space has a largest open X. Hence the finite-support directed system has a terminal object and

$$\mathcal{A}_{\mathrm{finqloc}}(X) = \mathcal{A}(X) \cong M_p(\mathbb{C}).$$

For $X = \operatorname{Spec}(\mathbb{Z}/6\mathbb{Z})$, all opens are finite and X is again terminal. The finite-spectrum quasi-local algebra is

$$\mathcal{A}_{\mathrm{finqloc}}(X) = M_2(\mathbb{C}) \otimes M_3(\mathbb{C}) \cong M_6(\mathbb{C}).$$

The proper open-local subalgebras are the two commuting singleton algebras

$$M_2(\mathbb{C}) \otimes \mathbf{1}_3, \qquad \mathbf{1}_2 \otimes M_3(\mathbb{C}),$$

and their common upper bound is the full open X.

26.8 Conclusion

For AQM-24 finite spectra, the algebraic finite-region convention is

$$\boxed{\mathcal{A}_{\mathrm{finqloc}}(X) = \varinjlim_{S \in \mathsf{F}(X)} \mathcal{A}(S) = \mathcal{A}(X)}$$

with S ranging over finite supports and with inclusions given by tensoring identities. The quasi-compact-open colimit is not the general affine finite-region construction, because affine X itself is quasi-compact and terminal. The closed-complement possibility is retained only as a separate closed-support construction, useful for vanishing loci and defects. Later infinite affine locality should cite the AQM-28 finite closed-support system, not a quasi-compact-open global colimit from this shard.

27 Presheaves, Cosheaves, and Observable Nets

27.1 Status and Source Anchors

Stacks Project `references/algebraic_geometry/stacks_project_sheaves.tex` defines presheaves in lines 73–123, sheaves of sets in lines 500–525, and category-valued sheaves by an equalizer diagram in lines 719–736. It also records that $U \mapsto \bigoplus_{x\in U} M_x$ is generally not a sheaf in lines 622–637.

Curry's cosheaf source `references/cosheaves/curry_1303_3255_source/ch_abstract_sheaves.tex` defines pre-sheaves and pre-cosheaves in lines 89–92, sheaves and cosheaves by cover limits and colimits in lines 102–118, and rewrites the cosheaf condition as a coequalizer in lines 184–218. The vector-space exact sequence form is in lines 243–259. Curry's category source `references/cosheaves/curry_1303_3255_source/ch_categories.tex` defines colimits in lines 337–358 and records coproducts and unions as colimits in lines 366–384.

27.2 Lamport Dependency Skeleton

$$
\begin{aligned}
D1 &: \ \mathsf{O}(X) \text{ is the category of open subsets ordered by inclusion.}\\
D2 &: \ \text{A presheaf is a functor } \mathsf{O}(X)^{op} \to \mathcal{C}.\\
D3 &: \ \text{A sheaf is a presheaf satisfying cover equalizer gluing.}\\
D4 &: \ \text{A precosheaf is a functor } \mathsf{O}(X) \to \mathcal{C}.\\
D5 &: \ \text{A cosheaf is a precosheaf satisfying cover coequalizer gluing.}\\
D6 &: \ U \mapsto E(U) = \textstyle\bigoplus_{x\in U} E_x \text{ is the label precosheaf.}\\
D7 &: \ U \mapsto \mathcal{A}(U) \text{ is the open-local observable assignment.}\\
L1 &: \ E \text{ satisfies the finite-cover cosheaf axiom in abelian groups.}\\
L2 &: \ \mathcal{A} \text{ is a precosheaf of unital } *\text{-algebras.}\\
L3 &: \ \mathcal{A} \text{ satisfies net additivity inside the ambient algebra.}\\
T1 &: \ \mathcal{A} \text{ is not an ordinary cosheaf of unital } *\text{-algebras.}\\
T2 &: \ \text{Quantization turns label cosheaf gluing into tensor-local net gluing.}
\end{aligned}
$$

27.3 Presheaves and Sheaves D1–D3

Let X be a topological space. The category $\mathsf{O}(X)$ has open sets $U \subset X$ as objects and a unique morphism $U \to V$ when $U \subset V$.

Let $\mathcal{C}$ be a category. A $\mathcal{C}$-valued presheaf on X is a contravariant functor

$$F : \mathsf{O}(X)^{op} \longrightarrow \mathcal{C}.$$

Equivalently, for $V \subset U$ it has restriction morphisms

$$\rho^U_V : F(U) \longrightarrow F(V)$$

with $\rho^U_U = \mathrm{id}$ and $\rho^U_W = \rho^V_W \rho^U_V$ for $W \subset V \subset U$.

Assume $\mathcal{C}$ has the products and equalizers needed for the displayed cover diagram. A presheaf F is a sheaf if for every open cover $U = \bigcup_i U_i$ the diagram

$$F(U) \longrightarrow \prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j)$$

is an equalizer. In sets this says that compatible local sections glue to a unique section over U.

27.4 Precosheaves and Cosheaves D4–D5

We use "precosheaf" and "copresheaf" synonymously. A $\mathcal{C}$-valued precosheaf on X is a covariant functor

$$G : \mathsf{O}(X) \longrightarrow \mathcal{C}.$$

For $V \subset U$ it has extension morphisms

$$r_{U,V} : G(V) \longrightarrow G(U)$$

with $r_{U,U} = \mathrm{id}$ and $r_{U,W} = r_{U,V} r_{V,W}$ for $W \subset V \subset U$.

Assume $\mathcal{C}$ has the coproducts and coequalizers needed for the displayed cover diagram. A precosheaf G is a cosheaf if for every open cover $U = \bigcup_i U_i$ the diagram

$$\coprod_{i,j} G(U_i \cap U_j) \rightrightarrows \coprod_i G(U_i) \longrightarrow G(U)$$

is a coequalizer. In an additive category such as abelian groups or vector spaces, the finite-cover form is the exact sequence

$$\bigoplus_{i,j} G(U_i \cap U_j) \longrightarrow \bigoplus_i G(U_i) \longrightarrow G(U) \longrightarrow 0.$$

The arrows identify data that have been counted twice on overlaps.

27.5 The Label Cosheaf D6–L1

At the finite residue-qudit level, each point $x \in X$ has a local Weyl label group $E_x = \kappa(x)^2$, regarded as an abelian group. Define

$$E(U) = \bigoplus_{x \in U} E_x.$$

For $V \subset U$, the extension map $E(V) \to E(U)$ is extension by zero outside V.

Lemma *L*1. For finite covers, E satisfies the cosheaf axiom in abelian groups.

Proof. Let $U = \bigcup_i U_i$ be a finite open cover. The map

$$\bigoplus_i E(U_i) \longrightarrow E(U)$$

adds the extension-by-zero images. It is surjective because every summand E_x with $x \in U$ lies in at least one U_i.

The only relations needed are overlap relations. If $x \in U_i \cap U_j$, then the two copies of E_x inside $E(U_i) \oplus E(U_j)$ have the same image in $E(U)$, and their difference is exactly in the image of $E(U_i \cap U_j)$. Conversely, if a finite sum maps to zero in $E(U)$, then for each point x the sum of its finitely many copies over the index set $\{i : x \in U_i\}$ is zero. This index set is connected through overlaps, because every pair i, j in it has $x \in U_i \cap U_j$. Hence the pointwise zero relation is generated by the pairwise overlap relations. Since the direct sum over points separates these pointwise calculations, the displayed sequence is exact.

27.6 The Observable Precosheaf D7–L3

Let $\mathcal{A}(U)$ be the open-local observable algebra from AQM-24. For $V \subset U$, define

$$\iota_{V,U} : \mathcal{A}(V) \hookrightarrow \mathcal{A}(U), \qquad a \mapsto a \otimes \mathbf{1}_{U \setminus V}.$$

Lemma $L2$. $U \mapsto \mathcal{A}(U)$ is a precosheaf of unital $*$-algebras.

Proof. The identity and composition laws are exactly Lemma $L1$ of AQM-24:

$$\iota_{U,U} = \mathrm{id}, \qquad \iota_{V,W}\iota_{U,V} = \iota_{U,W}$$

for $U \subset V \subset W$. Thus $\mathcal{A}$ is a covariant functor from $\mathsf{O}(X)$ to unital $*$-algebras.

Lemma $L3$. For finite unions,

$$\mathcal{A}(U \cup V) = \mathcal{A}(U) \vee \mathcal{A}(V),$$

where $\vee$ denotes the unital $*$-subalgebra generated by the two images inside $\mathcal{A}(U \cup V)$. If $U \cap V = \emptyset$, the two images commute and the generated algebra is the tensor product.

Proof. Every tensor factor of $\mathcal{A}(U \cup V)$ belongs to U or to V. Hence the images of $\mathcal{A}(U)$ and $\mathcal{A}(V)$ generate all local matrix factors in the union. If $U \cap V = \emptyset$, the two images act on disjoint tensor factors, so elementary tensors commute factor by factor. The generated algebra is then the tensor product of the two local algebras.

27.7 Why This Is Not an Ordinary Cosheaf T1

Theorem $T1$. The observable precosheaf $\mathcal{A}$ is not, in general, a cosheaf valued in the category of unital $*$-algebras with ordinary categorical colimits.

Proof. It is enough to use $X = \mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$. Let

$$U = \{(2)\}, \qquad V = \{(3)\}.$$

Then $U \cap V = \emptyset$, $U \cup V = X$, and AQM-24 gives

$$\mathcal{A}(U) = M_2, \qquad \mathcal{A}(V) = M_3, \qquad \mathcal{A}(X) = M_2 \otimes M_3.$$

Since $\mathcal{A}(\emptyset) = \mathbb{C}$, the cosheaf coequalizer for this cover is the coproduct of unital $*$-algebras amalgamated over the common unit:

$$M_2 *_{\mathbb{C}} M_3.$$

This coproduct is not $M_2 \otimes M_3$. Indeed, if it were the tensor product with the canonical embeddings, then every pair of unital $*$-homomorphisms $M_2 \to B$ and $M_3 \to B$ would have commuting images, because the two canonical tensor factors commute.

Take $B = M_6(\mathbb{C})$, written as 3×3 blocks in a 2×2 matrix. Let

$$f(a) = a \otimes I_3, \qquad g(b) = \begin{bmatrix} b & 0 \\ 0 & VbV^* \end{bmatrix},$$

where $V \in M_3(\mathbb{C})$ swaps the first two basis vectors. These are unital $*$-homomorphisms. For the matrix units $e_{12} \in M_2$ and $E_{11} \in M_3$,

$$f(e_{12})g(E_{11}) = \begin{bmatrix} 0 & E_{22} \\ 0 & 0 \end{bmatrix}, \qquad g(E_{11})f(e_{12}) = \begin{bmatrix} 0 & E_{11} \\ 0 & 0 \end{bmatrix}.$$

These are unequal. Therefore the universal object for arbitrary unital $*$-homomorphisms cannot be the tensor product. The cosheaf colimit in unital $*$-algebras is too free: it does not impose locality commutation.

27.8 Quantization and the Correct Dictionary T2

Theorem $T2$**.** The correct next-layer dictionary is:

$$\begin{aligned} E &: \text{ a cosheaf-like label object in abelian groups or vector spaces,} \\ \mathcal{A} &: \text{ a precosheaf/local net of observable algebras,} \\ & \quad \text{ with additivity and locality imposed inside the ambient algebra.} \end{aligned}$$

Proof. Lemma $L1$ proves that labels glue by the expected additive colimit. Lemmas $L2$ and $L3$ prove that observable algebras form a covariant local net and satisfy the physical additivity relation

$$\mathcal{A}(U \cup V) = \mathcal{A}(U) \vee \mathcal{A}(V).$$

Theorem $T1$ proves that this additivity is not the ordinary cosheaf axiom in unital $*$-algebras: categorical coproducts give free products, whereas quantized independent labels give commuting tensor factors. Thus quantization does not preserve the raw cosheaf colimit from labels to observable algebras; it turns additive label gluing into tensor-local algebra gluing.

27.9 Conclusion

The open-local observable assignment is naturally a precosheaf:

$$U \subset V \quad \Rightarrow \quad \mathcal{A}(U) \hookrightarrow \mathcal{A}(V).$$

It should not be called an ordinary cosheaf of unital $*$-algebras unless the target category or gluing notion is changed. The label layer E is the cosheaf-like layer. The observable layer is the AQFT-style net: isotone, local on disjoint opens, and additive by generated subalgebras.

28 State Presheaf of the Quasi-Local Algebra

28.1 Status and Source Anchors

This shard adds the state side of AQM-26. Derezinski records that normal states on $B(\mathcal{H})$ are given by trace-class functionals $\psi(A) = \mathrm{Tr}(\gamma A)$, with positive trace-one operators γ called density matrices, in `references/canonical_fields/derezinski_math_ph_0511030_source/derezinski.tex`, lines 3141–3149. Gottesman records the finite quantum-information convention that density matrices are positive trace-one operators and that a subsystem density matrix is obtained by tracing out the remaining degrees of freedom in `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 404–411.

28.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & \mathrm{State}(A) = \{\omega : A \to \mathbb{C} \mid \omega(1) = 1,\ \omega(a^*a) \geq 0\}. \\
D2 & : & \text{For } A = \mathrm{End}(\mathcal{H}),\ \omega(a) = \mathrm{Tr}(\rho a), \quad \rho \geq 0,\ \mathrm{Tr}\,\rho = 1. \\
D3 & : & \mathsf{S}(U) = \mathrm{State}(\mathcal{A}(U)). \\
L1 & : & U \subset V \Rightarrow \mathsf{S}(V) \to \mathsf{S}(U),\ \omega \mapsto \omega \circ \iota_{U,V},\ \text{is a state restriction.} \\
T1 & : & \mathsf{S} \text{ is a presheaf of convex sets.} \\
T2 & : & \mathsf{S} \text{ is not generally separated, hence not a sheaf.} \\
T3 & : & \text{Compatible overlapping marginals need not glue to a global state.} \\
T4 & : & \mathrm{State}(\mathcal{A}_{\mathrm{qloc}}(X)) \cong \varprojlim_U \mathrm{State}(\mathcal{A}(U)) \text{ for the algebraic filtered colimit.}
\end{array}$$

28.3 State Spaces D1–D3

Let A be a unital $*$-algebra. A state on A is a linear functional $\omega : A \to \mathbb{C}$ such that

$$\omega(1) = 1, \qquad \omega(a^*a) \geq 0 \quad \text{for all } a \in A.$$

For a finite-dimensional Hilbert space $\mathcal{H}$ and $A = \mathrm{End}(\mathcal{H})$, we use the density-matrix representation

$$\omega_\rho(a) = \mathrm{Tr}(\rho a), \qquad \rho \geq 0, \quad \mathrm{Tr}\,\rho = 1.$$

The state space is convex: if ω_0, ω_1 are states and $0 \leq t \leq 1$, then $t\omega_0 + (1-t)\omega_1$ is again normalized and positive.

For the open-local observable net $U \mapsto \mathcal{A}(U)$, define

$$\mathsf{S}(U) = \mathrm{State}(\mathcal{A}(U)).$$

28.4 Restriction Is a Presheaf L1–T1

Lemma $L1$. If $U \subset V$, then

$$\rho^V_U : \mathsf{S}(V) \longrightarrow \mathsf{S}(U), \qquad \rho^V_U(\omega) = \omega \circ \iota_{U,V}$$

is a well-defined affine map.

Proof. The inclusion $\iota_{U,V}$ is a unital $*$-homomorphism by AQM-24. Thus

$$\rho^V_U(\omega)(1) = \omega(\iota_{U,V}(1)) = \omega(1) = 1.$$

For $a \in \mathcal{A}(U)$,

$$\rho^V_U(\omega)(a^*a) = \omega(\iota_{U,V}(a^*a)) = \omega(\iota_{U,V}(a)^*\iota_{U,V}(a)) \geq 0.$$

Hence $\rho_U^V(\omega)$ is a state. Affineness follows from linearity of composition with $\iota_{U,V}$.

When $V = U \sqcup W$, the same restriction is the density-matrix partial trace. If ω_ρ is a state on $\mathcal{A}(U) \otimes \mathcal{A}(W)$, then the restricted state on $\mathcal{A}(U)$ is represented by

$$\rho_U = \mathrm{Tr}_W(\rho),$$

because $\mathrm{Tr}(\rho(a \otimes 1_W)) = \mathrm{Tr}(\rho_U a)$.

Theorem $T1$. S is a presheaf of convex sets on X.

Proof. For $U \subset V \subset W$,

$$\rho_U^W(\omega) = \omega \circ \iota_{U,W} = \omega \circ \iota_{V,W} \circ \iota_{U,V} = \rho_U^V(\rho_V^W(\omega)),$$

using functoriality of ι from AQM-24. Also $\rho_U^U = \mathrm{id}$. The restriction maps are affine by Lemma $L1$, so this is a contravariant functor from opens to convex sets.

28.5 Failure of Separatedness T2

Theorem $T2$. S is not generally a sheaf. In fact it is not generally separated: two global states can have the same restrictions to an open cover.

Proof. Use $X = \mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$, with the two singleton opens

$$U = \{(2)\}, \qquad V = \{(3)\}, \qquad X = U \sqcup V.$$

Then $\mathcal{A}(X) = M_2 \otimes M_3$. Choose bases $|0\rangle, |1\rangle$ of $\mathbb{C}^2$ and $|0\rangle, |1\rangle, |2\rangle$ of $\mathbb{C}^3$, and set

$$|\psi\rangle = \frac{|0,0\rangle + |1,1\rangle}{\sqrt{2}}, \qquad \rho = |\psi\rangle\langle\psi|.$$

Its restrictions are

$$\rho_U = \frac{I_2}{2}, \qquad \rho_V = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2}.$$

Now define the product density matrix

$$\sigma = \rho_U \otimes \rho_V.$$

The density matrices ρ and σ have the same restrictions to U and to V by construction. They are not equal: ρ has the off-diagonal matrix entry $|0,0\rangle\langle 1,1|/2$, while σ has no such entry. Thus the two global states agree on the cover $\{U, V\}$ but differ globally. The uniqueness part of the sheaf condition fails.

28.6 Failure of Existence for Overlapping Marginals T3

Theorem $T3$. Even when local states agree on an overlap, a global extending state need not exist.

Proof. Consider a discrete three-point finite spectrum with qubit factors labelled A, B, C. For example, one may take $\mathrm{Spec}(\mathbb{F}_2 \times \mathbb{F}_2 \times \mathbb{F}_2)$. Let

$$U = \{A, B\}, \qquad V = \{B, C\}, \qquad U \cap V = \{B\}.$$

On AB, take the Bell state

$$\rho_{AB} = |\Phi_{AB}\rangle\langle\Phi_{AB}|, \qquad |\Phi_{AB}\rangle = \frac{|0,0\rangle + |1,1\rangle}{\sqrt{2}}.$$

On BC, take

$$\rho_{BC} = |\Phi_{BC}\rangle\langle\Phi_{BC}|, \qquad |\Phi_{BC}\rangle = \frac{|0,0\rangle + |1,1\rangle}{\sqrt{2}}.$$

Both restrict to $I_2/2$ on the overlap B.

Assume a density matrix θ_{ABC} extends both. Since $\mathrm{Tr}_C\,\theta_{ABC} = \rho_{AB}$ is the rank-one projection $P_{AB} = |\Phi_{AB}\rangle\langle\Phi_{AB}|$, positivity forces θ_{ABC} to be supported on $\mathrm{ran}(P_{AB}) \otimes \mathbb{C}^2_C$. Indeed,

$$\mathrm{Tr}\big(((1 - P_{AB}) \otimes 1_C)\theta_{ABC}\big) = \mathrm{Tr}((1 - P_{AB})\rho_{AB}) = 0,$$

and a positive operator with zero trace is zero on that positive part. Therefore

$$\theta_{ABC} = P_{AB} \otimes \tau_C$$

for some density matrix τ_C. Its BC-marginal is then

$$\mathrm{Tr}_A(\theta_{ABC}) = \frac{I_2}{2} \otimes \tau_C,$$

which is a product state. It cannot equal the pure entangled Bell state ρ_{BC}. Hence the compatible pair (ρ_{AB}, ρ_{BC}) does not glue to a state on ABC.

28.7 Global States as an Inverse Limit T4

Let $\mathsf{K}(X)$ be the directed poset of quasi-compact opens used in AQM-25, and let

$$\mathcal{A}_{\mathrm{qloc}}(X) = \varinjlim_{U \in \mathsf{K}(X)} \mathcal{A}(U)$$

be the algebraic quasi-local $*$-algebra. Since the structure maps are injective, we may regard $\mathcal{A}_{\mathrm{qloc}}(X)$ as the algebraic union of the local algebras.

Theorem *T*4. There is a natural affine bijection

$$\mathrm{State}(\mathcal{A}_{\mathrm{qloc}}(X)) \cong \varprojlim_{U \in \mathsf{K}(X)} \mathrm{State}(\mathcal{A}(U)).$$

Proof. Given a state ω on $\mathcal{A}_{\mathrm{qloc}}(X)$, restrict it to each $\mathcal{A}(U)$. These restrictions are compatible by functoriality of restriction, so they define an element of the inverse limit.

Conversely, let $(\omega_U)_U$ be a compatible family of local states. If $a \in \mathcal{A}_{\mathrm{qloc}}(X)$, choose U and $a_U \in \mathcal{A}(U)$ representing it, and define

$$\omega(a) = \omega_U(a_U).$$

This is independent of the representative: if a_U and a_V represent the same quasi-local element, choose $W \supset U, V$; their images in $\mathcal{A}(W)$ agree, and compatibility gives the same value. Linearity follows after moving two representatives to a common W. Normalization is $\omega(1) = 1$. Positivity holds because if $a \in \mathcal{A}(U)$, then $a^*a \in \mathcal{A}(U)$ and

$$\omega(a^*a) = \omega_U(a^*a) \geq 0.$$

The two constructions are inverse to each other.

28.8 Conclusion

The state side reverses the observable-net arrows:

$$U \subset V \quad \Rightarrow \quad \mathrm{State}(\mathcal{A}(V)) \longrightarrow \mathrm{State}(\mathcal{A}(U)).$$

Thus local quantum states form a presheaf of convex sets. They do not form a sheaf in general: uniqueness fails because marginals do not determine correlations, and existence fails because compatible overlapping marginals may not have a joint quantum state. The correct quasi-local replacement is the inverse-limit statement of Theorem $T4$, not an ordinary sheaf gluing axiom.

29 Closed-Point Affine Arithmetic Fields

29.1 Status and Source Anchors

This shard records the convention needed before treating $\operatorname{Spec}\mathbb{F}_q[t]$ and $\operatorname{Spec}\mathbb{F}_q[x,y]$. The finite-spectrum rule of AQM-23 and AQM-24 attached one finite Weyl system to each point because all residue fields in those examples were finite. For infinite affine schemes over $\mathbb{F}_q$, not every point has finite residue field. The finite qudit layer is therefore indexed by closed points only.

The topology and residue-field definitions are source-local from the Stacks Project snapshot: `references/algebraic_geometry/stacks_project_algebra.tex`, lines 120–158, 250–274, 2972–2983, 3104–3117, and 3310–3350. The Nullstellensatz finite-residue statement is in the same file, lines 6520–6665. The finite Weyl laws use Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313, and the Pauli symplectic commutator source used in AQM-22.

29.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k=\mathbb{F}_q,\ R \text{ a finite-type } k\text{-algebra},\ X=\operatorname{Spec}R. \\
D2 & : & |X|_{\mathrm{cl}}=\operatorname{Max}R, \quad d(x)=[\kappa(x):k] \text{ for } x\in|X|_{\mathrm{cl}}. \\
L1 & : & \kappa(x) \text{ is finite and } |\kappa(x)|=q^{d(x)} \text{ for closed } x. \\
D3 & : & S\Subset|X|_{\mathrm{cl}} \text{ is a finite closed support.} \\
D4 & : & E(S)=\bigoplus_{x\in S}\kappa(x)^2, \quad \mathcal{H}(S)=\bigotimes_{x\in S}\ell^2(\kappa(x)). \\
L2 & : & E(S) \text{ has a finite Weyl projective representation } W_S. \\
D5 & : & C(U)=U\cap|X|_{\mathrm{cl}},\ E(U)=\bigoplus_{x\in C(U)}\kappa(x)^2,\ \mathcal{A}(U)=\varinjlim_{S\Subset C(U)}\mathcal{A}(S). \\
T1 & : & U\mapsto\mathcal{A}(U) \text{ is an open-local net.} \\
T2 & : & \operatorname{State}(\mathcal{A}(U))\cong\varprojlim_{S\Subset C(U)}\operatorname{State}(\mathcal{A}(S)). \\
T3 & : & \text{Regular-function profiles are generally not quasi-local operators.}
\end{array}$$

29.3 Closed Finite-Residue Sites D1–L1

Let $k=\mathbb{F}_q$, $q=p^r$, and let R be a finite-type k-algebra. Put $X=\operatorname{Spec}R$. A closed point of X means a maximal ideal $x=\mathfrak{m}\subset R$. Its residue field is

$$\kappa(x)=R_{\mathfrak{m}}/\mathfrak{m}R_{\mathfrak{m}}\cong R/\mathfrak{m}.$$

The last isomorphism holds because $\mathfrak{m}$ is maximal.

Lemma $L1$. If $x\in|X|_{\mathrm{cl}}$, then $\kappa(x)$ is a finite field. More precisely, if $d(x)=[\kappa(x):k]$, then

$$|\kappa(x)|=q^{d(x)}.$$

Proof. The Stacks Nullstellensatz source says that for a maximal ideal in a finite type k-algebra, the residue field is a finite extension of k. Hence $\kappa(x)$ is a $d(x)$-dimensional vector space over the q-element field k, so it has $q^{d(x)}$ elements.

Non-closed points are not discarded from X; they remain topological points. They are simply not finite Weyl sites in this convention unless an additional infinite-residue representation convention is later fixed.

29.4 Finite Closed Supports D3–L2

For a finite subset $S \Subset |X|_{\mathrm{cl}}$, define

$$E(S) = \bigoplus_{x \in S} \kappa(x)^2.$$

Write $e_x = (q_x, p_x)$. Define the local alternating form

$$\omega_x(e_x, f_x) = p_x q'_x - q_x p'_x$$

and the finite-support form

$$\Omega_S(e, f) = (\omega_x(e_x, f_x))_{x \in S}.$$

This is not a form into one field when the residue degrees vary. The representation uses the associated alternating bicharacter. Let

$$\psi_x(a) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{\kappa(x)/\mathbb{F}_p}(a)\right).$$

Set

$$B_S(e, f) = \prod_{x \in S} \psi_x(p_x q'_x - q_x p'_x).$$

Let

$$\mathcal{H}_x = \ell^2(\kappa(x)), \qquad \mathcal{H}(S) = \bigotimes_{x \in S} \mathcal{H}_x.$$

On the basis $|a\rangle$, $a \in \kappa(x)$, define

$$T_x(q)|a\rangle = |a + q\rangle, \qquad R_x(p)|a\rangle = \psi_x(pa)|a\rangle, \qquad W_x(q, p) = T_x(q)R_x(p).$$

For $e \in E(S)$, put $W_S(e) = \bigotimes_{x \in S} W_x(e_x)$.

Lemma *L2.* For $e, f \in E(S)$,

$$W_S(e)W_S(f) = c_S(e, f)W_S(e + f), \qquad c_S(e, f) = \prod_{x \in S} \psi_x(p_x q'_x),$$

and

$$W_S(e)W_S(f) = B_S(e, f)W_S(f)W_S(e).$$

The complex span

$$\mathcal{A}(S) = \operatorname{span}_{\mathbb{C}}\{W_S(e) : e \in E(S)\}$$

is

$$\mathcal{A}(S) \cong \bigotimes_{x \in S} \operatorname{End}(\mathcal{H}_x).$$

Proof. At one site,

$$R_x(p)T_x(q')|a\rangle = \psi_x(p(a + q'))|a + q'\rangle = \psi_x(pq')T_x(q')R_x(p)|a\rangle.$$

Thus $W_x(e_x)W_x(f_x) = \psi_x(p_x q'_x)W_x(e_x + f_x)$, and dividing by the same formula with e_x, f_x reversed gives the local commutator $\psi_x(p_x q'_x - q_x p'_x)$. Tensoring over the finite set S gives the two displayed laws. The local Weyl source says the finite-field operators $T^a R^b$ form an operator basis; tensor products of bases are bases of tensor products.

29.5 Open-Local Algebra D5–T1

For an open set $U \subset X$, define its closed-point support

$$C(U) = U \cap |X|_{\mathrm{cl}}.$$

The field-label group on U is the finite-support direct sum

$$E(U) = \bigoplus_{x \in C(U)} \kappa(x)^2 = \varinjlim_{S \Subset C(U)} E(S).$$

The observable algebra is the algebraic quasi-local algebra over finite closed supports:

$$\mathcal{A}(U) = \varinjlim_{S \Subset C(U)} \mathcal{A}(S) = \mathrm{span}_{\mathbb{C}}\{W_U(e) : e \in E(U)\}.$$

If $S \subset T$, the structure map is

$$a \longmapsto a \otimes \mathbf{1}_{\mathcal{H}(T \setminus S)}.$$

Theorem $T1$. $U \mapsto \mathcal{A}(U)$ is an open-local net: it is isotone, local on disjoint closed supports, and additive for open covers.

Proof. If $U \subset V$, then $C(U) \subset C(V)$. The finite-support inclusions therefore induce a unital injective $*$-homomorphism $\mathcal{A}(U) \hookrightarrow \mathcal{A}(V)$. If two labels have disjoint closed supports, then every local commutator factor in $B(e, f)$ is 1, so their Weyl operators commute. If $U = \bigcup_i U_i$ and $e \in E(U)$, the support of e is finite. Assign each support point to one U_i that contains it. Then e is a finite sum of labels $e_i \in E(U_i)$, and $W_U(e)$ is a scalar times a product of the corresponding $W_{U_i}(e_i)$. Thus $\mathcal{A}(U)$ is generated by the images of the $\mathcal{A}(U_i)$.

29.6 States T2

A state on a unital $*$-algebra is a normalized positive linear functional. For finite S, states on $\mathcal{A}(S)$ are density matrices on $\mathcal{H}(S)$. For an open U, define

$$\mathsf{S}(U) = \mathrm{State}(\mathcal{A}(U)).$$

Theorem $T2$. There is a natural affine bijection

$$\mathrm{State}(\mathcal{A}(U)) \cong \varprojlim_{S \Subset C(U)} \mathrm{State}(\mathcal{A}(S)).$$

Proof. The algebra $\mathcal{A}(U)$ is the filtered algebraic colimit of the finite matrix algebras $\mathcal{A}(S)$. Restricting a state to each finite $\mathcal{A}(S)$ gives a compatible inverse-limit family. Conversely, a compatible family (ω_S) defines $\omega(a) = \omega_S(a)$ whenever $a \in \mathcal{A}(S)$. If a has two finite-support representatives, move both to a common larger finite support; compatibility gives the same value. Linearity and positivity are checked in a common finite support. The two constructions are inverse.

If local density matrices ρ_x are chosen for all closed $x \in C(U)$, then

$$\rho_S = \bigotimes_{x \in S} \rho_x$$

defines a compatible product state. Taking each ρ_x to be the normalized identity gives the product tracial state, whose value on $W_U(e)$ is 0 for $e \neq 0$, because some nontrivial local Weyl operator has trace 0.

29.7 Regular Profiles Are Not Local Operators T3

For $a, b \in R$, reduction modulo each closed point gives a closed-point field profile

$$e_{a,b}(x) = (a \bmod x,\ b \bmod x) \in \kappa(x)^2.$$

This is a genuine arithmetic profile over all closed points. It is an element of $E(U)$, however, only when it has finite support in $C(U)$. Otherwise the formal product $W(e_{a,b})$ is an infinite-volume field symbol, not an element of the algebraic quasi-local algebra $\mathcal{A}(U)$.

Thus the present finite-residue theory has two layers:

$$\text{quasi-local Weyl observables} \quad \text{and} \quad \text{formal regular-function field profiles.}$$

A later analytic or representation-theoretic convention may relate them, but the current report does not silently identify them.

30 The Arithmetic Field on $\operatorname{Spec}\mathbb{F}_q[t]$

30.1 Status and Source Anchors

This shard instantiates AQM-28 for $R_1 = k[t]$, $k = \mathbb{F}_q$. The spectrum and $D(f)$ notation are sourced from the Stacks Project snapshot, `references/algebraic_geometry/stacks_project_algebra.tex`, lines 2972–2983 and 3104–3117. Residue fields are sourced from lines 250–274 and 3310–3350. The polynomial-ring classification below is proved directly from the Euclidean property of $k[t]$. The Weyl layer is the finite closed-point construction of AQM-28.

30.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & R_1 = k[t], \quad X_1 = \operatorname{Spec} R_1. \\
L1 & : & X_1 = \{\eta = (0)\} \cup \{x_\pi = (\pi) : \pi \text{ monic irreducible}\}. \\
L2 & : & \kappa(\eta) = k(t), \quad \kappa(x_\pi) = k[t]/(\pi), \quad |\kappa(x_\pi)| = q^{\deg \pi}. \\
L3 & : & D(g) = \emptyset \text{ if } g = 0, \text{ and for } g \neq 0 \text{ it omits exactly the closed points dividing } g. \\
D2 & : & E_1(U) = \bigoplus_{x_\pi \in U} \kappa(x_\pi)^2, \quad \mathcal{A}_1(U) = \varinjlim_{S \Subset C(U)} \mathcal{A}(S). \\
T1 & : & \mathcal{A}_1(U) \text{ is the open-local Weyl algebra on closed points of } U. \\
T2 & : & \operatorname{State}(\mathcal{A}_1(U)) \text{ is the inverse limit of finite-volume density-matrix states.} \\
T3 & : & \text{Nonzero polynomial field profiles have infinite closed-point support.}
\end{array}
$$

30.3 Points of the Affine Line L1–L2

Let $\mathcal{P}_k$ be the set of monic irreducible polynomials in $k[t]$. The zero ideal is prime because $k[t]$ is a domain; write

$$\eta = (0)$$

for the generic point.

Lemma $L1$. The remaining prime ideals are exactly

$$x_\pi = (\pi), \qquad \pi \in \mathcal{P}_k.$$

Proof. Since $k[t]$ is Euclidean, every nonzero ideal has a unique monic generator. Let $\mathfrak{p} \neq (0)$ be prime and write $\mathfrak{p} = (g)$ with g monic. If $g = ab$ with nonconstant a, b, then $ab \in \mathfrak{p}$, so primality gives $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, contradicting minimal degree of the monic generator g. Thus g is irreducible. Conversely, if π is irreducible, then $k[t]/(\pi)$ is a field, so (π) is maximal and therefore prime.

Lemma $L2$.

$$\kappa(\eta) = k(t), \qquad \kappa(x_\pi) = k[t]/(\pi).$$

If $d = \deg \pi$, then $\kappa(x_\pi)$ has q^d elements. Thus the closed point x_π carries one q^d-level finite-field qudit.

Proof. The residue field at the zero prime is the fraction field of $k[t]$, namely $k(t)$. For the maximal ideal (π), localization does not change the quotient residue field, so $\kappa(x_\pi) = k[t]/(\pi)$. The residue classes of $1, t, \ldots, t^{d-1}$ form a k-basis, hence the quotient has q^d elements.

The generic point η has infinite residue field $k(t)$. It is not a finite Weyl site under the closed-point convention of AQM-28.

30.4 Zariski Opens L3

Let $g \in k[t]$. If $g = 0$, then $D(g) = \emptyset$. If

$$g = u \prod_{i=1}^{r} \pi_i^{n_i}$$

with $u \in k^\times$ and distinct monic irreducibles π_i, then

$$D(g) = \{\eta\} \cup \{x_\pi : \pi \in \mathcal{P}_k,\ \pi \notin \{\pi_1, \ldots, \pi_r\}\}.$$

Derivation. The generic prime (0) contains no nonzero g, so $\eta \in D(g)$. The closed prime (π) contains g iff π divides g. Therefore $D(g)$ is exactly the displayed set. In particular every nonempty open contains η, and its closed-point support is cofinite in $\mathcal{P}_k$.

For a closed point x_π,

$$\{x_\pi\} = V(\pi), \qquad X_1 \setminus \{x_\pi\} = D(\pi).$$

Thus closed singletons are not open, but their complements are standard opens.

30.5 Weyl-Heisenberg Field Net D2–T1

For an open $U \subset X_1$, define

$$C_1(U) = U \cap \{x_\pi : \pi \in \mathcal{P}_k\}.$$

The affine-line field-label group is

$$E_1(U) = \bigoplus_{x_\pi \in C_1(U)} \kappa(x_\pi)^2.$$

For $D(g)$ with $g \neq 0$,

$$E_1(D(g)) = \bigoplus_{\pi \nmid g} \big(k[t]/(\pi)\big)^2.$$

The corresponding algebra is

$$\mathcal{A}_1(D(g)) = \varinjlim_{S \Subset \{\pi : \pi \nmid g\}} \bigotimes_{\pi \in S} M_{q^{\deg \pi}}(\mathbb{C}).$$

For $g = 1$, this is the global algebra

$$\mathcal{A}_1(X_1) = \varinjlim_{S \Subset \mathcal{P}_k} \bigotimes_{\pi \in S} M_{q^{\deg \pi}}(\mathbb{C}).$$

Theorem $T1$. The assignment $U \mapsto \mathcal{A}_1(U)$ is the open-local Weyl-Heisenberg net on $\operatorname{Spec} k[t]$.

Proof. This is AQM-28 applied to $R_1 = k[t]$. If $U \subset V$, then $C_1(U) \subset C_1(V)$, so finite closed supports in U are finite closed supports in V. The algebra inclusion tensors by identities on the extra closed points. Additivity for open covers follows because every Weyl label has finite closed support.

At a single closed point x_π, let $K_\pi = k[t]/(\pi)$. For $a, b \in K_\pi$, write

$$X_\pi(a) = W_\pi(a, 0), \qquad Z_\pi(b) = W_\pi(0, b).$$

Then

$$Z_\pi(b) X_\pi(a) = \psi_\pi(ab) X_\pi(a) Z_\pi(b),$$

while operators at distinct closed points commute.

30.6 States T2

For every open U,

$$\mathrm{State}(\mathcal{A}_1(U)) \cong \varprojlim_{S \Subset C_1(U)} \mathrm{State}\left(\bigotimes_{x_\pi \in S} M_{q^{\deg \pi}}(\mathbb{C}) \right).$$

The restriction maps are partial traces over omitted closed points. The product tracial state is obtained by taking the normalized trace at every closed point; it sends every nontrivial finite-support Weyl operator to 0.

The state presheaf is not separated for the Zariski topology. Let

$$x_0 = (t), \qquad x_1 = (t-1), \qquad U = D(t), \quad V = D(t-1).$$

Then $U \cup V = X_1$. On the two q-dimensional factors x_0, x_1, take the maximally entangled vector

$$|\Phi\rangle = q^{-1/2} \sum_{a \in k} |a, a\rangle.$$

Let $\rho = |\Phi\rangle\langle\Phi|$, and let $\sigma = (I_q/q) \otimes (I_q/q)$. Tensor both with the product tracial state on all other closed points. The two global states restrict equally to U and to V, because each restriction traces out one of x_0, x_1. They differ globally, since $\rho \neq \sigma$ on the two-site subalgebra.

30.7 Polynomial Field Profiles T3

A pair of polynomials $a, b \in k[t]$ defines a closed-point profile

$$e_{a,b}(x_\pi) = (a \bmod \pi,\ b \bmod \pi).$$

There are infinitely many monic irreducibles in $k[t]$: if $\pi_1, \ldots, \pi_n$ were all of them, then $P = \pi_1 \cdots \pi_n$ is nonconstant and $P + 1$ has an irreducible divisor which divides none of the π_i, a contradiction. If $a \neq 0$, then $a \bmod \pi = 0$ only for the finitely many irreducible divisors of a. Hence a is nonzero at infinitely many closed points. The same holds for b. Therefore

$$e_{a,b} \in E_1(X_1) \iff a = b = 0.$$

Thus the coordinate function t, and every nonzero polynomial profile, is a formal infinite-support field profile. Its finite restrictions

$$e_{a,b}|_S \in \bigoplus_{x_\pi \in S} \kappa(x_\pi)^2$$

give finite-volume Weyl operators, but the full profile does not define an element of the algebraic quasi-local algebra.

30.8 Conclusion

Spec $k[t]$ produces a countable closed-point array of finite-field qudits: one q^d-level qudit for each monic irreducible polynomial of degree d. Nonempty Zariski opens remove finitely many closed points and always contain the generic point, but the generic point has residue field $k(t)$ and is not a finite Weyl site in the current convention. The quasi-local observables are finite closed-support Weyl operators; polynomial fields are formal infinite-volume profiles until a completion convention is added.

31 The Arithmetic Field on $\operatorname{Spec}\mathbb{F}_q[x,y]$

31.1 Status and Source Anchors

This shard instantiates AQM-28 for $R_2 = k[x,y]$, $k = \mathbb{F}_q$. The Stacks Project source describes $\operatorname{Spec} k[x,y]$ in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 4889–4927: besides (0), height-one primes are generated by irreducible polynomials, and non-principal primes are maximal of the displayed quotient form. The finite residue field statement for maximal ideals is the Nullstellensatz source, lines 6520–6665. Weyl and state conventions are those of AQM-28.

31.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & R_2 = k[x,y], \quad X_2 = \operatorname{Spec} R_2. \\
L1 & : & X_2 \text{ has } (0),\ \text{height-one points } (f),\ \text{and closed maximal ideals } \mathfrak{m}. \\
L2 & : & \kappa(0) = k(x,y),\ \kappa((f)) = \operatorname{Frac}(k[x,y]/(f)),\ \kappa(\mathfrak{m})/k \text{ finite}. \\
D2 & : & C_2(U) = U \cap \operatorname{Max} R_2, \quad E_2(U) = \bigoplus_{\mathfrak{m}\in C_2(U)} \kappa(\mathfrak{m})^2. \\
T1 & : & \mathcal{A}_2(U) = \varinjlim_{S \Subset C_2(U)} \bigotimes_{\mathfrak{m}\in S} M_{q^{d(\mathfrak{m})}}(\mathbb{C}). \\
T2 & : & \text{The rational-point truncation is a finite } q^2\text{-site subtheory, not the full Spec theory.} \\
T3 & : & \operatorname{State}(\mathcal{A}_2(U)) \text{ is an inverse limit over finite closed supports.} \\
T4 & : & \text{Coordinate profiles such as } x, y \text{ are formal infinite-support fields.}
\end{array}$$

31.3 Point Types L1–L2

Let $X_2 = \operatorname{Spec} k[x,y]$.

Lemma $L1$**.** The points of X_2 have the following forms.

$$\begin{array}{ll}
\text{generic point:} & (0), \\
\text{curve-generic points:} & (f), \quad f \in k[x,y] \text{ irreducible}, \\
\text{closed points:} & \mathfrak{m} \subset k[x,y] \text{ maximal.}
\end{array}$$

Proof. The Stacks example cited above records that (0) is prime, that a principal ideal generated by an irreducible polynomial is prime because $k[x,y]$ is a UFD, and that the remaining non-principal primes are maximal after reducing to a prime of a one-variable polynomial ring over a finite algebraic extension of k. This is exactly the displayed trichotomy.

Lemma $L2$**.** The residue fields are

$$\kappa((0)) = k(x,y),$$

$$\kappa((f)) = \operatorname{Frac}(k[x,y]/(f)) \quad \text{for irreducible } f,$$

and, for a closed point $\mathfrak{m}$,

$$\kappa(\mathfrak{m}) = k[x,y]/\mathfrak{m} \quad \text{is a finite extension of } k.$$

If $d(\mathfrak{m}) = [\kappa(\mathfrak{m}) : k]$, then $|\kappa(\mathfrak{m})| = q^{d(\mathfrak{m})}$.

Proof. The first two formulas are the definition of residue field as the fraction field of $R/\mathfrak{p}$. The maximal-ideal formula follows because localizing at a maximal ideal does not change the quotient residue field. The Nullstellensatz source gives that this residue field is finite over k, and the cardinality follows from its k-vector-space dimension.

The generic point and curve-generic points are not finite Weyl sites in the current convention. The generic residue $k(x, y)$ is infinite. If $f \in k[x]$ is irreducible, then

$$k[x, y]/(f) \cong (k[x]/(f))[y],$$

so $\kappa((f)) \cong (k[x]/(f))(y)$, also infinite. If f has positive degree in y, then $k[x] \hookrightarrow k[x, y]/(f)$, since a nonzero polynomial in $k[x]$ cannot be divisible by such an f. Hence the quotient and its fraction field are infinite. After swapping x and y if necessary, this covers every irreducible f.

31.4 Closed Points as Finite Qudit Sites

A rational point $(a, b) \in k^2$ gives the maximal ideal

$$\mathfrak{m}_{a,b} = (x - a, y - b),$$

with residue field k. Hence it is one q-level site.

Closed points of larger degree are equally concrete. Let $\pi(x) \in k[x]$ be monic irreducible of degree d, let $K = k[x]/(\pi)$, and choose $\beta \in K$ represented by $\widetilde{\beta}(x) \in k[x]$. Then

$$\mathfrak{m}_{\pi,\beta} = (\pi(x),\, y - \widetilde{\beta}(x))$$

is maximal, with residue field K. It contributes one q^d-level finite-field qudit.

Derivation. The quotient by $(\pi(x), y - \widetilde{\beta}(x))$ sends x to the class of x in K and y to β, giving a quotient isomorphic to K. Since K is a field, the ideal is maximal.

31.5 Zariski Opens and Closed-Point Supports

For $h \in k[x, y]$,

$$D(h) = \{\mathfrak{p} : h \notin \mathfrak{p}\}.$$

The closed-point support used by the quantum net is

$$C_2(D(h)) = \{\mathfrak{m} \in \operatorname{Max} R_2 : h \notin \mathfrak{m}\}.$$

Equivalently, if $\bar{x}, \bar{y}$ are the residue classes in $\kappa(\mathfrak{m})$, then

$$\mathfrak{m} \in C_2(D(h)) \quad \Longleftrightarrow \quad h(\bar{x}, \bar{y}) \neq 0.$$

Examples:

$$C_2(D(x)) = \{\mathfrak{m} : x \notin \mathfrak{m}\},$$

so $D(x)$ removes the closed points on the vertical line $x = 0$. The open

$$D(x) \cup D(y) = X_2 \setminus V(x, y)$$

removes only the rational origin (x, y) from the closed-point support.

31.6 Weyl-Heisenberg Net T1

For an open $U \subset X_2$, define

$$E_2(U) = \bigoplus_{\mathfrak{m} \in C_2(U)} \kappa(\mathfrak{m})^2.$$

For a finite support $S \Subset C_2(U)$,

$$\mathcal{A}(S) = \bigotimes_{\mathfrak{m} \in S} \operatorname{End}_{\mathbb{C}} \ell^2(\kappa(\mathfrak{m})) \cong \bigotimes_{\mathfrak{m} \in S} M_{q^{d(\mathfrak{m})}}(\mathbb{C}).$$

The open-local algebra is

$$\mathcal{A}_2(U) = \varinjlim_{S \Subset C_2(U)} \mathcal{A}(S) = \operatorname{span}_{\mathbb{C}}\{W_U(e) : e \in E_2(U)\}.$$

Theorem *T1.** $U \mapsto \mathcal{A}_2(U)$ is the affine-plane open-local Weyl-Heisenberg net.

Proof. This is AQM-28 with $R = R_2$. Inclusion of opens gives inclusion of closed supports, hence tensor-by-identity inclusions of finite matrix algebras and an inclusion of their filtered colimits. The Weyl commutator at a closed point $\mathfrak{m}$ is

$$Z_{\mathfrak{m}}(b) X_{\mathfrak{m}}(a) = \psi_{\mathfrak{m}}(ab) X_{\mathfrak{m}}(a) Z_{\mathfrak{m}}(b),$$

and different closed points commute.

31.7 Rational-Point Truncation T2

Let

$$S_{\text{rat}} = \{\mathfrak{m}_{a,b} : a, b \in k\}.$$

This finite support has q^2 closed points, each with residue field k. The associated finite subtheory is

$$E(S_{\text{rat}}) = \bigoplus_{(a,b) \in k^2} k^2,$$

$$\mathcal{H}(S_{\text{rat}}) = \bigotimes_{(a,b) \in k^2} \ell^2(k), \qquad \mathcal{A}(S_{\text{rat}}) \cong \bigotimes_{(a,b) \in k^2} M_q(\mathbb{C}).$$

This is the finite k-rational-point truncation, matching the finite-set construction on k^2. It is not the full $\operatorname{Spec} k[x, y]$ theory, because the full closed-point set also contains all non-rational closed points of degrees $d > 1$.

31.8 States T3

For every open U,

$$\operatorname{State}(\mathcal{A}_2(U)) \cong \varprojlim_{S \Subset C_2(U)} \operatorname{State}(\mathcal{A}(S)).$$

The transition maps are finite partial traces.

The Zariski state presheaf is not separated. Let

$$x_0 = (x, y), \qquad x_1 = (x - 1, y),$$

and let $U = X_2 \setminus \{x_0\}$, $V = X_2 \setminus \{x_1\}$. These are open and cover X_2. On the two rational q-level sites x_0, x_1, use the same maximally entangled density matrix and product density matrix as in AQM-29, tensoring with the product tracial state on every other closed point. The two global states agree after restriction to U and to V, but they differ on the two-site algebra. Thus local marginals on a Zariski cover do not determine global correlations.

31.9 Coordinate Field Profiles T4

The coordinate pair $(x, 0)$ defines the closed-point profile

$$\mathfrak{m} \longmapsto (x \bmod \mathfrak{m}, 0).$$

It is nonzero at every closed point not lying on the line $x = 0$. Hence it has infinite support: the line $x = 1$ has the same closed points as $\operatorname{Spec} k[y]$, and AQM-29 proves there are infinitely many of them. Likewise $(0, y)$ has infinite support, using the line $y = 1$. Their finite-volume restrictions define Weyl operators on finite closed supports, but the full coordinate profiles are not elements of $\mathcal{A}_2(X_2)$.

The curve-generic point (x) is also not a replacement for the missing operator: its residue field is $k(y)$, not a finite qudit field. It records the generic point of the divisor $x = 0$, while the finite Weyl theory records finite closed-point excitations on or off that divisor.

31.10 Conclusion

$\operatorname{Spec} k[x, y]$ has three geometric point types. The finite arithmetic quantum field constructed here uses only the closed points as finite qudit sites, with one $q^{d(\mathfrak{m})}$-level qudit at a closed point of degree $d(\mathfrak{m})$. Rational points give a finite q^2-site truncation, but the full closed-point quasi-local algebra is the filtered tensor product over all closed points of all degrees. Generic and curve-generic points remain essential to the Zariski topology and to divisor geometry, but they are not finite Weyl sites under the current convention.

32 Weyl-Heisenberg Representation Status over Rings and LCA Systems

32.1 Status and Source Anchors

This shard records the source status needed before defining a Weyl-Heisenberg system for $k(t)$. The LCA baseline is Prasad's `references/heisenberg_weil/prasad_0912_0574_source/SvN.tex`, lines 95–107 and 164–190. The modern LCA quantum-kinematical source is `references/heisenberg_weil/beny_crann_lee_park_youn_2204_08162_source/main.tex`, lines 305–328, 411–444, and 461–473. The locally compact ring source is `references/heisenberg_weil/bekka_2502_00387_source/CCR-GeneralRings-v5.tex`, lines 287–350, 862–881, 997–1038, and 1048–1150. The finite Frobenius-ring Pauli sources are `references/heisenberg_weil/gluesing_luerssen_pllaha_1710_09884_source/StabCodesFrob5.tex`, lines 224–249, 286–323, 353–355, 412–423, and 469–504, and `references/heisenberg_weil/sidana_kashyap_2202_00248_source/EAQECCs_over_rings_v15.tex`, lines 103–145 and 180–197. The finite generalized-Heisenberg source is `references/heisenberg_weil/solomon_2501_00650_source/TotslipVersion5_Aug_25.tex`, lines 315–318, 904–981, 1127–1136, 1637–1649, and 1677–1718. Finite Clifford/Weil splitting results are registered from Lysenko, Trias, Hashimoto–Horibe–Hayashi, Korbelar–Tolar, and Galindo in `references/heisenberg_weil/SOURCES.md`.

32.2 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: \text{An LCA Weyl system uses } F\times\widehat{F}, \text{ not } F\times F \text{ unless } F\simeq\widehat{F}.\\
L1 &: \text{For an LCA group } F,\ W(x,\gamma)=T_xM_\gamma \text{ is the canonical irreducible Weyl representation.}\\
D2 &: \text{For a locally compact ring } R,\ \text{a character } \lambda \text{ defines CCR by } \lambda(a\cdot b).\\
L2 &: \text{Bekka's strong ring Stone-von Neumann theorem requires symmetry and } R\simeq\widehat{R}.\\
D3 &: \text{For a finite Frobenius ring } R,\ X(a),Z(b) \text{ give a Pauli/error basis.}\\
L3 &: \text{Commutation is controlled by the character-symplectic pairing.}\\
D4 &: \text{Finite Clifford/Weil splitting is an additional structure, not automatic from the Pauli basis.}\\
T1 &: \text{For } K=k(t),\ \text{the honest analytic uniqueness theorem applies after choosing a full LCA phase space.}
\end{aligned}$$

32.3 LCA Weyl Systems D1–L1

Let F be a locally compact abelian group. Its Pontryagin dual $\widehat{F}$ is the group of continuous characters $F\to\mathbb{T}$. The canonical Weyl phase space is

$$G_F=F\times\widehat{F}.$$

For $(x,\gamma),(x',\gamma')\in G_F$, define

$$\sigma_{\mathrm{can}}((x,\gamma),(x',\gamma'))=\gamma(x').$$

This is the source convention of Beny–Crann–Lee–Park–Youn, lines 461–465.

The representation space is $L^2(F)$. For $f\in L^2(F)$,

$$(T_xf)(y)=f(y-x),\qquad (M_\gamma f)(y)=\gamma(y)f(y).$$

Define

$$W(x,\gamma)=T_xM_\gamma.$$

Then

$$\begin{aligned}
W(x,\gamma)W(x',\gamma')&=T_xM_\gamma T_{x'}M_{\gamma'}\\
&=\gamma(x')T_{x+x'}M_{\gamma+\gamma'}\\
&=\sigma_{\mathrm{can}}((x,\gamma),(x',\gamma'))W(x+x',\gamma+\gamma').
\end{aligned}$$

Thus W is a projective representation with multiplier σ_{can}. The commutator bicharacter is

$$\Delta((x,\gamma),(x',\gamma')) = \gamma(x')\overline{\gamma'(x)}.$$

Prasad's Stone-von Neumann-Mackey theorem, lines 181–190, states that the canonical representation on $L^2(F)$ is irreducible and that any continuous unitary representation of the corresponding Heisenberg group with the standard central character decomposes as copies of it. This is the clean theorem we may use when the phase space is $F \times \widehat{F}$.

32.4 Locally Compact Rings D2–L2

Let R be a unital second-countable locally compact ring. Let $\lambda \in \widehat{R}$ be a unitary character of the additive group of R. For $d \geq 1$, Bekka defines a pair of unitary representations U, V of R^d to satisfy

$$U(a)V(b) = \lambda(a \cdot b)V(b)U(a), \qquad a \cdot b = \sum_{i=1}^{d} a_i b_i.$$

The Schrödinger pair on $L^2(R^d)$ is

$$(U_{\mathrm{Schr}}(a)f)(x) = f(x+a), \qquad (V_{\mathrm{Schr}}(b)f)(x) = \lambda(x \cdot b)f(x).$$

The strong theorem in Bekka, lines 319–350, assumes

$$\lambda(ab) = \lambda(ba)$$

and that

$$\nabla_\lambda : R \longrightarrow \widehat{R}, \qquad a \longmapsto (x \mapsto \lambda(xa))$$

is a topological group isomorphism. Under these hypotheses, every separable CCR pair decomposes into copies of the Schrödinger pair. The source lists local fields, finite fields, finite products of finite fields, $\mathbb{Z}/N$, matrix algebras over local fields, and adeles as examples where the relevant self-duality hypotheses hold in the stated forms.

The warning for $k(t)$ appears already in the same source: for a countably infinite ring with the discrete topology, line 1014 records that no character can satisfy the isomorphism condition. Therefore a bare discrete $K = k(t)$ cannot be treated as a self-dual locally compact ring in this strong theorem. It may still support an explicit algebraic representation, but the LCA uniqueness theorem must be applied to $K_{\mathrm{disc}} \times \widehat{K_{\mathrm{disc}}}$, to a local completion, or to an adelic completion.

32.5 Finite Frobenius Rings D3–L3

Let R be a finite commutative Frobenius ring with generating character χ. On the Hilbert space with basis $\{v_x : x \in R^n\}$, define

$$X(a)v_x = v_{x+a}, \qquad Z(b)v_x = \chi(b \cdot x)v_x$$

for $a, b, x \in R^n$, where $b \cdot x = \sum_i b_i x_i$. For $P = X(a)Z(b)$ and $P' = X(a')Z(b')$, the cited sources compute

$$PP' = \chi(b \cdot a')X(a+a')Z(b+b')$$

and hence

$$PP' = \chi(b \cdot a' - b' \cdot a)P'P.$$

The symplectic inner product used for stabilizer theory is

$$\langle (a,b),(a',b')\rangle_s = b\cdot a' - b'\cdot a.$$

The Pauli group is obtained by adjoining the necessary scalar roots of unity. In characteristic 2, this scalar group is larger than the characteristic roots; this is exactly the finite-ring version of the even-phase caveat.

Thus finite Frobenius-ring literature strongly supports the Pauli/error-basis and stabilizer layer: generating characters identify the ring with its character module, the X/Z operators are unitary, and commutation is controlled by the character-symplectic form. This is not, by itself, a full Clifford or Weil representation classification over every commutative Frobenius ring.

32.6 Finite Generalized Heisenberg and Clifford Status D4

Solomon's generalized Heisenberg group starts from finite abelian groups A, B, C and a bilinear pairing $\lambda : A\times B \to C$:

$$h(a,b,c)h(a',b',c') = h(a+a', b+b', c+c'+\lambda(a,b')).$$

The commutator is

$$[h(a,b,c),h(a',b',c')] = h(0,0,\lambda(a,b')-\lambda(a',b)).$$

If A, B are finite modules over a commutative ring R and λ is R-balanced, the quotient by the center inherits R-module structure. A character $p : C \to \mathbb{C}^\times$ gives the left Schrödinger representation on functions on A:

$$(h(a,b,c)\cdot f)(x) = p(\lambda(x,b)+c)f(x+a).$$

This is a direct finite algebraic analogue of translations and modulations.

The Clifford/Weil layer is subtler. Lysenko constructs canonical representations for finite Heisenberg groups of odd exponent, explicitly leaving the even-order case open in the cited introduction. Trias constructs Weil modules over coefficient rings for finite or local nonarchimedean fields of characteristic not 2. Hashimoto–Horibe–Hayashi, Korbelar–Tolar, and Galindo analyze projective Clifford extensions and splitting. Galindo's 2026 result states that for a finite abelian group A, the Clifford extension splits iff $4 \nmid |A|$. This is important status information: the obstruction is not a defect of notation but a genuine finite-even phenomenon.

32.7 Conclusion for $k(t)$ T1

The sourced status is therefore:

LCA groups/local fields:	full Stone-von Neumann uniqueness is available.
Locally compact rings:	strong uniqueness requires $R \simeq \widehat{R}$ via λ.
Finite Frobenius rings:	Pauli/stabilizer theory is robust.
Finite Clifford/Weil:	splitting and even-order issues are genuine extra data.

For $K = k(t)$, the next shard must therefore distinguish: the algebraic discrete K-label Heisenberg group; the full discrete-LCA system $K_{\text{disc}} \times \widehat{K_{\text{disc}}}$; and the local or adelic completion where K becomes arithmetic input inside a self-dual LCA configuration space.

33 Weyl-Heisenberg Representations for $k(t)$

33.1 Status and Source Anchors

This shard uses convention (z). The LCA uniqueness theorem is sourced from Prasad, `references/heisenberg_weil/prasad_0912_0574_source/SvN.tex`, lines 164–190. The canonical $F \times \widehat{F}$ Weyl system is sourced from Beny–Crann–Lee–Park–Youn, `references/heisenberg_weil/beny_crann_lee_park_youn_2204_08162_source/main.tex`, lines 461–473. The locally compact ring theorem and the warning about countably infinite discrete rings are sourced from Bekka, `references/heisenberg_weil/bekka_2502_00387_source/CCR-GeneralRings-v5.tex`, lines 319–350, 862–881, and 1014. The local and adelic field background is registered in `references/heisenberg_weil/SOURCES.md` from Poonen and Kudla. The algebraic $k(t)$ calculations below are direct derivations from the displayed definitions.

33.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & K = k(t), \quad \tau : k \to \mathbb{F}_p \text{ nonzero}, \quad \lambda_K(f) = \tau([t^{-1}]f). \\
L1 & : & (a,b) \mapsto \chi_K(ab) \text{ is a nondegenerate pairing on } K. \\
D2 & : & H_K = \mu_p \times K \times K \text{ with cocycle } \chi_K(ba'). \\
L2 & : & H_K \text{ is a class-two group with center } \mu_p. \\
D3 & : & T_a|\xi\rangle = |\xi + a\rangle, \quad M_b|\xi\rangle = \chi_K(b\xi)|\xi\rangle. \\
T1 & : & \pi(z,a,b) = zT_aM_b \text{ is a unitary irreducible representation of } H_K \text{ on } \ell^2(K). \\
T2 & : & \text{Stone-von Neumann uniqueness requires the full LCA dual or a local/adelic completion.} \\
E1 & : & k = \mathbb{F}_3 \text{ gives explicit } \omega\text{-commutation examples.}
\end{array}$$

33.3 The Coefficient Character D1

Let $k = \mathbb{F}_q$ have characteristic p, and let $K = k(t)$. Fix a nonzero $\mathbb{F}_p$-linear functional

$$\tau : k \longrightarrow \mathbb{F}_p.$$

For $f \in K$, expand f as a Laurent series at infinity:

$$f = \sum_{n \le N} c_n t^n, \qquad c_n \in k.$$

Only finitely many positive powers occur. Define

$$[t^{-1}]f = c_{-1}, \qquad \lambda_K(f) = \tau([t^{-1}]f).$$

The associated additive character is

$$\chi_K(f) = \exp\left(\frac{2\pi i}{p}\lambda_K(f)\right).$$

The trace $\mathrm{Tr}_{k/\mathbb{F}_p}$ is a standard choice for τ when a trace convention is wanted. The proofs below require only that $\tau \neq 0$.

Lemma $L1$**.** The pairing

$$B : K \times K \longrightarrow \mathbb{C}^{\times}, \qquad B(a,b) = \chi_K(ab)$$

is nondegenerate in each variable.

Proof. Let $a \in K$ be nonzero. Since $\tau \neq 0$, choose $c \in k$ with $\tau(c) \neq 0$. Put

$$b = \frac{c\,t^{-1}}{a} \in K.$$

Then $ab = c\,t^{-1}$, so

$$\lambda_K(ab) = \tau(c) \neq 0.$$

Thus $\chi_K(ab) \neq 1$. This proves that no nonzero a is killed by all right pairings. The same argument with the variables interchanged proves nondegeneracy in the second variable, because multiplication in K is commutative.

33.4 The Algebraic Heisenberg Group D2–L2

Let $\mu_p = \{z \in \mathbb{C} : z^p = 1\}$. Define

$$H_K = \mu_p \times K \times K$$

with multiplication

$$(z,a,b)(z',a',b') = \bigl(zz'\chi_K(ba'),\ a+a',\ b+b'\bigr).$$

Here a is the translation coordinate and b is the modulation coordinate.

Associativity. Let $h = (z,a,b)$, $h' = (z',a',b')$, and $h'' = (z'',a'',b'')$. The central factor in $(hh')h''$ is

$$zz'z''\chi_K(ba')\chi_K((b+b')a'') = zz'z''\chi_K(ba' + ba'' + b'a'').$$

The central factor in $h(h'h'')$ is

$$zz'z''\chi_K(b'a'')\chi_K(b(a'+a'')) = zz'z''\chi_K(b'a'' + ba' + ba'').$$

These are equal by additivity of χ_K. Therefore the multiplication is associative.

Identity and inverse. The identity is $(1,0,0)$. A direct multiplication gives

$$(z,a,b)^{-1} = \bigl(z^{-1}\chi_K(ba), -a, -b\bigr).$$

Commutator and center. From the multiplication law,

$$(1,a,b)(1,a',b') = \chi_K(ba')(1, a+a', b+b')$$

and

$$(1,a',b')(1,a,b) = \chi_K(b'a)(1, a+a', b+b').$$

Hence the commutator is the central element

$$[(1,a,b),(1,a',b')] = \bigl(\chi_K(ba' - b'a), 0, 0\bigr).$$

If (z,a,b) is central, then this phase is 1 for all a', b'. Taking $b' = 0$ gives $\chi_K(ba') = 1$ for all a', hence $b = 0$ by $L1$. Taking $a' = 0$ then gives $\chi_K(-b'a) = 1$ for all b', hence $a = 0$. Thus

$$Z(H_K) = \mu_p \times \{0\} \times \{0\}.$$

33.5 The Schrödinger Representation D3–T1

Let $\ell^2(K)$ have orthonormal basis $\{|\xi\rangle : \xi \in K\}$. Define

$$T_a|\xi\rangle = |\xi + a\rangle, \qquad M_b|\xi\rangle = \chi_K(b\xi)|\xi\rangle.$$

Both are unitary: T_a permutes the basis and M_b is diagonal with unit-modulus diagonal entries. Moreover,

$$\begin{aligned} M_bT_a|\xi\rangle &= M_b|\xi + a\rangle \\ &= \chi_K(b\xi + ba)|\xi + a\rangle \\ &= \chi_K(ba)T_aM_b|\xi\rangle. \end{aligned}$$

Therefore

$$T_aM_bT_{a'}M_{b'} = \chi_K(ba')T_{a+a'}M_{b+b'}.$$

The map

$$\pi_K : H_K \longrightarrow U(\ell^2(K)), \qquad \pi_K(z, a, b) = zT_aM_b$$

is a unitary representation with central character

$$\pi_K(z, 0, 0) = zI.$$

Irreducibility. Let A be a bounded operator on $\ell^2(K)$ commuting with all T_a and M_b. Write

$$A_{\xi,\eta} = \langle\xi|A|\eta\rangle.$$

Commutation with M_b gives

$$\chi_K(b\xi)A_{\xi,\eta} = \chi_K(b\eta)A_{\xi,\eta}$$

for all b, ξ, η. If $\xi \neq \eta$, then $L1$ gives b such that $\chi_K(b(\xi - \eta)) \neq 1$, so $A_{\xi,\eta} = 0$. Thus A is diagonal. Write $A_{\xi,\xi} = d_\xi$. Commutation with T_a gives

$$d_{\xi+a} = d_\xi$$

for all a, ξ. Since the additive group of K acts transitively on itself by translations, all d_ξ are equal. Thus the commutant of $\pi_K(H_K)$ is $\mathbb{C}I$. If a closed subspace were invariant, its orthogonal projection would commute with the representation, hence would be 0 or I. Therefore π_K is irreducible.

33.6 Conclusion

The concrete rational-function answer is:

$$H_{\mathbb{F}_q(t)} = \mu_p \times k(t) \times k(t), \qquad (z, a, b)(z', a', b') = (zz'\chi_K(ba'), a + a', b + b'),$$

with irreducible Schrödinger representation on $\ell^2(k(t))$. The analytic Stone-von Neumann answer is separate and is recorded in the following shard.

34 Full-Dual and Local Models for $k(t)$

34.1 Status and Source Anchors

This shard completes AQM-32. The full-dual LCA statement uses Prasad, `references/heisenberg_weil/prasad_0912_0574_source/SvN.tex`, lines 95–107 and 164–190, and Beny–Crann–Lee–Park–Youn, `references/heisenberg_weil/beny_crann_lee_park_youn_2204_08162_source/main.tex`, lines 461–473. The warning about countably infinite discrete rings uses Bekka, `references/heisenberg_weil/bekka_2502_00387_source/CCR-GeneralRings-v5.tex`, line 1014. Local and adelic background is registered in `references/heisenberg_weil/SOURCES.md` from Poonen's Tate notes and Kudla's local theta notes.

34.2 Lamport Dependency Skeleton

$D1$: K_{disc} is $K = k(t)$ with the discrete topology.
$L1$: $K_{\text{disc}} \not\cong \widehat{K_{\text{disc}}}$ as LCA groups.
$D2$: $G_{\text{full}} = K_{\text{disc}} \times \widehat{K_{\text{disc}}}$ is the full LCA Weyl phase space.
$L2$: The algebraic K-modulations form a dense subgroup of $\widehat{K_{\text{disc}}}$.
$D3$: $K_\infty = k((t^{-1}))$ is the local completion at infinity.
$T1$: Stone-von Neumann uniqueness belongs to G_{full} or to local/adelic LCA completions.
$E1$: $k = \mathbb{F}_3$ gives explicit ω-commutation examples.

34.3 The Full Discrete-Dual Model D1–L2

The algebraic representation of AQM-32 is an algebraic Schrödinger representation theorem for the concrete group H_K. It is not the full LCA Stone-von Neumann theorem. The reason is structural: if K is given the discrete topology, then its Pontryagin dual $\widehat{K_{\text{disc}}}$ is compact. Since K is countably infinite, K is not isomorphic to $\widehat{K_{\text{disc}}}$ as an LCA group. Bekka records the same obstruction for countably infinite discrete rings.

The full LCA Weyl system with configuration group K_{disc} is

$$G_{\text{full}} = K_{\text{disc}} \times \widehat{K_{\text{disc}}}.$$

It acts on $\ell^2(K)$ by

$$W(a, \gamma) = T_a M_\gamma, \qquad M_\gamma |\xi\rangle = \gamma(\xi)|\xi\rangle.$$

Prasad's theorem applies to this G_{full}. The algebraic H_K is the restriction to the subgroup of characters

$$\gamma_b(\xi) = \chi_K(b\xi), \qquad b \in K.$$

By AQM-32 $L1$, $b \mapsto \gamma_b$ is injective.

Density derivation. Let $\Gamma_K = \{\gamma_b : b \in K\} \subset \widehat{K_{\text{disc}}}$. Its annihilator in the double dual K_{disc} is

$$\Gamma_K^\perp = \{\xi \in K : \gamma_b(\xi) = 1 \text{ for all } b \in K\}.$$

By nondegeneracy of $(b, \xi) \mapsto \chi_K(b\xi)$, this annihilator is $\{0\}$. Pontryagin duality identifies the annihilator of the closure $\overline{\Gamma_K}$ with $\Gamma_K^\perp$. Hence $\overline{\Gamma_K} = \widehat{K_{\text{disc}}}$. The algebraic model therefore uses a dense rational-function family of modulations, while the full LCA theorem uses every continuous character of K_{disc}.

34.4 Local and Adelic Completions D3–T1

For analytic arithmetic quantum fields, the more natural locally compact configuration spaces are local completions such as

$$K_\infty = k((t^{-1}))$$

or, later, the adele ring $\mathbb{A}_K$. A nontrivial continuous additive character identifies a local field with its Pontryagin dual, so the canonical Weyl system has phase space $K_\infty \times K_\infty$ after that identification. The Schrödinger representation then lives on

$$L^2(K_\infty).$$

For the adelic theory, a residue character built from a differential gives the standard self-dual adelic setting; K embeds diagonally as arithmetic input, not as a replacement for the full dual phase group.

Thus there are three honest objects:

$\ell^2(K)$:	discrete algebraic rational-function representation;
$\ell^2(K_{\mathrm{disc}})$ with $\widehat{K}$:	full LCA discrete-dual Stone-von Neumann system;
$L^2(K_\infty)$ or $L^2(\mathbb{A}_K)$:	local or global analytic arithmetic system.

They should not be collapsed into one notation.

34.5 Concrete Examples for $k = \mathbb{F}_3$ E1

Let $k = \mathbb{F}_3$, $K = \mathbb{F}_3(t)$, $\tau = \mathrm{id}$, and

$$\omega = \exp(2\pi i/3).$$

Then

$$\chi_K(f) = \omega^{[t^{-1}]f}.$$

For $a = 1$ and $b = t^{-1}$,

$$M_{t^{-1}} T_1 = \omega\, T_1 M_{t^{-1}}.$$

Indeed, $ba = t^{-1}$, whose t^{-1}-coefficient is 1.

For $a = t$ and $b = t^{-1}$,

$$M_{t^{-1}} T_t = T_t M_{t^{-1}},$$

because $ba = 1$ has t^{-1}-coefficient 0. For $a = t$ and $b = t^{-2}$,

$$M_{t^{-2}} T_t = \omega\, T_t M_{t^{-2}},$$

because $ba = t^{-1}$.

On basis vectors,

$$T_{t+1}|t^{-1}\rangle = |t + 1 + t^{-1}\rangle,$$

$$M_{t^{-1}}|1\rangle = \omega|1\rangle, \qquad M_{t^{-1}}|t\rangle = |t\rangle,$$

and

$$M_{t^{-2}}|t\rangle = \omega|t\rangle.$$

Thus the rational-function Heisenberg group already contains infinitely many explicit qutrit-like commutation relations, indexed by rational functions rather than by closed points. This is a different layer from the closed-point quasi-local field of $\operatorname{Spec} \mathbb{F}_3[t]$, where each monic irreducible π contributes one finite qudit $\ell^2(\mathbb{F}_3[t]/(\pi))$.

34.6 Conclusion

The concrete rational-function construction gives an irreducible unitary representation on $\ell^2(k(t))$. Stone-von Neumann uniqueness belongs to a larger or completed LCA object: the full discrete-dual system, a local completion, or an adelic completion. The next arithmetic-field step should choose which completion is physically intended before importing Fourier transform, Gaussian, or Weil-representation statements.

35 Polynomial-Quotient Arithmetic Fields over $\mathbb{F}_3$

35.1 Status and Source Anchors

This shard is a checked finite affine example. The source-local notions are Spec, prime ideals, residue fields, and Zariski opens from `references/algebraic_geometry/stacks_project_algebra.tex`, lines 120–158, 250–274, 2972–2983, and 3104–3117. The polynomial-ring prime classification and finite residue-field count reuse the direct derivation of AQM-29. The finite Weyl representation is the closed-point construction of AQM-28, with the polynomial-quotient convention fixed in Convention (aa). The coordinate ring is written $R_p = \mathbb{F}_3[t]/(p)$.

35.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & k = \mathbb{F}_3,\ p \in k[t] \text{ monic nonzero},\ R_p = k[t]/(p),\ X_p = \operatorname{Spec} R_p. \\
D2 & : & p = \prod_{i=1}^{r} \pi_i^{e_i}, \quad \pi_i \text{ distinct monic irreducibles},\ K_i = k[t]/(\pi_i). \\
L1 & : & X_p = \{x_i = (\pi_i)/(p) : 1 \le i \le r\},\ \kappa(x_i) = K_i,\ \text{and } X_p \text{ is finite discrete.} \\
L2 & : & R_p \cong \prod_i k[t]/(\pi_i^{e_i}), \quad (R_p)_{\mathrm{red}} \cong \prod_i K_i. \\
D3 & : & E_p = \bigoplus_i K_i^2, \quad \mathcal{H}_p = \bigotimes_i \ell^2(K_i). \\
T1 & : & E_p \text{ has an explicit projective unitary Weyl representation } W_p \text{ on } \mathcal{H}_p. \\
T2 & : & \operatorname{span}\{W_p(e) : e \in E_p\} = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_p),\ \text{hence the representation is irreducible.} \\
T3 & : & R_p \text{ field profiles factor through } (R_p)_{\mathrm{red}}.
\end{array}
$$

35.3 The Spectrum L1–L2

Assume first that $p \neq 1$, so $R_p \neq 0$. Factor

$$p = \prod_{i=1}^{r} \pi_i^{e_i}$$

in $k[t]$, where the π_i are distinct monic irreducibles and $e_i \geq 1$. Put $d_i = \deg \pi_i$ and $K_i = k[t]/(\pi_i)$.

Lemma $L1$. The points of X_p are exactly

$$x_i = (\pi_i)/(p), \qquad 1 \le i \le r,$$

and

$$\kappa(x_i) = K_i, \qquad |K_i| = 3^{d_i}.$$

Every point is closed, and every subset of X_p is open.

Proof. A prime ideal of R_p has inverse image a prime ideal $\mathfrak{q} \subset k[t]$ containing (p). Since $p \in \mathfrak{q}$ and $p \neq 0$, the prime $\mathfrak{q}$ is nonzero. By the AQM-29 Euclidean-domain derivation, $\mathfrak{q} = (\pi)$ for a monic irreducible π. The containment $(p) \subset (\pi)$ is equivalent to $\pi \mid p$, so $\pi = \pi_i$ for a unique i. Conversely, each $(\pi_i)/(p)$ is prime because

$$R_p/x_i \cong k[t]/(\pi_i) = K_i$$

is a field. This also proves the residue-field formula, since x_i is maximal. The residue classes $1, t, \ldots, t^{d_i-1}$ form a k-basis of the quotient, so $|K_i| = 3^{d_i}$. Since every point is maximal, every singleton is closed. A finite space in which every singleton is closed is discrete, because the complement of a singleton is a finite union of closed singletons.

Lemma *L*2. There is a ring isomorphism

$$R_p \cong \prod_{i=1}^{r} k[t]/(\pi_i^{e_i}),$$

and the reduced quotient is

$$(R_p)_{\mathrm{red}} = R_p/\sqrt{0} \cong k[t]/(\pi_1 \cdots \pi_r) \cong \prod_{i=1}^{r} K_i.$$

Proof. If $i \neq j$, then $\pi_i^{e_i}$ and $\pi_j^{e_j}$ are coprime. Bezout therefore gives $a_{ij}\pi_i^{e_i} + b_{ij}\pi_j^{e_j} = 1$. The Chinese remainder construction gives a surjective map

$$k[t]/(p) \longrightarrow \prod_i k[t]/(\pi_i^{e_i}),$$

and its kernel is the intersection of the pairwise coprime ideals $(\pi_i^{e_i})$, equal to their product (p).

For the nilradical, a class $f \bmod p$ is nilpotent iff $p \mid f^N$ for some N. This happens iff every π_i divides f, equivalently $\pi_1 \cdots \pi_r \mid f$. Hence $\sqrt{0} = (\pi_1 \cdots \pi_r)/(p)$. Quotienting by this ideal gives $k[t]/(\pi_1 \cdots \pi_r)$, and another Chinese remainder step gives $\prod_i K_i$.

If $p = 1$, then $R_p = 0$, $\operatorname{Spec} R_p = \emptyset$, $E_p = 0$, and the observable algebra is the empty tensor product $\mathbb{C}$.

35.4 The Weyl-Heisenberg Representation D3–T2

For each closed point x_i, define the additive character

$$\psi_i(a) = \exp\!\left(\frac{2\pi i}{3}\operatorname{Tr}_{K_i/\mathbb{F}_3}(a)\right).$$

Let

$$\mathcal{H}_i = \ell^2(K_i), \qquad E_i = K_i^2.$$

On the basis $|s\rangle$, $s \in K_i$, set

$$T_i(a)|s\rangle = |s+a\rangle, \qquad R_i(b)|s\rangle = \psi_i(bs)|s\rangle, \qquad W_i(a,b) = T_i(a)R_i(b).$$

Lemma *L*3. For $a, b, a', b' \in K_i$,

$$W_i(a,b)W_i(a',b') = \psi_i(ba')W_i(a+a', b+b')$$

and

$$W_i(a,b)W_i(a',b') = \psi_i(ba' - b'a)W_i(a',b')W_i(a,b).$$

Proof. For $s \in K_i$,

$$R_i(b)T_i(a')|s\rangle = \psi_i(b(s+a'))|s+a'\rangle = \psi_i(ba')T_i(a')R_i(b)|s\rangle.$$

Substituting this into $T_i(a)R_i(b)T_i(a')R_i(b')$ gives the first formula. Dividing this product law by the same law with the two labels reversed gives the commutator formula.

For $J \subset \{1, \ldots, r\}$, write

$$E(J) = \bigoplus_{i \in J} K_i^2, \qquad \mathcal{H}(J) = \bigotimes_{i \in J} \mathcal{H}_i, \qquad \mathcal{A}(J) = \bigotimes_{i \in J} \operatorname{End}_{\mathbb{C}}(\mathcal{H}_i).$$

The empty tensor product is $\mathbb{C}$. For the global system, put

$$E_p = E(\{1, \dots, r\}), \qquad \mathcal{H}_p = \mathcal{H}(\{1, \dots, r\}).$$

For $e = ((a_i, b_i))_i \in E_p$, define

$$W_p(e) = \bigotimes_i W_i(a_i, b_i).$$

Theorem $T1$**.** The maps $W_p(e)$ form a projective unitary representation of the additive group E_p. Explicitly,

$$W_p(e)W_p(f) = c_p(e, f)W_p(e + f), \qquad c_p(e, f) = \prod_i \psi_i(b_i a_i'),$$

and

$$W_p(e)W_p(f) = B_p(e, f)W_p(f)W_p(e),$$

where

$$B_p(e, f) = \prod_i \psi_i(b_i a_i' - b_i' a_i).$$

Equivalently, the finite Heisenberg group

$$H_p^{\mathrm{WH}} = \mu_3 \times E_p$$

with multiplication

$$(\zeta, e)(\eta, f) = (\zeta\eta c_p(e, f), e + f)$$

has a unitary representation $\rho_p(\zeta, e) = \zeta W_p(e)$.

Proof. Each $T_i(a)$ is a permutation unitary and each $R_i(b)$ is a diagonal unitary, so $W_i(a, b)$ and $W_p(e)$ are unitary. Tensoring the one-site formula of $L3$ over all i gives the displayed multiplier and commutator. The multiplication law for H_p^{WH} is exactly the statement that the multiplier defect has been placed in the central μ_3-coordinate, so ρ_p is an ordinary unitary representation.

Theorem $T2$**.**

$$\mathrm{span}_{\mathbb{C}}\{W_p(e) : e \in E_p\} = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_p).$$

Hence ρ_p is irreducible on $\mathcal{H}_p$. Moreover

$$\dim_{\mathbb{C}} \mathcal{H}_p = \prod_i 3^{d_i} = 3^{\sum_i d_i} = 3^{\deg(\pi_1 \cdots \pi_r)}.$$

Proof. AQM-28 proves that the one-site finite-field Weyl operators span $\mathrm{End}(\mathcal{H}_i)$. Tensor products of spanning sets span the tensor product algebra, giving the first formula. A subspace invariant under all $W_p(e)$ is invariant under their complex span, hence under the full matrix algebra on $\mathcal{H}_p$. The only such subspaces are 0 and $\mathcal{H}_p$. The dimension formula follows from $\dim \ell^2(K_i) = |K_i| = 3^{d_i}$.

35.5 Coordinate-Ring Fields and Local Nets T3

The coordinate ring maps to residues by

$$\mathrm{ev} : R_p \longrightarrow \prod_i K_i, \qquad f \bmod p \longmapsto (f \bmod \pi_i)_i.$$

By $L2$, $\ker(\mathrm{ev}) = \sqrt{0}$. Thus a pair $(f, g) \in R_p^2$ defines the Weyl label

$$e_{f,g} = ((f \bmod \pi_i,\ g \bmod \pi_i))_i \in E_p,$$

and nilpotent changes in f or g do not change $e_{f,g}$. The coordinate-ring field operator is therefore

$$W_p(f, g) := W_p(e_{f,g}),$$

with product law

$$W_p(f, g)W_p(f', g') = \left(\prod_i \psi_i((g \bmod \pi_i)(f' \bmod \pi_i))\right) W_p(f + f', g + g').$$

The commutator phase is

$$\prod_i \psi_i((g \bmod \pi_i)(f' \bmod \pi_i) - (g' \bmod \pi_i)(f \bmod \pi_i)).$$

Since X_p is discrete, every open set is $U_J = \{x_i : i \in J\}$. The open-local algebra is

$$\mathcal{A}(U_J) = \mathcal{A}(J) = \bigotimes_{i\in J} \mathrm{End}_{\mathbb{C}}(\ell^2(K_i)),$$

and $J \subset J'$ gives the inclusion $a \mapsto a \otimes \mathbf{1}$. States on $\mathcal{A}(U_J)$ are density matrices on $\mathcal{H}(J)$, and restriction along $J \subset J'$ is partial trace over $J' \setminus J$.

35.6 Conclusion

For $R_p = \mathbb{F}_3[t]/(p)$, the complete residue-qudit arithmetic field is the tensor product of one finite-field Weyl system for each distinct irreducible factor of p. The Weyl-Heisenberg representation is explicit on $\bigotimes_i \ell^2(\mathbb{F}_3[t]/(\pi_i))$, spans the full matrix algebra, and is irreducible. Repeated factors of p are not ignored: they are scheme-theoretic nilpotent thickening data in the coordinate ring. They are invisible to the present residue-qudit Weyl labels because all such labels factor through the reduced quotient. AQM-36 adds the nilpotent-sensitive thickened Weyl layer.

36 The $t^n = 1$ Arithmetic Field over $\mathbb{F}_3$

36.1 Status and Source Anchors

This shard specializes AQM-34 to

$$R_n = \mathbb{F}_3[t]/(t^n - 1).$$

The input facts are the direct polynomial derivations of AQM-29 and AQM-34: irreducible factors of $k[t]$ give closed points, residue fields are the corresponding finite quotients, and the Weyl representation is the tensor product of the one-site finite-field Weyl systems from AQM-28. No new external source is used in this specialization.

36.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & n \geq 1, \quad R_n = \mathbb{F}_3[t]/(t^n - 1), \quad X_n = \operatorname{Spec} R_n. \\
D2 & : & n = 3^s m, \quad s = v_3(n), \quad 3 \nmid m. \\
L1 & : & t^n - 1 = (t^m - 1)^{3^s}, \quad t^m - 1 \text{ is squarefree.} \\
L2 & : & \text{Closed Weyl sites are the irreducible factors of } t^m - 1. \\
L3 & : & \text{The degrees of these factors are determined by } \operatorname{ord}_d(3) \text{ for } d \mid m. \\
D3 & : & E_n = \bigoplus_{\pi \mid t^m - 1} K_\pi^2, \quad \mathcal{H}_n = \bigotimes_{\pi \mid t^m - 1} \ell^2(K_\pi). \\
T1 & : & W_n \text{ is the complete projective unitary Weyl representation on } \mathcal{H}_n. \\
T2 & : & \dim_{\mathbb{C}} \mathcal{H}_n = 3^m, \quad \mathcal{A}_n \cong M_{3^m}(\mathbb{C}).
\end{array}$$

36.3 Factorization in Characteristic 3 L1–L3

Let $n = 3^s m$, with $s = v_3(n)$ and $3 \nmid m$. In characteristic 3, Frobenius gives

$$t^n - 1 = (t^m)^{3^s} - 1 = (t^m - 1)^{3^s}.$$

The derivative of $t^m - 1$ is mt^{m-1}, and $m \neq 0$ in $\mathbb{F}_3$. Since t does not divide $t^m - 1$, the greatest common divisor of $t^m - 1$ and mt^{m-1} is 1. Thus $t^m - 1$ is squarefree. Consequently the closed points of X_n are exactly the irreducible factors of $t^m - 1$, while each such point is thickened in R_n to multiplicity 3^s.

For $d \mid m$, $d > 1$, let

$$o_d = \operatorname{ord}_d(3).$$

A primitive d-th root α lies in $\mathbb{F}_{3^r}$ exactly when $\alpha^{3^r} = \alpha$, equivalently $d \mid 3^r - 1$. Hence the degree of α over $\mathbb{F}_3$ is o_d. The Frobenius orbit of α has o_d elements, so the $\varphi(d)$ primitive d-th roots produce

$$\frac{\varphi(d)}{o_d}$$

irreducible factors of degree o_d. For $d = 1$, the factor is $t - 1$. The sum of all factor degrees is

$$\sum_{d \mid m} \varphi(d) = m.$$

36.4 The Complete Weyl-Heisenberg Representation T1–T2

Let $\mathcal{P}_n$ be the set of monic irreducible factors of $t^m - 1$. For $\pi \in \mathcal{P}_n$, write

$$K_\pi = \mathbb{F}_3[t]/(\pi), \qquad d_\pi = \deg \pi.$$

The closed point is $x_\pi = (\pi)/(t^n - 1)$, and its residue field is $\kappa(x_\pi) = K_\pi$.

Define

$$E_n = \bigoplus_{\pi \in \mathcal{P}_n} K_\pi^2, \qquad \mathcal{H}_n = \bigotimes_{\pi \in \mathcal{P}_n} \ell^2(K_\pi).$$

For $a, b, s \in K_\pi$, set

$$\psi_\pi(c) = \exp\left(\frac{2\pi i}{3} \operatorname{Tr}_{K_\pi/\mathbb{F}_3}(c)\right),$$

$$T_\pi(a)|s\rangle = |s + a\rangle, \qquad R_\pi(b)|s\rangle = \psi_\pi(bs)|s\rangle, \qquad W_\pi(a, b) = T_\pi(a)R_\pi(b).$$

For $e = ((a_\pi, b_\pi))_\pi \in E_n$, put

$$W_n(e) = \bigotimes_{\pi \in \mathcal{P}_n} W_\pi(a_\pi, b_\pi).$$

Theorem $T1$**.** For $e = ((a_\pi, b_\pi))$ and $f = ((a'_\pi, b'_\pi))$,

$$W_n(e)W_n(f) = \left(\prod_{\pi \in \mathcal{P}_n} \psi_\pi(b_\pi a'_\pi)\right) W_n(e + f),$$

and

$$W_n(e)W_n(f) = \left(\prod_{\pi \in \mathcal{P}_n} \psi_\pi(b_\pi a'_\pi - b'_\pi a_\pi)\right) W_n(f)W_n(e).$$

The central extension

$$H_n^{\mathrm{WH}} = \mu_3 \times E_n, \qquad (\zeta, e)(\eta, f) = (\zeta\eta c_n(e, f), e + f),$$

with

$$c_n(e, f) = \prod_{\pi \in \mathcal{P}_n} \psi_\pi(b_\pi a'_\pi),$$

is represented unitarily by $\rho_n(\zeta, e) = \zeta W_n(e)$.

Proof. This is AQM-34 applied to $p = t^n - 1$. The one-site identity

$$R_\pi(b)T_\pi(a') = \psi_\pi(ba')T_\pi(a')R_\pi(b)$$

gives the multiplier, and tensoring over all factors gives the displayed laws.

Theorem $T2$**.**

$$\operatorname{span}_{\mathbb{C}}\{W_n(e) : e \in E_n\} = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_n),$$

so the representation is irreducible, and

$$\dim_{\mathbb{C}} \mathcal{H}_n = \prod_{\pi \in \mathcal{P}_n} 3^{d_\pi} = 3^m, \qquad \mathcal{A}_n \cong M_{3^m}(\mathbb{C}).$$

Proof. At each π, the finite-field Weyl operators span $\operatorname{End}(\ell^2(K_\pi))$. Tensor products span the tensor-product matrix algebra. Any invariant subspace is therefore invariant under the full matrix algebra and is 0 or $\mathcal{H}_n$. The dimension formula uses $\sum_{\pi \in \mathcal{P}_n} d_\pi = \deg(t^m - 1) = m$.

The coordinate ring has 3^n elements, but the current residue-qudit Hilbert space has dimension 3^m. They agree exactly when $s = 0$, i.e. when $3 \nmid n$. If $s > 0$, the nilradical is generated by the class of $t^m - 1$, and coordinate-ring field labels factor through

$$R_n/(t^m - 1) \cong \mathbb{F}_3[t]/(t^m - 1).$$

36.5 Open Sets and States

Because X_n is a finite set of closed points, it is discrete. For $J \subset \mathcal{P}_n$,

$$U_J = \{x_\pi : \pi \in J\}$$

is open, and

$$\mathcal{A}(U_J) = \bigotimes_{\pi \in J} \mathrm{End}_{\mathbb{C}}(\ell^2(K_\pi)).$$

The inclusion $J \subset J'$ sends a to $a \otimes \mathbf{1}_{J' \setminus J}$. States on $\mathcal{A}(U_J)$ are density matrices on $\bigotimes_{\pi \in J} \ell^2(K_\pi)$, and restriction is partial trace over omitted factors.

36.6 Small Cases

For $n = 1$, $t^n - 1 = t - 1$. There is one $\mathbb{F}_3$ site:

$$\mathcal{H}_1 \cong \mathbb{C}^3, \qquad \mathcal{A}_1 \cong M_3(\mathbb{C}).$$

For $n = 2$,

$$t^2 - 1 = (t-1)(t+1),$$

so there are two $\mathbb{F}_3$ sites and

$$\mathcal{H}_2 \cong \mathbb{C}^3 \otimes \mathbb{C}^3, \qquad \mathcal{A}_2 \cong M_9(\mathbb{C}).$$

For $n = 3$,

$$t^3 - 1 = (t-1)^3.$$

The scheme is a triple thickening of the point $t = 1$, but the residue-qudit layer is one qutrit. If $u = t - 1$, then $u^3 = 0$ in R_3, and u has zero residue at the unique Weyl site.

For $n = 4$,

$$t^4 - 1 = (t-1)(t+1)(t^2+1),$$

and $t^2 + 1$ has no root in $\mathbb{F}_3$. Thus

$$\mathcal{H}_4 \cong \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^9, \qquad \mathcal{A}_4 \cong M_{81}(\mathbb{C}).$$

On the $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2+1)$ factor, Frobenius sends $u \mapsto u^3 = -u$, so

$$\mathrm{Tr}_{\mathbb{F}_9/\mathbb{F}_3}(a + bu) = 2a$$

for $a, b \in \mathbb{F}_3$. This determines the explicit phase character on that 9-level site.

For $n = 6$, $n = 3 \cdot 2$, so $m = 2$ and

$$t^6 - 1 = (t^2-1)^3 = (t-1)^3(t+1)^3.$$

There are still exactly two residue Weyl sites, both $\mathbb{F}_3$-sites:

$$\mathcal{H}_6 \cong \mathbb{C}^3 \otimes \mathbb{C}^3, \qquad \mathcal{A}_6 \cong M_9(\mathbb{C}),$$

although the coordinate ring has 3^6 elements.

36.7 Conclusion

The $t^n = 1$ example makes the characteristic-3 issue explicit. The residue-qudit field depends on the squarefree part $t^m - 1$, where $m = n/3^{v_3(n)}$. The full coordinate ring still records the repeated factors of $t^n - 1$ as nilpotents, but in the residue-qudit layer those nilpotents have zero residues and therefore no separate residue Weyl operators. AQM-36 adds the separate nilpotent-sensitive layer.

37 Nilpotent Thickening Weyl Fields

37.1 Status and Source Anchors

This shard adds the nilpotent-sensitive layer proposed after AQM-34 and AQM-35. The finite Frobenius-ring source anchors are `references/heisenberg_weil/gluesing_luerssen_pllaha_1710_09884_source/StabCodesFrob5.tex`, lines 224–249 for Frobenius rings and generating characters, lines 286–323 for $X(a)$ and $Z(b)$, lines 353–355 for the multiplication law, and lines 436–452 for the error-basis statement. Sidana–Kashyap, `references/heisenberg_weil/sidana_kashyap_2202_00248_source/EAQECCs_over_rings_v15.tex`, lines 103–110 and 180–197, gives the same local Frobenius-ring Pauli framework. The trace pairing input for residue fields is the Stacks Project snapshot `references/algebraic_geometry/stacks_project_algebra.tex`, lines 45278–45284. The explicit generating-character proof below is local to this shard.

37.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & k=\mathbb{F}_3,\ \pi\in k[t] \text{ monic irreducible},\ e\geq 1,\ A=k[t]/(\pi^e).\\
D2 & : & \mathfrak{m}=(\pi)/(\pi^e),\ K=A/\mathfrak{m}=k[t]/(\pi),\ d=\deg\pi.\\
L1 & : & A \text{ has unique } \pi\text{-adic expansions and ideals } \mathfrak{m}^r.\\
D3 & : & \lambda(a)=\mathrm{Tr}_{K/k}(a_{e-1}) \text{ from the top } \pi\text{-coefficient},\quad \chi(a)=\exp(2\pi i\lambda(a)/3).\\
L2 & : & \chi \text{ is generating: } b\neq 0\Rightarrow (a\mapsto\chi(ba)) \text{ is nontrivial}.\\
D4 & : & \mathcal{H}_A=\ell^2(A),\quad E_A=A^2,\quad W_A(q,p)=T(q)R(p).\\
T1 & : & W_A \text{ is a projective unitary Weyl representation with multiplier } \chi(pq').\\
T2 & : & \{W_A(q,p):(q,p)\in A^2\} \text{ is an operator basis of } \mathrm{End}_{\mathbb{C}}(\mathcal{H}_A).\\
T3 & : & \mathfrak{m}^r \text{ position jets pair with } \mathfrak{m}^{e-1-r} \text{ momentum jets in the associated graded.}\\
T4 & : & \text{For } R_p=\mathbb{F}_3[t]/(p), \text{ the thickened layer has } \dim\mathcal{H}=3^{\deg p}.
\end{array}
$$

37.3 The Local Artin Ring L1–L2

Let

$$A=A_{\pi,e}=k[t]/(\pi^e),\qquad \mathfrak{m}=(\pi)/(\pi^e),\qquad K=k[t]/(\pi).$$

Every class $a\in A$ has a unique expansion

$$a=a_0+a_1\pi+\cdots+a_{e-1}\pi^{e-1},$$

where each a_j is represented by a polynomial of degree $< d=\deg\pi$. This follows by repeated Euclidean division by the monic polynomial π.

Lemma $L1$. The ideals of A are exactly

$$A=\mathfrak{m}^0\supset\mathfrak{m}\supset\cdots\supset\mathfrak{m}^{e-1}\supset\mathfrak{m}^e=0.$$

Every nonzero $a\in A$ has a unique valuation $v(a)=r$, $0\leq r<e$, with $a=\pi^r u$ for a unit $u\in A^\times$.

Proof. If $a\neq 0$, take the least r for which the coefficient a_r is nonzero. Then $a=\pi^r u$, where the residue of u in K is $a_r\neq 0$, so u is a unit. Let $I\subset A$ be a nonzero ideal and choose an element of minimal valuation r. Multiplying by the inverse of its unit part gives $\pi^r\in I$, hence $\mathfrak{m}^r\subset I$. Minimality of r gives $I\subset\mathfrak{m}^r$, so $I=\mathfrak{m}^r$.

Define

$$\lambda(a)=\mathrm{Tr}_{K/k}(a_{e-1}),\qquad \chi(a)=\exp\left(\frac{2\pi i}{3}\lambda(a)\right).$$

Here $a_{e-1}\in K$ is the top π-coefficient. For $e=1$ this is the residue-field character used in AQM-34.

Lemma $L2$. If $b \neq 0$ in A, then the additive character

$$a \longmapsto \chi(ba)$$

is nontrivial. Hence the map $b \mapsto (a \mapsto \chi(ba))$ identifies A with the additive character group $\widehat{A}$.

Proof. Write $b = \pi^r u$ with $u \in A^\times$. Let $\bar{u}$ be the residue of u in K. Since K is finite, Frobenius $x \mapsto x^3$ is injective and hence surjective; thus K is perfect and K/k is finite separable. The cited Stacks trace-pairing source says $(x, y) \mapsto \mathrm{Tr}_{K/k}(xy)$ is nondegenerate. Choose $\bar{c} \in K$ with $\mathrm{Tr}_{K/k}(\bar{u}\bar{c}) \neq 0$, and lift it to $c \in A$ of π-degree 0. Set

$$a = \pi^{e-1-r}c.$$

Then $ba = \pi^{e-1}uc$ modulo π^e, so the top coefficient of ba is $\bar{u}\bar{c}$. Therefore $\chi(ba) \neq 1$. The map $A \to \widehat{A}$ is injective, and both finite groups have cardinality $|A|$, so it is an isomorphism.

37.4 The Thickened Weyl Representation T1–T2

Let

$$\mathcal{H}_A = \ell^2(A)$$

with basis $|s\rangle$, $s \in A$. For $q, p \in A$, define

$$T(q)|s\rangle = |s + q\rangle, \qquad R(p)|s\rangle = \chi(ps)|s\rangle, \qquad W_A(q, p) = T(q)R(p).$$

Theorem $T1$. For $q, p, q', p' \in A$,

$$W_A(q, p)W_A(q', p') = \chi(pq')W_A(q + q', p + p')$$

and

$$W_A(q, p)W_A(q', p') = \chi(pq' - p'q)W_A(q', p')W_A(q, p).$$

Thus $\mu_3 \times A^2$, with multiplication

$$(\zeta, q, p)(\eta, q', p') = (\zeta\eta\chi(pq'), q + q', p + p'),$$

acts unitarily on $\mathcal{H}_A$ by $(\zeta, q, p) \mapsto \zeta W_A(q, p)$.

Proof. $T(q)$ is a permutation unitary and $R(p)$ is diagonal unitary. Also

$$R(p)T(q')|s\rangle = \chi(p(s + q'))|s + q'\rangle = \chi(pq')T(q')R(p)|s\rangle.$$

Substituting this identity into $T(q)R(p)T(q')R(p')$ gives the product law. The commutator law follows by comparing with the same product law in reversed order.

Theorem $T2$.

$$\mathrm{span}_{\mathbb{C}}\{W_A(q, p) : (q, p) \in A^2\} = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_A).$$

Consequently the thickened Weyl representation is irreducible.

Proof. Use the Hilbert–Schmidt inner product on $\mathrm{End}(\mathcal{H}_A)$. If $q \neq q'$, then $T(q' - q)$ has no fixed basis vector, so $\mathrm{Tr}(W_A(q, p)^* W_A(q', p')) = 0$. If $q = q'$ but $p \neq p'$, then the trace is

$$\sum_{s \in A} \chi((p' - p)s),$$

which is 0 because $s \mapsto \chi((p' - p)s)$ is a nontrivial character by L2. Thus the $|A|^2$ Weyl operators are pairwise orthogonal and nonzero. Since $\dim_{\mathbb{C}} \mathrm{End}(\ell^2(A)) = |A|^2$, they form a basis. Any invariant subspace is invariant under this full matrix algebra, hence is 0 or $\mathcal{H}_A$.

37.5 Jet Interpretation T3

The filtration

$$A \supset \mathfrak{m} \supset \cdots \supset \mathfrak{m}^{e-1} \supset 0$$

is the algebraic jet filtration. Its associated graded ring is

$$\operatorname{gr}_{\mathfrak{m}} A \cong K[\varepsilon]/(\varepsilon^e),$$

with $\mathfrak{m}^r/\mathfrak{m}^{r+1} \cong K\varepsilon^r$.

Lemma *L*3. For $0 \le r \le e$, the annihilator of $\mathfrak{m}^r$ under the pairing

$$A \times A \longrightarrow \mu_3, \qquad (x, y) \longmapsto \chi(xy)$$

is $\mathfrak{m}^{e-r}$.

Proof. If $y \in \mathfrak{m}^{e-r}$, then $xy \in \mathfrak{m}^e = 0$ for all $x \in \mathfrak{m}^r$, so y annihilates $\mathfrak{m}^r$. Conversely, if $y \notin \mathfrak{m}^{e-r}$, write $y = \pi^s u$ with $s < e - r$. Then $x = \pi^{e-1-s}c$ lies in $\mathfrak{m}^r$ for any $c \in A$ of degree 0. The proof of L2 chooses c so that $\chi(xy) \neq 1$. Hence y does not annihilate $\mathfrak{m}^r$.

Thus a position jet in $\mathfrak{m}^r/\mathfrak{m}^{r+1}$ pairs, in the associated graded, with a momentum jet in $\mathfrak{m}^{e-1-r}/\mathfrak{m}^{e-r}$. In particular, the ordinary residue value $A/\mathfrak{m}$ is dual to the socle momentum layer $\mathfrak{m}^{e-1}$, not to a new closed point.

The old residue qutrit embeds as the zeroth-position/socle-momentum subsystem: choose lifts $\tilde{q}, \tilde{p} \in A$ of $q, p \in K$, and use

$$q \longmapsto \tilde{q}, \qquad p \longmapsto \pi^{e-1}\tilde{p}.$$

On the subspace spanned by basis vectors with only degree-0 coefficient, these operators reproduce the residue-field Weyl law.

37.6 Global Polynomial Quotients T4

For

$$p = \prod_{i=1}^{r} \pi_i^{e_i},$$

set $A_i = \mathbb{F}_3[t]/(\pi_i^{e_i})$. By Chinese remainders,

$$R_p = \mathbb{F}_3[t]/(p) \cong \prod_i A_i.$$

The thickened arithmetic field is

$$\mathcal{H}_p^{\text{thick}} = \bigotimes_i \ell^2(A_i) \cong \ell^2(R_p), \qquad E_p^{\text{thick}} = \bigoplus_i A_i^2 \cong R_p^2.$$

Its Weyl operators are tensor products of the local W_{A_i}, equivalently translations and phase multiplications on $\ell^2(R_p)$ using the product character. The operator-basis theorem tensors over i, so

$$\operatorname{span}\{W(e) : e \in E_p^{\text{thick}}\} = \operatorname{End}(\mathcal{H}_p^{\text{thick}}).$$

Finally,

$$\dim_{\mathbb{C}} \mathcal{H}_p^{\text{thick}} = \prod_i |A_i| = \prod_i 3^{e_i \deg \pi_i} = 3^{\deg p} = |R_p|.$$

This is the sense in which thickenings add jet degrees of freedom at existing closed points.

37.7 Conclusion

Nilpotent thickenings should manifest as local jet degrees of freedom. They do not create additional Zariski points. The reduced theory records point values over residue fields; the thickened theory quantizes the local Artin rings and recovers the full coordinate-ring size. The phase character detects the socle, so the jet filtration is symplectically paired from the outside in: order r position jets pair with order $e-1-r$ momentum jets.

38 Thickened $t^n = 1$ Examples

38.1 Status and Dependencies

This shard is the example layer for AQM-36. It uses the factorization of $t^n - 1$ from AQM-35 and the nilpotent-sensitive Weyl theorem of AQM-36. No new external source is used.

38.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & R_n = \mathbb{F}_3[t]/(t^n-1), \quad n = 3^s m, \quad 3 \nmid m. \\
L1 & : & t^n - 1 = (t^m-1)^{3^s}, \quad t^m - 1 \text{ squarefree.} \\
D2 & : & A_\pi = \mathbb{F}_3[t]/(\pi^{3^s}) \text{ for } \pi \mid t^m - 1. \\
T1 & : & \mathcal{H}_n^{\text{thick}} = \bigotimes_{\pi \mid t^m-1} \ell^2(A_\pi), \quad \dim \mathcal{H}_n^{\text{thick}} = 3^n. \\
T2 & : & \mathcal{H}_n^{\text{red}} = \bigotimes_{\pi \mid t^m-1} \ell^2(\mathbb{F}_3[t]/(\pi)), \quad \dim \mathcal{H}_n^{\text{red}} = 3^m. \\
E1 & : & n = 3 \text{ gives one 27-dimensional third-order jet qutrit.} \\
E2 & : & n = 6 \text{ gives two third-order jet qutrits, dimension } 3^6.
\end{array}$$

38.3 General $t^n = 1$ Thickening T1–T2

AQM-35 proves that if $n = 3^s m$ with $3 \nmid m$, then

$$t^n - 1 = (t^m - 1)^{3^s}$$

and $t^m - 1$ is squarefree. Let $\mathcal{P}_m$ be the set of monic irreducible factors of $t^m - 1$. For each $\pi \in \mathcal{P}_m$, define

$$A_\pi = \mathbb{F}_3[t]/(\pi^{3^s}), \qquad K_\pi = \mathbb{F}_3[t]/(\pi).$$

The reduced layer of AQM-35 uses $\ell^2(K_\pi)$. The thickened layer of AQM-36 uses $\ell^2(A_\pi)$.

Theorem $T1$. The nilpotent-sensitive Hilbert space is

$$\mathcal{H}_n^{\text{thick}} = \bigotimes_{\pi \in \mathcal{P}_m} \ell^2(A_\pi),$$

and

$$\dim_{\mathbb{C}} \mathcal{H}_n^{\text{thick}} = 3^n.$$

Proof. Since $|A_\pi| = 3^{3^s \deg \pi}$,

$$\dim \mathcal{H}_n^{\text{thick}} = \prod_{\pi \in \mathcal{P}_m} 3^{3^s \deg \pi} = 3^{3^s \sum_{\pi \in \mathcal{P}_m} \deg \pi}.$$

Because $t^m - 1$ is squarefree and has degree m, the sum of the degrees of its irreducible factors is m. Therefore

$$\dim \mathcal{H}_n^{\text{thick}} = 3^{3^s m} = 3^n.$$

Theorem $T2$**.** The reduced Hilbert space has dimension

$$\dim_{\mathbb{C}} \mathcal{H}_n^{\mathrm{red}} = 3^m.$$

Thus the thickened/reduced dimension ratio is

$$\frac{\dim \mathcal{H}_n^{\mathrm{thick}}}{\dim \mathcal{H}_n^{\mathrm{red}}} = 3^{n-m}.$$

Proof. AQM-35 gives

$$\mathcal{H}_n^{\mathrm{red}} = \bigotimes_{\pi \in \mathcal{P}_m} \ell^2(K_\pi),$$

so

$$\dim \mathcal{H}_n^{\mathrm{red}} = \prod_{\pi \in \mathcal{P}_m} 3^{\deg \pi} = 3^m.$$

The ratio follows from $T1$.

38.4 Example $n = 3$

Let $u = t - 1$. Then

$$R_3 = \mathbb{F}_3[t]/(t^3 - 1) \cong \mathbb{F}_3[u]/(u^3).$$

The reduced layer sees one closed point and one qutrit:

$$\mathcal{H}_3^{\mathrm{red}} \cong \mathbb{C}^3.$$

The thickened layer is

$$\mathcal{H}_3^{\mathrm{thick}} = \ell^2(\mathbb{F}_3[u]/(u^3)), \qquad \dim \mathcal{H}_3^{\mathrm{thick}} = 27.$$

Write

$$a = a_0 + a_1 u + a_2 u^2, \qquad \lambda(a) = a_2, \qquad \chi(a) = \exp(2\pi i a_2/3).$$

For $q, p \in R_3$,

$$T(q)|s\rangle = |s + q\rangle, \qquad R(p)|s\rangle = \chi(ps)|s\rangle, \qquad W(q,p) = T(q)R(p).$$

If

$$q = \sum_{j=0}^{2} q_j u^j, \qquad p = \sum_{j=0}^{2} p_j u^j,$$

then the commutator phase of (q, p) with (q', p') is

$$\exp\left(\frac{2\pi i}{3} \sum_{r+s=2} (p_r q'_s - p'_r q_s) \right).$$

Thus the value coefficient pairs with the u^2-momentum coefficient, the first jet pairs with the first-jet momentum coefficient, and the u^2 position coefficient pairs with the value momentum coefficient.

38.5 Example $n = 6$

Since $6 = 3 \cdot 2$,

$$t^6 - 1 = (t^2 - 1)^3 = (t-1)^3(t+1)^3.$$

There are two closed points, but each is third-order thickened. Let

$$A_+ = \mathbb{F}_3[t]/((t-1)^3), \qquad A_- = \mathbb{F}_3[t]/((t+1)^3).$$

Then

$$\mathcal{H}_6^{\text{thick}} = \ell^2(A_+) \otimes \ell^2(A_-) \cong \mathbb{C}^{27} \otimes \mathbb{C}^{27},$$

so $\dim \mathcal{H}_6^{\text{thick}} = 3^6$. The reduced layer is

$$\mathcal{H}_6^{\text{red}} \cong \mathbb{C}^3 \otimes \mathbb{C}^3,$$

with dimension 3^2.

38.6 Conclusion

For $t^n = 1$, the reduced layer depends only on the squarefree part $t^m - 1$, while the thickened layer recovers the full coordinate-ring cardinality. The nilpotent variables are therefore genuine local jet modes, not additional closed points.

39 Lagrangian Stabilizer Descent

39.1 Status and Source Anchors

This shard is a checked derivation from the already registered finite Weyl and local-net layers. AQM-08 and AQM-22 source and derive the prime-field Weyl commutator and the criterion that stabilizer labels commute exactly when their symplectic pairing vanishes. AQM-26 proves that the Weyl label assignment is the cosheaf-like additive layer and that the observable assignment is a local net. AQM-27 proves that quantum states do not form a sheaf because local marginals do not determine global correlations.

The new convention is `CONVENTIONS.md` (ac). We work here in the finite vector-space case over an odd prime field $k = \mathbb{F}_p$, with the explicit counterexample over $k = \mathbb{F}_3$. Mixed residue fields and finite Frobenius modules are not imported into the dimension statements in this shard.

39.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & E = k_X^n \oplus k_Z^n, \quad \omega((x,z),(x',z')) = z\cdot x' - x\cdot z'. \\
D2 & : & W(e) = T_x R_z \text{ on } \mathcal{H} = \ell^2(k^n), \quad W(e)W(f) = c(e,f)W(e+f). \\
D3 & : & L \le E \text{ is isotropic if } \omega(L,L)=0, \text{ and Lagrangian if } L = L^\perp. \\
D4 & : & \alpha : L \to U(1) \text{ is a phase if } \alpha(e+f) = \alpha(e)\alpha(f)c(e,f)^{-1}. \\
L1 & : & \{W(e) : e \in E\} \text{ is a basis of } \operatorname{End}(\mathcal{H}). \\
L2 & : & \text{A phased isotropic } L \text{ gives a commuting stabilizer group.} \\
T1 & : & \text{A phased Lagrangian } L \text{ gives a rank-one full stabilizer state.} \\
D5 & : & \operatorname{Loc}_U(L;\mathcal{U}) = \sum_i (L \cap E(U_i)), \quad D_U(L;\mathcal{U}) = L/\operatorname{Loc}_U(L;\mathcal{U}). \\
T2 & : & D_U(L;\mathcal{U}) = 0 \iff L \text{ is generated by stabilizers supported on } \mathcal{U}. \\
T3 & : & \text{Disjoint products of local Lagrangians have zero defect.}
\end{array}$$

39.3 Finite Weyl Stabilizers D1–L2

Let $k = \mathbb{F}_p$ with p odd. Put

$$E = k_X^n \oplus k_Z^n, \qquad \mathcal{H} = \ell^2(k^n),$$

and write $e = (x, z)$. With a nontrivial additive character $\chi : k \to U(1)$, define

$$T_x|s\rangle = |s+x\rangle, \qquad R_z|s\rangle = \chi(z\cdot s)|s\rangle, \qquad W(x,z) = T_x R_z.$$

As in AQM-22,

$$W(e)W(f) = \chi(z\cdot x')W(e+f), \qquad W(e)W(f) = \chi(\omega(e,f))W(f)W(e).$$

Thus the multiplier in D2 is $c(e,f) = \chi(z\cdot x')$.

Lemma $L1$. The operators $\{W(e) : e \in E\}$ form a basis of $\operatorname{End}(\mathcal{H})$.

Proof. There are $|E| = p^{2n}$ such operators, and $\dim_{\mathbb{C}} \operatorname{End}(\mathcal{H}) = p^{2n}$. It is enough to prove linear independence. For $e = (x, z)$ and $f = (x', z')$, the trace of $W(e)^*W(f)$ is zero if $x \ne x'$, because the resulting translation has no fixed basis vector. If $x = x'$ but $z \ne z'$, the translation cancels and the trace is

$$\sum_{s\in k^n} \chi((z'-z)\cdot s) = 0$$

by nontriviality of the character on the nonzero linear functional $(z' - z)\cdot s$. If $e = f$, the trace is p^n. The Weyl operators are therefore orthogonal and nonzero for the Hilbert-Schmidt pairing, hence linearly independent.

Lemma *L*2. Let $L \leq E$ be isotropic and let $\alpha : L \to U(1)$ satisfy D4. Then

$$S_e = \alpha(e)^{-1} W(e), \qquad e \in L,$$

is an ordinary unitary representation of the additive group L. In particular the S_e commute.

Proof. Using D4 and the Weyl product law,

$$\begin{aligned} S_e S_f &= \alpha(e)^{-1}\alpha(f)^{-1} c(e,f) W(e+f) \\ &= \alpha(e+f)^{-1} W(e+f) \\ &= S_{e+f}. \end{aligned}$$

The operators $W(e)$ are unitary, and the phases have unit modulus, so each S_e is unitary. Since L is an Abelian additive group, $S_e S_f = S_{e+f} = S_{f+e} = S_f S_e$.

39.4 Full Stabilizers Are Lagrangian T1

Theorem *T*1. If $L \leq E$ is Lagrangian and phased by α, then the common $+1$-eigenspace of the stabilizer group $\{S_e : e \in L\}$ is one dimensional. Equivalently, a phased Lagrangian is a full stabilizer state at the label level.

Proof. Because E is a $2n$-dimensional nondegenerate symplectic vector space, $\dim L + \dim L^\perp = 2n$. Thus $L = L^\perp$ implies $\dim L = n$, hence $|L| = p^n$. The group average

$$P_L = \frac{1}{|L|} \sum_{e \in L} S_e$$

is the orthogonal projection onto the common $+1$-eigenspace. Indeed,

$$P_L^2 = \frac{1}{|L|^2} \sum_{e,f \in L} S_{e+f} = \frac{1}{|L|} \sum_{g \in L} S_g = P_L,$$

because every $g \in L$ has exactly $|L|$ presentations $g = e + f$. Also $P_L^* = |L|^{-1} \sum_e S_{-e} = P_L$, and $S_f P_L = P_L = P_L S_f$ by the substitution $e \mapsto f + e$. Conversely, if $S_f v = v$ for all $f \in L$, then $P_L v = v$.

Its trace is

$$\mathrm{Tr}(P_L) = \frac{1}{|L|} \sum_{e \in L} \alpha(e)^{-1} \,\mathrm{Tr}(W(e)).$$

By Lemma *L*1's trace calculation, $\mathrm{Tr}(W(e)) = 0$ for $e \neq 0$ and $\mathrm{Tr}(W(0)) = p^n$. Therefore

$$\mathrm{Tr}(P_L) = \frac{p^n}{p^n} = 1.$$

An orthogonal projection has trace equal to its rank, so the stabilizer space is one dimensional.

The same calculation also shows maximality among Weyl observables. If an operator $A = \sum_{a \in E} \lambda_a W(a)$ commutes with every S_e, then each nonzero coefficient can occur only for labels $a \in L^\perp = L$, because the commutator factor is $\chi(\omega(a,e))$. Thus the centralizer of the stabilizer group inside the Weyl basis is spanned by the same labels.

39.5 Descent Defect D5–T2

Now let U be a finite region or finite open in the residue-qudit net, and let $\mathcal{U} = \{U_i\}$ be a finite cover. The label cosheaf of AQM-26 includes each $E(U_i)$ into $E(U)$ by extension by zero. For a global stabilizer label space $L \le E(U)$, define

$$\mathrm{Loc}_U(L;\mathcal{U}) = \sum_i \bigl(L \cap E(U_i)\bigr) \le L$$

and

$$D_U(L;\mathcal{U}) = L/\,\mathrm{Loc}_U(L;\mathcal{U}).$$

This quotient is the stabilizer descent defect.

Theorem $T2$. For a fixed global phased stabilizer $L \le E(U)$, the following are equivalent:

(1) $D_U(L;\mathcal{U}) = 0$,
(2) $L = \sum_i (L \cap E(U_i))$,
(3) $\{S_e : e \in L\}$ is generated by the stabilizers supported in the U_i.

Proof. The equivalence of (1) and (2) is exactly the definition of the quotient defect. If (2) holds, write any $e \in L$ as $e = e_1 + \cdots + e_m$ with each $e_j \in L \cap E(U_{i_j})$. Lemma $L2$ gives

$$S_e = S_{e_1+\cdots+e_m} = S_{e_1} \cdots S_{e_m},$$

so the global stabilizer is generated by local stabilizers. Conversely, if the stabilizer group is generated by supported stabilizers, then passing to projective labels gives every $e \in L$ as a sum of elements from $L \cap E(U_i)$, so (2) holds.

Therefore vanishing defect is the label-level local-to-global condition. It is the generation half of a cosheaf condition for a stabilizer subobject of the label cosheaf. Phase gluing is additional character compatibility on overlaps; in the disjoint examples below there is no overlap condition.

39.6 The Product Case T3

Theorem $T3$. Let $U = U_1 \sqcup U_2$, so

$$E(U) = E(U_1) \oplus E(U_2)$$

as an orthogonal symplectic direct sum. If $L_i \le E(U_i)$ is Lagrangian for $i = 1, 2$, then

$$L = L_1 \oplus L_2 \le E(U)$$

is Lagrangian and $D_U(L; \{U_1, U_2\}) = 0$. With product phases, the stabilizer projection is $P_{L_1} \otimes P_{L_2}$.

Proof. Orthogonality of the direct sum gives

$$L^\perp = L_1^\perp \oplus L_2^\perp = L_1 \oplus L_2 = L,$$

so L is Lagrangian. Also

$$L \cap E(U_1) = L_1, \qquad L \cap E(U_2) = L_2,$$

hence $\mathrm{Loc}_U(L; \{U_1, U_2\}) = L$ and the defect vanishes. The group average over $L_1 \oplus L_2$ factors as the product of the two group averages, which is $P_{L_1} \otimes P_{L_2}$.

39.7 Conclusion

A Lagrangian subspace is the label-level form of a full stabilizer state. A vanishing stabilizer descent defect is the separate condition that this full stabilizer is generated from local pieces of a chosen cover. In cosheaf language, E is the additive label cosheaf; a stabilizer choice has the local-to-global generation property only when its labels have zero descent defect. The next shard gives the explicit two-qutrit counterexample showing that the Lagrangian condition alone does not force this.

40 Two-Qutrit Stabilizer Descent Counterexample

40.1 Status and Source Anchors

This shard is a checked finite calculation using the convention and theorems of AQM-38. It is the promised counterexample to the overly strong claim that choosing a global Lagrangian stabilizer is the same thing as satisfying a local-to-global principle. The Weyl commutation convention is the same prime-field convention as AQM-08, AQM-22, and `CONVENTIONS.md` (l).

40.2 Lamport Dependency Skeleton

$$\begin{array}{rcl} D1 & : & U = A \sqcup B, \quad E(U) = \mathbb{F}_3^2 \oplus \mathbb{F}_3^2. \\ D2 & : & \omega(e, e') = z_A x'_A - x_A z'_A + z_B x'_B - x_B z'_B. \\ D3 & : & \ell_1 = (1, 0, -1, 0), \quad \ell_2 = (0, 1, 0, 1), \quad L = \langle \ell_1, \ell_2 \rangle. \\ L1 & : & L \text{ is Lagrangian.} \\ L2 & : & L \cap E(A) = 0 \text{ and } L \cap E(B) = 0. \\ T1 & : & D_U(L; \{A, B\}) \cong L \neq 0. \\ T2 & : & \text{The stabilizer state has maximally mixed one-site marginals.} \\ T3 & : & \text{A global Lagrangian is not the same as stabilizer descent.} \end{array}$$

40.3 The Label Calculation D1–T1

Let $U = A \sqcup B$, with one qutrit on each site. Write labels as

$$(x_A, z_A, x_B, z_B) \in \mathbb{F}_3^2 \oplus \mathbb{F}_3^2$$

and use the direct-sum symplectic form

$$\omega(e, e') = z_A x'_A - x_A z'_A + z_B x'_B - x_B z'_B.$$

Define

$$\ell_1 = (1, 0, -1, 0), \qquad \ell_2 = (0, 1, 0, 1), \qquad L = \langle \ell_1, \ell_2 \rangle.$$

Lemma $L1$. The subspace L is Lagrangian.

Proof. The only nontrivial generator pairing is

$$\omega(\ell_1, \ell_2) = 0 \cdot 0 - 1 \cdot 1 + 0 \cdot 0 - (-1) \cdot 1 = -1 + 1 = 0.$$

Same-generator pairings vanish because ω is alternating, so L is isotropic. The ambient symplectic vector space has dimension 4, and ℓ_1, ℓ_2 are independent, so $\dim L = 2$. Since the form is nondegenerate, $\dim L + \dim L^\perp = 4$, so $\dim L^\perp = 2$. Isotropy gives $L \subseteq L^\perp$, and equal dimensions give $L = L^\perp$. Hence L is Lagrangian.

Lemma $L2$.

$$L \cap E(A) = 0, \qquad L \cap E(B) = 0.$$

Proof. A general element of L is

$$a\ell_1 + b\ell_2 = (a, b, -a, b).$$

If it is supported only on A, then $-a = 0$ and $b = 0$, hence $a = b = 0$. If it is supported only on B, then $a = 0$ and $b = 0$. These are exactly the displayed intersections.

Theorem $T1$**.** For the disjoint cover $\{A, B\}$,

$$D_U(L; \{A, B\}) \cong L.$$

In particular the defect has $\mathbb{F}_3$-dimension 2.

Proof. By Lemma $L2$,

$$\mathrm{Loc}_U(L; \{A, B\}) = (L \cap E(A)) + (L \cap E(B)) = 0.$$

The definition from AQM-38 gives

$$D_U(L; \{A, B\}) = L/\,\mathrm{Loc}_U(L; \{A, B\}) = L.$$

40.4 The Stabilizer State T2

Theorem $T2$**.** With the standard qutrit Weyl lift, the stabilizer labels L define a full global stabilizer state, but no nonzero local stabilizer label on either single site is inherited from L.

Proof. The two generator operators are

$$W(\ell_1) = X_A X_B^{-1}, \qquad W(\ell_2) = Z_A Z_B.$$

The Weyl multiplier is trivial on all of L, because

$$(b, b) \cdot (a', -a') = 0$$

for $a\ell_1 + b\ell_2$ and $a'\ell_1 + b'\ell_2$. Thus the phase $\alpha = 1$ satisfies the AQM-38 phase condition. The two displayed operators commute by Lemma $L1$. The vector

$$|\Phi_-\rangle = \frac{1}{\sqrt{3}} \sum_{s \in \mathbb{F}_3} |s, -s\rangle$$

is fixed by both operators. Indeed, $X_A X_B^{-1}$ sends $|s, -s\rangle$ to $|s+1, -s-1\rangle = |s+1, -(s+1)\rangle$, only reindexing the sum. Also $Z_A Z_B$ multiplies $|s, -s\rangle$ by $\chi(s - s) = 1$. Since L is Lagrangian, AQM-38 Theorem $T1$ says the common stabilizer space is one-dimensional, so this is the full stabilizer state.

The one-site density matrix on A is

$$\mathrm{Tr}_B(|\Phi_-\rangle\langle\Phi_-|) = \frac{1}{3} \sum_{s \in \mathbb{F}_3} |s\rangle\langle s| = I_3/3,$$

and the same computation gives $I_3/3$ on B. Thus the one-site restrictions are maximally mixed. This agrees with Lemma $L2$: the inherited one-site stabilizer label spaces are trivial, although the reduced density matrices have all one-site operators in their commutants.

40.5 Corrected Local-to-Global Statement T3

Theorem $T3$**.** Choosing a global phased Lagrangian is not equivalent to local-to-global gluing. The proved label-level statement used here is:

$$\begin{gathered}\text{global phased Lagrangian } L \le E(U) \text{ with AQM-38 phase } \alpha \\ \Longrightarrow \\ \text{full global stabilizer state,}\end{gathered}$$

with the converse requiring a separate stabilizer-state classification theorem. whereas

$$\begin{gathered}\text{fixed global phased Lagrangian generated from } \mathcal{U} \\ \Longleftrightarrow \\ \text{its labels satisfy } D_U(L;\mathcal{U}) = 0 \text{ and phases are restrictions of } \alpha.\end{gathered}$$

Independently chosen local phased stabilizers on an overlapping cover require additional overlap compatibility of the eigencharacters with the Weyl multiplier; no such phase-descent theorem is proved here.

Proof. The first implication is AQM-38 Theorem $T1$. AQM-38 Theorem $T2$ proves that $D_U(L;\mathcal{U}) = 0$ is exactly local generation of the global labels. For a fixed global phased stabilizer (L, α), the local phases are just the restrictions of α. AQM-38 Theorem $T3$ proves that product local stabilizer states satisfy the condition. Theorem $T1$ above proves that the Lagrangian condition alone does not imply it.

40.6 Conclusion

The broad equivalence must be rejected in this form. A phased Lagrangian gives a full global stabilizer state by AQM-38; the converse is not used here. A zero descent defect is the extra condition that the stabilizer labels are generated from the local pieces of a chosen cover. Nonzero defect is the stabilizer-label analogue of the state-presheaf phenomenon from AQM-27: local restrictions can miss global correlations.

41 Product-Field Spectra and n-Qudit Stabilizers

41.1 Status and Source Anchors

This shard studies $X = \operatorname{Spec} R$ for

$$R = k^n = k \times \cdots \times k, \qquad k = \mathbb{F}_q.$$

It is a finite reduced spectrum example in the residue-qudit convention of AQM-23, AQM-24, and AQM-28. The Weyl operator formulas use the same finite field sources as AQM-22 and AQM-23: Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313 for finite-field shift and phase operators, and Bombin–Martin–Delgado, `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 1208–1259 for qudit Paulis and symplectic commutators.

The new product-field convention is `CONVENTIONS.md` (ad). The key point is that $\operatorname{Spec}(k^n)$ has n points, not q^n points.

41.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k = \mathbb{F}_q,\ R = k^n,\ e_i = (0, \ldots, 1, \ldots, 0). \\
L1 & : & \operatorname{Spec} R = \{\mathfrak{m}_i = \ker(\mathrm{pr}_i)\}_{i=1}^n. \\
L2 & : & \kappa(\mathfrak{m}_i) = k, \quad D(e_i) = \{\mathfrak{m}_i\}. \\
D2 & : & U \subset X \text{ corresponds to } I_U \subset \{1, \ldots, n\}. \\
D3 & : & E(U) = \bigoplus_{i \in I_U} k^2 = k_X^{I_U} \oplus k_Z^{I_U}. \\
D4 & : & \mathcal{H}(U) = \bigotimes_{i \in I_U} \ell^2(k). \\
L3 & : & W_U(x, z) = T_x R_z \text{ satisfies the finite-field Weyl law.} \\
T1 & : & \operatorname{span}\{W_U(e) : e \in E(U)\} = \bigotimes_{i \in I_U} M_q(\mathbb{C}). \\
D5 & : & U \subset V \text{ gives extension by zero and tensoring by identities.} \\
T2 & : & X = \operatorname{Spec}(k^n) \text{ gives the ambient } n\text{-qudit Weyl net.} \\
T3 & : & \text{Stabilizer codes are extra isotropic labels plus phases.}
\end{array}$$

41.3 The Product-Field Spectrum D1–L2

Let $R = k^n$. Write $\mathrm{pr}_i : R \to k$ for the i-th projection and $e_i \in R$ for the idempotent with 1 in the i-th coordinate and 0 elsewhere.

Lemma $L1$. The prime ideals of R are exactly

$$\mathfrak{m}_i = \ker(\mathrm{pr}_i) = k \times \cdots \times k \times 0 \times k \times \cdots \times k.$$

Hence $\operatorname{Spec} R$ has n points.

Proof. First, $R/\mathfrak{m}_i \cong k$, so each $\mathfrak{m}_i$ is maximal and therefore prime.

Conversely, let $\mathfrak{p} \subset R$ be prime. Since

$$e_i e_j = 0 \quad (i \neq j), \qquad 1 = e_1 + \cdots + e_n,$$

at most one idempotent e_i can be outside $\mathfrak{p}$: if $e_i, e_j \notin \mathfrak{p}$ for $i \neq j$, primeness would be contradicted by $e_i e_j = 0 \in \mathfrak{p}$. At least one e_i is outside $\mathfrak{p}$, because otherwise $1 = \sum_i e_i \in \mathfrak{p}$. Let this unique index be i. Then $e_j \in \mathfrak{p}$ for $j \neq i$. Since $e_i R \cong k$ is a field and $\mathfrak{p} \cap e_i R$ is a proper ideal, it is 0. Thus $\mathfrak{p}$ is exactly the direct product of the $j \neq i$ factors and 0 in the i-th factor, namely $\mathfrak{m}_i$.

Lemma *L*2. For every i,

$$\kappa(\mathfrak{m}_i) \cong k, \qquad D(e_i) = \{\mathfrak{m}_i\}.$$

Consequently $X = \operatorname{Spec} R$ is a finite discrete topological space.

Proof. The quotient $R/\mathfrak{m}_i$ is k, so the residue field at $\mathfrak{m}_i$ is k. The standard open $D(e_i)$ consists of prime ideals not containing e_i. By the proof of Lemma $L1$, $e_i \notin \mathfrak{m}_i$, while $e_i \in \mathfrak{m}_j$ for $j \neq i$. Hence $D(e_i) = \{\mathfrak{m}_i\}$. Since every singleton is open, every subset is open.

41.4 The Residue-Qudit Field D2–L3

Let $U \subset X$, and let $I_U \subset \{1, \ldots, n\}$ be the corresponding index set. Define

$$E(U) = \bigoplus_{i \in I_U} \kappa(\mathfrak{m}_i)^2 = k_X^{I_U} \oplus k_Z^{I_U}.$$

Write $e = (x, z)$, with $x, z : I_U \to k$. Let

$$\chi(a) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{k/\mathbb{F}_p}(a)\right)$$

be the standard additive character. Define

$$\Omega_U(e, f) = \sum_{i \in I_U} (z_i x_i' - x_i z_i') \in k$$

and the commutator bicharacter

$$B_U(e, f) = \chi(\Omega_U(e, f)).$$

The Hilbert space is

$$\mathcal{H}(U) = \bigotimes_{i \in I_U} \ell^2(k) \cong \ell^2(k^{I_U}).$$

On the basis $|s\rangle$, $s : I_U \to k$, define

$$T_x|s\rangle = |s + x\rangle, \qquad R_z|s\rangle = \chi\left(\sum_{i \in I_U} z_i s_i\right)|s\rangle,$$

and

$$W_U(x, z) = T_x R_z.$$

Lemma *L*3. For $e = (x, z)$ and $f = (x', z')$,

$$W_U(e)W_U(f) = \chi\left(\sum_{i \in I_U} z_i x_i'\right) W_U(e + f),$$

and

$$W_U(e)W_U(f) = B_U(e, f)W_U(f)W_U(e).$$

Proof. Translations add and phases add:

$$T_x T_{x'} = T_{x+x'}, \qquad R_z R_{z'} = R_{z+z'}.$$

Also

$$\begin{aligned} R_z T_{x'}|s\rangle &= R_z|s+x'\rangle \\ &= \chi\left(\sum_i z_i(s_i+x'_i)\right)|s+x'\rangle \\ &= \chi\left(\sum_i z_i x'_i\right) T_{x'} R_z|s\rangle. \end{aligned}$$

Multiplying $T_x R_z T_{x'} R_{z'}$ gives the first formula. Dividing this formula by the same formula with e and f exchanged gives the commutator factor $\chi(\Omega_U(e,f))$.

41.5 Operator Algebra T1–T2

Theorem $T1$.

$$\mathrm{span}_{\mathbb{C}}\{W_U(e) : e \in E(U)\} = \mathrm{End}(\mathcal{H}(U)) \cong \bigotimes_{i\in I_U} M_q(\mathbb{C}).$$

Proof. There are $|E(U)| = q^{2|I_U|}$ Weyl operators and

$$\dim_{\mathbb{C}} \mathrm{End}(\mathcal{H}(U)) = (q^{|I_U|})^2 = q^{2|I_U|}.$$

It is enough to prove linear independence. If $e=(x,z)$ and $f=(x',z')$, the trace of $W_U(e)^*W_U(f)$ is zero when $x \neq x'$, because the resulting translation has no fixed basis vector. If $x=x'$ but $z \neq z'$, the trace is

$$\sum_{s\in k^{I_U}} \chi\left(\sum_i (z'_i - z_i)s_i\right) = 0,$$

because $z'-z$ is a nonzero linear functional and χ is nontrivial. If $e=f$, the trace is $q^{|I_U|}$. Thus the Weyl operators are orthogonal and nonzero for the Hilbert-Schmidt pairing, hence form a basis. The tensor-product matrix identification follows from $\mathcal{H}(U) = \bigotimes_{i\in I_U} \ell^2(k)$.

Local net D5. If $U \subset V$, include $E(U)$ in $E(V)$ by extension by zero. The operator inclusion is

$$\mathcal{A}(U) \longrightarrow \mathcal{A}(V), \qquad a \longmapsto a \otimes \mathbf{1}_{I_V\setminus I_U}.$$

The identity and composition laws are immediate from tensoring with identity operators on the missing tensor factors. Since X is discrete, this is the usual tensor-factor local net over all subsets of $\{1,\dots,n\}$.

Theorem $T2$. For $X = \mathrm{Spec}(k^n)$, the residue-qudit arithmetic field is the ambient standard n-qudit Weyl/Pauli net for q-level qudits:

$$E(X) = k_X^n \oplus k_Z^n, \qquad \mathcal{H}(X) = \ell^2(k)^{\otimes n}, \qquad \mathcal{A}(X) \cong M_{q^n}(\mathbb{C}).$$

Proof. By Lemma $L1$, X has exactly the n points $\mathfrak{m}_1,\dots,\mathfrak{m}_n$. By Lemma $L2$, every residue field is k, and the topology is discrete. Therefore the full open X has label space

$$E(X) = \bigoplus_{i=1}^n k^2 = k_X^n \oplus k_Z^n$$

and Hilbert space $\bigotimes_{i=1}^n \ell^2(k)$. Lemma $L3$ is exactly the finite-field multi-qudit Weyl law, and Theorem $T1$ gives the full matrix algebra on n q-level tensor factors.

41.6 Stabilizer Comparison T3

Theorem *T*3. The ring k^n supplies the ambient n-qudit Weyl/Pauli system. A stabilizer code is additional data: an isotropic additive subgroup

$$L \leq E(X)$$

for the bicharacter B_X, together with compatible phases/eigenvalues. If one restricts L to be k-linear and requires $\Omega_X(L, L) = 0$ in k, one gets the usual k-linear q-ary stabilizer subclass. Allowing $\mathbb{F}_p$-additive L with

$$\operatorname{Tr}_{k/\mathbb{F}_p} \Omega_X(L, L) = 0$$

gives the standard finite-field additive stabilizer convention.

Proof. By Lemma $L3$, two Weyl operators commute exactly when

$$B_X(e, f) = \chi(\Omega_X(e, f)) = 1.$$

Thus an additive subgroup L gives commuting projective stabilizer labels exactly when $B_X(e, f) = 1$ for all $e, f \in L$, equivalently $\operatorname{Tr}_{k/\mathbb{F}_p} \Omega_X(e, f) = 0$ for all $e, f \in L$. The phases/eigenvalues are not determined by R; as in AQM-22 and AQM-38, they are a separate lift of the commuting projective labels to actual operators with chosen joint eigenvalues.

If L is a k-linear subspace and the stronger equality $\Omega_X(L, L) = 0$ holds in k, then the trace condition holds automatically. This is the k-linear stabilizer subclass. Conversely, for general $q = p^r$, the trace condition may hold for an $\mathbb{F}_p$-linear additive subgroup that is not closed under multiplication by k. Such subgroups are part of the standard additive finite-field stabilizer formalism but are not k-linear subspaces. When $q = p$ is prime, these two notions coincide.

41.7 Conclusion

For $R = k^n$, with $X = \operatorname{Spec} R$, the hoped-for statement is correct with the standard stabilizer caveat. The product ring is the coordinate ring of a finite discrete n-point k-scheme. Its residue-qudit arithmetic field is exactly the ambient n q-level qudit Weyl/Pauli system. Stabilizer codes do not come for free from the ring: they are later choices of isotropic label subgroups and compatible phases.

42 Cech Cohomology and Quantum Descent

42.1 Status and Source Anchors

This shard adds the cover-complex layer requested after AQM-26–AQM-40. Stacks Cohomology, `references/algebraic_geometry/stacks_project_cohomology.tex`, lines 868–924, 926–946, 2438–2517, and 4071–4165, supplies the Cech complex, H^0 sheaf criterion, refinement colimit, and ordered Cech complex. Curry, `references/cosheaves/curry_1303_3255_source/ch_abstract_sheaves.tex`, lines 246–259 and 270–335, supplies the dual vector-space exactness and Cech homology convention. The quantum input is local: AQM-26 for labels and observable nets, AQM-27 for states, and AQM-38–AQM-39 for stabilizer descent. The new convention is `CONVENTIONS.md` (ae).

42.2 Lamport Dependency Skeleton

$D1$: $\mathcal{U} = \{U_i\}_{i\in I}$ is a finite ordered cover of U.
$D2$: $\check{C}^p(\mathcal{U}, F) = \prod_{i_0<\cdots<i_p} F(U_{i_0\cdots i_p})$.
$L1$: $d^2 = 0$ for the Cech differential.
$D3$: $\check{H}^p(\mathcal{U}, F) = \ker d^p / \operatorname{im} d^{p-1}$.
$L2$: $\check{H}^0(\mathcal{U}, F)$ is compatible local section data.
$D4$: $\check{H}^p(U, F) = \varinjlim_{\mathcal{U}} \check{H}^p(\mathcal{U}, F)$ when refinements are declared.
$D5$: $\check{C}_p(\mathcal{U}, G) = \bigoplus_{i_0<\cdots<i_p} G(U_{i_0\cdots i_p})$ for a precosheaf G.
$T1$: $\check{H}_0(\mathcal{U}, E) \cong E(U)$, $\quad \check{H}_p(\mathcal{U}, E) = 0$ $(p > 0)$.
$D6$: $\check{Z}^0(\mathcal{U}, \mathsf{S})$ is the set of compatible local states.
$T2$: $\mathsf{S}(U) \to \check{Z}^0(\mathcal{U}, \mathsf{S})$ may fail injectivity and surjectivity.
$D7$: $L_{\le U}(W) = L \cap E(W)$ for a global stabilizer $L \le E(U)$.
$T3$: $D_U(L; \mathcal{U})$ is the cokernel of stabilizer Cech H_0 generation.
$P1$: $\mathcal{A}$ has an observable-net comparison map, not ordinary abelian Cech cohomology.

42.3 Standard Cover Cohomology D1–D4

Let X be a topological space and let $\mathcal{U} = \{U_i\}_{i\in I}$ be a finite open cover of $U \subset X$. Fix a total order on I, and write

$$U_{i_0\cdots i_p} = U_{i_0} \cap \cdots \cap U_{i_p}.$$

For an abelian presheaf F, define

$$\check{C}^p(\mathcal{U}, F) = \prod_{i_0<\cdots<i_p} F(U_{i_0\cdots i_p}).$$

If $c \in \check{C}^p(\mathcal{U}, F)$, set

$$(dc)_{i_0\cdots i_{p+1}} = \sum_{j=0}^{p+1} (-1)^j \, c_{i_0\cdots \widehat{i_j} \cdots i_{p+1}}\big|_{U_{i_0\cdots i_{p+1}}}.$$

Lemma $L1$. The differential satisfies $d^2 = 0$.

Proof. Fix $i_0 < \cdots < i_{p+2}$. In $(d^2c)_{i_0\cdots i_{p+2}}$, the term obtained by deleting i_a and i_b, with $a < b$, appears exactly twice. The two restriction maps are equal by functoriality of the presheaf. The corresponding signs are

$$(-1)^a(-1)^{b-1} \quad \text{and} \quad (-1)^b(-1)^a.$$

Their sum is $(-1)^{a+b-1} + (-1)^{a+b} = 0$. Summing over all pairs (a, b) gives $d^2c = 0$.

Thus

$$\check{H}^p(\mathcal{U}, F) = \ker\bigl(d : \check{C}^p \to \check{C}^{p+1}\bigr) / \operatorname{im}\bigl(d : \check{C}^{p-1} \to \check{C}^p\bigr)$$

is well-defined.

Lemma $L2$. $\check{H}^0(\mathcal{U}, F)$ is the group of compatible local sections:

$$\check{H}^0(\mathcal{U}, F) = \left\{ (s_i) \in \prod_i F(U_i) : s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \text{ for all } i, j \right\}.$$

If F is a sheaf, the restriction map $F(U) \to \check{H}^0(\mathcal{U}, F)$ is bijective.

Proof. Since $\check{C}^{-1} = 0$, $\check{H}^0 = \ker(d : \check{C}^0 \to \check{C}^1)$. The displayed formula for d says exactly that all pairwise restrictions agree. The second sentence is precisely the sheaf gluing axiom, as in Stacks Cohomology, lines 926–946.

When a cover-independent group is needed, we follow Stacks Cohomology, lines 2438–2517, and define

$$\check{H}^p(U, F) = \varinjlim_{\mathcal{U}} \check{H}^p(\mathcal{U}, F),$$

over open covers ordered by refinement. In the arithmetic-field questions below, the fixed cover is part of the physical question, so we keep the cover in the notation unless this direct limit is explicitly invoked.

42.4 Dual Cech Homology for Labels D5–T1

Let G be a covariant precosheaf of abelian groups or vector spaces. Define

$$\check{C}_p(\mathcal{U}, G) = \bigoplus_{i_0 < \cdots < i_p} G(U_{i_0 \cdots i_p}).$$

For $s \in G(U_{i_0 \cdots i_p})$, put

$$\partial s = \sum_{j=0}^{p} (-1)^j r_{U_{i_0 \cdots \widehat{i_j} \cdots i_p}, U_{i_0 \cdots i_p}}(s),$$

where r is the covariant extension map. The same sign cancellation as in Lemma $L1$ gives $\partial^2 = 0$, hence $\check{H}_p(\mathcal{U}, G)$.

For the residue-qudit label layer, Weyl sites are the finite-residue sites $C(W) = W \cap |X|_{\mathrm{cl}}$ of AQM-28, or all points in finite all-closed spectra. Thus $E(W) = \bigoplus_{x \in C(W)} E_x$ is the finite-support direct sum, with extension by zero along inclusions. Assume $\mathcal{U}$ is finite.

Theorem $T1$. For every finite cover $\mathcal{U}$ of U,

$$\check{H}_0(\mathcal{U}, E) \cong E(U), \qquad \check{H}_p(\mathcal{U}, E) = 0 \quad (p > 0).$$

Proof. For a Weyl site $x \in C(U)$, let $I_x = \{i \in I : x \in U_i\}$. This set is nonempty because $\mathcal{U}$ covers U. Since direct sums distribute over the finite Cech chain groups,

$$\check{C}_p(\mathcal{U}, E) \cong \bigoplus_{x \in C(U)} \left(\bigoplus_{i_0 < \cdots < i_p,\ i_j \in I_x} E_x \right).$$

The differential preserves the summand indexed by x, because all extension maps are the identity on the E_x component. Therefore the x-summand is E_x tensored with the simplicial chain complex of the full simplex on the vertex set I_x.

Choose $a \in I_x$. The cone operator $h(\sigma) = [a, \sigma]$, with the orientation sign determined by inserting a into the ordered simplex, satisfies $\partial h + h\partial = \mathrm{id} - \epsilon$, where ϵ is the augmentation to one copy of E_x in degree 0. Thus the full simplex has homology E_x in degree 0 and 0 in higher degrees. Taking the direct sum over x gives the displayed result.

So the finite-support label cosheaf carries no higher cover obstruction: every label is assembled pointwise from the cover, and higher cycles are simplex boundaries.

42.5 Quantum Counterparts D6–P1

States. For the state presheaf $\mathsf{S}(W) = \mathrm{State}(\mathcal{A}(W))$, define only the zeroth compatible-family object

$$\check{Z}^0(\mathcal{U}, \mathsf{S}) = \left\{ (\omega_i) \in \prod_i \mathsf{S}(U_i) : \omega_i|_{U_i \cap U_j} = \omega_j|_{U_i \cap U_j} \right\}.$$

There is a natural restriction map

$$\rho_{\mathcal{U}} : \mathsf{S}(U) \longrightarrow \check{Z}^0(\mathcal{U}, \mathsf{S}).$$

Theorem $T2$. $\rho_{\mathcal{U}}$ is not generally injective and not generally surjective.

Proof. Non-injectivity is AQM-27 Theorem $T2$: on $\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$, an entangled state and a product state have the same one-factor restrictions. Non-surjectivity is AQM-27 Theorem $T3$: the AB and BC Bell marginals agree on B, but no ABC state has both as marginals.

Thus the state-side quantum counterpart is the marginal comparison map. Failure of injectivity measures hidden global correlations. Failure of surjectivity measures incompatible marginal data. These are not cohomology groups yet, because S is a convex set-valued presheaf, not an abelian presheaf.

Stabilizers. Let $L \leq E(U)$ be a global stabilizer label subspace. Define the supported sublabel precosheaf on opens $W \subset U$ by

$$L_{\leq U}(W) = L \cap E(W).$$

The augmentation $\check{C}_0(\mathcal{U}, L_{\leq U}) \to L$ sends a tuple of local stabilizer labels to its sum in L. Its image is

$$\mathrm{Loc}_U(L; \mathcal{U}) = \sum_i (L \cap E(U_i)).$$

The boundary from $L \cap E(U_i \cap U_j)$ maps to zero under this augmentation, because the two copies are the same global label with opposite signs. Hence the augmentation factors through $\check{H}_0(\mathcal{U}, L_{\leq U})$.

Theorem $T3$. The stabilizer descent defect of AQM-38 is the cokernel of this zeroth Cech homology generation map:

$$\mathrm{coker}\big(\check{H}_0(\mathcal{U}, L_{\leq U}) \to L\big) \cong L / \mathrm{Loc}_U(L; \mathcal{U}) = D_U(L; \mathcal{U}).$$

Proof. The image of $\check{C}_0 \to L$ is exactly $\mathrm{Loc}_U(L; \mathcal{U})$. Passing through the quotient $\check{H}_0$ does not change the image, because boundaries already map to zero. The cokernel is therefore the displayed quotient. AQM-38 Theorem $T2$ proves that its vanishing is equivalent to local generation of the stabilizer labels; AQM-39 gives the two-qutrit nonzero-defect example.

Observable nets. For a finite cover, let $\operatorname{colim}_{\mathcal{U}}^{*\mathrm{alg}}$ denote the ordinary colimit of the cover diagram of unital $*$-algebras. The observable net supplies a canonical comparison map

$$\pi_{\mathcal{U}} : \operatorname{colim}_{\mathcal{U}}^{*\mathrm{alg}} \mathcal{A} \longrightarrow \mathcal{A}(U).$$

AQM-26 proves that this ordinary algebra colimit is not the physical gluing: for disjoint pieces it is a free-product gluing, while the Weyl net glues by commuting tensor-local subalgebras inside the ambient algebra. Therefore the observable quantum counterpart is the comparison map $\pi_{\mathcal{U}}$: its image is the additive generated algebra, and its kernel records the extra locality and commutation relations imposed by the physical representation. Without abelianization or another declared linearization, this is not an ordinary Cech cohomology group.

42.6 Conclusion

Classical Cech cohomology is the abelian cover-complex layer. In AQF it splits into label Cech homology, stabilizer cokernels, state marginal maps, and observable-net colimit comparison maps. Only the first two are cohomology or homology groups at this stage.

43 Gaussian and Clifford Dynamics

43.1 Status and Source Anchors

This shard adds the first dynamics layer for the finite residue-qudit arithmetic fields. The new convention is `CONVENTIONS.md` (af). Gross `references/symplectic_qecc/gross_0602001_source/poswig.tex` is now registered locally: lines 340–453 define odd-dimensional Weyl operators, the symplectic form, the Weyl composition law, the trace, and the Heisenberg representation; lines 458–537 define the Clifford group and state its symplectic/metaplectic structure theorem; lines 794–802 give Clifford covariance of Wigner functions; lines 846–903 define stabilizer states from maximally isotropic subspaces; lines 935–976 compute their Wigner functions; lines 202–214 state the discrete Hudson theorem; and lines 1633–1639 state that positivity-preserving unitaries are Clifford. AQM-22, AQM-24, and AQM-40 supply the finite Weyl nets on arithmetic spectra. When $k = \mathbb{F}_{p^r}$, the intrinsic algebraic statements below use the k-symplectic form, while the Gross-Hudson state dictionary is imported after choosing an $\mathbb{F}_p$-basis and viewing $\ell^2(k^m) \cong \ell^2(\mathbb{F}_p^{rm})$.

43.2 Lamport Dependency Skeleton

$$
\begin{aligned}
D1 &: (E,\Omega) \text{ is a finite nondegenerate symplectic } k\text{-space}, \ \operatorname{char} k \neq 2.\\
D2 &: W_E(e)W_E(f) = \chi(\tfrac{1}{2}\Omega(e,f))W_E(e+f), \quad W_E(e)^* = W_E(-e).\\
D3 &: \mathcal{A}(E) = \operatorname{span}_{\mathbb{C}}\{W_E(e) : e \in E\}.\\
L1 &: \{W_E(e) : e \in E\} \text{ is a basis of } \mathcal{A}(E).\\
D4 &: S : E \to F \text{ is symplectic if } \Omega_F(Se,Sf) = \Omega_E(e,f).\\
L2 &: \text{A symplectic morphism from nondegenerate } E \text{ is injective.}\\
T1 &: \alpha_S(W_E(e)) = W_F(Se) \text{ is a unital } *\text{-monomorphism.}\\
T2 &: S \in \mathrm{Sp}(E) \text{ gives a } *\text{-automorphism of } \mathcal{A}(E).\\
T3 &: \text{For odd qudits, } S \text{ has a projective Clifford lift } U_S.\\
D5 &: \alpha_{a,S} = \mathrm{Ad}_{W(a)U_S} \text{ is affine Clifford dynamics.}\\
T4 &: \text{Gross-Hudson identifies pure nonnegative-Wigner states with stabilizer states.}
\end{aligned}
$$

43.3 Half-Weyl Systems D1–L1

Let $k = \mathbb{F}_q$ have odd characteristic and let $\chi : k \to U(1)$ be a nontrivial additive character. Let (E,Ω) be a finite-dimensional nondegenerate symplectic k-vector space. A half-Weyl system for E is a map $W_E : E \to U(\mathcal{H}_E)$ such that

$$W_E(e)W_E(f) = \chi\left(\frac{1}{2}\Omega(e,f)\right)W_E(e+f), \qquad W_E(e)^* = W_E(-e).$$

Here $1/2 \in k$ is defined because $\operatorname{char} k \neq 2$. Set

$$\mathcal{A}(E) = \operatorname{span}_{\mathbb{C}}\{W_E(e) : e \in E\}.$$

In the residue-qudit examples this is the full finite matrix algebra on $\mathcal{H}_E$. Gross's trace calculation for odd qudits, `poswig.tex`, lines 437–453, is the local source for the corresponding orthogonality statement.

Lemma $L1$. The operators $W_E(e)$ are linearly independent. Hence they form a basis of $\mathcal{A}(E)$ by definition of the span.

Proof. In the concrete residue-qudit representation, the trace pairing satisfies

$$\mathrm{Tr}\big(W_E(e)^* W_E(f)\big) = \dim(\mathcal{H}_E)\,\delta_{e,f},$$

as in the finite Weyl trace computation already used in AQM-22 and AQM-38. If $\sum_e c_e W_E(e) = 0$, multiply on the left by $W_E(f)^*$ and take the trace. The displayed orthogonality gives $\dim(\mathcal{H}_E)c_f = 0$. Since f was arbitrary, all c_f vanish.

43.4 Symplectic Morphisms and Algebra Maps D4–T2

Let (E, Ω_E) and (F, Ω_F) be finite nondegenerate symplectic k-spaces. A symplectic morphism is a k-linear map

$$S : E \to F$$

such that

$$\Omega_F(Se, Sf) = \Omega_E(e, f) \qquad (e, f \in E).$$

Lemma $L2$**.** Every symplectic morphism $S : E \to F$ with E nondegenerate is injective.

Proof. If $Se = 0$, then for every $f \in E$,

$$\Omega_E(e, f) = \Omega_F(Se, Sf) = 0.$$

Nondegeneracy of Ω_E gives $e = 0$.

Theorem $T1$**.** For every symplectic morphism $S : E \to F$, the formula

$$\alpha_S(W_E(e)) = W_F(Se)$$

extends uniquely to a unital $*$-monomorphism $\alpha_S : \mathcal{A}(E) \to \mathcal{A}(F)$.

Proof. Lemma $L1$ makes the linear extension unique. It is multiplicative because

$$\begin{aligned}
\alpha_S(W_E(e)W_E(f)) &= \chi\left(\frac{1}{2}\Omega_E(e, f)\right) W_F(S(e+f)) \\
&= \chi\left(\frac{1}{2}\Omega_F(Se, Sf)\right) W_F(Se+Sf) \\
&= W_F(Se)W_F(Sf) = \alpha_S(W_E(e))\alpha_S(W_E(f)).
\end{aligned}$$

It preserves the involution since

$$\alpha_S(W_E(e)^*) = \alpha_S(W_E(-e)) = W_F(-Se) = W_F(Se)^*.$$

It is unital because $S0 = 0$. Finally, S is injective by Lemma $L2$, so the labels Se are distinct; Lemma $L1$ for F then implies that the images $W_F(Se)$ are linearly independent. Thus α_S has zero kernel.

Theorem $T2$**.** If $S : E \to E$ is a symplectic automorphism, then α_S is a $*$-automorphism of $\mathcal{A}(E)$. Moreover

$$\alpha_S \alpha_T = \alpha_{ST}.$$

Proof. The first statement follows from Theorem $T1$ applied to S and S^{-1}. The composition law follows on the Weyl basis:

$$\alpha_S\alpha_T(W_E(e)) = \alpha_S(W_E(Te)) = W_E(STe) = \alpha_{ST}(W_E(e)).$$

Thus a discrete one-parameter Gaussian or quasi-free dynamics is simply a homomorphism $t \mapsto S_t \in \mathrm{Sp}(E)$, with $\alpha_t = \alpha_{S_t}$.

43.5 Clifford Implementations T3–D5

For odd qudits, Gross's Clifford structure theorem `poswig.tex`, lines 458–537, states that every symplectic S has a unitary $\mu(S)$, unique projectively, such that

$$\mu(S)W_E(e)\mu(S)^* = W_E(Se),$$

and that $S \mapsto \mu(S)$ is a projective representation. Therefore the observable automorphism α_S is implemented by a Clifford unitary. This is the unitary version of Theorem $T2$.

Allowing a displacement $a \in E$, define

$$U_{a,S} = W_E(a)\mu(S).$$

Then

$$\begin{aligned} U_{a,S}W_E(e)U_{a,S}^* &= W_E(a)W_E(Se)W_E(-a) \\ &= \chi(\Omega_E(a, Se))W_E(Se). \end{aligned}$$

The second equality uses the half-Weyl law twice:

$$\frac{1}{2}\Omega(a, Se) + \frac{1}{2}\Omega(a + Se, -a) = \Omega(a, Se).$$

These affine Clifford maps are the finite odd-qudit analogue of affine Gaussian phase-space dynamics. They move Wigner functions by affine symplectic transformations, as sourced by Gross's Clifford covariance theorem, `poswig.tex`, lines 794–802.

43.6 Local Nets and Support

For a finite open U, the arithmetic-field label space $E(U)$ is an orthogonal direct sum of local residue label spaces. The inclusion $E(U) \hookrightarrow E(V)$ for $U \subset V$, given by extension by zero, is a symplectic morphism. Theorem $T1$ therefore recovers the open-local observable inclusion

$$\mathcal{A}(U) \hookrightarrow \mathcal{A}(V)$$

from AQM-24 as a Gaussian morphism.

A global symplectic automorphism $S : E(X) \to E(X)$ gives a global dynamics on $\mathcal{A}(X)$. It is a dynamics of the open-local net only if it respects the support structure being used. For example, if $S(E(U)) = E(U)$ for each open U under consideration, then every $\mathcal{A}(U)$ is invariant. If instead $S(E(U)) = E(\varphi(U))$ for a declared open-set permutation φ, then S gives a covariant net dynamics $\mathcal{A}(U) \to \mathcal{A}(\varphi(U))$. Otherwise it is a nonlocal global automorphism, not an automorphism of the chosen local net.

43.7 Gaussian States and Gross-Hudson T4

Gross defines stabilizer states from maximally isotropic subspaces and Weyl eigenvalue equations, `poswig.tex`, lines 846–903. He proves that their Wigner functions are normalized indicators of affine Lagrangian subspaces, lines 935–976. His discrete Hudson theorem, lines 202–214, says that for odd d, a pure state in $L^2(\mathbb{Z}_d^n)$ has nonnegative Wigner function if and only if it is a stabilizer state; for full support, this is equivalent to a quadratic-phase form

$$\psi(q) \propto \exp\left(\frac{2\pi i}{d}(q\theta q + xq)\right), \qquad \theta^T = \theta.$$

In the present report, this fixes the finite odd-qudit meaning of "Gaussian state": pure Gaussian states are stabilizer states, equivalently phased Lagrangian label data. Clifford dynamics sends them to Gaussian states because it sends Lagrangian subspaces to Lagrangian subspaces and transports the phase character by conjugation. Gross's positivity-preservation theorem, lines 1633–1639, gives the converse in Wigner language: a unitary preserving nonnegative Wigner states is Clifford.

43.8 Conclusion

Dynamics in the finite odd residue-qudit AQF layer is therefore label-first. Symplectic morphisms give the functorial algebra maps; symplectic automorphisms give Gaussian/quasi-free automorphisms; Clifford unitaries are their projective Hilbert-space implementations; and Gross-Hudson identifies the pure finite Gaussian states with stabilizer states. Even characteristic and mixed-residue dynamics remain separate convention problems.

44 Finite-Field Scales and Continuum Limits

44.1 Status and Source Anchors

This shard adds finite-field extension rank as the first arithmetic "scale" parameter. The closed-point Weyl net is AQM-28 and the dynamics language is AQM-42. The scheme facts use the Stacks Project snapshot: `references/algebraic_geometry/stacks_project_algebra.tex`, lines 250–274, 2972–2983, 3104–3117, 3310–3350, 6520–6665, and 45278–45284. Frobenius is sourced from `references/algebraic_geometry/stacks_project_varieties.tex`, lines 7005–7112. The finite-field trace formula also appears in `references/heisenberg_weil/sidana_kashyap_2202_00248_source/EAQECCs_over_rings_v15.tex`, lines 789–801.

44.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k=\mathbb{F}_q,\ k_r=\mathbb{F}_{q^r}\subset \bar{k},\ k_r\subset k_s \iff r\mid s. \\
D2 & : & X_r = X_0\times_{\operatorname{Spec} k}\operatorname{Spec} k_r,\ \pi_{s,r}: X_s\to X_r. \\
D3 & : & \mathcal{A}_r(U) \text{ is the AQM-28 closed-point Weyl net on } X_r. \\
L1 & : & \text{A closed point } x\in X_r \text{ splits over } k_s \text{ through } B_x=\kappa(x)\otimes_{k_r} k_s. \\
L2 & : & \text{The naive diagonal label map multiplies the trace commutator by } n=s/r. \\
L3 & : & \text{A trace-one element } c_x\in B_x \text{ preserves the non-half Weyl multiplier.} \\
T1 & : & \text{Trace-normalized local embeddings induce monomorphisms of open-local Weyl nets.} \\
T2 & : & \text{Prime-to-}p\text{ scales have canonical coherent embeddings.} \\
T3 & : & X_0=\operatorname{Spec}\mathbb{F}_3 \text{ gives } \varinjlim_{3\nmid r} M_{3^r}(\mathbb{C}). \\
T4 & : & \text{Frobenius is an } \mathbb{F}_3\text{-linear trace-Weyl action } W(q,p)\mapsto W(q^3,p^3).
\end{array}$$

44.3 Scales and Closed-Point Nets D1–L1

Fix an algebraic closure $\bar{k}$ of $k=\mathbb{F}_q$, and let $k_r=\mathbb{F}_{q^r}$ be the unique subfield of $\bar{k}$ with q^r elements. Then $k_r\subset k_s$ exactly when $r\mid s$. For an affine k-scheme $X_0=\operatorname{Spec} R$, define the rank-$r$ scale

$$X_r = X_0\times_{\operatorname{Spec} k}\operatorname{Spec} k_r = \operatorname{Spec}(R\otimes_k k_r).$$

If $r\mid s$, base change along $k_r\subset k_s$ gives $\pi_{s,r}: X_s\to X_r$.

For an open $U\subset X_r$, $\mathcal{A}_r(U)$ denotes the AQM-28 closed-point Weyl algebra:

$$\mathcal{A}_r(U) = \varinjlim_{S\Subset U\cap|X_r|_{\mathrm{cl}}} \bigotimes_{x\in S} \operatorname{End}_{\mathbb{C}} \ell^2(\kappa(x)).$$

Thus a scale change should compare $\mathcal{A}_r(U)$ with $\mathcal{A}_s(\pi_{s,r}^{-1}U)$.

Let $x\in|X_r|_{\mathrm{cl}}$, put $K=\kappa(x)$, and write $B_x=K\otimes_{k_r} k_s$. Since finite fields are separable, $B_x\cong\prod_{y\mapsto x}\kappa(y)$, where y ranges over the closed points of X_s above x. This is the finite algebraic form of site splitting at finer scale.

44.4 The Trace Obstruction L2

The one-site character at x is

$$\psi_x(a) = \exp\!\left(\frac{2\pi i}{p}\operatorname{Tr}_{K/\mathbb{F}_p}(a)\right).$$

The naive diagonal map sends $(q, p_0) \in K^2$ to $\Delta_x(q, p_0) = ((q_y, p_{0,y}))_{y \mapsto x}$, where $q_y, p_{0,y}$ are the images in $\kappa(y)$. Let $a = p_0 q' - q p_0'$. The fine-scale commutator phase of the diagonal images is

$$\prod_{y \mapsto x} \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{\kappa(y)/\mathbb{F}_p}(a_y)\right).$$

The exponent is

$$\operatorname{Tr}_{B_x/\mathbb{F}_p}(a \otimes 1) = \operatorname{Tr}_{K/\mathbb{F}_p}\big(\operatorname{Tr}_{B_x/K}(a \otimes 1)\big).$$

As a K-algebra, B_x has dimension $n = s/r$, and $a \otimes 1$ acts on it as scalar multiplication by a. Hence

$$\operatorname{Tr}_{B_x/K}(a \otimes 1) = na,$$

so the fine phase is the coarse phase raised to the n-th power:

$$\prod_{y \mapsto x} \psi_y(a_y) = \psi_x(a)^n.$$

Therefore the naive diagonal label map is Weyl-compatible for all labels only when $n \equiv 1 \pmod p$. The geometry is canonical, but the Weyl phases are not automatically compatible.

44.5 Trace-Normalized Embeddings L3–T1

Choose $c_x = (c_y)_{y \mapsto x} \in B_x$ with $\operatorname{Tr}_{B_x/K}(c_x) = 1$. Such an element exists: for a finite separable field extension the trace pairing is nondegenerate, so the trace map is a nonzero K-linear map to the one-dimensional K-space K, hence is surjective; products are handled componentwise.

Define

$$\tau_{s,r,x}(q, p_0) = ((q_y, c_y p_{0,y}))_{y \mapsto x}.$$

The AQM-28 closed-point Weyl law is the non-half product law

$$W(q, p_0) W(q', p_0') = \psi_x(p_0 q') W(q + q', p_0 + p_0'),$$

so multiplier preservation must be checked before claiming an algebra map. For $e = (q, p_0)$, $f = (q', p_0')$, trace transitivity gives

$$\sum_y \operatorname{Tr}_{\kappa(y)/\mathbb{F}_p}(c_y p_{0,y} q_y') = \operatorname{Tr}_{K/\mathbb{F}_p}\big(\operatorname{Tr}_{B_x/K}(c_x(p_0 q' \otimes 1))\big) = \operatorname{Tr}_{K/\mathbb{F}_p}(p_0 q'),$$

Thus the actual Weyl multiplier is preserved; the adjoint scalar follows by setting $q' = q$. For $a = p_0 q' - q p_0'$, the same calculation gives

$$\sum_y \operatorname{Tr}_{\kappa(y)/\mathbb{F}_p}\big(c_y(p_{0,y} q_y' - q_y p_{0,y}')\big) = \operatorname{Tr}_{K/\mathbb{F}_p}(a),$$

Thus $\tau_{s,r,x}$ also preserves the character-valued Weyl commutator. It is injective because $c_x \neq 0$, so some component c_y is nonzero.

Taking the direct sum of $\tau_{s,r,x}$ over the finite support of a label gives

$$\tau_{s,r,U} : E_r(U) \to E_s(\pi_{s,r}^{-1} U).$$

The Weyl relations are preserved on every finite support, so

$$\alpha_{s,r,U}(W_r(e)) = W_s(\tau_{s,r,U} e)$$

extends to a unital $*$-monomorphism

$$\alpha_{s,r,U} : \mathcal{A}_r(U) \hookrightarrow \mathcal{A}_s(\pi_{s,r}^{-1} U).$$

For $U \subset V$, these maps commute with the open-local inclusions because both constructions are defined pointwise on closed supports.

44.6 Canonical and Noncanonical Continuum Systems T2

If $p \nmid n = s/r$, then

$$c_x = n^{-1} \cdot 1_{B_x}$$

is canonical and has trace 1, because $\mathrm{Tr}_{B_x/K}(1_{B_x}) = n$. These choices are coherent: for $r \mid s \mid t$, the normalizing factors multiply as

$$(t/r)^{-1} = (s/r)^{-1}(t/s)^{-1}.$$

Hence the scales r with $p \nmid r$, ordered by divisibility, form a canonical directed subsystem of Weyl nets. Its algebraic continuum net is the filtered colimit of these nets.

For the full tower of all r, the current trace-character convention requires extra data. If $p \mid n$, then n^{-1} does not exist. A trace-one element still exists, but it is not canonical in the above data. The report therefore does not call the full all-rank tower canonical until a coherent trace-one system, or an alternative compatible-character convention, has been fixed.

44.7 The Point over $\mathbb{F}_3$ T3

Let

$$X_0 = \operatorname{Spec} \mathbb{F}_3, \qquad k_r = \mathbb{F}_{3^r}.$$

Then $X_r = \operatorname{Spec} k_r$ has one closed point and

$$E_r = k_r^2, \qquad \mathcal{H}_r = \ell^2(k_r), \qquad \mathcal{A}_r \cong M_{3^r}(\mathbb{C}).$$

The character is

$$\psi_r(a) = \exp\!\left(\frac{2\pi i}{3}\,\mathrm{Tr}_{k_r/\mathbb{F}_3}(a)\right).$$

For $r \mid s$, put $n = s/r$. The naive inclusion $(q,p) \mapsto (q,p)$ gives

$$\psi_s(pq' - qp') = \psi_r(pq' - qp')^n.$$

Already $r = 1, s = 2$ fails: $\mathrm{Tr}_{\mathbb{F}_9/\mathbb{F}_3}(1) = 2$, so the phase of 1 becomes $\exp(4\pi i/3)$, not $\exp(2\pi i/3)$. The canonical corrected map is

$$S_{s,r}(q,p) = (q, n^{-1}p)$$

whenever $3 \nmid n$. Consequently the canonical prime-to-3 continuum algebra of the arithmetic point is

$$\mathcal{A}_\infty^{(3')} = \varinjlim_{\substack{r \geq 1 \\ 3 \nmid r}} M_{3^r}(\mathbb{C}),$$

with transition maps $W_r(q,p) \mapsto W_s(q, n^{-1}p)$. This is an algebraic colimit: every element is represented at a finite extension rank. No Hilbert space $\ell^2(\bar{\mathbb{F}}_3)$ or C*-completion is part of the definition yet.

For the first missing full-tower step, $r = 1, s = 3$, the naive trace is zero on $\mathbb{F}_3$ because $\mathrm{Tr}_{\mathbb{F}_{27}/\mathbb{F}_3}(1) = 3 = 0$. A trace-one element exists but is a choice. For any chosen $\theta \in \mathbb{F}_{27}$ with $\mathrm{Tr}(\theta) = 1$, $(q,p) \mapsto (q, \theta p)$ is a valid embedding $\mathcal{A}_1 \hookrightarrow \mathcal{A}_3$. Choosing θ is extra structure, not a canonical consequence of $k \subset k_3$.

44.8 Frobenius on the Point Net T4

At scale r, define
$$F_r : E_r \to E_r, \qquad F_r(q,p) = (q^3, p^3).$$
The trace is Frobenius-invariant:
$$\mathrm{Tr}_{k_r/\mathbb{F}_3}(a^3) = \sum_{i=0}^{r-1} a^{3^{i+1}} = \sum_{i=0}^{r-1} a^{3^i} = \mathrm{Tr}_{k_r/\mathbb{F}_3}(a).$$

The map F_r is $\mathbb{F}_3$-linear and k_r-semilinear, not a k_r-linear symplectic automorphism in the sense of AQM-42 when $r > 1$. Since $\mathrm{Tr}((pq')^3) = \mathrm{Tr}(pq')$, it preserves the non-half multiplier and commutator, so it induces
$$\Phi_r(W_r(q,p)) = W_r(q^3, p^3).$$
This is an $\mathbb{F}_3$-linear trace-Weyl algebra automorphism of order r, Clifford after choosing an $\mathbb{F}_3$-basis, not a k_r-linear Gaussian map. For $k_2 = \mathbb{F}_3[u]/(u^2+1)$, $u^3 = -u$, so
$$\Phi_2(W(a+bu, c+du)) = W(a-bu, c-du).$$

The prime-to-3 embeddings are Frobenius-equivariant. Indeed
$$S_{s,r}F_r(q,p) = (q^3, n^{-1}p^3) = F_s S_{s,r}(q,p),$$
because $n^{-1} \in \mathbb{F}_3$ is fixed by Frobenius. Thus Frobenius acts on the canonical prime-to-3 continuum algebra $\mathcal{A}_\infty^{(3')}$. In contrast, a noncanonical trace-one choice for a 3-divisible step is generally moved by Frobenius, so full-tower Frobenius equivariance is additional structure.

44.9 Conclusion

Finite-field extension rank gives a precise arithmetic scale system. Base change of schemes is canonical, and closed points split by the finite etale algebra $\kappa(x) \otimes_{k_r} k_s$. The Weyl net sees an extra phase constraint: absolute trace characters force trace-normalized embeddings. Therefore canonical continuum limits exist immediately on the prime-to-p subtower; the full tower exists only after coherent trace-one or compatible character data are added. For the arithmetic point over $\mathbb{F}_3$, the canonical continuum object currently justified by the report is the prime-to-3 algebraic colimit of the matrix algebras $M_{3^r}(\mathbb{C})$, with Frobenius acting by the compatible $\mathbb{F}_3$-linear trace-Weyl automorphisms $W(q,p) \mapsto W(q^3,p^3)$, not by k_r-linear Gaussian maps.

45 Trace Gauges for Full Continuum Towers

45.1 Status and Source Anchors

AQM-43 proved that pairwise trace-one elements repair the Weyl phase obstruction for finite-field scale embeddings. This shard proves that the pairwise choices can be made coherently for the whole divisibility tower, and records exactly what is gauge-like about the arbitrariness. The source anchors are Stacks Algebra, `references/algebraic_geometry/stacks_project_algebra.tex`, lines 45278–45284 for nondegenerate separable trace pairing, and Stacks Varieties, `references/algebraic_geometry/stacks_project_varieties.tex`, lines 7005–7112 for Frobenius. All coherence statements below are local derivations from those facts.

45.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k_r = \mathbb{F}_{q^r} \subset \bar{k}, \quad r \mid s \Rightarrow k_r \subset k_s. \\
D2 & : & \text{A trace density is } a_r \in k_r \text{ with } \mathrm{Tr}_{k_s/k_r}(a_s) = a_r. \\
L1 & : & \mathrm{Tr}_{k_s/k_r} : k_s \to k_r \text{ is surjective.} \\
T1 & : & \text{A coherent trace-density system } (a_r)_r \text{ with } a_1 = 1 \text{ exists.} \\
D3 & : & c_{s,r} = a_s/a_r. \\
L2 & : & \mathrm{Tr}_{k_s/k_r}(c_{s,r}) = 1, \quad c_{t,r} = c_{t,s}c_{s,r}. \\
T2 & : & (c_{s,r}) \text{ gives coherent full-tower Weyl-net embeddings.} \\
T3 & : & \text{The choice of } (a_r) \text{ is trace gauge data.} \\
T4 & : & p \mid s/r \text{ forbids a relative-Frobenius-fixed trace-one scalar.}
\end{array}$$

45.3 Coherent Trace Densities D1–T1

Fix $k = \mathbb{F}_q$ and $k_r = \mathbb{F}_{q^r} \subset \bar{k}$. A coherent trace-density system is a family $a_r \in k_r$ such that $a_1 = 1$ and, whenever $r \mid s$,

$$\mathrm{Tr}_{k_s/k_r}(a_s) = a_r.$$

This condition forces $a_r \neq 0$, because

$$\mathrm{Tr}_{k_r/k}(a_r) = a_1 = 1.$$

Lemma $L1$. For $r \mid s$, the trace map

$$\mathrm{Tr}_{k_s/k_r} : k_s \to k_r$$

is surjective.

Proof. Finite fields are perfect, hence k_s/k_r is separable. The sourced separable trace pairing is nondegenerate:

$$(x, y) \longmapsto \mathrm{Tr}_{k_s/k_r}(xy).$$

If the trace map itself were zero, then $\mathrm{Tr}_{k_s/k_r}(1 \cdot y) = 0$ for every y, contradicting nondegeneracy with $x = 1$. Thus the trace is a nonzero k_r-linear map from k_s to the one-dimensional k_r-space k_r, so it is surjective.

Theorem $T1$. Coherent trace-density systems exist.

Proof. Let $N_m = m!$. Start with $a_{N_1} = a_1 = 1$. Having chosen a_{N_m}, Lemma $L1$ gives $a_{N_{m+1}} \in k_{N_{m+1}}$ with

$$\mathrm{Tr}_{k_{N_{m+1}}/k_{N_m}}(a_{N_{m+1}}) = a_{N_m}.$$

For an arbitrary $r \geq 1$, choose m with $r \mid N_m$, and define

$$a_r = \mathrm{Tr}_{k_{N_m}/k_r}(a_{N_m}).$$

This is independent of m: if $n \geq m$, then transitivity of trace and the recursive construction give

$$\mathrm{Tr}_{k_{N_n}/k_r}(a_{N_n}) = \mathrm{Tr}_{k_{N_m}/k_r}\big(\mathrm{Tr}_{k_{N_n}/k_{N_m}}(a_{N_n})\big) = \mathrm{Tr}_{k_{N_m}/k_r}(a_{N_m}).$$

If $r \mid s$, choose m with $s \mid N_m$. Then

$$\mathrm{Tr}_{k_s/k_r}(a_s) = \mathrm{Tr}_{k_s/k_r}\big(\mathrm{Tr}_{k_{N_m}/k_s}(a_{N_m})\big) = \mathrm{Tr}_{k_{N_m}/k_r}(a_{N_m}) = a_r.$$

Thus $(a_r)_r$ is coherent.

45.4 Coherent Weyl Embeddings D3–T2

Given a coherent trace-density system, define

$$c_{s,r} = a_s/a_r \in k_s$$

for $r \mid s$, where a_r is embedded in k_s.

Lemma $L2$.

$$\mathrm{Tr}_{k_s/k_r}(c_{s,r}) = 1, \qquad c_{t,r} = c_{t,s}c_{s,r} \quad (r \mid s \mid t).$$

Proof. Since $a_r \in k_r^\times$,

$$\mathrm{Tr}_{k_s/k_r}(a_s/a_r) = a_r^{-1}\,\mathrm{Tr}_{k_s/k_r}(a_s) = 1.$$

The multiplicative identity is immediate:

$$c_{t,s}c_{s,r} = (a_t/a_s)(a_s/a_r) = a_t/a_r = c_{t,r}.$$

Let $X_0 = \operatorname{Spec} R$ be affine over k, and put $X_r = X_0 \times_k k_r$. If $x \in |X_r|_{\mathrm{cl}}$ has residue field $K = \kappa(x)$, then the points of X_s above x are encoded by

$$B_{s,r,x} = K \otimes_{k_r} k_s.$$

Use $1 \otimes c_{s,r} \in B_{s,r,x}$ as the trace-normalizing scalar. Its trace to K is 1: choose a k_r-basis of k_s; the multiplication matrix of $c_{s,r}$ has trace $\mathrm{Tr}_{k_s/k_r}(c_{s,r}) = 1$, and base change of that matrix from k_r to K leaves the trace equal to 1.

Theorem $T2$. The coherent trace-density system gives coherent monomorphisms of all open-local Weyl nets

$$\alpha_{s,r,U} : \mathcal{A}_r(U) \hookrightarrow \mathcal{A}_s(\pi_{s,r}^{-1}U), \qquad r \mid s.$$

Proof. AQM-43 proves that any trace-one normalizing scalar gives a Weyl-compatible monomorphism at one transition. The preceding paragraph gives trace-one scalars at every closed point, and Lemma $L2$ gives

$$(1 \otimes c_{t,s})(1 \otimes c_{s,r}) = 1 \otimes c_{t,r}.$$

Therefore the label embeddings compose strictly, hence the induced Weyl-net monomorphisms compose strictly.

45.5 Gauge Meaning and Frobenius T3–T4

The family (a_r), equivalently $(c_{s,r})$, is called a trace gauge. It is not extra geometry of X_0; it is extra presentation data needed to compare Weyl phases across scales. The prime-to-p subsystem has the canonical gauge $c_{s,r} = (s/r)^{-1}$. The full tower has many gauges, by Theorem $T1$, but no canonical one is singled out by the finite-field embeddings alone.

Frobenius acts on gauges. If $c_{s,r}$ has trace 1, then

$$\mathrm{Tr}_{k_s/k_r}(c_{s,r}^q) = \mathrm{Tr}_{k_s/k_r}(c_{s,r})^q = 1.$$

Thus Frobenius sends one trace gauge to another. For $p \mid s/r$, however, there is no trace-one scalar fixed by the relative Frobenius group. If $c \in k_s$ is fixed by $\mathrm{Gal}(k_s/k_r)$, then $c \in k_r$, and

$$\mathrm{Tr}_{k_s/k_r}(c) = (s/r)c = 0$$

in characteristic p. This cannot equal 1.

For $k = \mathbb{F}_3$, the first such case is $\mathbb{F}_3 \subset \mathbb{F}_{27}$. A trace-one c exists, but it cannot lie in $\mathbb{F}_3$. Hence the full tower supports Frobenius only gauge-covariantly unless an additional Frobenius-compatible gauge convention is supplied. This is the precise sense in which the arbitrariness is gauge freedom rather than a mathematical obstruction to the continuum limit.

46 The Arithmetic Quantum Field on $\operatorname{Spec}\mathbb{Z}$

46.1 Status and Source Anchors

This shard extends `CONVENTIONS.md` (u)–(y) and (ai). Source-local scheme facts are from `references/algebraic_geometry/stacks_project_algebra.tex`: lines 120–158, 250–274, 2972–2983, 3104–3160, 3251–3298, 3310–3342, and 4657–4703 cover prime ideals, residue fields, Spec, Zariski opens, functoriality, quasi-compact opens, and generic points. AQM-28 is the finite-field finite-support pattern; this is the separate convention (ai) $\operatorname{Spec}\mathbb{Z}$ closed-prime extension.

46.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & X = \operatorname{Spec}\mathbb{Z}, \quad \eta = (0), \quad x_\ell = (\ell). \\
L1 & : & \operatorname{Prime}(\mathbb{Z}) = \{0\} \cup \{(\ell) : \ell \text{ prime}\}. \\
L2 & : & \kappa(\eta) = \mathbb{Q}, \quad \kappa(x_\ell) = \mathbb{F}_\ell. \\
L3 & : & \text{Nonempty opens are } \{\eta\} \cup \{x_\ell : \ell \notin S\} \text{ with } S \text{ finite.} \\
D2 & : & E_{\mathrm{cl}}(U) = \bigoplus_{x_\ell \in U} \mathbb{F}_\ell^2, \quad \mathcal{A}_{\mathrm{cl}}(U) = \varinjlim_{S \Subset U_{\mathrm{cl}}} \bigotimes_{\ell \in S} M_\ell(\mathbb{C}). \\
T1 & : & U \mapsto \mathcal{A}_{\mathrm{cl}}(U) \text{ is the closed-prime open-local AQF.} \\
D3 & : & \mathcal{H}^{\Omega}_{\mathrm{cl}} = \ell^2(\bigoplus_\ell \mathbb{F}_\ell). \\
L4 & : & \eta \text{ is a } \mathbb{Q}\text{-Weyl sector, not a finite qudit.} \\
T2 & : & \operatorname{End}_{\mathrm{Sch}}(\operatorname{Spec}\mathbb{Z}) = \{\mathrm{id}\}. \\
D4 & : & \alpha_n = \prod_\ell^{\mathrm{loc}} \operatorname{Ad} T_\ell(n \bmod \ell). \\
T3 & : & \alpha : \mathbb{Z} \curvearrowright \mathcal{A}_{\mathrm{cl}}(X) \text{ is faithful and extends to } \prod_\ell \mathbb{F}_\ell. \\
T4 & : & \alpha_n,\ n \neq 0, \text{ is not implemented in the } |0\rangle \text{ sector.}
\end{array}$$

46.3 The Point Set and Topology D1–L3

Every ideal of $\mathbb{Z}$ has the form (n). If (n) is prime, then $\mathbb{Z}/(n)$ is a domain by the source-local prime-ideal criterion. This happens exactly for $n = 0$ or for $n = \pm\ell$ with ℓ a rational prime. Hence

$$X = \operatorname{Spec}\mathbb{Z} = \{\eta\} \cup \{x_\ell : \ell \text{ prime}\}, \qquad \eta = (0), \quad x_\ell = (\ell).$$

The point η is generic because $\overline{\{\eta\}} = V(0) = X$, using the Stacks closure formula $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$.

Lemma $L2$. The residue fields are

$$\kappa(\eta) = \mathbb{Q}, \qquad \kappa(x_\ell) = \mathbb{F}_\ell.$$

Proof. By the Stacks residue-field definition, $\kappa(\mathfrak{p})$ is the fraction field of $\mathbb{Z}/\mathfrak{p}$. For $\mathfrak{p} = (0)$ this is $\operatorname{Frac}(\mathbb{Z}) = \mathbb{Q}$. For $\mathfrak{p} = (\ell)$ this is $\operatorname{Frac}(\mathbb{Z}/\ell\mathbb{Z}) = \mathbb{F}_\ell$.

Lemma $L3$. The nonempty Zariski opens of X are exactly

$$U_S = X \setminus \{x_\ell : \ell \in S\} = \{\eta\} \cup \{x_\ell : \ell \notin S\}, \qquad S \Subset \{\text{rational primes}\}.$$

Proof. Closed subsets are $V(n)$. If $n = 0$, then $V(n) = X$. If $n = \pm 1$, then $V(n) = \emptyset$. If $n \neq 0, \pm 1$, then $x_\ell \in V(n)$ exactly when $\ell \mid n$, and $\eta \notin V(n)$. Thus every proper nonempty closed subset is a finite set of closed prime points. Taking complements gives the displayed opens.

46.4 Closed-Prime Qudit Field D2–T1

For a prime ℓ, attach

$$E_\ell = \mathbb{F}_\ell^2, \qquad \mathcal{H}_\ell = \ell^2(\mathbb{F}_\ell), \qquad \mathcal{A}_\ell = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_\ell) \cong M_\ell(\mathbb{C}).$$

Write a label $e_\ell = (q_\ell, r_\ell)$. The local Weyl operators are

$$T_\ell(q)|s\rangle = |s+q\rangle, \qquad R_\ell(r)|s\rangle = \exp(2\pi i\, rs/\ell)|s\rangle, \qquad W_\ell(q,r) = T_\ell(q)R_\ell(r).$$

For a finite set S of primes,

$$E(S) = \bigoplus_{\ell\in S} \mathbb{F}_\ell^2, \quad \mathcal{H}(S) = \bigotimes_{\ell\in S} \mathcal{H}_\ell, \quad \mathcal{A}(S) = \bigotimes_{\ell\in S} \mathcal{A}_\ell.$$

The Weyl commutator is

$$W_S(e)W_S(f) = \prod_{\ell\in S} \exp(2\pi i(r_\ell q'_\ell - q_\ell r'_\ell)/\ell)\, W_S(f)W_S(e),$$

by the one-site calculation of AQM-28.

For an open U, let $U_{\mathrm{cl}} = U \cap \{x_\ell\}$, set $E_{\mathrm{cl}}(U) = \bigoplus_{x_\ell\in U_{\mathrm{cl}}} \mathbb{F}_\ell^2$ with finite support, and define

$$\mathcal{A}_{\mathrm{cl}}(U) = \varinjlim_{S\Subset U_{\mathrm{cl}}} \mathcal{A}(S), \qquad a \mapsto a\otimes \mathbf{1} \text{ for } S \subset S'.$$

Theorem *T*1. The assignment $U \mapsto \mathcal{A}_{\mathrm{cl}}(U)$ is the closed-prime open-local arithmetic quantum field on $\mathrm{Spec}\,\mathbb{Z}$. In particular,

$$\mathcal{A}_{\mathrm{cl}}(X) = \varinjlim_{S\Subset\{\ell\}} \bigotimes_{\ell\in S} M_\ell(\mathbb{C}),$$

the algebraic infinite tensor product of one ℓ-level qudit for each rational prime ℓ.

Proof. This repeats AQM-28's finite-support construction on finite-residue closed prime points, as a separate extension beyond AQM-28's finite-field scope. If $U \subset V$, then $U_{\mathrm{cl}} \subset V_{\mathrm{cl}}$, so extension by identity gives a unital injective $*$-homomorphism $\mathcal{A}_{\mathrm{cl}}(U) \hookrightarrow \mathcal{A}_{\mathrm{cl}}(V)$. Disjoint finite supports commute because their Weyl commutator has no nontrivial local factor. If $U = \bigcup_i U_i$, every label has finite support, so each support point may be assigned to some U_i; the same additive proof as AQM-28 shows that $\mathcal{A}_{\mathrm{cl}}(U)$ is generated by the images of the $\mathcal{A}_{\mathrm{cl}}(U_i)$.

46.5 Incomplete Tensor Representation D3

The algebra $\mathcal{A}_{\mathrm{cl}}(X)$ is representation-independent. To get a Hilbert space, choose reference vectors $\Omega_\ell = |0\rangle \in \mathcal{H}_\ell$. Let

$$\Gamma_{\mathrm{fin}} = \bigoplus_\ell \mathbb{F}_\ell = \{s = (s_\ell) : s_\ell = 0 \text{ for all but finitely many } \ell\}.$$

Define

$$\mathcal{H}_{\mathrm{cl}}^{\Omega} = \ell^2(\Gamma_{\mathrm{fin}}), \qquad |s\rangle = \bigotimes_\ell |s_\ell\rangle$$

with the understood convention that all but finitely many tensor factors are $|0\rangle$. A finite-support algebra $\mathcal{A}(S)$ acts on the coordinates in S and fixes the others. These actions are compatible, so $\mathcal{H}_{\mathrm{cl}}^{\Omega}$ represents $\mathcal{A}_{\mathrm{cl}}(X)$. Changing the reference product state changes this Hilbert-space realization; it is not extra geometry of $\mathrm{Spec}\,\mathbb{Z}$.

46.6 The Generic Point L4

The generic point has residue field $\mathbb{Q}$, so it is not a finite qudit. If one includes it, the separate rational Weyl sector is

$$E_\eta = \mathbb{Q}^2, \qquad \mathcal{H}_\eta = \ell^2(\mathbb{Q}), \qquad \chi_{\mathbb{Q}}(a) = \exp(2\pi i a).$$

With

$$T_{\mathbb{Q}}(q)|s\rangle = |s+q\rangle, \qquad R_{\mathbb{Q}}(r)|s\rangle = \chi_{\mathbb{Q}}(rs)|s\rangle,$$

the same calculation gives $R_{\mathbb{Q}}(r)T_{\mathbb{Q}}(q) = \chi_{\mathbb{Q}}(rq)T_{\mathbb{Q}}(q)R_{\mathbb{Q}}(r)$. The pairing is nondegenerate: if $q \neq 0$, choose $r = (2q)^{-1}$; if $r \neq 0$, choose $q = (2r)^{-1}$.

Because every nonempty open contains η, set $\mathcal{A}_\eta = \mathrm{span}_{\mathbb{C}}\{T_{\mathbb{Q}}(q)R_{\mathbb{Q}}(r)\}$ as an algebraic Weyl span; no C*-completion, adelic replacement, or Stone–von Neumann uniqueness is claimed here. The generic-sector local rule is

$$\mathcal{A}_{\mathrm{gen}}(U) = \begin{cases} \mathbb{C}, & U = \emptyset, \\ \mathcal{A}_\eta \otimes \mathcal{A}_{\mathrm{cl}}(U), & U \neq \emptyset. \end{cases}$$

Thus the generic sector is present in every nonempty local algebra. It is a global generic-fibre layer, not another prime-local qudit.

46.7 No Scheme Translation $z \mapsto z+1$ on X T2

Theorem $T2$. Every scheme endomorphism of $\operatorname{Spec}\mathbb{Z}$ is the identity. Therefore there is no nontrivial scheme-theoretic translation $z \mapsto z+1$ on $\operatorname{Spec}\mathbb{Z}$.

Proof. By affine functoriality, scheme endomorphisms of $\operatorname{Spec}\mathbb{Z}$ correspond contravariantly to unital ring endomorphisms $\varphi : \mathbb{Z} \to \mathbb{Z}$. Such a φ satisfies $\varphi(1) = 1$, hence $\varphi(n) = n\varphi(1) = n$ for all $n \in \mathbb{Z}$. Thus $\varphi = \mathrm{id}$, and the induced scheme map is the identity.

Consequently the expression $z \mapsto z+1$ belongs to $\mathbb{A}^1_{\mathbb{Z}} = \operatorname{Spec}\mathbb{Z}[z]$, where z is a coordinate. It does not define a self-map of the base scheme $\operatorname{Spec}\mathbb{Z}$.

46.8 Internal Integer Translation D4–T4

There is nevertheless a canonical internal residue translation on the closed-prime qudit algebra. For $n \in \mathbb{Z}$, define at prime ℓ

$$U_\ell(n) = T_\ell(n \bmod \ell).$$

For finite support S, set

$$\alpha_n^S(a) = \left(\bigotimes_{\ell \in S} U_\ell(n)\right) a \left(\bigotimes_{\ell \in S} U_\ell(n)\right)^*.$$

These maps are compatible with $a \mapsto a \otimes \mathbf{1}$, hence define an automorphism α_n of $\mathcal{A}_{\mathrm{cl}}(X)$. On one Weyl operator,

$$\alpha_n(W_\ell(q,r)) = \exp(-2\pi i\, nr/\ell)\, W_\ell(q,r).$$

Theorem $T3$**.** The maps $n \mapsto \alpha_n$ define a faithful action of $\mathbb{Z}$ on $\mathcal{A}_{\rm cl}(X)$. On $\mathcal{A}(S)$ the action factors through $\mathbb{Z}/N_S\mathbb{Z}$, where $N_S = \prod_{\ell \in S} \ell$. Therefore it extends to the squarefree compact residue product $\prod_\ell \mathbb{F}_\ell$. The induced $\widehat{\mathbb{Z}}$-action factors through coordinatewise reduction $\widehat{\mathbb{Z}} \to \prod_\ell \mathbb{F}_\ell$.

Proof. The group law follows from $T_\ell(n)T_\ell(m) = T_\ell(n+m)$. If α_n is the identity, then for every prime ℓ and every $r \in \mathbb{F}_\ell$ the factor $\exp(-2\pi i n r/\ell)$ is 1. Hence $n \equiv 0 \pmod{\ell}$ for every ℓ, so $n = 0$. On finite S, only the residues $n \bmod \ell$ for $\ell \in S$ matter, which is the same as $n \bmod N_S$ by the Chinese remainder theorem. The same finite-coordinate dependence gives the continuous extension to the product of residue fields.

Theorem $T4$**.** For $n \neq 0$, the automorphism α_n is not implemented by a unitary in the reference-vector incomplete tensor representation $\mathcal{H}_{\rm cl}^{\Omega}$.

Proof. Let $P_\ell(a)$ be the rank-one projection onto $|a\rangle$. If a unitary U implemented α_n, then

$$UP_\ell(0)U^* = P_\ell(n \bmod \ell).$$

Since $P_\ell(0)\Omega = \Omega$, the vector $U\Omega$ would satisfy

$$P_\ell(n \bmod \ell)U\Omega = U\Omega$$

for every prime ℓ. For $n \neq 0$, $n \bmod \ell \neq 0$ for all but finitely many ℓ. But $\mathcal{H}_{\rm cl}^{\Omega}$ has basis indexed by configurations that are zero at all but finitely many primes. No nonzero vector can be supported only on configurations with $s_\ell = n \bmod \ell$ for infinitely many ℓ. Hence $U\Omega = 0$, impossible for a unitary.

46.9 Speculative Directions

The checked surprise is that the scheme has no translation, while the quantum closed-prime field has a faithful arithmetic translation action by residues. Finite experiments see only $\mathbb{Z}/N_S\mathbb{Z}$, while the infinite algebra sees the squarefree residue quotient $\prod_\ell \mathbb{F}_\ell$; any $\widehat{\mathbb{Z}}$-action factors through it. The global translation by 1 is an automorphism of the quasi-local algebra but not a unitary in the finite-excitation vacuum sector: a large arithmetic symmetry, not a local observable.

Source-gated questions now become visible. Should the generic $\mathbb{Q}$ sector be replaced by an adelic self-dual sector? Can a sourced C*-dynamical system with prime-indexed dimensions and $\log \ell$ energy weights recover Euler-product or zeta data without inserting them by hand? Is the nonimplementability of α_1 the first arithmetic superselection sector in this programme? These are questions, not conclusions.

47 Affine-Line Symmetries and Weyl-Net Automorphisms

47.1 Status and Source Anchors

This shard extends AQM-29. It fixes the word "symmetry" to mean a base-preserving k-scheme automorphism of $\mathbf{A}^1_k = \operatorname{Spec} k[x]$, where $k = \mathbb{F}_q$. The affine-space definition is source-local from the Stacks Project snapshot `references/algebraic_geometry/stacks_project_constructions.tex`, lines 733–758. The affine-scheme anti-equivalence and morphism-to-affine theorem are in `references/algebraic_geometry/stacks_project_schemes.tex`, lines 841–963. Spectrum, $D(f)$, and functoriality are in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 2972–2983 and 3123–3160. The finite Weyl operators use Ashikhmin–Knill, `references/symplectic_qecc/ashikhmin_knill_0005008_source/n9.tex`, lines 249–313. Odd-characteristic Clifford language is grounded by Gross, `references/symplectic_qecc/gross_0602001_source/poswig.tex`, lines 340–537.

47.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & k=\mathbb{F}_q, \quad X=\mathbf{A}^1_k=\operatorname{Spec} k[x].\\
D2 & : & \operatorname{Aut}_k(X) \text{ means automorphisms over } \operatorname{Spec} k.\\
L1 & : & \operatorname{Aut}_k(X)=\{g_{a,b}: x\mapsto ax+b,\ a\in k^\times,\ b\in k\}.\\
D3 & : & x_\pi=(\pi), \quad K_\pi=k[x]/(\pi), \quad \pi \text{ monic irreducible.}\\
L2 & : & g_{a,b}(x_\pi)=x_{\pi^g}, \quad \pi^g(X)=a^{\deg\pi}\pi((X-b)/a).\\
D4 & : & \tau_{g,\pi}: K_{\pi^g}\to K_\pi, \quad \tau_{g,\pi}([X])=a[X]+b.\\
L3 & : & \tau_{g,\pi} \text{ is a } k\text{-field isomorphism and preserves absolute traces.}\\
D5 & : & E(U)=\bigoplus_{x_\pi\in U} K_\pi^2, \quad \mathcal{A}(U)=\operatorname{span}\{W_U(e): e\in E(U)\}.\\
D6 & : & S_g: E(U)\to E(gU) \text{ is residue pushforward by } \tau^{-1}.\\
T1 & : & S_g \text{ preserves the Weyl multiplier and commutator.}\\
T2 & : & \alpha_g^U(W_U(e))=W_{gU}(S_g e) \text{ is a covariant net } *\text{-isomorphism.}\\
T3 & : & \text{On finite supports, } \alpha_g \text{ has a tensor-permutation Clifford implementation.}
\end{array}
$$

47.3 The Base-Preserving Symmetry Group D1–L1

Let $k = \mathbb{F}_q$. By the Stacks affine-space source,

$$\mathbf{A}^1_k = \operatorname{Spec} k[x].$$

A base-preserving symmetry means an automorphism of this affine scheme over $\operatorname{Spec} k$. By the affine-scheme anti-equivalence, this is the same data as a k-algebra automorphism

$$\theta : k[x] \longrightarrow k[x].$$

Lemma $L1$. Every k-scheme automorphism of $\mathbf{A}^1_k$ is uniquely

$$g_{a,b} : x \longmapsto ax + b, \qquad a \in k^\times,\ b \in k.$$

The group law is

$$g_{a,b}g_{c,d} = g_{ac,ad+b}.$$

Proof. Let $\theta : k[x] \to k[x]$ be a k-algebra automorphism. Put $h = \theta(x)$. Since $k[x]$ is generated as a k-algebra by x, θ is determined by h. Let $\psi = \theta^{-1}$ and put $r = \psi(x)$. Then

$$x = (\theta\psi)(x) = \theta(r) = r(h).$$

If $\deg h = m$ and $\deg r = n$, then $\deg r(h) = mn$. Since $\deg x = 1$, we have $mn = 1$. Thus $m = n = 1$, so $h = ax + b$ with $a \neq 0$. Conversely, every polynomial $ax + b$ with $a \neq 0$ gives the inverse $x \mapsto a^{-1}(x - b)$. Composition of functions gives the displayed law.

This proves the user's proposed group exactly in the base-preserving sense. Absolute automorphisms that are allowed to move the coefficient field are a different question and are not used in this shard.

47.4 Closed Points and Residue Fields D3–L3

By AQM-29, the closed points of X are

$$x_\pi = (\pi), \qquad \pi \in k[x] \text{ monic irreducible.}$$

The residue field at x_π is

$$K_\pi = \kappa(x_\pi) = k[x]/(\pi).$$

Let $g = g_{a,b}$. Its coordinate pullback is

$$\theta_g(f)(x) = f(ax + b).$$

Lemma *L*2. If π is monic irreducible of degree d, then

$$g(x_\pi) = x_{\pi^g}, \qquad \pi^g(X) = a^d \pi((X - b)/a).$$

The polynomial π^g is monic and irreducible.

Proof. Let $\alpha = [x] \in K_\pi$. The point image is the prime $\theta_g^{-1}((\pi))$. A polynomial f lies in that prime exactly when

$$f(a\alpha + b) = 0 \quad \text{in } K_\pi.$$

Thus the image closed point is the minimal polynomial over k of $\beta = a\alpha + b$. The displayed π^g is monic, because the leading coefficient is $a^d a^{-d} = 1$, and $\pi^g(\beta) = 0$. Since $\alpha = a^{-1}(\beta - b)$, the fields $k(\alpha)$ and $k(\beta)$ are equal. Hence β has degree d, so π^g is its irreducible minimal polynomial.

Definition *D*4. The residue pullback at x_π is

$$\tau_{g,\pi} : K_{\pi^g} \longrightarrow K_\pi, \qquad \tau_{g,\pi}([X]) = a[X] + b.$$

Lemma *L*3. $\tau_{g,\pi}$ is a k-field isomorphism. For every $u \in K_{\pi^g}$,

$$\mathrm{Tr}_{K_{\pi^g}/\mathbb{F}_p}(u) = \mathrm{Tr}_{K_\pi/\mathbb{F}_p}(\tau_{g,\pi} u).$$

Proof. The relation $\pi^g(X) = 0$ is sent to $\pi^g(a\alpha + b) = 0$, so the displayed formula defines a homomorphism. It is onto because $\alpha = a^{-1}(\tau([X]) - b)$ lies in its image. The two finite fields have the same size q^d, hence the map is an isomorphism. As a k-linear isomorphism, it conjugates the k-linear multiplication map m_u on K_{π^g} to $m_{\tau u}$ on K_π. Conjugate linear maps have the same trace over k; composing with $\mathrm{Tr}_{k/\mathbb{F}_p}$ gives the absolute-trace identity.

47.5 The Weyl Label Pushforward D5–T1

For an open $U \subset X$, define

$$E(U) = \bigoplus_{x_\pi \in U} K_\pi^2$$

with finite support. A label is written

$$e = (q_\pi, p_\pi)_\pi.$$

At x_π, use

$$T_\pi(q)|s\rangle = |s+q\rangle, \qquad R_\pi(p)|s\rangle = \exp\!\left(\frac{2\pi i}{p_0}\operatorname{Tr}_{K_\pi/\mathbb{F}_{p_0}}(ps)\right)|s\rangle,$$

where $p_0 = \operatorname{char} k$, and $W_\pi(q,p) = T_\pi(q)R_\pi(p)$. For finite support, $W_U(e)$ is the tensor product of the local operators.

Definition $D6$. The pushforward

$$S_g : E(U) \longrightarrow E(gU)$$

is defined by

$$(S_g e)_{\pi^g} = \big(\tau_{g,\pi}^{-1}(q_\pi), \tau_{g,\pi}^{-1}(p_\pi)\big).$$

Theorem $T1$. S_g preserves the Weyl multiplier and commutator. In particular,

$$W_{gU}(S_g e) W_{gU}(S_g f)$$

has the same scalar multiplier as $W_U(e)W_U(f)$.

Proof. It is enough to check one closed point. Let $(Q,P) = \tau^{-1}(q,p)$ and $(Q',P') = \tau^{-1}(q',p')$. By Lemma $L3$,

$$\operatorname{Tr}_{K_{\pi^g}/\mathbb{F}_{p_0}}(PQ') = \operatorname{Tr}_{K_\pi/\mathbb{F}_{p_0}}(pq').$$

The same identity with P, Q and P', Q' exchanged gives preservation of

$$\operatorname{Tr}(PQ' - QP').$$

These are exactly the local multiplier and commutator exponents. Multiplying over the finite support proves the claim.

47.6 Net Automorphism and Clifford Implementation T2–T3

Theorem $T2$. For every $g \in \operatorname{Aut}_k(X)$ and open U, the rule

$$\alpha_g^U(W_U(e)) = W_{gU}(S_g e)$$

extends uniquely to a unital $*$-isomorphism

$$\alpha_g^U : \mathcal{A}(U) \longrightarrow \mathcal{A}(gU).$$

These maps are covariant for inclusions and satisfy

$$\alpha_g^{hU}\alpha_h^U = \alpha_{gh}^U.$$

Proof. The Weyl operators form a basis on every finite support, and $\mathcal{A}(U)$ is the union of the finite-support algebras. Theorem $T1$ shows that the assignment respects products and adjoints on each finite support, hence on the union. It is bijective because $S_{g^{-1}}$ is the inverse label map. If $U \subset V$, extension by zero commutes with pushforward of finite supports, so the square

$$\mathcal{A}(U) \to \mathcal{A}(V), \qquad \mathcal{A}(gU) \to \mathcal{A}(gV)$$

commutes. The residue maps τ compose because the coordinate pullbacks θ_g compose; therefore $S_g S_h = S_{gh}$, and the displayed group law follows on the Weyl basis.

Theorem $T3$**.** For a finite closed support S, the finite-volume map

$$\alpha_g^S : \mathcal{A}(S) \to \mathcal{A}(gS)$$

is implemented by a unitary tensor relabelling. If

$$\mathcal{H}(S) = \bigotimes_{x_\pi \in S} \ell^2(K_\pi),$$

define

$$U_g^S \left(\bigotimes_{x_\pi \in S} |s_\pi\rangle \right) = \bigotimes_{x_\pi g \in gS} |\tau_{g,\pi}^{-1}(s_\pi)\rangle.$$

Then

$$U_g^S W_S(e) (U_g^S)^* = W_{gS}(S_g e).$$

Proof. At one site, residue relabelling sends $|s\rangle$ to $|\tau^{-1}s\rangle$. It carries translation by q to translation by $\tau^{-1}q$. It carries the phase $\exp(2\pi i \ \mathrm{Tr}(ps)/p_0)$ to the phase with $\tau^{-1}p$ and $\tau^{-1}s$, by Lemma $L3$. Tensoring the one-site calculation over S proves the formula.

Thus the affine-line symmetry acts on the Weyl local net by symplectic finite-support label maps. In odd characteristic, Gross's Clifford structure theorem makes each finite-volume implementation a Clifford unitary. In even characteristic, the algebra automorphism above is still proved, but the half-Weyl Clifford convention is not imported.

48 The $\mathbb{F}_3$ Rational-Support Affine-Line Symmetry Example

48.1 Status and Source Anchors

This shard is the finite rational-support line-by-line example for

$$k = \mathbb{F}_3, \qquad X = \mathbf{A}^1_{\mathbb{F}_3} = \operatorname{Spec} \mathbb{F}_3[x].$$

It uses the definitions and proofs of AQM-29 and AQM-46. The full countable closed-point Hilbert representation is AQM-48. No new general claim is introduced here. The source anchors remain Stacks for Spec, affine space, affine-scheme functoriality, and standard opens; Ashikhmin–Knill for finite-field shift and phase operators; and Gross for the odd-qudit Clifford interpretation.

48.2 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: \; k = \mathbb{F}_3, \quad X = \operatorname{Spec} k[x]. \\
D2 &: \; G = \operatorname{Aut}_k(X) = \{g_{a,b} : x \mapsto ax + b\}. \\
L1 &: \; G = \{x, x+1, x+2, 2x, 2x+1, 2x+2\}. \\
D3 &: \; x_r = (x - r), \; r = 0, 1, 2, \text{ are the rational closed points.} \\
L2 &: \; g_{a,b}(x_r) = x_{ar+b}. \\
D4 &: \; S_{\text{rat}} = \{x_0, x_1, x_2\}, \quad E_{\text{rat}} = \mathbb{F}_3^3 \oplus \mathbb{F}_3^3. \\
L3 &: \; S_g(q_0, q_1, q_2, p_0, p_1, p_2) = (q_{g^{-1}0}, q_{g^{-1}1}, q_{g^{-1}2}, p_{g^{-1}0}, p_{g^{-1}1}, p_{g^{-1}2}). \\
T1 &: \; S_g \text{ preserves } \Omega(e, f) = \textstyle\sum_r (p_r q'_r - q_r p'_r). \\
T2 &: \; \alpha_g(W(e)) = W(S_g e) \text{ is a Weyl algebra automorphism.} \\
T3 &: \; U_g |s_0, s_1, s_2\rangle = |s_{g^{-1}0}, s_{g^{-1}1}, s_{g^{-1}2}\rangle \text{ implements } \alpha_g.
\end{aligned}$$

48.3 Step 1: The Space D1

The coordinate ring is

$$R = \mathbb{F}_3[x].$$

The affine line is

$$X = \operatorname{Spec} R.$$

The closed points of X are ideals (π), where π is monic irreducible in $\mathbb{F}_3[x]$. The residue field is

$$K_\pi = \mathbb{F}_3[x]/(\pi).$$

At a degree d point, K_π has 3^d elements, so the site carries one 3^d-level qudit.

48.4 Step 2: The Symmetry Group D2–L1

By AQM-46,

$$G = \operatorname{Aut}_{\mathbb{F}_3}(X) = \{g_{a,b} : x \mapsto ax + b, \; a \in \mathbb{F}_3^\times, \; b \in \mathbb{F}_3\}.$$

Since

$$\mathbb{F}_3^\times = \{1, 2\},$$

there are $2 \cdot 3 = 6$ symmetries:

name	(a,b)	map
e	$(1,0)$	x
τ	$(1,1)$	$x+1$
τ^2	$(1,2)$	$x+2$
ρ	$(2,0)$	$2x$
$\rho\tau$	$(2,1)$	$2x+1$
$\rho\tau^2$	$(2,2)$	$2x+2$

The group law is

$$g_{a,b}g_{c,d} = g_{ac,ad+b}.$$

For example,

$$\tau\rho = g_{1,1}g_{2,0} = g_{2,1}, \qquad \rho\tau = g_{2,0}g_{1,1} = g_{2,2}.$$

Thus the group is nonabelian and has six elements.

48.5 Step 3: Rational Closed Points D3–L2

For $r \in \mathbb{F}_3$, define

$$x_r = (x - r).$$

The residue field is

$$K_r = \mathbb{F}_3[x]/(x - r) \cong \mathbb{F}_3.$$

For $g_{a,b}$, AQM-46 gives

$$g_{a,b}(x_r) = x_{ar+b}.$$

The six rational-point permutations are

g	x_0	x_1	x_2
x	x_0	x_1	x_2
$x+1$	x_1	x_2	x_0
$x+2$	x_2	x_0	x_1
$2x$	x_0	x_2	x_1
$2x+1$	x_1	x_0	x_2
$2x+2$	x_2	x_1	x_0

At rational points the residue-field isomorphism is the identity map on $\mathbb{F}_3$, because every K_r is canonically $\mathbb{F}_3$ after evaluation at r.

48.6 Step 4: Quadratic Closed Points

The monic irreducible quadratics over $\mathbb{F}_3$ are

$$q_0 = x^2 + 1, \qquad q_1 = x^2 + x + 2, \qquad q_2 = x^2 + 2x + 2.$$

They are irreducible because none has a root in $\mathbb{F}_3$. For a quadratic π, the transformation rule of AQM-46 is

$$\pi^g(X) = a^2\pi((X - b)/a).$$

In $\mathbb{F}_3$, $a^2 = 1$ for $a = 1, 2$. The action is

g	q_0	q_1	q_2
x	q_0	q_1	q_2
$x+1$	q_1	q_2	q_0
$x+2$	q_2	q_0	q_1
$2x$	q_0	q_2	q_1
$2x+1$	q_1	q_0	q_2
$2x+2$	q_2	q_1	q_0

Example: for $g = x + 1$,

$$q_0^g(X) = q_0(X-1) = (X-1)^2 + 1 = X^2 + X + 2 = q_1.$$

48.7 Step 5: The Rational Three-Site Label Space D4

Restrict now to the finite rational closed support

$$S_{\mathrm{rat}} = \{x_0, x_1, x_2\}.$$

The label space is

$$E_{\mathrm{rat}} = \bigoplus_{r=0}^{2} \mathbb{F}_3^2 = \mathbb{F}_{3,q}^3 \oplus \mathbb{F}_{3,p}^3.$$

Write a label as

$$e = (q_0, q_1, q_2, p_0, p_1, p_2).$$

The symplectic form is

$$\Omega(e, f) = \sum_{r=0}^{2} (p_r q_r' - q_r p_r').$$

This is the direct sum of the three one-qutrit forms.

48.8 Step 6: The Symplectic Map L3–T1

For $g \in G$, the label pushforward is

$$(S_g e)_{g(r)} = e_r$$

on both the q-coordinates and the p-coordinates. Equivalently,

$$S_g(q_0, q_1, q_2, p_0, p_1, p_2) = (q_{g^{-1}0}, q_{g^{-1}1}, q_{g^{-1}2}, p_{g^{-1}0}, p_{g^{-1}1}, p_{g^{-1}2}).$$

For $\tau : x \mapsto x + 1$,

$$\tau^{-1}(0) = 2, \quad \tau^{-1}(1) = 0, \quad \tau^{-1}(2) = 1,$$

so

$$S_\tau(e) = (q_2, q_0, q_1, p_2, p_0, p_1).$$

For $\rho : x \mapsto 2x$,

$$\rho^{-1} = \rho, \qquad S_\rho(e) = (q_0, q_2, q_1, p_0, p_2, p_1).$$

Check.

$$\begin{aligned}\Omega(S_g e, S_g f) &= \sum_y \big((S_g p)_y (S_g q')_y - (S_g q)_y (S_g p')_y\big) \\ &= \sum_y \big(p_{g^{-1}y} q'_{g^{-1}y} - q_{g^{-1}y} p'_{g^{-1}y}\big) \\ &= \sum_r (p_r q'_r - q_r p'_r) = \Omega(e, f).\end{aligned}$$

Thus S_g is symplectic.

48.9 Step 7: The Weyl Operators T2

The Hilbert space is

$$\mathcal{H}_{\mathrm{rat}} = \ell^2(\mathbb{F}_3)^{\otimes 3}$$

with basis $|s_0, s_1, s_2\rangle$. At site r,

$$T_r(q)|s_r\rangle = |s_r + q\rangle, \qquad R_r(p)|s_r\rangle = \omega^{p s_r}|s_r\rangle, \qquad \omega = e^{2\pi i/3}.$$

For $e = (q_0, q_1, q_2, p_0, p_1, p_2)$,

$$W(e) = \bigotimes_{r=0}^{2} T_r(q_r) R_r(p_r).$$

The Weyl law is

$$W(e)W(f) = \omega^{\sum_r p_r q'_r} W(e+f).$$

Because S_g preserves the exponent, the formula

$$\alpha_g(W(e)) = W(S_g e)$$

defines an algebra automorphism. For τ,

$$\alpha_\tau(W(q_0, q_1, q_2, p_0, p_1, p_2)) = W(q_2, q_0, q_1, p_2, p_0, p_1).$$

48.10 Step 8: The Clifford Unit T3

Define

$$U_g|s_0, s_1, s_2\rangle = |s_{g^{-1}0}, s_{g^{-1}1}, s_{g^{-1}2}\rangle.$$

For τ,

$$U_\tau|s_0, s_1, s_2\rangle = |s_2, s_0, s_1\rangle.$$

For ρ,

$$U_\rho|s_0, s_1, s_2\rangle = |s_0, s_2, s_1\rangle.$$

These are permutation matrices, hence unitaries. Directly,

$$U_g T_r(q) U_g^* = T_{g(r)}(q), \qquad U_g R_r(p) U_g^* = R_{g(r)}(p).$$

Multiplying over $r = 0, 1, 2$ gives

$$U_g W(e) U_g^* = W(S_g e) = \alpha_g(W(e)).$$

Since 3 is odd, Gross's finite Clifford theorem applies: this finite rational-support implementation is a Clifford unitary. This is only a finite-volume truncation; AQM-48 gives the full countable closed-point net, its infinite Hilbert representation, and the generic-point caveat.

49 The Full $\mathbb{F}_3$ Affine-Line Net and Hilbert Action

49.1 Status and Source Anchors

This shard corrects the scope of AQM-47. The three-rational-point calculation there is a finite closed-support truncation. It is not the full local net on $\mathbf{A}^1_{\mathbb{F}_3}$. The full closed-point net uses every monic irreducible polynomial in $\mathbb{F}_3[x]$. AQM-28 supplies the closed-point quasi-local convention, AQM-29 supplies the point and open-set calculation for $\operatorname{Spec} k[t]$, AQM-45 supplies the reference-vector incomplete tensor representation pattern, and AQM-46 supplies the affine symmetry action.

49.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k = \mathbb{F}_3, \quad R = k[x], \quad X = \operatorname{Spec} R. \\
D2 & : & \eta = (0), \quad \mathcal{P} = \{\pi \in k[x] : \pi \text{ monic irreducible}\}. \\
L1 & : & X = \{\eta\} \cup \{x_\pi = (\pi) : \pi \in \mathcal{P}\}, \quad \mathcal{P} \text{ is countably infinite.} \\
D3 & : & K_\pi = R/(\pi), \quad \mathcal{H}_\pi = \ell^2(K_\pi), \quad \mathcal{A}_\pi = \operatorname{End}(\mathcal{H}_\pi). \\
D4 & : & E(U) = \bigoplus_{x_\pi \in U} K_\pi^2, \quad \mathcal{A}(U) = \varinjlim_{S \Subset C(U)} \bigotimes_{x_\pi \in S} \mathcal{A}_\pi. \\
L2 & : & \text{Every nonempty open } U \text{ has cofinite } C(U) \subset \mathcal{P}. \\
T1 & : & \mathcal{A}(X) \text{ is an algebraic infinite tensor product over } \mathcal{P}. \\
D5 & : & \Gamma_{\mathrm{fin}} = \bigoplus_{\pi \in \mathcal{P}} K_\pi, \quad \mathcal{H}^0 = \ell^2(\Gamma_{\mathrm{fin}}). \\
L3 & : & \mathcal{H}^0 \text{ is countably infinite-dimensional and represents } \mathcal{A}(X). \\
D6 & : & G = \{g_{a,b} : x \mapsto ax + b,\ a \in \mathbb{F}_3^\times,\ b \in \mathbb{F}_3\}. \\
D7 & : & (V_g s)_{\pi^g} = \tau_{g,\pi}^{-1}(s_\pi), \quad U_g|s\rangle = |V_g s\rangle. \\
T2 & : & U_g \mathcal{A}(U) U_g^* = \mathcal{A}(gU), \quad U_g W_U(e) U_g^* = W_{gU}(S_g e). \\
L4 & : & g(\eta) = \eta, \quad \eta \in U \text{ for every nonempty open } U. \\
T3 & : & \eta \text{ is not a finite-residue qudit site in this net.}
\end{array}$$

49.3 The Countable Closed-Point Set D1–L1

Let

$$R = \mathbb{F}_3[x], \qquad X = \operatorname{Spec} R.$$

The zero ideal is prime because R is a domain; write

$$\eta = (0).$$

For each monic irreducible $\pi \in R$, write

$$x_\pi = (\pi).$$

Lemma *L*1. The points of X are exactly

$$\{\eta\} \cup \{x_\pi : \pi \in \mathcal{P}\}.$$

The set $\mathcal{P}$ is countably infinite.

Proof. The point classification is AQM-29 with $k = \mathbb{F}_3$. Directly: $\mathbb{F}_3[x]$ is Euclidean, so every nonzero prime ideal has a monic generator. The generator must be irreducible, and every monic irreducible generates a maximal, hence prime, ideal.

For countability, R is the union over degrees $n \geq 0$ of the finite sets of polynomials of degree at most n. Hence R, and therefore $\mathcal{P}$, is countable. For infinitude, suppose $\pi_1, \ldots, \pi_m$ were all monic irreducibles. Then

$$F = \pi_1 \cdots \pi_m + 1$$

is nonconstant. Choose a monic nonconstant divisor ρ of F with least degree; then ρ is irreducible. No π_i divides F, because π_i divides $F-1$. Thus ρ is a new monic irreducible, a contradiction.

49.4 The Full Closed-Point Net D3–T1

At the closed point x_π, put

$$K_\pi = R/(\pi).$$

If $\deg \pi = d$, then K_π has 3^d elements. Define

$$\mathcal{H}_\pi = \ell^2(K_\pi), \qquad \mathcal{A}_\pi = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_\pi) \cong M_{3^d}(\mathbb{C}).$$

For a finite closed support $S \Subset \mathcal{P}$, set

$$\mathcal{A}(S) = \bigotimes_{\pi \in S} \mathcal{A}_\pi.$$

For an open $U \subset X$, define

$$C(U) = \{\pi \in \mathcal{P} : x_\pi \in U\}.$$

The label group and local algebra are

$$E(U) = \bigoplus_{\pi \in C(U)} K_\pi^2, \qquad \mathcal{A}(U) = \varinjlim_{S \Subset C(U)} \mathcal{A}(S),$$

where the transition maps are $a \mapsto a \otimes \mathbf{1}$.

Lemma *L*2. If $U \neq \emptyset$, then $C(U)$ is cofinite in $\mathcal{P}$. If $U = \emptyset$, then $C(U) = \emptyset$.

Proof. Every closed subset of $\operatorname{Spec} R$ has the form $V(I)$. Since R is a principal ideal domain, $I = (h)$. Thus every open is $D(h)$. If $h = 0$, then $D(h) = \emptyset$. If $h \neq 0$, then

$$x_\pi \notin D(h) \iff h \in (\pi) \iff \pi \mid h.$$

A nonzero polynomial has only finitely many monic irreducible divisors. Hence the closed points omitted by $D(h)$ are finite.

Theorem *T*1. The global closed-point algebra is

$$\mathcal{A}(X) = \varinjlim_{S \Subset \mathcal{P}} \bigotimes_{\pi \in S} M_{3^{\deg \pi}}(\mathbb{C}).$$

It is an algebraic infinite tensor product over all closed points. The rational three-site algebra of AQM-47 is only the finite subalgebra for

$$S_{\mathrm{rat}} = \{x, x-1, x-2\} \subset \mathcal{P}.$$

Proof. For $X = D(1)$, $C(X) = \mathcal{P}$, so the displayed formula is Definition *D*4. Since $\mathcal{P}$ is infinite by Lemma *L*1, this is not a finite matrix algebra. The rational support is a finite subset of $\mathcal{P}$, so $\mathcal{A}(S_{\mathrm{rat}})$ maps into the filtered colimit. It does not equal the colimit, because any irreducible $\pi \notin S_{\mathrm{rat}}$ contributes the nontrivial one-site algebra $\mathcal{A}_\pi$.

49.5 The Infinite Hilbert Representation D5–L3

The algebraic net above is defined before choosing any Hilbert-space completion. To get the standard reference representation, choose the reference vector $|0\rangle_\pi \in \mathcal{H}_\pi$ at every closed point. Define

$$\Gamma_{\text{fin}} = \bigoplus_{\pi\in\mathcal{P}} K_\pi = \{s = (s_\pi)_\pi : s_\pi = 0 \text{ for all but finitely many } \pi\}.$$

Set

$$\mathcal{H}^0 = \ell^2(\Gamma_{\text{fin}}), \qquad |s\rangle = \bigotimes_{\pi\in\mathcal{P}} |s_\pi\rangle,$$

with all but finitely many tensor factors equal to $|0\rangle_\pi$.

Lemma *L*3. $\mathcal{H}^0$ is countably infinite-dimensional, and it represents $\mathcal{A}(X)$.

Proof. The set of finite subsets of the countable set $\mathcal{P}$ is countable. For a fixed finite support S, the set $\prod_{\pi\in S} K_\pi$ is finite. Thus Γ_{fin}, a countable union of finite sets, is countable. It is infinite because the configurations supported at one closed point with value 1 are distinct as π varies. Hence $\ell^2(\Gamma_{\text{fin}})$ has a countably infinite orthonormal basis.

For finite S, $\mathcal{A}(S)$ acts on the coordinates in S and fixes all other coordinates. If $S \subset T$, this action agrees with the action of $a \otimes \mathbf{1}_{\mathcal{H}(T\setminus S)}$. Therefore the compatible finite-support actions define a representation of the filtered colimit $\mathcal{A}(X)$.

49.6 The Full Affine Symmetry Implementation D6–T2

Let

$$G = \mathrm{Aut}_{\mathbb{F}_3}(X) = \{g_{a,b} : x \mapsto ax + b,\ a \in \mathbb{F}_3^\times,\ b \in \mathbb{F}_3\}.$$

For $\pi \in \mathcal{P}$, AQM-46 gives

$$g(x_\pi) = x_{\pi^g}, \qquad \pi^g(X) = a^{\deg\pi}\pi((X-b)/a),$$

and a residue-field isomorphism

$$\tau_{g,\pi} : K_{\pi^g} \to K_\pi, \qquad \tau_{g,\pi}([X]) = a[X] + b.$$

Define a map on finite-support configurations by

$$(V_g s)_{\pi^g} = \tau_{g,\pi}^{-1}(s_\pi).$$

Because $\tau_{g,\pi}^{-1}(0) = 0$, V_g sends finite-support configurations to finite-support configurations. It is bijective, with inverse $V_{g^{-1}}$. Hence

$$U_g|s\rangle = |V_g s\rangle$$

defines a unitary on $\mathcal{H}^0$.

Theorem *T*2. For every open $U \subset X$ and every $e \in E(U)$,

$$U_g W_U(e) U_g^* = W_{gU}(S_g e).$$

Consequently

$$U_g \mathcal{A}(U) U_g^* = \mathcal{A}(gU),$$

and $g \mapsto U_g$ is a unitary implementation of the full closed-point net action in the zero-reference representation.

Proof. At one closed point, the basis relabelling $|s\rangle \mapsto |\tau_{g,\pi}^{-1}s\rangle$ carries translation by q to translation by $\tau_{g,\pi}^{-1}q$. It carries the phase labelled by p to the phase labelled by $\tau_{g,\pi}^{-1}p$, because AQM-46 proves absolute-trace preservation for $\tau_{g,\pi}$. Thus the formula holds on a one-site Weyl operator. Tensoring over the finite support of e proves the displayed Weyl formula. The Weyl operators span each finite-support algebra, so the algebra statement follows. The maps V_g compose because the residue maps $\tau_{g,\pi}$ compose, again by AQM-46; hence the U_g compose on the basis of $\mathcal{H}^0$.

49.7 The Generic Point L4–T3

Lemma *L*4. The generic point $\eta = (0)$ is fixed by every $g \in G$, and every nonempty open $U \subset X$ contains η.

Proof. The coordinate pullback of g is an automorphism of the domain $\mathbb{F}_3[x]$, so the inverse image of the zero ideal is the zero ideal. Thus $g(\eta) = \eta$. By Lemma *L*2, every nonempty open is $D(h)$ with $h \neq 0$. Since $h \notin (0)$, the point η lies in $D(h)$.

Theorem *T*3. The generic point is not a tensor factor of the finite-residue Weyl net above.

Proof. Its residue field is

$$\kappa(\eta) = \mathbb{F}_3(x),$$

the fraction field of $\mathbb{F}_3[x]$. This field is infinite. Convention (y) attaches finite Weyl sites only to closed points, whose residue fields are finite. Therefore η affects the topology of opens, but it contributes no factor K_η^2, no finite qudit $\ell^2(K_\eta)$, and no coordinate of Γ_{fin} in the closed-point net.

If one adds the rational-function sector of convention (z), an additional choice of additive character is required. The closed-point rule $(q,p) \mapsto (\tau^{-1}q, \tau^{-1}p)$ must not be copied blindly. For the default functional λ that extracts the coefficient of x^{-1} at infinity, let $\rho : x \mapsto 2x$. Then $\tau_\rho^{-1}(x^{-1}) = 2x^{-1}$, so

$$\lambda(1 \cdot x^{-1}) = 1, \qquad \lambda(\tau_\rho^{-1}(1)\tau_\rho^{-1}(x^{-1})) = 2.$$

The multiplier changes. Thus a generic-sector affine action needs a separate dual-label or character-invariance convention before it can be called part of the Weyl-net symmetry theorem.

50 Regular Functions, Supports, and Weyl Labels on the Affine Line

50.1 Status and Source Anchors

This shard fixes the current meaning of "regular function" in the closed-point Weyl-net layer. It refines AQM-28 and AQM-29. The source-local facts are Stacks: affine-scheme global sections in `references/algebraic_geometry/stacks_project_schemes.tex`, lines 670–719; Spec, $D(f)$, and $V(I)$ in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 2972–3023 and 3104–3211; residue fields in the same file, lines 250–274 and 3310–3350. Finite-field Weyl operators are AQM-28, sourced from Ashikhmin–Knill as recorded there. The Artin-ring layer is AQM-36.

50.2 Lamport Dependency Skeleton

$D1$: $k = \mathbb{F}_Q, \quad R = k[x], \quad X = \operatorname{Spec} R, \quad \mathcal{P} = \{\pi \in R : \pi \text{ monic irreducible}\}.$
$D2$: $x_\pi = (\pi), \quad K_\pi = R/(\pi).$
$D3$: $h \neq 0, \quad D(h) = \{\mathfrak{p} : h \notin \mathfrak{p}\}, \quad \Gamma(D(h), \mathcal{O}_X) = R_h.$
$D4$: $\mathrm{ev}_U(F,G)_\pi = (F \bmod \pi, G \bmod \pi) \in K_\pi^2, \quad x_\pi \in U.$
$L1$: A nonzero regular-function pair on nonempty $D(h)$ has cofinite closed support in $D(h)$.
$T1$: $\mathrm{ev}_U(F,G) \in E(U)$ iff $(F,G) = (0,0)$.
$D5$: $Z_{\mathrm{cl}}(h) = \{x_\pi : \pi \mid h\}, \quad \mathrm{rad}(h) = \prod_{\pi \mid h} \pi.$
$L2$: $Z_{\mathrm{cl}}(h)$ is finite and depends only on $\mathrm{rad}(h)$.
$D6$: $E_h^{\mathrm{red}} = \bigoplus_{\pi \mid h} K_\pi^2, \quad W_h^{\mathrm{red}}(F,G) = W_{Z_{\mathrm{cl}}(h)}((F \bmod \pi, G \bmod \pi)_\pi).$
$L3$: $W_h^{\mathrm{red}}(F,G)$ depends only on $F, G \bmod \mathrm{rad}(h)$.
$T2$: Every finite closed-support Weyl label is represented by polynomial pairs modulo the squarefree support
$D7$: $h = \prod_i \pi_i^{e_i}$ also defines the thickened rings $R/(\pi_i^{e_i})$.
$T3$: Multiplicities e_i are invisible in E_h^{red} and visible only in the Artin-ring layer.

50.3 Regular Functions as Residue Profiles D1–T1

Let $k = \mathbb{F}_Q$, $R = k[x]$, and $X = \operatorname{Spec} R$. By the Stacks affine source,

$$\Gamma(X, \mathcal{O}_X) = R.$$

More generally, on the nonempty standard open $D(h)$, $h \neq 0$, the regular functions are the localized ring

$$\Gamma(D(h), \mathcal{O}_X) = R_h.$$

For $h = 1$, this is the global case.

Let $U = D(h) \neq \emptyset$. A closed point of U is x_π with $\pi \nmid h$. For $F = u/h^m \in R_h$, define

$$F(x_\pi) = (u \bmod \pi)(h \bmod \pi)^{-m} \in K_\pi.$$

This is well-defined because $h \bmod \pi \neq 0$ in the field K_π. For a pair $(F,G) \in R_h^2$, define the residue profile

$$\mathrm{ev}_U(F,G)_\pi = (F(x_\pi), G(x_\pi)) \in K_\pi^2.$$

Lemma $L1$. If $(F,G) \neq (0,0)$ in R_h^2, then the set of closed points

$$\{\pi : \pi \nmid h,\ \mathrm{ev}_U(F,G)_\pi \neq (0,0)\}$$

is cofinite in $\{\pi : \pi \nmid h\}$.

Proof. Write $F = u/h^m$ and $G = v/h^n$. The profile is zero at $x_\pi \in U$ exactly when $u \bmod \pi = 0$ and $v \bmod \pi = 0$. This means that π divides both u and v. If $(F,G) \neq (0,0)$, then at least one of u, v is nonzero after localization. A nonzero polynomial has only finitely many monic irreducible divisors. Hence only finitely many $\pi \nmid h$ can make both residues zero.

Theorem $T1$. Under the current closed-point Weyl-net convention,

$$\mathrm{ev}_U(F,G) \in E(U) \quad \Longleftrightarrow \quad (F,G) = (0,0).$$

Proof. By AQM-28,

$$E(U) = \bigoplus_{x_\pi \in U} K_\pi^2$$

means finite closed support. If $(F,G) = (0,0)$, the profile has empty support. If $(F,G) \neq (0,0)$, Lemma $L1$ gives cofinite support in the infinite closed-point set of U; AQM-29 proves that $D(h)$ omits only finitely many closed points from the infinite set $\mathcal{P}$. Hence the profile is not finite-support and is not an element of $E(U)$.

Thus a regular function is a classical global or local residue profile. It is not, by itself, a quasi-local Weyl observable.

50.4 Polynomials as Finite Closed Supports D5–L2

Now let $h \in R$ be nonzero and factor

$$h = c\prod_{i=1}^{r} \pi_i^{e_i}, \qquad c \in k^\times, \quad e_i \geq 1,$$

with distinct monic irreducibles π_i. Define

$$Z_{\mathrm{cl}}(h) = \{x_{\pi_i} : 1 \leq i \leq r\}, \qquad \mathrm{rad}(h) = \prod_{i=1}^{r} \pi_i.$$

Lemma $L2$. $Z_{\mathrm{cl}}(h)$ is finite and depends only on $\mathrm{rad}(h)$, not on the exponents e_i.

Proof. A closed point x_π lies in $V(h)$ exactly when $h \in (\pi)$, which is equivalent to $\pi \mid h$. The factorization of a nonzero polynomial has only finitely many irreducible factors. Replacing h by $\mathrm{rad}(h)$ preserves the same divisibility relation $\pi \mid h$.

This is the first quantum use of a polynomial: it selects a finite closed support. It does not yet choose a Weyl operator.

50.5 Reduced Smearing over a Support D6–T2

Define the reduced support label space

$$E_h^{\mathrm{red}} = \bigoplus_{\pi \mid h} K_\pi^2.$$

For polynomial pairs $(F, G) \in R^2$, define the reduced smeared label

$$e_h^{\mathrm{red}}(F,G) = ((F \bmod \pi,\ G \bmod \pi))_{\pi | h} \in E_h^{\mathrm{red}}$$

and the reduced smeared Weyl operator

$$W_h^{\mathrm{red}}(F,G) = W_{Z_{\mathrm{cl}}(h)}(e_h^{\mathrm{red}}(F,G)).$$

Lemma $L3$. $e_h^{\mathrm{red}}(F,G)$, hence $W_h^{\mathrm{red}}(F,G)$, depends only on the classes of F and G modulo $\mathrm{rad}(h)$.

Proof. The label at x_π is obtained by reducing modulo π. Two polynomials have the same reduction modulo every $\pi \mid h$ iff their difference is divisible by every $\pi \mid h$. Since these irreducibles are pairwise coprime, this is equivalent to divisibility by $\mathrm{rad}(h)$.

Theorem $T2$. For every finite closed support $S = \{x_{\pi_1}, \dots, x_{\pi_r}\}$ and every label

$$e = ((q_i, p_i))_{i=1}^r \in \bigoplus_{i=1}^r K_{\pi_i}^2,$$

there exist polynomials $F, G \in R$ with

$$F \bmod \pi_i = q_i, \qquad G \bmod \pi_i = p_i \quad \text{for every } i.$$

Thus distinct closed-point qudits have no finite compatibility condition in the reduced Weyl net.

Proof. The ideals (π_i) are pairwise coprime. The Chinese remainder theorem therefore gives a surjection

$$R \longrightarrow \prod_{i=1}^r R/(\pi_i) = \prod_{i=1}^r K_{\pi_i}.$$

Choose F mapping to $(q_i)_i$ and G mapping to $(p_i)_i$. These polynomials have the required residues. Since every finite label is obtained this way, the qudit factors indexed by distinct irreducibles are independent inside the finite-support Weyl algebra.

50.6 Multiplicity and the Artin-Ring Layer D7–T3

The same polynomial $h = \prod_i \pi_i^{e_i}$ may also be read as a closed subscheme with multiplicity. Then the local rings

$$A_i = R/(\pi_i^{e_i})$$

are finite local Artin rings supported at the same closed points x_{π_i}. AQM-36 quantizes this thickened layer by

$$E_h^{\mathrm{thick}} = \bigoplus_i A_i^2, \qquad \mathcal{H}_h^{\mathrm{thick}} = \bigotimes_i \ell^2(A_i).$$

Theorem $T3$. The reduced support operator $W_h^{\mathrm{red}}(F,G)$ forgets all multiplicities e_i. Multiplicities become quantum degrees of freedom only after replacing the residue fields K_{π_i} by the Artin rings A_i.

Proof. By Lemma $L3$, the reduced label only uses F, G modulo $\mathrm{rad}(h) = \prod_i \pi_i$. Therefore h and $\mathrm{rad}(h)$ give the same reduced label space, the same Hilbert space $\bigotimes_i \ell^2(K_{\pi_i})$, and the same reduced Weyl operators. AQM-36 instead uses $R/(\pi_i^{e_i})$, whose cardinality is $Q^{e_i \deg \pi_i}$. These larger rings record nilpotent jet coefficients at the same closed point. They are not additional Zariski points.

50.7 Conclusion

In the present framing, regular functions have three separate roles:

$$\begin{array}{rcl} \text{profile} & : & (F,G) \in \Gamma(U,\mathcal{O}_X)^2 \mapsto \mathrm{ev}_U(F,G),\ \text{usually infinite support;} \\ \text{support selector} & : & h \neq 0 \mapsto Z_{\mathrm{cl}}(h),\ \text{a finite closed support;} \\ \text{smeared operator} & : & (h,F,G) \mapsto W_h^{\mathrm{red}}(F,G),\ \text{or } W_h^{\mathrm{thick}}(F,G) \text{ after Artin thickening.} \end{array}$$

Only the last line is a Weyl observable, and only after a finite support or a thickened finite scheme has been chosen.

51 Regular Functions on $\mathbb{F}_3[x]$ Line by Line

51.1 Status and Source Anchors

This is the explicit $k = \mathbb{F}_3$ companion to AQM-49. It uses AQM-29 for the point set of $\operatorname{Spec} \mathbb{F}_3[x]$, AQM-28 for finite-support Weyl operators, AQM-34 for polynomial quotients, and AQM-36 for Artin thickenings.

51.2 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: k = \mathbb{F}_3, \quad R = k[x], \quad X = \operatorname{Spec} R.\\
D2 &: x_r = (x-r) \quad (r = 0,1,2), \quad q_0 = x^2+1.\\
L1 &: x \text{ as a function has infinite closed support.}\\
L2 &: x^{-1} \in \Gamma(D(x), \mathcal{O}_X) \text{ has infinite support in } D(x).\\
D3 &: h = x(x-1), \quad Z_{\mathrm{cl}}(h) = \{x_0, x_1\}.\\
T1 &: W_h^{\mathrm{red}}(F,G) \text{ is a two-qutrit operator determined by values at 0 and 1.}\\
T2 &: \text{Every two-qutrit label on } Z_{\mathrm{cl}}(h) \text{ is represented by polynomial pairs modulo } h.\\
D4 &: q_0 = x^2+1, \quad K_{q_0} = k[\alpha]/(\alpha^2+1).\\
T3 &: Z_{\mathrm{cl}}(q_0) \text{ gives one 9-level qudit.}\\
D5 &: m = x^2 q_0^3.\\
T4 &: m \text{ has reduced Hilbert dimension 27 and thickened dimension } 3^8.
\end{aligned}$$

51.3 Step 1: The Space and Its First Closed Points

Let

$$R = \mathbb{F}_3[x], \qquad X = \operatorname{Spec} R.$$

The closed points are $x_\pi = (\pi)$ for monic irreducible π. The rational closed points are

$$x_0 = (x), \qquad x_1 = (x-1), \qquad x_2 = (x-2).$$

Their residue fields are all $\mathbb{F}_3$. The three monic irreducible quadratics are

$$q_0 = x^2+1, \qquad q_1 = x^2+x+2, \qquad q_2 = x^2+2x+2.$$

For q_0, write

$$K_{q_0} = \mathbb{F}_3[x]/(q_0) = \mathbb{F}_3[\alpha], \qquad \alpha^2 = -1 = 2.$$

51.4 Step 2: A Global Regular Function Is Usually Infinite Support

Take the regular function $F = x \in R$. At a closed point x_π,

$$F(x_\pi) = x \bmod \pi \in K_\pi.$$

At rational points,

$$F(x_0) = 0, \qquad F(x_1) = 1, \qquad F(x_2) = 2.$$

At x_{q_0},

$$F(x_{q_0}) = \alpha \in K_{q_0}.$$

Lemma $L1$. The profile of x is nonzero at every closed point except x_0.

Proof. $x \bmod \pi = 0$ iff $x \in (\pi)$, equivalently $\pi \mid x$. Since x is monic irreducible, this happens only for $\pi = x$. Thus all closed points except x_0 lie in the support.

Therefore x is not a quasi-local Weyl label. It is a classical residue profile with cofinite support.

51.5 Step 3: A Local Regular Function Still Has Infinite Support

Let

$$U = D(x) = X \setminus \{x_0\}.$$

The function x^{-1} is regular on U. Its residue value at $x_\pi \in U$ is

$$x^{-1}(x_\pi) = (x \bmod \pi)^{-1} \in K_\pi.$$

At rational points in U,

$$x^{-1}(x_1) = 1, \qquad x^{-1}(x_2) = 2,$$

because $2^{-1} = 2$ in $\mathbb{F}_3$. At q_0,

$$x^{-1}(x_{q_0}) = \alpha^{-1} = 2\alpha,$$

since $\alpha(2\alpha) = 2\alpha^2 = 4 = 1$.

Lemma $L2$. The profile of x^{-1} is nonzero at every closed point of $D(x)$.

Proof. For $x_\pi \in D(x)$, $\pi \nmid x$. Hence $x \bmod \pi \neq 0$ in the field K_π, so it has an inverse. An inverse is never zero.

Thus local regularity on $D(x)$ does not make x^{-1} a finite-support Weyl label.

51.6 Step 4: A Polynomial as a Finite Support Selector

Now take

$$h = x(x-1).$$

The closed zero set is

$$Z_{\mathrm{cl}}(h) = \{x_0, x_1\}.$$

The reduced support Hilbert space is

$$\mathcal{H}_h^{\mathrm{red}} = \ell^2(\mathbb{F}_3) \otimes \ell^2(\mathbb{F}_3),$$

so it has dimension 9. The reduced label space is

$$E_h^{\mathrm{red}} = \mathbb{F}_3^2 \oplus \mathbb{F}_3^2.$$

51.7 Step 5: A Concrete Smeared Two-Qutrit Operator

Choose

$$F = 1, \qquad G = x + 1.$$

At x_0,

$$(F, G) = (1, 1).$$

At x_1,

$$(F, G) = (1, 2).$$

Therefore

$$e_h^{\mathrm{red}}(F,G) = ((1,1),(1,2)).$$

If $T_r(a)$ and $R_r(b)$ are the qutrit shift and phase at x_r, then

$$W_h^{\mathrm{red}}(F,G) = T_0(1)R_0(1) \otimes T_1(1)R_1(2).$$

This is a genuine quasi-local observable because the support has two closed points.

51.8 Step 6: No Compatibility Between the Two Qutrits

Prescribe the label

$$q_0 = 2, \quad q_1 = 1, \qquad p_0 = 1, \quad p_1 = 0.$$

We seek $F, G \in \mathbb{F}_3[x]$ with

$$F(0) = 2, \quad F(1) = 1, \qquad G(0) = 1, \quad G(1) = 0.$$

Take

$$F = 2 + 2x, \qquad G = 1 + 2x.$$

Then

$$F(0) = 2, \quad F(1) = 2 + 2 = 1,$$

and

$$G(0) = 1, \quad G(1) = 1 + 2 = 0.$$

Theorem $T2$**.** Every label in $\mathbb{F}_3^2 \oplus \mathbb{F}_3^2$ is obtained by polynomial pairs modulo $x(x-1)$.

Proof. The ideals (x) and $(x-1)$ are coprime, since

$$1 = x - (x-1).$$

The Chinese remainder theorem gives

$$\mathbb{F}_3[x]/(x(x-1)) \cong \mathbb{F}_3[x]/(x) \times \mathbb{F}_3[x]/(x-1) \cong \mathbb{F}_3 \times \mathbb{F}_3.$$

Apply this once to the position labels and once to the momentum labels.

51.9 Step 7: A Quadratic Support Is One Nine-Level Qudit

Take

$$h = q_0 = x^2 + 1.$$

Then

$$Z_{\mathrm{cl}}(h) = \{x_{q_0}\}.$$

The residue field is

$$K_{q_0} = \mathbb{F}_3[\alpha]/(\alpha^2 + 1),$$

which has 9 elements. Thus the reduced Hilbert space is

$$\mathcal{H}_{q_0}^{\mathrm{red}} = \ell^2(K_{q_0}).$$

Choose

$$F = x + 1, \qquad G = 2x.$$

Then the reduced label is

$$(F \bmod q_0,\ G \bmod q_0) = (\alpha + 1,\ 2\alpha) \in K_{q_0}^2.$$

The smeared operator is the one-site 9-level Weyl operator

$$W_{q_0}^{\mathrm{red}}(F, G) = W_{q_0}(\alpha + 1, 2\alpha).$$

51.10 Step 8: Repeated Factors and Artin Data

Take

$$m = x^2 q_0^3.$$

Its reduced closed support is

$$Z_{\mathrm{cl}}(m) = \{x_0, x_{q_0}\}.$$

Therefore the reduced Hilbert dimension is

$$\dim \mathcal{H}_m^{\mathrm{red}} = |\mathbb{F}_3| \cdot |K_{q_0}| = 3 \cdot 9 = 27.$$

The exponents 2 and 3 have disappeared.

In the thickened layer, the local Artin rings are

$$A_0 = \mathbb{F}_3[x]/(x^2), \qquad A_q = \mathbb{F}_3[x]/(q_0^3).$$

Their sizes are

$$|A_0| = 3^2, \qquad |A_q| = 3^{3 \deg q_0} = 3^6.$$

Thus

$$\dim \mathcal{H}_m^{\mathrm{thick}} = |A_0| \cdot |A_q| = 3^8 = 6561.$$

Theorem *T*4. For $m = x^2 q_0^3$, the reduced theory sees two closed-point qudits, while the thickened theory sees jet degrees of freedom of total Hilbert dimension $3^{\deg m} = 3^8$.

Proof. The reduced theory uses only the distinct irreducible factors x and q_0, so its dimension is $3^1 \cdot 3^2 = 27$. The thickened theory uses the Artin rings $R/(x^2)$ and $R/(q_0^3)$, whose sizes are 3^2 and 3^6. Tensoring the two local Hilbert spaces gives dimension 3^8. This equals $3^{\deg m}$, since $\deg m = 2 + 3 \cdot 2 = 8$.

51.11 Conclusion

For $\mathbf{A}^1_{\mathbb{F}_3}$, a regular function such as x or x^{-1} on $D(x)$ is a residue profile with infinite closed support. A nonzero polynomial h becomes finite only when it is used as an equation: it selects $Z_{\mathrm{cl}}(h)$. A Weyl observable then needs labels on that finite support. Repeated factors of h are invisible in the reduced support net and become visible exactly in the Artin-ring thickened layer.

52 Affine Symmetry Action on Regular Functions

52.1 Status and Source Anchors

This shard extends AQM-46 and AQM-49. The group $\mathrm{Aut}_k(\mathbf{A}^1_k)$, its action on closed points, and the residue-field isomorphisms $\tau_{g,\pi}$ are those of AQM-46. Regular functions, residue profiles, reduced supports, and reduced smearing are those of AQM-49. The Artin-ring layer is AQM-36. No new external source is used; the new assertions below are checked algebraic consequences of those shards.

52.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k = \mathbb{F}_q, \quad R = k[x], \quad X = \operatorname{Spec} R, \quad g = g_{a,b} : x \mapsto ax + b. \\
D2 & : & \theta_g(f)(x) = f(ax+b), \quad \sigma_g = \theta_{g^{-1}}, \quad F^g = \sigma_g(F). \\
L1 & : & \sigma_g : R_h \to R_{\sigma_g(h)} \text{ is an isomorphism and } gD(h) = D(\sigma_g(h)). \\
D3 & : & \mathrm{Prof}(U) = \prod_{x_\pi \in U} K_\pi^2. \\
D4 & : & (S_g^\Pi e)_{\pi^g} = (\tau_{g,\pi}^{-1} q_\pi, \tau_{g,\pi}^{-1} p_\pi). \\
T1 & : & \mathrm{ev}_{gU}(F^g, G^g) = S_g^\Pi \, \mathrm{ev}_U(F, G). \\
T2 & : & \text{The affine action preserves the distinction between profiles and finite Weyl labels.} \\
D5 & : & h^g = \sigma_g(h), \quad W_h^{\mathrm{red}}(F,G) = W_{Z_{\mathrm{cl}}(h)}(e_h^{\mathrm{red}}(F,G)). \\
T3 & : & \alpha_g(W_h^{\mathrm{red}}(F,G)) = W_{h^g}^{\mathrm{red}}(F^g, G^g). \\
D6 & : & A_{\pi,e} = R/(\pi^e), \quad A_{\pi^g,e} = R/((\pi^g)^e). \\
T4 & : & \sigma_g \text{ induces } A_{\pi,e} \simeq A_{\pi^g,e} \text{ and preserves the multiplicity } e.
\end{array}$$

52.3 Pushforward of Regular Functions D1–L1

Let $g = g_{a,b}$, with $a \in k^\times$ and $b \in k$. Its pullback on the coordinate ring is

$$\theta_g : R \longrightarrow R, \qquad \theta_g(f)(x) = f(ax+b).$$

For regular functions we use the covariant, or pushforward, notation

$$F^g = g_\# F := \sigma_g(F), \qquad \sigma_g = \theta_{g^{-1}}.$$

Thus, as a function on points,

$$(g_\# F)(y) = F(g^{-1}y).$$

Lemma $L1$**.** If $h \neq 0$, then σ_g extends to a ring isomorphism

$$\sigma_g : R_h \longrightarrow R_{\sigma_g(h)}$$

by

$$\sigma_g(u/h^m) = \sigma_g(u)/\sigma_g(h)^m.$$

Moreover

$$g(D(h)) = D(\sigma_g(h)).$$

Proof. The map σ_g is a k-algebra automorphism of R, with inverse θ_g. It sends the multiplicative set $\{1, h, h^2, \ldots\}$ bijectively to $\{1, \sigma_g(h), \sigma_g(h)^2, \ldots\}$, so it extends uniquely to the displayed localization isomorphism.

For a point $y \in X$,

$$y \in g(D(h)) \iff g^{-1}y \in D(h) \iff h(g^{-1}y) \neq 0 \iff \sigma_g(h)(y) \neq 0.$$

This is exactly $y \in D(\sigma_g(h))$.

Composition has the expected order:

$$(gh)_\# = g_\# h_\#,$$

because $(gh)^{-1} = h^{-1}g^{-1}$.

52.4 Residue Profiles D3–T2

For an open U, define the full residue-profile set

$$\mathrm{Prof}(U) = \prod_{x_\pi \in U} K_\pi^2.$$

The finite-support Weyl label group $E(U)$ of AQM-46 and AQM-49 is the subgroup of $\mathrm{Prof}(U)$ with finite closed support.

Define

$$S_g^\Pi : \mathrm{Prof}(U) \longrightarrow \mathrm{Prof}(gU)$$

by

$$(S_g^\Pi e)_{\pi^g} = \big(\tau_{g,\pi}^{-1}(q_\pi), \tau_{g,\pi}^{-1}(p_\pi)\big),$$

where $e_\pi = (q_\pi, p_\pi)$. On finite-support profiles this is the finite Weyl-label map S_g from AQM-46.

Theorem $T1$**.** For $F, G \in \Gamma(U, \mathcal{O}_X)$,

$$\mathrm{ev}_{gU}(F^g, G^g) = S_g^\Pi \, \mathrm{ev}_U(F, G).$$

Proof. It is enough to check one closed point $x_\pi \in U$. Let $x_{\pi^g} = g(x_\pi)$. The residue map $\tau_{g,\pi} : K_{\pi^g} \to K_\pi$ is induced by θ_g. Therefore, for every $H \in \Gamma(gU, \mathcal{O}_X)$,

$$\tau_{g,\pi}(H(x_{\pi^g})) = (\theta_g H)(x_\pi).$$

Apply this to $H = F^g = \sigma_g(F)$. Since $\theta_g \sigma_g = \mathrm{id}$,

$$\tau_{g,\pi}(F^g(x_{\pi^g})) = F(x_\pi).$$

Thus $F^g(x_{\pi^g}) = \tau_{g,\pi}^{-1}(F(x_\pi))$. The same argument applies to G, proving the displayed profile identity.

Theorem $T2$**.** The affine action never turns a nonzero regular-function profile into a finite Weyl label, and never turns the zero profile into a nonzero one.

Proof. The map S_g^Π is a bijection of closed-point profiles. It sends the zero profile to the zero profile. It sends a finite-support profile to a finite-support profile because $x_\pi \mapsto x_{\pi^g}$ is a bijection of closed points. It sends an infinite-support profile to an infinite-support profile for the same reason. AQM-49 proves that a nonzero regular-function pair on a nonempty standard open has cofinite, hence infinite, closed support. Therefore the profile/operator distinction is invariant under g.

52.5 Reduced Smeared Operators D5–T3

For a nonzero polynomial h, put

$$h^g = \sigma_g(h).$$

This polynomial need not be monic. Multiplying it by a nonzero scalar does not change $D(h^g)$, $Z_{\mathrm{cl}}(h^g)$, or the reduced Weyl support. If π is monic irreducible, then $\pi \mid h$ exactly when $\pi^g \mid h^g$, so

$$g(Z_{\mathrm{cl}}(h)) = Z_{\mathrm{cl}}(h^g).$$

Theorem $T3$. For $F, G \in R$,

$$\alpha_g\Big(W_h^{\mathrm{red}}(F,G)\Big) = W_{h^g}^{\mathrm{red}}(F^g, G^g).$$

Proof. At the source point $x_\pi \in Z_{\mathrm{cl}}(h)$, the reduced label is

$$(F \bmod \pi,\ G \bmod \pi) \in K_\pi^2.$$

At the target point x_{π^g}, Theorem $T1$ gives the label

$$(F^g \bmod \pi^g,\ G^g \bmod \pi^g) = \big(\tau_{g,\pi}^{-1}(F \bmod \pi), \tau_{g,\pi}^{-1}(G \bmod \pi)\big).$$

Thus

$$e_{h^g}^{\mathrm{red}}(F^g, G^g) = S_g e_h^{\mathrm{red}}(F,G).$$

AQM-46 proves

$$\alpha_g(W(e)) = W(S_g e)$$

on finite supports. Substituting the displayed labels proves the theorem.

Consequently the rule is simple: move the support by g, and precompose the regular functions by g^{-1}.

52.6 Artin-Ring Transport D6–T4

Let $h = \prod_i \pi_i^{e_i}$. AQM-49 records that the reduced theory sees only the distinct π_i, while the thickened layer uses

$$A_{\pi_i,e_i} = R/(\pi_i^{e_i}).$$

Theorem $T4$. For each (π, e), the automorphism σ_g induces a ring isomorphism

$$\bar\sigma_{g,\pi,e} : R/(\pi^e) \longrightarrow R/((\pi^g)^e).$$

It sends the class of F to the class of F^g, and it preserves the exponent e.

Proof. AQM-46 gives

$$\pi^g = a^{\deg \pi}\sigma_g(\pi).$$

Hence the ideals $(\sigma_g(\pi)^e)$ and $((\pi^g)^e)$ are equal, because they differ by a unit. Therefore σ_g descends to the displayed quotient isomorphism. The exponent remains e because an isomorphism sends the e-th power of the source ideal to the e-th power of the target ideal.

This is the canonical thickened support transport. A thickened Weyl automorphism also needs a character statement. If the target generating character is transported through $\bar\sigma_{g,\pi,e}$, the same proof as AQM-36 gives Weyl covariance. If one insists on separately chosen default top-coefficient characters at source and target, a dual momentum-label correction must be supplied before claiming character preservation.

52.7 Conclusion

The affine symmetry action on regular functions is:

$$(h, F, G) \longmapsto (h^g, F^g, G^g) = (\sigma_g(h), \sigma_g(F), \sigma_g(G)).$$

On profiles it is the product residue relabelling S_g^Π. On finite reduced Weyl smearings it is exactly the local-net automorphism of AQM-46:

$$\alpha_g W_h^{\mathrm{red}}(F,G) = W_{h^g}^{\mathrm{red}}(F^g, G^g).$$

Thus the symmetry does not add compatibility conditions between closed-point qudits. It only permutes the closed supports and relabels each residue field by the induced field isomorphism.

53 The $\mathbb{F}_3$ Regular-Function Symmetry Action Line by Line

53.1 Status and Source Anchors

This is the $k = \mathbb{F}_3$ companion to AQM-51. It uses AQM-46 for the affine symmetry group, AQM-48 for the full closed-point Hilbert action, AQM-49 for regular functions and reduced smearing, and AQM-50 for the concrete regular-function examples.

53.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & R = \mathbb{F}_3[x],\ X = \operatorname{Spec} R,\ G = \{g_{a,b} : x \mapsto ax + b\}. \\
D2 & : & g_\# F = F(a^{-1}(x-b)). \\
L1 & : & g_\# x = a^{-1}(x-b) \text{ for all six } g. \\
D3 & : & \tau : x \mapsto x+1, \quad \rho : x \mapsto 2x. \\
T1 & : & \operatorname{ev}(g_\# F) = S_g^{\Pi} \operatorname{ev}(F) \text{ at rational points.} \\
D4 & : & h = x(x-1), \quad F = 1, \quad G = x+1. \\
T2 & : & \alpha_\tau W_h^{\mathrm{red}}(F,G) = W_{\tau_\# h}^{\mathrm{red}}(\tau_\# F, \tau_\# G). \\
D5 & : & q_0 = x^2+1, \quad q_1 = x^2+x+2. \\
T3 & : & \tau_\# q_0 = q_1 \text{ and quadratic labels transform by } \tau_{g,\pi}^{-1}. \\
D6 & : & m = x^2 q_0^3. \\
T4 & : & \tau_\# m = (x-1)^2 q_1^3, \text{ preserving reduced and thickened dimensions.}
\end{array}$$

53.3 Step 1: The Six Symmetries D1–L1

For $k = \mathbb{F}_3$,

$$G = \operatorname{Aut}_{\mathbb{F}_3}(X) = \{g_{a,b} : x \mapsto ax + b,\ a \in \{1,2\},\ b \in \mathbb{F}_3\}.$$

By AQM-51,

$$g_\# F = F \circ g^{-1}, \qquad g_\# F(x) = F(a^{-1}(x-b)).$$

Since $1^{-1} = 1$ and $2^{-1} = 2$ in $\mathbb{F}_3$,

g	(a,b)	$g_\# x$
x	$(1,0)$	x
$x+1$	$(1,1)$	$x-1 = x+2$
$x+2$	$(1,2)$	$x-2 = x+1$
$2x$	$(2,0)$	$2x$
$2x+1$	$(2,1)$	$2(x-1) = 2x+1$
$2x+2$	$(2,2)$	$2(x-2) = 2x+2$

Lemma $L1$. The table gives $g_\# x = g^{-1}(x)$ in every case.

Proof. The inverse of $g_{a,b}$ is $g_{a^{-1},-a^{-1}b}$. Therefore $g_\# x = a^{-1}(x-b)$. Substituting the six pairs (a,b) gives the table.

53.4 Step 2: Translation and Reflection D3

Write

$$\tau : x \mapsto x+1, \qquad \rho : x \mapsto 2x.$$

Then $\tau_\# F = F(x-1)$ and $\rho_\# F = F(2x)$. For $F = x$,

$$\tau_\# x = x - 1, \qquad \rho_\# x = 2x.$$

For the local regular function $x^{-1} \in \Gamma(D(x), \mathcal{O}_X)$,

$$\tau_\#(x^{-1}) = (x-1)^{-1} \in \Gamma(D(x-1), \mathcal{O}_X), \rho_\#(x^{-1}) = (2x)^{-1} = 2x^{-1} \in \Gamma(D(x), \mathcal{O}_X).$$

The open $D(x)$ moves to $D(x-1)$ under τ, because x_0 moves to x_1. It is fixed by ρ, because x_0 is fixed by ρ.

53.5 Step 3: Rational-Point Profile Check T1

The rational closed points are $x_r = (x-r)$, $r = 0, 1, 2$. The translation τ sends $x_r \mapsto x_{r+1}$. At rational points the residue fields are all $\mathbb{F}_3$, and the residue-field isomorphism is the identity after evaluation.

Check $F = x$. The source value at x_r is r. The transformed function is $\tau_\# F = x - 1$. At x_{r+1}, $(\tau_\# F)(x_{r+1}) = (r+1) - 1 = r$. Line by line,

source	F	target	$\tau_\# F$
x_0	0	x_1	$1-1=0$
x_1	1	x_2	$2-1=1$
x_2	2	x_0	$0-1=2$

This is exactly $\mathrm{ev}(\tau_\# F) = S_\tau^\Pi \, \mathrm{ev}(F)$.

For x^{-1} on $D(x)$, the source points are x_1, x_2. They move to x_2, x_0. The values are

source	x^{-1}	target	$(x-1)^{-1}$
x_1	1	x_2	$(2-1)^{-1} = 1$
x_2	2	x_0	$(0-1)^{-1} = 2$

Thus local regular functions satisfy the same covariance rule.

53.6 Step 4: A Reduced Two-Qutrit Smearing D4–T2

Use the AQM-50 support and labels

$$h = x(x-1), \qquad F = 1, \qquad G = x + 1.$$

The support is $Z_{\mathrm{cl}}(h) = \{x_0, x_1\}$. The source labels are $x_0 : (1,1)$ and $x_1 : (1,2)$. Under τ,

$$\tau_\# h = (x-1)(x-2), \qquad \tau_\# F = 1, \qquad \tau_\# G = x.$$

The target support is $Z_{\mathrm{cl}}(\tau_\# h) = \{x_1, x_2\}$. At x_1 and x_2, the target labels are $(1,1)$ and $(1,2)$.

Theorem $T2$.

$$\alpha_\tau W^{\mathrm{red}}_{x(x-1)}(1, x+1) = W^{\mathrm{red}}_{(x-1)(x-2)}(1, x).$$

Proof. The source label $((1,1),(1,2))$ at (x_0, x_1) has just been carried to the same ordered values at (x_1, x_2). AQM-46 says $\alpha_\tau W(e) = W(S_\tau e)$, and AQM-51 identifies $S_\tau e$ with the reduced label of $(1, x)$ on the transformed support.

53.7 Step 5: Quadratic Closed Point D5–T3

Let $q_0 = x^2 + 1$ and $q_1 = x^2 + x + 2$. For τ,

$$\tau_{\#} q_0 = q_0(x-1) = (x-1)^2 + 1 = x^2 + x + 2 = q_1.$$

Let $K_{q_0} = \mathbb{F}_3[\alpha]$, $\alpha^2 = 2$, and $K_{q_1} = \mathbb{F}_3[\beta]$, $\beta^2 + \beta + 2 = 0$. The residue pullback is

$$\tau_{\tau,q_0} : K_{q_1} \to K_{q_0}, \qquad \beta \longmapsto \alpha + 1.$$

Hence $\tau_{\tau,q_0}^{-1}(\alpha) = \beta - 1 = \beta + 2$.

Use the AQM-50 labels

$$F = x + 1, \qquad G = 2x$$

on q_0. The source label is $(\alpha + 1,\ 2\alpha)$. The transformed functions are $\tau_{\#} F = x$ and $\tau_{\#} G = 2x + 1$, giving $(\beta,\ 2\beta + 1)$ at q_1.

Theorem *T*3.

$$(\beta,\ 2\beta + 1) = \big(\tau_{\tau,q_0}^{-1}(\alpha + 1), \tau_{\tau,q_0}^{-1}(2\alpha)\big).$$

Proof. Since $\tau_{\tau,q_0}^{-1}(\alpha) = \beta + 2$,

$$\tau_{\tau,q_0}^{-1}(\alpha + 1) = \beta, \qquad \tau_{\tau,q_0}^{-1}(2\alpha) = 2(\beta + 2) = 2\beta + 1.$$

These are exactly the target residues of x and $2x + 1$ at q_1.

53.8 Step 6: Repeated Factors D6–T4

Take the AQM-50 repeated-factor polynomial $m = x^2 q_0^3$. Under τ,

$$\tau_{\#} m = (x-1)^2 q_1^3.$$

The reduced source support is $\{x_0, x_{q_0}\}$, and the reduced target support is $\{x_1, x_{q_1}\}$. The reduced Hilbert dimension is unchanged:

$$3 \cdot 9 = 27.$$

The thickened source rings are $\mathbb{F}_3[x]/(x^2)$ and $\mathbb{F}_3[x]/(q_0^3)$. The target rings are $\mathbb{F}_3[x]/((x-1)^2)$ and $\mathbb{F}_3[x]/(q_1^3)$. The map $F \mapsto F(x-1)$ gives ring isomorphisms from source to target. Therefore the thickened dimension is also unchanged:

$$3^2 \cdot 3^6 = 3^8 = 6561.$$

Theorem *T*4. The affine action preserves the exponents 2 and 3. It moves the closed support and the thickened Artin rings, but it does not turn a repeated factor into extra closed points.

Proof. The equality $\tau_{\#} m = (x-1)^2 q_1^3$ shows that exponents are carried with their irreducible factors. The reduced support keeps only the distinct factors, giving one 3-level site and one 9-level site before and after translation. The thickened rings keep the powers, so their sizes remain 3^2 and 3^6. Thus multiplicity is preserved as Artin-ring thickness, not converted into a larger closed-point set.

53.9 Step 7: The Full Hilbert-Space Statement

The full closed-point Hilbert representation from AQM-48 is

$$\mathcal{H}^0 = \ell^2\left(\bigoplus_{\pi} K_\pi\right),$$

where π runs over all monic irreducibles and vectors have finite closed support. The affine unitary satisfies

$$U_g W_h^{\mathrm{red}}(F, G) U_g^* = W_{g_\# h}^{\mathrm{red}}(g_\# F, g_\# G)$$

for every finite support selector h and polynomial labels F, G.

There is no operator $W(\mathrm{ev}(x))$ in the algebraic finite-support Weyl net, because AQM-50 proves that x has infinite closed support. The symmetry acts on that infinite profile by relabelling it, but the Weyl operator exists only after choosing a finite support such as h, or after adding a new infinite-volume completion convention.

54 Minimal Fermion Geometry Catalogue

54.1 Status and Source Anchors

This shard is a definition catalogue for later fermion constructions. It does not assert that ordinary arithmetic quantum fields automatically contain fermions. The CAR input is AQM-19 and `references/canonical_fields/SOURCES.md`. Supergeometric odd variables, reduced spaces, parity reversal, and Clifford quantization are sourced from `references/supergeometry/SOURCES.md`. Kähler differentials use `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33853, 33856–33866, 34103–34120, 34310–34315, and 34392–34415. Spin structures as square roots of the canonical line bundle use `references/spin_geometry/SOURCES.md`. Logarithmic differentials use `references/log_geometry/SOURCES.md`. The nilpotent Weyl layer is AQM-36. The identity $\Gamma(\mathbf{P}^1, \mathcal{O}) = k$ is sourced by `references/algebraic_geometry/SOURCES.md`.

54.2 Lamport Dependency Skeleton

$$
\begin{array}{rcl}
D1 & : & \text{One complex fermion mode is the one-mode CAR representation.} \\
L1 & : & D1 \text{ satisfies } c^2 = (c^*)^2 = 0,\ \{c, c^*\} = I. \\
D2 & : & \text{A supercommutative } k\text{-algebra is } \mathbb{Z}/2\text{-graded.} \\
L2 & : & \operatorname{char} k \neq 2 \text{ and } \theta \text{ odd imply } \theta^2 = 0. \\
D3 & : & \mathsf{Spt}_k = (\operatorname{Spec} k, k[\theta]) \text{ is the algebraic superpoint.} \\
D4 & : & \Pi\Omega^1_{A/k} \text{ is the cotangent parity-shift mode source.} \\
L3 & : & \Omega^1_{k/k} = 0. \\
L4 & : & \Omega^1_{k[\epsilon]/(\epsilon^2)/k} \cong k\,\mathrm{d}\epsilon \text{ when } \operatorname{char} k \neq 2. \\
D5 & : & (C, D, S, \mu : S^{\otimes 2} \cong \Omega^1_C(\log D)) \text{ is a log-spin datum.} \\
L5 & : & (\mathbf{P}^1_k, \{0, \infty\}, \mathcal{O}, \mathrm{d}\log x) \text{ is a log-spin datum.} \\
L6 & : & \mathbb{F}_3\text{-vector spaces do not canonically scalar-extend to } \mathbb{C}\text{-Hilbert spaces.} \\
T1 & : & \operatorname{Spec} k \text{ has no forced fermion in this catalogue.} \\
T2 & : & \mathsf{Spt}_{\mathbb{F}_3},\ \Pi\Omega^1_{\mathbb{F}_3[\epsilon]/(\epsilon^2)/\mathbb{F}_3}, \text{ and } (\mathbf{P}^1_{\mathbb{F}_3}, \{0, \infty\}, \mathcal{O}) \text{ give one mode each.}
\end{array}
$$

54.3 The One-Mode CAR D1–L1

Let

$$\mathcal{H}_1 = \mathbb{C}|0\rangle \oplus \mathbb{C}|1\rangle.$$

Define linear maps $c, c^* : \mathcal{H}_1 \to \mathcal{H}_1$ by

$$c|0\rangle = 0, \quad c|1\rangle = |0\rangle, \quad c^*|0\rangle = |1\rangle, \quad c^*|1\rangle = 0.$$

The catalogue phrase "one fermion mode" means this CAR representation, or a unitarily isomorphic copy. For r modes it means the antisymmetric Fock space $\bigwedge^\bullet \mathbb{C}^r$, as in AQM-19.

Lemma $L1$. The operators c, c^* satisfy

$$c^2 = (c^*)^2 = 0, \qquad cc^* + c^*c = I_{\mathcal{H}_1}.$$

Proof. Both c^2 and $(c^*)^2$ kill $|0\rangle$ and $|1\rangle$, so they are zero. Also

$$(cc^* + c^*c)|0\rangle = c|1\rangle + 0 = |0\rangle$$

and

$$(cc^* + c^*c)|1\rangle = 0 + c^*|0\rangle = |1\rangle.$$

The two basis vectors span $\mathcal{H}_1$, hence $cc^* + c^*c = I$.

54.4 Odd Variables and the Superpoint D2–D3

Assume char $k \neq 2$ in this subsection. A supercommutative k-algebra is a $\mathbb{Z}/2$-graded algebra

$$A = A_{\bar{0}} \oplus A_{\bar{1}}$$

such that homogeneous elements satisfy

$$uv = (-1)^{|u||v|} vu.$$

Elements of $A_{\bar{0}}$ are even; elements of $A_{\bar{1}}$ are odd.

Lemma *L*2. If $\theta \in A_{\bar{1}}$, then $\theta^2 = 0$.

Proof. Supercommutativity gives $\theta\theta = (-1)\theta\theta$. Hence $2\theta^2 = 0$. Since $2 \in k^\times$, multiplication by 2 is injective, so $\theta^2 = 0$.

Define

$$k[\theta] = k \oplus k\theta, \qquad |\theta| = \bar{1}, \qquad \theta^2 = 0.$$

The algebraic superpoint is the locally ringed graded object

$$\mathsf{Spt}_k = (\operatorname{Spec} k, k[\theta]).$$

Its reduced ordinary point is obtained by quotienting by the odd ideal:

$$k[\theta]/(\theta) \cong k.$$

For $k = \mathbb{F}_3$, the ungraded ring underlying $k[\theta]$ is isomorphic to the even dual-number ring $k[\epsilon]/(\epsilon^2)$, by $\theta \mapsto \epsilon$. The difference is the parity assignment, not the ungraded multiplication table.

The CAR quantization convention for Spt_k assigns the odd line $k\theta$ one complex fermion mode *D*1. This is a catalogue convention: the standard Hilbert-space CAR is complex, while the odd geometric line is a k-vector space.

54.5 Cotangent Parity Shift D4–L4

Let A be a commutative k-algebra. The module of Kähler differentials $\Omega^1_{A/k}$ is the universal target for k-derivations $A \to M$, as recorded in the Stacks source. The cotangent parity-shift catalogue entry is

$$\Pi\Omega^1_{A/k},$$

meaning that the module $\Omega^1_{A/k}$ is treated as odd. If $\dim_k \Omega^1_{A/k} = r < \infty$, the standard complex CAR realization uses r complex fermion modes.

Lemma *L*3. $\Omega^1_{k/k} = 0$.

Proof. Every k-derivation $D : k \to M$ kills all elements of k by definition. Thus the functor $M \mapsto \operatorname{Der}_k(k, M)$ is the zero functor. The zero module represents this functor, so the universal module $\Omega^1_{k/k}$ is zero.

Lemma *L4.* Let $A = k[\epsilon]/(\epsilon^2)$ and assume $\operatorname{char} k \neq 2$. Then

$$\Omega^1_{A/k} \cong k\,\mathrm{d}\epsilon.$$

Proof. Let $S = k[t]$ and $I = (t^2)$, so $A = S/I$. The Stacks polynomial-ring lemma gives $\Omega^1_{S/k} = S\,\mathrm{d}t$. The quotient exact sequence for differentials gives

$$I/I^2 \longrightarrow \Omega^1_{S/k} \otimes_S A \longrightarrow \Omega^1_{A/k} \longrightarrow 0,$$

where $t^2 \mapsto \mathrm{d}(t^2) = 2t\,\mathrm{d}t$. Therefore

$$\Omega^1_{A/k} \cong A\,\mathrm{d}\epsilon/(2\epsilon\,\mathrm{d}\epsilon).$$

Since $2 \in k^\times$, this is

$$A\,\mathrm{d}\epsilon/(\epsilon\,\mathrm{d}\epsilon) \cong (A/(\epsilon))\,\mathrm{d}\epsilon \cong k\,\mathrm{d}\epsilon.$$

For $k = \mathbb{F}_3$, the even thickened point $\operatorname{Spec}(\mathbb{F}_3[\epsilon]/(\epsilon^2))$ has two distinct catalogue layers:

$$\ell^2(\mathbb{F}_3[\epsilon]/(\epsilon^2))$$

is the 9-dimensional even Artin-Weyl layer of AQM-36, while $\Pi\Omega^1_{A/k} \cong \mathbb{F}_3\,\mathrm{d}\epsilon$ supplies one optional fermion mode by the cotangent parity-shift convention.

54.6 Log-Spin Datum D5–L5

Let C be a smooth curve over k, and let D be an effective divisor for which the logarithmic one-forms $\Omega^1_C(\log D)$ are defined. A log-spin datum in this catalogue is

$$(C, D, S, \mu), \qquad \mu : S^{\otimes 2} \xrightarrow{\sim} \Omega^1_C(\log D),$$

where S is a line bundle on C. Its algebraic mode source is

$$\Gamma(C, S).$$

If $\dim_k \Gamma(C, S) = r < \infty$, the standard complex CAR realization uses r complex fermion modes. Atiyah's source supports the square-root spin dictionary over compact complex curves; the logarithmic variant is local.

Lemma *L5.* Let $C = \mathbf{P}^1_k$ with affine coordinate x, and let

$$D = \{0, \infty\}.$$

Then $\Omega^1_C(\log D) \cong \mathcal{O}_C$, generated by

$$\omega = \mathrm{d}\log x.$$

Consequently $(C, D, \mathcal{O}_C, \mu)$, with $\mu(1 \otimes 1) = \omega$, is a log-spin datum.

Proof. On $U_0 = \operatorname{Spec} k[x]$, the divisor point 0 has local equation x. The Stacks log source says logarithmic one-forms are generated by ordinary forms and $\mathrm{d}\log x = x^{-1}\mathrm{d}x$. Since $\mathrm{d}x = x\,\mathrm{d}\log x$, the module is generated by $\mathrm{d}\log x$ on U_0. On $U_\infty = \operatorname{Spec} k[u]$, with $u = x^{-1}$, the divisor point ∞ has local equation u, and

$$\mathrm{d}\log x = \mathrm{d}\log(u^{-1}) = -\mathrm{d}\log u.$$

Thus the same form generates the logarithmic module on U_∞. The two descriptions agree on $U_0 \cap U_\infty$, so they glue to a global basis of $\Omega^1_C(\log D)$. Therefore $\Omega^1_C(\log D) \cong \mathcal{O}_C$. The map $\mu : \mathcal{O}_C^{\otimes 2} \to \Omega^1_C(\log D)$ sending $1 \otimes 1$ to ω is an isomorphism.

54.7 Finite-Field Scalar Caveat L6

Lemma *L*6. There is no unital field homomorphism $\mathbb{F}_3 \to \mathbb{C}$.

Proof. If $\iota : \mathbb{F}_3 \to \mathbb{C}$ were unital, then

$$0 = \iota(0) = \iota(1 + 1 + 1) = 1 + 1 + 1 = 3,$$

which is false in $\mathbb{C}$.

Therefore an $\mathbb{F}_3$-vector-space mode source V does not canonically become a complex one-particle Hilbert space by scalar extension. The catalogue records $r = \dim_{\mathbb{F}_3} V$ and then uses a chosen complex Hilbert space $\mathbb{C}^r$ for the standard CAR realization. A genuinely $\mathbb{F}_3$-linear Clifford theory is a different algebraic layer and is not defined here.

54.8 Catalogue Consequences T1–T2

Theorem *T*1. The ordinary point $\operatorname{Spec} k$ has no forced fermion mode in this catalogue.

Proof. The superpoint entry requires an odd structure sheaf $k[\theta]$; the ordinary structure sheaf k has no odd part. The cotangent parity-shift entry gives no mode by Lemma *L*3. The log-spin entry requires a curve, a divisor, and a square-root line bundle; none is part of the ordinary point $\operatorname{Spec} k$. Thus no catalogue mechanism above assigns a nonzero fermion mode to $\operatorname{Spec} k$ without extra data.

Theorem *T*2. For $k = \mathbb{F}_3$, the following are the minimal one-mode entries:

geometric datum	mode source	$\mathbb{F}_3$-dimension	complex CAR Hilbert space
$\mathsf{Spt}_{\mathbb{F}_3}$	$\mathbb{F}_3\theta$	1	$\mathcal{H}_1$
$\operatorname{Spec}(\mathbb{F}_3[\epsilon]/(\epsilon^2))$ with $\Pi\Omega^1$	$\mathbb{F}_3\, \mathrm{d}\epsilon$	1	$\mathcal{H}_1$
$(\mathbf{P}^1_{\mathbb{F}_3}, \{0, \infty\}, \mathcal{O})$	$\Gamma(\mathbf{P}^1, \mathcal{O}) = \mathbb{F}_3$	1	$\mathcal{H}_1$

The ordinary closed point $\operatorname{Spec} \mathbb{F}_3$ contributes zero fermion modes.

Proof. The superpoint row is Definition *D*3. The dual-number row is Lemma *L*4. The log-spin row is Lemma *L*5 plus the cited projective-space cohomology calculation $\Gamma(\mathbf{P}^1, \mathcal{O}) = \mathbb{F}_3$. Lemma *L*6 explains why each row determines the standard complex CAR Hilbert space only through the dimension-one mode count. The last sentence is Theorem *T*1 with $k = \mathbb{F}_3$.

55 Derivative Dynamics from Kähler Differentials over $\mathbb{F}_3$

55.1 Status and Source Anchors

This shard is a checked definition-and-example shard plus one labelled proposal. The definitions of derivation, Kähler differentials, their universal property, the quotient exact sequence, polynomial differentials, and the de Rham complex are sourced from the Stacks algebra snapshot `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33872, 34103–34120, 34310–34340, and 34388–34524. Tangent space by dual numbers and the cotangent-tangent duality are sourced from `references/algebraic_geometry/stacks_project_varieties.tex`, lines 2915–3019. Residue fields are sourced from `references/algebraic_geometry/stacks_project_algebra.tex`, lines 250–278. AQM-08 and AQM-09 supply the CSS/stabilizer dictionary, AQM-36 supplies the even Artin-Weyl layer for dual numbers, and AQM-53 supplies the cotangent parity-shift comparison. The convention for this shard is `CONVENTIONS.md` (ao).

55.2 Lamport Dependency Skeleton

$$\begin{array}{lcl}
D1 & : & k = \mathbb{F}_3, \quad A \text{ is a finite commutative } k\text{-algebra.} \\
D2 & : & \mathrm{Der}_k(A, M) \text{ is the } A\text{-module of } k\text{-derivations } A \to M. \\
D3 & : & \Omega^1_{A/k} \text{ represents } M \mapsto \mathrm{Der}_k(A, M). \\
D4 & : & T^*_{A/k,x} = \Omega^1_{A/k} \otimes_A \kappa(x), \quad T_{A/k,x} = \mathrm{Hom}_A(\Omega^1_{A/k}, \kappa(x)). \\
D5 & : & A \xrightarrow{d_0} \Omega^1_{A/k} \xrightarrow{d_1} \Omega^2_{A/k} \to \cdots \text{ is the algebraic de Rham complex.} \\
P1 & : & \text{The de Rham CSS test puts qudits on } C^1 = \Omega^1_{A/k}. \\
L1 & : & \Omega^1_{\mathbb{F}_3/\mathbb{F}_3} = 0. \\
L2 & : & \Omega^1_{\mathbb{F}_3[\epsilon]/(\epsilon^2)/\mathbb{F}_3} \cong \mathbb{F}_3\, d\epsilon. \\
T1 & : & \text{For } A = \mathbb{F}_3[\epsilon]/(\epsilon^2), \text{ the de Rham CSS test has one } X\text{-check and no logical qudit.}
\end{array}$$

55.3 Definitions D1–D5

Let $k \to A$ be a ring map and let M be an A-module. A k-derivation is an additive map $D : A \to M$ satisfying

$$D(c) = 0 \quad (c \in k), \qquad D(ab) = aD(b) + bD(a).$$

The Stacks source denotes the resulting A-module by $\mathrm{Der}_k(A, M)$.

The module of Kähler differentials $\Omega^1_{A/k}$, with universal derivation $d : A \to \Omega^1_{A/k}$, is characterized by the functorial identity

$$\mathrm{Hom}_A(\Omega^1_{A/k}, M) \cong \mathrm{Der}_k(A, M).$$

Thus $\Omega^1_{A/k}$ is the algebraic cotangent object before any quantization choice.

If $x \in \mathrm{Spec}\, A$, with residue field $\kappa(x)$, the cotangent fibre is

$$T^*_{A/k,x} = \Omega^1_{A/k} \otimes_A \kappa(x),$$

and the tangent space is

$$T_{A/k,x} = \mathrm{Hom}_A(\Omega^1_{A/k}, \kappa(x)).$$

This is the derivative geometry missing from the bare Weyl-Heisenberg kinematics: Weyl labels give conjugate translations and phases, while Ω^1 records which first-order directions are geometric.

The algebraic de Rham complex is

$$A = \Omega^0_{A/k} \xrightarrow{d_0} \Omega^1_{A/k} \xrightarrow{d_1} \Omega^2_{A/k} \xrightarrow{d_2} \cdots ,$$

where $\Omega^i_{A/k} = \wedge^i_A \Omega^1_{A/k}$. The Stacks construction gives $d_{i+1}d_i = 0$. For finite k-algebras, each term is a finite k-vector space in the examples below.

55.4 Calculable Dynamical Proposals P1

The first proposal to test is the *de Rham CSS test*. Choose finite k-bases for

$$C^0 = A, \qquad C^1 = \Omega^1_{A/k}, \qquad C^2 = \Omega^2_{A/k}.$$

Put qudits on C^1. With the basis dot pairing, form

$$R_X = \mathrm{im}(d_0) \subset C^1, \qquad R_Z = \mathrm{im}(d_1^T) \subset C^1.$$

Since $d_1 d_0 = 0$, every element of R_X is orthogonal to every element of R_Z. By AQM-08, this is exactly the CSS isotropy condition. After choosing compatible phases, this gives commuting stabilizer equations. This is a candidate dynamical principle: exact one-forms and coexact one-forms become local checks, and the encoded-qudit count is controlled by the finite de Rham cohomology

$$H^1_{\mathrm{dR}}(A/k) = \ker d_1 / \operatorname{im} d_0.$$

The full CSS logical Pauli module is larger than this cohomology half: with the chosen finite bases and dot pairing it is the symplectic reduction

$$L^\perp/L \cong \left(\ker d_1 / \operatorname{im} d_0\right) \oplus \left(\ker d_0^T / \operatorname{im} d_1^T\right),$$

matching the cohomology-plus-dual/homology convention of AQM-08 and `CONVENTIONS.md` (l).

This test deliberately does not claim that the ring alone chooses all phases, a Hamiltonian normalization, or a metric. Three further calculable branches are therefore left as proposals:

cotangent-Weyl test:	quantize $T^*_{A/k,x}$ or $\Omega^1_{A/k}$ directly;
Artin-jet comparison:	compare Ω^1 with the even A-Weyl layer of AQM-36;
Hodge/Laplacian test:	add a pairing or trace density and form $d^*d + dd^*$.

The last branch is least canonical because d^* requires extra metric or pairing data.

55.5 Example 1: The Field $A = \mathbb{F}_3$ L1

Let $A = k = \mathbb{F}_3$. A k-derivation $D : k \to M$ must annihilate every element of the image of k. Since the image is all of A,

$$D(0) = D(1) = D(2) = 0.$$

Therefore $\mathrm{Der}_k(k, M) = 0$ for every k-module M. By the universal property,

$$\mathrm{Hom}_k(\Omega^1_{k/k}, M) = 0 \quad \text{for all } M,$$

so

$$\Omega^1_{k/k} = 0.$$

The unique point is $x = (0)$, with $\kappa(x) = k$. Hence

$$T^*_{k/k,x} = 0 \otimes_k k = 0, \qquad T_{k/k,x} = \mathrm{Hom}_k(0, k) = 0.$$

The de Rham complex begins and ends as

$$k \longrightarrow 0 \longrightarrow 0.$$

The de Rham CSS test has no C^1 site. Thus a reduced one-point field contributes no intrinsic derivative stabilizer. This agrees with AQM-40's caveat: the ambient Weyl system may exist, but stabilizer data do not come for free from a geometrically discrete point.

55.6 Example 2: $A = \mathbb{F}_3[\epsilon]/(\epsilon^2)$ L2–T1

Let

$$A = k[\epsilon]/(\epsilon^2), \qquad k = \mathbb{F}_3.$$

Write $A = k \oplus k\epsilon$, with $\epsilon^2 = 0$. Present A as

$$S/I, \qquad S = k[t], \quad I = (t^2),$$

and let ϵ be the image of t. The Stacks polynomial-ring lemma gives

$$\Omega^1_{S/k} = S\,dt.$$

The quotient exact sequence gives

$$I/I^2 \longrightarrow \Omega^1_{S/k} \otimes_S A \longrightarrow \Omega^1_{A/k} \longrightarrow 0,$$

where $t^2 \in I$ maps to

$$d(t^2) \otimes 1 = 2t\,dt \otimes 1 = 2\epsilon\,d\epsilon.$$

In $\mathbb{F}_3$, the scalar 2 is invertible. Thus the relation is

$$\epsilon\,d\epsilon = 0.$$

Consequently

$$\Omega^1_{A/k} \cong A\,d\epsilon/(\epsilon\,d\epsilon) \cong (A/(\epsilon))\,d\epsilon \cong k\,d\epsilon.$$

The universal derivation is explicit:

$$d(a + b\epsilon) = b\,d\epsilon, \qquad a, b \in k.$$

The spectrum has one prime, $\mathfrak{m} = (\epsilon)$, and its residue field is

$$\kappa(\mathfrak{m}) = A/\mathfrak{m} \cong k.$$

Therefore the cotangent fibre is

$$T^*_{A/k,\mathfrak{m}} = \Omega^1_{A/k} \otimes_A k \cong k\,d\epsilon,$$

and the tangent space is one-dimensional:

$$T_{A/k,\mathfrak{m}} = \mathrm{Hom}_A(\Omega^1_{A/k}, k) \cong k.$$

The tangent vector v with $v(d\epsilon) = 1$ corresponds to the derivation

$$D_v(a + b\epsilon) = b.$$

Since $\Omega^1_{A/k}$ is generated by one element, its exterior square is zero:

$$\Omega^2_{A/k} = 0.$$

Thus the de Rham complex is

$$A \xrightarrow{d_0} k\,d\epsilon \longrightarrow 0, \qquad d_0(a + b\epsilon) = b\,d\epsilon.$$

The kernel is k, the image is all of $k\,d\epsilon$, and hence

$$H^0_{\mathrm{dR}}(A/k) = k, \qquad H^1_{\mathrm{dR}}(A/k) = 0.$$

Now apply the de Rham CSS test. Choose the basis $d\epsilon$ of $C^1 = k\,d\epsilon$. Then

$$R_X = \operatorname{im} d_0 = k\,d\epsilon, \qquad R_Z = 0.$$

The Pauli label space for the single C^1-qutrit is $k \oplus k$, and the candidate stabilizer label subspace is

$$L = (R_X \oplus 0) = k \oplus 0.$$

It is isotropic because all X-type labels commute with each other. Its rank is 1, so the corresponding all-+1 stabilizer space has dimension

$$3^{1-1} = 1.$$

This is the first toy calculation where the nilpotent first-order geometry selects a nontrivial stabilizer equation. It is not yet a general dynamical law; it is a checked finite test case for convention (ao).

56 More $\mathbb{F}_3$ Examples for the de Rham CSS Prototype

56.1 Status, Sources, and Common Rule

This shard is a follow-up calculation for the finite chain-complex prototype in AQM-54 and convention `CONVENTIONS.md` (ao), not the canonical tangent-cotangent Weyl quantisation. The prototype chooses a finite $C^1 = \Omega^1_{A/k}$ basis, attaches auxiliary qudits to it, and tests whether the de Rham differential selects commuting Pauli operators. All rings below are finite-dimensional coordinate rings over

$$k = \mathbb{F}_3.$$

The derivative input is the module of Kähler differentials $\Omega^1_{A/k}$; the prototype uses auxiliary C^1-qutrits and selects CSS support spaces

$$R_X = \mathrm{im}(d_0 : C^0 \to C^1), \qquad R_Z = \mathrm{im}(d_1^T : C^{2*} \to C^{1*}),$$

with $d_1 d_0 = 0$ giving the CSS commutation check. The source anchors are the same as AQM-54: Stacks algebra lines 33793–33872, 34103–34120, 34310–34340, and 34388–34524. AQM-08 supplies the CSS stabilizer dictionary. The common one-variable calculation is this. Let $A = k[t]/(f)$, $\tau = t \bmod (f)$. For $S = k[t]$, the polynomial differential rule gives $\Omega^1_{S/k} = S\,dt$. The quotient exact sequence sends f to $df = f'(t)\,dt$, hence

$$\Omega^1_{A/k} \cong A\,d\tau/(f'(\tau)d\tau).$$

Because $\Omega^1_{A/k}$ is generated by one symbol in all one-variable examples, $\Omega^2_{A/k} = 0$ there and $R_Z = 0$.

56.2 Lamport Registers for the Five Examples

$$\begin{array}{lcl} E0 & : & A_0 = k[t]/(t) \cong k. \\ E1 & : & A_1 = k[t]/(t^2-1), \quad \text{the split quadratic two-point ring.} \\ E2 & : & A_2 = k[t]/(t^2+1), \quad \text{the irreducible quadratic field point.} \\ E3 & : & A_3 = k[t]/(t^2), \quad \text{the double point at 0.} \\ E4 & : & A_4 = k[u,v]/(u^2,v^2), \quad \text{the quadratic fat plane point.} \end{array}$$

The examples are ordered so that $E0, E1, E2$ test the reduced squarefree cases, while $E3, E4$ test nilpotent quadratic directions.

56.3 E0: The Linear Point

Let $A_0 = k[t]/(t)$, and let $\tau = 0$ be the image of t. Here $f = t$, $f' = 1$. The common rule gives

$$\Omega^1_{A_0/k} = A_0\,d\tau/(1 \cdot d\tau) = 0.$$

Thus

$$C^1 = 0, \qquad R_X = 0, \qquad R_Z = 0.$$

The de Rham CSS test selects no qutrit and no stabilizer generator. This is the same result as AQM-54 Example 1, now recorded as the linear hypersurface case $t = 0$.

56.4 E1: The Split Quadratic Two-Point Ring

Let

$$A_1 = k[t]/(t^2-1), \qquad \tau = t \bmod (t^2-1).$$

The relation is $\tau^2 = 1$. The defining polynomial and derivative are

$$f = t^2 - 1, \qquad f' = 2t.$$

In A_1,

$$(2\tau)(2\tau) = 4\tau^2 = \tau^2 = 1,$$

because $4 = 1$ in k. Hence 2τ is a unit. The quotient relation is therefore

$$2\tau\, d\tau = 0,$$

and multiplication by the inverse of 2τ gives

$$d\tau = 0.$$

Consequently

$$\Omega^1_{A_1/k} = 0.$$

The de Rham CSS data are

$$C^1 = 0, \qquad R_X = 0, \qquad R_Z = 0.$$

Thus the split reduced quadratic coordinate ring contributes two reduced points but no intrinsic derivative qutrit. In this test, reduced squarefree finite geometry has kinematics from the Weyl layer but no derivative stabilizer selected by Ω^1.

56.5 E2: The Irreducible Quadratic Field Point

Let

$$A_2 = k[t]/(t^2+1), \qquad \tau = t \bmod (t^2+1).$$

Since $0^2+1 = 1$, $1^2+1 = 2$, and $2^2+1 = 5 = 2$ in k, the polynomial t^2+1 has no k-root. The ring A_2 is the quadratic residue-field extension represented by the closed point t^2+1. Its relation is

$$\tau^2 = -1 = 2.$$

Again

$$f = t^2 + 1, \qquad f' = 2t.$$

In A_2,

$$(2\tau)\tau = 2\tau^2 = 2 \cdot 2 = 4 = 1.$$

Thus 2τ is a unit, now with inverse τ. The quotient relation

$$2\tau\, d\tau = 0$$

forces

$$d\tau = 0, \qquad \Omega^1_{A_2/k} = 0.$$

Therefore

$$C^1 = 0, \qquad R_X = 0, \qquad R_Z = 0.$$

The irreducible quadratic closed point behaves like the split squarefree quadratic for this derivative test: separability kills first differentials.

56.6 E3: The Quadratic Double Point

Let

$$A_3 = k[t]/(t^2), \qquad \epsilon = t \bmod (t^2).$$

Then $A_3 = k \oplus k\epsilon$ and $\epsilon^2 = 0$. Here

$$f = t^2, \qquad f' = 2t.$$

The common rule gives

$$\Omega^1_{A_3/k} = A_3\, d\epsilon/(2\epsilon\, d\epsilon).$$

Since 2 is a unit in k, this is

$$\Omega^1_{A_3/k} = A_3\, d\epsilon/(\epsilon\, d\epsilon) \cong k\, d\epsilon.$$

The universal derivation is computed on the basis $1, \epsilon$:

$$d_0(a + b\epsilon) = b\, d\epsilon.$$

Thus

$$C^1 = k\, d\epsilon, \qquad R_X = \operatorname{im} d_0 = k\, d\epsilon.$$

Because C^1 has one generator, $C^2 = \Omega^2_{A_3/k} = 0$, so

$$R_Z = 0.$$

In the prototype, with one auxiliary qutrit labelled by $e_1 = d\epsilon$, the selected stabilizer is

$$X(e_1).$$

There is no Z-check. The stabilizer rank is 1 on one qutrit, so the all-+1 stabilizer space has dimension $3^{1-1} = 1$.

56.7 E4: The Quadratic Fat Plane Point

Let

$$A_4 = k[u, v]/(u^2, v^2),$$

where u, v also denote the residue classes. A k-basis is

$$1, \quad u, \quad v, \quad uv,$$

with $u^2 = v^2 = 0$. Start from $S = k[U, V]$. The polynomial differential rule gives

$$\Omega^1_{S/k} = S\, dU \oplus S\, dV.$$

The quotient ideal is (U^2, V^2). The two differential relations are

$$d(U^2) = 2U\, dU, \qquad d(V^2) = 2V\, dV.$$

Passing to A_4, and using that 2 is a unit, gives

$$u\, du = 0, \qquad v\, dv = 0.$$

Therefore

$$\Omega^1_{A_4/k} = (A_4\, du \oplus A_4\, dv)/(u\, du,\ v\, dv).$$

The $A_4\,du$ part has basis $du, v\,du$, since $u\,du = 0$ and hence $uv\,du = v(u\,du) = 0$. The $A_4\,dv$ part has basis $dv, u\,dv$, since $v\,dv = 0$ and hence $uv\,dv = u(v\,dv) = 0$. Set

$$e_1 = du, \quad e_2 = v\,du, \quad e_3 = dv, \quad e_4 = u\,dv.$$

Thus $C^1 = \Omega^1_{A_4/k} \cong k^4$, so the prototype has four auxiliary qutrits. The derivation $d_0 : A_4 \to C^1$ is fixed on the basis:

$$\begin{aligned} d_0(1) &= 0, \\ d_0(u) &= e_1, \\ d_0(v) &= e_3, \\ d_0(uv) &= u\,dv + v\,du = e_4 + e_2. \end{aligned}$$

Hence

$$R_X = \langle e_1,\ e_3,\ e_2 + e_4 \rangle_k.$$

Now compute $C^2 = \Omega^2_{A_4/k}$. It is generated by

$$\omega = du \wedge dv.$$

Wedging $u\,du = 0$ with dv gives $u\omega = 0$, and wedging $v\,dv = 0$ with du gives $v\omega = 0$. Hence

$$C^2 = \Omega^2_{A_4/k} \cong k\,\omega.$$

The de Rham differential $d_1 : C^1 \to C^2$ is

$$\begin{aligned} d_1(e_1) &= d(du) = 0, \\ d_1(e_2) &= d(v\,du) = dv \wedge du = -\omega = 2\omega, \\ d_1(e_3) &= d(dv) = 0, \\ d_1(e_4) &= d(u\,dv) = du \wedge dv = \omega. \end{aligned}$$

With the ordered bases e_1, e_2, e_3, e_4 and ω, the matrix of d_1 is

$$\begin{bmatrix} 0 & 2 & 0 & 1 \end{bmatrix}.$$

Thus the Z-support selected by the transpose is

$$R_Z = \langle 2e_2 + e_4 \rangle_k.$$

The CSS orthogonality is visible without abbreviation:

$$e_1 \cdot (2e_2 + e_4) = 0, \quad e_3 \cdot (2e_2 + e_4) = 0, \quad (e_2 + e_4) \cdot (2e_2 + e_4) = 2 + 1 = 0.$$

Therefore the selected stabilizer generators on the four qutrits are

$$X_1, \qquad X_3, \qquad X_2X_4, \qquad Z_2^2 Z_4.$$

Their rank is 4, so the all-+1 stabilizer space has dimension

$$3^{4-4} = 1.$$

56.8 End Summary: Prototype Qudits and Stabilizers

For C^1-basis vector e_i, write $X_i = X(e_i)$ and $Z_i = Z(e_i)$. The de Rham CSS test gives:

example	C^1-qutrits	X-checks from R_X	Z-checks from R_Z
$E0:\ k[t]/(t)$	none	none	none
$E1:\ k[t]/(t^2-1)$	none	none	none
$E2:\ k[t]/(t^2+1)$	none	none	none
$E3:\ k[t]/(t^2)$	$e_1 = d\epsilon$	X_1	none
$E4:\ k[u,v]/(u^2,v^2)$	$e_1 = du,\ e_2 = vdu,\ e_3 = dv,\ e_4 = udv$	$X_1,\ X_3,\ X_2X_4$	$Z_2^2 Z_4$

Thus this prototype detects nilpotent or singular first-order algebra in C^1 and is silent on these squarefree finite reduced examples. The canonical tangent-Weyl reading must instead pass through the pointwise spaces $T_PX \oplus T_P^*X$ and their Weyl representations.

57 $\mathbb{F}_2$ Qubit Examples for the de Rham CSS Prototype

57.1 Status, Sources, and Characteristic-Two Warning

This shard continues the finite chain-complex prototype of AQM-54 and AQM-55 with $k = \mathbb{F}_2$. Its qubits are auxiliary C^1-basis qubits, not the Hilbert carriers obtained by quantising the pointwise $T_P X \oplus T_P^* X$ label spaces. The source anchors are unchanged: Stacks algebra lines 33793–33872, 34103–34120, 34310–34340, and 34388–34524. AQM-08 supplies the CSS/stabilizer dictionary, and `CONVENTIONS.md` (ao) names the `de_rham_css_test`. The characteristic-two edge case is immediate from the polynomial rule:

$$d(t^2) = 2t\,dt = 0.$$

Thus a square-zero equation $t^2 = 0$ imposes no relation on $t\,dt$. This is the opposite of the $\mathbb{F}_3$ dual-number calculation in AQM-54, where 2 is invertible and $\epsilon\,d\epsilon = 0$. A squarefree control still behaves normally: for $k[t]/(t^2 + t)$, $d(t^2 + t) = dt$, hence $\Omega^1 = 0$. The examples below deliberately use nilpotent quadratic relations, because those are where the qubit CSS test sees structure.

57.2 Lamport Registers

$$\begin{array}{lcl}
F0 & : & A_0 = k[t]/(t^6), \quad 6 \text{ qubits.} \\
F1 & : & A_1 = k[t]/(t^{10}), \quad 10 \text{ qubits.} \\
F2 & : & A_2 = k[x, y]/(x^2, y^2, xy), \quad 5 \text{ qubits.} \\
F3 & : & A_3 = k[x, y]/(x^2, y^2), \quad 8 \text{ qubits.} \\
F4 & : & A_4 = k[x, y, z]/(x, y, z)^2, \quad 9 \text{ qubits.}
\end{array}$$

For a prototype C^1-basis vector e_i, write $X_i = X(e_i)$ and $Z_i = Z(e_i)$. All stabilizers below have $+1$ phase; the ring calculation fixes only the support spaces.

57.3 F0: The Length-Six Fat Line

Let $A_0 = k[t]/(t^6)$, and write $\tau = t \bmod (t^6)$. The k-basis is

$$1, \tau, \tau^2, \tau^3, \tau^4, \tau^5.$$

The quotient rule gives

$$\Omega^1_{A_0/k} = A_0\,d\tau/(d(\tau^6)).$$

Since $d(\tau^6) = 6\tau^5\,d\tau = 0$ in k, no relation is imposed. Set

$$e_i = \tau^{i-1} d\tau, \qquad 1 \le i \le 6.$$

Then $C^1 = \Omega^1_{A_0/k} \cong k^6$, so the prototype has six auxiliary qubits. The universal derivation is

$$d_0(\tau^j) = j\,\tau^{j-1} d\tau,$$

so only odd j contribute:

$$d_0(\tau) = e_1, \quad d_0(\tau^3) = e_3, \quad d_0(\tau^5) = e_5.$$

Hence

$$R_X = \langle e_1, e_3, e_5 \rangle, \qquad R_Z = 0,$$

because a cyclic one-form module has $\Omega^2 = 0$. The selected stabilizers are

$$X_1, \qquad X_3, \qquad X_5.$$

The rank is 3 on 6 qubits, leaving $6 - 3 = 3$ logical qubits.

57.4 F1: The Length-Ten Fat Line

Let $A_1 = k[t]/(t^{10})$, with $\tau = t \bmod (t^{10})$. Its basis is

$$1, \tau, \tau^2, \tau^3, \tau^4, \tau^5, \tau^6, \tau^7, \tau^8, \tau^9.$$

Again

$$d(\tau^{10}) = 10\tau^9 \, d\tau = 0,$$

so $\Omega^1_{A_1/k} = A_1 \, d\tau$. Set

$$e_i = \tau^{i-1} d\tau, \qquad 1 \le i \le 10.$$

Thus $C^1 \cong k^{10}$, giving ten auxiliary prototype qubits. The same parity calculation gives

$$d_0(\tau) = e_1, \quad d_0(\tau^3) = e_3, \quad d_0(\tau^5) = e_5, \quad d_0(\tau^7) = e_7, \quad d_0(\tau^9) = e_9,$$

while $d_0(1), d_0(\tau^2), d_0(\tau^4), d_0(\tau^6), d_0(\tau^8)$ vanish. Therefore

$$R_X = \langle e_1, e_3, e_5, e_7, e_9 \rangle, \qquad R_Z = 0.$$

The selected stabilizers are

$$X_1, \qquad X_3, \qquad X_5, \qquad X_7, \qquad X_9.$$

The stabilizer rank is 5 on 10 qubits, so the code space has $10 - 5 = 5$ logical qubits.

57.5 F2: The Two-Variable Square-Zero Point

Let

$$A_2 = k[x, y]/(x^2, y^2, xy).$$

The basis is $1, x, y$. The square relations give no differential relation:

$$d(x^2) = 0, \qquad d(y^2) = 0.$$

The mixed relation gives

$$d(xy) = x \, dy + y \, dx = 0.$$

Thus $x \, dy = y \, dx$. A C^1-basis is

$$e_1 = dx, \quad e_2 = x \, dx, \quad e_3 = dy, \quad e_4 = y \, dy, \quad e_5 = x \, dy = y \, dx.$$

The prototype has five auxiliary qubits. The derivation on the basis of A_2 is

$$d_0(1) = 0, \qquad d_0(x) = e_1, \qquad d_0(y) = e_3,$$

so

$$R_X = \langle e_1, e_3 \rangle.$$

Let $\omega = dx \wedge dy$. Wedging $x \, dy + y \, dx = 0$ with dx and dy gives $x\omega = 0$ and $y\omega = 0$, so $C^2 = k\omega$. Now

$$d_1(e_1) = d_1(e_2) = d_1(e_3) = d_1(e_4) = 0, \qquad d_1(e_5) = \omega.$$

Hence $R_Z = \langle e_5 \rangle$. The selected stabilizers are

$$X_1, \qquad X_3, \qquad Z_5.$$

The rank is 3 on 5 qubits, leaving 2 logical qubits.

57.6 F3: The Two-Variable Quadratic Fat Point

Let

$$A_3 = k[x,y]/(x^2,y^2).$$

The basis is $1, x, y, xy$. Since $d(x^2) = d(y^2) = 0$, the module $\Omega^1_{A_3/k}$ is freely generated over A_3 by dx, dy. Use

$$\begin{aligned} &e_1 = dx, \quad e_2 = x\,dx, \quad e_3 = y\,dx, \quad e_4 = xy\,dx, \\ &e_5 = dy, \quad e_6 = x\,dy, \quad e_7 = y\,dy, \quad e_8 = xy\,dy. \end{aligned}$$

Thus the prototype has eight auxiliary qubits. The derivation is

$$d_0(x) = e_1, \qquad d_0(y) = e_5, \qquad d_0(xy) = x\,dy + y\,dx = e_6 + e_3.$$

Therefore

$$R_X = \langle e_1, e_5, e_3 + e_6 \rangle.$$

Let $\omega = dx \wedge dy$; then $C^2 = A_3\omega$ has basis $\omega, x\omega, y\omega, xy\omega$. The nonzero d_1-values are

$$d_1(e_3) = \omega, \quad d_1(e_4) = x\omega, \quad d_1(e_6) = \omega, \quad d_1(e_8) = y\omega.$$

Hence

$$R_Z = \langle e_3 + e_6,\ e_4,\ e_8 \rangle.$$

The shared support $e_3 + e_6$ is not a mistake: in characteristic two its self-dot-product is $1 + 1 = 0$. The selected stabilizers are

$$X_1, \qquad X_5, \qquad X_3X_6, \qquad Z_3Z_6, \qquad Z_4, \qquad Z_8.$$

The rank is 6 on 8 qubits, leaving 2 logical qubits.

57.7 F4: The Three-Variable Square-Zero Tangent Space

Let

$$A_4 = k[x,y,z]/(x,y,z)^2.$$

The basis is $1, x, y, z$. The relations $x^2 = y^2 = z^2 = xy = xz = yz = 0$ give

$$x\,dy = y\,dx, \qquad x\,dz = z\,dx, \qquad y\,dz = z\,dy,$$

while the square relations again give no $x\,dx$, $y\,dy$, or $z\,dz$ relation. Choose

$$\begin{aligned} &e_1 = dx, && e_2 = dy, && e_3 = dz, \\ &e_4 = x\,dx, && e_5 = y\,dy, && e_6 = z\,dz, \\ &e_7 = x\,dy = y\,dx, && e_8 = x\,dz = z\,dx, && e_9 = y\,dz = z\,dy. \end{aligned}$$

Thus $C^1 \cong k^9$, so the prototype has nine auxiliary qubits. The image of d_0 is

$$R_X = \langle e_1, e_2, e_3 \rangle.$$

For C^2, set

$$\omega_{xy} = dx \wedge dy, \quad \omega_{xz} = dx \wedge dz, \quad \omega_{yz} = dy \wedge dz.$$

The wedged mixed relations make the coefficient terms compatible; the only extra coefficient direction needed below is

$$\eta = x\omega_{yz} = y\omega_{xz} = z\omega_{xy}.$$

Indeed, wedging $x\,dy + y\,dx = 0$, $x\,dz + z\,dx = 0$, and $y\,dz + z\,dy = 0$ with dx, dy, dz gives

$$\begin{aligned} x\omega_{xy} &= 0, \quad y\omega_{xy} = 0, \quad x\omega_{yz} = y\omega_{xz}, \\ x\omega_{xz} &= 0, \quad z\omega_{xz} = 0, \quad x\omega_{yz} = z\omega_{xy}, \\ y\omega_{yz} &= 0, \quad z\omega_{yz} = 0, \quad y\omega_{xz} = z\omega_{xy}. \end{aligned}$$

The d_1-values relevant for R_Z are

$$d_1(e_7) = \omega_{xy}, \qquad d_1(e_8) = \omega_{xz}, \qquad d_1(e_9) = \omega_{yz},$$

and $d_1(e_i) = 0$ for $1 \le i \le 6$. Therefore

$$R_Z = \langle e_7, e_8, e_9 \rangle.$$

The selected stabilizers are

$$X_1, \qquad X_2, \qquad X_3, \qquad Z_7, \qquad Z_8, \qquad Z_9.$$

The rank is 6 on 9 qubits, leaving 3 logical qubits.

57.8 End Summary: Prototype Qubits and de Rham CSS Stabilizers

ex.	A	$\#C^1$	X-checks	Z-checks
$F0$	$k[t]/(t^6)$	6	X_1, X_3, X_5	none
$F1$	$k[t]/(t^{10})$	10	X_1, X_3, X_5, X_7, X_9	none
$F2$	$k[x,y]/(x^2, y^2, xy)$	5	X_1, X_3	Z_5
$F3$	$k[x,y]/(x^2, y^2)$	8	X_1, X_5, X_3X_6	Z_3Z_6, Z_4, Z_8
$F4$	$k[x,y,z]/(x,y,z)^2$	9	X_1, X_2, X_3	Z_7, Z_8, Z_9

The prototype qubit lesson is sharper than the odd-characteristic lesson: in characteristic two, quadratic nilpotent equations can leave more global one-form directions alive. These C^1-qubits are not, without further fibrewise Weyl analysis, the canonical tangent-Weyl qubits of the scheme.

58 Entanglement Inside the $\mathbb{F}_2$ de Rham CSS Prototype

58.1 Status and Local Test for Entanglement

This shard continues the finite C^1-basis CSS prototype of AQM-54–AQM-56. Its qubits and Bell sectors are conditional on the displayed prototype stabilizers; they are not canonical tangent-Weyl consequences of $\operatorname{Spec} A$. It uses the same source anchors: `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33872, 34103–34120, 34310–34340, and 34388–34524 for Kähler differentials and the de Rham complex, AQM-08 for the CSS/stabilizer dictionary, and `CONVENTIONS.md` (ao) for the de Rham CSS test. The new claims in this shard are direct solutions of the displayed prototype stabilizer equations. For a displayed C^1 basis label u, the notation X_u and Z_u means the corresponding support Pauli, and all listed stabilizer equations below use the added all-+1 phase convention. The de Rham calculation itself fixes only the support spaces R_X and R_Z. The local entanglement test used below is the following. If two qubits p, q are constrained by

$$X_pX_q = +1, \qquad Z_pZ_q = +1,$$

and no other displayed generator touches them, then their joint state is

$$|\Phi^+\rangle_{pq} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Indeed, $Z_pZ_q = +1$ removes the $|01\rangle$ and $|10\rangle$ amplitudes, while $X_pX_q = +1$ makes the $|00\rangle$ and $|11\rangle$ amplitudes equal. The reduced density matrix on either qubit is

$$\operatorname{diag}(1/2, 1/2),$$

so the state is not a product state. This is the only entanglement criterion used in the examples.

58.2 Example C0: The Eight-Qubit Square Fat Point

Let

$$A_0 = \mathbb{F}_2[x, y]/(x^2, y^2).$$

Use the $C^1 = \Omega^1_{A_0/\mathbb{F}_2}$ basis

$$\begin{aligned} &e_1 = dx, \quad e_2 = x\,dx, \quad e_3 = y\,dx, \quad e_4 = xy\,dx, \\ &e_5 = dy, \quad e_6 = x\,dy, \quad e_7 = y\,dy, \quad e_8 = xy\,dy. \end{aligned}$$

This is the eight auxiliary-qubit example from AQM-56. The de Rham CSS images are

$$R_X = \langle e_1,\ e_5,\ e_3 + e_6\rangle, \qquad R_Z = \langle e_3 + e_6,\ e_4,\ e_8\rangle.$$

Thus the selected stabilizer generators are

$$X_1, \qquad X_5, \qquad X_3X_6, \qquad Z_3Z_6, \qquad Z_4, \qquad Z_8.$$

The six generators are independent and commute; independence is visible from the private supports $1, 5, 4, 8$ plus the common two-qubit support $3, 6$. Therefore the code parameters are

$$[[8,\ 8-6,\ d]] = [[8, 2, d]].$$

The pair $3, 6$ is forced to be

$$|\Phi^+\rangle_{3,6}.$$

The remaining pinned qubits are

$$|+\rangle_1, \quad |+\rangle_5, \quad |0\rangle_4, \quad |0\rangle_8.$$

Qubits 2 and 7 are untouched by every generator, so a code basis is

$$|+\rangle_1 \, |\alpha\rangle_2 \, |\Phi^+\rangle_{3,6} \, |0\rangle_4 \, |+\rangle_5 \, |\beta\rangle_7 \, |0\rangle_8,$$

where $\alpha, \beta \in \{0, 1\}$. The logical operators may be chosen as

$$\bar{X}_1 = X_2, \quad \bar{Z}_1 = Z_2, \qquad \bar{X}_2 = X_7, \quad \bar{Z}_2 = Z_7.$$

Hence $d = 1$. Inside the prototype this is an entangled stabilizer code, but not an error-correcting code.

58.3 Example C1: A Ten-Qubit Fat Rectangle

Let

$$A_1 = \mathbb{F}_2[x, y]/(x^3, y^2).$$

Since $d(x^3) = x^2 dx$ and $d(y^2) = 0$, the dx-part has coefficients $1, x, y, xy$, while the dy-part has coefficients $1, x, x^2, y, xy, x^2y$. Use

$$\begin{aligned} &a_0 = dx, && a_1 = x\,dx, && b_0 = y\,dx, && b_1 = xy\,dx, \\ &c_0 = dy, && c_1 = x\,dy, && c_2 = x^2 dy, && d_0 = y\,dy, \\ &d_1 = xy\,dy, && d_2 = x^2 y\,dy. \end{aligned}$$

There are ten auxiliary prototype qubits. The d_0-image is

$$d(x) = a_0, \quad d(y) = c_0, \quad d(xy) = b_0 + c_1, \quad d(x^2y) = c_2,$$

so

$$R_X = \langle a_0, \ c_0, \ b_0 + c_1, \ c_2 \rangle.$$

For C^2, write $\omega = dx \wedge dy$. The relation $x^2 dx = 0$ implies $x^2 \omega = 0$, so the target basis used for the transpose is

$$\omega, \quad x\omega, \quad y\omega, \quad xy\omega.$$

Against this basis, the nonzero d_1-values are

$$d_1(b_0) = \omega, \quad d_1(c_1) = \omega, \quad d_1(b_1) = x\omega, \quad d_1(d_1) = y\omega.$$

The displayed rows have image rank 3; the $xy\omega$ row is zero. Thus

$$R_Z = \langle b_0 + c_1, \ b_1, \ d_1 \rangle.$$

The selected stabilizers are

$$X_{a_0}, \ X_{c_0}, \ X_{b_0} X_{c_1}, \ X_{c_2}, \qquad Z_{b_0} Z_{c_1}, \ Z_{b_1}, \ Z_{d_1}.$$

Their rank is $4 + 3 = 7$, so the code is

$$[[10, \ 10 - 7, \ d]] = [[10, 3, d]].$$

The pair b_0, c_1 is forced to be a Bell pair. The pinned single-qubit states are

$$|+\rangle_{a_0}, \quad |+\rangle_{c_0}, \quad |+\rangle_{c_2}, \quad |0\rangle_{b_1}, \quad |0\rangle_{d_1}.$$

The untouched qubits a_1, d_0, d_2 give logical operators

$$X_{a_1}, Z_{a_1}, \qquad X_{d_0}, Z_{d_0}, \qquad X_{d_2}, Z_{d_2}.$$

Therefore $d = 1$. The prototype code has a real Bell sector but no single-qubit error protection.

58.4 Example C2: An Eighteen-Qubit Fat Rectangle

Let

$$A_2 = \mathbb{F}_2[x,y]/(x^5, y^2).$$

The relation $d(x^5) = x^4 dx$ removes $x^4 dx$ and $x^4 y\, dx$, while $d(y^2) = 0$. Use the 18 basis one-forms

$$\begin{aligned} a_i &= x^i dx \ (0 \le i \le 3), \quad b_i = x^i y\, dx \ (0 \le i \le 3), \\ c_i &= x^i dy \ (0 \le i \le 4), \quad d_i = x^i y\, dy \ (0 \le i \le 4). \end{aligned}$$

The nonzero d_0-images needed for R_X are

$$\begin{array}{rclrcl} d(x) &=& a_0, & d(x^3) &=& a_2, \\ d(y) &=& c_0, & d(xy) &=& b_0 + c_1, \\ d(x^2 y) &=& c_2, & d(x^3 y) &=& b_2 + c_3, \\ d(x^4 y) &=& c_4. & & & \end{array}$$

Hence

$$R_X = \langle a_0, a_2, c_0, c_2, c_4, b_0 + c_1, b_2 + c_3 \rangle.$$

Again let $\omega = dx \wedge dy$, with $x^4 \omega = 0$. The target basis used for the transpose is

$$\omega, x\omega, x^2\omega, x^3\omega, y\omega, xy\omega, x^2 y\omega, x^3 y\omega.$$

Against this basis, the nonzero d_1-rows are

$$\begin{array}{rclrcl} d_1(b_0) &=& \omega, & d_1(c_1) &=& \omega, \\ d_1(b_1) &=& x\omega, & d_1(b_2) &=& x^2\omega, \\ d_1(c_3) &=& x^2\omega, & d_1(b_3) &=& x^3\omega, \\ d_1(d_1) &=& y\omega, & d_1(d_3) &=& x^2 y\omega. \end{array}$$

The displayed row image has rank 6; the $xy\omega$ and $x^3 y\omega$ rows are zero. Thus

$$R_Z = \langle b_0 + c_1,\ b_1,\ b_2 + c_3,\ b_3,\ d_1,\ d_3 \rangle.$$

The selected stabilizers are therefore

$$\begin{aligned} &X_{a_0}, X_{a_2}, X_{c_0}, X_{c_2}, X_{c_4}, X_{b_0} X_{c_1}, X_{b_2} X_{c_3}, \\ &Z_{b_0} Z_{c_1}, Z_{b_1}, Z_{b_2} Z_{c_3}, Z_{b_3}, Z_{d_1}, Z_{d_3}. \end{aligned}$$

The rank is $7 + 6 = 13$, so this is an

$$[[18,\ 18 - 13,\ d]] = [[18, 5, d]]$$

CSS code. The two disjoint pairs b_0, c_1 and b_2, c_3 are forced to be

$$|\Phi^+\rangle_{b_0,c_1} \otimes |\Phi^+\rangle_{b_2,c_3}.$$

The untouched qubits

$$a_1, \quad a_3, \quad d_0, \quad d_2, \quad d_4$$

give five weight-one logical X/Z pairs, so $d = 1$. This example reaches the requested 15+-qubit scale and shows the same diagnosis: the de Rham CSS test naturally creates entangled stabilizer sectors, but these fat rectangles still carry unprotected differential logicals.

58.5 Summary

The three prototype examples have parameters

ring	qubits	Bell sectors	$[[n,k,d]]$
$\mathbb{F}_2[x,y]/(x^2,y^2)$	8	$(y\,dx,\ x\,dy)$	$[[8,2,1]]$
$\mathbb{F}_2[x,y]/(x^3,y^2)$	10	$(y\,dx,\ x\,dy)$	$[[10,3,1]]$
$\mathbb{F}_2[x,y]/(x^5,y^2)$	18	$(y\,dx,\ x\,dy),\ (x^2y\,dx,\ x^3dy)$	$[[18,5,1]]$

So the first prototype qubit-code conclusion is mixed. Entanglement is exactly derivable from the displayed C^1-CSS stabilizers. However, the same characteristic-two pathology that creates many global one-form qubits also leaves weight-one logical operators. These are prototype entangled stabilizer codes, not canonical tangent-Weyl dynamics and not useful distance codes.

59 Product Rings and the Formal Real Affine Line

59.1 Status and Source Anchors

This shard answers two follow-up questions about convention `CONVENTIONS.md` (ao). Kähler differentials, polynomial differentials, and the de Rham complex use the Stacks algebra snapshot `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33872, 34310–34340, and 34388–34524. Tangent space by dual numbers and cotangent-tangent duality use `references/algebraic_geometry/stacks_project_varieties.tex`, lines 2915–3019. The product-ring Weyl layer is AQM-40 and convention (ad). The real CCR representation comparison is the finite-dimensional CCR layer of AQM-19, sourced there from Derezinski and Bekka.

59.2 Part I: Product Rings Have Trivial Tangent-Weyl Labels

Let k be a field and

$$A = k^n = ke_1 \oplus \cdots \oplus ke_n,$$

where e_i is the idempotent with 1 in the i-th factor and 0 elsewhere. Let M be any A-module and let $D : A \to M$ be a k-derivation. We first prove $D(e_i) = 0$ without assuming anything about the characteristic. Since $e_i^2 = e_i$,

$$D(e_i) = D(e_i^2) = e_iD(e_i) + e_iD(e_i).$$

Multiplying by e_i gives

$$e_iD(e_i) = 2e_iD(e_i),$$

so $e_iD(e_i) = 0$. Multiplying the displayed derivation equation by $1 - e_i$ gives

$$(1 - e_i)D(e_i) = 0.$$

Adding the two pieces gives $D(e_i) = 0$.

Every element of A is $a = \sum_i a_ie_i$ with $a_i \in k$. Because D is a k-derivation, $D(a_i) = 0$, and hence

$$D(a) = \sum_i a_iD(e_i) = 0.$$

Thus $\mathrm{Der}_k(A, M) = 0$ for every M. By the universal property,

$$\Omega^1_{A/k} = 0.$$

Consequently

$$A \longrightarrow 0 \longrightarrow 0$$

is the de Rham complex, and the de Rham CSS test has

$$C^1 = 0, \qquad R_X = 0, \qquad R_Z = 0.$$

So k^n has zero tangent-cotangent Weyl label space in exactly the same sense as the single field k. This does not erase the older residue-Weyl comparison layer. By AQM-40,

$$\mathrm{Spec}(k^n) = \{x_1, \dots, x_n\}, \qquad \kappa(x_i) = k,$$

and the residue-Weyl prototype is the ambient n-qudit system

$$\mathcal{H} = \ell^2(k)^{\otimes n}, \qquad \mathcal{A} = \bigotimes_{i=1}^{n} \operatorname{End}_{\mathbb{C}} \ell^2(k).$$

The conclusion is therefore sharp: the corrected tangent-cotangent recipe sees points and residue fields, then finds $T_{x_i}X = T^*_{x_i}X = 0$ and hence no nontrivial geometric Weyl operators. The displayed residue-Weyl qudits are a separate comparison layer, and the de Rham CSS prototype selects no stabilizer and no Hamiltonian.

59.3 Part II: Cotangent Data on $\operatorname{Spec}\mathbb{R}[x]$

Let

$$A = \mathbb{R}[x], \qquad X = \operatorname{Spec} A.$$

The polynomial differential source gives

$$\Omega^1_{A/\mathbb{R}} = A\,dx = \mathbb{R}[x]\,dx.$$

The universal derivation is the formal derivative:

$$d(f(x)) = f'(x)\,dx.$$

The algebraic de Rham complex begins

$$\mathbb{R}[x] \xrightarrow{d_0} \mathbb{R}[x]\,dx \longrightarrow 0.$$

Since every polynomial $g(x)$ has the polynomial antiderivative

$$\int g(x)\,dx$$

over $\mathbb{R}$, the map d_0 is onto $\mathbb{R}[x]\,dx$. Its kernel is the constants. Hence

$$H^0_{\mathrm{dR}}(\mathbb{R}[x]/\mathbb{R}) = \mathbb{R}, \qquad H^1_{\mathrm{dR}}(\mathbb{R}[x]/\mathbb{R}) = 0.$$

This is only a formal algebraic calculation; it is not an analytic statement about domains of differential operators.

Now compute fibres. At the real closed point

$$a \in \mathbb{R}, \qquad \mathfrak{m}_a = (x-a),$$

the residue field is $\kappa(\mathfrak{m}_a) = \mathbb{R}$. The cotangent fibre is

$$T^*_{X/\mathbb{R},a} = \Omega^1_{A/\mathbb{R}} \otimes_A \mathbb{R} = \mathbb{R}\,dx_a.$$

The tangent space is the dual line

$$T_{X/\mathbb{R},a} = \operatorname{Hom}_{\mathbb{R}}(\mathbb{R}\,dx_a, \mathbb{R}) \cong \mathbb{R}.$$

The tangent vector v with $v(dx_a) = 1$ corresponds to the derivation

$$D_v(f) = f'(a).$$

More generally $c\,v$ corresponds to $D_{cv}(f) = cf'(a)$.

There are also non-real closed points. If

$$\pi(x) = x^2 + bx + c$$

is irreducible over $\mathbb{R}$, then

$$\kappa(\pi) = \mathbb{R}[x]/(\pi) \cong \mathbb{C},$$

and

$$T^*_{X/\mathbb{R},\pi} \cong \mathbb{C}\,dx_\pi, \qquad T_{X/\mathbb{R},\pi} \cong \mathrm{Hom}_{\mathbb{C}}(\mathbb{C}\,dx_\pi, \mathbb{C}).$$

The generic point has residue field $\mathbb{R}(x)$, so its cotangent fibre is

$$T^*_{X/\mathbb{R},\eta} \cong \mathbb{R}(x)\,dx.$$

59.4 Operator-Label Reading

The cotangent construction says that the algebraic phase space of the real affine line is the total cotangent bundle

$$T^*X = \operatorname{Spec}\operatorname{Sym}_A(\Omega^1_{A/\mathbb{R}}) \cong \operatorname{Spec}\mathbb{R}[x,p].$$

Here x is the position coordinate and p is the fibre-linear momentum coordinate dual to dx. With the tangent-cotangent ordering fixed in convention (ao), $E_a = T_aX \oplus T_a^*X$, the corresponding algebraic symplectic form is

$$dx \wedge dp.$$

At a real point, the corrected pointwise Weyl label space is therefore

$$E_a = T_aX \oplus T_a^*X \cong \mathbb{R}\,\partial_x|_a \oplus \mathbb{R}\,dx_a.$$

In the usual global affine coordinate this is the one-dimensional CCR label space with labels $(q,p) \in \mathbb{R}^2$, symplectic form

$$\Omega((q,p),(q',p')) = p'q - pq'.$$

This is the tangent-cotangent sign of AQM-60, not the opposite sign used by the older AQM-19 comparison display $W(q,p) = T(q)R(p)$. To compare with that Schrodinger display, either reverse the translation/modulation order or identify the AQM-19 display momentum with $-p$. After making such a sign bridge, the irreducible regular projective representation has a Hilbert carrier. In the standard analytic realization on $L^2(\mathbb{R})$, position is multiplication by x and the cotangent label p is represented formally with the corresponding sign convention by

$$i\,\frac{d}{dx}.$$

The word "formally" is essential here. The cotangent calculation supplies the label space for Weyl operators; the representation theorem supplies Hilbert carriers. It does not by itself choose a stabilizer family, self-adjoint domains, a Hamiltonian such as $-d^2/dx^2 + V(x)$, boundary conditions, or an equation of motion.

The de Rham CSS proposal is different. Applied literally to $\mathbb{R}[x]$, it is infinite-dimensional and outside the finite-qudit scope. Moreover d_0 is surjective in the full polynomial complex, so any finite-dimensional truncation would need its own truncation rule before one could infer a concrete set of C^1 checks. The physical particle-on-a-line quantisation therefore belongs to the tangent-cotangent Weyl representation layer plus further analytic dynamics, not to the finite de Rham CSS stabilizer prototype.

60 The Finite-Field Circle: Tangent-Weyl Labels

60.1 Status and Source Anchors

Let $k = \mathbb{F}_q$ and

$$A_q = k[x, y]/(x^2 + y^2 - 1), \qquad C_q = \operatorname{Spec} A_q.$$

This shard uses Stacks algebra lines 34103–34120 for quotient differentials, 34310–34340 for polynomial differentials, and 34388–34524 for the de Rham complex. The closed-point finite-residue Weyl layer is convention (y) and AQM-28–AQM-30. The product-ring comparison for finite rational truncations is AQM-40 and AQM-58.

60.2 Odd Characteristic: Geometry and Differentials

Assume q is odd. In

$$S = k[x, y], \qquad f = x^2 + y^2 - 1, \qquad A_q = S/(f),$$

the quotient exact sequence gives

$$\Omega^1_{A_q/k} = (A_q\, dx \oplus A_q\, dy)/(2x\, dx + 2y\, dy).$$

Because $2 \in k^\times$, this is equivalently

$$x\, dx + y\, dy = 0.$$

At a k-rational point $P = (a, b)$, the cotangent fibre is

$$T^*_{C_q/k,P} = (k\, dx \oplus k\, dy)/(a\, dx + b\, dy).$$

Since $a^2 + b^2 = 1$, the pair (a, b) is not $(0, 0)$. Thus the relation has rank 1, and

$$\dim_k T^*_{C_q/k,P} = 1, \qquad \dim_k T_{C_q/k,P} = 1.$$

Concrete bases are:

$$\begin{aligned} &b \neq 0: \quad T^*_P = k\, dx, \quad dy = -(a/b)\, dx, \\ &b = 0: \quad a = \pm 1, \quad T^*_P = k\, dy, \quad dx = 0. \end{aligned}$$

The same calculation over any closed point P, with residue field $\kappa(P)$, gives a one-dimensional cotangent fibre over $\kappa(P)$.

60.3 Odd Characteristic: Rational Points as q Varies

The point $(1, 0)$ lies on C_q. Draw a line of slope $t \in k$ through it:

$$y = t(x - 1).$$

Substitute into $x^2 + y^2 = 1$:

$$x^2 + t^2(x - 1)^2 - 1 = (x - 1)\big((1 + t^2)x + (1 - t^2)\big).$$

One intersection is $x = 1$. The second, when $1 + t^2 \neq 0$, is

$$x = \frac{t^2 - 1}{t^2 + 1}, \qquad y = t(x - 1) = \frac{-2t}{t^2 + 1}.$$

This parametrizes all rational points except $(1,0)$, unless $1+t^2=0$, in which case the line passes through a missing point at infinity and gives no second affine point.

Therefore
$$|C_q(k)| = 1 + \#\{t \in k : 1+t^2 \neq 0\}.$$
If -1 is not a square in k, then $1+t^2$ never vanishes and
$$|C_q(k)| = q+1.$$
If -1 is a square, there are two roots of $1+t^2$, and
$$|C_q(k)| = q-1.$$
Equivalently, for odd q,
$$|C_q(\mathbb{F}_q)| = q - \chi_q(-1),$$
where χ_q is the quadratic character on $k^\times$. The same calculation over the extension field $\mathbb{F}_{q^m}$ gives
$$a_m(q) := |C_q(\mathbb{F}_{q^m})| = q^m - \chi_{q^m}(-1).$$
If -1 is already a square in k, then $\chi_{q^m}(-1) = 1$ for every m, so $a_m(q) = q^m - 1$. If -1 is not a square in k, then it becomes a square exactly over even-degree extensions, so
$$a_m(q) = q^m - (-1)^m.$$

60.4 Even Characteristic: The Double Line

Assume q is even. Then
$$x^2 + y^2 - 1 = (x+y+1)^2.$$
Thus
$$A_q = k[x,y]/((x+y+1)^2)$$
is nonreduced. Its reduced coordinate ring is
$$A_{q,\mathrm{red}} = k[x,y]/(x+y+1) \cong k[x],$$
so the topological closed points are those of an affine line. Rationally,
$$C_q(k) = \{(a,b) \in k^2 : a+b+1 = 0\},$$
and hence
$$|C_q(k)| = q.$$
Over $\mathbb{F}_{q^m}$, the same reduced line has
$$a_m(q) = q^m$$
points. The derivative pathology is sharper. The relation is $g^2 = 0$, where $g = x+y+1$, but
$$d(g^2) = 2g\,dg = 0$$
in characteristic 2. Hence the quotient differential relation imposes nothing:
$$\Omega^1_{A_q/k} = A_q\,dx \oplus A_q\,dy.$$
At a rational point P, the residue field kills the nilpotent g, but the cotangent fibre remains
$$T^*_{C_q/k,P} = k\,dx \oplus k\,dy,$$
of dimension 2. The reduced line would have the relation $dx+dy=0$ and one-dimensional cotangent fibre. The unreduced circle therefore has the same closed-point Weyl sites as a line but a thicker cotangent geometry.

60.5 Closed Points and Tangent-Weyl Representation Data

The complete finite-residue tangent-Weyl data are not only the rational-point set. Let $|C_q|_{\text{cl}}$ be the closed points of C_q. For each closed point P, put

$$K_P = \kappa(P), \qquad V_P = T_P C_q, \qquad V_P^* = T_P^* C_q, \qquad E_P = V_P \oplus V_P^*.$$

The Weyl-Heisenberg step assigns projective representations

$$\rho_P : E_P \to PU(\mathcal{H}_{\rho_P}),$$

so elements of E_P label Weyl operators on the carrier $\mathcal{H}_{\rho_P}$. If $m_P = \dim_{K_P} V_P$, then the finite carrier dimension is $|K_P|^{m_P}$.

For odd q, every closed point is smooth on the conic, so $m_P = 1$. A degree-d point has $|K_P| = q^d$, label space $K_P \oplus K_P^*$, and Weyl carrier dimension q^d. In characteristic two for the unreduced double line, the calculation above gives $m_P = 2$, hence a degree-d point has carrier dimension q^{2d}. The reduced line would instead have $m_P = 1$.

Let $N_d(q)$ be the number of degree-d closed points. If

$$a_m(q) = |C_q(\mathbb{F}_{q^m})|,$$

then every degree-d closed point contributes d points over $\mathbb{F}_{q^m}$ exactly when $d \mid m$. Hence

$$a_m(q) = \sum_{d|m} d\, N_d(q), \qquad N_d(q) = \frac{1}{d} \sum_{r|d} \mu(r)\, a_{d/r}(q).$$

Thus the complete tangent-Weyl data contain $N_d(q)$ degree-d sites. The local Hilbert carrier dimension is q^d in odd characteristic and q^{2d} for the unreduced characteristic-two circle.

The rational tangent-Weyl truncation keeps only $N_1(q) = |C_q(k)|$ sites:

$$\dim \mathcal{H}_{\text{rat}}^{\text{geom}}(C_q) = \prod_{P \in C_q(k)} q^{m_P}.$$

For odd q, this is $q^{q-\chi_q(-1)}$. For the unreduced even circle, it is q^{2q}. The older residue-Weyl truncation would give q^q in even characteristic; that is a comparison layer, not the tangent-Weyl reading.

60.6 Increasing q: First Values

q	$\operatorname{char} k$	$\lvert C_q(k)\rvert$	$\dim \mathcal{H}_{\text{rat}}^{\text{geom}}$
2	2	2	2^4
3	odd, -1 nonsquare	4	3^4
4	2	4	4^8
5	odd, -1 square	4	5^4
7	odd, -1 nonsquare	8	7^8
8	2	8	8^{16}
9	odd, -1 square	8	9^8
11	odd, -1 nonsquare	12	11^{12}
13	odd, -1 square	12	13^{12}

This table is only the rational finite tangent-Weyl truncation. The complete closed-point data also have degree-d sites for every $d \geq 2$, counted by the Möbius formula above.

60.7 Derivative Dynamics Versus Rational Truncation

There is a subtle but important distinction. If one replaces the rational point set $C_q(k)$ by its coordinate algebra of functions

$$k^{C_q(k)},$$

then AQM-58 applies: this product ring has

$$\Omega^1_{k^{C_q(k)}/k} = 0.$$

So the rational finite set has trivial de Rham CSS prototype data.

The scheme $C_q = \operatorname{Spec} A_q$ is different. For odd q, it is a smooth affine curve with one-dimensional cotangent fibre at every closed point. For even q, it is a nonreduced double line with two-dimensional cotangent fibres before reduction. Thus the corrected closed-point Weyl data are built from the residue fields together with the fibre dimensions of $T_P C_q \oplus T_P^* C_q$. The differential geometry supplies operator labels; a stabilizer-selection principle is still required before any dynamics or code subspace is selected.

61 Tangent-Weyl Operator Principle and de Rham CSS Prototype

61.1 Status and Source Anchors

This shard corrects the interpretation of AQM-54–AQM-59. Kähler differentials are sourced from `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33872 and 34310–34340. Tangent spaces are sourced from `references/algebraic_geometry/stacks_project_varieties.tex`, lines 2915–3019. Finite Weyl operators use AQM-28. The stabilizer/CSS isotropy dictionary is AQM-08. The tangent-cotangent carrier convention is `CONVENTIONS.md` (ao), with the relative/intrinsic warning in `CONVENTIONS.md` (av). The real CCR comparison is AQM-19.

The corrected principle is not "put a Hilbert space on the tangent space" and not a separate geometric intermediate step. The principle is: compute the pointwise symplectic label spaces, quantise those label spaces by Weyl-Heisenberg projective unitary representations, and only then ask for a selection principle choosing commuting Weyl operators.

61.2 Lamport Dependency Skeleton

$$
\begin{aligned}
D1 &: k \to A \text{ is fixed}, \quad X = \operatorname{Spec} A, \quad P \in X, \quad K = \kappa(P).\\
D2 &: V_P^* = T^*_{A/k,P}X = \Omega^1_{A/k} \otimes_A K.\\
D3 &: V_P = T_{A/k,P}X = \operatorname{Hom}_K(V_P^*, K).\\
D4 &: E_P = V_P \oplus V_P^*.\\
D5 &: \omega((v,\alpha),(w,\beta)) = \beta(v) - \alpha(w).\\
D6 &: \rho_P : E_P \to PU(\mathcal{H}_{\rho_P}) \text{ is a Weyl-Heisenberg projective representation.}\\
D7 &: e \in E_P \text{ labels a Weyl operator } W_P(e) \text{ on } \mathcal{H}_{\rho_P}.\\
D8 &: E_S = \textstyle\bigoplus_{P\in S} E_P, \quad \rho_S = \bigotimes_{P\in S} \rho_P \text{ for finite } S.\\
D9 &: L \subset E_S \text{ isotropic, meaning trivial Weyl commutator} \Rightarrow \{W(e) : e \in L\} \text{ commutes.}\\
D10 &: \text{A compatible phased-stabilizer character on } L \text{ defines a joint eigenspace.}\\
D11 &: C^0 = A \to C^1 = \Omega^1_{A/k} \to C^2 = \Omega^2_{A/k} \text{ is a separate finite CSS prototype.}\\
T1 &: D1\text{–}D10 \text{ are the tangent-Weyl operator workflow.}\\
T2 &: D11 \text{ and residue-Weyl constructions are comparison/prototype layers.}
\end{aligned}
$$

61.3 The Local Symplectic Label Space

Let $V = V_P = T_{A/k,P}X$ and $V^* = V_P^* = T^*_{A/k,P}X$, all relative to the fixed base map $k \to A$. Define

$$E_P = V \oplus V^*$$

and

$$\omega((v,\alpha),(w,\beta)) = \beta(v) - \alpha(w).$$

The form is bilinear and alternating because

$$\omega((v,\alpha),(v,\alpha)) = \alpha(v) - \alpha(v) = 0.$$

It is nondegenerate: if $v \neq 0$, choose $\beta \in V^*$ with $\beta(v) \neq 0$; if $\alpha \neq 0$, choose $w \in V$ with $\alpha(w) \neq 0$. Pairing with $(0,\beta)$ or $(w,0)$ gives a nonzero value. Hence E_P is the local symplectic label space.

In coordinates $V = K\,\partial_x$, $V^* = K\,dx$, this convention gives

$$\omega((q,p),(q',p')) = p'q - pq'.$$

Thus it agrees with the $dx \wedge dp$ sign used in AQM-58 after that shard's repair. It is opposite to the display obtained from the residue-Weyl ordering $W(q,p) = T(q)R(p)$; comparison with that ordering requires either reversing the translation/modulation order or replacing the display momentum by its negative.

For finite K, the Weyl-Heisenberg step constructs projective unitary representations of E_P. With central character

$$\psi_K(a) = \exp\left(\frac{2\pi i}{p}\operatorname{Tr}_{K/\mathbb{F}_p}(a)\right),$$

the irreducible finite representation has carrier dimension

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = |K|^{\dim_K V}.$$

This is where Hilbert carriers enter the recipe: they are carriers of the Weyl-Heisenberg projective representation. They are not assigned directly to $T_{A/k,P}X$.

61.4 Stabilizer Selection Is the Isotropic Step

A stabilizer principle chooses label data

$$L \subset E_S = \bigoplus_{P\in S} E_P$$

whose Weyl commutator bicharacter is trivial. For a common residue field this is the usual vanishing of the summed symplectic pairing; for mixed finite residue fields it is the componentwise product of the local trace-character commutators. This isotropy condition is exactly what is needed for the corresponding Weyl operators to commute after the central lifts are fixed. After the selection principle also fixes a compatible phased-stabilizer character on L, the selected stabilizer subspace is the joint eigenspace inside the carrier of ρ_S.

Thus the role previously being described by the wrong geometric word is played by the stabilizer-selection principle itself. A maximal isotropic choice is a Lagrangian-type stabilizer choice, but it is still a choice of commuting Weyl operators, not an extra canonical step between geometry and representation.

61.5 Comparison Layers

The older residue-Weyl comparison attaches to a finite residue field K the label shape $K_q \oplus K_p$ and a carrier such as $\ell^2(K)$. This is a valid comparison layer from AQM-28–AQM-30, but it uses only K. It does not see whether P is smooth, singular, reduced, nonreduced, tangent-rich, or tangent-zero.

The finite de Rham CSS prototype is also separate. It uses

$$C^0 = A \xrightarrow{d_0} C^1 = \Omega^1_{A/k} \xrightarrow{d_1} C^2 = \Omega^2_{A/k}$$

when these are finite-dimensional k-vector spaces, chooses a C^1-basis, and forms

$$R_X = \operatorname{im} d_0, \qquad R_Z = \operatorname{im} d_1^T.$$

The equation $d_1 d_0 = 0$ proves CSS isotropy for that auxiliary chain-complex construction. It is useful, but it is not the canonical pointwise tangent-Weyl quantisation of X.

61.6 Example I: Product Fields

Let

$$A = k^n = ke_1 \oplus \cdots \oplus ke_n.$$

AQM-58 proves that every k-derivation $D : A \to M$ vanishes. Hence

$$\Omega^1_{A/k} = 0.$$

At the i-th point P_i,

$$V^*_{P_i} = 0, \qquad V_{P_i} = 0, \qquad E_{P_i} = 0.$$

The tangent-Weyl representation is therefore trivial at every point: there are no nontrivial geometric Weyl operator labels to select. The older residue-Weyl comparison would attach one k-level carrier per residue point, but that is not the corrected tangent-cotangent construction. The de Rham CSS prototype also selects no stabilizer because $C^1 = 0$.

61.7 Example II: The Odd Finite-Field Circle

Let $k = \mathbb{F}_q$ with q odd and

$$C = \operatorname{Spec} k[x, y]/(x^2 + y^2 - 1).$$

For a rational point $P = (a, b)$, AQM-59 computes

$$V^*_P = (k\,dx \oplus k\,dy)/(a\,dx + b\,dy).$$

The tangent space is

$$V_P = \{(u, v) \in k^2 : au + bv = 0\}.$$

Since $a^2 + b^2 = 1$, both spaces are one-dimensional. Hence

$$E_P = V_P \oplus V^*_P$$

is a two-dimensional symplectic k-label space, and the finite Weyl-Heisenberg carrier has dimension q. For

$$C(\mathbb{F}_3) = \{(1, 0), (2, 0), (0, 1), (0, 2)\},$$

the rational tangent-Weyl support has four local carriers of dimension 3. This is still only kinematics. A code or state appears only after a principle selects an isotropic subset of the four-point label space

$$\bigoplus_{P \in C(\mathbb{F}_3)} (V_P \oplus V^*_P).$$

61.8 Example III: The Formal Real Affine Line

Let

$$A = \mathbb{R}[x], \qquad X = \operatorname{Spec} A.$$

AQM-58 computes

$$\Omega^1_{A/\mathbb{R}} = A\,dx.$$

At the real point $P_a = (x - a)$,

$$V^*_{P_a} = \mathbb{R}\,dx_a, \qquad V_{P_a} = \mathbb{R}\,\partial_x|_a.$$

The local operator-label space is

$$E_{P_a} = \mathbb{R}\,\partial_x|_a \oplus \mathbb{R}\,dx_a,$$

with symplectic form

$$\omega((r\partial_x, s\,dx), (r'\partial_x, s'\,dx)) = s'r - sr'.$$

The Weyl/CCR representation theory supplies Hilbert carriers for this label space; the usual $L^2(\mathbb{R})$ realization is an analytic model, not a stabilizer-selection principle. It does not choose the free-particle Hamiltonian, a potential, a domain, boundary conditions, or a distinguished commuting operator family.

61.9 Operational Conclusion

The corrected workflow is:

1. Choose a base map $k \to A$ and set $X = \operatorname{Spec} A$.
2. Compute points P and residue fields $\kappa(P)$.
3. Compute the relative $T_{A/k,P}X$ and $T^*_{A/k,P}X$.
4. Build $E_P = T_{A/k,P}X \oplus T^*_{A/k,P}X$.
5. Quantise E_P by Weyl-Heisenberg projective representations.
6. Use a separate principle to select isotropic Weyl labels.
7. Choose compatible stabilizer phases and take the joint eigenspace.

AQM-61–AQM-63 use this workflow. AQM-55–AQM-57 remain useful only as finite C^1-CSS prototype calculations unless they are explicitly pushed through the pointwise tangent-Weyl representation layer.

62 Tangent-Cotangent Weyl Kinematics: Recipe and First Examples

62.1 Status and Source Anchors

This shard starts the kinematical recipe named in convention (ao) as `tangent_cotangent_weyl_kinematics`. Kähler differentials and cotangent fibres are sourced from `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33872 and 34310–34340. Tangent spaces by derivations and dual numbers are sourced from `references/algebraic_geometry/stacks_project_varieties.tex`, lines 2915–3019. Finite Weyl operators and additive characters are the AQM-28 convention, sourced there from Ashikhmin–Knill. AQM-60 separates this kinematics from the finite `de_rham_css_test`.

62.2 Recipe

Let $X = \operatorname{Spec} A$ be a k-scheme presented by a k-algebra, and let $P \in X$ be a finite-residue point with

$$K = \kappa(P).$$

The cotangent and tangent spaces are

$$T_P^*X = \Omega^1_{A/k} \otimes_A K, \qquad T_PX = \operatorname{Hom}_K(T_P^*X, K).$$

Set

$$E_P^{\mathrm{geom}} = T_PX \oplus T_P^*X$$

with the canonical symplectic form

$$\omega((v,\alpha),(w,\beta)) = \beta(v) - \alpha(w).$$

For finite K, use

$$\psi_K(a) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{K/\mathbb{F}_p}(a)\right).$$

Quantisation means choosing the Weyl-Heisenberg projective representation

$$\rho_P : E_P^{\mathrm{geom}} \to PU(\mathcal{H}_{\rho_P})$$

with central character ψ_K. Elements $(w,\alpha) \in E_P^{\mathrm{geom}}$ label Weyl operators $W_P(w,\alpha)$ on the carrier $\mathcal{H}_{\rho_P}$. The standard coordinate display of this representation is

$$T(w)|v\rangle = |v+w\rangle, \qquad R(\alpha)|v\rangle = \psi_K(\alpha(v))|v\rangle.$$

If $\dim_K T_PX = m$, then representation theory gives

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = |K|^m.$$

After choosing a K-basis of T_PX, the carrier has the size of m qudits of local dimension $|K|$.

For a finite set S of finite-residue points, define

$$E_S^{\mathrm{geom}} = \bigoplus_{P\in S} E_P^{\mathrm{geom}}, \qquad \rho_S^{\mathrm{geom}} = \bigotimes_{P\in S} \rho_P^{\mathrm{geom}}.$$

This is kinematics only. No stabilizer, Hamiltonian, or dynamics is selected in this shard.

62.3 Presentation Rule

For a presentation

$$A = k[x_1, \dots, x_n]/(f_1, \dots, f_r)$$

and a K-valued point $P = (a_1, \dots, a_n)$, a tangent vector is

$$v = (v_1, \dots, v_n) \in K^n$$

such that

$$\sum_i \frac{\partial f_j}{\partial x_i}(P) v_i = 0 \qquad \text{for every } j.$$

The cotangent space is the dual quotient generated by dx_i modulo the linear relations

$$\sum_i \frac{\partial f_j}{\partial x_i}(P) dx_i = 0.$$

62.4 Examples 1–5

1. The point $\operatorname{Spec}\mathbb{F}_3$. Let $A = k = \mathbb{F}_3$. There is one point P, $K = k$, and

$$\Omega^1_{k/k} = 0.$$

Hence

$$T^*_P X = 0, \qquad T_P X = 0, \qquad E_P^{\mathrm{geom}} = 0.$$

The Weyl-Heisenberg representation carrier is one-dimensional:

$$\mathcal{H}_{\rho_P} \cong \mathbb{C},$$

because $E_P^{\mathrm{geom}} = 0$. There is no nontrivial local Weyl label in this geometric kinematics.

2. The product k^2. Let $A = k^2$ with $k = \mathbb{F}_3$. AQM-58 proves

$$\Omega^1_{A/k} = 0.$$

There are two points P_1, P_2, both with residue field $K = k$, but at each point

$$T^*_{P_i} X = 0, \qquad T_{P_i} X = 0.$$

Thus

$$E_{P_i}^{\mathrm{geom}} = 0, \qquad \mathcal{H}_{\rho_{P_i}} \cong \mathbb{C}.$$

This differs from residue-Weyl kinematics, which would attach one three-level residue qudit to each point.

3. Dual numbers over $\mathbb{F}_3$. Let

$$A = k[\epsilon]/(\epsilon^2), \qquad k = \mathbb{F}_3,$$

and let $P = (\epsilon)$. Since

$$d(\epsilon^2) = 2\epsilon\, d\epsilon,$$

the fibre at P kills ϵ, so

$$T_P^*X = k\, d\epsilon, \qquad T_P X = k\, \partial_\epsilon.$$

Therefore

$$E_P^{\text{geom}} = k\, \partial_\epsilon \oplus k\, d\epsilon, \qquad \omega((r,s),(r',s')) = s'r - sr'.$$

The Weyl-Heisenberg carrier has dimension 3:

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3.$$

The labels $a\partial_\epsilon + b\, d\epsilon$ become Weyl operators $W_P(a,b)$ on this carrier.

4. The cubic fat point in characteristic three. Let

$$A = k[t]/(t^3), \qquad k = \mathbb{F}_3, \qquad P = (t).$$

Here

$$d(t^3) = 3t^2 dt = 0,$$

so the relation contributes no first-differential relation. Thus

$$T_P^*X = k\, dt, \qquad T_P X = k\, \partial_t.$$

Again

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3.$$

The length-three nilpotent thickening is invisible to the first tangent dimension beyond saying that the Weyl carrier has one qutrit's dimension.

5. Two independent dual directions. Let

$$A = k[u,v]/(u^2, v^2), \qquad k = \mathbb{F}_3, \qquad P = (u,v).$$

The relations give $2u\, du = 0$ and $2v\, dv = 0$. At P, both u and v vanish, so

$$T_P^*X = k\, du \oplus k\, dv, \qquad T_P X = k\, \partial_u \oplus k\, \partial_v.$$

The label space is $k^2 \oplus (k^2)^*$, and the Weyl-Heisenberg carrier has dimension

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3^2 = 9.$$

The four coordinates in E_P^{geom} label Weyl operators; no commuting subset has yet been selected.

Examples 6–10 continue in AQM-62, including affine-line, affine-plane, crossing, and finite-field circle supports.

63 More Tangent-Cotangent Weyl Kinematical Examples

63.1 Status and Source Anchors

This shard continues AQM-61 and uses the same local sources. The operative recipe is:

$$T_P^*X = \Omega^1_{A/k} \otimes_A \kappa(P), \qquad T_PX = \mathrm{Hom}_{\kappa(P)}(T_P^*X, \kappa(P)),$$

The full symplectic label space is

$$E_P^{\mathrm{geom}} = T_PX \oplus T_P^*X.$$

The Weyl-Heisenberg step chooses a projective representation

$$\rho_P : E_P^{\mathrm{geom}} \to PU(\mathcal{H}_{\rho_P}).$$

For a finite rational support S, use

$$E_S^{\mathrm{geom}} = \bigoplus_{P\in S} E_P^{\mathrm{geom}}, \qquad \rho_S = \bigotimes_{P\in S} \rho_P.$$

63.2 Examples 6–10

6. The affine line over $\mathbb{F}_3$. Let $X = \mathrm{Spec}\, k[x]$, $k = \mathbb{F}_3$. At the rational point $P_a = (x - a)$,

$$T^*_{P_a}X = k\, dx, \qquad T_{P_a}X = k\, \partial_x.$$

Thus each rational point has a one-dimensional tangent label direction and

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_{P_a}} = 3.$$

For the rational support $S = \mathbb{A}^1(k) = \{0, 1, 2\}$,

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_S} = 3^3.$$

For a closed point of degree d, $K = \mathbb{F}_{3^d}$, $T_PX = K$, and

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3^d.$$

7. The affine plane over $\mathbb{F}_3$. Let $X = \mathrm{Spec}\, k[x, y]$, $k = \mathbb{F}_3$. At any rational point $P = (a, b)$,

$$T_P^*X = k\, dx \oplus k\, dy, \qquad T_PX = k\, \partial_x \oplus k\, \partial_y.$$

Hence

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3^2 = 9.$$

For the rational support $S = k^2$, there are nine sites, each with carrier dimension 9, so

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_S} = 9^9 = 3^{18}.$$

8. The crossing $xy = 0$ over $\mathbb{F}_3$. Let

$$A = k[x, y]/(xy), \qquad k = \mathbb{F}_3.$$

At a rational point $P = (a, b)$ on $xy = 0$, the tangent equation is

$$b\,u + a\,v = 0.$$

At the origin this imposes no equation, so

$$T_{(0,0)}X = k^2, \qquad \dim_{\mathbb{C}} \mathcal{H}_{\rho_{(0,0)}} = 9.$$

At the four other rational points

$$(1, 0), (2, 0), (0, 1), (0, 2)$$

the tangent space is one-dimensional, so each carrier has dimension 3. For all rational points,

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_{X(k)}} = 9 \cdot 3^4 = 3^6.$$

9. The circle over $\mathbb{F}_3$. Let

$$C = \operatorname{Spec} k[x, y]/(x^2 + y^2 - 1), \qquad k = \mathbb{F}_3.$$

The rational points are

$$(1, 0), (2, 0), (0, 1), (0, 2).$$

At $P = (a, b)$, the tangent equation is

$$a\,u + b\,v = 0,$$

because 2 is invertible in k. Each point is smooth and has $\dim_k T_P C = 1$. Therefore each site has

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3.$$

For the rational support,

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_{C(k)}} = 3^4.$$

10. The circle over $\mathbb{F}_5$. Let

$$C = \operatorname{Spec} \mathbb{F}_5[x, y]/(x^2 + y^2 - 1).$$

Since -1 is a square in $\mathbb{F}_5$, AQM-59 gives four rational points:

$$(1, 0), (4, 0), (0, 1), (0, 4).$$

At $P = (a, b)$, the tangent equation is again

$$a\,u + b\,v = 0,$$

so every rational tangent space is one-dimensional over $\mathbb{F}_5$. Consequently

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 5$$

at each rational point, and

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_{C(\mathbb{F}_5)}} = 5^4.$$

63.3 End Summary

The Weyl-Heisenberg carrier dimension is determined by tangent dimension, not by the residue field alone:

local tangent dimension m	$\dim_{\mathbb{C}} \mathcal{H}_{\rho_P}$	qudit reading over K
0	1	no nontrivial local Weyl label
1	$\lvert K\rvert$	one $\lvert K\rvert$-level carrier dimension
2	$\lvert K\rvert^2$	two $\lvert K\rvert$-level carrier dimensions.

Dynamics remains separate. The de Rham CSS construction may later suggest commuting Weyl operators, but this shard does not impose them.

64 Evaluated de Rham Constraints on Tangent-Weyl Kinematics

64.1 Status and Source Anchors

This shard refines the compatibility question left open by AQM-60–AQM-62. The tangent-cotangent kinematics first constructs the full symplectic label space and its Weyl-Heisenberg projective representation. Exact one-forms from AQM-54 give a possible stabilizer-selection rule only after they are evaluated as cotangent labels at finite-residue points. The old finite C^1-CSS rule remains a prototype; the fibrewise rule below is the part compatible with the corrected tangent-Weyl principle.

64.2 Definition: Evaluated Exact-One-Form Constraints

Let $k \to A$ be the fixed base map, let $X = \operatorname{Spec} A$, and let S be a finite set of finite-residue points. All tangent and cotangent spaces below are relative to this base. Put

$$K_P = \kappa(P), \qquad V_P = T_{A/k,P}X, \qquad V_P^* = T^*_{A/k,P}X$$

and

$$V_S = \bigoplus_{P \in S} V_P, \qquad V_S^* = \bigoplus_{P \in S} V_P^*, \qquad E_S = V_S \oplus V_S^*.$$

Let

$$\rho_S : E_S \to PU(\mathcal{H}_{\rho_S})$$

be the Weyl-Heisenberg projective representation. For mixed finite-residue supports, E_S is an additive, equivalently prime-field-linear, label group with componentwise local trace characters; in the all-k-rational examples below it reduces to an ordinary k-vector space. For $\omega \in \Omega^1_{A/k}$, define its fibrewise evaluation

$$\mathrm{ev}^1_S(\omega) = \big((\omega \otimes 1)_P\big)_{P \in S} \in V_S^*.$$

The evaluated exact-one-form label space is

$$L_{\mathrm{dR}}(S) = \mathrm{ev}^1_S(dA) \subset 0 \oplus V_S^* \subset E_S.$$

Since $0 \oplus V_S^*$ is isotropic, the Weyl operators

$$\{W(0,\alpha) : \alpha \in L_{\mathrm{dR}}(S)\}$$

commute after the central phases are fixed. A stabilizer eigenspace is defined only after choosing a compatible phased-stabilizer character on $L_{\mathrm{dR}}(S)$, in the sense of `CONVENTIONS.md` (ac). The standard coordinate display below uses the compatible all-$+1$ choice.

For explicit finite-field calculations one may use the standard coordinate display. With $+1$ phases, define

$$A_S^{\mathrm{dR}} = \{v \in V_S : \prod_{P \in S} \psi_{K_P}(\alpha_P(v_P)) = 1 \text{ for every } \alpha \in L_{\mathrm{dR}}(S)\}.$$

In that coordinate display, the selected subspace is

$$\mathcal{H}_S^{\mathrm{dR}} = \ell^2(A_S^{\mathrm{dR}}).$$

When all points are $k = \mathbb{F}_p$-rational, this is the simpler condition

$$\sum_{P \in S} \alpha_P(v_P) = 0 \qquad \text{for all } \alpha \in L_{\mathrm{dR}}(S).$$

64.3 Why This Is Compatible but Not Yet Full CSS

The construction is compatible because exact one-forms evaluate to cotangent vectors, and cotangent vectors are labels in $T_{A/k,P}X \oplus T^*_{A/k,P}X$. No basis of $\Omega^1_{A/k}$ is used as a list of physical qubits.

It is not yet the old full de Rham CSS rule. A tangent-type label $w \in V_S$ commutes with all evaluated exact one-form labels exactly when

$$\alpha(w) = 0 \qquad \text{for every } \alpha \in L_{\mathrm{dR}}(S).$$

Thus a choice of tangent labels commuting with $L_{\mathrm{dR}}(S)$ is extra data. Possible sources are a metric, Hodge star, action functional, correspondence, or a deliberately chosen subspace of the annihilator. Without such data, the output is only the cotangent operator family and its stabilizer subspace.

64.4 Example 1: A Reduced Point

Let $A = k = \mathbb{F}_3$. Then $\Omega^1_{k/k} = 0$, $V_P = 0$, and

$$E_P = 0, \qquad \dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 1.$$

Also $dA = 0$, so

$$L_{\mathrm{dR}} = 0.$$

The evaluated de Rham constraint does nothing, as it should.

64.5 Example 2: Dual Numbers over F3

Let

$$A = k[\epsilon]/(\epsilon^2), \qquad k = \mathbb{F}_3, \qquad P = (\epsilon).$$

AQM-61 gives

$$V_P = k\,\partial_\epsilon, \qquad V_P^* = k\,d\epsilon, \qquad \dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3.$$

For $f = a + b\epsilon$,

$$df = b\,d\epsilon.$$

Therefore

$$L_{\mathrm{dR}} = k\,d\epsilon = V_P^*.$$

For $v = r\partial_\epsilon$, $d\epsilon(v) = r$. The annihilator condition is $br = 0$ for all $b \in k$, hence $r = 0$. Thus the selected stabilizer subspace is one-dimensional; in the coordinate display it is

$$\mathcal{H}_P^{\mathrm{dR}} = \mathbb{C}|0\rangle.$$

The old AQM-54 X-check is a Fourier-dual presentation of this prototype, not the canonical tangent-coordinate reading.

64.6 Example 3: The Rational Affine Line over F3

Let $X = \operatorname{Spec} k[x]$, $k = \mathbb{F}_3$, and take rational support

$$S = \{0, 1, 2\}.$$

AQM-62 gives $V_S = k^3$ and

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_S} = 3^3.$$

The exact one-form $df = f'(x)\,dx$ evaluates to

$$(f'(0), f'(1), f'(2)) \in k^3 = V_S^*.$$

The three functions x, x^2, x^5 give

f	$(f'(0), f'(1), f'(2))$
x	$(1, 1, 1)$
x^2	$(0, 2, 1)$
x^5	$(0, 2, 2)$.

The determinant of these three rows is $2 \in \mathbb{F}_3^\times$, so

$$L_{\mathrm{dR}}(S) = V_S^*.$$

Consequently the stabilizer subspace is one-dimensional; in coordinates it is

$$\mathcal{H}_S^{\mathrm{dR}} = \mathbb{C}|0, 0, 0\rangle.$$

64.7 Example 4: Two Independent Dual Directions

Let

$$A = k[u, v]/(u^2, v^2), \qquad k = \mathbb{F}_3, \qquad P = (u, v).$$

AQM-61 computes

$$V_P = k\,\partial_u \oplus k\,\partial_v, \qquad \dim_{\mathbb{C}} \mathcal{H}_{\rho_P} = 3^2 = 9.$$

The exact forms du and dv occur as $d(u)$ and $d(v)$. Hence

$$L_{\mathrm{dR}} = k\,du \oplus k\,dv = V_P^*.$$

The annihilator is zero, so the stabilizer subspace is one-dimensional; in coordinates it is $\mathbb{C}|0, 0\rangle$.

64.8 Example 5: The Crossing over F3

Let $A = k[x, y]/(xy)$ and take the rational support. AQM-62 gives one two-dimensional tangent at the origin and four one-dimensional tangents away from it:

$$V_S \cong k^6, \qquad \dim_{\mathbb{C}} \mathcal{H}_{\rho_S} = 3^6.$$

A function on the crossing is represented by an x-axis polynomial and a y-axis polynomial with the same constant term. Its differential along the x-axis gives arbitrary derivative values at $x = 0, 1, 2$, by Example 3. Its differential along the y-axis gives arbitrary derivative values at $y = 0, 1, 2$, again by Example 3. These six derivatives are exactly the six cotangent coordinates dual to V_S. Hence

$$L_{\mathrm{dR}}(S) = V_S^*.$$

The stabilizer subspace is one-dimensional; in coordinates it is

$$\mathbb{C}|0, 0, 0, 0, 0, 0\rangle.$$

64.9 Example 6: The Circle over F3

Let

$$C = \operatorname{Spec} k[x,y]/(x^2+y^2-1), \qquad k = \mathbb{F}_3.$$

Use the rational point order

$$(1,0),(2,0),(0,1),(0,2).$$

AQM-62 gives $V_S \cong k^4$ and

$$\dim_{\mathbb{C}} \mathcal{H}_{\rho_S} = 3^4.$$

Use tangent bases $(0,1)$ at the first two points and $(1,0)$ at the last two. The following exact forms evaluate to cotangent rows:

f	$(df_P(\tau_P))_{P\in S}$
y	$(1,1,0,0)$
x	$(0,0,1,1)$
xy	$(1,2,1,2)$
xy^3	$(0,0,1,2)$.

The determinant of this 4×4 row matrix is 2, so the rows span

$$V_S^*.$$

Thus $L_{\mathrm{dR}}(S) = V_S^*$, and the stabilizer subspace is one-dimensional; in coordinates it is $\mathbb{C}|0,0,0,0\rangle$.

64.10 Conclusion

The compatibility result is positive but austere:

$$dA \quad \rightsquigarrow \quad L_{\mathrm{dR}}(S) \subset \bigoplus_{P\in S} T^*_{A/k,P}X$$

is a canonical cotangent-label construction inside the tangent-cotangent Weyl projective representation, after the base $k \to A$ has been fixed. The examples show that using all exact forms on a finite rational support often gives

$$L_{\mathrm{dR}}(S) = V_S^*$$

and therefore a one-dimensional stabilizer subspace. A nontrivial dynamical principle must restrict the allowed exact forms, add an action/Hamiltonian, or choose additional tangent-type constraints from a justified source. The old de Rham CSS test is therefore compatible only after this fibrewise reinterpretation, and it remains a prototype rather than a final dynamics.

65 Quantum-System Association Meta-Plan

65.1 Status

This is a planning shard. It records a user-directed investigation programme, not a mathematical result. Its detailed working document is `docs/quantum_system_associations/PLAN.md`. The persistent work items are the Beads chain `aqm-qsa01` through `aqm-qsa23`.

The programme asks for a catalogue of ways to associate quantum kinematical systems to an object X. The guiding convention is to avoid treating this as a canonical functorial quantisation claim. Each association workflow must record the structure placed on X, the extra choices made, the quantum carrier or observable algebra produced, and the evidence supporting any comparison with another workflow.

65.2 Reserved Shard Sequence

Each item below is intended to become one report shard. A review shard means that earlier report material already contains much of the evidence, but the new shard must still cite the exact local anchors and explain the result in the language of this meta-plan.

Step	Planned shard	Kind	Topic
1	AQM-65	New	Association vocabulary and non-functoriality
2	AQM-66	New	Test object catalogue
3	AQM-67	New	Structureless finite set: first association
4	AQM-68	Review/New	Structureless finite set: tensor sites
5	AQM-69	New	Structureless finite set: direct-sum particle
6	AQM-70	New	AND/OR many-particle grammar
7	AQM-71	Review	Finite symplectic field labels on a set
8	AQM-72	Review/New	Bosonic and fermionic association layer
9	AQM-73	Gap	Fusion-category endpoint
10	AQM-74	New	Single-particle reduction
11	AQM-75	Review/New	What counts as a point
12	AQM-76	New	Cotangent-based first association
13	AQM-77	New	Intrinsic versus relative cotangent data
14	AQM-78	Review/New	Weyl-Heisenberg kinematics from finite phase spaces
15	AQM-79	Review	Geometric field association $\mathrm{Hom}(X, V)$
16	AQM-80	New	Finite ring example matrix
17	AQM-81	Review/New	Residue-field site association for $\mathrm{Spec}\, R$
18	AQM-82	Review/New	Nilpotent-sensitive association
19	AQM-83	Review	Product rings and tensor factorization
20	AQM-84	Review/New	Infinite ring boundary cases
21	AQM-85	New	Agreement and inequivalence table
22	AQM-86	New	Degenerate and over-restrictive cases
23	AQM-87	New	Open problems and next catalogue targets

65.3 Evidence Contract

The detailed plan fixes the first standard object families: bare finite sets, finite examples over $\mathbb{F}_3$, product rings, nilpotent Artin examples, selected mixed-characteristic examples such as $\mathbb{Z}/6\mathbb{Z}$, and infinite boundary cases already present in the lab book. Any example not yet sourced or checked remains a question until the relevant source, convention, derivation, or run exists.

Future comparison shards must distinguish at least four statuses: agreement proved by an explicit map or invariant, inequivalence proved by a concrete obstruction, degenerate collapse, and blocked/unknown due to missing convention or source. This shard itself proves none of those comparison statuses.

66 Quantum-System Association Vocabulary

66.1 Status and Local Anchors

This is a planning and convention shard only. It introduces no substantive equivalence theorem, no sourced mathematical classification, and no new representation-theoretic assertion. Its local anchors are `CONVENTIONS.md` convention `(ap)`, the meta-plan shard `AQM-64-QUANTUM-SYSTEM-ASSOCIATION-META-PLAN` (§65), and Step 01 of `docs/quantum_system_associations/PLAN.md`.

66.2 Association, Not Canonical Quantisation

In this programme, a *quantum-system association* is a recorded workflow which starts from specified structure on an object X, adds explicitly named choices, and returns kinematical quantum data. Possible output slots include a Hilbert carrier, observable algebra, Weyl–Heisenberg projective representation, Fock sector, or stabilizer subspace, but the chosen slot must be named in the relevant shard.

This language deliberately does not assert a canonical functorial quantisation. A later shard may study functorial behaviour for a specified workflow only after recording the objects, morphisms, choices, and evidence. Until then, the word "association" means a reproducible construction proposal with an evidence status, not a theorem that one universal quantisation functor exists.

66.3 Local Shorthand

The following terms are local programme labels.

`Quant1`, `Quant2`. Placeholders for the first and second association workflows in a comparison. They carry no fixed mathematical content until a shard expands them into explicit inputs, choices, and outputs.

field association. An association whose planned kinematical output is organized as field-like data over points, opens, supports, sections, or labels. The term is an umbrella for comparison planning, not a claim that all such workflows are instances of one established construction.

single-particle sector. A comparison label for the part of a workflow in which the carrier represents one alternative among sites or modes rather than independent coexistence of all sites. A later shard must say whether this means a direct sum, a selected Fock sector, or another explicitly recorded operation.

association equivalence. An evidence status for two association workflows after a concrete object X, all needed choices, and a checkable comparison map or invariant have been recorded. Without that data, "equivalence" remains a question or proposal.

degenerate case. A case in which a workflow collapses, forgets the intended structure, becomes too small or too large for the intended comparison, or reduces to an earlier case. Calling a case degenerate requires the same local evidence discipline as any other comparison status.

66.4 Dynamics Deferred

The kinematical output of a quantum-system association does not include dynamics by default. Hamiltonians, actions, time evolutions, Witten-index statements, and dynamical selection principles are later choices unless a future convention and shard explicitly add them.

66.5 Evidence Rule for Later Comparisons

Every future comparison word such as “agrees”, “reduces to”, “is equivalent to”, “is inequivalent to”, or “degenerates” must cite a local source, local derivation, test, run artifact, or explicit gap label. The comparison must name the object X, the structure placed on X, the choices made by each workflow, the kinematical outputs being compared, and the map or invariant used for the comparison. Otherwise the statement must remain an open question in the report.

67 Quantum-System Association Test Object Catalogue

67.1 Status and Anchor Contract

This is a catalogue and planning shard for Step 02 of `docs/quantum_system_associations/PLAN.md`. It records the first objects to use in later quantum-system association comparisons. It does not prove that two workflows agree, disagree, reduce to one another, or are equivalent.

The vocabulary and evidence rules are those of `CONVENTIONS.md` convention `(ap)` and AQM-65. The main local anchors are AQM-14 and `runs/2026-05-24-arithmetic-quantum-fields/data/arithmetic_quantum_field_examples.csv` for finite function-space checks; AQM-23, AQM-34, AQM-36, and AQM-40 for finite-ring examples; `runs/2026-05-26-finite-ring-database/README.md` and indexed CSV review exports under `runs/2026-05-26-finite-ring-database/data/` for finite-ring helper status; and AQM-29, AQM-30, and AQM-45 for infinite boundary examples.

The status summaries in the tables are row-local prose, not comparison tokens from convention `(au)`: *available anchor* means a precise object and at least one local workflow anchor already exist; *open question* means a proposed object or workflow still needs a local source, derivation, or convention; *helper gap* and *blocked helper* mean a local helper or workflow records an explicit missing-evidence obstruction.

67.2 Workflows to Evaluate

The later comparison shards should evaluate only named workflows. The current short list is:

finite-set carrier, tensor, direct-sum workflows	AQM-67–AQM-70 planned,
finite symplectic field labels	AQM-14; full CSV paths in the table below,
residue-field sites	AQM-22, AQM-23, AQM-28, AQM-40,
nilpotent-sensitive or thickened Frobenius layer	AQM-36 and AQM-37,
tangent-cotangent Weyl kinematics	AQM-58, AQM-61, AQM-62,
infinite-boundary quasi-local/generic layers	AQM-29, AQM-30, AQM-45.

The finite-ring database helpers are evidence about the current helper scope, not a populated catalogue: `runs/2026-05-26-finite-ring-database/README.md` records that `data/finite_rings.sqlite` is schema-only, with no populated or audited ring rows, while the CSV files are in-memory MVP review exports.

67.3 Finite Sets and Finite Function Spaces

Object	Status	Planned workflows	Local anchor or needed evidence
$X_0 = \emptyset$.	question	Finite-set carrier, tensor-site, direct-sum, and all-map field-label edge case.	Step 03–06 planned derivations must fix empty direct sum, empty tensor product, and Gram-choice conventions.
$X_1 = \{*\}$.	open question; available anchor	Structureless one-point workflows are planned; one finite-residue site is already available after choosing $X = \operatorname{Spec} \mathbb{F}_3$.	AQM-22 for the prime-field one-site anchor; Step 03 for the structureless set carrier.
$X_2 = \{1, 2\}$.	available	All-map F_3^2-valued field labels; later finite-set carrier, tensor, and direct-sum comparisons.	AQM-14 and the row `finite_two_points_all_maps` in `runs/2026-05-24-arithmetic-quantum-f` `data/arithmetic_quantum_field_examples.csv`.
$X_3 = \{1, 2, 3\}$.	available	Same workflows as X_2, with checked all-map field labels.	AQM-14 and the row `finite_set_3_full` in `runs/2026-05-24-arithmetic-quantum-f` `data/arithmetic_quantum_field_examples.csv`.
$X_n = \{1, \ldots, n\}$.	question	General finite-set carrier, tensor-site, direct-sum, and all-map field-label family.	PLAN.md Step 02 fixes the family; AQM-67–AQM-70 still need the general derivations.
$\mathbb{F}_3$, $\mathbb{F}_3^2$, the parabola $y = x^2$, and $G_m(\mathbb{F}_3)$ as finite point sets with chosen scalar function spaces.	available	Finite symplectic field-label tests with radicals and reduced Weyl labels.	AQM-14 and the named rows in `runs/2026-05-24-arithmetic-quantum-f` `data/arithmetic_quantum_field_examples.csv`; no comparison with structureless carriers is asserted here.

67.4 Finite Rings over $\mathbb{F}_3$

Let $k = \mathbb{F}_3$. The rows below keep the presentation visible. No row silently replaces a nonreduced or product presentation by another object.

Object	Status	Planned workflows	Local anchor or needed evidence
$R = k$.	available	Residue-field site, finite-set one-point comparison, underlying-set comparison, and tangent-cotangent kinematics.	AQM-22, AQM-61, and row `F_3` in `runs/2026-05-26-finite-ring-database` `data/ring_quantization_summary.csv`.
$R = k^n$, especially k^2 and the MVP row $k \times k$.	available	Product-ring residue sites, tensor-site comparison, and tangent-cotangent zero-label comparison.	AQM-40, AQM-58, AQM-61, and row `F_3xF_3` in `runs/2026-05-26-finite-ring-database` `data/ring_quantization_summary.csv`.
$R = k[\epsilon]/(\epsilon^2)$.	available anchor; helper gap	Residue-qudit layer, nilpotent-sensitive layer, tangent-cotangent Weyl kinematics, and finite-ring helper rows.	AQM-34 and AQM-36 anchor report workflows; AQM-61 anchors tangent-cotangent. The finite-ring MVP residue helper is blocked by `missing_certified_maximal_ideal` while the thickened-Frobenius row is available.
$R = k[t]/(t^3)$.	available	Cubic fat-point tangent-cotangent and thickened-Weyl tests.	AQM-36 and AQM-61; database MVP coverage would need a new presentation row before use there.
$R_p = k[t]/(p)$ for monic p, with first targets $p = t^2 - 1, t^2 + 1, t^3 - 1, t^6 - 1$.	available	Residue-qudit quotient workflow and nilpotent-sensitive workflow when p has repeated factors.	AQM-34 gives the general quotient recipe; AQM-35 and AQM-37 give the $t^n = 1$ specializations.
$?\, R = k[x, y]/I$ for a two-direction local Artin standard example.	question	Candidate object for comparing residue, thickened, and tangent-cotangent workflows.	The ideal I must be fixed by a local source or exact derivation before this becomes a standard row. AQM-61 already treats $k[u, v]/(u^2, v^2)$ for tangent-cotangent kinematics only.

67.5 Mixed-Characteristic and Database-MVP Rows

Object	Status	Planned workflows	Local anchor or needed evidence
$\mathbb{Z}/6\mathbb{Z}$.	available anchor; helper gap	Residue-field site workflow and finite-ring database helper status; possible thickened-Frobenius comparison later.	AQM-23 and row `Z/6Z` in `runs/2026-05-26-finite-ring-database` `data/ring_quantization_summary.csv` make the residue row available. The thickened-Frobenius helper is blocked by a missing certified generating character.
$\mathbb{F}_2 \times \mathbb{F}_3$.	available	Product presentation to track separately from its canonical MVP representative.	`runs/2026-05-26-finite-ring-database` `data/ring_presentations.csv` records a certified-isomorphic merge to the $\mathbb{Z}/6\mathbb{Z}$ canonical row; later comparisons must name which presentation is being used.
$\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/8\mathbb{Z}$, $\mathbb{Z}/9\mathbb{Z}$.	blocked	Residue and thickened-Frobenius database-helper rows.	`runs/2026-05-26-finite-ring-database` `data/ring_quantization_obstruction.csv` records missing certified maximal-ideal decomposition for the residue helper and missing generating character for the thickened helper.
$\mathbb{F}_2[\epsilon]/(\epsilon^2)$.	blocked	Even dual-number finite-ring helper row; possible comparison with $k = \mathbb{F}_2$ conventions later.	The MVP quantisation rows are blocked by missing residue decomposition and missing generating character; no report comparison is asserted.
The one-element zero ring.	blocked helper; layer not applicable	Database edge case only until layer semantics are fixed.	CONVENTIONS.md `(an)` and `runs/2026-05-26-finite-ring-database` `data/ring_quantization_summary.csv` plus `runs/2026-05-26-finite-ring-database` `data/ring_quantization_obstruction.csv` record `not_applicable_until_layer_sema` no Hilbert-space claim is made.

67.6 Infinite Boundary Objects

Object	Status	Planned workflows	Local anchor or needed evidence
Spec $\mathbb{F}_3[t]$.	available anchor; open question	Closed-point finite-residue quasi-local net; finite rational-support truncations; formal regular-function profiles; generic-point boundary.	AQM-29, AQM-48, AQM-49, and AQM-50. Infinite-product, completion, and generic-sector comparisons remain later questions.
Spec $\mathbb{F}_3[x, y]$.	available anchor; open question	Closed-point finite-residue net and rational 3^2-point truncation; tangent-cotangent rational-support checks.	AQM-30 and AQM-62. Full comparison with two-direction local Artin limits requires a fixed Artin standard object.
Spec $\mathbb{Z}$.	available anchor; open question	Closed-prime residue-qudit net, incomplete tensor representation after a reference-vector choice, and generic $\mathbb{Q}$ boundary sector.	AQM-45. Adelic, local-field, and full generic-sector completion conventions are not fixed here.

67.7 Catalogue Gaps to Carry Forward

The finite-ring SQLite artifact is not a populated or audited catalogue; the current evidence is the schema-only run README plus in-memory MVP CSV review exports. Database-consuming QSA shards must keep that boundary visible.

Several MVP quantisation rows are intentionally blocked. Residue rows for $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/8\mathbb{Z}$, $\mathbb{Z}/9\mathbb{Z}$, and dual-number examples need certified maximal-ideal and residue-field data inside the database helper. Thickened-Frobenius rows need certified generating characters or a recorded policy.

The two-direction local Artin standard object is not frozen by this shard. The presentation $k[u, v]/(u^2, v^2)$ is available as an AQM-61 tangent-cotangent example, but using it as the standard finite-ring comparison object still requires a local source or exact derivation for the intended finite-ring and thickened-Weyl workflows.

68 Structureless Finite-Set First Association

68.1 Status and Local Anchors

This is QSA Step 03 from `docs/quantum_system_associations/PLAN.md`. It is a local derivation from `CONVENTIONS.md` conventions `(ap)` and `(aq)`, together with the planning contract in AQM-64, the vocabulary rule in AQM-65, and the finite-set rows in AQM-66. It uses no external source and introduces no theorem about canonical Hilbert-space inner products.

The result is deliberately narrow. The input is a bare finite set X, with no order, topology, measure, field-valued function space, site Hilbert spaces, symplectic label space, or dynamics. The output is only a formal complex carrier with basis labels indexed by X.

68.2 First Carrier

Under convention `(aq)`, the first structureless finite-set association is

$$X \longmapsto \mathbb{C}\langle X\rangle := \bigoplus_{x\in X} \mathbb{C}e_x.$$

The Step 03 shorthand $\mathbb{C}^X$ means this same formal carrier in this finite-set context. The symbols e_x are basis labels, not vectors with a preassigned length, norm, phase convention, or orthogonality relation.

An element is displayed as

$$v = \sum_{x\in X} a_x e_x, \qquad a_x \in \mathbb{C}.$$

This is the whole association at this stage:

$$\mathsf{FSet}_1(X) = \big(\mathbb{C}\langle X\rangle, \{e_x\}_{x\in X}\big).$$

It does not return an adjoint, a $*$-observable algebra, a Hamiltonian, a tensor-product site system, a direct-sum particle interpretation, or Weyl–Heisenberg data.

68.3 Checked Edge Cases

The checks in this subsection are just the finite bookkeeping consequences of the displayed definition and convention `(aq)`.

Empty set. For $X = \emptyset$,

$$\mathbb{C}\langle\emptyset\rangle = \bigoplus_{x\in\emptyset} \mathbb{C}e_x = 0.$$

There are no basis labels and no Gram entries. Treating this zero carrier as a Hilbert-space edge case, or comparing it with an empty tensor product, is not part of this shard.

One-point set. For $X = \{*\}$,

$$\mathbb{C}\langle\{*\}\rangle = \mathbb{C}e_*, \qquad v = ae_*.$$

The association records one formal basis label. A one-by-one Gram entry such as G_{**} would be an added choice; it is not supplied by the bare point.

n-point set. For a displayed finite set $X = \{x_1, \dots, x_n\}$,

$$\mathbb{C}\langle X\rangle = \mathbb{C}e_{x_1} \oplus \cdots \oplus \mathbb{C}e_{x_n}.$$

The listing $x_1, \dots, x_n$ is only a display choice. It lets us write coordinates $(a_1, \dots, a_n)$ for $\sum_{i=1}^n a_i e_{x_i}$, but the ordered coordinate display is not extra structure returned by the bare-set association.

68.4 Gram-Matrix Caveat

An inner product is not determined in this workflow. If a later construction supplies a Gram array $G = (G_{xy})_{x,y\in X}$, then one may write

$$\langle e_x, e_y\rangle_G = G_{xy}$$

as notation for that added data. This shard does not derive G from X, does not select the identity matrix, and does not classify which arrays are acceptable inner products. The data for a metric version would be (X, G), not the bare set X.

Consequently, this is a structureless finite-set workflow, not a canonical Hilbert-space inner-product theorem. Words such as orthonormal, unitary, adjoint, and Hilbert completion remain unavailable here unless a later shard records the missing Gram or Hilbert convention.

68.5 Comparison Boundary

AQM-66 lists the finite-set carrier, tensor-site, direct-sum, and all-map field-label workflows as separate comparison targets. This shard makes only the first carrier workflow available. It proves no agreement, reduction, or equivalence with the tensor-site systems planned for AQM-68, the direct-sum particle systems planned for AQM-69, or the finite symplectic field-label review planned for AQM-71.

In particular, special comparisons using one-dimensional internal spaces in a future tensor or direct-sum workflow remain planned checks. They are not proved here, even in the one-point or n-point cases.

69 Structureless Finite-Set Tensor-Site Association

69.1 Status and Local Anchors

This is QSA Step 04 from `docs/quantum_system_associations/PLAN.md`. It is a local finite-dimensional construction using `CONVENTIONS.md` conventions `(ap)` and `(ar)`, with AQM-64–AQM-66 as the planning and catalogue anchors.

The input is not a bare finite set alone. A bare finite set X supplies site labels only. The tensor-site workflow starts after one adds per-site quantum data. The earlier AQM-67 workflow instead returned only the formal carrier $\mathbb{C}\langle X\rangle$ from convention `(aq)`; it did not include inner products, adjoints, observable algebras, or tensor factors.

69.2 Tensor-Site Input Data

There are two finite-dimensional input versions.

Hilbert-site input. For every $x \in X$, choose a finite-dimensional complex Hilbert space $\mathcal{H}_x$. Then set

$$\mathcal{A}_x = \mathrm{End}_{\mathbb{C}}(\mathcal{H}_x).$$

Algebra-site input. Alternatively, choose directly a finite-dimensional unital $*$-algebra $\mathcal{A}_x$ for every $x \in X$. This gives an observable-algebra workflow, but it does not choose a Hilbert carrier unless a representation is added separately.

These choices are part of the data. They are not recovered from the structureless set X. An ordering of a finite subset may be used only to write tensor products; the set itself does not supply a preferred order.

69.3 Finite Tensor Products

For $S \subset X$, define in the Hilbert-site version

$$\mathcal{H}(S) = \bigotimes_{x\in S} \mathcal{H}_x, \qquad \mathcal{A}(S) = \bigotimes_{x\in S} \mathcal{A}_x \subseteq \mathrm{End}_{\mathbb{C}}(\mathcal{H}(S)).$$

The empty tensor products are

$$\mathcal{H}(\emptyset) = \mathbb{C}, \qquad \mathcal{A}(\emptyset) = \mathbb{C}.$$

In the algebra-site version, keep the same formula for $\mathcal{A}(S)$ as the finite algebraic tensor product over $\mathbb{C}$.

If $X = \{x_1, \ldots, x_n\}$ is listed for display and $\dim \mathcal{H}_{x_i} = d_i$, then

$$\dim \mathcal{H}(X) = \prod_{i=1}^{n} d_i, \qquad \dim \mathcal{A}(X) = \prod_{i=1}^{n} d_i^2$$

in the full-matrix Hilbert-site version. This is the elementary finite tensor bookkeeping for the chosen factors, not a claim that X determines the numbers d_i.

69.4 Local Observable Inclusions

If $S \subset T \subset X$, the tensor-site inclusion is

$$\iota_{S,T} : \mathcal{A}(S) \longrightarrow \mathcal{A}(T), \qquad a \longmapsto a \otimes \mathbf{1}_{T \setminus S}.$$

It preserves multiplication, the unit, and adjoints because each operation is computed factorwise. For $S \subset T \subset U$,

$$\iota_{T,U} \circ \iota_{S,T} = \iota_{S,U}$$

by the associativity of the finite tensor product and by tensoring identities on exactly the missing factors. Thus subsets of X, ordered by inclusion, carry a covariant finite local-observable system after the site algebras have been chosen.

69.5 Boundary With AQM-67

The Step 03 carrier and the Step 04 tensor-site system answer different questions. AQM-67 gives

$$\mathbb{C}\langle X \rangle = \bigoplus_{x \in X} \mathbb{C} e_x$$

with basis labels but no Hilbert or observable structure. This shard gives

$$(X, \{\mathcal{H}_x\}_{x \in X}) \longmapsto \bigotimes_{x \in X} \mathcal{H}_x$$

or its observable-algebra variant. The first construction has additive bookkeeping in the labels; the second has tensor-product bookkeeping for independent coexisting sites.

Even the one-dimensional-site edge case is not an identification of workflows. If all $\mathcal{H}_x = \mathbb{C}$, then $\bigotimes_x \mathcal{H}_x \cong \mathbb{C}$, while AQM-67 has one formal generator per element of X. Any later comparison with $\mathbb{C}\langle X \rangle$ must name the extra choices and the comparison map.

69.6 Review Anchors and Caution

The existing arithmetic shards already contain important tensor-site examples, but each has extra structure beyond a bare finite set.

AQM-14 starts from a finite point set together with a chosen finite symplectic target V, a chosen field-label subspace $E \leq \mathrm{Map}(X, V)$, and the radical quotient before forming its Weyl observable algebra. This is a field-label workflow, not a theorem that every bare finite set has canonical qudit factors.

AQM-22 proves the uniform prime-field case with $X = \mathbb{F}_p$ as a finite set only after choosing all maps $E = \mathrm{Map}(X, \mathbb{F}_p^2)$ and the Weyl operators on $\ell^2(\mathbb{F}_p^X)$. The same shard separates this from $X = \operatorname{Spec} \mathbb{F}_p$, which is the one-site spectrum.

AQM-24 and AQM-25 use finite-spectrum or quasi-compact-open Zariski locality after residue fields have supplied local factors $\ell^2(\kappa(x))$ and algebras $\mathrm{End}(\ell^2(\kappa(x)))$. Their tensor-by-identity inclusions are anchors for the local-net formula above, but their sites are residue-field sites, not arbitrary sites canonically attached to a bare set.

AQM-40 proves the product-field spectrum case $\mathrm{Spec}(k^n)$: the spectrum has n discrete residue-field points and the resulting residue-qudit arithmetic field is the ambient n-qudit Weyl net. That is a uniform finite-qudit anchor for the particular arithmetic object k^n, not a universal equivalence between structureless finite sets and all tensor-site systems.

69.7 Conclusion

The QSA Step 04 output is available only for the enhanced input

$$(X, \{\mathcal{H}_x\}_{x\in X}) \quad \text{or} \quad (X, \{\mathcal{A}_x\}_{x\in X}).$$

After this data is supplied, finite tensor products and tensor-by-identity local observable inclusions are formal finite-dimensional bookkeeping. The bare set alone still supplies only labels; it does not select the site Hilbert spaces, observable algebras, Weyl labels, stabilizer data, or dynamics.

70 Structureless Finite-Set Direct-Sum Particle Association

70.1 Status and Local Anchors

This is QSA Step 05 from `docs/quantum_system_associations/PLAN.md`. It is a local finite-dimensional construction using `CONVENTIONS.md` conventions `(ap)`, `(aq)`, `(ar)`, and `(as)`, with AQM-64–AQM-66 as the planning and catalogue anchors. AQM-67 is the formal finite-set carrier comparison point, and AQM-68 is the tensor-site comparison point.

The input is not a bare finite set alone. A bare finite set X supplies only labels for alternatives. The direct-sum particle workflow starts after one adds an internal finite-dimensional complex carrier, or a finite-dimensional complex Hilbert space, at each site. This shard uses no canonical boson or fermion field claim from AQM-19. AQM-23 is used only as a cautionary local anchor: direct sums in label bookkeeping and Hilbert-space direct sums must not be interchanged with tensor-product residue-qudit systems without an explicit workflow.

70.2 Input Data

There are two finite-dimensional input versions.

Carrier-site input. For every $x \in X$, choose a finite-dimensional complex vector space K_x. The data are the labelled family $\{K_x\}_{x\in X}$, not merely the set X.

Hilbert-site input. For every $x \in X$, choose a finite-dimensional complex Hilbert space $\mathcal{H}_x$. This includes the inner product on each internal carrier.

The word "internal" only records what is carried while the alternative site label is x. It does not add dynamics, hopping amplitudes, a Hamiltonian, a preferred observable algebra, or a tensor-factor interpretation.

70.3 Direct-Sum Carrier

Under convention `(as)`, the carrier-site version is

$$K_\oplus(X) = \bigoplus_{x\in X} K_x.$$

An element is a tuple $\psi = (\psi_x)_{x\in X}$ with $\psi_x \in K_x$. Since X is finite, no support or completion issue is present. The canonical injections and projections are denoted

$$j_x : K_x \longrightarrow K_\oplus(X), \qquad p_x : K_\oplus(X) \longrightarrow K_x,$$

with $p_x j_y = 0$ for $x \neq y$ and $p_x j_x = \mathrm{id}_{K_x}$. If a displayed listing $X = \{x_1, \dots, x_n\}$ is chosen and $\dim_{\mathbb{C}} K_{x_i} = d_i$, then

$$\dim_{\mathbb{C}} K_\oplus(X) = \sum_{i=1}^{n} d_i.$$

This dimension count is finite direct-sum bookkeeping for the chosen carriers, not a number determined by the bare set.

In the Hilbert-site version,

$$\mathcal{H}_\oplus(X) = \bigoplus_{x\in X} \mathcal{H}_x$$

with inner product

$$\langle \psi, \phi \rangle_{\oplus} = \sum_{x \in X} \langle \psi_x, \phi_x \rangle_{\mathcal{H}_x}.$$

The orthogonality of distinct summands belongs to this chosen Hilbert direct sum. It is not inherited from the bare finite set and was not supplied by AQM-67.

70.4 Edge Cases

Empty set. For $X = \emptyset$,

$$K_{\oplus}(\emptyset) = 0, \qquad \mathcal{H}_{\oplus}(\emptyset) = 0$$

in the corresponding versions. There are no site labels and no internal carriers to add. This differs from the empty tensor-site convention in AQM-68, where the empty tensor product is $\mathbb{C}$.

One-point set. For $X = \{*\}$,

$$K_{\oplus}(X) = K_*, \qquad \mathcal{H}_{\oplus}(X) = \mathcal{H}_*.$$

Thus the one-point direct-sum workflow returns exactly the chosen internal carrier at the single site. A one-dimensional result appears only if that internal carrier was chosen to be one-dimensional.

70.5 Boundary With Tensor-Site Coexistence

AQM-68 records the tensor-site workflow

$$(X, \{\mathcal{H}_x\}_{x \in X}) \longmapsto \bigotimes_{x \in X} \mathcal{H}_x.$$

That construction represents independent coexisting site factors. The present workflow represents alternatives by

$$(X, \{\mathcal{H}_x\}_{x \in X}) \longmapsto \bigoplus_{x \in X} \mathcal{H}_x.$$

For displayed finite dimensions $d_i = \dim \mathcal{H}_{x_i}$, the tensor-site carrier has dimension $\prod_i d_i$, while the direct-sum carrier has dimension $\sum_i d_i$. The empty cases also differ:

$$\bigotimes_{x \in \emptyset} \mathcal{H}_x = \mathbb{C}, \qquad \bigoplus_{x \in \emptyset} \mathcal{H}_x = 0.$$

These are different workflows even when the same labelled family of site Hilbert spaces is available.

This shard also does not select an observable algebra. If a later step needs one, the full algebra $\mathrm{End}_{\mathbb{C}}(\bigoplus_x \mathcal{H}_x)$ and the block-preserving algebra $\bigoplus_x \mathrm{End}_{\mathbb{C}}(\mathcal{H}_x)$ are different choices and must be named. The tensor-site algebra from AQM-68 is instead $\bigotimes_x \mathrm{End}_{\mathbb{C}}(\mathcal{H}_x)$.

70.6 One-Dimensional Internal Spaces and AQM-67

The comparison with AQM-67 is available only after extra choices are stated. Assume that every carrier K_x is a one-dimensional complex vector space and choose a nonzero vector $u_x \in K_x$ for each $x \in X$. Then

$$\Phi : \mathbb{C}\langle X \rangle \longrightarrow \bigoplus_{x \in X} K_x, \qquad \Phi(e_x) = j_x(u_x),$$

is a vector-space isomorphism. In the empty case this is the unique map $0 \to 0$. In the one-point case it is the chosen identification $\mathbb{C}e_* \cong K_*$.

Without the chosen vectors u_x, the bare set does not identify the formal generator e_x with a vector in K_x. If the K_x are Hilbert lines, then chosen unit vectors u_x make Φ compatible with the identity Gram convention on $\mathbb{C}\langle X\rangle$. That Hilbert comparison is still a comparison of enhanced data: AQM-67 did not supply inner products, unit vectors, or orthonormality from the set alone.

70.7 Conclusion

The QSA Step 05 output is the alternatives carrier

$$\bigoplus_{x\in X} K_x \quad \text{or} \quad \bigoplus_{x\in X} \mathcal{H}_x$$

after the internal carriers or Hilbert spaces have been supplied. It is not the formal bare-set carrier of AQM-67 unless one-dimensional internal carriers and basis vectors are chosen, and it is not the tensor-site coexistence system of AQM-68.

71 AND/OR Many-Particle Grammar

71.1 Status and Local Anchors

This is QSA Step 06 from `docs/quantum_system_associations/PLAN.md`. It is a local finite-dimensional construction using `CONVENTIONS.md` conventions `(ap)`, `(ar)`, `(as)`, and `(at)`, with AQM-64–AQM-66 as the planning and catalogue anchors. AQM-68 supplies the finite tensor/coexistence operation, and AQM-69 supplies the finite direct-sum/alternative operation.

The shard uses no external source and proves no canonical bosonic, fermionic, para, Fock, or fusion-category theorem. AQM-23 is used only as the local caution that direct sums and tensor products live in different layers unless a workflow explicitly relates them. AQM-19 is used only as a boundary marker for later symmetric and antisymmetric field enhancements; no canonical-field equivalence from that shard is imported here.

71.2 Finite Grammar

The input is a finite list of already chosen finite-dimensional complex carriers. These may be the site Hilbert spaces $\mathcal{H}_x$ from AQM-68 or the internal carriers K_x from AQM-69. The bare finite set still supplies only labels; the carriers are extra data.

A grammar expression is built by the rules

$$E ::= 0 \mid \mathbf{1} \mid K \mid \mathrm{AND}(E, F) \mid \mathrm{OR}(E, F),$$

where K ranges over the chosen carrier atoms, 0 is the zero carrier, and $\mathbf{1}$ is the tensor unit $\mathbb{C}$. Its carrier $\mathcal{V}(E)$ is defined recursively by

$$\begin{aligned}
\mathcal{V}(0) &= 0, \\
\mathcal{V}(\mathbf{1}) &= \mathbb{C}, \\
\mathcal{V}(K) &= K, \\
\mathcal{V}(\mathrm{AND}(E, F)) &= \mathcal{V}(E) \otimes_{\mathbb{C}} \mathcal{V}(F), \\
\mathcal{V}(\mathrm{OR}(E, F)) &= \mathcal{V}(E) \oplus \mathcal{V}(F).
\end{aligned}$$

Thus AND means independent coexistence in the finite tensor product sense of AQM-68, while OR means alternatives in the finite direct-sum sense of AQM-69. Parentheses are part of the written grammar. Associativity and distributivity may be used as finite vector-space isomorphisms after the expression is fixed, but they are not new structure coming from the bare set.

If the atoms are Hilbert spaces, the same recursion uses the finite Hilbert tensor product and finite Hilbert direct sum. If the atoms are only complex carriers, the output is only a complex vector space. No observable algebra is selected by this grammar alone; $\mathrm{End}(\mathcal{V}(E))$, a block-preserving direct-sum algebra, and a tensor-site local algebra would be different later choices.

71.3 Particle-Count Sectors

When every carrier atom is declared to be one particle and $\mathbf{1}$ is declared to be the zero-particle carrier, the finite grammar has a recursive particle-count grading. Write $\mathcal{V}_n(E)$ for the degree-n sector. The definitions are

$$\begin{array}{rclcrcll}
\mathcal{V}_0(\mathbf{1}) & = & \mathbb{C}, & & \mathcal{V}_n(\mathbf{1}) & = & 0 & (n \neq 0), \\
\mathcal{V}_1(K) & = & K, & & \mathcal{V}_n(K) & = & 0 & (n \neq 1), \\
\mathcal{V}_n(0) & = & 0 & \text{for all } n. & & & &
\end{array}$$

The operations are

$$\mathcal{V}_n(\mathrm{OR}(E,F)) = \mathcal{V}_n(E) \oplus \mathcal{V}_n(F)$$

and

$$\mathcal{V}_n(\mathrm{AND}(E,F)) = \bigoplus_{i+j=n} \mathcal{V}_i(E) \otimes_{\mathbb{C}} \mathcal{V}_j(F).$$

Because the parse tree is finite, only finitely many sectors are nonzero, and

$$\mathcal{V}(E) = \bigoplus_{n\geq 0} \mathcal{V}_n(E)$$

is a finite direct sum. The integer n is a grammar count of declared one-particle atoms. It is not a Hamiltonian, a dynamics, a statistics rule, or an infinite-particle completion.

71.4 Basic Finite Sectors

The AQM-69 one-particle alternatives carrier is recovered by the expression

$$P_X = \mathrm{OR}_{x\in X} K_x, \qquad \mathcal{V}(P_X) = \bigoplus_{x\in X} K_x.$$

If all K_x are declared one-particle atoms, then $\mathcal{V}_1(P_X) = \mathcal{V}(P_X)$ and all other sectors vanish.

A finite site-option expression is

$$G_X = \mathrm{AND}_{x\in X}(\mathbf{1}\,\mathrm{OR}\,K_x).$$

After choosing a display order for the finite tensor product, finite distributivity gives

$$\mathcal{V}_n(G_X) \cong \bigoplus_{\substack{S\subset X\\ |S|=n}} \bigotimes_{x\in S} K_x.$$

This expression allows each displayed site factor to choose either the zero-particle alternative $\mathbf{1}$ or the one-particle alternative K_x. That site-option rule is part of the expression, not a consequence of the bare finite set.

By contrast, if

$$P_X^{\mathrm{AND}\,m} = \underbrace{P_X\,\mathrm{AND}\cdots\mathrm{AND}\,P_X}_{m\text{ factors}},$$

then

$$\mathcal{V}(P_X^{\mathrm{AND}\,m}) \cong \bigoplus_{(x_1,\ldots,x_m)\in X^m} K_{x_1}\otimes\cdots\otimes K_{x_m},$$

and the whole carrier lies in degree m. The m tensor factors are grammar slots. They are not symmetrized, antisymmetrized, or quotiented, and the expression does not forbid two slots from choosing the same site.

A finite truncated many-sector expression can be written as

$$T_{\leq M}(X) = \mathbf{1}\,\mathrm{OR}\,P_X\,\mathrm{OR}\,P_X^{\mathrm{AND}\,2}\,\mathrm{OR}\cdots\mathrm{OR}\,P_X^{\mathrm{AND}\,M}.$$

It has sectors $0,\ldots,M$ by construction. This is still a finite direct-sum/tensor expression, not an infinite Fock completion.

71.5 Boundary With Enhancements

The grammar above is elementary vector-space bookkeeping. It does not choose statistics.

For symmetric or antisymmetric sectors, one would need extra data and a new rule: an action of the permutation group on tensor slots, plus a choice of invariants, coinvariants, projectors, or quotients and compatible Hilbert structure. None of that is supplied by AND and OR alone. AQM-19 belongs to that later canonical-field boundary; this shard does not claim that $T_{\leq M}(X)$ is a bosonic or fermionic Fock construction.

Para-statistics would likewise require a separately recorded permutation-sector or algebraic selection rule. Fusion-category and Levin-Wen/string-net variants require their own objects, tensor product, direct-sum or fusion rules, associativity data, and any pivotal, spherical, or braiding conventions needed by the model. Those choices are not hidden in this finite carrier grammar and are deferred to later QSA shards.

Infinite-particle constructions are also out of scope. The expression $\bigoplus_{m\geq 0} P_X^{\mathrm{AND}\,m}$, any Hilbert direct-sum completion of it, and any infinite tensor product require a separate completion convention. Step 06 only records finite expressions and their finite particle-count sectors.

71.6 Conclusion

QSA Step 06 adds a syntax for combining the AQM-68 tensor-site operation and the AQM-69 direct-sum particle operation:

$$\mathrm{AND} = \otimes_{\mathbb{C}}, \qquad \mathrm{OR} = \oplus.$$

With one-particle atoms and the vacuum unit $\mathbf{1} = \mathbb{C}$, this syntax identifies finite particle-count sectors by recursion. It remains separate from statistics, Fock completions, and fusion-category enhancements.

72 Finite Symplectic Field Labels on a Set

72.1 Status and Local Anchors

This is QSA Step 07 from `docs/quantum_system_associations/PLAN.md`. It is a review shard, not a new construction shard. It adds no new examples, no new source claims, and no new run artifacts.

The vocabulary and evidence discipline are those of `CONVENTIONS.md` convention `(ap)` and AQM-65. The finite field-label convention is `CONVENTIONS.md` convention `(q)`. The main report anchors are AQM-14, especially its dependencies D1–D4, lemmas L1–L5, and Weyl step L6; the exact $\mathbb{F}_3$ run bundle `runs/2026-05-24-arithmetic-quantum-fields/`; and AQM-22, especially its all-map prime-field construction D3–D6 and theorems T1–T2. The finite-support variant is the closed-point finite-residue version recorded in AQM-28 D5–T1 and `CONVENTIONS.md` convention `(y)`.

72.2 Association Data

The finite-set version starts with a finite point set X, a finite symplectic vector space (V, ω_V) over $\mathbb{F}_q$, and a chosen $\mathbb{F}_q$-linear subspace

$$E \leq \mathrm{Map}(X, V).$$

As recorded in convention `(q)` and AQM-14 D1–D4, the neutral function-space notation for a set is $\mathrm{Map}(X, V)$. The word Hom is reserved for a separately specified structure-preserving class of maps.

Thus this association is not attached to a bare set alone. The set supplies labels for points; the workflow also needs the target V, the subspace E, and the pointwise pairing weights. In the existing checked run all weights are 1.

For infinite closed-point layers over affine schemes, the finite-qudit variant uses finite support rather than all maps. AQM-28 defines, for an open $U \subset X$,

$$E(U) = \bigoplus_{x \in U \cap |X|_{\mathrm{cl}}} \kappa(x)^2$$

as a finite-support direct sum over closed points. Regular-function profiles may have infinite support there, so AQM-28 keeps them separate from quasi-local Weyl operators.

72.3 Pointwise Pairing and Radical

For $E \leq \mathrm{Map}(X, V)$, AQM-14 and convention `(q)` use the pointwise alternating form

$$\Omega_E(\phi, \psi) = \sum_{x \in X} w_x \, \omega_V(\phi(x), \psi(x)).$$

AQM-14 L1–L2 check the pointwise vector-space and bilinear alternating-form properties. AQM-14 L3 records nondegeneracy for the all-map case when ω_V is nondegenerate and all weights are nonzero.

For a chosen subspace E, the radical must still be checked:

$$\mathrm{rad}(E) = \{\phi \in E : \Omega_E(\phi, \psi) = 0 \text{ for all } \psi \in E\}.$$

AQM-14 L4–L5 and convention `(q)` fix the rule:

$$E_{\mathrm{red}} = E/\,\mathrm{rad}(E).$$

If the radical is zero, E itself is the Weyl label space. If the radical is nonzero, the honest finite symplectic Weyl label space is the quotient E_{red}, not the unreduced E.

In the checked $\mathbb{F}_3$ scalar-function examples, AQM-14 uses $V = \mathbb{F}_3^2$ with

$$\omega((q,p),(q',p')) = pq' - qp',$$

and a scalar subspace $U \le \mathrm{Map}(X, \mathbb{F}_3)$ gives

$$E = Uq \oplus Up, \qquad \Omega((q,p),(q',p')) = B(p,q') - B(q,p'),$$

where $B(f,g) = \sum_x f(x)g(x)$.

72.4 Weyl Labels and Qudit Carriers

After reduction, AQM-14 L6 chooses a symplectic basis

$$E_{\mathrm{red}} \cong \mathbb{F}_q^n \oplus \mathbb{F}_q^n$$

and represents the labels on

$$\mathcal{H} = \mathbb{C}^{\mathbb{F}_q^n} \cong \ell^2(\mathbb{F}_q^n)$$

by shifts, phases, and Weyl operators

$$T_q|a\rangle = |a+q\rangle, \qquad R_p|a\rangle = \chi(p \cdot a)|a\rangle, \qquad W(q,p) = T_q R_p.$$

The resulting observable algebra is the complex span of these Weyl operators, as stated in AQM-14 after L6.

AQM-22 is the prime-field qudit comparison. It separates the one-point spectrum $X_{\mathrm{sp}} = \mathrm{Spec}\,\mathbb{F}_p$ from the p-element set $X_{\mathrm{el}} = \mathbb{F}_p$. For X_{el} and

$$E_{\mathrm{el}} = \mathrm{Map}(X_{\mathrm{el}}, \mathbb{F}_p^2),$$

AQM-22 T1 identifies the generated Weyl operators with the ambient odd-p p-qudit Pauli group, one qudit for each element of $\mathbb{F}_p$. AQM-22 T2 records that stabilizer codes are extra data: one must choose an isotropic label subspace and compatible phases.

72.5 Checked $\mathbb{F}_3$ Rows

The exact run `runs/2026-05-24-arithmetic-quantum-fields/` was produced by `scripts/bridges/arithmetic_quantum_fields.jl`. Its README states that the run checks pointwise symplectic function spaces, radicals, reduced Weyl labels, Hilbert dimensions, and observable-algebra dimensions. Its schema is the `arithmetic_quantum_field_examples.csv` entry in `data/SCHEMA.md`.

The table below is only a review of existing CSV rows in `runs/2026-05-24-arithmetic-quantum-fields/data/arithmetic_quantum_field_examples.csv`. It is not a new computation.

CSV row	$\|X\|$	$\dim U$	$\dim \mathrm{rad}$	$\|E_{\mathrm{red}}\|$	$\dim \mathcal{H}$
`finite_two_points_all_maps`	2	2	0	81	9
`finite_set_3_full`	3	3	0	729	27
`vector_space_F3_affine`	3	2	2	9	3
`affine_plane_F3_linear`	9	2	4	1	1
`parabola_F3_degree_le_1`	3	3	0	729	27

The companion basis file `runs/2026-05-24-arithmetic-quantum-fields/data/arithmetic_quantum_field_bases.csv` records the scalar basis values and scalar Gram rows behind those summary rows. For example, the row group `vector_space_F3_affine` records basis labels $1, t$ and Gram rows `0 0` and `0 2`; the row group `parabola_F3_degree_le_1` records basis labels $1, x, y$ and Gram rows `0 0 2`, `0 2 0`, and `2 0 2`.

72.6 Boundary

This shard closes only the review requested by QSA Step 07. It does not extend the example list beyond the existing AQM-14/AQM-22 material and the 2026-05-24 CSV rows. Any new finite set, field space, weighting, radical calculation, qudit carrier count, or comparison with another association workflow would require a new local source, derivation, or run row before it can be stated as report evidence.

73 Bosonic and Fermionic Association Layer

73.1 Status and Local Anchors

This is QSA Step 08 from `docs/quantum_system_associations/PLAN.md`. It is a review shard. It adds no new examples, no new source claims, no new run artifacts, and no comparison theorem with the finite AND/OR grammar of AQM-70.

The vocabulary and evidence discipline are those of `CONVENTIONS.md` convention `(ap)` and AQM-65. The canonical boson/fermion comparison convention is `CONVENTIONS.md` convention `(s)`. The main report anchors are AQM-19 for finite CCR/CAR field-label tests, AQM-20 for the abstract Grassmann displacement ray algebra and Cahill–Glauber representation, and AQM-21 for the unitarity, displacement, ray-product, and ordinary Hilbert-space boundary derivations.

The active source manifest is `references/canonical_fields/SOURCES.md`. For the bosonic CCR layer it registers Derezinski's `references/canonical_fields/derezinski_math_ph_0511030_source/derezinski.tex` at lines 1078–1135 for Weyl-form CCR, 1197–1284 for the Schrodinger representation and finite-dimensional Stone–von Neumann statement, and 2251–2301 for bosonic Fock CCR fields; it also registers Bekka's `references/canonical_fields/bekka_2502_00387_source/CCR-GeneralRings-v5.tex` at lines 220–231, 253–273, 287–350, and 603–633 for the finite real or local-field CCR uniqueness layer. For the fermionic CAR layer it registers Derezinski's `references/canonical_fields/derezinski_math_ph_0511030_source/derezinski.tex` at lines 1437–1524 for CAR definitions, 1612–1696 for finite-dimensional CAR uniqueness, 1803–1964 and 2098–2131 for Fock and creation/annihilation operators, and 2629–2662 for fermionic Fock CAR fields.

For the Grassmann displacement layer, the active Cahill–Glauber source is the official arXiv e-print extraction `references/canonical_fields/cahill_glauber_physics_9808029v1_source/fxxx.tex`. The active locators are lines 323–335 for CAR and vacuum, 337–365 for Grassmann variables and adjunction, 603–643 for nilpotency and parity of Grassmann functions, 367–390 for $D(\gamma)$, 391–414 for displacement identities, 416–464 for the ray-product formula, 815–833 for completeness, and 835–854 for the caveat that physical quantum operators do not themselves involve Grassmann variables. The historical APS/marker chain recorded in the manifest is source-integrity provenance only and is not used as active evidence here.

The same canonical-field manifest registers Keyl–Schlingemann's `references/canonical_fields/keyl_schlingemann_0906_2929_source/fidelity_arXiv.tex` at lines 253–260, 270–355, 441–492, and 496–531 for GAR as a twisted Grassmann/CAR framework and for the caveat that the Grassmann phase-space right hand side is not an ordinary complex scalar. The supergeometry manifest `references/supergeometry/SOURCES.md` registers Witten's `references/supergeometry/witten_1209_2199_source/supernotes.tex` at lines 652–660 for odd variables quantized by Clifford anticommutation relations and lines 671–704 for the ΠTM parity-reversal and Clifford-action dictionary. This shard uses those supergeometry locators only as background anchors for the parity/Clifford boundary; it does not import a new finite-field supergeometry example.

73.2 Catalogue Slots

In QSA language, the locally available boson/fermion layers are the following review entries.

Layer	Added input data	Kinematical output	Status
Finite bosonic CCR field	Finite site set X, real symplectic target $V_B = \mathbb{R}_q \oplus \mathbb{R}_p$, and $E_B = \mathrm{Map}(X, V_B)$ with the summed pointwise form.	Regular irreducible Weyl/CCR representation, Schrodinger model $L^2(\mathbb{R}^{\|X\|})$, and equivalent finite-mode bosonic Fock model.	reviewed from AQM-19 T1
Finite fermionic CAR field	Finite site set X, finite-dimensional complex internal Hilbert space K, $Z = \mathrm{Map}(X, K)$, and $Y_F = \mathrm{Re}(Z \oplus \overline{Z})$ with positive scalar product.	CAR/Clifford field operators on the antisymmetric Fock space $\Gamma_{\mathrm{a}}(Z)$.	reviewed from AQM-19 T2
Grassmann-Weyl displacement calculus	Finite CAR mode set I, Grassmann coefficient algebra, and the graded Grassmann-extended CAR algebra.	Unitary elements $D(\gamma)$ in the extended algebra with Grassmann-valued ray product and CAR displacement identities.	reviewed from AQM-20/21

The entries are not bare finite-set associations. Each begins only after the displayed extra data have been supplied.

73.3 Finite Bosonic Association

AQM-19 D1–D4 and convention `(s)` place a finite bosonic field in the same broad field-label slot as AQM-14, but with real symplectic labels rather than finite-field labels. The label space is

$$E_B = \mathrm{Map}(X, \mathbb{R}_q \oplus \mathbb{R}_p),$$

and the form is the finite sum

$$\Omega_B((q,p),(q',p')) = \sum_{x\in X}(p_x q'_x - p'_x q_x).$$

AQM-19 L1 checks that this is a finite-dimensional real symplectic space. AQM-19 L2 then invokes the locally registered Stone–von Neumann sources to identify an irreducible regular Weyl-form representation with the Schrodinger representation on $L^2(\mathbb{R}^{|X|})$, and AQM-19 T1 records the finite bosonic field comparison. This is a sourced finite kinematical CCR association, not a dynamics statement.

73.4 Finite Fermionic CAR Association

AQM-19 D5–D7 use different input data for standard finite fermions. After choosing a finite-dimensional complex internal one-particle Hilbert space K, set

$$Z = \mathrm{Map}(X, K), \qquad Y_F = \mathrm{Re}(Z \oplus \overline{Z}).$$

The scalar product is

$$\alpha((z,\bar z),(w,\bar w)) = \mathrm{Re}\langle z, w\rangle_Z.$$

On the antisymmetric Fock space

$$\Gamma_{\mathrm{a}}(Z) = \bigoplus_k \wedge^k Z,$$

AQM-19 defines $\phi(z, \bar{z}) = a^*(z) + a(z)$, proves the exterior-algebra CAR in L3, and proves

$$\{\phi(y), \phi(y')\} = 2\alpha(y, y')I$$

in L4. AQM-19 T2 therefore records the finite standard fermionic canonical field as a CAR/Clifford association. It is not obtained by taking the bosonic real symplectic map-space recipe and replacing the target by Grassmann variables.

73.5 Grassmann-Weyl Displacement Review

The Grassmann layer is available only inside a Grassmann-extended CAR algebra. AQM-20 defines a finite CAR algebra generated by $a_i, a_i^\dagger$, a Grassmann coefficient algebra generated by γ_i, γ_i^*, and the graded product algebra in which the Grassmann generators anticommute with the odd CAR generators. It then defines

$$K(\gamma) = \sum_i (a_i^\dagger \gamma_i - \gamma_i^* a_i), \qquad D(\gamma) = e^{K(\gamma)}.$$

AQM-21 proves $K(\gamma)^\dagger = -K(\gamma)$, so $D(\gamma)$ is unitary in the extended algebra; proves that conjugation displaces the CAR generators by $a_i \mapsto a_i + \gamma_i$ and $a_i^\dagger \mapsto a_i^\dagger + \gamma_i^*$; and proves the ray product

$$D(\alpha)D(\beta) = D(\alpha + \beta) \exp\left(\frac{1}{2}\sum_i (\beta_i^* \alpha_i - \alpha_i^* \beta_i)\right).$$

AQM-21 T2 is the boundary: the multiplier is an even Grassmann unit, not a complex $U(1)$ phase, so this is not an ordinary scalar-projective unitary representation on a complex Hilbert space unless the Grassmann coefficient algebra has first been included and tracked.

73.6 Boundary With QSA Grammar

AQM-70 supplies a finite AND/OR carrier grammar: AND $= \otimes_{\mathbb{C}}$ and OR $= \oplus$. That grammar records finite vector-space bookkeeping and explicitly defers symmetric, antisymmetric, para, Fock, and fusion-category enhancements. This shard does not identify the bosonic CCR or fermionic CAR layers with that grammar.

The finite bosonic entry requires a real symplectic map space plus a regular CCR/Weyl representation. The finite fermionic entry requires a positive CAR/Clifford label space and antisymmetric Fock representation. The Grassmann displacement entry requires the extended CAR coefficient algebra and has a Grassmann-valued multiplier. No map, quotient, invariant, or local derivation currently compares these outputs with AQM-70 as an agreement or equivalence result.

73.7 Gaps

The following variants are not results in the current catalogue.

- **Parafermions and para-statistics.** No parafermion source locator, convention, mode algebra, permutation-sector rule, or checked example is registered for this QSA step. Convention `(at)` already marks para-statistics as a separate enhancement, so any parafermion association is a gap.
- **Fusion-category and Levin-Wen/string-net variants.** Convention `(h)` leaves the Levin-Wen fusion-category gauge, pivotal/spherical structure, and F/R-data unfixed, and the QSA plan assigns this to Step 09. No fusion-category endpoint is asserted here.

- **Finite AND/OR equivalence.** AQM-70 and this shard have no recorded comparison map or invariant. Any claim that CCR/CAR/Fock layers agree with, reduce to, or are generated by the finite AND/OR grammar remains unsupported.
- **Arithmetic Grassmann target replacement.** AQM-19 L5 rejects only the strict scalar-projective target-replacement proposal: Grassmann variables do support fermionic phase-space calculus, but inside GAR or Grassmann-extended CAR algebra rather than as an ordinary finite-field or real symplectic Weyl target. No new arithmetic Grassmann association is introduced here.

73.8 Conclusion

QSA Step 08 places the existing canonical-field work into the association catalogue as three reviewed layers: real symplectic CCR fields, CAR/Clifford fermion fields, and Grassmann-Weyl displacement calculus in the extended CAR algebra. The unsupported variants remain explicit gaps, and no equivalence with the elementary finite AND/OR grammar is claimed.

74 Fusion-Category Endpoint

74.1 Status and Local Anchors

This is QSA Step 09 from `docs/quantum_system_associations/PLAN.md`. It is a gap shard, not a technical fusion-category shard. It adds no source-local definition of a fusion category, no example, no fusion-rule calculation, no F-symbol or R-symbol equation, no Levin-Wen/string-net Hamiltonian, and no comparison theorem with earlier QSA workflows.

The local anchors are the source queue in AQM-04, which still asks for Levin-Wen string-net models and categorical input data beyond the toric stabilizer/CSS layer; the QSA meta-plan AQM-64; the vocabulary and evidence rules of AQM-65 and `CONVENTIONS.md` convention `(ap)`; the finite AND/OR deferral in AQM-70 and convention `(at)`; and the boson/fermion gap statement in AQM-72. Convention `(h)` explicitly leaves Levin-Wen fusion-category gauge, pivotal/spherical structure, and F/R-data unfixed.

The relevant local source manifest is `references/toric_code/SOURCES.md`. It currently registers sources for the toric-code stabilizer and CSS boundary-map conventions used earlier in the report. This shard does not treat that manifest as a source for fusion-category, Levin-Wen, or string-net technical data.

74.2 Intended Role in the Catalogue

The intended role of this endpoint is to reserve a future association family for categorical kinematical data. Earlier QSA shards cover formal carriers, tensor-site coexistence, direct-sum alternatives, finite AND/OR carrier syntax, finite symplectic Weyl labels, and reviewed CCR/CAR layers. Those shards also state that para, statistics, Fock, and fusion-category enhancements are separate choices.

The future fusion-category family is therefore a planned catalogue slot for workflows in which the input is not merely a set of sites or finite-dimensional carriers. It must include a registered categorical source, the category-side data selected from that source, any lattice or local-constraint data used to build a kinematical carrier, and the exact evidence status of each output. Until those items exist locally, the endpoint remains a blocked/gap row in the QSA catalogue.

74.3 Required Source and Convention Decisions

Before any technical statement is added under this endpoint, the following decisions must be sourced or recorded as conventions.

Decision slot	Required local record before use
Hom spaces	The source and notation for hom spaces, their ground field or coefficient setting, finite-dimensionality assumptions if any, basis choices if any, and the meaning of adjoints or duals if those are used.
Simple objects	The source's convention for simple objects, the chosen label set, ordering of labels, unit label, dual labels, and whether representatives or isomorphism classes are being used in local tables.
Tensor products and fusion rules	The tensor-product convention, unit convention, decomposition convention, and the exact source or checked computation behind any fusion-rule table.
Associator and F-symbol data	The associator convention, F-symbol notation, gauge or basis convention, normalization choices, and the source or local check for any equation using that data.
Pivotal or spherical structure	Whether pivotal, spherical, trace, dimension, or orientation-sensitive data are required, and which source/convention fixes them.
Braiding and R-symbols	Whether braiding is part of the selected workflow. If it is, the source, gauge, label order, and R-symbol convention must be recorded before any braiding-dependent statement appears.
Lattice, Levin-Wen, or string-net data	The lattice or cellulation convention, orientation and boundary convention, where labels live, what local data are attached to vertices/edges/faces, and which source fixes the model input.
Hilbert carrier and local constraints	The carrier basis convention, inner-product convention, local-constraint convention, and evidence that the resulting kinematical subspace or algebra is the intended output.

These are requirement slots, not imported definitions. A later shard may split them across several conventions if the selected sources use incompatible normalizations.

74.4 Explicit Deferrals

The following are not claimed anywhere in this shard.

- No Levin-Wen or string-net technical construction is developed.
- No fusion-rule table is computed or imported.
- No F-symbol equation, R-symbol equation, gauge transformation, or coherence check is stated.
- No Hamiltonian, local projector, commuting-projector result, ground space, excitation, or topological order claim is stated.
- No quantum-double, Drinfeld-center, or category-of-excitations claim is stated.
- No comparison is made between a future fusion-category workflow and the AQM-68 tensor-site workflow, the AQM-70 finite AND/OR grammar, or the AQM-72 CCR/CAR layers.

Each item remains blocked until a local source manifest, convention entry, and either a faithful local derivation or reproducible check support it.

74.5 Unblock Checklist

To turn this endpoint into a technical association family, a later session should first acquire and register the relevant fusion-category and Levin-Wen/string-net sources, then add the convention entries required above. Only after that should it introduce examples, tables, equations, Hamiltonian terms, projectors, or comparison maps. The first technical shard should be able to point every displayed datum to a local source locator, checked local derivation, or run artifact.

74.6 Conclusion

QSA Step 09 closes as a gap record. Fusion categories are kept in the association catalogue as a future endpoint, but the report currently has only the source queue and convention warnings needed to say what is missing. The technical Levin-Wen/string-net and categorical claims are deliberately deferred.

75 Single-Particle Reduction

75.1 Status and Local Anchors

This is QSA Step 10 from `docs/quantum_system_associations/PLAN.md`. It is a local finite-dimensional comparison shard. It adds no external source, no new run artifact, and no general theorem that field, tensor, CCR, CAR, or Fock workflows reduce to one bare finite-set carrier.

The conventions are `CONVENTIONS.md (ap)` for comparison language, `(aq)` for the formal carrier $\mathbb{C}\langle X\rangle$, `(ar)` for tensor-site data, `(as)` for direct-sum particle data, and `(at)` for the finite AND/OR carrier grammar. The local report anchors are AQM-67 through AQM-73. AQM-23 and AQM-40 are used only as cautions that direct sums in label bookkeeping and Hilbert-space tensor products are different layers. AQM-19 and AQM-72 are used only for the recorded finite CAR/Fock boundary language.

75.2 Meaning of Single-Particle Sector Here

In this shard, a *single-particle sector* is not a universal operation. It means one of the following named finite carriers.

Direct-sum particle. For the AQM-69 carrier

$$K_{\oplus}(X) = \bigoplus_{x\in X} K_x,$$

the single-particle sector is the whole alternatives carrier. It represents one alternative site label together with its internal carrier.

AND/OR grammar degree one. For an AQM-70 grammar expression E whose atoms are declared one-particle and whose tensor unit $\mathbf{1} = \mathbb{C}$ is declared zero-particle, the single-particle sector is

$$\mathcal{V}_1(E).$$

This is the degree-one summand from the recursive grading in AQM-70.

Site-option expression. If one uses the AQM-70 site-option expression

$$G_X = \mathrm{AND}_{x\in X}(\mathbf{1}\,\mathrm{OR}\,K_x),$$

then its single-particle sector is the degree-one summand

$$\mathcal{V}_1(G_X) \cong \bigoplus_{\substack{S\subset X\\ |S|=1}} \bigotimes_{x\in S} K_x.$$

The isomorphism is the finite distributivity bookkeeping already recorded in AQM-70.

An arbitrary AQM-68 tensor-site carrier

$$\mathcal{H}(X) = \bigotimes_{x\in X} \mathcal{H}_x$$

has no single-particle sector unless extra data choose a vacuum line and an excitation carrier at each site. The tensor product by itself is coexistence, not one alternative among sites.

75.3 Direct-Sum and Grammar Maps

Let $P_X = \mathrm{OR}_{x\in X} K_x$ be the AQM-70 expression whose carrier is the AQM-69 direct sum. With the same displayed finite parse order,

$$\mathcal{V}(P_X) = \bigoplus_{x\in X} K_x, \qquad \mathcal{V}_1(P_X) = \mathcal{V}(P_X).$$

Thus the reduction map

$$R_P : K_\oplus(X) \longrightarrow \mathcal{V}_1(P_X)$$

is the identity after both sides are read as the same finite direct sum. In the empty case this is the unique map $0 \to 0$.

For the site-option expression G_X, choose a display order only to write the finite tensor product. Define

$$R_G : K_\oplus(X) \longrightarrow \mathcal{V}_1(G_X)$$

on the x-summand by

$$R_G(j_x v) = 1 \otimes \cdots \otimes 1 \otimes v \otimes 1 \otimes \cdots \otimes 1, \qquad v \in K_x,$$

where v occupies the K_x alternative inside $(\mathbf{1}\,\mathrm{OR}\,K_x)$, and every other site occupies the $\mathbf{1} = \mathbb{C}$ alternative with scalar 1.

The inverse is the projection from the degree-one distributive expansion onto the singleton summands, followed by the identification of the singleton tensor $\bigotimes_{y\in\{x\}} K_y$ with K_x. Therefore R_G is a finite vector-space isomorphism between the named carriers. If the K_x are Hilbert spaces and the AQM-70 Hilbert recursion is used, this is also the corresponding Hilbert direct-sum identification of the degree-one sector; it does not identify the full site-option carrier with its degree-one sector.

75.4 When a Tensor-Site Carrier Has Such a Sector

The AQM-68 tensor-site carrier can be compared with the site-option expression only after extra choices are made. Suppose each site Hilbert space is given as

$$\mathcal{H}_x \cong \mathbb{C}\Omega_x \oplus K_x$$

with a chosen unit vector Ω_x and a chosen inclusion of K_x. Then the map

$$\Theta : \bigoplus_{x\in X} K_x \longrightarrow \bigotimes_{x\in X} \mathcal{H}_x$$

is

$$\Theta(j_x v) = \Omega_{x_1} \otimes \cdots \otimes \Omega_{x_{i-1}} \otimes v \otimes \Omega_{x_{i+1}} \otimes \cdots \otimes \Omega_{x_n}$$

for a displayed order $X = \{x_1, \dots, x_n\}$ and $x = x_i$. Its image is the chosen one-excitation subspace. This is a map for the enhanced tensor-site data $(\mathcal{H}_x, \Omega_x, K_x)_{x\in X}$, not for the bare AQM-68 tensor-site carrier.

Without the decomposition $\mathcal{H}_x \cong \mathbb{C}\Omega_x \oplus K_x$, there is no named degree-one subspace. The dimension obstruction is already visible for displayed dimensions $d_x = \dim \mathcal{H}_x$:

$$\dim \bigotimes_x \mathcal{H}_x = \prod_x d_x, \qquad \dim \bigoplus_x \mathcal{H}_x = \sum_x d_x.$$

The empty cases also differ: the empty tensor product is $\mathbb{C}$, while the empty direct sum is 0.

75.5 Comparison With the AQM-67 Formal Carrier

The AQM-67 carrier is

$$\mathbb{C}\langle X\rangle = \bigoplus_{x\in X} \mathbb{C}e_x$$

with no inner product or Gram matrix. A comparison with the direct-sum particle carrier is available only in the one-dimensional internal case.

Assume every K_x is one-dimensional and choose a nonzero vector $u_x \in K_x$. Then

$$\Phi : \mathbb{C}\langle X\rangle \longrightarrow K_\oplus(X), \qquad \Phi(e_x) = j_x(u_x),$$

is a vector-space isomorphism. Composing with R_G gives the corresponding map into the degree-one sector of the site-option expression:

$$R_G \circ \Phi : \mathbb{C}\langle X\rangle \longrightarrow \mathcal{V}_1(G_X).$$

If the K_x are Hilbert lines, chosen unit vectors u_x and an added identity Gram convention on $\mathbb{C}\langle X\rangle$ make Φ isometric. The Hilbert statement uses enhanced data. AQM-67 alone supplies neither the inner product nor the unit vectors, so it does not canonically reduce to the direct-sum particle carrier.

75.6 Field and Fock Boundaries

AQM-71 finite symplectic field labels and the residue-qudit examples in AQM-23 and AQM-40 do not automatically reduce to the carriers above. In those shards, orthogonal direct sums of phase labels quantize to tensor products of Hilbert carriers and full matrix algebras. AQM-23 explicitly rules out replacing the qubit-tensor-qutrit Hilbert space by $\mathbb{C}^2 \oplus \mathbb{C}^3$, and AQM-40 separates the ambient n-qudit tensor system from extra stabilizer labels and phases. A single-excitation subspace in such a tensor system would require a chosen vacuum/reference vector, chosen one-site excitation carriers, and a checked comparison map.

The finite fermionic CAR layer in AQM-19 has one local carrier map at the one-particle level. There $Z = \mathrm{Map}(X, K)$ and $\Gamma_{\mathrm{a}}(Z) = \bigoplus_k \wedge^k Z$. The degree-one Fock summand is $\wedge^1 Z = Z$, and the explicit finite map

$$E_Z : Z \longrightarrow \bigoplus_{x\in X} K, \qquad z \longmapsto (z_x)_{x\in X}$$

has inverse $(v_x)_x \mapsto (x \mapsto v_x)$. This identifies only the one-particle carrier with the AQM-69 direct-sum carrier for the uniform choice $K_x = K$. It does not reduce the CAR algebra, creation and annihilation operators, antisymmetric higher sectors, Grassmann-Weyl displacement calculus, or Grassmann-valued multipliers to AQM-67 or AQM-69.

No comparable general reduction is claimed for the AQM-72 CCR/Fock review layer. The bosonic CCR review uses a real symplectic label space and a regular Weyl representation, while the Grassmann-Weyl review works inside a Grassmann-extended CAR algebra with non-ordinary scalar multipliers. This shard records no local map or invariant identifying those full structures with the finite AND/OR grammar or the bare finite-set carrier.

75.7 Obstruction Summary

The successful reductions above are only maps between named carriers under stated choices. The main obstructions are:

internal carriers	$\dim K_x > 1$ prevents comparison with AQM-67 without forgetting data,
tensor coexistence	$\prod_x d_x$ and $\sum_x d_x$ are different carrier counts,
statistics	$symmetric, antisymmetric, para, andfusionrulesarenotinAQM - 70$,
phases	$Weyl, stabilizer, CAR, andGrassmannmultipliersareextraalgebraicdata$,
Gram choices	$Hilbertorisometriccomparisonsneedinnerproductsandunitchoices.$

Thus QSA Step 10 closes only as a finite carrier-map inventory. It does not collapse the tensor-site, field-label, CCR/CAR, Fock, or future fusion-category association families to one universal single-particle workflow.

76 What Counts As A Point

76.1 Status and Local Anchors

This is QSA Step 11 from `docs/quantum_system_associations/PLAN.md`. It is a review/new synthesis shard: it adds no source, no run artifact, and no new representation convention. Its role is to make the point-selection choice explicit before later geometric association shards use points as quantum sites.

The governing comparison language is `CONVENTIONS.md` convention `(ap)` and AQM-65. The point conventions reviewed here are `CONVENTIONS.md (t)` for the prime-field underlying-set versus $\operatorname{Spec}\mathbb{F}_p$ distinction, `(u)` for finite affine spectra with residue qudits, `(y)` for closed points of affine finite-type $\mathbb{F}_q$-schemes, and `(ai)` for $\operatorname{Spec}\mathbb{Z}$. The local report anchors are AQM-22, AQM-23, AQM-28, AQM-29, AQM-30, and AQM-45, with AQM-66 and AQM-71 used only for QSA catalogue language.

76.2 Point-Selection Choices

The table records choices already visible in the report. A row is a point convention only for the object class named in that row; it is not a universal rule for every object called X.

Choice	Object class where used	Local quantum role	Finite-system status
Underlying set of elements	Bare finite sets and selected finite algebraic sets; in AQM-22, the field $\mathbb{F}_p$ viewed as a p-element set.	Indexes all-map field labels such as $\mathrm{Map}(\mathbb{F}_p, \mathbb{F}_p^2)$, or the structureless finite-set carriers of AQM-67–AQM-71 after extra data are supplied.	Finite when the underlying set and target/carrier choices are finite. In AQM-22 this gives p qudits for the p-element set, not the one-point spectrum.
Rational points $X(k)$	Schemes or varieties over a fixed field k, especially finite-field examples when a finite rational support is intentionally selected.	Gives finite truncations such as the q^2-site rational-support subtheory inside AQM-30. It can also feed the finite-set field-label workflow after a target V, subspace E, and pairing are chosen.	Finite for the finite rational supports used in the report. For an affine finite-type presentation $R = k[x_1, \ldots, x_n]/I$ with $k = \mathbb{F}_q$, the local finite-tuple derivation places $X(k)$ inside the finite set k^n. It is still only a truncation when non-rational closed points also exist, as in AQM-30.
Closed points	Finite spectra with finite residue fields in AQM-23 and convention `(u)`; affine finite-type $\mathbb{F}_q$-schemes in AQM-28–AQM-30 and convention `(y)`; closed prime points of $\operatorname{Spec}\mathbb{Z}$ in AQM-45 and convention `(ai)`.	Each selected closed point x contributes the finite residue field $\kappa(x)$, the label space $\kappa(x)^2$, and the finite local carrier $\ell^2(\kappa(x))$. Finite closed supports quantize by tensor products of the local matrix algebras.	Finite for a finite closed support, and finite globally when the selected closed-point set is finite, as in $\operatorname{Spec}(\mathbb{Z}/6\mathbb{Z})$. Infinite closed-point sets give algebraic quasi-local filtered colimits, not one finite Hilbert space.
Generic points	Generic points in the infinite boundary examples: $(0) \in \operatorname{Spec}\mathbb{F}_q[t]$, $(0) \in \operatorname{Spec}\mathbb{F}_q[x, y]$, and $\eta = (0) \in \operatorname{Spec}\mathbb{Z}$. AQM-30 also records curve-generic points.	They control topology, closures, divisor geometry, and open-set behaviour. AQM-45 records an optional separate $\mathbb{Q}$-Weyl sector for $\operatorname{Spec}\mathbb{Z}$, but not as a finite qudit site.	Not finite Weyl sites under the current closed-point conventions. Generic and curve-generic residue fields in AQM-29, AQM-30, and AQM-45 are infinite or rational-function fields; analytic completions and adelic replacements are gaps.
All scheme points	The full topological space $\lvert X\rvert$ of an affine scheme, used for Zariski opens and for stating the geometry in AQM-28–AQM-30 and AQM-45.	Provides the ambient topology and inclusion order. It is not automatically the quantum-site set unless every selected point also has a recorded finite or otherwise represented residue-sector convention.	Finite only in special finite-spectrum cases already covered by the closed point row. For $\operatorname{Spec}\mathbb{F}_q[t]$, $\operatorname{Spec}\mathbb{F}_q[x, y]$, and $\operatorname{Spec}\mathbb{Z}$, all scheme points include non-closed generic-type points, so this row is not a finite kinematical system by default.

76.3 Closed-Point Finite Weyl Review

The finite Weyl-site convention already has three local forms.

First, AQM-23 and convention `(u)` treat a finite affine spectrum with finite residue fields by assigning one local finite-field phase space to each prime point. For $\mathrm{Spec}(\mathbb{Z}/6\mathbb{Z})$, the two points have residue fields $\mathbb{F}_2$ and $\mathbb{F}_3$. The label group is the orthogonal direct sum $\mathbb{F}_2^2 \oplus \mathbb{F}_3^2$, while the Hilbert carrier is the tensor product $\ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3)$. AQM-23 proves that the observable algebra is the full matrix algebra $M_6(\mathbb{C})$, not a Hilbert direct-sum algebra.

Second, AQM-28 and convention `(y)` extend this to affine schemes $X = \mathrm{Spec}\, R$ of finite type over $\mathbb{F}_q$. The finite Weyl sites are the closed points $x \in |X|_{\mathrm{cl}}$. The local residue field $\kappa(x)$ is finite by the Nullstellensatz locator cited in AQM-28, so

$$E_x = \kappa(x)^2, \qquad H_x = \ell^2(\kappa(x)), \qquad A_x = \mathrm{End}_{\mathbb{C}}(H_x)$$

are finite. For an open U, the label group is the finite-support direct sum over $U \cap |X|_{\mathrm{cl}}$, and the observable algebra is the filtered colimit over finite closed supports.

Third, AQM-45 and convention `(ai)` use the same finite-residue rule for the closed prime points $x_\ell = (\ell)$ of $\mathrm{Spec}\, \mathbb{Z}$, with $\kappa(x_\ell) = \mathbb{F}_\ell$. The global closed-prime algebra is the algebraic infinite tensor product over finite prime supports. A Hilbert-space realization needs an added representation choice, such as the reference-vector incomplete tensor representation recorded in AQM-45.

76.4 Finite Kinematical Boundary

The point-selection rule gives a finite kinematical system only after both the indexing family and the per-point quantum data are finite.

- A finite underlying set with finite carriers or finite symplectic target data gives finite carriers after the extra data are named. AQM-22 is the prime-field example where $\mathbb{F}_p$ as a set gives p qudit sites.
- A finite rational-point support gives a finite truncation. AQM-30 uses the k-rational points of the affine plane as a finite q^2-site subtheory, not as the full closed-point theory.
- A finite closed support gives a finite tensor product of residue-field qudits. If the whole selected closed-point set is finite, as in AQM-23, the global system is finite.
- Infinite closed-point sets, as in AQM-29, AQM-30, and AQM-45, give algebraic quasi-local systems through filtered finite supports. They are not single finite Hilbert spaces.
- Generic points, curve-generic points, and other infinite-residue points do not become finite Weyl sites without a separate representation convention.

76.5 Gaps Carried Forward

This shard does not choose a generic-point representation convention. The generic residue fields $k(t)$, $k(x, y)$, curve-function fields, and $\mathbb{Q}$ are outside the finite closed-point Weyl rule. AQM-45 records one optional algebraic $\mathbb{Q}$-sector for $\mathrm{Spec}\, \mathbb{Z}$, but that does not settle local-field, adelic, C^*-completion, or analytic Stone–von Neumann hypotheses for the other infinite-residue cases.

All later QSA shards that say "point" must therefore name the row above: underlying element, rational point, closed point, generic point, or all scheme point. If the row is not the closed finite-residue row, the shard must also state the extra carrier, residue, completion, support, or truncation choice before making any finite Weyl or Hilbert-space claim.

77 Cotangent-Based First Association

77.1 Status and Local Anchors

This is QSA Step 12 from `docs/quantum_system_associations/PLAN.md`. It is a new synthesis/derivation shard: it adds no external source, no run artifact, and no finite-ring database claim. Its role is to place the already developed tangent-cotangent Weyl workflow into the QSA catalogue as a geometric first-association workflow.

The governing comparison language is AQM-65 and `CONVENTIONS.md` convention `(ap)`. The point-selection discipline is AQM-75. The technical convention is `CONVENTIONS.md (ao)`, and the local technical anchors are AQM-58 through AQM-63. The Kähler differential and tangent-space source locators are inherited from AQM-58–AQM-61; this shard does not introduce a new literature layer.

77.2 Workflow Slot in the QSA Catalogue

The cotangent-based first association is the following geometric workflow. It starts from a geometric presentation, not from a bare finite set:

$$X = \operatorname{Spec} A$$

over a named base field or base ring already fixed for the differential calculation. It then chooses a point convention in the sense of AQM-75. For the finite Weyl part of this shard, the selected points must be finite-residue points; rational supports are allowed only when they are explicitly named finite truncations, and closed finite-residue supports are the current full finite-site convention for affine finite-type $\mathbb{F}_q$-schemes.

For each selected finite-residue point x, put

$$K_x = \kappa(x).$$

Convention `(ao)` attaches the cotangent fibre

$$V_x^* = T^*_{X/k,x} = \Omega^1_{A/k} \otimes_A K_x$$

and the tangent space

$$V_x = T_{X/k,x} = \operatorname{Hom}_{K_x}(V_x^*, K_x).$$

The geometric phase-label space is

$$E_x^{\mathrm{geom}} = V_x \oplus V_x^*.$$

The symplectic form is the canonical tangent-cotangent pairing

$$\omega_x((v,\alpha),(w,\beta)) = \beta(v) - \alpha(w).$$

This is the required symplectic datum before any Weyl representation is used. It is bilinear and alternating by direct cancellation. It is nondegenerate when V_x is finite-dimensional over K_x: a nonzero tangent vector is detected by some cotangent vector, and a nonzero cotangent vector is detected by some tangent vector. This is the local derivation already recorded in AQM-60.

If K_x is finite and $m_x = \dim_{K_x} V_x < \infty$, convention `(ao)` applies the finite Weyl-Heisenberg step to

$$E_x^{\mathrm{geom}}$$

using the additive character

$$\psi_{K_x}(a) = \exp\left(\frac{2\pi i}{p}\operatorname{Tr}_{K_x/\mathbb{F}_p}(a)\right).$$

The resulting kinematical output is a projective representation

$$\rho_x^{\mathrm{geom}} : E_x^{\mathrm{geom}} \to PU(\mathcal{H}_x^{\mathrm{geom}})$$

with carrier dimension

$$\dim_{\mathbb{C}} \mathcal{H}_x^{\mathrm{geom}} = |K_x|^{m_x}.$$

For a finite support S, the phase labels and representations combine as

$$E_S^{\mathrm{geom}} = \bigoplus_{x\in S} E_x^{\mathrm{geom}}, \qquad \rho_S^{\mathrm{geom}} = \bigotimes_{x\in S} \rho_x^{\mathrm{geom}}.$$

Thus the first association output is not a direct assignment of a Hilbert space to the set of points. It is the sequence

$$\begin{aligned}\text{chosen points} \quad \rightsquigarrow \quad \text{cotangent/tangent fibres} \quad \rightsquigarrow \quad \text{symplectic phase labels}\\ \rightsquigarrow \quad \text{Weyl-Heisenberg carriers when finite.}\end{aligned}$$

77.3 Relation to Earlier Association Families

This workflow is different from the structureless finite-set first association of AQM-67. AQM-67 starts with a bare finite set and produces the formal carrier $\mathbb{C}\langle X\rangle$, with any Gram matrix marked as extra data. Here the input includes a scheme presentation, a base for differentials, and a point convention. The output is controlled by $\Omega^1_{A/k}$, not by cardinality alone.

It is also different from the residue-Weyl field-label workflow reviewed in AQM-71. The residue-Weyl layer uses the residue field shape

$$K_x^{(q)} \oplus K_x^{(p)}$$

at a site. The cotangent-first layer uses

$$T_{X/k,x} \oplus T^*_{X/k,x}.$$

Convention `(ao)` records that these agree only after extra choices in special cases, for example at a smooth curve point where $\dim_{K_x} T_{X/k,x} = 1$ and a tangent basis identifies the geometric label space with K_x^2. Product rings, higher-dimensional smooth points, singular points, and nonreduced points show that the two workflows are not the same catalogue entry.

77.4 Anchored Example Inventory

The existing shards supply the following checked or locally derived examples. This table is an inventory of local anchors, not a finite-ring-wide cotangent database.

Object and point choice	Local cotangent-first output	Anchor and status
k^n with its n spectrum points	$\Omega^1_{k^n/k} = 0$, so every $E^{\mathrm{geom}}_x = 0$ and every geometric carrier is one-dimensional.	AQM-58 proves the vanishing by derivations; AQM-60 and AQM-61 record the trivial tangent-Weyl reading. This is a geometric zero case, not the residue-qudit system of AQM-40.
$\operatorname{Spec} \mathbb{F}_3$	One point, zero cotangent and tangent spaces, and no nontrivial geometric Weyl label.	AQM-61 Example 1.
$\operatorname{Spec} \mathbb{F}_3[\epsilon]/(\epsilon^2)$ at its closed point	One tangent direction and one cotangent direction; the finite carrier has dimension 3.	AQM-61 Example 3. This is first-derivative geometry, not the Artin-ring thickened Weyl layer.
$\operatorname{Spec} \mathbb{F}_3[t]/(t^3)$ at its closed point	One tangent direction in characteristic 3; the carrier has dimension 3.	AQM-61 Example 4. The first tangent dimension does not record the whole length-three thickening.
$\operatorname{Spec} \mathbb{F}_3[u, v]/(u^2, v^2)$ at its closed point	Two tangent directions; the carrier has dimension $3^2 = 9$.	AQM-61 Example 5.
$\mathbb{A}^1_{\mathbb{F}_3}$, rational support	Each rational point has one tangent direction; the support carrier has dimension 3^3.	AQM-62 Example 6. Closed degree-d points have residue field $\mathbb{F}_{3^d}$ and carrier dimension 3^d.
$\mathbb{A}^2_{\mathbb{F}_3}$, rational support	Each rational point has two tangent directions; the support carrier has dimension 3^{18}.	AQM-62 Example 7.
The crossing $xy = 0$ over $\mathbb{F}_3$, rational support	The origin has two tangent directions and the four other rational points have one; the support carrier has dimension 3^6.	AQM-62 Example 8.
Finite-field circle $x^2 + y^2 = 1$	Odd-characteristic smooth points have one tangent direction; the $\mathbb{F}_3$ rational support carrier has dimension 3^4, and the $\mathbb{F}_5$ rational support carrier has dimension 5^4.	AQM-59 computes the cotangent fibres and point counts; AQM-62 Examples 9–10 record the Weyl carrier dimensions. AQM-59 also records the characteristic-two nonreduced tangent thickening.

77.5 What AQM-63 Adds and What It Does Not Add

AQM-63 records a compatible cotangent-label selection test:

$$L_{\mathrm{dR}}(S) = \mathrm{ev}^1_S(dA) \subset 0 \oplus \bigoplus_{x \in S} T^*_{X/k,x}.$$

Because the cotangent summand is isotropic, these labels define commuting Weyl operators after phases are fixed. This is a local compatibility result inside the cotangent-first kinematics.

For QSA Step 12, this does not turn the workflow into dynamics. AQM-63 also records that tangent-type labels, Hamiltonians, action functionals, metrics, Hodge identifications, and other dynamical or CSS-completion choices remain extra data. Therefore the present shard uses AQM-63 only to say that exact one-forms can be read as optional cotangent/momentum constraints inside the already built phase-label space.

77.6 Gaps Carried to Later Steps

The intrinsic-versus-relative cotangent problem is deliberately not settled here. Every displayed $T^*_{X/k,x}$ uses the base appearing in convention `(ao)` and in the anchor shard. Step 13 must compare local/intrinsic and relative cotangent choices for $\operatorname{Spec} R$ with the base ring named in each example.

No finite-ring-wide cotangent table is asserted. In particular, this shard does not claim populated database-backed cotangent rows, helper coverage, or finite-ring example-matrix completeness. Those rows remain gaps pending the separate finite-ring cotangent work item `aqm-qsa-frcot01` and the later QSA finite-ring matrix steps.

Finally, AQM-75 still controls point selection. Generic points, infinite residue fields, analytic completions, and all-scheme-point indexing do not become finite Weyl systems by this shard. If K_x is not finite, or if the representation category is not fixed, the output stops at the geometric phase-label space and the missing representation choice must be labelled as a gap.

78 Intrinsic Versus Relative Cotangent Data

78.1 Status and Local Anchors

This is QSA Step 13 from `docs/quantum_system_associations/PLAN.md`. It is a new derivation/comparison shard: it adds no external source, no run artifact, and no populated finite-ring cotangent table. Its purpose is to compare two legitimate first-order inputs for a cotangent-first quantum-system association.

The governing vocabulary is AQM-65 and `CONVENTIONS.md` conventions `(ap)` and `(av)`. Point-selection discipline is AQM-75, and the tangent-cotangent Weyl workflow is AQM-76 and convention `(ao)`. Kähler differentials and their universal property use `references/algebraic_geometry/stacks_project_algebra.tex`, lines 33793–33872. Surjective base maps give zero relative differentials by the same source, lines 33883–33893. Quotient and polynomial differential calculations use lines 34103–34120 and 34310–34340. Cotangent fibres and tangent duality use `references/algebraic_geometry/stacks_project_varieties.tex`, lines 2987–3019. The comparison with $\mathfrak{m}_x/(\mathfrak{m}_x^2 + \mathfrak{m}_s\mathcal{O}_{X,x})$ uses the same file, lines 3058–3094, and the warning that $\mathfrak{m}/\mathfrak{m}^2$ and relative embedding dimension can differ uses lines 10758–10797.

78.2 Two Cotangent Choices

Let $X = \operatorname{Spec} A$, choose a point x, and write

$$A_x \text{ for the local ring,} \qquad \mathfrak{m}_x \subset A_x, \qquad K = \kappa(x) = A_x/\mathfrak{m}_x.$$

The intrinsic local cotangent object is

$$C_x^{\mathrm{loc}} = \mathfrak{m}_x/\mathfrak{m}_x^2.$$

It depends on the local ring at x, not on a chosen base.

For a named base ring map $B \to A$, the relative Kähler cotangent object is

$$C_x^{\mathrm{rel}}(B) = \Omega^1_{A/B} \otimes_A K.$$

It depends on the morphism $X \to \operatorname{Spec} B$. In the residue-field-equal or separable algebraic situation covered by the Stacks tangent-space locator, this relative fibre is the local quotient

$$\mathfrak{m}_x/\big(\mathfrak{m}_x^2 + \mathfrak{m}_s\mathcal{O}_{X,x}\big),$$

where s is the image of x in $\operatorname{Spec} B$. Thus base directions may be removed by the relative construction.

Each choice feeds AQM-76 separately. Put

$$V_x^{\mathrm{loc}} = \operatorname{Hom}_K(C_x^{\mathrm{loc}}, K), \qquad E_x^{\mathrm{loc}} = V_x^{\mathrm{loc}} \oplus C_x^{\mathrm{loc}},$$

and

$$V_x^{\mathrm{rel}}(B) = \operatorname{Hom}_K(C_x^{\mathrm{rel}}(B), K), \qquad E_x^{\mathrm{rel}}(B) = V_x^{\mathrm{rel}}(B) \oplus C_x^{\mathrm{rel}}(B).$$

Both use the canonical tangent-cotangent symplectic form

$$\omega((v,\alpha),(w,\beta)) = \beta(v) - \alpha(w).$$

If K is finite and $m = \dim_K C$ for the chosen cotangent object C, then the finite Weyl carrier dimension is $|K|^m$. Therefore changing the cotangent object can change V_x, V_x^*, the symplectic label dimension $2m$, and the finite carrier size.

78.3 Agreement Examples over $\mathbb{F}_3$

1. The field $\mathbb{F}_3$. Base ring: $B = k = \mathbb{F}_3$. Point convention: the unique closed point of $X = \operatorname{Spec} k$. The local ring is k, so

$$\mathfrak{m}_x = 0, \qquad C_x^{\mathrm{loc}} = 0.$$

The relative differential module is $\Omega^1_{k/k} = 0$, as in AQM-61 Example 1, so

$$C_x^{\mathrm{rel}}(k) = 0.$$

Both choices give $E_x = 0$ and a one-dimensional Weyl carrier.

2. Product fields k^n. Base ring: $B = k = \mathbb{F}_3$. Point convention: all spectrum points x_i of $X = \operatorname{Spec}(k^n)$, equivalently the idempotent factors. At each point, the local ring is k, so $C_{x_i}^{\mathrm{loc}} = 0$. AQM-58 proves by derivations that

$$\Omega^1_{k^n/k} = 0,$$

hence

$$C_{x_i}^{\mathrm{rel}}(k) = 0.$$

The intrinsic and relative cotangent-first readings agree and vanish at every point. This is still not the residue-qudit workflow of AQM-40.

3. Dual numbers. Base ring: $B = k = \mathbb{F}_3$. Point convention: the unique closed point $x = (\epsilon)$ of

$$A = k[\epsilon]/(\epsilon^2).$$

The local maximal ideal is $\mathfrak{m} = (\epsilon)$, and $\mathfrak{m}^2 = 0$, so

$$C_x^{\mathrm{loc}} = k\,\bar{\epsilon}.$$

For the relative calculation, the quotient differential sequence gives

$$\Omega^1_{A/k} = A\,d\epsilon/(2\epsilon\,d\epsilon).$$

Tensoring with $K = k$ kills ϵ, so

$$C_x^{\mathrm{rel}}(k) = k\,d\epsilon.$$

The two cotangent objects agree in dimension and under the first-order identification $\bar{\epsilon} \leftrightarrow d\epsilon$. Thus $m = 1$, E_x has K-dimension 2, and the carrier dimension is 3, as in AQM-61 Example 3.

4. The cubic fat point in characteristic three. Base ring: $B = k = \mathbb{F}_3$. Point convention: the unique closed point $x = (t)$ of

$$A = k[t]/(t^3).$$

Here $\mathfrak{m} = (t)$ and $\mathfrak{m}^2 = (t^2)$, hence

$$C_x^{\mathrm{loc}} = k\,\bar{t}.$$

Since $d(t^3) = 3t^2dt = 0$ in characteristic 3, AQM-61 computes

$$C_x^{\mathrm{rel}}(k) = k\,dt.$$

Again $m = 1$, so the intrinsic and relative phase-label spaces have dimension 2 over k and finite carrier dimension 3. Neither first-order cotangent choice records the full length-three thickening.

5. Two independent dual directions over k. Base ring: $B = k = \mathbb{F}_3$. Point convention: the unique closed point $x = (u, v)$ of

$$A = k[u, v]/(u^2, v^2).$$

The maximal ideal is $\mathfrak{m} = (u, v)$, and $\mathfrak{m}^2 = (uv)$, because $u^2 = v^2 = 0$. Therefore

$$C_x^{\mathrm{loc}} = k\,\bar{u} \oplus k\,\bar{v}.$$

The quotient differential calculation gives

$$\Omega^1_{A/k} = (A\,du \oplus A\,dv)/(2u\,du,\ 2v\,dv).$$

After tensoring with $K = k$, both u and v vanish, so

$$C_x^{\mathrm{rel}}(k) = k\,du \oplus k\,dv.$$

The choices agree with $m = 2$. Hence E_x has dimension 4 over k, and the finite Weyl carrier dimension is $3^2 = 9$, as in AQM-61 Example 5.

78.4 Difference Examples

6. The same two-direction ring relative to one base direction. Base ring:

$$B = k[u]/(u^2), \qquad k = \mathbb{F}_3,$$

with $B \to A = k[u, v]/(u^2, v^2)$ sending u to u. Point convention: the unique closed point $x = (u, v)$ of A, mapping to $s = (u)$ of B. The intrinsic cotangent is still

$$C_x^{\mathrm{loc}} = k\,\bar{u} \oplus k\,\bar{v}.$$

Relatively, $A = B[v]/(v^2)$. Thus

$$\Omega^1_{A/B} = A\,dv/(2v\,dv),$$

and at x

$$C_x^{\mathrm{rel}}(B) = k\,dv.$$

Equivalently, the local quotient formula removes the base direction:

$$\mathfrak{m}_x/\big(\mathfrak{m}_x^2 + (u)\mathcal{O}_{X,x}\big) \cong k\,\bar{v}.$$

So the intrinsic phase-label space has dimension 4 over k and carrier dimension 9, while the relative phase-label space over B has dimension 2 over k and carrier dimension 3. This is an inequivalence of cotangent choices, not a contradiction.

7. A mixed-characteristic quotient. Base ring: $B = \mathbb{Z}$. Point convention: the unique closed point $x = (2)$ of

$$A = \mathbb{Z}/4\mathbb{Z}.$$

The local ring is A, the maximal ideal is $\mathfrak{m} = (2)$, and $\mathfrak{m}^2 = (4) = 0$. Therefore

$$C_x^{\mathrm{loc}} = \mathfrak{m}/\mathfrak{m}^2 \cong \mathbb{F}_2.$$

But the base map $\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ is surjective, so the Stacks surjective-map lemma gives

$$\Omega^1_{A/\mathbb{Z}} = 0, \qquad C_x^{\mathrm{rel}}(\mathbb{Z}) = 0.$$

The intrinsic choice gives $m = 1$, local symplectic label dimension 2 over $\mathbb{F}_2$, and a carrier of dimension 2 under the finite Weyl step. The relative-over-$\mathbb{Z}$ choice gives $m = 0$, $E_x = 0$, and a one-dimensional carrier. The same calculation works for $\mathbb{Z}/9\mathbb{Z}$ with $x = (3)$, replacing $\mathbb{F}_2$ by $\mathbb{F}_3$.

78.5 Implication for Cotangent-First QSA

AQM-76 remains correct only after the cotangent object is named. The local phase-label row is not simply "the cotangent of $\operatorname{Spec} A$"; it is either

$$E_x^{\mathrm{loc}} = V_x^{\mathrm{loc}} \oplus C_x^{\mathrm{loc}}$$

or

$$E_x^{\mathrm{rel}}(B) = V_x^{\mathrm{rel}}(B) \oplus C_x^{\mathrm{rel}}(B)$$

for a specific base B. Agreement examples allow the shorthand used in AQM-61, because both choices give the same finite dimensions there. The relative $B = k[u]/(u^2)$ and $B = \mathbb{Z}$ examples show why later shards must not infer Weyl carrier sizes from the local ring alone or from Ω^1 without recording the base.

This shard does not assert populated database-backed cotangent rows, helper coverage, finite-ring-wide tables, de Rham CSS completions, Hamiltonians, or generic/infinite-residue representation choices. Those remain separate gaps, including the finite-ring cotangent work item `aqm-qsa-frcot01`.

79 Weyl-Heisenberg Kinematics From Finite Phase Spaces

79.1 Status and Local Anchors

This is QSA Step 14 from `docs/quantum_system_associations/PLAN.md`. It is a review/new synthesis shard: it adds no source manifest, no convention entry, no run artifact, and no new all-finite-ring theorem. Its purpose is to isolate the common representation-theoretic step used after earlier association workflows have already produced a finite phase-label space.

The governing vocabulary is AQM-65 and `CONVENTIONS.md` convention `(ap)`. The finite-field arithmetic anchors are AQM-22, AQM-28, AQM-60, AQM-76, and the finite symplectic field-label review AQM-71. The ring-status anchor is AQM-31. The source manifests are `references/symplectic_qecc/SOURCES.md` and `references/heisenberg_weil/SOURCES.md`. In particular, AQM-31 cites Prasad, Beny–Crann–Lee–Park–Youn, Bekka, Lysenko, Sidana–Kashyap, Gluesing-Luerssen–Pllaha, Solomon, and the finite Clifford/Weil splitting sources at the line locators recorded there.

79.2 The Project's Local Weyl Step

The Weyl-Heisenberg step begins only after a finite label object and its commutator pairing have already been fixed. In the finite-field vector-space case, the local display form is

$$E = Q \oplus P, \qquad \omega((q,p),(q',p')) = p(q') - p'(q),$$

where Q is a finite-dimensional vector space over a finite field K, and $P = Q^*$ or a chosen identified dual. With

$$\psi_K(a) = \exp\left(\frac{2\pi i}{p} \operatorname{Tr}_{K/\mathbb{F}_p}(a)\right),$$

the carrier is

$$\mathcal{H} = \ell^2(Q).$$

For $a \in Q$, define translations and phases by

$$T(q)|a\rangle = |a+q\rangle, \qquad R(p)|a\rangle = \psi_K(p(a))|a\rangle,$$

and set

$$W(q,p) = T(q)R(p).$$

Then

$$W(q,p)W(q',p') = \psi_K(p(q'))W(q+q',p+p')$$

and

$$W(e)W(f) = \psi_K(\omega(e,f))W(f)W(e).$$

This is the local finite-field construction used in AQM-22 for prime-field qudit Paulis, AQM-28 for closed finite-residue sites, AQM-60 for tangent-cotangent labels, and AQM-76 for the cotangent-first QSA workflow. If an earlier workflow produces an abstract finite symplectic vector space rather than an already polarized $Q \oplus Q^*$, a choice of symplectic basis or polarization is presentation data for writing this Schrodinger model.

This step is kinematical. It supplies a projective representation, Weyl operators, and the full finite matrix observable algebra on the carrier. It does not choose a Hamiltonian, time evolution, de Rham CSS stabilizer, state, Lagrangian, or code subspace. AQM-22 T2, AQM-60, AQM-63, and AQM-76 all keep stabilizer or de Rham selections as later choices.

79.3 Case 1: Prime-Field and Finite-Field Vector Spaces

The prime-field case is fully local in the report. AQM-22 uses

$$E_{\mathrm{el}} = \mathrm{Map}(\mathbb{F}_p, \mathbb{F}_p^2)$$

and proves that the operators $T_\xi R_\zeta$ generate the ambient odd-p p-qudit Pauli group on the carrier $\ell^2(\mathbb{F}_p^{\mathbb{F}_p})$. It also records the $p = 2$ phase edge: the projective label space and commutator criterion remain the same, but the conventional Hermitian Pauli group uses fourth roots of unity.

AQM-28 extends the same finite-field Weyl step to closed points of affine finite-type $\mathbb{F}_q$-schemes. A closed point x contributes

$$E_x = \kappa(x)^2, \qquad \mathcal{H}_x = \ell^2(\kappa(x)),$$

and finite supports use direct sums of labels and tensor products of carriers. AQM-60 and AQM-76 use the same construction after replacing the residue label K^2 by the geometric tangent-cotangent label

$$T_{X/k,x} \oplus T^*_{X/k,x}.$$

For $m = \dim_{\kappa(x)} T_{X/k,x}$, the finite carrier dimension is $|\kappa(x)|^m$, as recorded in convention `(ao)` and reviewed in AQM-76.

The sourced classification status is narrower than the construction. The finite-field Pauli/error-basis laws are anchored in `references/symplectic_qecc/SOURCES.md`: Ashikhmin–Knill `n9.tex:249-313`, Gottesman `Thesis.tex:1627-1681`, and Bombin–Martin–Delgado `HomologicalErrorCorrection6.tex:1206-125`
For odd finite-dimensional Weyl systems, Gross is registered at `poswig.tex:340-453` for Weyl operators and at `poswig.tex:1966-2042` for a discrete Stone-von Neumann/Clifford proof. Thus odd finite-field Clifford/Gaussian statements may be used only in that sourced range. The even case keeps the phase caveats already recorded in AQM-22, AQM-31, and convention `(af)`.

79.4 Case 2: Finite Frobenius-Ring Pauli and Stabilizer Cases

AQM-31 records the finite Frobenius-ring status from Gluesing-Luerssen–Pllaha and Sidana–Kashyap. For a finite commutative Frobenius ring R with generating character χ, the sourced Pauli operators on $\ell^2(R^n)$ are

$$X(a)v_x = v_{x+a}, \qquad Z(b)v_x = \chi(b \cdot x)v_x,$$

with

$$X(a)Z(b)X(a')Z(b') = \chi(b \cdot a')X(a + a')Z(b + b')$$

and commutator controlled by

$$b \cdot a' - b' \cdot a.$$

The local locators are `references/heisenberg_weil/gluesing_luerssen_pllaha_1710_09884_source/StabCodesFrob5.tex`, lines 224–249, 286–323, 353–355, 412–423, and 469–504, and `references/heisenberg_weil/sidana_kashyap_2202_00248_source/EAQECCs_over_rings_v15.tex`, lines 103–145 and 180–197.

The project already has one explicit Artin/Frobenius local proof in AQM-36. For $A = \mathbb{F}_3[t]/(\pi^e)$, it defines the top-coefficient character, proves it is generating, constructs $W_A(q, p)$ on $\ell^2(A)$, proves the projective product and commutator laws, and proves that the Weyl operators form an operator basis of $\mathrm{End}_{\mathbb{C}}(\ell^2(A))$. This supports the nilpotent-sensitive thickened layer in conventions `(ab)` and `(an)`.

The boundary is important. These sources support Pauli/error bases, character-symplectic commutation, and stabilizer self-orthogonality over the stated finite Frobenius-ring scope. They do not, in the current project, provide a complete Clifford/Weil representation classification for every finite commutative Frobenius ring, and they do not classify projective representations over all finite rings.

79.5 Case 3: Sourced Finite Generalized Heisenberg Cases

AQM-31 also records Solomon's generalized finite Heisenberg source. The local support is `references/heisenberg_weil/solomon_2501_00650_source/TotslipVersion5_Aug_25.tex`, lines 315–318, 904–981, 1127–1136, 1637–1649, and 1677–1718. In that setting, finite abelian groups A, B, C and a bilinear pairing

$$\lambda : A \times B \to C$$

define a Heisenberg group with product

$$h(a, b, c)h(a', b', c') = h(a + a', b + b', c + c' + \lambda(a, b')).$$

A character $p : C \to \mathbb{C}^{\times}$ gives the Schrodinger representation on functions on A:

$$(h(a, b, c) \cdot f)(x) = p(\lambda(x, b) + c)f(x + a).$$

This is a legitimate sourced finite Heisenberg kinematics whenever the chosen finite groups, pairing, and character are exactly in that sourced scope or are checked locally against it. It is broader than finite fields in shape, but it is not a license to treat every finite ring label module as classified. The QSA catalogue may use it as a finite generalized Heisenberg slot only after recording the concrete A, B, C, λ, p data.

79.6 Case 4: General Finite-Ring Questions and Gaps

For a general finite commutative ring R, the project does not currently claim a complete Weyl-Heisenberg, Stone-von Neumann, Clifford, or Weil classification. The following remain gaps unless a later shard adds local sources, conventions, or checked derivations.

- If R is not equipped with a certified generating character, the translation/phase recipe $R^n \oplus R^n$ does not yet have the Frobenius-ring self-duality input used in AQM-31 and AQM-36.
- If R is Frobenius, the Pauli/error-basis and stabilizer layer is sourced as above, but full Clifford/Weil splitting or classification is additional structure. AQM-31 records finite Clifford splitting subtleties from Hashimoto–Horibe–Hayashi, Korbelar–Tolar, and Galindo; it does not turn them into a universal finite-ring classification.
- Even-order and 4-divisibility phenomena are genuine source-backed warnings, not cosmetic phase choices. AQM-31 records Lysenko's odd-exponent scope and Galindo's finite-abelian splitting criterion from `references/heisenberg_weil/SOURCES.md`.
- Finite-ring database rows in convention `(an)` are layer-labelled. A residue row, a thickened-Frobenius row, and a blocked quantisation row are different evidence states; none is a hidden all-finite-ring theorem.

79.7 Conclusion for the QSA Catalogue

QSA Step 14 supplies a common final kinematical move:

$$\text{finite phase labels with commutator data} \quad \rightsquigarrow \quad \text{Weyl-Heisenberg projective representation and observable algebr}$$

For finite fields and the prime-field qudit cases, this move is already used through AQM-22, AQM-28, AQM-60, AQM-71, and AQM-76. For finite Frobenius rings, it is available at the Pauli/stabilizer and checked Artin-Weyl level described by AQM-31 and AQM-36. For Solomon-style finite generalized Heisenberg groups, it is available when the finite groups, pairing, and central character are explicitly supplied. General finite-ring classification, full Clifford/Weil classification, and dynamics remain questions, not conclusions.

80 Geometric Field Association $\mathrm{Hom}(X, V)$

80.1 Status and Local Anchors

Status: Review. This is QSA Step 15 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, run artifact, or general functorial quantisation theorem. Its purpose is to place the existing geometric field-space shards into the QSA catalogue.

The title uses $\mathrm{Hom}(X, V)$ as the prompt label for this branch. In the local conventions, this is deliberately not the default object. By `CONVENTIONS.md` convention `(q)`, a finite set uses $\mathrm{Map}(X, V)$; the word Hom is reserved for a specified structure-preserving class of maps. The QSA vocabulary and non-functoriality rule are AQM-65 and convention `(ap)`. The point-choice boundary is AQM-75, and the finite phase-space representation boundary is AQM-78.

The field-space anchors reviewed here are AQM-14–AQM-18, AQM-24–AQM-30, and AQM-49–AQM-52. The relevant conventions are `(q)` for finite map spaces and radicals, `(r)` for projective sheaf fields, `(v)`–`(w)` for open-local observable nets and their presheaf/cosheaf distinction, `(y)` for closed-point finite-residue supports, and `(ak)`–`(al)` for regular-function profiles, reduced smearings, and affine action.

80.2 Choice Table

Every row below is constrained by a convention or prior shard; no row is a universal $\mathrm{Hom}(X, V)$ rule.

Field-space choice	How X constrains it	Operator/Weyl status	Radical or support condition	Anchor
`Review:` all maps on a finite set	Only the finite point labels of X are used; extra structure is ignored unless a subspace is chosen.	`Checked` finite Weyl labels after the pointwise form is reduced.	Use $E \le \mathrm{Map}(X, V)$ and quotient by $\mathrm{rad}(E)$.	AQM-14, AQM-71, `(q)`
`Review:` structured finite subspaces	Linear, affine, or polynomial structure on X may choose a smaller evaluated subspace $E \le \mathrm{Map}(X, V)$.	`Checked` only for the local $\mathbb{F}_3$ rows already run.	The same radical test applies; evaluation rank is not the same as symplectic rank.	AQM-14, arithmetic-field run bundle
`Review:` finite closed supports	For affine finite-type $\mathbb{F}_q$-schemes, only closed points are finite residue sites; non-closed points still control topology.	`Checked/Review` quasi-local Weyl net over finite supports.	Labels have finite closed support: $\bigoplus_{x\in U\cap\lvert X\rvert_{\mathrm{cl}}} \kappa(x)^2$.	AQM-28–AQM-30, `(y)`
`Review:` rational truncations	A selected finite set $X(k)$ or finite rational support is used instead of all closed points.	`Checked/Review` finite subtheory, not the full scheme theory.	Finite by choice of support; non-rational closed points are omitted.	AQM-16, AQM-30, AQM-75
`Review:` projective sheaf sections	Projective geometry makes regular maps to affine targets too small, so a sheaf $\mathcal{L}$ is chosen.	`Checked` for $\mathbf{P}^1_{\mathbb{F}_3}$ and $\mathcal{O}(d)$, $0 \le d \le 4$.	$E = H^0(X, \mathcal{L}) \otimes V$, evaluated on chosen finite points, then quotiented by the radical.	AQM-15–AQM-18, `(r)`
`Review:` regular-function profiles	For $X = \operatorname{Spec} R$, $R = \mathbb{F}_q[x]$, $U = D(s)$ gives $\Gamma(U, \mathcal{O}_X) = R_s$; a separate $h \neq 0$ may select $Z_{\mathrm{cl}}(h)$.	`Gap` for nonzero R_s^2 profiles as finite operators; `Checked` for reduced polynomial smearings.	Localized profiles evaluate only where $\pi \nmid s$; finite support $Z_{\mathrm{cl}}(h)$ uses $P, Q \in R$ modulo $\mathrm{rad}(h)$.	AQM-49–AQM-52, `(ak)`–`(al)`
`Review:` observable nets	Zariski opens and closed supports control which finite labels are allowed in each local algebra.	`Checked/Review` open-local isotone net; not an ordinary cosheaf of unital $*$-algebras.	Finite-support labels glue additively; observable algebras glue by generated commuting tensor-local subalgebras.	AQM-24–AQM-27, `(v)`–`(w)`

80.3 Finite Maps and Closed-Point Fields

Review. The finite-set model of AQM-14 starts with a finite point set X, a finite symplectic target (V, ω_V), and a chosen $\mathbb{F}_q$-linear subspace

$$E \le \mathrm{Map}(X, V).$$

This is not an unsourced $\mathrm{Hom}(X, V)$. AQM-14 D1–D4 and convention `(q)` fix the pointwise form

$$\Omega_E(\phi, \psi) = \sum_{x\in X} w_x\, \omega_V(\phi(x), \psi(x)),$$

with all weights equal to 1 in the checked runs. AQM-14 L4–L5 and AQM-71 record the required reduction

$$E_{\mathrm{red}} = E/\operatorname{rad}(E).$$

Thus the structure of X constrains the field space only when the shard has already named a structure-sensitive subspace, such as the linear, affine, or bounded-degree polynomial examples evaluated in AQM-14. The finite Weyl step applies to E_{red}, not to an arbitrary unreduced display space.

Review. The closed-point affine branch changes the point choice. By AQM-28 and convention `(y)`, if

$$X = \operatorname{Spec} R$$

is affine of finite type over $\mathbb{F}_q$, the finite Weyl sites are the closed points $x \in |X|_{\mathrm{cl}}$. Each such point has finite residue field $\kappa(x)$, local label space $\kappa(x)^2$, carrier $\ell^2(\kappa(x))$, and local matrix algebra. On an open U,

$$E(U) = \bigoplus_{x \in U \cap |X|_{\mathrm{cl}}} \kappa(x)^2$$

means finite closed support. AQM-29 applies this to $\operatorname{Spec} \mathbb{F}_q[t]$, where each monic irreducible of degree d gives one q^d-level qudit. AQM-30 applies it to $\operatorname{Spec} \mathbb{F}_q[x, y]$, where rational points give only a finite q^2-site truncation of the full closed-point theory.

Gap. Generic and curve-generic points are not finite Weyl sites in this branch. AQM-29 and AQM-30 keep them as topological and geometric points, but no finite-residue field operator is attached to them. AQM-75 records this as a point-selection boundary, and AQM-78 does not remove it: AQM-78 starts only after a finite phase-label object has already been fixed.

80.4 Sheaf Sections and Regular Functions

Review. AQM-15 and convention `(r)` record why a projective geometric field space is not obtained by silently reading $\operatorname{Hom}(X, V)$ as regular maps to an affine vector space. For a proper geometrically integral projective variety, the local Stacks locator in AQM-15 gives only constant global regular functions. The projective replacement used in the report is instead

$$E(X, \mathcal{L}, V) = H^0(X, \mathcal{L}) \otimes_{\mathbb{F}_q} V,$$

with the sheaf $\mathcal{L}$ chosen as part of the datum.

AQM-16 checks this for

$$X = \mathbf{P}^1_{\mathbb{F}_3}, \qquad \mathcal{L} = \mathcal{O}(d), \qquad V = \mathbb{F}_3^2,$$

using the rational-point order and evaluation convention in `(r)`. The $d = 1$ row has independent evaluation rows but total radical for the all-weights-one scalar pairing, while $d = 4$ has a one-dimensional scalar evaluation kernel and a two-dimensional field radical. These are checked examples of the same rule: sections first, chosen evaluations second, radical quotient third, Weyl labels fourth.

Review. AQM-17 and AQM-18 refine the finite point value: the stalk is the local field germ, while the fiber or residue value is what the finite Weyl pairing samples. For $\mathcal{O}(d) \otimes \mathbb{F}_3^2$, the stalk reduces to $V = \mathbb{F}_3^2$ only after a chosen local frame is reduced modulo the maximal ideal.

Review. The affine-line regular-function branch in AQM-49–AQM-52 is a structure-sensitive replacement for a vague Hom-space. Put $R = \mathbb{F}_q[x]$. On $U = D(s)$, $s \neq 0$, a pair $(F, G) \in R_s^2$ gives a residue profile at closed points $x_\pi \in U$, equivalently $\pi \nmid s$:

$$\operatorname{ev}_U(F, G)_\pi = (F(x_\pi), G(x_\pi)) \in \kappa(x_\pi)^2.$$

Denominators are invertible there; this localized datum is not evaluated at closed points with $\pi \mid s$. By AQM-49 and convention `(ak)`, a nonzero profile on a nonempty standard open has cofinite closed support in that open, so it is formal infinite-volume data, not a quasi-local Weyl operator.

Checked. A separate nonzero polynomial $h \in R$ may instead select the finite closed support

$$Z_{\mathrm{cl}}(h) = \{x_\pi : \pi \mid h\}.$$

AQM-49's reduced smearing over this support uses polynomial labels $(P, Q) \in R^2$, equivalently classes modulo $\mathrm{rad}(h)$:

$$W_h^{\mathrm{red}}(P, Q) = W_{Z_{\mathrm{cl}}(h)}\big((P \bmod \pi,\ Q \bmod \pi)_{\pi \mid h}\big).$$

It does not evaluate arbitrary localized elements where denominators vanish. AQM-50 checks this over $\mathbb{F}_3$, including rational two-qutrit supports, a quadratic 9-level site, and repeated factors invisible to the reduced net. AQM-51–AQM-52 and convention `(al)` check that affine symmetries move profiles, supports, and reduced smearings compatibly while preserving the profile/operator distinction.

80.5 Observable Nets and the AQM-78 Boundary

Review. The observable-net layer is not a new restriction of $\mathrm{Hom}(X, V)$; it records where finite labels are supported. AQM-24 and convention `(v)` use the open-local assignment

$$U \longmapsto \mathcal{A}(U),$$

with $U \subset V$ giving tensor-by-identity inclusions. The closed-complement assignment is valid only as a named closed-support layer, because its arrows are contravariant in opens.

AQM-25 builds the algebraic quasi-local algebra as the filtered colimit over quasi-compact opens. AQM-26 and convention `(w)` separate the cosheaf-like label layer

$$E(U) = \bigoplus_{x \in U} E_x$$

from the observable layer, an isotone local net of unital $*$-algebras. AQM-27 records the state-side reversal as a presheaf of states. These net statements support finite-support observables; they do not create infinite-volume Weyl operators for regular-function profiles.

Review. AQM-78 begins after one of the rows above has produced a finite phase label with commutator data. It supplies the Weyl-Heisenberg projective representation and observable algebra for finite phase spaces. Thus Step 15 does not claim that every geometric $\mathrm{Hom}(X, V)$ has been quantized. It says only that the report already contains several locally constrained field-space choices, and that AQM-78 is the common finite Weyl boundary once the label object is finite and the radical/support issues have been resolved.

80.6 Next-Step Gaps

Next step	What this shard supplies	Remaining condition	Status
QSA Step 16: finite-ring example matrix	Rows can cite whether a field-space entry is all-map, finite-support closed-point, regular-function profile, rational truncation, or sheaf-section style.	Each finite-ring row still needs its own point choice, residue/cotangent or function-space convention, and local evidence; no populated finite-ring database claim is imported here.	`Gap`
QSA Step 17: residue-field site association	AQM-23 and AQM-28–AQM-30 already show variable finite residue fields and finite-support direct sums.	A finite-ring-wide residue-site table must not infer maximal ideals, residue fields, or tensor factors without certified local decomposition evidence.	`Question`
QSA Step 20: infinite boundary cases	AQM-29, AQM-30, AQM-45, and AQM-49 separate finite supports, quasi-local algebras, regular-function profiles, and generic-point caveats.	Infinite-volume field operators, generic-point sectors, local-field or adelic completions, and C^*- or Hilbert-space completions still require separate conventions and sources.	`Gap`

81 Finite Ring Example Matrix

81.1 Status and Local Anchors

Status: Synthesis matrix. This is QSA Step 16 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, script, run bundle, generated data, populated database row, or finite-ring classification claim. Its purpose is to put the standard finite-ring examples into one gap-aware comparison table before the dedicated residue-site and nilpotent-sensitive shards.

The local anchors are AQM-23, AQM-34–AQM-37, AQM-40, AQM-58, and AQM-76–AQM-79. The finite-ring database boundary is the PRD `docs/finite_commutative_ring_database_prd.md`, the implementation and population plans, convention `(an)`, the schema `data/SCHEMA.md`, and `runs/2026-05-26-finite-ring-database/README.md`. That run bundle records a schema-only SQLite smoke build with one `build_run` row and zero `ring` or `presentation` rows. Its CSV files are in-memory MVP review exports; they do not populate or audit `finite_rings.sqlite`.

The open blockers are part of the table. General finite-ring residue decomposition and deterministic site ordering remain blocked by `aqm-qsa-frres01`. Database-backed intrinsic or relative cotangent columns remain blocked by `aqm-qsa-frcot01`. Two-direction local Artin standardization remains blocked by `aqm-qsa-artin01` except for the local cotangent derivation in AQM-77. Quotient-collapse examples beyond the existing $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{F}_2 \times \mathbb{F}_3$ anchor remain blocked by `aqm-qsa-collapse01`.

81.2 Status Tokens

The table uses the report-planning labels of convention `(au)`. They are not CSV sentinel rows, do not change `data/SCHEMA.md`, and do not alter the `#`-prefixed sentinel comments in the finite-ring review exports. Here `available` means that a named local shard or review export supplies the row; `blocked` means that a named source, convention, derivation, helper, or bead is missing; `unknown` means the matrix has no local determination yet; `not_applicable` means the workflow is outside the row's stated semantics; and `degenerate` marks a locally recorded collapse or forgotten-structure case.

81.3 Points, Residues, and Cotangent Data

Ring row	Points	Residue data	Cotangent-first data
$\mathbb{F}_3$	`available`: one Spec-point by AQM-22 and convention `(u)`.	`available`: residue field $\mathbb{F}_3$, one qutrit site.	`available`: AQM-77 gives $C_x^{\mathrm{loc}} = C_x^{\mathrm{rel}} = 0$; the cotangent-first carrier is one-dimensional.
$\mathbb{F}_3 \times \mathbb{F}_3$	`available`: two discrete product-spectrum points by AQM-40 and convention `(ad)`.	`available`: two $\mathbb{F}_3$ residue sites, hence two qutrit factors in the residue workflow.	`degenerate`: AQM-58 and AQM-77 give $\Omega^1_{\mathbb{F}_3^2/\mathbb{F}_3} = 0$; the cotangent-first labels vanish at both points.
$\mathbb{Z}/6\mathbb{Z}$	`available`: AQM-23 derives the two points (2) and (3).	`available`: residues $\mathbb{F}_2$ and $\mathbb{F}_3$, with label group $\mathbb{F}_2^2 \oplus \mathbb{F}_3^2$.	`unknown`: no local cotangent comparison row is recorded here; database-backed cotangent data are blocked by `aqm-qsa-frcot01`.
$\mathbb{F}_3[\epsilon]/(\epsilon^2)$	`available`: the polynomial-quotient/local-Artin point is the unique closed point in AQM-34 and AQM-77.	`available`: local one $\mathbb{F}_3$ residue site; the database residue helper row is still `blocked` by `aqm-qsa-frres01`.	`available`: AQM-77 gives one intrinsic and relative $\mathbb{F}_3$-cotangent direction over $\mathbb{F}_3$, carrier dimension 3.
$\mathbb{F}_3[t]/(t^3)$	`available`: one closed point, a triple thickening of $t = 0$, by AQM-34 and AQM-77.	`available`: one $\mathbb{F}_3$ residue site; the reduced residue layer is one qutrit.	`available`: AQM-77 gives one first-order cotangent direction in characteristic 3; this does not record the full length-three thickening.
$\mathbb{F}_3[t]/(t^n - 1)$	`available`: AQM-35 uses the irreducible factors of the squarefree part t^m-1, $m = n/3^{v_3(n)}$.	`available`: each factor π contributes $\mathbb{F}_3[t]/(\pi)$; repeated factors do not add residue sites.	`unknown`: no uniform cotangent table is claimed for the family in this matrix.
$\mathbb{F}_3[u, v]/(u^2, v^2)$	`available`: local AQM-77 cotangent example only at (u, v); standard Artin-row status is `blocked`.	`available`: local residue $\mathbb{F}_3$ in AQM-77; no database residue row or standardized finite-ring catalogue row is claimed.	`available`: over $\mathbb{F}_3$, AQM-77 gives two cotangent directions and carrier dimension 9; relative to $\mathbb{F}_3[u]/(u^2)$, it gives one direction.
$\mathbb{Z}/4\mathbb{Z}$	`available`: AQM-77 local cotangent comparison only at (2).	`available`: local residue $\mathbb{F}_2$ in AQM-77; the database residue helper row is `blocked`.	`available`: AQM-77 gives $C_x^{\mathrm{loc}} \cong \mathbb{F}_2$ but $C_x^{\mathrm{rel}}(\mathbb{Z}) = 0$.
$\mathbb{F}_2[x]/(x^2 + x)$ candidate	`blocked`: possible quotient-collapse row only.	`blocked`: do not import product points before `aqm-qsa-collapse01` supplies a local derivation or certificate.	`not_applicable`: no cotangent comparison is made before the row is accepted as checked.

81.4 Field Labels, Nilpotents, and Run Artifacts

Ring row	Field-label data	Nilpotent-sensitive data	Run artifact or gap marker
$\mathbb{F}_3$	`available`: one-site finite phase labels; the underlying-set $\mathbb{F}_3$ all-map system is a separate AQM-22/AQM-71 object.	`not_applicable` as a nilpotent-thickening example; the database thickened helper still records a blocked policy row.	Finite-ring review CSV: residue row `available`; SQLite is schema-only, not populated.
$\mathbb{F}_3 \times \mathbb{F}_3$	`available`: AQM-40 gives the product-field residue label net; bare two-point all-map rows are separate finite-set examples.	`not_applicable` as a nilpotent-thickening example; no thickened product policy is imported.	Finite-ring review CSV: residue row `available`; thickened row `blocked`; no database audit claim.
$\mathbb{Z}/6\mathbb{Z}$	`available`: mixed-residue label direct sum in AQM-23; a single $\mathbb{F}_3$-linear field-label reading is `not_applicable`.	`blocked`: no certified generating character or thickened policy is recorded for this row.	`degenerate`: the review export records the $\mathbb{F}_2 \times \mathbb{F}_3$ presentation merged to the $\mathbb{Z}/6\mathbb{Z}$ representative; no extra collapse example is used.
$\mathbb{F}_3[\epsilon]/(\epsilon^2)$	`available`: reduced coordinate-ring field profiles forget ϵ, as in AQM-34.	`available`: AQM-36 gives the top-coefficient Artin-Weyl layer on $\ell^2(R)$, dimension 9.	Finite-ring review CSV: residue row `blocked`, thickened row `available`; SQLite has no ring rows.
$\mathbb{F}_3[t]/(t^3)$	`available`: reduced field profiles factor through $\mathbb{F}_3$; nilpotent t-classes vanish at the residue site.	`available`: AQM-36/AQM-37 give $\ell^2(\mathbb{F}_3[t]/(t^3))$ of dimension 27.	Local report derivation only; no MVP database presentation or populated SQLite row is claimed.
$\mathbb{F}_3[t]/(t^n - 1)$	`available`: AQM-35 field labels depend on the squarefree residue part in the reduced layer.	`available`: AQM-37 gives thickened dimension 3^n and reduced dimension 3^m.	Local report family only; database population and audit remain pending.
$\mathbb{F}_3[u,v]/(u^2,v^2)$	`unknown`: no field-label or residue-site matrix row beyond the AQM-77 cotangent derivation is standardized.	`blocked`: two-direction Artin standardization and any thickened Weyl row remain under `aqm-qsa-artin01`.	No run artifact; use AQM-77 only for cotangent data.
$\mathbb{Z}/4\mathbb{Z}$	`unknown`: no field-label row is standardized in this matrix.	`blocked`: finite-ring review CSV records missing certified generating character; no thickened claim is imported.	Finite-ring review CSV: residue and thickened rows `blocked`; AQM-77 only supports the cotangent comparison.
$\mathbb{F}_2[x]/(x^2 + x)$ candidate	`blocked`: possible `degenerate` quotient-collapse case is not asserted as checked.	`unknown`: no nilpotent-sensitive or reduced-row conclusion is used.	Blocked by `aqm-qsa-collapse01`; quotient-helper local-core status is not a report-level collapse proof.

81.5 Interpretation

The matrix separates four workflows that are easy to conflate. The residue-site workflow uses closed or maximal points and residue fields, as in AQM-23, AQM-34, AQM-35, and AQM-40. The cotangent-first workflow uses intrinsic or relative first-order cotangent data, as in AQM-76 and AQM-77. The field-label workflow uses finite maps, finite closed supports, or coordinate-ring profiles, as reviewed in AQM-79. The nilpotent-sensitive workflow uses local Artin or Frobenius-ring data, as in AQM-36 and AQM-37.

The table also records the current database boundary. The 2026-05-26 finite-ring run is useful evidence about the schema-only smoke build and the in-memory MVP review-export status of selected rows. It is not a populated or audited finite-ring database, and no row above treats it as one.

Step 17 must therefore supply certified residue decompositions and site orderings before extending the residue workflow across finite rings. Step 18 must keep reduced residue sites separate from nilpotent-sensitive Artin or Frobenius data, and it must not use the two-direction or quotient-collapse candidate rows until their blocker beads have local evidence.

82 Residue-Field Site Association for $\operatorname{Spec} R$

82.1 Status and Local Anchors

Status: Review/New for sourced finite cases only. This is QSA Step 17 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, script, run bundle, generated data, populated database row, or general finite-ring algorithm. It packages the residue-site workflow already present in conventions `(u)`, `(y)`, `(aa)`, `(ad)`, and `(an)` and in AQM-23, AQM-34, AQM-35, AQM-40, AQM-75, AQM-78, and AQM-80.

The finite-ring database boundary is unchanged. The local anchors are convention `(an)` and the run-bundle README

`runs/2026-05-26-finite-ring-database/README.md`.

They record a schema-only SQLite smoke build plus in-memory MVP review exports. The SQLite artifact is not a populated or audited finite-ring database. The in-memory residue helper is deliberately limited to source-backed cases and does not certify a general maximal-ideal or residue-field decomposition. General finite-ring maximal/residue decomposition and deterministic site ordering remain blocked by `aqm-qsa-frres01`.

82.2 Workflow for Selected Finite Residue Sites

The input is not an arbitrary ring with an implicit site algorithm. The input is a finite displayed site set

$$S = \{x_1, \ldots, x_r\}$$

of prime or maximal points of $\operatorname{Spec} R$, together with the locally anchored residue field

$$K_x = \kappa(x)$$

for each $x \in S$. The site set may be all of $\operatorname{Spec} R$ in a finite sourced spectrum such as AQM-23, AQM-34, AQM-35, or AQM-40; otherwise it is a selected finite support under the point-choice discipline of AQM-75.

The default residue-site target is the one-degree finite-field phase space

$$E_x = K_x^2 = K_{x,X} \oplus K_{x,Z}, \qquad \omega_x((q,p),(q',p')) = pq' - p'q.$$

The local Weyl carrier and observable algebra are

$$\mathcal{H}_x = \ell^2(K_x), \qquad \mathcal{A}_x = \operatorname{End}_{\mathbb{C}}(\mathcal{H}_x).$$

This is the one-site finite-field Weyl step reviewed in AQM-78. If a later row supplies a different finite-dimensional K_x-vector space Q_x, the same sourced finite-field Weyl step may use $E_x = Q_x \oplus Q_x^\vee$ and $\ell^2(Q_x)$. The point of Step 17 is that this target is allowed to vary with the residue field K_x; it is not required to be a vector space over a single common residue field.

For the finite support S, form the phase-label group and carrier

$$E(S) = \bigoplus_{x \in S} E_x, \qquad \mathcal{H}(S) = \bigotimes_{x \in S} \mathcal{H}_x, \qquad \mathcal{A}(S) = \bigotimes_{x \in S} \mathcal{A}_x.$$

The direct sum is a direct sum of finite abelian phase groups with its orthogonal local commutator pairings. A chosen ordering of S may be used to display the finite tensor product; by convention `(ar)` it is display data unless a separate convention fixes an order. Empty support gives the zero label group and the empty tensor product $\mathbb{C}$.

82.3 Mixed Residues Are Not One Common-Field Space

AQM-23 is the warning example. For

$$R = \mathbb{Z}/6\mathbb{Z}$$

the two points are (2) and (3), with residues $\mathbb{F}_2$ and $\mathbb{F}_3$. Step 17 therefore gives

$$E(S) = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2, \qquad \mathcal{H}(S) = \ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3).$$

This is a direct sum of finite abelian phase groups and a tensor product of local carriers. It is not an $\mathbb{F}_3$-vector space, not an $\mathbb{F}_2$-vector space, and not a Hilbert direct sum $\mathbb{C}^2 \oplus \mathbb{C}^3$. A reduction to a common prime-field abelian group may be introduced only after recording the prime-field pairing and character used for that reduction.

82.4 Sourced Example Table

Object	Residue sites	Step 17 carrier	Boundary marker
$\mathbb{F}_3$	`available`: one Spec-point with residue $\mathbb{F}_3$, by AQM-22 and convention `(u)`.	$E = \mathbb{F}_3^2$, $\mathcal{H} = \ell^2(\mathbb{F}_3)$, one qutrit residue site.	This is the residue-spectrum point, not the three-point underlying-set model of AQM-22/AQM-71.
$\mathbb{F}_3 \times \mathbb{F}_3$	`available`: AQM-40 and convention `(ad)` give two discrete points, both with residue $\mathbb{F}_3$.	$E = \mathbb{F}_3^2 \oplus \mathbb{F}_3^2$, $\mathcal{H} = \ell^2(\mathbb{F}_3)^{\otimes 2}$.	Stabilizer codes are later choices of isotropic labels and phases, not forced by the product ring.
$\mathbb{Z}/6\mathbb{Z}$	`available`: AQM-23 gives (2) and (3) with residues $\mathbb{F}_2$ and $\mathbb{F}_3$.	$E = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2$, $\mathcal{H} = \ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3)$.	Mixed residues are not one common-field vector space. The observable algebra is $M_2(\mathbb{C}) \otimes M_3(\mathbb{C})$.
$\mathbb{F}_3[t]/(p)$	`available` for monic nonzero $p = \prod_i \pi_i^{e_i}$ in AQM-34: one site $x_i = (\pi_i)/(p)$ per distinct irreducible factor, with $K_i = \mathbb{F}_3[t]/(\pi_i)$.	$E = \bigoplus_i K_i^2$, $\mathcal{H} = \bigotimes_i \ell^2(K_i)$; if $p = 1$, this is the empty support case.	Repeated exponents e_i are nilpotent thickening data and do not add residue sites.
$\mathbb{F}_3[t]/(t^n - 1)$ reduced layer	`available`: AQM-35 writes $n = 3^s m$, $3 \nmid m$, and uses the irreducible factors of the squarefree part $t^m - 1$.	$E = \bigoplus_{\pi \mid t^m-1} K_\pi^2$, $\mathcal{H} = \bigotimes_{\pi \mid t^m-1} \ell^2(K_\pi)$, dimension 3^m.	AQM-37 gives the separate thickened dimension 3^n; Step 17 uses only the reduced residue layer.
$\mathbb{Z}/4\mathbb{Z}$	`blocked`: database-backed residue-site row. AQM-77 locally uses the point (2) and residue $\mathbb{F}_2$ for cotangent comparison, but the finite-ring residue helper does not certify the row.	No general Step 17 database-backed residue-site row is imported here.	Keep as a local cotangent anchor only until `aqm-qsa-frres01` supplies certified decomposition/site ordering.
$\mathbb{F}_3[\epsilon]/(\epsilon^2)$	`available`: local AQM-34/AQM-80 anchor records one reduced residue site with residue $\mathbb{F}_3$; the database residue helper row remains `blocked`.	Reduced Step 17 carrier is the one qutrit $\ell^2(\mathbb{F}_3)$ when that local point is selected.	The $\ell^2(\mathbb{F}_3[\epsilon]/(\epsilon^2))$ Artin-Weyl layer is nilpotent-sensitive AQM-36/Step 18, not an extra residue point.

82.5 What This Supplies to Later Steps

Step 17 supplies a precise reduced residue-site input to later QSA shards: a selected finite set of prime or maximal points, the residue field at each site, the local finite-field phase data over that residue field, and the direct-sum/tensor-product assembly over finite supports. It also supplies the mixed-residue warning needed before any stabilizer, dynamics, Clifford, or common-prime-field comparison: those are later choices and require their own pairings, phases, and sources.

The blocked items are unchanged. There is no general algorithm here for all finite commutative rings, no populated/audited finite-ring database evidence, and no deterministic site ordering beyond the sourced displays. Reduced residue sites remain separate from nilpotent-sensitive thickened layers: AQM-36 and AQM-37 quantize local Artin rings as jet degrees of freedom at the same closed points, and QSA Step 18 will compare that layer without collapsing nilpotents into extra residue sites.

83 Nilpotent-Sensitive Association

83.1 Status and Local Anchors

Status: Review/New for local thickened examples only. This is QSA Step 18 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, script, run bundle, generated data, populated database row, or general finite-ring representation theorem. Its job is to compare the reduced residue-site association of AQM-81 with the nilpotent-sensitive Artin/Frobenius-ring Weyl layer already established in AQM-31, AQM-36, and AQM-37.

The governing conventions are `(u)`, `(y)`, `(aa)`, and `(ab)` for residue and polynomial-quotient Weyl layers, convention `(an)` for the finite-ring database boundary, and convention `(au)` for report-planning status tokens. AQM-31 records the sourced finite Frobenius-ring Pauli/stabilizer status and the need for a generating character. AQM-36 proves the top-coefficient character construction for $\mathbb{F}_3[t]/(\pi^e)$, proves the corresponding Weyl product and operator-basis statements, and interprets the nilpotent filtration as jet data. AQM-37 applies that result to the $t^n = 1$ family. AQM-80 and AQM-81 record the finite-ring matrix and reduced residue-site workflows that this shard compares against.

The blockers are part of the conclusion. Arbitrary finite-ring maximal/residue decomposition and deterministic site ordering remain blocked by `aqm-qsa-frres01`. Two-direction local Artin examples remain blocked by `aqm-qsa-artin01`, except for the cotangent comparison in AQM-77. Certified generating characters beyond the local AQM-36 polynomial chain-ring examples remain blocked; the finite-ring database helper scope in convention `(an)` is MVP source-backed only and is not a general Frobenius-ring recognizer.

83.2 Reduced Sites Versus Thickened Carriers

For a monic nonzero polynomial

$$p = \prod_i \pi_i^{e_i} \in \mathbb{F}_3[t],$$

AQM-34 and convention `(aa)` give one reduced residue site

$$x_i = (\pi_i)/(p)$$

for each distinct irreducible factor π_i. The reduced residue carrier at that site is

$$\mathcal{H}_i^{\mathrm{red}} = \ell^2(K_i), \qquad K_i = \mathbb{F}_3[t]/(\pi_i),$$

with phase labels K_i^2. The exponent e_i does not create e_i residue sites. It is scheme-theoretic thickening data that is invisible to the reduced closed-point Weyl labels because coordinate-ring field profiles factor through the reduced quotient, as recorded in AQM-34 and AQM-35.

The nilpotent-sensitive carrier keeps the local Artin factor

$$A_i = \mathbb{F}_3[t]/(\pi_i^{e_i})$$

at the same closed point x_i. AQM-36 and convention `(ab)` therefore use

$$\mathcal{H}_i^{\mathrm{thick}} = \ell^2(A_i), \qquad E_i^{\mathrm{thick}} = A_i \times A_i.$$

Globally this gives

$$\mathcal{H}_p^{\mathrm{thick}} = \bigotimes_i \ell^2(A_i), \qquad E_p^{\mathrm{thick}} = \bigoplus_i A_i^2,$$

equivalently the Weyl system on $\ell^2(\mathbb{F}_3[t]/(p))$ through the Chinese-remainder display used in AQM-36. This is not a replacement for the reduced layer. The reduced layer is the zeroth-order residue-site association of AQM-81; the thickened layer adds local jet/internal degrees at the same selected points.

83.3 Extra Data Beyond Residue Fields

The extra datum is not merely the list of residue fields. For the AQM-36 local ring

$$A = \mathbb{F}_3[t]/(\pi^e), \qquad K = A/(\pi),$$

one first expands

$$a = a_0 + a_1\pi + \cdots + a_{e-1}\pi^{e-1}$$

with coefficients represented in K. The thickened Weyl layer then uses the top-coefficient functional

$$\lambda(a) = \mathrm{Tr}_{K/\mathbb{F}_3}(a_{e-1}), \qquad \chi(a) = \exp(2\pi i\lambda(a)/3).$$

AQM-36 proves locally that this χ is generating: for $b \neq 0$, the character $a \mapsto \chi(ba)$ is nontrivial. That proof is the step that turns the additive group of A into its character group for the Weyl construction. Without such a certified generating character, the translation/phase recipe over $A\times A$ has not met the finite Frobenius-ring self-duality input reviewed in AQM-31.

Once the character is fixed, AQM-36 defines

$$T(q)|s\rangle = |s+q\rangle, \qquad R(p)|s\rangle = \chi(ps)|s\rangle, \qquad W_A(q,p) = T(q)R(p)$$

on $\ell^2(A)$, and proves

$$W_A(q,p)W_A(q',p') = \chi(pq')W_A(q+q',p+p')$$

with commutator phase $\chi(pq'-p'q)$. AQM-36 also proves that these Weyl operators span

$$\mathrm{End}_{\mathbb{C}}(\ell^2(A)).$$

Step 18 may therefore treat this local chain-ring layer as available, but it may not import a complete Weyl, Clifford, or projective-representation classification for arbitrary finite commutative rings.

83.4 Dual Numbers as the Minimal Test

For

$$A = \mathbb{F}_3[\epsilon]/(\epsilon^2)$$

there is one closed point with residue field $\mathbb{F}_3$. The reduced residue-site workflow of AQM-81 gives

$$\mathcal{H}^{\mathrm{red}} = \ell^2(\mathbb{F}_3),$$

one qutrit. The nilpotent class ϵ reduces to zero at that site; it is not an additional point.

The AQM-36 thickened workflow writes $a = a_0 + a_1\epsilon$ and uses

$$\lambda(a) = a_1, \qquad \chi(a) = \exp(2\pi i a_1/3).$$

Its carrier is

$$\mathcal{H}^{\mathrm{thick}} = \ell^2(\mathbb{F}_3[\epsilon]/(\epsilon^2)), \qquad \dim_{\mathbb{C}}\mathcal{H}^{\mathrm{thick}} = 9.$$

Thus the comparison is not "one qutrit plus a second point". It is one reduced qutrit versus a single thickened Artin site with an additional first-order internal coordinate, certified by the top-coefficient generating character proof of AQM-36.

83.5 The $t^n = 1$ Review

AQM-35 and AQM-37 use

$$R_n = \mathbb{F}_3[t]/(t^n - 1), \qquad n = 3^s m, \quad 3 \nmid m.$$

AQM-35 proves

$$t^n - 1 = (t^m - 1)^{3^s}$$

and uses the irreducible factors of the squarefree part $t^m - 1$ as reduced residue Weyl sites. The reduced Hilbert dimension is

$$\dim \mathcal{H}_n^{\mathrm{red}} = 3^m.$$

AQM-37 applies AQM-36 to the thickened factors

$$A_\pi = \mathbb{F}_3[t]/(\pi^{3^s}), \qquad \pi \mid t^m - 1,$$

and proves

$$\mathcal{H}_n^{\mathrm{thick}} = \bigotimes_{\pi \mid t^m - 1} \ell^2(A_\pi), \qquad \dim \mathcal{H}_n^{\mathrm{thick}} = 3^n.$$

The concrete examples in AQM-37 make the distinction visible. For $n = 3$,

$$\mathbb{F}_3[t]/(t^3 - 1) \cong \mathbb{F}_3[u]/(u^3)$$

has one reduced qutrit site but a 27-dimensional third-order thickened carrier. For $n = 6$, the reduced layer has two qutrit sites, while the thickened layer uses two third-order Artin carriers and has dimension 3^6. In both cases the nilpotent variables are jet degrees at the already listed closed points, not new points.

83.6 Boundary Table

Object	Reduced residue-site layer	Thickened layer	Status boundary
$\mathbb{F}_3$	`available`: one residue $\mathbb{F}_3$ site by AQM-22 and AQM-81.	`not_applicable`: no nilpotent thickening in this row.	No Artin data beyond the field.
$\mathbb{F}_3[\epsilon]/(\epsilon^2)$	`available`: local one $\mathbb{F}_3$ residue site; database helper coverage remains blocked as in AQM-80/AQM-81.	`available`: AQM-36 top-coefficient character; carrier $\ell^2(A)$, dimension 9.	Nilpotent ϵ is a jet/internal degree at the unique point.
$\mathbb{F}_3[t]/(t^3)$ or $\mathbb{F}_3[t]/((t-1)^3)$	`available`: local one $\mathbb{F}_3$ residue site by AQM-34, AQM-35, and AQM-81.	`available`: AQM-36/AQM-37 chain-ring carrier of dimension 27.	Third-order thickening, not three closed points.
$\mathbb{F}_3[t]/(t^n - 1)$	`available`: AQM-35 reduced dimension 3^m, with $n = 3^s m$, $3 \nmid m$.	`available`: AQM-37 thickened dimension 3^n.	Only this locally derived family is covered.
$\mathbb{F}_3[u, v]/(u^2, v^2)$	`available`: AQM-77 cotangent comparison only; no standardized residue matrix row is claimed.	`blocked`: no two-direction Artin-Weyl generating-character policy is registered.	Blocked by `aqm-qsa-artin01`.
$\mathbb{Z}/4\mathbb{Z}$ and other MVP rings	`blocked`: database-backed residue rows; local AQM-77 cotangent anchors are separate.	`blocked`: no certified generating character or thickened policy is imported.	Convention `(an)` keeps helper scope MVP source-backed only.

83.7 Conclusion for the QSA Catalogue

The nilpotent-sensitive association is a separate finite Frobenius/Artin-ring workflow, not a correction to the reduced residue-site workflow. Its local input is a thickened ring such as $\mathbb{F}_3[t]/(\pi^e)$, a certified generating character, and the resulting translation/phase Weyl representation on $\ell^2(A)$. The reduced workflow uses residue fields and finite closed points; nilpotents add local jet or internal degrees at those same points or remain `blocked`.

Therefore Step 18 supplies only the locally established comparison: dual-number and $t^n = 1$ thickenings differ from their reduced residue-site models by certified Artin-Weyl jet data. It does not settle arbitrary finite rings, two-direction Artin examples, certified generating characters beyond the AQM-36 family, Clifford/Weil classification, dynamics, or database-backed finite-ring coverage.

84 Product Rings and Tensor Factors

84.1 Status and Local Anchors

Status: Review shard. This is QSA Step 19 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, script, run bundle, generated data, populated database row, or finite-ring decomposition theorem. It reviews exactly the product-ring evidence already in AQM-40, AQM-58, AQM-68, AQM-80, AQM-81, and AQM-82.

The governing conventions are `(ad)` for product-field spectra, `(an)` for finite-ring database and helper scope, `(ar)` for structureless finite-set tensor sites, and `(au)` for QSA table status tokens. The product-field anchor is

$$R = k^n, \qquad k = \mathbb{F}_q,$$

where AQM-40 proves that $\mathrm{Spec}(k^n)$ is the finite discrete n-point spectrum with residue field k at every point. The main comparison anchor is AQM-58: the same product ring has

$$\Omega^1_{k^n/k} = 0,$$

so the cotangent-first tangent-Weyl layer can vanish even when the residue workflow has n qudit tensor factors.

84.2 Selected Workflow: Product-Field Residue Sites

In the selected product-field residue-site workflow, the ring supplies more than a bare set but less than a universal quantum model. For

$$R = k^n$$

the idempotents and projections select the points

$$x_i = \ker(\mathrm{pr}_i), \qquad i = 1, \dots, n,$$

with $\kappa(x_i) = k$. Convention `(ad)` and AQM-40 then give

$$E(X) = \bigoplus_{i=1}^{n} k^2, \qquad \mathcal{H}(X) = \bigotimes_{i=1}^{n} \ell^2(k), \qquad \mathcal{A}(X) = \bigotimes_{i=1}^{n} \mathrm{End}_{\mathbb{C}}\, \ell^2(k).$$

Thus $\mathbb{F}_3^n$ gives n qutrit residue factors, not 3^n spectrum points. Open subsets are just subsets of the n points, and the local observable inclusions are tensoring by identities on the missing residue factors.

This is a tensor factorization statement for this residue-site association only. It does not identify product rings with all first-association, cotangent-first, field-label, nilpotent-sensitive, stabilizer, Clifford, or dynamics workflows. AQM-40 already separates the ambient n-qudit Weyl net from the later choice of an isotropic stabilizer subgroup and phases.

84.3 Comparison With Structureless Tensor Sites

AQM-68 starts with a bare finite set X and only then adds site data

$$\{\mathcal{H}_x\}_{x\in X} \quad \text{or} \quad \{\mathcal{A}_x\}_{x\in X}.$$

After those choices, it forms

$$\mathcal{H}(S) = \bigotimes_{x\in S} \mathcal{H}_x, \qquad \mathcal{A}(S) = \bigotimes_{x\in S} \mathcal{A}_x.$$

The product-field workflow supplies the following extra data that the bare finite set did not supply.

- The site set is not arbitrary: it is the spectrum $\{x_i = \ker(\mathrm{pr}_i)\}$ of the displayed product ring.
- The topology is discrete for k^n, with singleton opens $D(e_i)$.
- Each site has a residue field $\kappa(x_i) = k$, hence a local finite-field phase space k^2.
- The local carrier is the particular Weyl carrier $\ell^2(\kappa(x_i))$, and the local algebra is its full endomorphism algebra.

The analogy with AQM-68 is therefore precise but limited: both workflows use finite tensor products and tensor-by-identity inclusions once the local factors are known. They do not have the same input. The structureless finite set allows arbitrary chosen Hilbert spaces; the product-field residue-site workflow fixes the local Weyl factors from the residue fields.

84.4 Mixed Residue Warning

AQM-23 and AQM-81 supply the warning example

$$R = \mathbb{Z}/6\mathbb{Z}.$$

The residue-site workflow has two points, with residues

$$\mathbb{F}_2, \qquad \mathbb{F}_3,$$

and hence

$$E = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2, \qquad \mathcal{H} = \ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3).$$

This is still an independent tensor-factor residue workflow, but it is not one common $\mathbb{F}_3$-linear or $\mathbb{F}_2$-linear phase space. Any common-prime-field reduction, stabilizer comparison, or Clifford comparison would require its own pairing and phase convention.

This shard does not promote $\mathbb{Z}/6\mathbb{Z}$ to a general quotient-collapse theorem. AQM-80 keeps quotient-collapse examples beyond the existing $\mathbb{Z}/6\mathbb{Z}$ anchor blocked by `aqm-qsa-collapse01`.

84.5 Cotangent Edge Cases

Product-ring tensor factors are highly workflow-dependent. The clearest edge case is AQM-58. For k^n, every k-derivation out of k^n vanishes, so

$$\Omega^1_{k^n/k} = 0.$$

In the tangent-cotangent Weyl workflow of conventions `(ao)` and `(av)`, the local tangent and cotangent labels at each product point therefore vanish. The cotangent-first geometric Weyl carrier for those directions is the zero-label, one-dimensional case; it is not the n-qudit residue carrier of AQM-40.

The single field row already shows the same distinction in smaller form. $\mathrm{Spec}(\mathbb{F}_3)$ has one residue-site qutrit in AQM-81, while its relative cotangent over $\mathbb{F}_3$ is zero in AQM-77 and AQM-80. The product field repeats this zero-cotangent phenomenon n times while the residue-site layer still has n local Weyl factors.

For mixed or nonreduced finite rings, the table in AQM-80 is the current boundary. $\mathbb{Z}/6\mathbb{Z}$ has the mixed-residue tensor carrier above, but no database-backed cotangent comparison row is imported here. $\mathbb{Z}/4\mathbb{Z}$ and dual-number examples have local cotangent or nilpotent-sensitive anchors in AQM-77 and AQM-82, but they are not product field tensor-factor examples in this Step 19 review.

84.6 Database and Helper Boundary

Convention `(an)` records a manual `finite_ring_product` constructor for the finite-ring database implementation: it preserves argument order, concatenates factor coordinate blocks and identity vectors, copies factor multiplication into each block, and sets cross-factor products to zero. This is a data-layout convention for MVP constructors, not a classification theorem for arbitrary finite rings.

The same convention limits the in-memory residue-quantisation helper to source-backed cases already fixed in the lab book: prime fields, the $\mathbb{Z}/6\mathbb{Z}$ row from AQM-23, and explicit finite product fields from AQM-40. The SQLite smoke build remains schema-only, and the review CSVs are in-memory MVP exports. No populated or audited finite-ring database is claimed.

The active gaps therefore remain:

`aqm-qsa-frres01`. Arbitrary finite-ring residue decomposition and site ordering.

`aqm-qsa-frcot01`. Database-backed cotangent rows.

`aqm-qsa-collapse01`. Quotient-collapse examples beyond the $\mathbb{Z}/6\mathbb{Z}$ anchor.

`aqm-qsa-artin01`. Unsupported thickened or multi-direction Artin layers.

In particular, a product of thickened local rings may have a displayed tensor product after each local Artin/Frobenius layer has its own certified generating character, but Step 19 does not assert such a general policy.

84.7 Conclusion

Product rings give independent tensor factors only in a selected and well-anchored sense. For k^n, the product-field residue-site association supplies n finite Weyl sites, n local matrix algebras, and the usual tensor-by-identity local net. This matches the form of the structureless finite-set tensor-site construction only after the product ring has supplied the sites and residue-field Weyl factors.

The same ring may look degenerate in a different association workflow: cotangent-first labels vanish for k^n, stabilizers and dynamics remain extra choices, mixed residues are not a single common-field phase space, and nilpotent-sensitive product layers require separate Artin/Frobenius evidence. Those boundaries are part of the QSA catalogue rather than exceptions to a universal product-ring tensor-factor rule.

85 Infinite Ring Boundary Cases

85.1 Status and Local Anchors

Status: Review/New boundary shard. This is QSA Step 20 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, script, run bundle, generated data, or analytic completion. Its purpose is to mark where the finite association catalogue touches infinite spectra.

The controlling conventions are `(v)`–`(w)` for open-local observable nets, `(y)` for closed-point finite-residue sites on finite-type affine $\mathbb{F}_q$-schemes, `(z)` for rational-function Weyl sectors over $K = \mathbb{F}_q(t)$, `(ai)` for the $\operatorname{Spec}\mathbb{Z}$ closed-prime layer, and `(ak)`–`(al)` for regular-function profiles and finite smearings. The report anchors are AQM-25, AQM-29, AQM-30, AQM-45, AQM-48, AQM-75, and AQM-79.

85.2 Boundary Vocabulary

Every infinite construction below is assigned one of the following statuses.

Finite-support. A finite subset S of the selected closed points is quantized by an ordinary finite tensor product of local matrix algebras.

Algebraic quasi-local. The local algebra is the filtered colimit over finite closed supports inside an open set. No Hilbert-space completion or C^*-completion is part of this status.

Incomplete tensor representation. A Hilbert representation is chosen by fixing a reference vector at every closed point, so basis vectors have finite excitation support relative to that reference.

Formal infinite-volume. A regular-function profile or similar global label has infinite closed support. Its finite restrictions exist, but the full profile is not an element of the algebraic quasi-local Weyl algebra.

Generic-sector proposal. A generic or curve-generic residue field is given a separate algebraic Weyl sector only after a character and representation convention are stated. It is not a finite closed-point site.

Completion proposal. A local-field, adelic, profinite, analytic, or C^*-completed object is named as a possible later layer, not as a present report construction.

This vocabulary is only a status partition for Step 21. It does not assert agreement or inequivalence between workflows.

85.3 $\operatorname{Spec}\mathbb{F}_q[t]$

AQM-29 is the affine-line boundary example. The scheme has the generic point $\eta = (0)$, with residue field $\mathbb{F}_q(t)$, and closed points

$$x_\pi = (\pi)$$

indexed by monic irreducible polynomials. A closed point of degree d has residue field $\mathbb{F}_q[t]/(\pi)$ and contributes one q^d-level finite-field Weyl site.

The finite-support construction is the finite tensor product over a selected finite set S of closed points:

$$\mathcal{A}(S) = \bigotimes_{x_\pi \in S} \operatorname{End}_{\mathbb{C}} \ell^2(\mathbb{F}_q[t]/(\pi)).$$

For an open U, the algebraic quasi-local construction is

$$\mathcal{A}(U) = \varinjlim_{S \Subset U \cap |X|_{\mathrm{cl}}} \mathcal{A}(S).$$

This is the status of the closed-point field net. It is not a single finite Hilbert space.

AQM-48 records an incomplete tensor representation in the case $q = 3$: after choosing $|0\rangle_\pi$ at every closed point, the configuration basis is indexed by

$$\Gamma_{\mathrm{fin}} = \bigoplus_\pi \mathbb{F}_3[x]/(\pi).$$

This Hilbert-space representation is extra data. It is not forced by $\operatorname{Spec} \mathbb{F}_3[x]$ alone.

AQM-29 and convention `(ak)` also record the formal infinite-volume boundary. A nonzero polynomial pair gives a closed-point residue profile with infinite support. Its finite restrictions define finite-volume Weyl operators, but the full regular-function profile is not in the algebraic quasi-local algebra until a separate infinite-product or completion convention is fixed.

The generic point is not a finite Weyl site. Convention `(z)` gives a separate rational-function Weyl layer for $K = \mathbb{F}_q(t)$ only after an additive character and representation convention are chosen. The LCA/local completion alternatives in `(z)` remain completion proposals here.

85.4 $\operatorname{Spec} \mathbb{F}_q[x, y]$

AQM-30 is the affine-plane boundary example. Its point types are the generic point, curve-generic points (f), and closed maximal ideals. Only the closed maximal ideals are finite-residue Weyl sites under convention `(y)`. If a closed point $\mathfrak{m}$ has residue degree $d(\mathfrak{m})$, then the local carrier is

$$\ell^2(\kappa(\mathfrak{m})), \qquad |\kappa(\mathfrak{m})| = q^{d(\mathfrak{m})}.$$

Finite-support and algebraic quasi-local statuses are exactly as in the affine-line case, with finite subsets of

$$U \cap \operatorname{Max} \mathbb{F}_q[x, y].$$

The rational-point support

$$\{(x - a, y - b) : a, b \in \mathbb{F}_q\}$$

is a finite-support truncation with q^2 q-level sites. It is not the full $\operatorname{Spec} \mathbb{F}_q[x, y]$ theory, because higher-degree closed points are omitted.

Coordinate profiles such as $(x, 0)$ and $(0, y)$ have formal infinite-volume status in AQM-30. Their finite restrictions are legitimate finite-support labels, but the full profiles are not quasi-local Weyl operators.

The generic point and curve-generic points are generic-sector or completion questions, not finite Weyl sites. Their residue fields are rational-function fields or function fields of curves. No local-field completion, Pontryagin-dual phase space, adelic replacement, C^*-completion, or generic-point Hilbert-space convention is fixed for them in this shard.

85.5 Spec $\mathbb{Z}$

AQM-45 and convention `(ai)` give the integer boundary example:

$$X = \operatorname{Spec}\mathbb{Z}, \qquad \eta = (0), \qquad x_\ell = (\ell).$$

The closed prime x_ℓ has residue field $\mathbb{F}_\ell$, so the finite-support closed-prime layer uses

$$\mathcal{A}(S) = \bigotimes_{\ell \in S} M_\ell(\mathbb{C})$$

for a finite set S of rational primes.

For an open U, the algebraic quasi-local closed-prime algebra is

$$\mathcal{A}_{\mathrm{cl}}(U) = \varinjlim_{S \Subset U \cap \{x_\ell\}} \mathcal{A}(S).$$

This is the algebraic infinite tensor product over finite prime supports when $U = X$. It is not a canonical Hilbert-space representation.

AQM-45 also records an incomplete tensor representation after choosing $|0\rangle_\ell \in \ell^2(\mathbb{F}_\ell)$ at every prime. The basis is indexed by finite-support residue configurations

$$\bigoplus_\ell \mathbb{F}_\ell.$$

This representation is useful for local finite-excitation calculations, but it is not a C^*-completion and it does not implement every profinite-type automorphism by a unitary in that sector.

The generic point has residue field $\mathbb{Q}$. AQM-45 records an optional algebraic $\mathbb{Q}$-Weyl sector with carrier $\ell^2(\mathbb{Q})$, but it is present in every nonempty open and is a global generic-fibre layer, not a prime-local qudit. Adelic, local-completion, and analytic Stone–von Neumann versions remain completion proposals until separate conventions and sources are added.

85.6 Representation and Completion Gaps

The missing choices for infinite boundary cases are now explicit.

- Infinite residue fields do not inherit the finite closed-point $\ell^2(\kappa(x))$ qudit convention. They need a character, dual-label space, and representation convention.
- Generic points and curve-generic points do not become finite Weyl sites by default. If included, they are generic-sector proposals.
- Analytic or local completions require locally compact groups, Pontryagin duals, continuity hypotheses, and a sourced Stone–von Neumann–Mackey statement before uniqueness language is allowed.
- C^*-completions of the algebraic quasi-local algebras require a norm, completion, and state-continuity convention.
- Adelic and profinite sectors require their own topology, restricted product, measure, and representation conventions. The profinite residue action in AQM-45 is not the same as an adelic Hilbert-space model.
- Formal infinite-volume regular-function profiles require a completion or infinite-product field-operator convention before they become operators.

85.7 Step 21 Handoff

Step 21 may use this shard as a status classifier. Rows for $\operatorname{Spec} \mathbb{F}_q[t]$, $\operatorname{Spec} \mathbb{F}_q[x, y]$, and $\operatorname{Spec} \mathbb{Z}$ should distinguish finite-support truncations, algebraic quasi-local closed-point algebras, incomplete tensor representations, formal infinite-volume profiles, generic-sector proposals, and completion proposals. This shard deliberately does not decide final agreement, inequivalence, or degeneracy claims. Those require the explicit maps, invariants, or obstructions demanded by the Step 21 synthesis.

86 Agreement and Inequivalence Table

86.1 Status and Local Anchors

Status: New gap-first synthesis shard. This is QSA Step 21 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, script, run bundle, generated data, populated database row, or report-level classification theorem. Its role is to aggregate the local comparison evidence already recorded in AQM-65–AQM-84.

The comparison vocabulary is AQM-65 and `CONVENTIONS.md` conventions `(ap)` and `(au)`. The finite-set workflow anchors are AQM-67–AQM-74. Point, cotangent, finite phase, field-space, finite-ring, residue-site, nilpotent-sensitive, product-ring, and infinite-boundary anchors are AQM-75–AQM-84. Earlier arithmetic-field and Weyl anchors are cited only through those review shards unless named below.

86.2 Reading Rule

The entries below are not a final equivalence table. The token `agreement` is used only when a prior shard gives a named comparison map, identity of assembled data after choices, or checked invariant for the selected workflows. The token `inequivalence` is used only when a prior shard records a local obstruction for the selected workflows. Dimension equality is never used as evidence of agreement. A dimension mismatch may be listed only as one obstruction among the locally recorded differences.

`available` means that a workflow has a local anchor but has not been compared as an agreement or inequivalence result in this shard. `degenerate` marks locally recorded collapse or forgotten-structure cases. `blocked`, `unknown`, and `not_applicable` keep their meanings from convention `(au)`. AQM-84 boundary categories such as *generic-sector proposal* and *completion proposal* are prose classifiers for witness and gap columns, not status tokens in this table. They are not finite Weyl-site claims.

86.3 Core Comparison Ledger

Comparison row	Local status	Witness or obstruction	Anchor
Bare finite-set carrier versus one-dimensional direct-sum particle carrier	`agreement`, after choices only.	Choose one-dimensional K_x and nonzero vectors u_x; the map $e_x \mapsto j_x(u_x)$ is the recorded vector-space isomorphism. Hilbert agreement also needs unit vectors and a Gram convention.	AQM-69, AQM-74
Full tensor-site carrier versus direct-sum particle carrier	`inequivalence` for the named full carriers.	The workflows represent coexistence and alternatives. The empty cases are $\mathbb{C}$ and 0, and the full carrier counts are tensor-product versus direct-sum counts for the same chosen local factors.	AQM-68, AQM-69, AQM-74
AND/OR alternatives and site-option degree-one sector versus direct-sum particle carrier	`agreement`, for the named degree-one sector.	AQM-74 gives R_P as the identity on the direct-sum expression and R_G as the explicit map into the one-excitation sector of the site-option expression. No full Fock or statistics comparison follows.	AQM-70, AQM-74
Formal finite-set carrier versus finite symplectic field labels	`available`.	AQM-71 has checked finite field-label rows after choosing V, E, a pairing, radical quotient, and Weyl step. AQM-67 has only formal basis labels; comparison remains `unknown` because no map compares the full outputs.	AQM-67, AQM-71
Fermionic one-particle carrier versus uniform direct-sum particle carrier	`agreement`, for that carrier only.	AQM-74 records $E_Z : Z = \mathrm{Map}(X, K) \to \bigoplus_x K$ and its inverse. The CAR algebra, exterior higher sectors, and Grassmann-Weyl multiplier do not reduce by this map.	AQM-72, AQM-74
CCR/CAR/Fock layers versus finite AND/OR grammar	`blocked`.	AQM-72 states that no local map, quotient, invariant, or derivation compares the full CCR/CAR or Grassmann-Weyl outputs with the finite grammar; comparison remains `unknown`.	AQM-70, AQM-72
Finite phase labels to Weyl-Heisenberg kinematics	`available`.	AQM-78 supplies the common kinematical step after a finite commutator label object is fixed. This is not an agreement between earlier workflows; it is a shared downstream representation step.	AQM-78

Comparison row	Local status	Witness or obstruction	Anchor
Product-field residue sites versus structureless tensor-site assembly	`agreement`, at the assembly layer only.	For $R = k^n$, AQM-83 records that the residue workflow supplies the sites, fields, carriers $\ell^2(k)$, and local algebras; after those choices the finite tensor products and tensor-by-identity inclusions match AQM-68's tensor-site bookkeeping.	AQM-68, AQM-83
Product-field residue sites versus product-field cotangent-first labels	`inequivalence`.	For k^n, AQM-83 cites AQM-58 and AQM-77: residue sites give n finite Weyl factors, while the relative cotangent labels over k vanish at every product point, so the cotangent output is `degenerate`.	AQM-58, AQM-77, AQM-83
Intrinsic versus relative cotangent choices	`agreement` in named rows and `inequivalence` in named rows.	They agree for $\mathbb{F}_3$, k^n, dual numbers over k, and the listed k-Artin examples. They differ for the two-direction ring relative to $k[u]/(u^2)$, and for $\mathbb{Z}/4\mathbb{Z}$ over $\mathbb{Z}$.	AQM-77
Reduced residue-site layer versus nilpotent-sensitive Artin layer	`inequivalence`, for the local thickened examples.	AQM-82 records one reduced residue site for dual numbers but the thickened carrier $\ell^2(\mathbb{F}_3[\epsilon]/(\epsilon^2))$ after a top-coefficient generating character. For $n = 3^s m$ with $3 \nmid m$, the $t^n = 1$ family has reduced dimension 3^m and thickened dimension 3^n.	AQM-36, AQM-37, AQM-81, AQM-82
Infinite closed-point finite supports versus quasi-local, incomplete tensor, or formal infinite-volume layers	`available`.	AQM-84 classifies finite supports, algebraic quasi-local colimits, reference-vector Hilbert representations, and formal infinite-volume profiles as separate boundary categories. Their comparison is `blocked`; formal infinite-volume operators still need a completion convention.	AQM-84
Algebraic generic-fibre layers versus completed infinite boundary layers	`available`.	The named algebraic sectors are locally anchored: convention `(z)` records the algebraic $K = \mathbb{F}_q(t)$ Weyl layer, and AQM-45 records the optional $\mathbb{Q}$-generic sector for $\operatorname{Spec}\mathbb{Z}$. Completion comparisons remain `blocked`; these are global generic-fibre layers, not closed-point finite sites or analytic/local, adelic, profinite, C^*, or infinite-product completions.	AQM-45, AQM-84, convention `(z)`

86.4 Finite-Ring Status Snapshot

Object row	Established comparison status	Blocked or unknown comparison	Anchor
$\mathbb{F}_3$	`available`: one residue qutrit site; cotangent-first labels over $\mathbb{F}_3$ have `degenerate` zero output.	Underlying three-point all-map system is a separate finite-set object, not an agreement with $\operatorname{Spec}\mathbb{F}_3$.	AQM-22, AQM-77, AQM-80, AQM-81
$\mathbb{F}_3^n$	`agreement` with tensor-site assembly after residue data are supplied; `inequivalence` with cotangent-first residue over $\mathbb{F}_3$.	No stabilizer, dynamics, or universal product-ring equivalence is asserted.	AQM-40, AQM-58, AQM-83
$\mathbb{Z}/6\mathbb{Z}$	`available`: mixed residue carrier $\ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3)$.	Common-field vector-space readings are `not_applicable`; cotangent and thickened comparisons are `unknown` or `blocked` by row; further quotient-collapse rows are blocked.	AQM-23, AQM-80, AQM-81, AQM-83
$\mathbb{F}_3[\epsilon]/(\epsilon^2)$ and $\mathbb{F}_3[t]/(t^3)$	`inequivalence`: reduced residue site versus thickened Artin carrier; `available`: first-order cotangent rows.	Database-backed residue and general Artin policies remain blocked outside the locally proved examples.	AQM-77, AQM-80, AQM-81, AQM-82
$\mathbb{F}_3[t]/(t^n-1)$	`inequivalence`: AQM-82 records reduced dimension 3^m and thickened dimension 3^n for $n = 3^s m$, $3 \nmid m$, in the locally derived family.	No populated finite-ring database or arbitrary quotient comparison is imported.	AQM-35, AQM-37, AQM-82
$\mathbb{F}_3[u,v]/(u^2,v^2)$	`available`: cotangent comparison from AQM-77.	`blocked`: standard two-direction Artin thickened Weyl row and database row.	AQM-77, AQM-80, AQM-82
$\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/8\mathbb{Z}$, $\mathbb{Z}/9\mathbb{Z}$, and even dual-number MVP rows	`available` only for the local AQM-77 $\mathbb{Z}/4\mathbb{Z}$ cotangent contrast where stated.	`blocked`: certified residue decomposition, deterministic site ordering, and generating-character/thickened rows are missing in the current MVP scope.	AQM-66, AQM-77, AQM-80, AQM-81, AQM-82
$\operatorname{Spec}\mathbb{F}_q[t]$, $\operatorname{Spec}\mathbb{F}_q[x,y]$, and $\operatorname{Spec}\mathbb{Z}$	`available`: finite closed supports, algebraic quasi-local closed-point nets, selected incomplete tensor representations where recorded, and the algebraic generic sectors for $K = \mathbb{F}_q(t)$ and $\mathbb{Q}$ where separately anchored.	`blocked`: curve-generic sectors without a recorded convention and analytic/local, adelic, profinite, C^*, and infinite-volume operator completions. AQM-84 proposal categories are gap prose, not status tokens.	AQM-29, AQM-30, AQM-45, AQM-48, AQM-84

86.5 Open Blockers Carried Forward

`aqm-qsa-src02`. Source-acquisition or source-registration blocker carried by the QSA tracker: no comparison in this shard relies on a new unregistered source.

`aqm-qsa-frres01`. General finite-ring maximal/residue decomposition and deterministic site ordering remain blocked. AQM-81 covers only sourced finite cases.

`aqm-qsa-frcot01`. Database-backed intrinsic or relative cotangent rows remain blocked. AQM-77 is a local derivation shard, not a populated finite-ring cotangent table.

`aqm-qsa-collapse01.` Quotient-collapse examples beyond the $\mathbb{Z}/6\mathbb{Z}$ anchor remain blocked until a local derivation or certificate is recorded.

`aqm-qsa-artin01.` Two-direction and arbitrary Artin/Frobenius thickened Weyl layers remain blocked outside the AQM-36/AQM-37 polynomial chain-ring examples and the AQM-77 cotangent-only derivations.

86.6 Step 22 Handoff

Step 22 should catalogue failure modes rather than promote the table above to a theorem list. The immediate degenerate or over-restrictive cases are: zero cotangent labels for fields and product fields despite nontrivial residue-site carriers; reduced residue layers forgetting nilpotents; rational point truncations omitting higher-degree closed points; regular-function profiles becoming formal infinite-volume labels; and common-field readings failing for mixed residues such as $\mathbb{Z}/6\mathbb{Z}$.

Any new row must still satisfy the AQM-65 evidence rule: name the object, the workflows, the choices, the outputs compared, and the local map, invariant, or obstruction. Otherwise its status remains `blocked` or `unknown`.

87 Degenerate and Over-Restrictive Cases

87.1 Status and Local Anchors

Status: New failure-mode catalogue. This is QSA Step 22 from `docs/quantum_system_associations/PLAN.md`. It adds no source, convention entry, script, run bundle, generated data, populated database row, Beads update, or report-level classification theorem. It records concrete failure modes already visible in AQM-65–AQM-85.

The vocabulary and evidence rule are AQM-65 and `CONVENTIONS.md` convention `(ap)`. The status tokens are exactly those of convention `(au)`. The main anchors are AQM-67–AQM-71 for finite-set and radical examples, AQM-74 for single-particle comparison boundaries, AQM-77 for cotangent zero and cotangent-choice examples, AQM-80–AQM-83 for finite-ring, quotient, nilpotent, and product-ring rows, AQM-84 for infinite boundary statuses, and AQM-85 for the gap-first agreement ledger.

87.2 Reading Rule

The word `degenerate` is not a dismissal. It marks a concrete row in which a workflow collapses, forgets structure, becomes too small for an intended comparison, or reduces to an earlier row. Such rows stay in the catalogue because they are useful negative controls.

The token `blocked` is different. It means that required evidence, conventions, or tooling are missing. A blocked row is not a mathematical obstruction and is not evidence that the workflow fails. Likewise, `unknown` means not locally determined, and `not_applicable` means that the workflow is outside the row's stated semantics.

87.3 Too Little Structure

Bare finite set without a Gram choice. For a two-point set $X = \{a, b\}$, AQM-67 and convention `(aq)` produce only

$$\mathbb{C}\langle X\rangle = \mathbb{C}e_a \oplus \mathbb{C}e_b$$

with formal basis labels. No inner product, adjoint, observable algebra, Hamiltonian, tensor factorization, or Weyl label space is returned. The identity Gram matrix would be extra data. Therefore any Hilbert-space or unitary comparison from the bare set alone is `blocked` by missing Gram data, not proved false.

Empty input versus zero output. AQM-67 gives

$$\mathbb{C}\langle\emptyset\rangle = 0.$$

Here $\emptyset$ is the `empty` input and 0 is the `zero` carrier output. AQM-69 gives the same zero carrier for the empty direct sum, whereas AQM-68 gives the empty tensor product $\mathbb{C}$. AQM-74 uses this as a concrete obstruction to silently identifying tensor-site coexistence with direct-sum alternatives.

87.4 Radicals and Quotient Label Spaces

Affine maps on $\mathbb{F}_3$. AQM-71 reviews the checked row displayed below.

```
vector_space_F3_affine
```

The row is in `runs/2026-05-24-arithmetic-quantum-fields/data/arithmetic_quantum_field_examples.csv`. The scalar space has basis $1, t$, the field phase dimension is 4, the symplectic radical has dimension 2, and the reduced phase dimension is 2. The honest Weyl label space is therefore the quotient $E/\operatorname{rad}(E)$, not the unreduced affine-function label space. Forgetting this quotient would overcount labels.

Linear coordinate functions on $\mathbb{F}_3^2$. The same AQM-71 review lists the checked row displayed below.

`affine_plane_F3_linear`

It has field phase dimension 4, symplectic rank 0, radical dimension 4, and reduced phase dimension 0. This is a concrete `degenerate` radical case: the selected linear coordinate-function space is totally radical for the all-weights-one pairing, so the reduced Weyl label group is trivial and the Hilbert carrier is one-dimensional. The row is not removed from the catalogue; it records that the chosen field space and pairing were too restrictive for nontrivial Weyl labels.

Projective-line sheaf sections. AQM-16 gives a second checked radical warning. For $\mathbb{P}^1_{\mathbb{F}_3}$ with $\mathcal{O}(1)$, the rational-point evaluation row has scalar pairing rank 0, symplectic rank 0, radical dimension 4, and reduced phase dimension 0 in `runs/2026-05-24-projective-line-sheaf-fields/data/projective_line_sheaf_field_summary.csv`. For $\mathcal{O}(4)$, the evaluation kernel and radical are nonzero but the reduced phase dimension is still 8. These examples show why the radical quotient is part of the workflow, not cleanup after the fact.

87.5 Zero Cotangent Labels

The field $\mathbb{F}_3$. AQM-77 records that for $X = \operatorname{Spec}\mathbb{F}_3$ over $\mathbb{F}_3$, both the intrinsic cotangent and the relative cotangent are zero:

$$C_x^{\mathrm{loc}} = C_x^{\mathrm{rel}} = 0.$$

The cotangent-first phase label is therefore zero and the Weyl carrier is one-dimensional. This is a `zero` output for that workflow. It is not the residue-site workflow, where AQM-81 has one qutrit site.

Product fields. For $R = k^n$ with $k = \mathbb{F}_3$, AQM-83 cites AQM-58 and AQM-77:

$$\Omega^1_{k^n/k} = 0$$

and the local cotangent labels vanish at every product point. The same ring has n residue-field Weyl factors in AQM-40 and AQM-83. Thus the cotangent-first row is `degenerate`/`zero` for the selected cotangent labels, while the residue-site row remains nontrivial. The catalogue must keep both rows.

87.6 Quotients and Collapsed Presentations

$\mathbb{Z}/6\mathbb{Z}$ versus the product presentation. AQM-23 computes the two residue fields $\mathbb{F}_2$ and $\mathbb{F}_3$ for $\operatorname{Spec}(\mathbb{Z}/6\mathbb{Z})$, and AQM-81/AQM-83 use the mixed-residue carrier

$$\ell^2(\mathbb{F}_2) \otimes \ell^2(\mathbb{F}_3).$$

AQM-80 records that the finite-ring review export treats the $\mathbb{F}_2 \times \mathbb{F}_3$ presentation as merged to the $\mathbb{Z}/6\mathbb{Z}$ representative. This is a concrete `degenerate` catalogue warning: the two presentations should not be counted as independent examples in this local evidence layer. It is not a general quotient-collapse theorem.

The $\mathbb{F}_2[x]/(x^2+x)$ candidate. AQM-80 lists $\mathbb{F}_2[x]/(x^2+x)$ only as a possible quotient-collapse row. It remains `blocked` by `aqm-qsa-collapse01`; the current report has no local derivation or certificate that may be promoted to a checked collapse row. This is a concrete unsupported finite-ring example, and its blocked status is part of the catalogue.

87.7 Nilpotents Forgotten by Reduced Layers

Dual numbers. For

$$A = \mathbb{F}_3[\epsilon]/(\epsilon^2),$$

AQM-82 records one reduced residue site with carrier $\ell^2(\mathbb{F}_3)$. The nilpotent class ϵ reduces to zero at that site. AQM-36 and AQM-82 also record the thickened Artin carrier

$$\ell^2(\mathbb{F}_3[\epsilon]/(\epsilon^2)), \qquad \dim_{\mathbb{C}} = 9,$$

after the top-coefficient generating character is fixed. The reduced row is `degenerate` if the intended question is nilpotent-sensitive; the thickened row is a separate available local Artin workflow.

The $t^n = 1$ family. AQM-82 reviews AQM-35/AQM-37 for

$$R_n = \mathbb{F}_3[t]/(t^n - 1), \qquad n = 3^s m, \quad 3 \nmid m.$$

The reduced residue layer has Hilbert dimension 3^m, while the locally proved thickened layer has dimension 3^n. The repeated factors are not extra closed points; they are thickened degrees at the existing reduced points. Removing this row would hide exactly the nilpotent-sensitive failure mode Step 22 is meant to catalogue.

87.8 Mixed Residues and Not-Applicable Readings

The $\mathbb{Z}/6\mathbb{Z}$ row is also a concrete `not_applicable` warning. AQM-23, AQM-80, AQM-81, and AQM-83 use the mixed label group

$$\mathbb{F}_2^2 \oplus \mathbb{F}_3^2.$$

A single $\mathbb{F}_3$-linear field-label reading is outside the row's semantics. This is not a blocked theorem about impossibility; it is the wrong workflow specification for the selected mixed-residue object.

Similarly, AQM-82 marks the nilpotent-sensitive workflow as `not_applicable` for the field row $\mathbb{F}_3$: there is no nilpotent thickening in that object. The status is different from the blocked thickened rows for unsupported finite rings.

87.9 Unsupported Infinite-Representation Cases

AQM-84 gives the concrete infinite boundary rows $\operatorname{Spec} \mathbb{F}_q[t]$, $\operatorname{Spec} \mathbb{F}_q[x, y]$, and $\operatorname{Spec} \mathbb{Z}$. Finite closed supports and algebraic quasi-local closed-point algebras are available there. The following are not available finite-site conclusions:

- generic and curve-generic residue fields as finite Weyl sites;
- local-field, adelic, profinite, analytic, or C^*-completed representations without the missing topology, character, norm, and representation conventions;

- formal infinite-volume regular-function profiles as actual quasi-local Weyl operators.

These are `blocked`, generic-sector proposal, formal infinite-volume, or completion-proposal statuses in the AQM-84 vocabulary. They are not mathematical no-go results.

87.10 Status-Token Checklist

The concrete distinctions to preserve in later shards are:

`empty`. The indexing input or support is empty, as in $X = \emptyset$ for AQM-67 or the empty finite support in a local net.

`zero`. The mathematical output is the zero carrier or zero label object, as in $\mathbb{C}\langle\emptyset\rangle = 0$ or the zero cotangent labels of $\operatorname{Spec}\mathbb{F}_3$.

`degenerate`. A concrete row collapses or forgets structure: total radical reduction, product-field zero cotangent labels, reduced residue layers forgetting nilpotents, or merged $\mathbb{Z}/6\mathbb{Z}$ product presentations.

`blocked`. Evidence, conventions, or tooling are missing, as for $\mathbb{F}_2[x]/(x^2 + x)$ quotient collapse, arbitrary finite-ring residue decomposition, database-backed cotangent rows, and infinite completions.

`unknown`. No local determination has been recorded, as in AQM-80's missing cotangent comparison row for $\mathbb{Z}/6\mathbb{Z}$.

`not_applicable`. The workflow is outside the row semantics, as for a common $\mathbb{F}_3$-linear reading of the mixed-residue $\mathbb{Z}/6\mathbb{Z}$ row or a nilpotent-sensitive thickening of the field $\mathbb{F}_3$.

87.11 Blockers Carried Forward

The open blockers from AQM-85 remain unchanged:

`aqm-qsa-src02`, `aqm-qsa-frres01`, `aqm-qsa-frcot01`,
`aqm-qsa-collapse01`, `aqm-qsa-artin01`.

This shard does not close or reinterpret any of them. Step 23 should turn these preserved failure modes into explicit open problems and next catalogue targets, not into unsupported equivalence or no-go theorems.

88 Open Problems and Next Catalogue Targets

88.1 Status and Local Anchors

Status: New gap/synthesis shard. This is QSA Step 23 of the plan in `docs/quantum_system_associations/`. It adds no source, convention entry, script, run bundle, generated data, Beads update, database row, representation classification, dynamics principle, or theorem. It closes the first QSA meta-plan by recording the next work that would be needed before further catalogue rows become evidence.

The governing vocabulary is AQM-65 and `CONVENTIONS.md` convention `(ap)`. The table tokens are those of convention `(au)`. The local QSA anchors are AQM-72 and AQM-73 for parafermion and fusion-category gaps, AQM-78 for finite Weyl-Heisenberg representation boundaries, AQM-80 for finite-ring database status, AQM-84 for infinite-ring and completion boundaries, AQM-85 for the agreement/inequivalence ledger, and AQM-86 for degenerate and over-restrictive cases.

The index-level anchors are the source-topic statuses in `INDEX.md` and the finite-ring database run from 2026-05-26. Those anchors say what exists locally; they are not imported as new mathematical sources in this shard.

88.2 Reading Rule

Every item below is an open question or a proposal. A proposed target is not a claim that the target is correct, natural, or unique. It is a record of missing evidence needed before a later shard may add a result row.

The word `blocked` keeps the AQM-85–AQM-86 meaning: a source, convention, computation, helper, audit, or run is missing. It is not a mathematical obstruction. Conversely, an `available` local anchor is not a final equivalence statement unless AQM-85's evidence rule supplies a specific object, workflows, choices, and a local map, invariant, or obstruction.

88.3 Prioritized Dependency Ledger

Pri.	Bucket	Open question or proposal	Next target must supply	Anchors
1	Source and convention acquisition	Question: which parafermion or fusion-category association should be the first technical extension beyond CCR, CAR, Grassmann-Weyl, and finite AND/OR grammar?	Acquire and register the missing parafermion or fusion-category source; record conventions for mode algebra or categorical data, including hom spaces, simple labels, tensor rules, associator data, pivotal/spherical data, and any lattice input before using them.	AQM-72, AQM-73
2	Finite-ring database, residue, and cotangent audits	Question: which standard finite-ring rows can become database-backed rather than local exceptions or review-export notes?	Produce a populated and audited finite-ring run; record deterministic residue-site ordering and database-backed intrinsic/relative cotangent computations before extending the AQM-80 matrix.	AQM-80, AQM-81
3	Quotient collapse and two-direction Artin rows	Question: which quotient presentations collapse to earlier examples, and which two-direction Artin rings support a checked thickened Weyl layer?	For quotient collapse, add a local derivation or certificate beyond the $\mathbb{Z}/6\mathbb{Z}$ anchor. For two-direction Artin examples, add the missing source or exact derivation, generating-character convention, and checked computation.	AQM-80, AQM-82, AQM-86
4	Finite Weyl and projective representation boundaries	Question: for which finite rings or finite abelian-group pairings can the Weyl-Heisenberg/projective-representation step be classified beyond the locally sourced finite-field, finite Frobenius-ring Pauli, and Solomon-style generalized-Heisenberg slots?	Name the finite label data, central character, commutator pairing, and source or derivation. If the claim is computational, add a run checking the projective product law, commutator law, irreducibility or decomposition invariant, and operator-basis count.	AQM-31, AQM-78
5	Single-particle, Fock, and statistics comparisons	Question: when do the AQM-74 one-particle maps extend to symmetric, antisymmetric, para, or full Fock sectors, and when do they fail?	Record the statistics convention, particle-number grading, completion or finite-sector cutoff, and an explicit comparison map or obstruction. For para-statistics, first acquire a source and mode-algebra convention.	AQM-70, AQM-72, AQM-74
6	Infinite, generic, analytic, and completion boundaries	Question: which topology or completion should turn finite-support closed point nets, generic sectors, or formal infinite-volume profiles into analytic quantum systems?	Choose and source the topology, measure or character, Pontryagin-dual data, norm or C^*-completion, adelic or profinite restricted product, and representation theorem before claiming a Hilbert-space model.	AQM-84
7	Dynamics selection	Question: what principle, if any, should select Hamiltonians, time evolutions, actions, or Witten-index-style data after a kinematical association has been fixed?	Add a dynamics convention naming the sign, domain, adjoint, grading, boundary/support condition, and source or checked invariant. Until then, Hamiltonians and time evolution remain outside QSA kinematics.	AQM-65, AQM-78
8	Noncommutative or nonunital extension boundary	Question: what should replace $\operatorname{Spec} R$, residue fields, cotangent data, and tensor-site assembly if the input ring is noncommutative or lacks a unit?	Acquire sources and record conventions for the chosen replacement of points and local data before adding examples. Possible prompts such as unitization, idempotent/corner data, or module-category input remain only prompts until sourced.	AQM-75–AQM-83

88.4 Blockers Carried Forward

The blockers below are carried forward without reinterpretation.

`aqm-qsa-src02`, `aqm-qsa-frres01`, `aqm-qsa-frcot01`,
`aqm-qsa-collapse01`, `aqm-qsa-artin01`.

`aqm-qsa-src02`. Question: which missing sources must be acquired before parafermions, fusion categories, Levin-Wen/string-net data, or related categorical association rows become technical report content? The next target must name the source topic, local manifest path, retrieval route, source locators, and the conventions that the source fixes.

`aqm-qsa-frres01`. Question: can the finite-ring database supply certified maximal/residue decompositions and deterministic site orderings for the standard finite-ring rows? The next target must supply a populated run, audit result, and report or schema hooks before using those rows as database-backed evidence.

`aqm-qsa-frcot01`. Question: can intrinsic and relative cotangent rows be computed and audited from the finite-ring database rather than by local hand derivations? The next target must name the base ring convention, cotangent algorithm, verification invariant, and run artifact.

`aqm-qsa-collapse01`. Question: which quotient presentations are checked collapses rather than new examples? The next target must provide a local derivation or certificate for each claimed collapse beyond the $\mathbb{Z}/6\mathbb{Z}$ anchor.

`aqm-qsa-artin01`. Question: which two-direction or arbitrary Artin/Frobenius examples have the generating-character and thickened Weyl data needed for a nilpotent-sensitive row? The next target must supply the source or exact derivation, the character convention, and a checked example.

88.5 Catalogue Hygiene Rules for the Next Round

- Proposal: every new row should begin by naming the object, the selected workflow, the extra choices, and the intended output before any comparison word is used.
- Proposal: every comparison target should declare whether it needs a missing source, convention, computation, or run; otherwise its status should remain `blocked` or `unknown`.
- Proposal: every finite-ring extension should state whether it is a residue-site row, cotangent-first row, reduced field-label row, nilpotent-sensitive row, quotient-collapse row, or database-audit row.
- Proposal: every infinite or analytic extension should state whether it is finite-support, algebraic quasi-local, incomplete tensor, generic-sector, formal infinite-volume, or completion status in the AQM-84 vocabulary.
- Proposal: dynamics should stay out of the QSA catalogue unless a later shard explicitly chooses and sources a dynamics-selection principle.

88.6 Close of the First QSA Pass

The first QSA pass has produced a controlled vocabulary, a standard test object list, finite-set workflows, finite symplectic field-label workflows, boson/fermion review rows, geometric point and cotangent workflows, finite-ring residue/nilpotent/product rows, infinite boundary statuses, an agreement/inequivalence ledger, and a failure-mode catalogue. This shard does not promote that material to a complete classification. It records the evidence still missing before the next catalogue pass can responsibly expand the association zoo.

89 Derivative Constraints on Residue Weyl-Heisenberg Fields

89.1 Status and Source Anchors

This shard replaces the rejected outside-in attempt. Its output is a constraint rule: derivative predicates on named regular coordinate families select and reject Weyl-Heisenberg elements.

The residue-site Weyl anchors are AQM-23, AQM-28, AQM-49, AQM-78, and AQM-81. Finite Weyl laws are sourced through `references/symplectic_qecc/SOURCES.md`: Ashikhmin–Knill source lines 249–313 and Bombin–Martin–Delgado source lines 1206–1259; finite Heisenberg sources are catalogued in `references/heisenberg_weil/SOURCES.md`. The algebraic source is the Stacks algebra snapshot registered in `references/algebraic_geometry/SOURCES.md`. Use locators 250–278, 2972–3023, 33793–33872, 34103–34120, and 34310–34340. The convention is `CONVENTIONS.md (aw)`.

89.2 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: B \to A \text{ finite commutative algebra; } S \subset \operatorname{Spec} A \text{ finite selected finite-residue support.}\\
D2 &: E_S = \textstyle\bigoplus_{x\in S} \kappa(x)^2, \quad \pi : \mathrm{WH}(E_S) \to E_S.\\
D3 &: \rho_S : \mathrm{WH}(E_S) \to PU(\mathcal{C}_S) \text{ is the residue Weyl field.}\\
D4 &: F \le A = \Gamma(\operatorname{Spec} A, \mathcal{O}) \text{ is a named regular coordinate family.}\\
D5 &: \lambda_{S,F}(q,p) = ((q \bmod x, p \bmod x))_{x\in S}.\\
D6 &: d : A \to \Omega^1_{A/B} \text{ and } P \subset \Omega^1_{A/B} \times \Omega^1_{A/B}.\\
D7 &: C_P(S,F) = \lambda_{S,F}\{(q,p) \in F^2 : (dq,dp) \in P\}.\\
L1 &: C_P(S,F) \text{ gives selected/rejected elements of } \mathrm{WH}(E_S).\\
L2 &: F, P \text{ additive } \Rightarrow C_P(S,F) \le E_S.\\
L3 &: \text{Representative-free derivative claims require fibre saturation.}\\
T1 &: \mathbb{Z}/6\mathbb{Z} \text{ has qubit-factor } \otimes \text{ qutrit-factor Weyl kinematics and no } d\text{-rejection.}\\
T2 &: \mathbb{F}_3[t]/(t^2(t-1)) \text{ has genuine shallow-depth } d\text{-rejections.}
\end{aligned}$$

89.3 Definitions D1–D7

For $x \in S$, put $K_x = \kappa(x)$ and

$$E_x = K_{xq} \oplus K_{xp}, \qquad \omega_x((q,p),(q',p')) = pq' - p'q.$$

The residue phase group is $E_S = \bigoplus_{x\in S} E_x$. Its Weyl-Heisenberg group is a central extension

$$1 \to Z \to \mathrm{WH}(E_S) \xrightarrow{\pi} E_S \to 0$$

with the product of the local finite-field commutator characters. The arithmetic quantum field at this layer is

$$\rho_S : \mathrm{WH}(E_S) \to PU(\mathcal{C}_S).$$

Write $W(e)$ for the projective Weyl operator labelled by $e \in E_S$.

A map family here is not an arbitrary set-map family. It is a named finite additive subgroup, or named finite B-submodule when available,

$$F \le A = \Gamma(\operatorname{Spec} A, \mathcal{O}).$$

Coordinate-pair maps are elements $(q,p) \in F^2$. Their residue phase labels are

$$\lambda_{S,F}(q,p) = ((q \bmod x, p \bmod x))_{x\in S} \in E_S.$$

Let $d : A \to \Omega^1_{A/B}$ be the universal derivation. A derivative predicate P is a displayed condition on (dq, dp). The basic predicate is

$$P_{\text{flat}} = \{(0, 0)\},$$

and, after a scalar/action μ is named,

$$P_\mu = \{(\xi, \eta) : \eta = \mu\xi\}.$$

The selected residue labels are

$$C_P(S, F) = \lambda_{S,F}\{(q, p) \in F^2 : (dq, dp) \in P\} \subseteq E_S.$$

Facts $L1$–$L3$. The selected and rejected Weyl-Heisenberg elements are

$$\mathrm{Sel}_P(S, F) = \pi^{-1}(C_P(S, F)), \qquad \mathrm{Rej}_P(S, F) = \pi^{-1}(E_S \setminus C_P(S, F)).$$

If F and P are additive, then $C_P(S, F)$ is an additive subgroup of E_S. A derivative predicate gives a representative-free yes/no condition on labels in $\lambda_{S,F}(F^2)$ exactly when it is constant on every fibre of $\lambda_{S,F}$.

Proof. The map π forgets only the central coordinate, so a Weyl-Heisenberg element is selected exactly when its phase label lies in $C_P(S, F)$. If F and P are additive, then $\{(q, p) \in F^2 : (dq, dp) \in P\}$ is a subgroup of F^2, and $\lambda_{S,F}$ is a homomorphism, so its image is a subgroup. Finally, two coordinate pairs with the same $\lambda_{S,F}$-value label the same projective Weyl operator; the derivative condition is label-intrinsic exactly when all representatives in that fibre give the same answer.

89.4 Breadth-First Test Discipline

A breadth-first test is a finite queue of named families

$$F_0 \subseteq F_1 \subseteq \cdots \subseteq A$$

and named predicates P. At each node one records

$$\lambda_{S,F_i}(F_i^2) \subseteq E_S, \qquad C_P(S, F_i) \subseteq E_S,$$

then translates $C_P(S, F_i)$ to selected/rejected Weyl-Heisenberg elements by Facts $L1$–$L3$. The derivative has an *internal bite* when

$$C_P(S, F_i) \subsetneq \lambda_{S,F_i}(F_i^2),$$

and an *ambient bite* when $C_P(S, F_i) \subsetneq E_S$. These are constraints on Weyl-Heisenberg elements, not informal diagnostics.

89.5 Example 1: $\mathbb{Z}/6\mathbb{Z}$ T1

Let $A = \mathbb{Z}/6\mathbb{Z}$, $B = \mathbb{Z}$, and

$$S = \{(2), (3)\}.$$

AQM-23 proves

$$\kappa(2) = \mathbb{F}_2, \qquad \kappa(3) = \mathbb{F}_3,$$

and the Chinese remainder map gives

$$A \cong \mathbb{F}_2 \times \mathbb{F}_3, \qquad a \mapsto (a \bmod 2, a \bmod 3).$$

Hence

$$E_S = \mathbb{F}_2^2 \oplus \mathbb{F}_3^2.$$

The residue Weyl field is the tensor product of a qubit-factor projective representation and a qutrit-factor projective representation, because the phase group is the orthogonal direct sum of the two local phase groups.

For the derivative, the quotient exact sequence over $B = \mathbb{Z}$ gives

$$\Omega^1_{A/\mathbb{Z}} = 0.$$

Thus $dq = dp = 0$ for every $q, p \in A$. With $F = A$, $\lambda_{S,A}(A^2) = E_S$, and therefore

$$C_{P_{\text{flat}}}(S, A) = E_S, \qquad \mathrm{Sel}_{P_{\text{flat}}}(S, A) = \mathrm{WH}(E_S), \qquad \mathrm{Rej}_{P_{\text{flat}}}(S, A) = \emptyset.$$

So d rejects no residue Weyl element in this example. The nontrivial content is the mixed-residue Weyl kinematics.

89.6 Example 2: Products and Dual Numbers

For $A = \mathbb{F}_3 \times \mathbb{F}_3$ over $B = \mathbb{F}_3$, write $e_0 = (1, 0)$, $e_1 = (0, 1)$. If $e^2 = e$, then

$$(2e - 1)de = 0, \qquad (2e - 1)^2 = 1,$$

so $de = 0$. Therefore every $ae_0 + be_1$ has zero derivative, and the full family $F = A$ gives $C_{P_{\text{flat}}}(S, A) = E_S$. Product idempotents separate sites without causing derivative rejection.

For

$$A = k[\epsilon]/(\epsilon^2), \qquad k = \mathbb{F}_3, \qquad S = \{(\epsilon)\},$$

AQM-54 computes

$$\Omega^1_{A/k} \cong k\, d\epsilon, \qquad d(a + b\epsilon) = b\, d\epsilon.$$

The residue map is $a + b\epsilon \mapsto a$, so $E_S = k^2$. With $F = A$, the flat predicate requires the ϵ-coefficients of q and p to vanish, but every residue pair is represented by constants. Hence

$$C_{P_{\text{flat}}}(S, A) = E_S.$$

The derivative sees the nilpotent coefficient, but the reduced residue Weyl-Heisenberg group has no corresponding rejection. If $F = k\epsilon$, then $\lambda_{S,F}(F^2) = \{0\}$, so only $W(0)$ is selected; this is a map-family support restriction.

89.7 Example 3: A Two-Site Nonreduced Breadth-First Test T2

Let

$$A = k[t]/(t^2(t - 1)), \qquad k = \mathbb{F}_3, \qquad S = \{x_0 = (t),\ x_1 = (t - 1)\}.$$

Both residues are k, so $E_S = k^2 \oplus k^2$. Write elements as

$$u = a + bt + ct^2.$$

The residues are

$$r_0(u) = a, \qquad r_1(u) = a + b + c.$$

For $f = t^2(t-1) = t^3 - t^2$, one has $f' = 3t^2 - 2t = t$ in k. Hence

$$\Omega^1_{A/k} \cong A\,dt/(t\,dt), \qquad d(a + bt + ct^2) = b\,dt.$$

First take

$$F_1 = \langle 1, t\rangle_k.$$

The scalar residue map is $a + bt \mapsto (a, a+b)$, so F_1 represents every scalar residue pair and $\lambda_{S,F_1}(F_1^2) = E_S$. For

$$e = ((q_0, p_0), (q_1, p_1)),$$

the flat predicate imposes $q_1 - q_0 = 0$ and $p_1 - p_0 = 0$. Thus

$$C_{P_{\text{flat}}}(S, F_1) = \{((q, p), (q, p)) : q, p \in k\}.$$

The selected Weyl-Heisenberg elements are the central preimage of this diagonal subgroup. The rejected elements are exactly those with

$$q_0 \neq q_1 \quad \text{or} \quad p_0 \neq p_1.$$

Only 3^2 of the 3^4 residue phase labels are selected.

For $P_\mu = \{dp = \mu\,dq\}$, $\mu \in k$, the selected labels are

$$C_{P_\mu}(S, F_1) = \{e \in E_S : p_1 - p_0 = \mu(q_1 - q_0)\}.$$

Thus 3^3 labels are selected and $3^4 - 3^3$ are rejected. This is a derivative constraint stated directly on Weyl-Heisenberg labels.

Now take the next family

$$F_2 = \langle 1, t, t^2\rangle_k = A.$$

The derivative still records only b, but the residue difference is

$$r_1(u) - r_0(u) = b + c.$$

For P_{flat}, set $b_q = b_p = 0$ and choose c_q, c_p to match any desired residue differences. Therefore

$$C_{P_{\text{flat}}}(S, F_2) = E_S.$$

For P_μ, choose b_q, set $b_p = \mu b_q$, and use c_q, c_p to match the two desired residue differences. Hence $C_{P_\mu}(S, F_2) = E_S$. The derivative bite at F_1 disappears at F_2.

89.8 Conclusion

The corrected rule is

$$(A, B, S, F, P) \rightsquigarrow C_P(S, F) \subseteq E_S \rightsquigarrow \pi^{-1}(C_P(S, F)) \subseteq \mathrm{WH}(E_S).$$

All derivative conclusions in this residue-site layer must end as selected or rejected Weyl-Heisenberg elements. The examples show no d-rejection for $\mathbb{Z}/6\mathbb{Z}$, no reduced residue rejection for dual numbers, and real shallow-depth bites for $\mathbb{F}_3[t]/(t^2(t-1))$ that disappear when the map family is enlarged.

90 Scheme-Incidence Sources for Code Complexes

90.1 Status and Source Anchors

This shard answers whether rings or schemes can give the complexes proposed after AQM-88. The answer is conditional: a bare finite ring gives finite residue Weyl sites, while a code complex needs extra selected functions, incidence relations, or group-law data whose boundary identity selects a commuting Weyl-Heisenberg subgroup. The CSS bridge is AQM-08. It proves that a finite chain complex

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0, \qquad \partial_1\partial_2 = 0,$$

selects the Weyl-label subgroup

$$L = (\operatorname{im}\delta^0 \oplus 0) + (0 \oplus \operatorname{im}\partial_2) \subset C^1 \oplus C_1.$$

AQM-10 fixes the Steane binary presentation; AQM-05 and AQM-06 validate the periodic square toric complex for $k = 4$; AQM-40 proves $\operatorname{Spec}(k^n)$; AQM-88 requires final selected/rejected Weyl-Heisenberg elements. The convention is `CONVENTIONS.md (ax)`. The only group-scheme material used here is the explicit constant finite case $G_k = \operatorname{Spec}(k^G)$. Stacks supplies affine-scheme anti-equivalence in `references/algebraic_geometry/stacks_project_schemes.tex`, lines 841–963, and finite products/fibre products in lines 986–1004. No general nonconstant group-scheme theory is imported.

90.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & \operatorname{Spec} R \text{ gives residue sites, not a complex.} \\
D2 & : & X_i = \operatorname{Spec}(k^{S_i}),\ I_{10}, I_{21} \text{ define } \partial_1, \partial_2. \\
L1 & : & \partial_1\partial_2 = 0 \Rightarrow L = (\operatorname{im}\delta^0 \oplus 0) + (0 \oplus \operatorname{im}\partial_2) \text{ is isotropic.} \\
T1 & : & \text{Steane, Reed-Muller, Shor, toric are separated below by evidence status.}
\end{array}$$

90.3 Why a Bare Finite Ring Is Not Enough

Finite rings have finitely many ideals, hence satisfy DCC and are Artinian. Stacks `references/algebraic_geometry/stacks_project_algebra.tex`, lines 12727–12805 and 14364–14444, records finite maximal ideals, maximal primes, and finite discrete spectrum after local decomposition. AQM-40 gives the concrete reduced case

$$\operatorname{Spec}(k^n) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}, \qquad \kappa(\mathfrak{m}_i) = k.$$

This gives $E = \bigoplus_i k^2$, but it does not supply edges, faces, or a boundary map. Stabilizer constraints are extra isotropic label choices. Thus the correct outside-in target is not $\operatorname{Spec} R \rightsquigarrow C_2 \to C_1 \to C_0$ without further data, but

$$\text{finite schemes plus named incidence/group structure} \rightsquigarrow C_2 \to C_1 \to C_0 \rightsquigarrow \pi^{-1}(L) \subset \mathrm{WH}(E).$$

90.4 Steane: Punctured Boolean Three-Space

Let $X_7 = \mathbb{F}_2^3 \setminus \{0\}$, ordered as

$$111,\ 110,\ 101,\ 100,\ 011,\ 010,\ 001.$$

The finite scheme is $X_{\mathrm{St}} = \mathrm{Spec}(\mathbb{F}_2^{X_7})$, equivalently Boolean affine three-space with the origin idempotent removed:

$$\mathbb{F}_2[x_1, x_2, x_3]/(x_i^2 + x_i,\ (1 + x_1)(1 + x_2)(1 + x_3)).$$

Evaluate the three coordinate functions on the displayed order. The rows are

$$x_1 = 1111000,$$
$$x_2 = 1100110,$$
$$x_3 = 1010101.$$

These are exactly the Steane rows g_1, g_2, g_3 in AQM-10. Put $D = \langle x_1, x_2, x_3\rangle \subset \mathbb{F}_2^{X_7}$. Each row has weight 4, and any two distinct rows overlap in 2 sites, so

$$D \subset D^{\perp}.$$

Therefore the selected Weyl-label subgroup

$$L_{\mathrm{St}} = (D \oplus 0) + (0 \oplus D)$$

is isotropic. AQM-10 then identifies $L_{\mathrm{St}}^{\perp}/L_{\mathrm{St}}$ with one binary symplectic logical pair.

The nontrivial selection principle is the choice of homogeneous degree-one functions $\langle x_1, x_2, x_3\rangle$, not all affine functions. Including the constant function would change the row space. Hence the ring/scheme source is promising but not yet a dynamical derivation of the selection rule.

90.5 Reed-Muller: Boolean Degree Filtration

Let

$$B_m = \mathbb{F}_2[x_1, \ldots, x_m]/(x_i^2 + x_i), \qquad X_m = \mathrm{Spec}\, B_m.$$

Then X_m is the finite Boolean m-cube and B_m is the full function ring on it. For $S \subset \{1, \ldots, m\}$, write

$$x_S = \prod_{i \in S} x_i.$$

The squarefree monomials form the function basis. Define the degree-filtered subspace

$$RM(r, m) = \langle x_S :\ |S| \le r\rangle \subset \mathbb{F}_2^{\mathbb{F}_2^m}.$$

This shard uses this displayed definition as the local Reed-Muller convention.

Use the pointwise binary pairing

$$\langle f, g\rangle = \sum_{a \in \mathbb{F}_2^m} f(a)g(a).$$

For monomials,

$$\langle x_S, x_T\rangle = \sum_a x_{S\cup T}(a) = \begin{cases} 1, & S \cup T = \{1, \ldots, m\}, \\ 0, & S \cup T \ne \{1, \ldots, m\}. \end{cases}$$

Indeed, $x_{S\cup T}$ is 1 on $2^{m-|S\cup T|}$ points, and this number is odd exactly when $|S \cup T| = m$.

Hence $RM(r, m)$ is orthogonal to $RM(s, m)$ whenever $r + s < m$. For $0 \le r < m$,

$$RM(r, m)^{\perp} = RM(m - r - 1, m),$$

because the inclusion follows from the monomial calculation and the dimensions sum to 2^m.

Therefore every pair $r_X + r_Z < m$ gives a checked CSS Weyl-label selection

$$L_{r_X,r_Z} = (RM(r_X, m) \oplus 0) + (0 \oplus RM(r_Z, m)).$$

The selected Weyl-Heisenberg elements are $\pi^{-1}(L_{r_X,r_Z})$. Any stronger Reed-Muller-code claim, for example a particular distance or transversal-gate property, still needs its own local source or run.

90.6 Shor: A Sourced Incidence Target, Not Yet a Natural Scheme

Gottesman's local source lists Shor's nine-qubit stabilizer table in `references/symplectic_qecc/gottesman_9705052_source/Thesis.tex`, lines 920–935. In binary CSS row notation, take $S_1 = \{1, \ldots, 9\}$ and

$$H_X = \begin{pmatrix} 111111000 \\ 111000111 \end{pmatrix},$$

$$H_Z = \begin{pmatrix} 110000000 \\ 101000000 \\ 000110000 \\ 000101000 \\ 000000110 \\ 000000101 \end{pmatrix}.$$

Each Z-row is supported inside one block of three. Each X-row is constant on any block it touches. Thus every mixed dot product is either 0 or 2, hence zero in $\mathbb{F}_2$:

$$H_X H_Z^T = 0.$$

This gives a finite incidence chain complex by setting

$$C_0 = \mathbb{F}_2^2, \quad C_1 = \mathbb{F}_2^9, \quad C_2 = \mathbb{F}_2^6, \quad \partial_1 = H_X, \quad \partial_2 = H_Z^T.$$

The selected Weyl labels are

$$L_{\text{Shor}} = (\operatorname{row} H_X \oplus 0) + (0 \oplus \operatorname{row} H_Z).$$

The rows have ranks 2 and 6, so L_{Shor} has rank 8 inside $\mathbb{F}_2^9 \oplus \mathbb{F}_2^9$.

This is a valid scheme-incidence object in convention `(ax)`: take $\operatorname{Spec}(\mathbb{F}_2^{S_i})$ for the check sets and encode the ones of H_X, H_Z as incidence relations. The open question is naturality. Bombin–Martin–Delgado support the homological-family statement locally in `references/toric_code/bombin_martin_delgado_0605094_source/HomologicalErrorCorrection6.tex`, lines 320–321 and 2488–2493; this shard still does not reconstruct their natural surface 2-complex.

90.7 Toric: Constant Group Scheme and Cayley Squares

Let $G = C_k \times C_k$ be the finite abelian group written additively, with generators $a = (1, 0)$ and $b = (0, 1)$. Over $\mathbb{F}_2$, the constant finite group scheme used here is $G_{\mathbb{F}_2} = \operatorname{Spec}(\mathbb{F}_2^G)$. As a bare scheme this is only the product-field spectrum with k^2 points. The group structure is extra morphism data. In coordinate functions $\delta_g \in \mathbb{F}_2^G$, multiplication is the affine-scheme morphism with pullback

$$m^{\#}(\delta_h) = \sum_{u+v=h} \delta_u \otimes \delta_v.$$

Unit and inverse are defined similarly. Thus the constant group scheme is a scheme, but not merely $\mathrm{Spec}(\mathbb{F}_2^G)$ with its extra structure forgotten.

Build the periodic square incidence complex:

$$S_0 = G, \qquad S_1 = G \times \{a, b\}, \qquad S_2 = G.$$

For $g \in G$, use edges $(g, a) : g \to g + a$ and $(g, b) : g \to g + b$. Over $\mathbb{F}_2$ signs disappear, so set

$$\partial_1(g, a) = g + (g + a), \qquad \partial_1(g, b) = g + (g + b),$$

and

$$\partial_2(g) = (g, a) + (g + a, b) + (g + b, a) + (g, b).$$

Then

$$\begin{aligned}\partial_1\partial_2(g) &= g + (g + a) + (g + a) + (g + a + b)\\ &\quad + (g + b) + (g + a + b) + g + (g + b)\\ &= 0.\end{aligned}$$

The cancellation is exactly the commuting-square identity $g + a + b = g + b + a$.

Put qubit Weyl labels on S_1. The toric selected label subgroup is

$$L_{\mathrm{tor}}(G, a, b) = (\operatorname{im} \delta^0 \oplus 0) + (0 \oplus \operatorname{im} \partial_2) \subset \mathbb{F}_2^{S_1} \oplus \mathbb{F}_2^{S_1}.$$

The selected Weyl-Heisenberg elements are $\pi^{-1}(L_{\mathrm{tor}})$. For $k = 4$, this gives $|S_0| = 16$, $|S_1| = 32$, and $|S_2| = 16$, matching the AQM-05/AQM-06 validated toric instance. The Haah/Laurent-polynomial layer is deferred to AQM-91 and is not used by this finite Cayley-incidence precursor.

Is this cheating? It is not cheating if the input is allowed to be a scheme with named group law and named generators. It is cheating if one says that the bare ring $\mathbb{F}_2^G$, or the bare scheme $\mathrm{Spec}(\mathbb{F}_2^G)$, canonically produces the toric code. The bare scheme only knows k^2 points. The toric complex uses three extra choices:

$$\text{group law on } G, \qquad \text{commuting generators } a, b, \qquad \text{Cayley-square face convention.}$$

These choices are substantial, but they are structured. They replace an arbitrary 32-qubit CSS matrix by a translation-invariant incidence complex whose CSS commutation follows from $a + b = b + a$.

90.8 Conclusion

The evidence separates the four routes: Steane is punctured Boolean three-space plus a degree-one selection rule; Reed-Muller is the Boolean cube with a degree filtration whose orthogonality is checked here, while distance and gate claims still need sources or runs; Shor has the displayed incidence target and sourced BMD family locator, but the natural surface complex is not reconstructed here; toric is the constant group scheme plus the finite Cayley-square complex, with the arithmetic status of the group/incidence input still to justify. In all four cases the final object is a selected Weyl-label subgroup L, hence $\pi^{-1}(L) \subset \mathrm{WH}(E)$.

91 Algebraic Torus Quant2 Test Case

91.1 Status and Source Anchors

This shard develops the first careful test case for the algebraic torus. Stacks supplies spectra, $D(f)$, localization, standard opens, and Kähler differentials in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 2972–3211, 33793–33872, and 34310–34340; affine morphisms and anti-equivalence in `references/algebraic_geometry/stacks_project_schemes.tex`, lines 841–963; and finite closed-point residues by the AQM-28 Nullstellensatz locator. The term $\mathrm{d}\log$ is anchored by `references/log_geometry/stacks_project_tag_0FMU.html`, lines 214–223. The Weyl step is AQM-14, AQM-28, AQM-78, and convention `(ay)`. CSS constraints are AQM-08 and AQM-89.

91.2 Lamport Dependency Skeleton

$$\begin{array}{rcl}
D1 & : & k=\mathbb{F}_q, \quad A_T=k[x,x^{-1},y,y^{-1}], \quad T=\operatorname{Spec}A_T. \\
D2 & : & T=D(xy)\subset \operatorname{Spec}k[x,y], \quad T(k)=(k^\times)^2. \\
D3 & : & E_T(U)=\bigoplus_{z\in U\cap |T|_{\mathrm{cl}}}\kappa(z)^2, \quad \rho_U:\mathrm{WH}(E_T(U))\to PU(\mathcal{C}_U). \\
L1 & : & \text{Nonzero Laurent monomials have infinite full closed support.} \\
D4 & : & \mathrm{d}\log x=x^{-1}dx, \quad \mathrm{d}\log y=y^{-1}dy. \\
L2 & : & \Omega^1_{A_T/k}=A_T\,\mathrm{d}\log x\oplus A_T\,\mathrm{d}\log y. \\
L3 & : & d(x^ay^b)=x^ay^b(\bar a\,\mathrm{d}\log x+\bar b\,\mathrm{d}\log y). \\
T1 & : & d\text{-flat monomial labels need fibre saturation on finite supports.} \\
D5 & : & \alpha,\beta\in k^\times, \quad G=\langle\alpha\rangle\times\langle\beta\rangle. \\
T2 & : & G\text{ gives a Cayley-square complex selecting edge Weyl labels.}
\end{array}$$

91.3 The Torus as a Scheme

Let $A_T=k[x,x^{-1},y,y^{-1}]$. Equivalently, $A_T=k[x,y]_{xy}$, so

$$T=\operatorname{Spec}A_T=D(xy)\subset\mathbb{A}^2_k.$$

Thus a prime of T is a prime of $k[x,y]$ not containing x or y. A k-rational point is a k-algebra map $A_T\to k$, hence is determined by nonzero images of x and y. Therefore

$$T(k)=(k^\times)^2.$$

For $a,b\in k^\times$, the corresponding maximal ideal is

$$\mathfrak{m}_{a,b}=(x-a,y-b)A_T, \qquad \kappa(\mathfrak{m}_{a,b})=k.$$

The torus group law is not a slogan. It is the affine morphism with

$$m^\#(x)=x\otimes x, \quad m^\#(y)=y\otimes y, \quad e^\#(x)=e^\#(y)=1, \quad i^\#(x)=x^{-1}, \quad i^\#(y)=y^{-1}.$$

The group axioms reduce, under affine anti-equivalence, to associativity, unit, and inverse identities for multiplication of Laurent monomials. This is the precise sense in which T is an algebraic group object used here.

91.4 Raw Closed-Point Quant2

For an open $U \subset T$, the raw arithmetic-field Quant2 label group is

$$E_T(U) = \bigoplus_{z \in U \cap |T|_{\mathrm{cl}}} \kappa(z)^2,$$

with finite support, and the arithmetic quantum field is

$$\rho_U : \mathrm{WH}(E_T(U)) \longrightarrow PU(\mathcal{C}_U).$$

This is exactly the AQM-28 closed-point construction applied to A_T. The rational support $S_{\mathrm{rat}} = T(k) = (k^\times)^2$ is a finite subtheory with $(q-1)^2$ residue sites, each of residue field k. It is not the full torus theory, because the closed points of T are infinite: for every monic irreducible $\pi(x) \neq x$, the ideal

$$(\pi(x), y-1)A_T$$

is a closed point with residue $k[x]/(\pi)$, and AQM-29 records infinitely many such π.

Lemma $L1$. Every Laurent monomial $x^a y^b$, $a, b \in \mathbb{Z}$, is nonzero at every closed point of T. Hence its full closed-point profile has infinite support.

Proof. The elements x and y are units in A_T. Their images in every residue field $\kappa(z)$ are therefore nonzero units, as are all Laurent monomials $x^a y^b$. Since the closed-point set is infinite, the support is infinite.

Thus Laurent monomials define formal full profiles. They define actual Weyl operators only after restriction to a finite closed support S, for example S_{rat} or $G \subset T(k)$.

91.5 The Derivation on Laurent Monomials

Let $d : A_T \to \Omega^1_{A_T/k}$ be the universal k-derivation. Define

$$\mathrm{d}\log x = x^{-1}dx, \qquad \mathrm{d}\log y = y^{-1}dy.$$

For every A_T-module M, a k-derivation $D : A_T \to M$ is determined freely by $D(x)$ and $D(y)$, because the inverse relation gives

$$D(x^{-1}) = -x^{-2}D(x), \qquad D(y^{-1}) = -y^{-2}D(y).$$

Conversely, arbitrary $u, v \in M$ define a derivation by the Laurent Leibniz rule with $D(x) = u$, $D(y) = v$. Hence

$$\Omega^1_{A_T/k} \cong A_T\,dx \oplus A_T\,dy = A_T\,\mathrm{d}\log x \oplus A_T\,\mathrm{d}\log y.$$

Lemma $L3$. For $a, b \in \mathbb{Z}$,

$$d(x^a y^b) = x^a y^b(\bar{a}\,\mathrm{d}\log x + \bar{b}\,\mathrm{d}\log y),$$

where $\bar{a}, \bar{b} \in k$ are the images of the integers.

Proof. The formula follows from the Leibniz rule for positive exponents. The inverse relations above give the negative-exponent case. Add the x- and y-parts.

Since $x^a y^b$ is a unit and $\mathrm{d}\log x, \mathrm{d}\log y$ are a basis,

$$d(x^a y^b) = 0 \iff a \equiv b \equiv 0 \pmod p.$$

Thus the derivative is highly nontrivial on Laurent families: it kills p-power monomials but not ordinary coordinate monomials.

91.6 Derivative Selection Is Not Automatically Label-Intrinsic

For a finite exponent set $M \subset \mathbb{Z}^2$, set

$$U_M = \langle x^a y^b : (a,b) \in M \rangle_k \subset A_T.$$

For a finite support $S \subset |T|_{\mathrm{cl}}$, the residue label map is

$$\lambda_{S,M} : U_M^2 \to E_T(S).$$

The flat derivative predicate selects

$$C_{\text{flat}}(S,M) = \lambda_{S,M}\{(q,p) \in U_M^2 : dq = dp = 0\} \subset E_T(S).$$

The selected Weyl-Heisenberg elements are exactly

$$\mathrm{Sel}_{\text{flat}}(S,M) = \pi^{-1}(C_{\text{flat}}(S,M)),$$

where $\pi : \mathrm{WH}(E_T(S)) \to E_T(S)$ is the label map.

The crucial warning is fibre saturation. Over $k = \mathbb{F}_5$, on the full rational support $S = T(k)$, the functions x and x^5 have the same residue profile because $a^5 = a$ for every $a \in \mathbb{F}_5$. But

$$d(x) = x \,\mathrm{d}\log x \neq 0, \qquad d(x^5) = 5x^5 \,\mathrm{d}\log x = 0.$$

Therefore the same Weyl label can have a non-flat representative and a flat representative if the family contains both x and x^5. By AQM-88, the image definition still gives a selected subset, but it is not a representative-free derivative predicate on labels unless the chosen family and predicate are checked to be fibre-saturated.

91.7 Cayley Squares from Torus Units

Choose units $\alpha, \beta \in k^\times$, with orders N_x, N_y, and set

$$G = \langle \alpha \rangle \times \langle \beta \rangle \subset T(k).$$

Write a point as $g = (u,v)$. Define

$$Xg = (\alpha u, v), \qquad Yg = (u, \beta v).$$

The two translations commute because multiplication in $k^\times$ commutes:

$$XYg = YXg = (\alpha u, \beta v).$$

Now form the finite incidence sets

$$S_0 = G, \qquad S_1 = G \times \{x, y\}, \qquad S_2 = G.$$

Over $\mathbb{F}_2$, define

$$\partial_1(g,x) = g + Xg, \qquad \partial_1(g,y) = g + Yg,$$

and

$$\partial_2(g) = (g,x) + (Xg,y) + (Yg,x) + (g,y).$$

Then

$$\begin{aligned}\partial_1\partial_2(g) &= g + Xg + Xg + XYg + Yg + YXg + g + Yg \\ &= XYg + YXg = 0.\end{aligned}$$

This is the torus group law showing up as a boundary identity.

Put qubit Weyl labels on the edge set S_1:

$$E_\square(G) = \mathbb{F}_2^{S_1}{}_q \oplus \mathbb{F}_2^{S_1}{}_p.$$

The CSS subgroup selected by the square complex is

$$L_\square(G) = (\operatorname{im}\partial_1^T \oplus 0) + (0 \oplus \operatorname{im}\partial_2) \subset E_\square(G).$$

By AQM-08, $\partial_1\partial_2 = 0$ is exactly the isotropy condition. The selected and rejected Weyl-Heisenberg elements are therefore

$$\mathrm{Sel}_\square(G) = \pi^{-1}(L_\square(G)), \qquad \mathrm{Rej}_\square(G) = \pi^{-1}(E_\square(G) \setminus L_\square(G)).$$

91.8 Worked Finite Examples

For $k = \mathbb{F}_3$, choose $\alpha = \beta = 2$. Both have order 2. Thus $G = T(\mathbb{F}_3)$, with

$$|S_0| = 4, \qquad |S_1| = 8, \qquad |S_2| = 4.$$

This is the 2×2 periodic square complex. It is already a valid edge Weyl-Heisenberg constraint system, though it is very small.

For $k = \mathbb{F}_5$, choose $\alpha = \beta = 2$. Since $2, 2^2 = 4, 2^3 = 3, 2^4 = 1$, both units have order 4. Hence

$$G = T(\mathbb{F}_5), \qquad |S_0| = 16, \quad |S_1| = 32, \quad |S_2| = 16.$$

The face at $g = (1,1)$ is

$$((1,1),x) + ((2,1),y) + ((1,2),x) + ((1,1),y),$$

whose boundary cancels because both paths end at $(2,2)$. This has the same size as the $k = 4$ toric instance validated in AQM-05 and AQM-06, now with provenance from torus group law plus the named unit 2.

For $k = \mathbb{F}_2$, $k^\times = \{1\}$. No nontrivial finite step is available on $T(k)$. The rational-support Cayley-square route therefore degenerates at $\mathbb{F}_2$, even though qubit edge labels may still be placed on a nontrivial torus coming from a larger field such as $\mathbb{F}_5$.

91.9 Conclusion

The algebraic torus gives a sharper test case than the constant finite group scheme. Raw Quant2 on T is the closed-point residue Weyl field of AQM-28. The rational support $T(k)$ is only a finite truncation. The derivation has canonical logarithmic coordinates and kills exactly the p-power Laurent directions in the monomial basis, but derivative constraints on finite rational Weyl labels are not automatically representative-free. Finally, after choosing two finite unit steps, the torus group law produces a square chain complex whose output is precisely a selected subgroup of the edge Weyl-Heisenberg group. This is the rigorous toric-code analogue to pursue next, not as a bare consequence of $\operatorname{Spec} A_T$, but as

$$(T,\ x, y,\ \alpha, \beta,\ \mathbb{F}_2\text{-edge labels}) \rightsquigarrow L_\square \subset E_\square \rightsquigarrow \pi^{-1}(L_\square) \subset \mathrm{WH}(E_\square).$$

92 Haah Laurent Modules and Torus Quant2

92.1 Status and Source Anchors

This shard is a comparison shard, not a new theorem about all arithmetic quantum fields. The local Haah source manifest `references/lattice_stabilizer_codes/SOURCES.md`, lines 17–29, records the free-module source and exact TeX locators. Lines 31–43 record the thesis source for the toric-code matrix, and lines 87–99 record the two-dimensional classification source. The algebraic-torus side is AQM-90 and conventions `(ay)`–`(az)`.

92.2 Lamport Dependency Skeleton

$$
\begin{aligned}
D1 &: A_T = k[x^{\pm 1}, y^{\pm 1}], \quad T = \operatorname{Spec} A_T.\\
D2 &: R_{\mathrm{tr}} = \mathbb{F}_p[X^{\pm 1}, Y^{\pm 1}] = \mathbb{F}_p[\mathbb{Z}^2].\\
D3 &: x, y \text{ are torus coordinates; } X, Y \text{ are translation operators.}\\
D4 &: G = \langle \alpha \rangle \times \langle \beta \rangle \subset T(k).\\
D5 &: R_{G,p} = \mathbb{F}_p[X^{\pm 1}, Y^{\pm 1}]/(X^{N_x} - 1, Y^{N_y} - 1).\\
D6 &: \text{For the AQM-90 comparison, } p = 2,\ R_G = R_{G,2}.\\
D7 &: P_G = R_G^4 \cong \mathbb{F}_2^{G\times\{x,y\}}{}_q \oplus \mathbb{F}_2^{G\times\{x,y\}}{}_p.\\
L1 &: \sigma_\square \text{ reproduces the AQM-90 square CSS subgroup.}\\
L2 &: \sigma_\square^\dagger \lambda_2 \sigma_\square = 0 \text{ is } \partial_1 \partial_2 = 0.\\
L3 &: d : A_T \to \Omega^1_{A_T/k} \text{ is not Haah's excitation map.}\\
T1 &: \text{Use finite Laurent matrices as Weyl-label selection principles.}
\end{aligned}
$$

92.3 The Same Laurent Ring in Two Different Roles

Haah's starting point is translation invariance. For prime-qudit labels over $\mathbb{F}_p$, a lattice with translation group $\mathbb{Z}^2$ gives the group algebra

$$R_{\mathrm{tr}} = \mathbb{F}_p[\mathbb{Z}^2] = \mathbb{F}_p[X^{\pm 1}, Y^{\pm 1}].$$

The exponents record lattice positions. A translation-invariant finite list of local Pauli or Weyl data is therefore a finite Laurent matrix. Haah records this as a map between free modules over the translation group algebra; the Pauli module is free over that ring, and commutation is encoded by the Laurent symplectic form.

AQM-90 starts with the split torus

$$A_T = k[x^{\pm 1}, y^{\pm 1}], \qquad T = \operatorname{Spec} A_T.$$

Here x, y are invertible coordinate functions, not shifts. Raw Quant2 is a closed-point residue Weyl field

$$E_T(U) = \bigoplus_{z \in U \cap |T|_{\mathrm{cl}}} \kappa(z)^2$$

with finite support. Thus the formal algebras are isomorphic as rings only after forgetting their roles. The comparison convention is deliberately notational: lower-case x, y are coordinates; upper-case X, Y are translation operators.

92.4 The Finite Periodic Bridge

Choose $\alpha, \beta \in k^\times$, with orders N_x, N_y, and set

$$G = \langle \alpha \rangle \times \langle \beta \rangle \subset T(k).$$

AQM-90 uses this G as a finite vertex set. The corresponding generic prime-qudit periodic translation quotient is

$$R_{G,p} = \mathbb{F}_p[X^{\pm 1}, Y^{\pm 1}]/(X^{N_x} - 1, Y^{N_y} - 1).$$

As an $\mathbb{F}_p$-vector space, $R_{G,p}$ has the basis $\{X^i Y^j : 0 \le i < N_x,\ 0 \le j < N_y\}$, hence one basis vector for each site of the chosen finite torus subgroup. Multiplication by X and Y is the shift by α and β.

This is the finite bridge. It is not the claim that R_{tr} is the coordinate ring of $\operatorname{Spec} A_T$. It is the finite periodic claim that the translation quotient acts on a site-indexed label module. The literal comparison with AQM-90's qubit edge module is the specialization $p = 2$, written $R_G = R_{G,2}$ below.

92.5 Recovering Haah's Toric Matrix from AQM-90

Now specialize to $p = 2$. Use qubit edge labels and order

$$P_G = R_G^4 = (q_x, q_y, p_x, p_y).$$

This is the same label space as

$$E_\square(G) = \mathbb{F}_2^{G\times\{x,y\}}{}_q \oplus \mathbb{F}_2^{G\times\{x,y\}}{}_p$$

from AQM-90. With the edge $e_x(g)$ running from g to Xg, the vertex coboundary at g touches $e_x(g)$ and $e_x(X^{-1}g)$, and also $e_y(g)$ and $e_y(Y^{-1}g)$. Therefore its Laurent column is

$$\begin{pmatrix} 1+\bar{X} \\ 1+\bar{Y} \\ 0 \\ 0 \end{pmatrix}.$$

The face boundary at g contains

$$e_x(g) + e_y(Xg) + e_x(Yg) + e_y(g),$$

so its phase-label column is

$$\begin{pmatrix} 0 \\ 0 \\ 1+Y \\ 1+X \end{pmatrix}.$$

Thus the AQM-90 square complex gives the Haah toric matrix

$$\sigma_\square = \left(\begin{array}{cc} 1+\bar{X} & 0 \\ 1+\bar{Y} & 0 \\ \hline 0 & 1+Y \\ 0 & 1+X \end{array}\right).$$

This is Haah's displayed two-dimensional toric-code matrix, modulo the orientation and row-order convention fixed in `(az)`.

92.6 Weyl-Heisenberg Selections

Let

$$\pi : \mathrm{WH}(P_G) \to P_G$$

be the label map. Haah's generating matrix selects the subgroup

$$L_{\mathrm{gen}}(G) = \operatorname{im} \sigma_\square \subset P_G,$$

so the selected generated Weyl-Heisenberg elements are

$$\mathrm{Sel}_{\mathrm{gen}}(G) = \pi^{-1}(L_{\mathrm{gen}}(G)).$$

This is exactly AQM-90's CSS subgroup

$$L_\square(G) = (\operatorname{im} \partial_1^T \oplus 0) + (0 \oplus \operatorname{im} \partial_2).$$

Haah also gives the excitation-map test. Let

$$\epsilon_\square = \sigma_\square^\dagger \lambda_2 : P_G \to R_G^2.$$

Then the labels commuting with all selected generators are

$$C_{\mathrm{comm}}(G) = \ker \epsilon_\square,$$

and the corresponding accepted and rejected Weyl-Heisenberg elements are

$$\mathrm{Sel}_{\mathrm{comm}}(G) = \pi^{-1}(C_{\mathrm{comm}}(G)), \qquad \mathrm{Rej}_{\mathrm{comm}}(G) = \pi^{-1}(P_G \setminus C_{\mathrm{comm}}(G)).$$

The matrix equation

$$\sigma_\square^\dagger \lambda_2 \sigma_\square = 0$$

is the Haah form of the AQM-90 identity $\partial_1 \partial_2 = 0$. Over the infinite translation ring R_{tr}, the corresponding toric map σ_{tr} has excitation map $\epsilon_{\mathrm{tr}} = \sigma_{\mathrm{tr}}^\dagger \lambda_2$, and the source-local toric exactness statement is

$$\ker \epsilon_{\mathrm{tr}} = \operatorname{im} \sigma_{\mathrm{tr}},$$

as recorded in the local Haah thesis locator `references/lattice_stabilizer_codes/haah_1305_6973_source/alg-theory.tex`, lines 1515–1567. On the finite periodic quotient R_G, this shard does not assert that equality without a local finite-periodic check. The object to record is instead the possible logical quotient

$$\ker \epsilon_\square / \operatorname{im} \sigma_\square.$$

For AQM purposes, the source-local exactness equation is a sharper anchor and proposal template than bare isotropy, while the finite-periodic quotient is a quantity to measure.

92.7 What Is Not Shared

The Kähler derivation in AQM-90 is

$$d : A_T \to \Omega^1_{A_T/k}, \qquad d(x^a y^b) = x^a y^b (\bar{a}\, d \log x + \bar{b}\, d \log y).$$

It is a coordinate-ring derivation. Haah's excitation map is instead

$$\epsilon = \sigma^\dagger \lambda,$$

a finite-difference symplectic map on translation labels. The factors $1 + X$, $1 + Y$, $1 + \bar{X}$, and $1 + \bar{Y}$ are boundary differences, not Kähler differentials. Therefore d should not be advertised as directly producing the toric-code boundary operator.

There is still a disciplined comparison to pursue: restrict coordinate families from A_T to finite supports G, separately build translation matrices over R_G, and ask when derivative-defined image predicates are compatible with Haah-style submodules. AQM-90 already warns that derivative predicates on finite residue profiles need fibre saturation. Haah's method adds the stronger requirement that stable constraints should preferably be finitely generated R_G-submodules.

92.8 Lessons for Arithmetic Quantum Field Dynamics

The first lesson is grammatical. A coarse dynamical principle should be a submodule or kernel inside a finite Weyl label module, not an informal family of functions. The output must be a selected or rejected subset of $\mathrm{WH}(P_G)$.

The second lesson is exactness discipline. AQM-90 verified the square boundary identity. Haah teaches us to ask for the pair

$$\operatorname{im}\sigma \subseteq \ker(\sigma^\dagger\lambda)$$

and then to measure, rather than erase, the finite-periodic quotient

$$\ker(\sigma^\dagger\lambda)/\operatorname{im}\sigma.$$

In future arithmetic-field shards this quotient is the natural place to record finite-periodic logical Weyl labels.

The third lesson is separation of roles. The algebraic torus supplies arithmetic points, residue fields, units, and logarithmic differentials. Haah supplies the Laurent-module constraint grammar for translation-invariant topological code layers. The productive bridge is

$$(T, \alpha, \beta) \rightsquigarrow G \rightsquigarrow R_G \rightsquigarrow \sigma \subset P_G \rightsquigarrow \pi^{-1}(\operatorname{im}\sigma) \text{ and } \pi^{-1}(\ker\sigma^\dagger\lambda).$$

This is the precise commonality we can use without collapsing the arithmetic torus into a bare periodic lattice.

93 Haah Data over Schemes: Permissive Module Grammar

93.1 Status and Source Anchors

Status: local definitions and proposals, with source-local Haah inputs. This shard implements convention `(ba)`: a pre-Haah datum is only an algebraic control package, a scheme Haah datum is its Spec(A)-interpretation through sheaves, and neither layer by itself gives Pauli operators, Hilbert spaces, locality, Hamiltonians, or codes (`CONVENTIONS.md`, lines 2462–2505). The affine-scheme and module-sheaf background is Stacks: Spec(R), $D(f)$, and residue fields are in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 273–278 and 2972–3023; associated module sheaves, global sections, localizations, stalks, and exactness of $M \mapsto \widetilde{M}$ are in `references/algebraic_geometry/stacks_project_schemes.tex`, lines 664–718.

The Haah source-local inputs are fixed by the lattice-stabilizer manifest `references/lattice_stabilizer_codes/SOURCES.md`, lines 17–43. In the free-module source, Haah encodes translation-invariant commuting Pauli Hamiltonians by maps between free modules over a Laurent/translation group algebra (`references/lattice_stabilizer_codes/haah_1204_1063_source/alg-code-hamiltonian-v6.tex`, lines 168–183). The Pauli module over $R = \mathbb{F}_2[\Lambda]$, the antipode, trace, commutation form, $P = R^{2q}$, and $\sigma^\dagger \lambda_q \sigma = 0$ criterion are source-local in the same file, lines 411–525 and 528–544. The generator-label map $\sigma : G \to P$ and excitation map $\epsilon = \sigma^\dagger \lambda_q$ are also in Haah's thesis (`references/lattice_stabilizer_codes/haah_1305_6973_source/alg-theory.tex`, lines 320–337). The complex condition $\epsilon \circ \sigma = 0$ and $\operatorname{im} \sigma \subseteq \ker \epsilon$ are source-local in `references/lattice_stabilizer_codes/haah_1204_1063_source/alg-code-hamiltonian-v6.tex`, lines 656–667. AQM-91 records the finite periodic bridge and warns that Haah's translation Laurent ring is not automatically the coordinate ring of an algebraic torus (AQM-91, lines 41–91 and 165–200).

93.2 The Source-Local Grammar to Generalize

Haah's source-local data have five algebraic pieces before any scheme generalization is attempted. First, there is a coefficient ring, in the source case a translation group algebra. Second, the Pauli label module P is free over that ring. Third, a generator-label module G maps to P by $\sigma : G \to P$. Fourth, a Laurent symplectic form λ_q, together with the dagger induced by transpose and the antipode, gives the excitation map $\epsilon = \sigma^\dagger \lambda_q$. Fifth, isotropy or commutation is the equation $\sigma^\dagger \lambda_q \sigma = 0$. The preceding paragraph cites the local source lines for each item.

Local proposal. The general lesson kept here is not "every Spec(A) is a Pauli lattice". The lesson is the module grammar

$$G \xrightarrow{\sigma} P, \qquad P \xrightarrow{\epsilon} G^\vee, \qquad \epsilon\sigma = 0,$$

plus the habit of recording the quotient $\ker \epsilon / \operatorname{im} \sigma$ when it is defined. All translation, finite support, Pauli, Hilbert, and Hamiltonian meanings are deliberately stripped away unless a later layer adds the extra data named in convention `(ba)`.

93.3 Pre-Haah Data over a Ring

Definition D1: permissive pre-Haah datum. Let A be a commutative ring. A *pre-Haah datum over A* is a tuple

$$\mathcal{D} = (G, P, \sigma, \lambda_{\text{opt}})$$

where G and P are specified A-modules, $\sigma : G \to P$ is an A-linear map, and λ_{opt} is either absent or is a specified A-linear map

$$\lambda : P \to P^\vee, \qquad P^\vee = \operatorname{Hom}_A(P, A).$$

When λ is absent, the datum records only the generated submodule $\operatorname{im}\sigma \subseteq P$. It has no excitation map and no defect quotient.

Definition D2: formed and isotropic datum. If λ is present, define the formal excitation map

$$\epsilon = \sigma^\vee \circ \lambda : P \longrightarrow G^\vee.$$

This is a local definition in the category of A-modules; it is the identity-involution analogue of Haah's source-local $\sigma^\dagger \lambda_q$. The formed datum is *isotropic* if

$$\epsilon\sigma = \sigma^\vee \lambda \sigma = 0.$$

Only in the formed isotropic case do we define the formal logical-defect module

$$\mathcal{L}_{\mathrm{alg}}(\mathcal{D}) = \ker(\epsilon)/\operatorname{im}(\sigma).$$

This quotient is an algebraic module diagnostic. It is not a physical logical operator group unless a finite Weyl-Haah or translation Haah datum supplies the missing representation and locality data, as required by `CONVENTIONS.md`, lines 2493–2505.

Definition D3: perfection and finite local bases. The word *symplectic* is reserved here for a chosen alternating $\lambda : P \to P^\vee$ that is an isomorphism. Matrix adjoints, local ranks, residue-fibre nondegeneracy, and base-changed duals require finite locally free or finite projective hypotheses. Stacks defines finite locally free modules in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 19004–19017, and records the equivalence with finite projective conditions in lines 19027–19053. The dual of a finite locally free sheaf is finite locally free, and tensoring with one is exact (`references/algebraic_geometry/stacks_project_cohomology.tex`, lines 13836–13845). The Hom-tensor comparison used when duals are moved through base change is guaranteed for finite projective modules in `references/algebraic_geometry/stacks_project_algebra.tex`, lines 19351–19357. Without these hypotheses, ϵ may still be a formal module map, but fibrewise matrix claims are not made here.

93.4 Scheme Haah Data over $\operatorname{Spec} A$

Definition D4: scheme interpretation. Let $X = \operatorname{Spec}(A)$. A *scheme Haah datum* is a pre-Haah datum over A viewed through the quasi-coherent sheaves

$$\widetilde{G}, \qquad \widetilde{P}, \qquad \widetilde{\sigma} : \widetilde{G} \to \widetilde{P},$$

and, when present, $\widetilde{\lambda} : \widetilde{P} \to \widetilde{P^\vee}$. On a standard open $D(f)$, the restricted modules are G_f and P_f; at $x = \mathfrak{p}$, the stalks are $G_\mathfrak{p}$ and $P_\mathfrak{p}$ by the Stacks affine-module-sheaf result cited above. The residue fibre used in this shard is the local definition

$$P(x) = P_\mathfrak{p} \otimes_{A_\mathfrak{p}} \kappa(\mathfrak{p}), \qquad G(x) = G_\mathfrak{p} \otimes_{A_\mathfrak{p}} \kappa(\mathfrak{p}).$$

The permissive scheme layer allows quasi-coherent modules when the intended operation is restriction, localization, stalks, support language, or residue fibres. It asks for coherent modules when a finite-equation or finite-support scheme-theoretic statement is intended: on locally Noetherian schemes, coherent is equivalent to quasi-coherent finite type/finitely presented in the sense recorded in `references/algebraic_geometry/stacks_project_coherent.tex`, lines 2224–2242, and quasi-coherent submodules and quotients of coherent modules remain coherent by lines 2328–2344. It asks for finite locally free modules when the operation needs local bases, ranks, perfect forms, duals compatible with fibres, or a matrix $\sigma^\vee \lambda \sigma$ whose entries have stable local meaning.

93.5 Restrictive Ladder

Layer	Added data	Allowed claim
Pre-Haah	A-modules G, P, map σ	$\operatorname{im}\sigma \subseteq P$ is an algebraic submodule.
Formed pre-Haah	$\lambda : P \to P^\vee$	$\epsilon = \sigma^\vee\lambda$, isotropy, and $\ker\epsilon/\operatorname{im}\sigma$ if isotropic.
Scheme Haah	sheaves on $X = \operatorname{Spec} A$	restriction, localization, stalks, residue fibres, support, and base-change questions.
Finite locally free	finite projective/local-basis hypotheses	matrix adjoints, perfect forms, fibrewise ranks, and dual-compatible comparisons.
Finite Weyl or translation	generating characters, finite self-duality, group algebra, antipode, trace, finite support	Weyl labels or Haah translation-stabilizer interpretations; these are outside this shard.

93.6 Examples at the Scheme-Module Layer

Field k. Local derivation. A field has exactly one prime ideal, (0), so Spec(k) has one point. For $k = \mathbb{F}_p$, AQM-22 gives this calculation and separates Spec($\mathbb{F}_p$) from the p-element underlying set (AQM-22, lines 48–69). A finite locally free pre-Haah datum is just finite-dimensional k-vector spaces G and P, a matrix σ, and optionally an alternating matrix λ. Then $\epsilon = \sigma^T\lambda$ and $\mathcal{L}_{\mathrm{alg}} = \ker\epsilon/\operatorname{im}\sigma$ are vector-space objects if $\sigma^T\lambda\sigma = 0$. The finite-field Weyl qudit statement belongs to AQM-22/AQM-81, not to this bare module example.

Product field ring k^n. AQM-40 proves that $R = k^n$ has points $\mathfrak{m}_i = \ker(\mathrm{pr}_i)$, residues k, and singleton opens $D(e_i)$ (AQM-40, lines 51–97); convention `(ad)` records that Spec(k^n) has n, not q^n, points for $k = \mathbb{F}_q$ (`CONVENTIONS.md`, lines 1012–1035). **Local derivation:** since the idempotents e_i are orthogonal and sum to 1, every R-module M decomposes as $\bigoplus_i e_iM$. Thus a scheme Haah datum over k^n is componentwise n field-level data after applying the idempotents. AQM-40, lines 232–277, warns that stabilizer codes are later choices of isotropic labels and phases; the product ring itself supplies no incidence complex or lattice locality.

Dual numbers $k[\varepsilon]/(\varepsilon^2)$. Stacks defines dual numbers as the k-module with basis $1, \varepsilon$ and $\varepsilon^2 = 0$ (`references/algebraic_geometry/stacks_project_varieties.tex`, lines 2918–2924). **Local derivation:** every prime contains the nilpotent ε, so the topological spectrum is the one point of the quotient k, while the structure sheaf still remembers the nilpotent direction. For a free module datum, write

$$\sigma = \sigma_0 + \varepsilon\sigma_1, \qquad \lambda = \lambda_0 + \varepsilon\lambda_1.$$

The isotropy equation expands to the zeroth-order condition $\sigma_0^T\lambda_0\sigma_0 = 0$ and the first-order condition

$$\sigma_1^T\lambda_0\sigma_0 + \sigma_0^T\lambda_1\sigma_0 + \sigma_0^T\lambda_0\sigma_1 = 0.$$

This is an algebraic deformation calculation only. For $\mathbb{F}_3[\varepsilon]/(\varepsilon^2)$, AQM-82 records that the reduced residue-site workflow has one $\mathbb{F}_3$ site, while the thickened Artin-Weyl layer needs the separate AQM-36 top-coefficient generating character (AQM-82, lines 128–156).

$\mathbb{Z}/6\mathbb{Z}$. AQM-23 proves the Chinese-remainder isomorphism $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{F}_2 \times \mathbb{F}_3$, computes the two prime ideals (2) and (3), and gives residues $\mathbb{F}_2$ and $\mathbb{F}_3$ (AQM-23, lines 57–100). Therefore a module-level Haah datum over $\mathbb{Z}/6\mathbb{Z}$ splits into its two residue/product components after the Chinese-remainder idempotents are chosen. The formed equations $\epsilon\sigma = 0$ and the algebraic defect module are componentwise equations over the two fields. AQM-23, lines 102–225, supplies a separate finite Weyl residue construction for this ring; this shard uses none of that as a consequence of the bare pre-Haah datum.

$\mathbb{F}_3[t]/(t^n - 1)$. AQM-35 writes $n = 3^s m$, $3 \nmid m$, proves

$$t^n - 1 = (t^m - 1)^{3^s},$$

and records that the closed residue Weyl sites are the irreducible factors of the squarefree part $t^m - 1$ (AQM-35, lines 27–54). At the present layer, take $A = R_n = \mathbb{F}_3[t]/(t^n - 1)$ and choose modules, σ, and possibly λ over the full nonreduced ring. The residue fibres of the scheme Haah datum see the quotients $K_\pi = \mathbb{F}_3[t]/(\pi)$, while σ, λ, $\ker \epsilon$, and $\mathcal{L}_{\mathrm{alg}}$ may still contain nilpotent information over R_n. AQM-35, lines 241–246, states that nilpotents are invisible to the reduced residue-qudit layer, and AQM-37, lines 34–65 and 171–174, treats the separate thickened Weyl layer. No periodic Haah lattice is obtained here unless a translation quotient, antipode, trace, and finite-support bridge like AQM-91 is added.

93.7 What This Layer Does Not Yet Give

- It does not turn a bare $\mathrm{Spec}(A)$ into sites of a lattice, a Pauli module, a Weyl representation, a Hilbert space, a Hamiltonian, locality, or a quantum code.
- It does not supply Haah's dagger unless the needed duality, finite locally free hypotheses, and, for translation data, antipode/trace conventions are named.
- It does not assert exactness $\ker \epsilon = \mathrm{im}\, \sigma$; outside source-local Haah examples or checked finite computations, $\ker \epsilon / \mathrm{im}\, \sigma$ is only a module to study.
- It does not quantize arbitrary finite rings. Finite Weyl-Haah data need finite self-duality, residue-field labels, or a certified Frobenius-ring generating character, exactly as convention `(ba)` requires.
- It does not identify coordinate rings with translation group algebras. That bridge is extra data, and AQM-91 is the local model for making it finite and explicit.

94 Finite-Ring Haah Layers: Residues, Thickenings, and Blockers

94.1 Status and Source Anchors

Status: local synthesis and definitions. This shard adds no source manifest, convention entry, script, run bundle, generated data, populated finite-ring database row, or all-finite-ring representation theorem. It uses the layer vocabulary fixed in convention `(ba)` and the finite-ring QSA boundaries in AQM-80–AQM-83. The Haah-side anchors are AQM-91, convention `(az)`, convention `(ba)`, and `references/lattice_stabilizer_codes/SOURCES.md`, lines 17–43 and 87–99, which register the Laurent-module Pauli, stabilizer, excitation-map, toric-code matrix, and two-dimensional classification sources. The finite-Weyl anchors are AQM-71, AQM-78, AQM-31, AQM-36, AQM-37, and `references/heisenberg_weil/SOURCES.md`, especially the finite Frobenius-ring and generating-character locators for Gluesing-Luerssen–Pllaha and Sidana–Kashyap.

The scheme-theoretic anchor is `references/algebraic_geometry/SOURCES.md`; its `references/algebraic_geometry/stacks_project_algebra.tex` entry records prime/maximal ideals, residue fields, spectra, Artinian finite-spectrum facts, and finite-discrete spectrum equivalences. The only all-finite-ring step used below is local: a finite ring has finitely many ideals, so descending ideal chains stabilize; with the cited Artinian-spectrum source, its prime spectrum is finite and all primes are maximal. If $x \in \operatorname{Spec} R$, then R/x is a finite domain, hence a field because multiplication by a nonzero element is injective and therefore surjective. This derives that $\kappa(x)$ is finite, but it does not compute ideals, order sites, or build a database row.

94.2 Layer 0: Bare Finite Spectrum

Let R be a finite commutative ring and $X = \operatorname{Spec} R$.

Local definition. The bare $\operatorname{Spec} R$ layer is the finite topological/scheme support X, with residue maps $R \to \kappa(x)$ when those points and fields have been locally derived or supplied. It may discuss supports, opens, specializations, residue fibres, and nilpotent thickenings in the scheme sense, following the Stacks anchors and AQM-81. It has no Pauli operators, Hilbert space, stabilizer Hamiltonian, translation action, or Haah generator map. Therefore it is not a Haah code.

This permissive layer is already enough to prevent a common mistake. A nilpotent class in R can be part of the local scheme structure while having zero image in every residue field. AQM-34, AQM-35, AQM-36, AQM-37, AQM-81, and AQM-82 record this distinction for polynomial quotients over $\mathbb{F}_3$: repeated factors change the thickening, not the number of reduced residue sites.

94.3 Layer 1: Pre-Haah Module Data

Local definition, from convention `(ba)`. A pre-Haah datum over R is an algebraic package (G, P, λ, σ), where G and P are specified R-modules, λ is a specified alternating or symplectic form when chosen, and $\sigma : G \to P$ is a specified generator map. If an adjoint convention is also part of the package, one may write $\epsilon = \sigma^\dagger \lambda$. The quotient $\ker(\epsilon)/\operatorname{im}(\sigma)$ is only a logical-defect module, not automatically a physical logical-Pauli group.

Proposal. For finite rings, this is the broadest Haah-like grammar: it keeps the lessons of Haah's matrix formalism, namely generator maps, commutation tests, and defect quotients, while refusing to add Weyl operators until a finite nondegenerate commutator pairing and phase character have been named. The module package may be interpreted sheaf-theoretically on $\operatorname{Spec} R$, but this still does not identify Zariski points with lattice translation sites. That separation is exactly the warning in AQM-91 and convention `(az)`: coordinate rings and translation group algebras can look like Laurent rings while playing different roles.

94.4 Layer 2: Residue Weyl-Haah Data

Local definition. Let $S \subset X$ be a selected finite set of locally certified points. Fix an integer $q \geq 1$. The residue Weyl-Haah phase module is

$$P_{\mathrm{res}}(S,q) = \bigoplus_{x \in S} \left(\kappa(x)^q \oplus (\kappa(x)^q)^\vee\right).$$

For $e_x = (a_x, \alpha_x)$ and $f_x = (b_x, \beta_x)$, use $\omega_x(e_x, f_x) = \alpha_x(b_x) - \beta_x(a_x)$ and assemble the orthogonal direct sum over S. The Weyl carrier is

$$\mathcal{H}_{\mathrm{res}}(S,q) = \bigotimes_{x \in S} \ell^2(\kappa(x)^q).$$

This is the finite-field Weyl step of AQM-78 applied sitewise to the residue-site workflow of AQM-81; it generalizes the finite-set symplectic label grammar reviewed in AQM-71.

This layer is still not a Haah code. It becomes a finite residue Weyl-Haah datum only after a selected generator map

$$\sigma_{\mathrm{res}} : G_{\mathrm{res}} \longrightarrow P_{\mathrm{res}}(S,q)$$

has been supplied and checked to be isotropic with respect to the displayed pairing, with phases handled by the local Weyl convention. AQM-22, AQM-40, and AQM-78 all make the same distinction: the ambient Weyl system is not yet a stabilizer code.

94.5 Layer 3: Thickened Frobenius Weyl-Haah Data

Local definition with blocker. A thickened finite-ring Weyl-Haah datum replaces the residue vector spaces by finite ring modules only when the required self-duality mechanism is already locally sourced: a finite Frobenius-ring generating character as in AQM-31 and `references/heisenberg_weil/SOURCES.md`, or the explicit top-coefficient chain-ring character proved in AQM-36 and used in AQM-37. For a certified finite ring or local factor A, the available model has $P_{\mathrm{thick}}(A,q) = A^q \oplus A^q$ and $\mathcal{H}_{\mathrm{thick}}(A,q) = \ell^2(A^q)$, with translations and phases defined through the chosen generating character.

For the AQM-36 chain rings

$$A = \mathbb{F}_3[t]/(\pi^e),$$

the character is the top-coefficient character

$$\chi(a) = \exp\bigl(2\pi i \ \mathrm{Tr}_{K/\mathbb{F}_3}(a_{e-1})/3\bigr), \qquad K = \mathbb{F}_3[t]/(\pi),$$

and AQM-36 proves it is generating, proves the Weyl product and commutator laws, and proves the operator-basis statement. AQM-37 applies this to $\mathbb{F}_3[t]/(t^n - 1)$.

The block is part of the definition. Arbitrary finite commutative rings are not promoted to this layer in this report. AQM-78 says that a general finite-ring Weyl step needs the pairing and character data first; AQM-82 and convention `(an)` record that the current finite-ring helpers emit `blocked` rows when a certified generating character or recognized policy is absent. In particular, this shard does not claim that every finite commutative ring is Frobenius and does not claim a nondegenerate $R^q \oplus R^q$ Weyl label system for arbitrary R.

94.6 Layer 4: Translation Haah Data

Local definition, from convention `(ba)`**.** A translation Haah datum adds a group algebra, an involution or antipode, and the identity-coefficient trace convention used by Haah's Laurent-polynomial formalism. This is the layer of finite Laurent matrices, local translation constraints, and excitation maps $\epsilon = \sigma^\dagger\lambda$. AQM-91 constructs the current finite periodic bridge: choose finite torus steps, form a quotient of the translation group algebra, put Weyl labels on the chosen edge module, and compare the AQM-90 square complex with Haah's toric generator matrix. The translation basis is group data. It is not the same thing as the Zariski point set of $\operatorname{Spec} R$ unless a separate finite support bridge identifies them.

94.7 Permissive-to-Restrictive Ledger

Layer	Added input	Output allowed	Blocker preserved
Bare $\operatorname{Spec} R$	Finite commutative ring and locally supplied spectrum/residue data.	Supports, opens, fibres, nilpotent thickenings.	No Pauli labels, no Hilbert space, no code.
Pre-Haah module	Specified R-modules G, P, optional λ, and $\sigma : G \to P$.	Formal $\epsilon = \sigma^\dagger\lambda$ and defect module.	No physical logical group without Weyl data.
Residue Weyl-Haah	Selected finite sites S, residue fields $\kappa(x)$, and finite-field duals.	$P_{\mathrm{res}}(S, q)$, $\mathcal{H}_{\mathrm{res}}(S, q)$, and possible isotropic residue generators.	Nilpotents are invisible except through their effect on residue sites.
Thickened Frobenius Weyl	Certified generating character or sourced Frobenius-ring mechanism.	$A^q \oplus A^q$ Weyl labels and $\ell^2(A^q)$ for certified A.	Arbitrary finite rings remain `blocked`.
Translation Haah	Group algebra, antipode/involution, identity-coefficient trace, local translation generators.	Laurent matrices, finite-difference excitation maps, periodic logical quotients.	Not supplied by $\operatorname{Spec} R$ alone.

94.8 Example Ledger

Ring or role	Residue layer	Thickened/Frobenius layer	Haah-style lesson
$\mathbb{F}_q$	One spectrum point and one residue field site; AQM-22, AQM-78, and AQM-81 give the finite-field Weyl carrier $\ell^2(\mathbb{F}_q^q)$ after q is chosen.	No nilpotent thickening in this row. Finite-field Pauli data are sourced, but stabilizers are extra choices.	Bare field geometry gives a site, not a generator matrix.
k^n with $k = \mathbb{F}_q$	AQM-40 and AQM-83 give n discrete residue sites with carrier $\ell^2(k)^{\otimes n}$ for $q = 1$.	No thickened layer is imported for the reduced product-field row.	Tensor factors come from product-spectrum residues; Haah translation still needs group action and incidence data.
$\mathbb{Z}/6\mathbb{Z}$ or mixed residues	AQM-23 and AQM-81 give residues $\mathbb{F}_2$ and $\mathbb{F}_3$, hence $\mathbb{F}_2^2 \oplus \mathbb{F}_3^2$ for $q = 1$.	AQM-80/AQM-82 keep the thickened-Frobenius row blocked without a certified generating character policy for this mixed row.	Mixed residue labels are a direct sum of finite abelian phase groups, not one common-field Laurent module.
$\mathbb{F}_3[\epsilon]/(\epsilon^2)$	AQM-81 records one reduced $\mathbb{F}_3$ residue site.	AQM-36/AQM-82 certify the top-coefficient character and the carrier $\ell^2(\mathbb{F}_3[\epsilon]/(\epsilon^2))$.	The nilpotent is an internal jet coordinate at the point, not an extra translation site.
$\mathbb{F}_3[t]/(t^n - 1)$	AQM-35 gives the reduced sites from the squarefree part $t^m - 1$, with reduced dimension 3^m.	AQM-37 gives the certified thickened dimension 3^n for the local chain-ring factors.	As a group-algebra quotient it may be used as translation data only after that role, antipode, trace, and generators are named.
Arbitrary finite ring R with no certified character	Bare $\operatorname{Spec} R$ and selected residue sites may be discussed once locally derived.	`blocked`: no nondegenerate $R^q \oplus R^q$ Weyl layer is asserted.	The pre-Haah module grammar is allowed, but physical Weyl or code claims wait for pairing, character, and generator evidence.

94.9 Conclusion

Finite rings teach a separation principle. $\operatorname{Spec} R$ supplies supports, residue fibres, opens, and nilpotent thickenings. Haah-like modules require additional algebraic control data: a generator map, a commutator form, an adjoint or excitation-map convention, and usually an isotropy check. Weyl operators require still more: finite-field residue duals, or a certified Frobenius/generating-character mechanism for thickened ring labels. Translation-invariant Haah codes require a further group-algebra role, antipode or involution, identity-coefficient trace, and local incidence or translation-generator data.

Thus the safe finite-ring bridge is layered rather than automatic:

$$\operatorname{Spec} R \rightsquigarrow \text{pre-Haah module data} \rightsquigarrow \text{residue or certified thickened Weyl data} \rightsquigarrow \text{translation Haah data only after a}$$

Any missing arrow is a blocker, not a convention silently filled in by the word “finite”.

95 Translation Haah Data versus Bare Spectrum

95.1 Status and Source Anchors

This shard is a local synthesis, not a new classification theorem. Its Haah anchors are `references/lattice_stabilizer_codes/SOURCES.md` and the local TeX locators `references/lattice_stabilizer_codes/haah_1204_1063_source/alg-code-hamiltonian-v6.tex` lines 169–183, 416–432, 514–544, and 634–667; `references/lattice_stabilizer_codes/haah_1305_6973_source/alg-theory.tex` lines 700–716 and 1515–1567; and `references/lattice_stabilizer_codes/haah_1812_11193_source/cph2d-arXiv3.tex` lines 321–365, 409–453, and 505–511. The torus comparison is AQM-90, AQM-91, and `CONVENTIONS.md (az)–(ba)`. The residue-spectrum and product-field warnings are AQM-40, AQM-81, AQM-36, AQM-37, and conventions `(ad)`, `(aa)`, and `(ab)`.

95.2 Lamport Dependency Skeleton

$$D1 : (G, P, \lambda, \sigma) \text{ is the permissive pre-Haah datum over } A.$$
$$D2 : \operatorname{Spec} A \text{ adds sheaf/support/fibre geometry but not sites.}$$
$$D3 : \text{finite Weyl data adds a self-dual label convention.}$$
$$D4 : R = \mathbb{F}_p[\Lambda],\ \bar{(\cdot)},\ \mathrm{tr},\ \dagger \text{ are the translation layer.}$$
$$D5 : P = R^{2q}, \quad \lambda_q = \begin{pmatrix} 0 & I_q \\ -I_q & 0 \end{pmatrix}, \quad \epsilon = \sigma^\dagger \lambda_q.$$
$$L1 : \sigma^\dagger \lambda_q \sigma = 0, \qquad L2 : \ker \epsilon = \operatorname{im} \sigma \text{ only when sourced.}$$

95.3 From Permissive to Restrictive

Following `CONVENTIONS.md (ba)`, the permissive layer is a *pre-Haah datum* (G, P, λ, σ) over a commutative ring A: modules, a chosen alternating or symplectic form when available, and a generator map. Here $\epsilon = \sigma^\dagger \lambda$ and $\ker \epsilon / \operatorname{im} \sigma$ are only local definitions of a logical-defect module, not a physical logical Pauli group, Hilbert space, or Hamiltonian.

A *scheme Haah datum* interprets the same package on $X = \operatorname{Spec} A$ by sheaves, supports, localization, residue fibres, and nilpotent thickenings. By convention `(ba)`, it still does not identify points of $\operatorname{Spec} A$ with translation sites. A *finite Weyl-Haah datum* adds finite self-duality for Weyl-Heisenberg labels; AQM-81 gives the residue-field case, while finite-ring versions require the Frobenius-ring or generating-character sources in `references/heisenberg_weil/SOURCES.md`.

The most restrictive layer is the *translation Haah datum*: choose an abelian translation group Λ and

$$R = \mathbb{F}_p[\Lambda],$$

for example $R = \mathbb{F}_p[X^{\pm 1}, Y^{\pm 1}]$ when $\Lambda = \mathbb{Z}^2$. Haah's source-local package uses Laurent coefficients for finite-support Pauli or qudit labels and exponents for translation positions (`references/lattice_stabilizer_codes/haah_1204_1063_source/alg-code-hamiltonian-v6.tex`, lines 169–183).

95.4 The Exact Translation Package

Let Λ be abelian. Haah's qubit source takes

$$P = R^{2q}, \qquad R = \mathbb{F}_2[\Lambda],$$

as the Pauli module for finite-support Paulis on $\Lambda\times\{1,\dots,q\}$, modulo phase factors; the prime-qudit two-dimensional source uses $R=\mathbb{F}_p[x^{\pm},y^{\pm}]$ (`references/lattice_stabilizer_codes/haah_1204_1063_source/alg-code-hamiltonian-v6.tex`, lines 416–420 and 532–544; `references/lattice_stabilizer_codes/haah_1812_11193_source/cph2d-arXiv3.tex`, lines 321–365).

The antipode is $\overline{g}=g^{-1}$, or $x\mapsto x^{-1}$, $y\mapsto y^{-1}$; $v^\dagger$ is transpose followed by this involution. The trace is the identity-coefficient functional

$$\operatorname{tr}\left(\sum_{g\in\Lambda} a_g g\right)=a_1,$$

not a field trace. These are the sourced conventions in `references/lattice_stabilizer_codes/haah_1204_1063_source/alg-code-hamiltonian-v6.tex`, lines 422–434 and 514–525, with the two-variable coefficient-of-1 bracket in `references/lattice_stabilizer_codes/haah_1812_11193_source/cph2d-arXiv3.tex`, lines 354–365 and 505–511.

A generator matrix $\sigma: R^t\to P$ selects a stabilizer submodule. With

$$\epsilon=\sigma^\dagger\lambda_q: P\to R^t,$$

generator commutation is $\sigma^\dagger\lambda_q\sigma=0$, giving the complex $R^t\xrightarrow{\sigma}P\xrightarrow{\epsilon}R^t$ and $\operatorname{im}\sigma\subseteq\ker\epsilon$ (`references/lattice_stabilizer_codes/haah_1204_1063_source/alg-code-hamiltonian-v6.tex`, lines 634–667). When topological order is sourced, the stronger exactness is $\ker\epsilon=\operatorname{im}\sigma$ (`references/lattice_stabilizer_codes/haah_1305_6973_source/alg-theory.tex`, lines 700–716; `references/lattice_stabilizer_codes/haah_1812_11193_source/cph2d-arXiv3.tex`, lines 409–453).

95.5 Why This Is Not Just $\operatorname{Spec} R$

The central separation is role, not notation. In AQM-90 the ring

$$A_T=k[x^{\pm1},y^{\pm1}]$$

is a coordinate ring for the split torus $T=\operatorname{Spec}A_T$. In AQM-91 and `CONVENTIONS.md (az)`, the ring

$$R_{\mathrm{tr}}=\mathbb{F}_p[X^{\pm1},Y^{\pm1}]$$

is a translation group algebra acting on finite-support labels. The same abstract Laurent algebra can be written in both places, but the basis $\{X^aY^b\}_{(a,b)\in\mathbb{Z}^2}$ is group-basis data, not the set of Zariski points of $\operatorname{Spec}R_{\mathrm{tr}}$.

AQM-90 proves a concrete warning: as a scheme, the torus has infinitely many closed points, while Laurent monomials have infinite full closed support and only become Weyl operators after restriction to a finite support. Haah's finite-support coefficients are instead indexed by translation exponents. Therefore no shard should read a point of $\operatorname{Spec}\mathbb{F}_p[\Lambda]$ as a Haah lattice site unless it has first named a finite-support bridge such as the AQM-91 periodic bridge.

95.6 Examples

1. The infinite Laurent translation algebra. For $R_{\mathrm{tr}}=\mathbb{F}_p[\mathbb{Z}^2]$, the translation basis is

$$\{X^aY^b:(a,b)\in\mathbb{Z}^2\}.$$

This is the coefficient basis for Haah finite-support labels. If the same ring is viewed as a coordinate ring, then $\operatorname{Spec}R_{\mathrm{tr}}$ is the split torus studied in AQM-90. The exponent pair (a,b) is not a point of this spectrum.

2. A finite periodic quotient. AQM-91 names the finite bridge. Choose finite orders N_x, N_y and set

$$R_{G,p} = \mathbb{F}_p[X^{\pm 1}, Y^{\pm 1}]/(X^{N_x} - 1, Y^{N_y} - 1) = \mathbb{F}_p[C_{N_x} \times C_{N_y}].$$

As an $\mathbb{F}_p$-vector space it has basis

$$\{X^i Y^j : 0 \le i < N_x,\ 0 \le j < N_y\}.$$

This is the finite periodic translation configuration module in AQM-91, with X and Y acting as shifts. It is not automatically the coordinate ring of the finite set of sites.

3. The AQM-91 qubit toric square. For the qubit toric comparison, AQM-91 specializes to $p = 2$ and writes

$$P_G = R_G^4 = (q_x, q_y, p_x, p_y).$$

With the AQM-90 edge orientation, the source-local Haah matrix is

$$\sigma_\square = \begin{pmatrix} 1 + X^{-1} & 0 \\ 1 + Y^{-1} & 0 \\ 0 & 1 + Y \\ 0 & 1 + X \end{pmatrix}.$$

Then $\epsilon_\square = \sigma_\square^\dagger \lambda_2$ and $\epsilon_\square \sigma_\square = 0$ give the Haah form of the AQM-90 square boundary identity. AQM-91 cites the infinite-ring toric exactness $\ker \epsilon_{\mathrm{tr}} = \operatorname{im} \sigma_{\mathrm{tr}}$ to `references/lattice_stabilizer_codes/haah_1305_6973_source/alg-theory.tex`, lines 1515–1567, and it deliberately does not assert the same finite-periodic equality without a local finite-periodic check.

4. The modular thickening $\mathbb{F}_3[C_3]$. This is a local derivation, with the nilpotent-sensitive comparison anchored by AQM-37. Let g generate C_3. In characteristic 3,

$$\mathbb{F}_3[C_3] \cong \mathbb{F}_3[g]/(g^3 - 1) = \mathbb{F}_3[g]/((g-1)^3).$$

With $u = g - 1$, this is $\mathbb{F}_3[u]/(u^3)$. AQM-37 records the same ring as $\mathbb{F}_3[t]/(t^3 - 1) \cong \mathbb{F}_3[u]/(u^3)$, with one reduced closed point and a thickened Hilbert layer $\ell^2(\mathbb{F}_3[u]/(u^3))$. The translation basis is still $\{1, g, g^2\}$. Thus the spectrum is a single third-order thickened point, while the translation module has three group-basis elements.

5. A split product warning. This is a local derivation using the product-field spectrum convention of AQM-40 and `CONVENTIONS.md (ad)`. Over $\mathbb{F}_3$,

$$\mathbb{F}_3[C_2] \cong \mathbb{F}_3[t]/(t^2 - 1) \cong \mathbb{F}_3[t]/(t-1) \times \mathbb{F}_3[t]/(t+1) \cong \mathbb{F}_3 \times \mathbb{F}_3.$$

Therefore its spectrum has two reduced product-field points. The translation basis is $\{1, t\}$, while the product idempotents

$$e_+ = 2(1+t), \qquad e_- = 2(1-t)$$

are the split character or Fourier basis in this displayed example. Passing to these idempotents is extra decomposition data, not the same as remembering the group-basis shifts used by a Haah finite-difference matrix.

95.7 Checklist for Reaching the Translation Haah Layer

A scheme proposal has reached the translation Haah layer only after naming or sourcing: an abelian translation group Λ or finite quotient; the group algebra $R = \mathbb{F}_p[\Lambda]$ or periodic quotient; the group basis and finite-support convention; the antipode, dagger, and identity-coefficient trace; $P = R^{2q}$ and λ_q; a finite generator matrix σ; the excitation map $\epsilon = \sigma^\dagger \lambda_q$ and checked isotropy $\sigma^\dagger \lambda_q \sigma = 0$; a sourced exactness claim or the recorded quotient $\ker \epsilon / \operatorname{im} \sigma$; a finite periodic or residue/Weyl bridge for finite Hilbert spaces; and an explicit warning for any Fourier, character, residue, or Spec-point decomposition added to the translation basis.

95.8 Right Lesson

Bare $\operatorname{Spec} A$ gives geometric supports, residue fields, localizations, and nilpotent thickenings in the conventions used by AQM-28, AQM-36, and AQM-81. It does not by itself give Haah lattice sites. Haah's lesson for this programme is more restrictive and more useful: once a chosen commutative ring is a translation group algebra or a controlled quotient of one, modules over that ring can carry symplectic duality, an antipode, an identity-coefficient trace, and finite-difference stabilizer maps. The physical code layer is the extra structured package

$$(R, \bar{(\cdot)}, \mathrm{tr}, P, \lambda_q, \sigma, \epsilon)$$

together with finite-support or finite-periodic interpretation. A scheme proposal may borrow this grammar, but it has not become Haah data until those translation choices are explicit and sourced.